Sūrya Siddhānta: Scientific Commentary

English translation of the Hindi prose in the supplied OCR edition. Sanskrit verses are preserved in their original script. Images are omitted. Mathematical labels are translated into English where legible.

> **Editorial note.** This is a machine-assisted translation of OCR text. Unresolved terms are romanized. Dates, figures, and formulas should be checked against the source scan before scholarly use.

SOURCE PAGE 1

Sūrya Siddhānta: Scientific Commentary, Parts I and II

Commentator: The late Mahāvīra Prasāda Śrīvāstava

Dr. Ratnakumari Swadhyaya Institute, Science Council Building, Allahabad 2

SOURCE PAGE 2

Introduction

Swargiya Mahabir Prasad Srivastava was born on Kartik Shukla 2, 1644, according to that, on October 18, 1887, in the village Bijhauli, tehsil Handia, district Allahabad. He passed the High School (1606) and Inter Examinations (1608) from the Kayastha school of Allahabad and the B.S.-C. examination (1610) from Moore College. Due to not being in good financial condition, he took a job. From Kartik month of year 1679, he started writing the 'Vigyanbhashya' of 'Surya-Siddhanta' which was continuously published in the monthly magazine of the Science Council. This ended in year 1667. The Science Council published it in book form in two parts.

He received the Mangalaprasad award and the Chunnulal award for this text.

Upon completion of the first edition of this text, it is being re-published at the request of the readers.

This book, equipped with the 'Vigyan Bhashya Tika', is a priceless treasure of ancient Indian literature and a glorious book of Hindi. 'Surya-Siddhanta' is useful and collectible for all libraries and inquirers.

SOURCE PAGE 3

Sūrya Siddhānta: Scientific Commentary, Volume I [Chapter on Mean Motions; Chapter on True Positions; Chapter on the Three Problems]

Commentator: The late Mahāvīra Prasāda Śrīvāstava

Dr. Ratnakumari Swadhyaya Institute, Allahabad

SOURCE PAGE 4

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SOURCE PAGE 5

Introduction

Many texts of ancient literature have been published on behalf of Dr. Ratnamukari Swadhyay Sansthan —the texts of Swami Satyaprakash and Dr. Usha Jyotishmati, such as Bakshali-Manuscript, Shulba-Sutras (in Sanskrit and English). In this tradition, we are publishing the Science Commentary of Surya-Siddhanta by Swargiya Shri Mahavir Prasad Srivastava. This honor has been received from the Science Council, Prayag, for which we are indebted to the officers of the council.

It would have been difficult for us to complete the arduous task of publication of this book if we had not received financial assistance from the respected Raisahab Chaudhary Pratap Singh ji and the trust established by him in Karnal. We are extremely grateful to Chaudhary Saheb. We are currently publishing the first part, which includes Madhyamadhikar, Spashtadhikar, and Triprashnadhikar. In the second part, this book will be completely finished.

SOURCE PAGE 6

Role

Swami Dayananda gave a comprehensive curriculum of study and teaching to graduates of the Vidya-Layas in 1874 in his immortal text Satyarth Prakash. This course has historical significance because before this arrangement by Rishi Dayananda, there was no comprehensive system of courses in this country. In 1858, three universities were opened in the country—Calcutta, Madras, and Bombay. In 1888, two more universities (after the death of Maharishi Dayananda) were established in Prayag and Lahore. Such examinations of Sanskrit, like that of Kashi, had not begun. Therefore, we can surely say that the entire curriculum method of Rishi Dayananda is the first attempt in this direction. After studying grammar, the Rishi said that he taught Manusmriti, Valmiki Ramayana, Mahabharata, Shad Darshanas, Brahmanas, Upavedas (Ayurveda, Dhanurveda, Gandharvaveda; Arthaveda-Shilpa etc.). Then he wrote that in two years, learn astrology, the theory of the sun, which includes algebra, numbers, geography, astronomy, and geology, as is.

The primary source of astrological subjects is the hymns of the Vedas. The name of Sukta 164 of the first Mandala of Rigveda is Asyavamiya Sukta. The mantra-darshita of this Sukta is Rishi Dirgatamas. This Sukta is the inspirational source of astrology-

Twelve are not the wheel that circulates in the mind of the wise.

O son of fire, do not be deceitful here for seven hundred and twenty-five.

(RV 1/164/11)

The cycle of the sun which has twelve rays, which constantly moves, never tires, does not age, does not die. It has 720 sons (360 days and 360 nights). In the Atharvaveda, they are in the nineteenth section of two Suktas (Suktas 7 and 8), whose sage is Gargya. In this Sukta, the calculation of 27 constellations is given. Taking inspiration from the Vedas, Lagadha Muni composed Vedanga Jyotish, whose verses are found in Rig Jyotish and Yuj-Jyotish. Lagadha's time is considered to be before 600 AD. There is another word for Jyotish for this, Kaal-Vidhan-Shastra (Yuj-Jyotish 3). Lagadha also used the word "Ganit" in this sense, and the glory of this part of Vedanga is expressed as follows—

SOURCE PAGE 7

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As the splendor of the crest of peacocks and the jewels of snakes are as such, so is the mathematics of the Vedangas.

Yuj: Jyotish, verse 4

The splendor of the crest in the body of the peacock and the glory of the jewel on the head of the snakes are the mathematics of the Vedangas, that is, the mathematics of astrology.

After the Vedangas, many principles of astrology were practiced in the field of Jyotish. Five of these principles are of special importance. Varahamihira discussed them comparatively in his famous work "Panchasiddtantika". The five principles are counted in the following verse—

Polish-Romak-Vasishtha-Sora-Patamahasru Panchasiddhanta:

Latdev analyzed the first two of these. According to Varahamihira, the calculation in the Polish theory is true, and the Romak theory is also almost the same. The true theory of sunlight, i.e., the solar theory or the sun theory, is even clearer than these two. The remaining two, Vasishtha and Patamaha, are false (misled). One edition of Panchasiddtantika was published by Professor Thibaut and Mahamahopadhyaya Sudhakar Dwivedi along with the English translation. The name of Sudhakar ji's Sanskrit commentary is “Panchasiddtantika Prakashika” (Prayag, 15 December 1888).

In the same tradition, Shri Mahavir Prasad Srivastava, an old councilor of the Science Council, Prayag, took up the arduous task of the science commentary of the Sun Theory. This lifetime work of his ended in 1640 AD. The work began in 1622, its first chapter (Madhyamadhikar) was published in 1624. The twelfth chapter was printed in 1631. Work stopped due to lack of funds. The last two chapters were printed in 1640. Thus, this commentary ended in Bhai Tritiya Samvat 1667 B.E. i.e., 1640 AD. This book was published in the science-monthly magazine in small parts, and some copies of the book were also prepared separately. This science commentary of the Sun Theory has been unavailable for the last about 20 years.

The Swadhyay Sansthan has attained this honor, by the immense generosity of the officers of the Science Council, Prayag, which obtained permission for its publication. The Swadhyay Sansthan has taken charge of the organization of the publication of precious books of ancient literature. We are publishing this book in two parts. The pictures included in the book were taken by the Block Science-Parishad

SOURCE PAGE 8

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Handed over after preparation to the institute. Shri Shri Krishna Srivastava, son of late Mahavir Prasad Srivastava, retired judge, took interest in this work. Dr. Shivgopal Mishra, General Secretary of the Science Council, personally and with eagerness did the proof-correction work of the entire text. The original Sanskrit text of the Sun's theory is given based on the edited version by Prof. Kripashankar Shukla, Lucknow University. We also received much help in the revision of the text from our elderly member, Shri Omkarnath Sharma ji. We are grateful to all of them.

Late Shri Srivastava ji had written a detailed introduction to the book (of 43 pages), which was published with the final section (1640). It is regrettable that this introduction could not be provided in its entirety in the current edition. We are giving some of the necessary parts here. Late Srivastava ji has provided very useful material on Indian astronomy in the book "Saral Vigyansagar" (1646) published by the Science Council Prayag. It would be best if it were published separately.

Late Shri Mahavir Prasad Srivastava had dedicated this science treatise to his late revered father. The words of dedication were— "In memory of my revered father, on whose feet I studied the first lesson of mathematics."

SOURCE PAGE 9

Role of the Commentator

[A brief excerpt of the introduction written by Sw. Shri Mahavir Prasad Srivastava]

Astrology is the principal part of the Vedas because from it the time of sacrifices is determined. Therefore, in ancient times in Bharata, the study-teaching of astrology was considered meritorious work. It also has another name, Kalavidhyashastra or Kalajnan. According to Kashyap Samhita, there were 18 Acharyas who were the originators of astrology, whose names are:-

1—Surya, 2—Pitamah, 3—Vyasa, 4—Vasishtha, 5—Atri, 6—Parashar, 7—Kashyap, 8—Narad, 9—Garg, 10—Marichi, 11—Manu, 12—Angira, 13—Lomash, 14—Paulish, 15—Chyavan, 16—Yavana, 17—Bhrigu and 18—Shaunak.

The principles and texts of these 18 names are unknown. Many are the doers of both Samhitas and Siddhantas, some only in one subject, and the text of some names is not available in both subjects.

The ancient principles that have been discussed since the time of Varahamihira are 1—Paulish, 2—Romak, 3—Vasishtha, 4—Saur, and 5—Patamah Siddhanta, which Varahamihira has collected in his book named Panch-Siddhantika (approximately 550 AD), in which he also states that the Paulish Siddhanta is clear, the Romak-Siddhanta is close to it, but the Surya-Siddhanta is the clearest, the other two are very corrupted. On page 2 of Panchsiddhantika-Prakashika Tika, M. M. Sudhakar Dwivedi quotes Suryarun-Samvad and says that the knowledge of the sages Gargadi which Maharshi Paulish said is the Paulish Siddhanta, which arose in the city of Romak due to Brahmashap, which the Sun God revealed to the Yavana caste of Romak is the Romak Siddhanta, which Vasishtha gave to his son Parashar is the Vasishtha Siddhanta, which the Sun gave to May Daitya is the Saur-Siddhanta, and which Brahma gave to his son Vasishtha is the Patamah-Siddhanta.

Where this Surya-Runa dialogue is taken from is not stated. According to this, the preceptors of both the Rumak theory and the solar theory are the Sun God, the first

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It was stated in Romaka Nagar that the Yavanas and the second demon whose place is unknown, but whom Western scholars consider to be of Greek (Yavanas) and Indian scholars of Assyria or Babylon. However, Brahmagupta (628 AD) considered only the Romaka theory as foreign (memory external), not the solar theory. Chapter 2-6 of the current solar theory states that at the end of Satya Yuga, an asura named Maya performed penance of the Sun God to learn astrology, and then he appointed his part-man for this task. This same part-man made the entire solar theory, which the Sun God himself had told the sages in previous ages, known to Maya-asura, from which the sages then obtained it.

If the scriptures, Brahmin texts, and Upanishads are studied impartially, it will be revealed that even in that prehistoric period, the experiences of sky-gazing are filled everywhere and much debate was held on them. If this were not the case, from where does the fact that they occur in different constellations of the Ayan biṃdu come, and how could they know the relationship between seasons and months and dates and constellations! Even in medieval India, there was no lack of wise and microscopic astrologers and mathematicians. By reading the texts of Aryabhata, Varahamihira, Brahmagupta, Manjal, Keshav, Ganesh, etc., they can know what subtle things they wrote about the events of the sky. Now, Shankar Balakrishna Dikshit, Prabodhchandra Sengupta, etc., have also proven through the comparative study of the texts of Eastern and Western astrologers that Indian astrology is completely independent of the astrology of Hipparchus or Ptolemy.

It is clear from Aryabhatiya that Aryabhata created the Aryabhatiya text by churning the ancient Indian astrology and by self-observing the Sun, Moon, planets, and constellations.1

G. R. Kaye writes, "There is no doubt that the boar-

> १ सदसज्ज्ञान समुद्रात् समुद्धृतं देवताप्रसादेन ।
> सज्ज्ञानोत्तर रत्नं मया निमग्नं स्वमतिनावा ॥४६॥
> आर्यभटीयं नाम्नापूर्व स्वायम्भुवं सदासद्यत् ।
> सुकृतायुषोः प्रणाशं कुरुते प्रति कञ्चुकं योऽस्यः ॥५०॥
> क्षितिरवियोगाद्दिनकृद्रवीन्दुयोगात् प्रसाधितश्चेन्दुः ।
> शशि तारा ग्रह योगात्तथैव ताराग्रहास्सर्वे ॥४८॥(आर्यभटीय गोलपाद)

SOURCE PAGE 11

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Mihir possessed the Siddhanta texts of the Yavanas and the manner of their explanation suggests that they were introducing new ideas or at least ideas and words that were not common in both in their texts, and it is certain that some of these ideas are later than those of Talmi. To refute this view, the strongest evidence I know is that Varahamihira showed ignorance of the motion of the ecliptic or the motion of the vernal equinox, although he knew that in his time it occurred at the beginning of the south motion of the constellation Cancer or in the Pushya constellation, and in ancient times it occurred at half of Ashlesha.1 If he had been familiar with the texts of Hipparchus or Talmi, he would have certainly accepted their writing on the precession of the equinoxes. Brahmagupta also did not pay attention to this, which shows that the matter of the precession of the equinoxes was not emphasized by medieval Indian astrologers, although he had experienced it through direct observation of the sky in ancient times that the constellations of Uttaraayana and Dakshinayana had changed because the

Maitrayini Upanishad mentions that Dakshinayana occurred when the sun remained from Magha to half of Dhanishtha and Uttaraayana occurred when the sun remained from Shravishtha-ardha to Ashlesha. In Vedanga Jyotisha, it is clearly stated that the beginning of Uttaraayana occurred when the sun came to the beginning of Shravistha. Varahamihira, while discussing this, state that in our time, Dakshinayana is in Punarvasu, and if it is different from this, one should determine it by observation2.

It is clear from the above numbers that the Indian value of the sidereal year is not more than 3 minutes 27 seconds larger than the actual sidereal year, and Brahmagupta's value is the closest to it. In fact, this value is about a quarter minute less than the value of the mean-centre year, so they are almost exactly the same. This is why the motion of the apogee of the sun is generally considered negligible in our texts. In comparison, there is more error in the values of the sidereal year of Meton and Babylon. The values of the sidereal year of Thaalmi and Kepler are six or seven and a half minutes more than the actual sidereal year. Therefore

> १. आश्लेषाद्धदासीद्यदा निवृत्तिः किलोष्णकिरणस्य ।
> युक्तमयनं तदासीत् साम्प्रतमयनं पुनर्वसुतः ॥२१॥
> २. वाराही संहिता आदित्य चार पृष्ठ १६, १७, विज्ञानभाष्य पृष्ठ ३३६ ।

SOURCE PAGE 12

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According to the various differing opinions of the year, the current Surya Siddhanta gives the day hour minute second more than the actual by 3 minutes 27 seconds. Aryabhatiya 365 61 12 36.56 more than the actual by 3 minutes 27 seconds. Brahma-sphuta 365 61 12 29.64,, 3,, 20 " Brahmagupta's theory 365 61 12 6,, 3 minutes more. The mean zodiacal year according to Lockyer is 365 66 6 6.3 actual zodiacal year "center-year" 365 66 13 48.3 sine-year" 365 5 48. 46.054 actual sine-year plus 6 min 26 sec sine-year according to Talmi*. 365 5 55 12 plus according to Copernicus X 365 5 55 58,,,, 7,, 48 " Chakshannavarsha according to Meton 365 68 57,,,, 4 8 " Babylonian 365 61 13 36.3,,,, 30 " × Heroes of Science, p. 63 § Encyclopaedia Britannica, History of Astronomy.

* Syntaxis, Karl Manitius’s edition, Vol. I, p. 146.

SOURCE PAGE 13

(R) SOURCE: (13)

It is proven that our astrologers are more pure and independent than the latitudes of Kepler and their latitudes, and therefore independent.

Similarly, if the apogees, nodes, and other polar coordinates are compared, it becomes apparent that the latitudes considered by our Acharyas are more accurate than reality in most cases. Therefore, it is proven that our astrology is not the shadow of Talmi or any other foreign astrology but is independent. The direction of these matters has been shown in various places in the Scientific Commentary.

The composition period of Sūrya Siddhānta—there is a great difference of opinion among scholars regarding this. The composition period stated in this book does not make sense by logic because if it had continued in the same way for so ancient a time, twenty-one lakh, sixty thousand years, then why does the question arise that when will the fourteen years of exile of the Pandavas be completed, in the court

of Maharaja-adhiraj Duryodhana in the Mahabharata era, where great scholars existed at that time? Even from Bhishma's answer, it is not understandable that he had knowledge of the astrological calculations of a text like Sūrya Siddhānta at that time. However, there is no doubt that the original form of this text is much earlier than Varahamihira, and it is possible that the form seen in Panchasiddhāntika is not the original form, but Varahamihira made some amendments to it. But it is surprising that Aryabhata (476 AD) did not mention it anywhere in his text, from which it is estimated that it was composed around the time of the composition of Aryabhatiya. The form of Sūrya Siddhānta given by Varahamihira is different from the current Sūrya Siddhānta, in which the matters of precession have been added from behind because if the discussion of precession had been in Sūrya Siddhānta before Varahamihira, it must have been included somewhere in Panchasiddhāntika and not been satisfied by writing only that the precession point was in the Nakshatra of Ayana in ancient times, now it is in Punarvasu. Even in the time of Brahmagupta (628-665 AD), the form of Sūrya Siddhānta did not have the matter of precession because in such a situation, Brahmagupta would have certainly discussed it. This point must have been found some time before Bhaskaracārya, but still not in that form as seen now because Bhaskaracārya himself has expressed doubt about its purity. Therefore, it is proven that the original form of the current Sūrya Siddhānta is the same as that given in Panchasiddhāntika and existed even before Varahamihira, or Varaha Sūrya Siddhānta is its modified form and later astrologers have periodically refined it.

SOURCE PAGE 14

In the beginning of Kali Yuga 3002 B.C. Kali Yuga 1000 In the beginning of Kali Yuga 2002 Kali Yuga 102 B.C. Kali Yuga 3000 Kali Yuga 102 B.C. Kali Yuga 3636 538 B.C. Kali Yuga 4102 1061 B.C. Art of the Zodiac. Art of the Zodiac. Art of the Zodiac. Art of the Zodiac. Art of the Zodiac. Art of the Zodiac. Art of the Zodiac. Art of the Zodiac. Art of the Zodiac. Mercury + 33 25 35 + 25 6 52 + 16 6 + 8 38 26 + 3 12 2945 Venus + 32 43 36 -24 37 31 -16 31 26 -8 25 14 4 3 636 Mars + 12 5 42 + 6 26 + 6 47 22 + 4 8 + 2 58 26 Jupiter - 17 2 53 -12 44 16 -8 25 36 -4 7 2 -1 414 Saturn + 20 56 3 + 15 43 20 + 10 27 37 -5 11 50 10 -1 25 88 Moon -5 52 41 -3 50 48 -27 -0 52 -0 30 -0 011 1067 Lunar Apogee -30 11 25 -23 6 36 -16 7 -4 5 58 -4 36 -0 163 Lunar Node +23 37 31 +17 56 21 +12 31 11 +7 3 +3 50 1188

Chapter on Mean Motions

SOURCE PAGE 15

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Having revealed this in the present form, for which assistance must have been taken from the texts or treatises of Aryabhata, Brahmagupta, etc., as Sengupta mahoday proves in the role of his Khandakhadya.

The opinion of Western scholars—Western scholars have greatly pondered the relationship with the time of composition of this, for which it is our utmost duty to thank them. Mr. Bentley has done a lot of work in this matter. He calculated the positions of the Sun, Moon, and other planets for different times according to the Sūrya Siddhānta and new observations, and fixed the time of composition of the Sūrya Siddhānta when the portions of the planets according to both calculations become equal. This will be clear from the bracket on page 14 where the inaccuracies of the planets are shown.

From this Bentley derives the result that the Sūrya Siddhānta was written in the late phase of the 11th century. In appearance, it seems very logical and there is no doubt that the author has shown great

novelty in it, but in reality, this method is not acceptable without proper examination. I understand sufficiently this point in the words of the publication committee of Burgess's translation:-

The times that Bentley has taken for the relationship of the other planets with the zero error are very close to the times we have calculated, which we have derived by calculating backward from the errors of 1860 AD, and which are given with the commentary of the 67th verse.

Upon comparing these two brackets, it will immediately be seen that Bentley did not derive his decision from the inaccuracies of the absolute positions of the planets, but from the comparison with the position of the Sun. However, he did not consider the inaccuracy of the Sun's position. The Hindu zodiac is sidereal and does not depend in any way on the motion of the Sun. Like other planets, the Sun was not at the position assumed in 3102 B.C., and therefore the motion of the Sun considered in this method is different from the reality because the sidereal year is considered 3 minutes larger. Therefore, why not consider the inaccuracy of the Sun and why consider the sidereal motion of the planets in comparison to this inaccurate motion? It is apparent that Bentley should have considered the position of the Sun completely and should have shown that the same result follows from the calculation of other planets, and if not, what was the cause of the inconsistency? Thus

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SOURCE PAGE 16

On the first of January in the year 1860 at midnight in Washington, the medium portion of the planets according to ancient astrology, the degrees of the celestial bodies are: Sun 66 18 21 96 18 21 100 5 6 Mercury 155 2 30 148 25 36 151 28 Venus 336 54 55 334 57 18 366 13 36 Mars 162 36 5 162 36 5 167 26 32 Jupiter 104 7 22 100 48 56 103 35 17 Saturn 128 17 11 133 14 46 137 10 10 Moon 6 4 6 6 4 6 12 41 23 Lunar Ascent 327 50 24 326 11 11 326 47 35 Lunar Eruption 312 26 51 310 50 38 312 48 10

SOURCE PAGE 17

According to Sūrya Siddhānta, the inaccuracies of the planets when they are absolute, when they are absolute, when they are relative, when they are E. degrees arcminutes arcseconds E. degrees arcminutes arcseconds E. degrees arcminutes arcseconds E. degrees arcminutes arcseconds E. degrees arcminutes arcseconds E. Sun —3 46 45 250 0 0..... — 3 46 45 250 0 0..... Mercury +3 34 10 2332 +7 20 55 3271 —3 2 41 1517 +0 44 5 1670 Venus +3 41 16 1222 +7 28 4 641 —1 16 18 2126 +2 30 27 1506 Mars —4 50 27 886 —1 3 42 1455 —4 50 27 886 —1 3 42 1455 1455 Jupiter +0 32 5 1571 +4 18 50 832 —2 46 21 4203 +1 0 24 1575 Saturn —8 52 56 666 —5 6 14 857 —3 55 21 1250 —0 8 36 1825 Moon —3 37 14 1115 +0 9 31 1067 —3 37 14 115 +0 6 31 1067 Lunar Altitude +1 2 46 1676 +4 46 34 1252 —0 36 24 1266 +3 10 21 1456 Lunar Longitude —0 18 16 1676 +3 28 26 1162 —1 57 32 2714 +1 46 13 1468

SOURCE PAGE 18

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By not doing so, a very important part of the entire calculation has been left out in our understanding from which a very important result emerges.

"But what is the real matter about our theory? We get that in it such polaritudes of the planets have been given whose positions, when tested according to the method above, it seems that these polaritudes were not determined with the idea that their actual celestial positions could be known at a specified time. Rather, they give such evidence that an attempt was made in the 10th or 11th century AD to know

the positions of the planets exactly in comparison to the position of the Sun. This is also doubtful in terms of time because there is a great variation in those times, in which error is considered zero. In the history of Sanskrit literature, this fact is as certain as anything that the Sūrya Siddhānta existed much earlier than that time. The instructions and quotations from other astronomical texts also show that there were many recensions of this book and we have also seen above (in shloka 6) that there are not exactly the same polaritudes given in the current text that were considered in the Sūrya Siddhānta. Therefore, what else could be that the calculation of the seed which was done for the revision in the 10th or 11th century was merely done by changing four or five shlokas in the original.

The time of the zero inaccuracy of the sun in our brackets is 250 AD. There is no need to emphasize the inaccuracy of this date, because it depends on the purity of the position of the sun which was determined first and then directed from that point which is called the initial point of the celestial circle. It was impossible to observe with the naked eye whether the center of the sun was when, according to the mean motion, the star Zeta Pisces was ten lunar phases before the conjunction of Revati, and it is clear that the Hindus have made big mistakes in the other points of the zodiac determined from this point, and if even a fraction of an error is made in determining the position of the sun, a difference of 425 years may occur in the time of zero inaccuracy.

SOURCE PAGE 19

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The time of Sūrya Siddhānta also comes around the time of the conjunction stars' polar longitudes given by Brahmagupta and others by comparison. Some polar longitudes are obtained in these. These are those which are integral in all three texts, some of which the authors themselves have purified. From their comparative study we can know the superior and inferior limit of the Sūrya Siddhānta period. In Varaha's Panchasiddhāntikā as well (according to Thibo), seven polar longitudes of the conjunction stars are given which we consider the polar longitudes of that ancient Sūrya Siddhānta which existed before Varaha (see pages 20-21).*

The constellations written below are chosen for the depiction of the ancient Sūrya Siddhānta time, for which the longitudes of Brahmagupta are greater than the polar longitudes of the Sūrya Siddhānta are shown:-

Ansha kala kṛttikā 4 48 Rohini 1 28 Punarvasu 5 3 Magha 3 0 Po Phaalguni 3 0 Chitra 3 0 Yoga 20 16 Madhyamamana 3 23

The mean of the excess of the polar longitudes of these six stars is equal to 3 degrees 23 arc. If an increase of one degree of longitude is considered to have happened in 72 years due to the motion of the ecliptic, then this increase must have happened in approximately 244 years. However, we know for certain from Brahmagupta's time that it is 628 AD. Therefore, the time of Sūrya Siddhānta comes 244 years before this or 384 AD. It can be understood in a gross sense as 400 AD. This is the superior limit of this time. To find the inferior limit, we should look at the polar longitudes of the stars written below whose longitudes are greater than those of Brahmagupta by 1 degree 15 arc.

*The Preface written by P.C. Sengupta of the second edition of the translation of Verghese's Sūrya Siddhānta

SOURCE PAGE 20

The polar longitude of the star Panch Siddhantika Brahmasphut Siddhanta Shishadhivridhi current Surya-Siddhanta from where was taken the degrees arcminutes degrees arcminutes degrees arcminutes degrees arcminutes degrees arcminutes Siddhānta where the polar was taken Ashwini... 8 0838383 on the tradition received Bharani... 200200200000,, Krittika 32 40 28 28 36 0 37 30 30 Brahmagupta Rohini 48 0462846 0463030 Brahmagupta Mrigashira... 63 062 063 063 0,, Ardra... 67 070 067 06720 New? Purvvasu 58 0633620630 Brahmagupta Pushya 67201060105201060,, Ashlesha 10740108011401060 Nutan Magha 1260126012801280 Brahmagupta Poo Phalguni... 1470136014401550 Brahmagupta Uo Phalguni... 155015401550

SOURCE PAGE 21

Hasta 180 50 170 0 173 0 170 0 Brahmagupta Chitra 50 183 0 184 20 186 0 Paramparaprapta Swati 166 0 167 20 196 0 Brahmagupta Vishakha 212 0 212 0 213 0 Nutan Janutava 224 0 232 0 224 0 Brahmagupta Vyoshtha 226 0 228 0 241 0 Paramparapta Mool 241 0 254 0 254 0 “ Purvabhadrada 254 0 267 0 260 0 Brahmagupta Uttarashadha 260 0 267 0 266 0 Naya Abhijit 265 0 267 10 280 0 “ Avana 278 0 283 10 290 0 Brahmagupta Sanishta 260 0 266 22 260 0 Brahmagupta Ritbhisha 320 0 313 20 322 0 “ Purvabhadrapada 326 0 327 0 326 0 “ Uttarabhadrapad 337 0 335 0 337 0 “ Revati 0 0 356 0 356 50 “

SOURCE PAGE 22

(22)

The art must have occurred 60 years after Brahmagupta. Thus, this lower limit is 628+60=718 AD.

Fractional art Ashlesha 1.0 Vishakha 1.0 Abhijit 1.0 Shravana 2.0 Yoga 5.0 Madhyama 15

Thus, it can be concluded that the composition of Sūrya Siddhānta took place from 400 AD to 725 AD. Bentley and Burgess also compared the polar stars in Sūrya Siddhānta with the proven polar stars from the modern astronomical measurements to determine the time of Sūrya Siddhānta in this way, but it does not seem so accurate, because the kind of error that the Sūrya Siddhāntakar made regarding the arcs, Brahmagupta and others must have also made. Thus, the upper limit of the period of composition can be extended further. However, the lower limit can be brought up to 1100 AD as proved by Bentley.

The view of Bentley is also supported by the matter of the motion of the ecliptic. In the time of Bhaskara-Charrya, the ecliptic motion written in Sūrya Siddhānta was not as much as in the current Sūrya Siddhānta. It is apparent that the correction of the ecliptic motion was made after Bhaskaraacharya. However, the time of Bhaskaraacharya is 1150 AD.

The author of Sūrya Siddhānta—it has been shown that the polar coordinates of Sūrya Siddhānta do not match the polar coordinates of the Greek astrologers Hipparchus, Talmi, or other Western country astrologers. Therefore, this treatise was not written based on the observations of those people. Its tradition is explained by Vivsvān, that is, by the Sun (see Chapter on Mean Motions verse 8-6), just as the tradition of the Yoga path described in the Bhagavad Gita is explained (see Bhagavad Gita Chapter 4 verse 1-2). Maya is not foreign. He was a resident of India and obtained this knowledge, and the sages also learned from him. This seems to be the secret of the story given in Sūrya Siddhānta. Due to the Brahmashāpa, the story of the Sun God arising in the luminous city is fabricated and is found only in one or two handwritten books.

SOURCE PAGE 23

(23)

is, therefore, it is projected which no one has accepted. If the Sūrya Siddhānta had been obtained by Trideśi, Brahmagupta would surely have written it, because for the Romak Siddhānta, which is undoubtedly foreign, he has written it clearly. There is no trace in the Prakashika ṭīkā of M.M.P. Sudhakar Dwivedi that gives the origin of the Surya-sūrvāda, so it cannot be said who wrote it and on what basis. The writing in Nityānanda1 that it was written when the 3600 years of Kali Yuga had passed, i.e., in the year 421 or 466 AD, when Aryabhata wrote his Aryabhatiya, is a delusion. Perhaps because of this delusion, Muniśvara also considered Aryabhata the author of the Sūrya Siddhānta. Alberuni's statement that it is a delusion that Latāchārya or Laṭsiṃha made it2 is because neither Varāhamihira nor Brahmagupta considered Latāchārya the author of the Sūrya Siddhānta. According to Varāhamihira, Latāchārya is an interpreter of the Romak and Polish theories. Latā's measurements3 are from the sunset time of Yavanpura. Thus, it is apparent that all the estimations regarding the time of composition and author of the Sūrya Siddhānta are not proven. Based on internal evidence, it can only be said that its growth started around 400 AD and ended by 1200 AD. Initially, Varāhamihira must have expanded (amended) it. Then other modifiers tried to make it Up-to-date by taking advantage of the observations of Aryabhata, Brahmagupta, Manjula, etc. This amendment continued until Rangnath ji wrote its commentary and bound its verses. Some more verses are found in Madhav Purohit's commentary, don't know if they were written on the basis of an old text or just added.

Prayag Bhratri-Dvitiya; 1967 B.S. Mahavir Prasad Srivastava

1 The composition time of Sūrya Siddhānta is established by the principles—the text is in the following —five Siddhāntikas, page 2.

2 As stated by Latācharye, in Yavanpura, the first chapter of the Pancha Siddhāntika is in the 3rd verse of the Sun.

3 That same chapter, verse 18.

SOURCE PAGE 24

First Chapter

Chapter on Mean Motions

(Brief Description)

[1 verse—Praise of God. 2-7 verses—To know Jyotishastra, Mayasura performed penance for the Sun God, the Sun God pleased and gave a boon, and created a man from his body. 8-6 verses—Tell the man from Mayasura that what the Sun God said to the sages before, is being said with some changes. 10 verses—Two types of time (1) infinite and finite, (2) cumulative. 11-20 verses—Units of time from Nimesha to Kalpa. 21-23 verses—Current age of Brahma. 24 verses—How much time was taken to create the universe from the beginning of Kalpa. 25-27 verses—The cause of the motion of constellations and planets. 28 verses—Units of angle measurement. 26-34 verses—How many revolutions of planets, their epicycles, the height of the Moon, the node, and the constellations occur in one Mahayuga. 35-36 verses—Relation between lunar and solar months. 37-36 verses—How many Savan days, Adhimasa, and

Tithis are in one Mahayuga. 40 verses—How many Savan days and Tithis are in one Kalpa. 41-44 verses—How many revolutions of the apogees and nodes of the planets are in one Kalpa. 45-47 verses—Time from the beginning of Kalpa to the end of Satya Yuga. 48-50 verses—The method of knowing how many days have passed since the beginning of creation. 51-52 verses—The method of knowing the day-lord, year-lord, and month-lord. 53-54 verses—The method of knowing the mean positions of the planets. 55 verse—The method of knowing the year (Sanvatsara) of Jupiter. 56-58 verses - What were the positions of the planets at the end of Satya Yuga. 56 verse—Relation between Vyasa and circumference and the magnitude of the earth's circumference. 60-61 verses—Knowing the magnitude of the latitude circle of a place and deriving the mean position of the planet from it. 62 verse—Which famous cities are on the mid-line of the Indian land. 63-65 verses—Knowing from lunar eclipse whether a certain place is how far east or west of the mid-line. 66 verse—When does the solar-precession occur. 67 verse—What is the position of the planets at a certain auspicious time. 68-70 verses—How many North or South can the light and shadow of the Moon and other planets move by their radiance.]

SOURCE PAGE 25

> अचिन्त्याव्यक्तरूपाय निर्गुणाय गुणात्मने ।
> समस्तजगदाधारमूर्तये ब्रह्मणे नमः ॥१॥

Translation—Salutations to the Brahman upon whom the form cannot be meditated upon nor manifested, who is without qualities yet from whom all qualities have arisen and who is the basis of the entire universe॥ 1॥

> अल्पावशिष्टे तु कृते मयो नाम महासुरः ।
> रहस्यं परमं पुण्यं जिज्ञासुर्ज्ञानमुत्तमम् ॥ २ ॥
> वेदाङ्गमग्र्यमखिलं ज्योतिषां गतिकारणम् ।
> आराधयन्विवस्वन्तं तपस्तेपे सुदुष्करम् ॥ ३ ॥

Translation—When some of the remaining parts of Satya Yuga were left, the great demon named Maya, who explained the motions of all the celestial bodies in all the Vedangas, and desired to know the supreme holy and mysterious knowledge, performed severe penance worshipping the Sun God. 2-3॥

Scientific Commentary—The explanations of Satya Yuga, Treta, Dwapara, and Kali Yuga are given in the 16th verse of this chapter.

There are 6 Vedangas—Shiksha, Chanda, Vyakarana, Nirukta, Kalpa, and Jyotisha. They help in understanding and explaining the Vedas, which is why they are called Vedangas. Bhaskaraacharya ji has shown the superiority of astrology among the Vedangas in this way—the word-science is the face of the Vedas, Jyotishastra is the eye, Nirukta is the ear, Kalpa is the hand, Shiksha is the nose, Chanda is the feet, therefore, just as the eye is superior in all limbs, Jyotishastra is superior in all Vedangas.

> तोषितस्तपसा तेन प्रीतस्तस्मै वरार्थिने ।
> ग्रहाणां चरितं प्रादान्मयाय सविता स्वयम् ॥ ४ ॥

Translation—Satisfied and pleased by his penance, the Sun God Himself gave the instruction of the movements of the planets, i.e., Jyotishastra, to the Maya who desired boons॥ 4॥

Scientific Commentary—Western astrologers call the celestial bodies that revolve around the Sun planets. According to this definition, Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune are these eight planets, of which the location of the last two planets

SOURCE PAGE 26

It has occurred within the last two hundred years and is not visible to the naked eye. The Moon revolves around the Earth, so it is a satellite. Satellites of other planets have also been seen by telescopic instruments. However, in our astrological texts, the Earth is not considered a planet but the Sun. The Moon is also placed in the category of planets. There is no discussion of Uranus and Neptune anywhere. Therefore, in our system, the Sun, Moon, Mars, Mercury, Jupiter, or Brihaspati, Venus, and Saturn are considered seven gross planets, and Rahu and Ketu are two subtle planets. The full description of these two subtle planets will be written in this chapter while describing the nodes of the Moon. The phenomena that occur in the sky due to the motions of these planets are described in Jyotishshastra, hence the second name of Jyotishshastra is mentioned as 'The History of the Planets' in this verse.

> विदितस्ते मया भावः तपसाऽऽराधितस्त्वहम्१ ।
> दद्यां कालाश्रयं ज्ञानं ज्योतिषां२ चरितं महत् ॥ ५ ॥
> न मे तेजस्सहः कश्चिदाख्यातुं नास्ति मे क्षणः ।
> मदंशः पुरुषोऽयं ते निश्शेषं कथयिष्यति ॥ ६ ॥

Translation—Lord Surya said that I have known your feeling and I am very satisfied with your penance, I instruct you about the great history of the planets, from which one can know the exact knowledge of time; but my radiance cannot be borne and I do not have time to give instruction, so this man, who is my part, will instruct you very well. 5-6

> इत्युक्ताऽन्तर्दधे देवस्तमादिश्यांशमात्मनः ।
> स पुमान्मयमाहेदं प्रणतं प्राञ्जलि स्थितम् ॥७॥*
> शृणुष्वैकमनाः पूर्वं यदुक्तं ज्ञानमुत्तमम् ।
> युगे युगे महर्षीणां स्वयमेव विवस्वता ॥८॥

SOURCE PAGE 27

> शास्त्रमाद्यं तदेवेदं यत्पूर्व प्राह भास्करः ।
> युगानां परिवर्तन कालभेदोऽत्र केवलम् ॥६॥

Translation—Thus the Sun God disappeared, and the Sun's aspect man, as per order, bowed humbly and joined his hands, said—Pay close attention and listen to this excellent knowledge which Lord Surya had told the great sages from time to time; the first scripture that Lord Surya taught is this original scripture; only some difference has occurred in time due to the changes of ages ॥7-6॥

Scientific Commentary—some people interpret the second verse of the ninth verse to mean that the scripture that the Sun God taught to the sages is that same scripture without any change, only a difference in the time of saying. But if this were the meaning, what was the need to say that there is a difference in time? What was said in the first verse was sufficient. Therefore, the more reasonable meaning of this verse is that if there is any difference in the Jyotiḥ-shastra previously stated and the one being stated at this time, it is due to time, and there is no essential difference. How can there be a

difference due to time? The reason is that the Jyotiḥ-shastra is an experimental science and some subtle errors remain in the experiment, which is called experimental error. This error accumulates in Jyotiḥ-shastra year by year and after hundreds of years it takes a very large form; therefore, it has to be revised from time to time, which is called seed-transformation. From this perspective, this sentence was said by Suryāṁśa Puruṣa, for which there are 21 verses named 'Bijopanayana' in the last chapter of the Surya Siddhānta, which Rangnath Ji and Pan. Indranarayan Dwivedi Ji have not considered as a commentary and have explained the reason for not considering it as a commentary, which is that how the seed-transformations are done by the Sun God in the scripture spoken by the Sun God. But Rangnath Ji, while explaining the 6th verse in his gūḍhārtha-prakāśikā commentary, also state that some difference occurs upon taking time. His sentences are as follows:

And also by the movements of time, if there are some celestial bodies, then in the time between the ages, the scriptures were revealed by the compassionate ones for the purpose of the planetary movements, as in the subsequent scriptures, the meaning of which is. Thus, I have deduced that the planetary movements established by the Sun-related scriptures, the Sun-related scriptures established by the planetary movements, and the Sun-related scriptures, without regard to the necessity of the plan, are the very things that are deduced. And also, in the time between the ages, the movements of the planets are shown, then in that time, the subsequent

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SOURCE PAGE 28

Those who are currently doing the following do not know the seed in the previous book. From the absence of the previous books, the dialogue of Surya-Yarshi is not seen. What is this proof?

There are many examples of intervals found by measuring time, which will be explained in due place in this commentary.

> लोकानामन्तकृत्कालः कालोन्यः कलनात्मकः ।
>
> स द्विधा स्थूल सूक्ष्मत्वान्मूर्तश्चामूर्त उच्यते ॥१०॥

One type destroys the world and the other is progressive, meaning it can be gone to. This is also of two types—(1) gross and (2) subtle. The gross can be measured, therefore it is called tangible, and the subtle cannot be measured, therefore it is called intangible॥10॥

Scientific Commentary—the imagination of the first type of time also cannot be there, because it is not known when it began nor is it known when it will end. It is continuous and vast; but the end of the world happens in between it or while it exists, Brahma is born, creates, and dissolves, but time remains. Therefore, it is called the destroyer and ender of the worlds. That is why death is also called time.

Only a small part of time can be known; even in that, the very small cannot be measured and is called abstract. The subtlety with which one measures will also give a new definition to abstract time; as will be shown in the explanation of the next verse.

> प्राणादिः कथितो मूर्तः त्रुट्याद्योऽमूर्तसंज्ञकः ।
> षड्भिःप्राणैः विनाड़ी स्यात्तत्षष्ट्या नाड़िका स्मृता ॥११॥
> नाड़ी षष्ट्या तु नाक्षत्रमहोरात्रं प्रकीर्तितम् ।
> तत्त्रिंशता भवेन्मासः सावनोऽर्कोदयैःस्मृतः ॥१२॥
> ऐन्दवस्तिथिभिः तद्वत्सङ्क्रान्त्या सौर उच्यते ।
> मासैर्दवादशभिवर्ष दिव्यं तदह उच्यते ॥१३॥

From Anuvāda-prāṇa to the higher time units, they are concrete

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SOURCE PAGE 29

They are called abstract from the error to the units below the life force. 6
There is one measure of life force (moment) and one pulse (clock) of 60 measures of life force. 11
One Nakshatra Ahuratra (a pair of day and night) of 60 pulses and one Nakshatra month of 30 Nakshatra Ahuratras.
In the same way, there is one Savan month of 30 Sawan days. 12
Similarly, there is one Chandra month of 30 lunar tithiyoṃ and the time from one transit to another transit is called the solar month. 12
A year of 12 months exists; which is called *divyadina* or the day of the gods.

Scientific Commentary—When a healthy person is sitting comfortably, the time it takes to inhale and exhale air (life air) naturally is called Pran. This is the smallest unit that could be measured at that time. There was no means to measure less time than this at that time; therefore, it was called abstract. Now such clocks are made by which the measurement of that unit is easy, which are called abstract. In one lunar day, 60 clocks = 60 X 60 shadows = 60 X 60 X 60 Pran or 21600 Pran exist. Similarly, in 1 day, 24 hours = 24 X 60 minutes = 24 X 60 X 60 seconds or 86400 seconds exist. Therefore, there are 4 seconds in 1 Pran. Everyone knows how easy it is to measure a second when the hand of the clock is attached to it. There are also such clocks by which the fifth or tenth part of 1 second can be easily known. However, the tenth part of 1 second is the same as the fortieth part of 1 Pran. Therefore, some units below Pran can also be called concrete nowadays.

Pran is also called Asu. In the Siddhānta-Shiromani of the famous astrologer Bhaskaraacharya ji, the second definition of Pran is given in the words of Chandas-shastra—the time taken to pronounce one Guru letter is called Pran ten times that time. A Guru letter is that which has a conjunct letter after it, such as Sanuswar, Visargant, Dirgha, and the short letter.

There is also a name for a unit of weighing, which is like four tole.

This word reveals that the 12 months that make up a year are the solar months. The year of Chandra, Nakshatra, or Sawan months is not.

SOURCE PAGE 30

The time taken for 1 pal or 4 tola of water to ascend in a special aperture of the instrument is called a pal.

Bhaskaracharya explained the error in this way. The time taken for the palak to fall is called a nimisha. Thirty parts of 1 nimisha and the hundredth part of 1 tatpar are called error. The relationship of the units above nimisha is as follows:-

18nimesha = 1kashtha

30kashtha = 1kala

30Art = 1hour

2ghatika = 1muhurta

30muhurt = 1day (Nakshatra)

Sūrya Siddhānta = 30 X 2 X 30 X 30 X 18nimesha = 67200nimesha

It is shown that in 1 day there are 21600 prana or 86400 seconds, so in 1 prana there are 672000 nimesha or 45 nimesha and 1 second has 113 nimesha.

Nakshatra Ahuratra—Nakshatra means star, star-group and the 27th part of the cycle on which the sun appears to revolve once in a year. Due to the daily motion of the Earth, all stars in the sky rise in the east, move towards the west, set in the west, and then rise in the east. Note down the time of rising of any star by looking at the clock and see when that star rises again. If the clock is correct, the time between these two risings is approximately 23 hours 56 minutes and 4 seconds. This is called Nakshatra Ahuratra or only Nakshatra day. This is always the same, it does not increase or decrease, if the very subtle motion of the stars is not considered. Therefore, astrologers calculate time from this.

Sawan day — the time from one sunrise of the sun to the next sunrise is called Sawan day. It is 4 minutes larger than Nakshatra day. The Sawan day's

1. Its special description will be made in the 13th chapter named Jyotishopanishat.

2. Siddhanta Shiromani Ganitadhyaya Madhyamadhikar, Kala Manayaadhyaya Shloka 16, 17.

SOURCE PAGE 31

The value is not the same. Therefore, the value of the mean Sāvan day is the time known by clocks.

The Indav Tithi or Chandra Tithi—the time when the moon reaches very close to the sun while revolving in the sky is the new moon. The time between one new moon and the next new moon is called the lunar month. Its mean value is 26.530587646 of the mean Sāvan day. After the new moon, the moon moves further towards the east from the sun, and when it moves 12 degrees further, the first tithi (pariva) passes. When the difference is between 12 and 24 degrees, it remains duija. When the moon is ahead of the sun from 24 to 36 degrees, it remains tīja. When the difference is between 178 and 180 degrees, it is the full moon. When the moon is ahead from 180 to 162 degrees, it is the 16th tithi or pariva (pratipada). Duija remains from 1620 to 2040, etc. After the full moon, the moon moves back by 2 hours (48 minutes) each day from sunset. The 14, 15 days from the full moon to the new moon are called the dark fortnight. The new moon is also called the 30th tithi, which is why 30 is written for the new moon in Panchangs.

Sārmāsa—the path along which the sun travels while revolving in the sky is called the ecliptic. Its twelfth part is called the Rāshi. The center of the solar system, at the time it enters one Rāshi and moves to another Rāshi, that time is the transit of the second Rāshi. The time between one transit and the next transit is called Sārmāsa. The 12 solar months are different in magnitude; the reason for this is that the motion of the sun is not always the same. When the motion of the sun is fast, it completes one Rāshi quickly and the solar month is shorter. Conversely, when the motion of the sun is slow, the solar month is larger.

The number of types of months is equal to the number of types of years; a lunar year of twelve lunar months, a Sawan year of twelve Sawan months, and a solar year of twelve solar months. Our astrologer tradition has believed this. However, the direct meaning of the second part of the 13th verse is that there is a year of 12 months, which is called the divine day. Therefore, the year of twelve months mentioned is not other months, only solar months. This proves that the Surya Siddhanta discusses only the solar year and considers the solar year as the year, not others.

SOURCE PAGE 32

Chapter on Mean Motions

It is stated that the North Pole of the Earth is the abode of the gods and the South Pole is the abode of demons. Therefore, the North Pole is called Deva-loka and the South Pole is called Asur-loka. At the time when the sun comes to the equator, day and night are equal. This phenomenon occurs only twice a year. The sun remains north of the equator for 6 months and south for 6 months. In the first half, the day is longer and the night is shorter in the northern hemisphere and the day is shorter and the night is longer in the southern hemisphere. In the second half, it is exactly the opposite. However, when the sun remains north of the equator, it is visible on the North Pole (on Mount Sumeru) for 6 months and not visible at the South Pole at this time. Therefore, this half is called the day of the gods and the night of the demons. When the sun remains south of the equator for 6 months, the gods are not seen at the North Pole and the demons are seen equally at the South Pole for 6 months. Therefore, this half is called the night of the gods and the day of the demons. Therefore, our 12 months are like one yuga of the gods or demons.

> सुरासुराणामन्योन्यमहोरात्रं विपर्ययात् ।
>
> षट् षष्टिसङ्गुणं दिव्यं वर्षमासुरमेव च ॥१४॥

Translation—What is the day of the gods is the night of the demons, and what is the night of the gods is called the day of the demons. This is called the divine or demon year, which is 60 times 6 times the yuga of the gods or demons.

Scientific Commentary—as if one imagines a year of one Sawan day, a year of Sawan. In the same way, a year of 360 divine days is considered a divine year. Its direct meaning is that our 360 years are a year of the gods.

> तद्द्वादशसहस्राणि चतुर्युगमुदाहृतम् ।
> सूर्याब्दसंख्यया द्वित्रिसागरैरयुताहतैः ॥१५॥
> सन्ध्यासन्ध्यांशसहितं विज्ञेयं तच्चतुर्युगम् ।
> कृतादीनां व्यवस्थेयं घर्मपादव्यवस्थया ॥१६॥
> युगस्य दशमो भागः चतुस्त्रिद्वयेक सङ्गुणः ।
> क्रमात्कृतयुगादीनां षष्ठांऽशः सन्ध्ययोः स्वकः ॥१७॥

Translation—These twelve thousand divine years are a Chaturyuga, the number of which is thirty-four lakh twenty thousand (432000) in solar years. In this, the years of Sandhya and Sandhyansh are also included. A Chaturyuga includes Satya Yuga, Treta, Dwapar, and

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SOURCE PAGE 33

> कलियुग चार युग होते हैं; जिनके मान धर्म के चरणों के अनुसार होते हैं । चतुर्युग के दसवें भाग का चार गुना सत्ययुग, तीन गुना त्रेता, दो गुना द्वापर और एक गुना कलियुग होता है । प्रत्येक युग के छठें भाग के समान उसकी दोनों संध्याएँ होती हैं ॥ १५-१७ ॥

Scientific Commentary—the 14th verse states that there is a divine year of 360 days for gods or asuras. The 13th verse states that one day of the gods is like one solar year, so it is clear that one year of the gods is equal to 360 solar years. According to the 15th verse, a Chaturyuga is a period of 12000 divine years or 12000 X 360 (i.e., 432000) solar years. Chaturyuga is also called Mahayuga. In one Mahayuga there are four Yugas: Satya Yuga; Treta, Dvapara, and Kali Yuga, which is why it is also called Chaturyuga. In Satya Yuga, Dharma has four stages, in Treta three stages, in Dvapara two stages, and in Kali Yuga one stage. Similarly, in one Mahayuga, Satya Yuga has four parts, Treta has three parts, Dvapara has two parts, and Kali Yuga has one part. Therefore,

Divine years in solar years, Satya Yuga, Treta, 4800 1728000, Dvapara, 3600 1296000, Kali Yuga, 2400 864000, Mahayuga 1200 432000

Both sandhyas of Satya Yuga are equal to the sixth part, so one sandhya (period) is equal to the twelfth part. The sandhya at the beginning of the Yuga and the sandhya at the end are called Adi Sandhya and Ant Sandhya. Their values are as follows:-

In divine years in solar years, the beginning and end sandhyas of Satya Yuga are 400 144000, of Treta 300 108000, of Dvapara 200 72000, of Kali Yuga 100 360000

Just as there are two sandhyas in the morning and evening in one Ahoraatra, there are two sandhyas in each Yuga, one at the beginning and one at the end.

SOURCE PAGE 34

> युगानां सप्ततिस्सैका मन्वन्तरमिहोच्यते ।
> कृताब्दसङ्ख्या तस्यान्ते सन्धिः प्रोक्तो जलप्लवः ॥१८॥
> ससन्धयस्ते मनवः कल्पे ज्ञेयाश्चतुर्दश ।
> कृतप्रमाणः कल्पादौ संधिः पञ्चदश स्मृताः ॥१९॥

Translation—There is a cycle of 71 Mahayugas, at the end of which there is an evening similar to Satya Yuga. In this evening, there is Jalplava. There is a Kalpa of 14 Manvantaras including the conjunction, at the beginning of which there is also an evening similar to Satya Yuga; therefore, in one Kalpa, there were 14 Manvantaras and 15 evenings similar to Satya Yuga.18-19

Scientific Commentary—Two evenings are considered in each Gug of the four Yugas; but only one evening is considered at the end of the Manvantara, which is equal to Satya Yuga. 1 Manvantara is equal to 71 Mahayugas, which is 71 imes 4320000 = 30672000 solar years. At the end of each Manvantara, there is an evening of 1728000 solar years, and at the beginning of the Kalpa, there is an evening of the same. Thus

1 K L P = 14 M N V N T R + 15 S T Y Y U G K E S M A N A S A N D H Y A E N = 14 imes 71 M H A Y U G + 15 S T Y Y U G = 664 M H A Y U G + 15 imes 40 M H A Y U G [where Satya Yuga = M H A Y U G imes 40] = 664 + 6 M H A Y U G = 1000 M H A Y U G

or = 1000 X 1200000 = 1200000000 divine years

or = 1000 X 4320000 = 4320000000 solar years

These values of Mahayuga or Manvantara are found in the Dharmashastras of Manusmriti etc.; but Aryabhata gave some different values of the Yugas in his Aryabhatiya. According to them, in 1 Kalpa there are 14 Manu and in 1 Manu there are 72 Chaturyugas, each Yuga being equal to Satya Yuga, Treta, Dwapara and Kali Yuga.

Thus it is clear that according to Aryabhata, in one Kalpa there are 14 imes 72 = 1008 Chaturyugas.

1. See the commentary on the 23rd verse.

SOURCE PAGE 35

> इत्थं युगसहस्रेण भूतसंहारकारकः ।
> कल्पो ब्राह्ममहः प्रोक्तं शर्बरी तस्य तावती ॥२०॥

Translation—Thus, there is a Kalpa of one thousand Yugas which is like one day of Brahma. There is a night of Brahma of the same time, in which the creation dissolves॥ 20॥

Scientific Commentary—Lord Krishna has shown a very good picture of Brahma's day and night in the eighth chapter of the Śrīmadbhagavatgītā:—

> "सहस्रयुगपर्यन्तमहर्यद् ब्रह्मणो विदुः ।
> रात्रिं युगसहस्रां तां तेऽहोरात्रविदो जनाः ॥ १७ ॥
> अव्यक्ताद् व्यक्तयः सर्वाः प्रभवन्त्यहरागमे ।
> रात्यागमे प्रलीयन्ते तत्रैवाव्यक्त संज्ञके ॥ १८ ॥
> भूतग्रामः स एवायं भूत्वा भूत्वा प्रलीयते ।
> रात्यागमेऽवशः पार्थ प्रभवत्यहरागमे ॥ १९ ॥

Meaning "(17) The men who know the night (tattvam) understand that (Krita, Treta, Dwapara, and Kali) these four Yugas are a Great Yuga of a thousand Yugas, and such a time is one day of Brahma, and such a night is one night of those thousand Yugas.

(18) ‘When the beginning of Brahma's day occurs, all (subtle) things are created from the unmanifest, and when night occurs, they merge back into the same unmanifest. (16) O Partha! This same community of ghosts repeatedly arises and disappears, meaning that whether there is a desire or not, as soon as night occurs, it merges, and as soon as day occurs (again), it is born.’*

> परमायुश्शतं तस्य तयाऽहोरात्रसङ्ख्यया ।
> आयुषोऽर्धमितं तस्य शेषात्कल्पोऽयमादिमः ॥ २१ ॥
> कल्पादस्माच्च मनवः षड् व्यतीतावस्ससंघयः ।
> वैवस्वतस्य च मनोः युगानां त्रिघनो गतः ॥ २२ ॥
> अष्टाविशाद्युगादस्माद्यातमेकं कृतं युगम् ।
> अतः कालं प्रसंख्याय सङ्ख्यामेकत्र पिण्डयेत् ॥ २३ ॥

Translation—(21) Brahma's age is one hundred years from those days and times. At this time, half of Brahma's age has passed, and this first Kalpa of the remaining half age

Gita Rahasya page 734. 735

SOURCE PAGE 36

is. (22) Six Manu have passed with the treaties of this Kalpa, and seven Manu have passed with the great Yugas of the universe, and (23) the Satya Yuga of the eighteen great Yugas has passed;
Therefore, these numbers should be collected for the calculation of time. (21-23)

Scientific Commentary—The age is considered to be one hundred years. The supreme age of man is one hundred solar years, the age of the deity is one hundred divine years, and one divine year is 360 solar years. Similarly, one should understand that the age of Brahma is one hundred Brahma years. One Brahma year is 360 Brahma days, and one Brahma day (Ahoratra) is two Kalpas or 2000 Mahayugas. From this calculation, 50 years of Brahma have passed, and the first day (Kalpa) of the twenty-first year has begun, in which 6 Manus, 27 Mahayugas, and the age of Satya Yuga of the 28th Mahayuga with their conjunctions have passed. It should be remembered that this is being said at the end of Satya Yuga; as revealed by 'Alpavashistetu krite' etc. in the second verse. From this calculation, the time from the beginning of the current Kalpa to the end of the Satya Yuga of the 28th Mahayuga is found to be:—

100 years Kalpa's beginning evening 17,28,000 6 Manvantaras = 6 × 30,67,20,000 1,84,03,20,000 6 Manvantaras' evenings = 6 × 17,28,000 1,03,68,000 27 Mahayugas of the seventh Manvantara = 27 × 43,20,000 11,66,40,000 28th Mahayuga's Satya Yuga 17,28,000

In this time, 5023 years of Kali Yuga have passed in 1676 AD; therefore, if one wants to know the time from the beginning of the Kalpa until now, one should add the 12,96,000 solar years of the end of Satya Yuga, the 8,64,000 solar years of Dvapara, and the 5023 years of Kali Yuga to the solar years of the end of Satya Yuga. In this way, the time from the beginning of the Kalpa until now is 1,97,26,46,023 solar years. The calculation of time in the mantra is done in this way, the time part of which is:-

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SOURCE PAGE 37

As per Aryabhata, from the beginning of the Kalpa to the beginning of the Kali Yuga, there are twenty-eight parts... in the present, in this current era, in this current lunar month, in this current fortnight, in this current day, in this current month, joined with the Moon... etc.

Time from the beginning of the Kalpa to the beginning of the Kali Yuga according to Aryabhata's view = 6 Manu + 27 Chaturyuga + 3/4 Chaturyuga†

> Calculation: 6 Manu + 27 Chaturyuga + 3/4 Chaturyuga = 432 + 27 + 3/4 Chaturyuga = 456 1/4 X 43,20,000 Solar Years = (460 - 1/4) X 43,20,000 = 1,68,72,00,000 — 10,80,000 Solar Years = 1,68,61,20,000 Solar Years.

If 5023 years are added to this, the number of solar years passed from the beginning of the Kalpa in 1676 will be obtained. Brahmagupta, Bhaskaraacharya, etc., did not accept Aryabhata's view. According to their view, the number of solar years from the beginning of the Kalpa till now is the same as according to the Sūrya Siddhānta.

> बीते हुए ६ मन्वन्तरों के नाम हैं—(१) स्वायम्भुव,(२) स्वारोचिष,(३) औत्तमि,(४) तामस,(५) रैवत और(६) चाक्षुष । वर्तमान मन्वन्तर का नाम वैवस्वत है । वर्तमान कल्प को श्वेत-वाराह-कल्प कहते हैं ।

> ग्रहक्षदेवदैत्यादिसृजतोस्य चराचरम् ।
> कृताद्रिवेदा दिव्याब्दाः शतघ्ना वेधसो गताः ॥२४॥

Translation-To create the universe of beings, celestial bodies, gods, demons, humans, animals, birds, mountains, trees, etc., Brahma took 47,400 divine years or 47,400 X 360 = 1,70,64,000 solar years. (Therefore, the entire creation was prepared after so much time from the beginning of the Kalpa)॥ 24॥

Scientific Commentary—the view of the Sūrya Siddhānta is that creation did not happen in the beginning of the Kalpa. For this, Brahma had to be assigned 1,70,64,000 solar years.

> †काहो मनवोढ(१४) मनुयुग श्ख(७२) गतास्ते च(६) मनुयुग छना(२७) च । कल्पादेर्युगपादा ग(३) च गुरु दिवसाच्च भारतात्पूर्वम् ॥ ३ ॥ आर्य-
> भटीय प्रथम पाद, बा० उदयनारायण सिंह द्वारा संपादित ।

SOURCE PAGE 38

The second Aryabhata also has this opinion; but it seems from the mathematics of Brahmagupta, Bhaskaraacharya etc., that they did not accept this opinion, because they have calculated from the beginning of the Kalpa to know the positions of the planets; but the Surya Siddhānta has left the time it took for the creation of the universe in planetary mathematics.

> पश्चाद्व्रजन्तोऽतिजवान्नक्षत्रैस्सततं ग्रहाः ।
> नीयमानाश्च लम्बन्ते तुल्यमेव स्वमार्गगाः ॥२५॥
> प्राग्गतित्वमतस्तेषां भगणैः प्रत्यहं गतिः ।
> परिणाहवशाद्भिन्नाः तद्वशाद्भानि भुञ्जते ॥२६॥
> शीघ्रगस्स्वर्क्षमल्पेन कालेन महताऽल्पगः ।
> तेषां तु परिवर्तन पौष्णान्ते भगणास्मृतः ॥२७॥

Translation—(25) The planets, moving with the swift constellations, always fall behind in their respective orbits by the same amount while moving towards the west; (26) Therefore, they appear to move towards the east and their daily motion also appears different according to the circumference of the orbits; Therefore, the lunar cycle also completes in this different time, that is, (27) in a short time of

swift movement and in a long time of slow movement. The cycle that is completed at the end of Revati is called bhagaṇa. ॥25-27॥

Scientific Commentary—the principle of the motion of the planets is explained in these three shlokas; therefore, these are of great importance. From these, it is revealed in brief what the ancient Indian astronomers thought about the planets. It is stated in the 25th shloka that all the stars seen in the sky are moving towards the west with the planets; but due to the very fast movement of the constellations, the planets lag behind and appear to move in the easterly direction. The speed of their movement towards the east is the same, but because the extent of their orbits is different, their motion also appears different. The secret of this will be revealed in the following figure—suppose that the inner circle in the given figure is 10 inches and the outer is 15 inches, and suppose that two ants moving at a speed of 1 inch per second orbit the inner and outer circles from the positions of x and y, which are in a straight line with the center; then it is clear that the ant moving on the outer circle will complete one orbit in 15 seconds and the ant moving on the inner circle will complete one orbit in 10 seconds. This proves that when moving with the same linear motion, the revolution of orbits of different sizes will take different times. But in the 27th shloka

SOURCE PAGE 39

It has been said that fast-moving planets complete their orbits in a short time and slow-moving planets in a long time. There seems to be some opposition here, but this is not an opposition; because the motion stated in the first verse is the planimetric motion and the motion in this verse is the angular (Angular velocity). One revolution is of 360 degrees; therefore, one inch of the outer circle makes an angle of 360/15 = 24° at the center and one inch of the inner circle makes an angle of 360/10 = 36°. Therefore, although the rectilinear motion of ants is the same at 1 inch per second, although their angular motion is different per second. The outer ant moves at 24° per second and the inner at 36°. Therefore, it is clear that the fast-moving ones will complete their orbits in less time and the slow-moving ones in more time.

In the 27th verse, the definition of revolution is also given. When the planet moves towards the east, starting from the end of the constellation Revati, and after completing one revolution, it comes back to the end of Revati, then it completes one revolution (constellation group which is 27). Therefore, revolution is also called a circuit.

SOURCE PAGE 40

According to this principle, it must be assumed that there is a special relationship between the distances of the planets and their periods of revolution. The farther a planet is, the longer its period of revolution (time taken to complete a revolution) is. This relationship is stated very briefly here. Its full explanation will be given in the twelfth chapter called Bhūgolāyādhya, comparing the principles of Indian and Western astrologers.

When the linear motions of all the planets are assumed to be equal, it can be easily proven what is the mutual relationship between the distances of the planets; because it is not difficult to know how many days each planet takes to complete one revolution. When it is known that Saturn takes a revolution in a coarse manner in 30 years and the Sun in 1 year and their linear motions are equal, then it is self-evident that the orbit of the Sun is 30 times the orbit of Saturn; because in 30 years the Sun travels 30 times its orbit and Saturn is able to complete its orbit only once. Therefore, the orbit of Saturn = 30 X

orbit of the Sun. That is, the distance of Saturn from the Earth is 30 times the distance of the Sun. Similarly, the distances of the other planets can also be known.

It has been known from the current observations that the relationship between the mutual distances of the planets is not so simple, nor are their linear motions the same. Now it is proven that the distance of Saturn from the Earth is almost 10 times the distance of the Sun from the Earth.

> विकलानां कला षष्ट्या तत् षष्ट्या भाग उच्यते ।
>
> तत्त्रिंशता भवेद्राशिः भगणो द्वादशेव ते ॥ २८ ॥

Translation—A skill of 60 zodiacs, a part or fraction of 60 zodiacs, a quantity of 30 parts or fractions, and a revolution of 12 zodiacs॥28॥

Scientific Commentary—these are the units for measuring angles. The entire zodiac cycle is called revolution. If twelve equal parts are made of this revolution, each part is called rashi. The thirtieth part of rashi is an degrees, the sixty-th part of degrees is arcminutes, and the sixty-th part of arcminutes is arcseconds. Revolution and rashi are used only for that celestial cycle in which the sun appears to revolve around the earth in its plane and other planets revolve somewhat shifted. However, degrees, arcminutes, and arcseconds are also used in measuring other angles. Nowadays, the way to write degrees in short is that a small circle slightly to the right of the digit indicating the magnitude of degrees.

SOURCE PAGE 41

They write, for writing the arts, they draw a crooked line slightly to the right of the number and a similar two slanted lines for the obliquity; for example, if one wants to write 5 degrees 16 minutes and 50 seconds, they write 5°16'50".

There is a proportional relationship in the units of measuring angles and time. The time it takes the Sun to complete one revolution is one year, the time it takes for one zodiac sign to move is one month, the time it takes for one degrees to move is one day, the time it takes for one arcminutes to move is approximately one hour, and the time it takes for one arcseconds to move is approximately the same as a moment.

> युगे सूर्यज्ञशुक्राणां खचतुष्करदार्णवाः ।
> कुजार्किगुरुशीघ्राणां भगणाः पूर्वयायिनाम् ॥ २६ ॥
> इन्दो रसाग्नि त्रित्रीषुसप्तभूधरमार्गणाः ।
> दस्त्रत्र्यष्टरसाङ्काक्षिलोचनानि कुजस्य तु ॥ ३० ॥
> बुधशीघ्रस्य शून्यर्तुखाद्रित्यङ्कनगेन्दवः ।
> बृहस्पतेः खदस्राक्षिवेदषड्वह्नयस्तथा ॥ ३१ ॥
> सितशीघ्रस्य षट्सप्तत्रियमाश्विखभूधराः ।
> शनेर्भुजङ्गःषट्पञ्चरसवेदनिशाकराः ॥३२ ॥
> चन्द्रोच्चस्याग्निशून्याश्विवसुसर्पार्णवा युगे ।
> वामं पातस्य वस्वग्नियमाश्विशिखिदस्रकाः ॥ ३३ ॥

Translation—(26) The Sun, Mercury, and Venus moving in the retrograde direction in one (Maha) yuga, the revolutions of Mars, Saturn, and Jupiter are also the same number of revolutions as the Moon's

5,77,53,336 revolutions, the 22,96,832 revolutions of Mars (31), 176,37,060 revolutions of Mercury, and 3,64,220 revolutions of Jupiter. (32) The revolutions of Venus are 70,22,376 revolutions, the revolutions of Saturn are 1,46,568 revolutions and (33) the revolutions of the Moon are 4,88,203 revolutions and the revolutions of the Moon moving towards the left (west) are 2,32,238 revolutions. 26-33॥

Scientific Commentary—It would be good to state here what is the old tradition for writing numbers among us. Some words have been fixed to write the numbers one, two, three, four, etc., in verse. Those or their synonyms are placed in order according to the rule 'ankānām vāmato gatiḥ', that is, the word indicating the number written in the unit place is written first, then the word indicating the tens place.

SOURCE PAGE 42

The word indicating the number being referred to, and the word indicating the place in the hundreds, are kept in sequence. For example, if 325 is said, the word indicating 5 is written first, such as Panch, Ishu, Margana, etc., and after it, the word indicating 2 is written, such as Dvi, Ashvi, Yama, etc., then the word indicating 3 is written, such as Tri, Agni, Shikhi, etc. In this way, 325 can be written as Panchaavashikhi or Ishuyamagni. This is the same procedure for writing numbers in Sūrya Siddhānta, Brahmasphuta-Siddhānta, and Siddhanta-Shiromani. The ways of writing numbers in the Aryabhatiya of the first Aryabhata and the Mahā-Siddhānta of the second Aryabhata are different from this.

The number of revolutions of the planets in one great age are given above according to the Sūrya Siddhānta. According to the theories of Aryabhata and Brahmagupta, there are some differences in the values of the great age revolutions, and the values of the revolutions that are known with the help of sūkṣmayaṃtroṃ nowadays do not match any theory but have only a slight variation. On the next page, we give a comparative view of the values of the Sūrya Siddhānta, the Brahmagupta-Siddhānta, and the modern revolutions, from which it will be apparent how little the difference is between the revolution periods calculated by our ancient astrologers and the revolution periods calculated by the modern sūkṣmayaṃtroṃ. The time taken for a planet to complete one revolution or orbit is called the revolution period. The method for calculating this is as follows according to the theory: divide the number of revolutions of that planet in one great age by the solar years of the great age, then 1 revolution period (in solar years) will be obtained. Now, you can write this either in decimal fractions or in Sawan days. The easiest way to calculate the revolution period in Sawan days is to divide the number of Sawan days in the great age by the great age revolution, the remainder is the number of Sawan days. Make the rest into hours, minutes, etc. Just as there are 60 minutes in 1 hour, so it can be imagined that there are 60 vipala in 1 pala and 60 prativipala in 1 vipala.

In these verses, the new words used are graha-shighra, chandrocch, and pata. To understand these words, we first need to know the imaginations of our sages whom they considered in relation to the motion of the planets. They considered the Earth to be stationary and believed that the Sun, Moon, planets, and constellations revolve around the Earth. However, this alone was not enough to accurately calculate the motions of the planets; therefore, they imagined the graha-shighras. They saw that the two planets Mercury and Venus remain around the Sun; therefore

SOURCE PAGE 43

According to the Sun Siddhanta, according to Brahmagupta's Siddhanta, according to modern discovery, day hour month leap day hour month leap day hour month leap day 365 15 31 31.4 365 15 30 22.5 365 15 22 56.87 Moon 27 16 1.6 27 16 0 25 27 16 17 58.866 Lunar Birth 3232 5 37 13.6 3232 44 2 45

3232 34 31 14.088 Lunar Eclipse 6764 23 56 23.5 6762 15 14 14.7 6768 16 44 24.000 Mercury 87 58 10 557 87 58 11 43.7 87 58 624.866 Venus 224 41 54 50.6 224 41 52 34.7 224 42 2 47.486 Mars 686 56 50 5.87 686 52 52 33.7 686 58 46 2.518 Jupiter 4332 16 14 20.6 4332 14 24 16.2 4332 35 51 7.46 Saturn 10765 46 23 4.1 10765 48 54 51.2 10756 13 10 57.46

— Quoted from Marathi Indian Jyotishastra pg. 203.

SOURCE PAGE 44

To know their position, one must first know where the sun is. Once the position of the sun is known, it is determined that Mercury will be about 281 degrees ahead or behind the sun, and Venus will be about 472 degrees ahead or behind the sun. This is why in the 26th verse, the great revolution of the sun, Mercury, and Venus is stated as equal. However, to know how much Mercury or Venus is ahead or behind the sun, it is not possible without knowing their epicycles or the positions of their epicycles. The period of revolution of their epicycles is the same as the time it takes for Mercury or Venus to revolve around the sun according to present opinion. Therefore, the period of revolution of the epicycles of Mercury or Venus should be understood as the time it takes for these constellations to revolve around the sun, not the zodiac, but around the sun. The epicycles of Mars, Jupiter, and Saturn are separate from the epicycles of the other two planets. Their period of revolution is considered the same as that of the sun. This means that the epicycles of Mars, Jupiter, and Saturn are the points that complete a revolution of the entire constellation cycle in 1 year. But the sun also completes one revolution of the constellation cycle in 1 year; therefore, the epicycles of Mars, Jupiter, and Saturn remain near the sun. Special discussion will be made about these epicycles in the second chapter.

Mandaucch or uchch—it is stated above that the moon completes 4,88,203 cycles in one Mahayuga, therefore, according to the theory of the Sun, one Bhagankal is 3232 Sawan days, 5 ghadi, 37 pal, and 1306 vipal. The moon's high is that point of the lunar orbit which is at the farthest distance from the other points of the lunar orbit than the Earth. When the moon is at this point, due to being very far, it appears extremely small in size and its motion is also very slow. In the lunar orbit, there is also a point at 1800° from the lunar high which is very close to the Earth. When the moon comes to this point, its motion becomes the most intense and due to being very close, its size also appears very large. Due to this irregular motion of the moon, it is not easily possible to tell what its position will be at any time. Above this also

1. The greatest distance of Mercury from the Sun is between 160° 12' and 280° 48'.

2. The greatest difference of Venus from the Sun is not more than 47°. (Outlines of Astronomy by Herschel pp. 281 and 291)

SOURCE PAGE 45

It is stated that the period of revolution of the Moon is 27.32167 mean Sawan days. The position that the Moon takes from this mean position is called the mean position. From this mean position, the Moon sometimes appears a little ahead and sometimes a little behind. The place where the Moon is seen directly is called the true position. The greatest difference between the mean position and the true position is 502’ 30”. The center of the Moon's orbit from the Earth is considered to be at the same difference of such angles of sine. And the Moon revolves around the Earth, revolving around this center. In figure 2, P is the center of the Earth, and C is the center of the Moon's orbit, and P is

Figure 2 Figure 3

The sine of 502’ 30”. The Moon revolves around the Earth while revolving on the circle U. It is clear that when the Moon is at U, it is at a very great distance from P, and when it is at N, it is very near. U is called apogee and N is called perigee. This point U

SOURCE PAGE 46

In the sky, a place does not remain fixed but keeps moving towards the east with a slow motion. The Moon completes one orbit in approximately 3232 Sawan days. The higher or apogee of other planets also move towards the east with even slower speeds. This imagination is not used nowadays. It has been proven by mathematics that the Moon and the Earth and other planets revolve around the Sun, and the path of revolution is not circular but elliptical. Before saying anything about this, it is necessary to state some properties of the ellipse. The ellipse is a figure (Figure 3). The upper axis is called the major axis and the lower axis is called the minor axis, and the point where these two axes meet is called the center of the ellipse. At the center, the minor axis and the major axis have two equal parts. There are two such points on the major axis at equal distances from the center, which if connected to any point P, A, or B of the ellipse, then P + A = B + BA = PN + PI. These two points are called the focus of the ellipse. If the Moon's orbit is assumed, then the position of the Earth will not be there. From point N, the distance of the Moon's orbit is maximum at point U and minimum at point B, so point U will be the higher or apogee of the Moon and point B will be the lower of the Moon. If the position of the apogee is known, the position of the lower can be easily known, as it always remains at 180° from the higher.

Similarly, the Earth, Mars, Mercury, Venus, etc., also revolve in an ellipse around the Sun, and the Sun remains at the focus of these orbits. Why the motion is very slow at a high place and very fast at a low place is due to the variation in the force of attraction. When the planet is at a high place, the force of attraction is very small due to the very large distance, which slows down the planet's motion, and when it is at a low place, the force of attraction is very large due to the very small distance, which makes the planet's motion very fast. Many rules have been known regarding this, which are famous by the name of Kepler's laws; these will be discussed in due place in the second chapter named Chapter on True Positions.

Node—the path on which the Sun moves and revolves the sky in 1 year is called the circle of declination. Similarly, the path on which the Moon moves and orbits the Earth is called the lunar orbit. The circle of declination and the lunar orbit are not in the same plane and are not parallel; therefore, these two orbits appear to meet at two points; just as the strings of two flying kites meet at one point even though they remain very far from each other.

SOURCE PAGE 47

It seems and due to the variation in the motions of those kites, this point does not appear in the same direction. These points are called the nodes of the Moon. The Moon moves in its orbit and remains north of the ecliptic for half the period of revolution and south of the ecliptic for half the period of revolution. When it reaches its node, it either moves towards the north or/or the south of the ecliptic. The node to which it moves towards the north is called the Ascending node and the node to which it moves towards the south is called the Descending node. The Ascending node is also called Rahu and the Descending node is also called Ketu. When the Moon is near these nodes at the time of the full moon or new moon, a lunar eclipse or solar eclipse occurs; that is why it is imagined that Rahu and Ketu are demons caused by the eclipse. Some people consider the tip of the Earth's shadow as Rahu and the tip of the Moon's shadow as Ketu; but this is an illusion.

The positions of these nodes are also not fixed but seem to shift towards the west. The time taken for them to complete one revolution while shifting towards the west is called their period of revolution. Similarly, one should understand the positions of other planets. The reason why they shift towards the west is explained by very subtle mathematics in Physical Astronomy, which will also be explained in this Scientific Commentary in the appropriate place. In Figure 4, if the arc is understood as the path of the sun, i.e., the ecliptic, and the arc is understood as the lunar orbit, then the points R and Q are called the nodes of the moon. When the moon moves ahead of the node in the direction of the arrow, it becomes north of the ecliptic and remains north up to the arc part, therefore the node is called the northern node. Upon reaching point Q, the moon moves south of the ecliptic, therefore Q is called the southern node. The Indian astrologers call the northern node of the moon Rahu and the southern node Ketu. The angle between the ecliptic and the lunar orbit is the angle between the planes of the ecliptic and the lunar orbit, which is approximately 50. This angle of deviation of the moon is discussed in the 68th verse of this chapter.

Here it can be asked how the revolutions given above were known and why there are differences in various opinions. The answer, according to the view of Bhaskaracharya ji, is as follows:-

By the knowledge of the language, by the knowledge of the region, by the knowledge of the sphere, the planets can be heard, by none other. The planets go in their own paths with mean revolutions and epicycle apogees.

SOURCE PAGE 48

The only proof is that. There are many defects in the writings of the learned teachers of mathematics due to the passage of time; then what is the proof? Then it is said that the mathematical section is the proof. The revolutions that are achieved by the means are the accepted ones. However, not. Because the knowledge of the man is possible only to the extent of the man's short life. In the process, the planets are measured by the instrument, up to the revolution. Thus, the revolution of the years are completed by the three-quarters. By the years of the apogee. Therefore, this is not the result of the man. Thus, the accountants, according to the accumulated knowledge, have recognized the advanced accountants to show the skill of the mathematics of the planets and other doubts, they write the books.

Meaning—but this method can only be known by one who has acquired proficiency in the special language of (astrology), knows the positions of constellations, and has heard well about geography and astronomy. The fact that the planets, apogee, epicycle apogee, and node, while moving along their own paths, revolve so much in one kalpa, is proven by Āgama, i.e., traditional knowledge. But due to the passage of more time, Āgamas have become numerous due to the mistakes of writers, teachers, and readers! Therefore, the question arises as to which Āgama should be considered as proof. If it is said that the Āgama which is proven correct according to mathematics should be taken as proof, and the revolutions derived from it should be accepted, this is also not right. Because even a very learned person may only be capable of knowing the method; but he cannot calculate the number of revolutions of the planets from the method. The reason is that man's lifespan is very short and to know the proof, the planet has to be measured daily until the revolution is complete. In this way, one revolution of Shani is completed in 30 years. The revolutions of the apogees are completed in many centuries. Therefore, this work is not suitable for a man. Therefore, wise calculators create other texts by accepting an Āgama that is considered correct at that time and which has been accepted by the calculators who have gained prestige, to show their skill in the mathematics and calculations of the planets, and to remove the

erroneous defects that have arisen by mistake. Their intention is that we should show skill in accurately calculating the positions of the planets, whatever the Āgama may be.

Siddhanta Shiromani Ganitadhyaya page 16-17 (Printed in Calcutta. Second edition)

SOURCE PAGE 49

In relation to the revolution of the Sun, Mercury, and Venus, Bhaskaraacharya says that the number of revolutions of the Sun is equal to the number of years in a Kalpa. Therefore, the period of the Sun's revolution is a year. Mercury and Venus remain sometimes a little ahead and sometimes a little behind the Sun, like a follower. Therefore, their revolutions are also the same as the Sun's revolution.

This method is given to know the period of the Sun's revolution—

Draw a circle on a plane and mark the directions in it. When the Sun is in the Uttarayana position, on the day when it rises a little south of the east, find the Sun by passing through the nail embedded in the center of the circle. After this, one should calculate the rising of the Sun for one year. The Sun will rise 365 times in a year. The last rising will be a little south of the rising place of the first day. Write the difference between these two. Find the Sun by rising on the second day. On this day, it will rise a little north of the east. Also know how much north it rises from the rising place of the previous day. Then, by the ratio, find out that when such a much north moves in 60 hours, then how much time has passed for the first difference. In this way, 15 hours 30 minutes 22 minutes 30 seconds and 365 Sawan days, the Sun rises in the same place as it did at the beginning of the year. Therefore, this time is the period of the Sun's revolution. Then, by the ratio, find out that when there are such Sawan days in 1 year, then how many Sawan days are there in 1 Kalpa years, etc.

What is used nowadays to know the vernal equinox is very much found in the method explained by Bhaskaraacharya ji. The difference is that Bhaskaraacharya ji asked to divide on the horizon circle, and nowadays it is divided on the meridian circle, so that no error can occur due to the lengthening and apparent deviation of light; the second difference is that the instruments of today are very subtle, but in the method explained by Bhaskaraacharya, it was done with a naked eye.

The derivation of the lunar revolution should also be known by dividing the spherical instrument in which the constellation circle, declination circle, obliquity circle, lunar orbit, planetary orbit, etc., remain, and knowing it. There does not seem to be a need to describe this in great detail. It is written only because ancient astronomers also tested the things given in the text by division and accepted what turned out to be correct.

SOURCE PAGE 50

Method of knowing the period of the moon's culmination

One should find the true motion by eclipsing the moon with the sphere instrument every day. On the day when the motion is least, there is no difference in the positions of the mean and true moon. This is the position of the moon's high point. Similarly, while observing daily, when the moon's motion becomes very small again, that same place should be considered the high point. This place remains some ahead of the first place. Knowing how far ahead it goes, one should calculate by the ratio how much the daily motion of the high point is and in how many days one revolution period is completed.

Method of knowing the period of lunar eclipse—while eclipsing the moon daily, one should see on which day the southward deviation of the moon becomes zero. At the time when the deviation is zero, the moon is at the position of the lunar eclipse. Similarly, when the southward deviation of the moon becomes zero in the second revolution, it should be understood that it has reached its node. The position of the node is some west of the first place; that is why it is said that the motion of the node is opposite, i.e., towards the west. Then one should know by the ratio that when the node moves this much in so many days, how much it will move in one day. This should be understood as the daily motion of the node. Similarly, one should also know how many revolutions occur in one kalpa.

Regarding the epicycle apogees of Mars, Jupiter, and Saturn—when they are ahead of the Sun, Saturn, Jupiter, or Mars, the planets remain some ahead of the mean position, and when the Sun is behind, the planets remain behind the mean position; therefore, the scholars imagined that the epicycle apogees of these three remain with the Sun and attract the planets towards themselves, i.e., towards the Sun; therefore, their revolution periods are the same as that of the Sun.

> भानामष्टाक्षिवस्वद्रित्रिद्विद्वयष्टशरेन्दवः ।
>
> भोदया भगणैः स्वैः स्वैरूनाः स्वस्वोदया युगे ॥३४॥

Translation—1 In one Great Age there are 1,58,22,37,828 revolutions of the constellations. Subtracting the Great Age revolution of a planet from the Great Age revolution of the constellation leaves the number of times that planet rises in the eastern horizon in one Great Age. ॥34॥

Scientific Commentary—In the commentary of the 12th verse of the science commentary, the definition of Nakshatra-Ahoratra is given. In one Nakshatra-Ahoratra, the stars move towards the west and a

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SOURCE PAGE 51

They complete the orbit. This orbit is called the sidereal revolution. Therefore, in one great age, the number of sidereal revolutions is equal to the number of sidereal-night.

The preceding part of the above shloka explains the method of knowing how many times a planet rises in the eastern horizon in one great age. Subtract the number of revolutions of the planet from the number of Nakshatra revolutions of a great age by the number of revolutions of the planet in one great age; the remaining number will be the number of times that planet will rise in the eastern horizon in one great age. Suppose we want to know how many times the sun rises in the eastern horizon in one great age. It is stated in the 26th shloka that the sun makes 43,20,000 revolutions in one great age. If this is subtracted from the Nakshatra revolutions of a great age of 1,58,22,37,828, then the remainder is 1,57,76,17,828. The sun rises in the eastern horizon 1,57,76,17,828 times in one great age. However, it is stated in the Scientific Commentary of the 12th shloka that the time between one rise and the next rise of the sun is called a Sawan day. Therefore, according to the 34th shloka, there are 1,57,76,17,828 Sawan days in one great age.

Similarly, the number of rises of the planets can also be known. Its derivation is as follows: If a star rises with the sun on a certain day, then the next day that star rises 3 minutes 56 seconds before the sun. Because in this time, the sun moves one degree towards the west. On the third day, that star will rise 2 times the time difference from the sun, that is 3 minutes 56 seconds earlier, i.e., 7 minutes 52 seconds before. On the fourth day, it will rise three times earlier than that, and on the 16th day, it will rise fifteen times earlier, i.e., 56 minutes and 1 minute less 1 hour earlier. In this way, after lagging behind, on the

361st day, i.e., after 360 sidereal days, that star will rise 24 minutes less, 24 hours earlier, and after 366 sidereal days, the whole 24 hours, i.e., 1 day earlier, when the sun and that star will be together again. Therefore, in the time it takes the constellation to complete 366 revolutions, the sun rises 1 time less and completes one revolution in that time. Therefore, the sun rises 366-1 times in one revolution period (in 1 solar year). Similarly, one should understand about the rises of other planets.

> भवन्ति शशिनो मासाः सूर्येन्दुभगणान्तरम् ।
>
> रविमासोनितास्ते तु शेषाः स्युरधिमासकाः ॥३५॥

the difference between the great revolutions of the Sun and the Moon

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SOURCE PAGE 52

> उतने ही चान्द्र मास एक महायुग में होते हैं। एक महायुग में जितने सौर मास होते हैं उनकी संख्या को महायुगीय चान्द्र मासों की संख्या से घटा देने पर शेष अधिमासों की संख्या होती है ॥३५॥

Sūrya Siddhānta—the time when the Sun and the Moon are in conjunction is called Amavasya. At this time, the Moon and the Sun are very close. The time from one Amavasya to the next Amavasya is called Chandra-masa. Therefore, if one wants to know how many Chandra-masas are in one Mahayuga, one should first know how many times the Sun and the Moon are in conjunction in one Mahayuga. For this, it is sufficient to find the difference in the Mahayuga revolutions of the Sun and the Moon. Because it is easily known that if two boys start revolving around a round field and if one boy makes 5 revolutions and the other 3, then if both start running from the same place at the same time, they will meet each other 5—3=2 times in an hour. For this, the example of the movement of the hands that tell the hours and minutes of a clock is very appropriate. At twelve o'clock, both hands remain together, meaning their conjunction remains. After this, both start revolving and meet for the first time at 1 hour and 561/9 minutes. The second time they meet at 2 hours and 10110/9 minutes, the third time at 3 hours and 1640/9 minutes, the fourth time at 4 hours and 2150/9 minutes, the fifth time at 5 hours and 2730/9 minutes, the sixth time at 6 hours and 3280/9 minutes, the seventh time at 7 hours and 3820/9 minutes, the eighth time at 8 hours and 4340/9 minutes, the ninth time at 6 hours and 4620/9 minutes, the tenth time at 10 hours and 5440/9 minutes, and the eleventh time exactly at twelve o'clock. For these eleven conjunctions, the minute hand had to make 12 revolutions and the hour hand had to make 1 revolution. Therefore, the number of conjunctions is the difference between their revolutions (12—1). Similarly, the number of Mahayuga Chandra-masas

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= Mean Lunar - revolution — Mean Sun - revolution = 5, 77, 53, 336 — 43, 20, 000 = 5, 34, 33, 336
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The calculation of months and years is done from the lunar month and the solar year. In one solar year there are 12 solar months and 365.25875 mean summer days, but 12 lunar months are equal to 354.36705 mean summer days; therefore

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SOURCE PAGE 53

Twelve lunar months in a solar year is 10.86170 mean Sawan days less than the solar year; therefore, this difference becomes equal to one lunar month in thirty-three months. In the solar year in which this difference becomes equal to one lunar month, there are 13 lunar months. Then one lunar month is left

in the name of Adhimasa or Malmasa. If this is not done, the festivals and celebrations to be observed according to the lunar months will begin to fall in different seasons like Muslim festivals. It is stated in the above shlokas that the number of lunar months to be left in the name of Adhimasa is equal to the number of lunar months out of the total number of solar months in one Mahayuga. Therefore, the number of Adhimasa in one Mahayuga.

= Mahayugiya lunar months—Mahayugiya solar months

```
= ( 5, 34, 33, 336 - 43, 20, 000 ) X 12
```

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= 15, 63, 336.
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> सावनाहानि चान्द्रेभ्यो द्युभ्यः प्रोज्झ्य तिथिक्षयाः ।
>
> उदयादुदयं भानोर्भूमिसानवनवासराः ॥ ३६ ॥

Translation—From the number of lunar days in one Mahayuga, subtracting the number of Sawan days of the Mahayuga, the number of days that decay, that is, which are not counted, is obtained. The time between one sunrise and the next sunrise of the sun is called Bhūmisāvan day.॥36॥

Scientific Commentary—There are 30 tithis in one lunar month, so if the number of great lunar months is multiplied by 30, then it will be known how many tithis are in one great age. It has already been stated that how many Sawan days are in one great age and in one Sawan day, the calculation of the same tithi is done; therefore, the number of tithis is not calculated as much as the number of Sawan days; therefore, these are called decayed or unmeasured tithis.

The latter part of this verse has been explained many times. Here only it is further explained that the Sawan day is also called Bhumisawan day.

> वसुद्वयष्टारिद्ररूपाङ् कसप्ताद्रितिथयो युगे ।
> चान्द्राः खाष्टखखव्योमखाग्निखर्तुनिशाकराः ॥३७॥
> षड्वह्नि त्रिहुताशाङ्कतिथयश्चाधिमासकाः ।
> तिथिक्षया यमार्थाश्विद् यष्टव्योमशराश्विनः ॥३८॥

SOURCE PAGE 54

> खचतुस्कसमुद्राष्टकुपञ्च रविमासकाः ।
> भवन्ति भोदया भानुभगणैरूनिताः कहाः ॥३६॥

Translation—(37) 1,57,76,17,828 Sawan days in one Great Yuga; 1,60,30,00,080 lunar days, i.e., dates; (38) 15,63,336 lunar months; 2,50,82,252 lunar days and (39) 5,18,40,000 solar months exist. The number of land Sawan days is obtained by subtracting the number of revolutions of the Sun from the rising of the Nakshatra.37-36॥

Scientific Commentary — The numbers given in these verses are examples of the three verses preceding them.

The Great Yuga numbers given above are not the same in all the theories. A little difference is found. The number of Sawan days in one Great Yuga is found in different theories, which can be easily known from the table below:-

According to the prevailing Surya Siddhanta, 1,57,76,17,828, according to the Panchasiddhant of Surya Siddhanta, 1,57,76,17,800, according to Aryabhatiya, 1,57,76,17,500, according to Brahmasphut Siddhanta, according to Siddhanta Shiromani, 1,57,76,16,450, according to Mahasiddhanta, 1,57,76,17,542

> अधिमासोनरात्र्यर्क्षचान्द्रसावनवासराः ।
> एते सहस्रगुणिताः कल्पे स्युर्भगणादयः ॥४०॥

Translation—Multiplying the numbers (mentioned) of lunar months, lunar days, Nakshatras, lunar and Sawan days, and the revolutions of the planets by one thousand yields the numbers of Kalpa॥ 40॥

Scientific Commentary—A Kalpa of 1000 Mahayugas is therefore, by multiplying the numbers of Mahayuga revolutions, etc., by 1000, the numbers of Kalpas can be known.

> प्राग्गतेस्सूर्यमन्दस्य कल्पे सप्ताष्टवह्नयः ।
> कौजस्य वेदखयमा बुधस्याष्टर्तुवह्नयः ॥४१॥
> खखरन्ध्राणि जैवस्य शौक्रस्यार्थगुणासवः ।
> गोऽग्नयश्शनिमंदस्य पातानामथ वामतः ॥४२॥

SOURCE PAGE 55

> मनुदस्त्रास्तु कौजस्य बौधस्याष्टाष्टसागराः ।
> कृताद्रिचन्द्रा जैवस्य त्रिखाङ्काश्च भृगोस्तथा ॥४३॥
> शनिपातस्य भगणाः कल्पे यमरसर्तवः ।
> भगणाः पूर्वमेवात्र प्रोक्ताः चन्द्रोच्चपातयोः ॥४४॥

Translation—(41) Moving towards the east in one Kalpa, the apogee of the Sun is 387 revolutions, the apogee of the Moon is 204 revolutions, the apogee of Mercury is 368 revolutions, (42) the apogee of Jupiter is 600 revolutions, the apogee of Venus is 535 revolutions and the apogee of Saturn is 36 revolutions. The motion of the nodes is to the west. In one Kalpa, the node of Mercury is 214 revolutions, the node of Mercury is 488 revolutions, the node of Jupiter is 174 revolutions, the node of Venus is 603 revolutions and (44) the node of Saturn is 662 revolutions. The number of the high and node revolutions of the Moon has already been stated in the first (33rd) verse.

Scientific Commentary—the apogees and the motion of the nodes of the planets are very subtle. None of them complete 1 revolution in 1 Mahayuga, therefore their numbers are given according to the Kalpa.

The discussion of what apogee and node are has been done along with the high and node of the Moon. There is no clear proof of how these institutions were known; but there is no doubt that this information must have been obtained from long-term observation. Some Western scholars say that this knowledge must have come to the Indian astrologers from the Greeks; but this is their illusion as seen in the table below. It is necessary to state here that the number of these revolutions is very different from the numbers known by the modern subtle instruments.

In modern texts, the annual motion of these apogees and nodes is given; therefore, the value of the annual motion can be calculated by comparing it with the annual motion of the Kalpa revolutions of the Sūrya Siddhānta. The annual motion is calculated as follows:—the number of revolutions in a Kalpa is divided by the solar-year of the Kalpa, a different number of revolutions is obtained. Multiplying this by 360, multiplying the fraction by 60, and multiplying the art by 60 yields the annual motion. For example, if the Sun makes 387 revolutions in one Kalpa, then in one year it

387 X 360 X 60 X 60 equals 1161 4320000000000000

Vikala will proceed.

SOURCE PAGE 56

In this way, the annual motion of the apogees and nodes of other planets can be known in arcminutes.

The sign of plus(+) before the digits in this table indicates that the motion is towards the east, and the sign of minus(-) before the digits indicates that the motion is towards the west.

The numbers given in the second column are given with respect to Aries, meaning they show how much the difference between the apogees and nodes of the planets from Aries (Vernal Equinox) changes per year.

But Aries is moving. It shifts towards the west by 50.26 arcminutes per year; therefore, if the annual difference between the apogees and nodes from Pisces is to be known, 50.26 arcminutes should be subtracted from the digits in the second column. By doing this, the differences that will come are the annual differences between the apogees and nodes from Pisces. By doing this, it is seen that the motion of Venus's apogee is towards the west, meaning Venus's apogee is shifting towards the west and not towards the east among the stars. In our practical astrological texts, the motion of Aries is considered 60 arcminutes. Because our year is 8 arcminutes more than the actual sidereal year; therefore, if the annual motion of the Vernal Equinox is taken as 60 arcminutes and 60 arcminutes is subtracted from the digits given in the second column, the difference that comes is what is written in the fourth column. The digits that appear in this column should be compared with the digits of the Sūrya Siddhānta.

The apogees of the planets according to modern subtle measures of the nodes or vernal equinox from the true Aries, from our theories, from the apogees of the Sūrya Siddhānta are 1 2 3 4 5, arcminutes arcminutes arcminutes arcminutes Ravi +61.5 +11.24 +1.5 +0.1161 Mangal +65.7 +15.46 +5.7 +0.0612

Taken from Loomis' Practical Astronomy

Indian Astronomy Text page 207

SOURCE PAGE 57

1 2 3 4 5 Mercury +56.1 +5.81 - 3.6 +0.104 Jupiter +56.6 +6.65 - 3.1 +0.27 Venus +47.0 - 3.24 - 13.0 +0.1605 Saturn +66.6 +16.31 + 6.6 +0.0117 Mars's sign +25.0 - 25.22 - 35.0 - 0.0642 Mercury, +40.2 - 10.07 - 16.8 -0.1464 Jupiter,, +34.3 - 15.60 - 25.7 -0.0522 Venus,, +26.7 - 20.50 - 30.3 -0.2706 Saturn,, +30.7 - 16.54 - 26.3 -0.1686

Chapter on Mean Motions

Chapter on True Positions

By combining the numbers of the 4th and 7th columns, it will be known how close our principles were to the actual situation and how far Talam was. The calculation made by Keropant is according to modern values, so from their calculation, the actual position of the meridians and parallels is revealed. This comparison also shows that our theoreticians knew all these values from independent experience and not from the Greeks or other countrymen, as some Western scholars believe.

> षण्मनूनां च संपिण्ड्य कालं तत्सन्धिभिस्सह ।
> कल्पादिसन्धिन सार्धं वैवस्वतमनोस्तथा ॥४५॥
> युगानां त्रिघनं यातं तथा कृतयुगं त्विदम् ।
> प्रोज्झ्य सृष्टेस्ततः कालं पूर्वोक्तं दिव्यसंख्यया ॥४६॥
> सूर्याब्दसंख्यया ज्ञेयाः कृतस्याऽन्ते गता अमी ।
> खचतुष्कयमाद्यग्निशरनन्दनिशाकराः ॥४७॥

The late

SOURCE PAGE 58

1 2 3 4 positions of apogees and nodes (according to Sūrya Siddhānta) in 205 degrees, calculation of positions of apogees and nodes (sine) 5 difference, calculation of the amount of degrees of art of the shadow of the sun according to the calculation of the amount of degrees of art of the eclipse, calculation of the amount of degrees of art of the eclipse of the sun Siddhānta 277 2715 + 082152530 -535 Mars, 48141 +150413632530 -66 Mercury, 74171026 -133571832 -6100 -3832 Jupiter, 503852116 +03851575110 -47 Venus, 621321646 -21114616181250 -23118 Saturn, 851272637 -835728457230 -545 Node of Mars, 186105 +156152602530 -656 Mercury, 101802044 -63402650100 -165 Jupiter, 253021641 -5462221210 -311 Venus, 234012646 -35420361250 -536 Saturn, 30133025 +0123728630 -8532

SOURCE PAGE 59

> वही(वर्तमान) सत्ययुग के अन्त तक सौर वर्षों में संख्या हुई जो
> १,६५,३७,२०,००० है ॥४५-४७॥

The science commentary—what has been said in the last 22nd and 23rd shlokas is repeated here. In the science commentary of these shlokas, the method for knowing the number of solar years from the beginning of the Kalpa to the end of the present Mahayuga Satya Yuga is explained, which is 1,67,07,84,000. If 1,70,64,000 solar years of the creation are subtracted from this, the remainder is 1,65,37,20,000. So many solar years have passed from the beginning of creation to the end of Satya Yuga.

> अत ऊर्ध्वममी युक्ता गतकालाब्दसङ्ख्यया ।
> मासीकृता युता मासैःमधु शुक्लादिभिर्गतैः ॥४८॥
> पृथक्स्थास्तेऽधिमासघ्नाः सूर्यमासविभाजिताः ।
> लब्धाधिमासकैर्युक्ता दिनीकृत्य दिनान्विताः ॥४९॥
> द्विष्ठास्तिथिक्षयाभ्यस्ताश्चान्द्रवासरभाजिताः ।
> लब्धोनरात्रि रहिता लङ्कायामार्धरात्रिकः ॥५०॥

Translation—(48) Add up the number of solar years that have passed after the end of Satya Yuga up to the end of Satya Yuga from the solar years mentioned above. The sum will be the number of solar years up to Ishtaka. Make this a month, that is, multiply by 12. Also add the number of months that have passed from Chaitra Shukla Pratipada to Ishtaka to the number of months. (46) Keep this number in two places, multiply it by the number of lunar months of Mahayuga and divide by the number of solar months of Mahayuga, the remainder will be the number of eras from the beginning of creation to Ishtaka. Add this remainder to the number of months kept in the second place. The sum is the number of lunar months from the beginning of creation to Ishtaka. Multiply this by 30 to make lunar days (tithis) and add the number of days that have passed in the current month up to Ishtaka, so that the number of days that have passed from the beginning of creation to Ishtaka will be known. (50) Keep the number of these days in two places. Multiply one by the number of Mahayuga decay-days and divide the product by the number of Mahayuga days, the remainder will be the number of decay-days from the beginning of creation to Ishtaka. Subtract this from the number of days kept in the second place, the remainder will be the number of Sawan-days up to the half-night of Lanka, which is one less than the number of half-nights. (48-50)

SOURCE PAGE 60

Scientific Commentary—When it is to be known what are the positions of the planets at a certain auspicious time, then it should first be known how many Sawan days have passed from the beginning of creation until that auspicious time. When the number of Sawan days is known, then the positions of the planets are easily known by Trairashi. Because all the planets were together at the beginning of creation and how many revolutions they make in one great age and how many Sawan days there are, this has also been stated; therefore, by knowing how many revolutions occur in the Sawan days of the great age, in the Sawan days up to the auspicious time, the position of the planet comes out. The method of knowing the number of Sawan days up to the auspicious time, which is called Ahargana, is explained in the above three shlokas.

Example—How many days passed from the beginning of creation until the spring Panchami (Magh Sudi 5) of Vikram Samvat 1676?

From the beginning of creation to the end of Satya Yuga = 1,65,37,20,000 solar years Treta,,,, = 12,96,000 solar years Dvapara,,,, = 8,64,000,, until the beginning of Chaitra Shukla 1 of 1676 AD = 5,023*,, Therefore, from the beginning of creation until the beginning of Chaitra Shukla 1 of 1676 AD = 1,65,58,85,023 years = 23,47,06,20,276 solar months* From the beginning of Magh Shukla 1 to the beginning of Magh Shukla 10 of Chaitra Shukla 1 = 10 lunar months, therefore from the beginning of creation until Magh Shukla 1 of 1676 = 23,47,06,20,286 mean months When there are 5,18,40,000 solar months in one great age, then there are 15,63,336 adhimas; therefore, the number of adhimas in 23,47,06,20,286 mean months is 23,47,06,20,286 X 15,63,336 = 72,13,84,726 adhimas 5,18,40,000 Therefore, the number of lunar months from the beginning of creation until Magh Shukla 1 of 1676 AD is 24,16,20,05,012 = 7,25,76,01,50,360 dates ∴ 7,25,76,01,50,365 dates up to Magh Sudi 5

* In the year 1676 AD, when the Aries solstice will occur, these will be called mean solar years or months.

SOURCE PAGE 61

But in one great age there were 5343336 lunar months and 2,50,82,252 intercalary days, therefore the number of intercalary days in 24,16,20,05,012 lunar months = 24,16,20,05,012 ×2,50,82,252/5343336

= 11,35,60,18,761

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5343336
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∴ Magha Sudi 5 from the beginning of creation

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Number of Sawan days = 7, 14, 40, 41, 31, 604
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Midnight before Magha Sudi 5

7,14,40,41,31,603

This rule for knowing the number of Sawan days for a certain time is very troublesome, and even a small mistake can lead to hours of wasted effort. Therefore, in practice, such large time calculations are not done, but tables and tables are made by which this calculation is done easily. Further in this book, the instruction on making calculations from the end of Satyug is also given. If one wants to perform the calculation in an easy manner, the method written below can be used:--

It has been stated that in one Mahayug there are 43,20,000 solar years and 1,57,76,17,828 Sawan days.

Therefore, in one solar year there are ↟rac(1,75,76,17,828)(43,20,000) Sawan-days, that is 365.2587564815 Sawan-days. To know the number of calculations for a certain time, find the calculation for the day of Aries transit in which it is. To do this, multiply the number of Sawan days of a year by the solar years from the beginning of creation to the desired year. In the above example, the day of Aries transit of 1676 AD is 1,95,58,85,023 solar years since the beginning of creation; therefore, 365.2587564815 X 1,95,58,85,023 Sawan days, that is 7,14,40,41,31,321.7700260745 Sawan days have passed. However, the true Aries transit occurs 2.1706644 Sawan days before the mean transit. Therefore, if 2.1706644 Sawan days are subtracted from the calculations up to the mean Aries transit, then the true Aries transit occurs at 7,14,40,41,31,319.5663-

*In the book, dates are mentioned instead of lunar months so that the calculation is pure; but multiplication and division are more, so lunar months are taken. It is possible that a day's error occurs, which can be corrected by finding the day of the week.

SOURCE PAGE 62

316745 Sawan days have passed. This means that after so many mean Sawan days from the beginning of creation, the Aries solstice occurred in Lanka in 1676 AD.

Therefore, the calculations up to midnight of the day of the Aries solstice were 7,14,40,41,31,320. Now it should be seen how many days after the Aries solstice the Spring Panchami fell.

For this, it is first necessary to know which Tithi was on the day of the Aries solstice.

There are 1 Chandramaasa equal to 26053058794607 Sawan days. Therefore, if the calculations of the Aries solstice are divided by these Sawan days, the remainder obtained will be the number of lunar months passed from the beginning of creation to the Aries solstice, and the remaining will be the Sawan days of the current lunar month. If this remainder is multiplied by 30 and then divided by the Sawan days of the lunar months, the obtained result will be the number of Tithis.

Doing this, the number of Tithis at the time of the Aries solstice comes to 1652576, which is 2 more than Vaishakh Badi from the Purnimanta calculation and 2 more than Chait Badi from the Amanta calculation.

Now it should be known how many Sawan days passed from Vaishakh Badi 2 to Magha Sudi 5. Therefore, it should first be seen how many tithis passed at this time. There are 6 lunar months from Vaishakh Badi 3 to Magha Badi 2, because no Malmas occurred this year, and there are 17 tithis from the beginning of Magha Badi 3 to Magha Sudi 5, that is, until the end of the fourth. Therefore, from Mesh Sankranti to the end of Magha Sudi 4, 6 × 30 + 17 tithis, that is, 287 tithis passed.

But 30 tithis = 26·53058794607 mean Sawan days; therefore 287 tithis

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= 287 X 26.53058794607 X 1 / 30 = 282.506261350736 = 282.51sawandinstulroopse
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But the arc of the half-night of Aries solstice = 7,14,40,41,31,320 therefore the arc from Makha Sudi 5 to

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7, 14, 40, 41, 31, 602.51
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That is to say, 7,14,40,41,31,603 Sawan days passed until the first half-night of Magha Sudhi 5, which is obtained from the first method. For this calculation, the digits up to the eleventh place of the decimal are taken because the multiplier (from creation

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SOURCE PAGE 63

The number of solar years till now is in billions. If one wants to find the age from the beginning of the leader or Kali Yuga for which there is an order ahead, then taking the number of days of Sawan months and solar years to seven decimal places will be sufficient.

Now it remains to check whether this number is pure or not. For this, it is sufficient to check which day will begin after so many days from creation. If the day comes out correctly or there is a difference of 1 day, then it should be understood that the age is correct, otherwise it is impure. Its procedure is given in the following verses.

> सावनो द्युगणस्सूर्याद्दिनमासाब्दपास्ततः ।
> सप्तभिः क्षयितश्शेषः सूर्याद्यो वासरेश्वरः ॥५१॥
> मासाब्ददिनसंख्याप्तौ द्वित्रिघ्नौ रूपसंयुतौ ।
> सप्तोद्धृतावशेषौ तु विज्ञेयो मासवर्षपौ ॥५२॥

Translation—(51) One should know by counting from the Sun the day-lord, month-lord, and year-lord from the number of Sawan days. Divide this number by 7, the remainder is the day-lord starting from the order of days of the Sun. (52) If this (number of Sawan days) is divided sequentially by the numbers of months and years, and the quotients are multiplied sequentially by two and three, and one is added to each product, and the sums are divided by 7, then the remainder is the month-lord and year-lord in the order of days of the Sun. (51-52)

Scientific Commentary—The name of the day is given on the name of the planet which is considered the lord of the first hour (hora) at the beginning of the day. The planet which is the lord of the first hour is also considered the lord of that day. Similarly, the day that falls at the beginning of the Sawan month is

considered the lord of that Sawan month, and the day that falls at the beginning of the Sawan year is considered the lord of that Sawan year. For example, the lord of the first hour of Sunday is Ravi, the lord of that day is Ravi, the lord of the month that starts with Sunday is Ravi, and the lord of the year that starts with Sunday is also Ravi.

*Sawan should not be considered as Shravan. The month of 30 Sawan days and the year of 12 Sawan months is called Sawan year.

SOURCE PAGE 64

To know which planet is the lord of which hour(hour), one should understand the sequence in which the lords of the hours change. Saturn is farther from the Earth than all the planets, then Jupiter is nearest to it, Mars is nearest to Jupiter, Sun is nearest to Mars, Venus is nearest to Sun, Mercury1 is nearest to Venus, and the Moon is nearest to Mercury. The lords of the hours change in this sequence. If the lord of the first hour is Saturn, then the lord of the second hour is Jupiter, the third is Mars, the fourth is the Sun, the fifth is Venus, the sixth is Mercury, the seventh is the Moon, the eighth is Saturn, and so on in sequence. However, the name of the day should be Saturday on the day when the lord of the first hour is Saturn. Therefore, the lord of the second hour of Saturday is Jupiter, the lord of the third hour is Mars, etc. In this way, the same sequence of lords restarts after every seven hours. Therefore, the lord of the 22nd hour of Saturday is Saturn, the 23rd is Jupiter, the 24th is Mars, and the lord of the hour after the 24th should be the Sun. But this 25th hour is the first hour of the next day whose lord is the Sun, so after Saturday comes Sunday. The lord of the second hour of Sunday is Venus, the third is Mercury, the fourth is the Moon, and so on in sequence, the lord of the 11th and 18th 25th hours is also the Moon. But the 25th hour is the first hour of the next day, so the next day falls under the name of the lord of this hour, which is Wednesday. In this way, the naming of other days has also been done2.

The reason why after Saturday comes Sunday and after Sunday comes Monday and after Monday comes Tuesday is now clear. In the order of the planets, Saturn is the fourth planet from the Sun, the Moon is the fourth planet from the Sun, Mars is the fourth planet from the Moon. Therefore, this rule has been established that the lord of each fourth planet, counted from Saturn, is the lord of the next day.

1. Earth is farther than Mercury and Venus, but in our astrological texts, Venus is considered farther. The reason for this is that the planet which is as far as much, completes its revolution in the same amount of time. This is also the belief of our astrologers, but they have taken this distance from Earth, and modern astrologers have taken it from the Sun.

2. This order of days is found in all countries. But the origin of their names, as done here, is worth finding out whether it is the same elsewhere or not.

SOURCE PAGE 65

The Maspati—if the first day of a Sawan month is Sunday, then the first day of the next Sawan month will be the 31st day from Sunday because the Sawan month is 30 days. But the 31st day from Sunday is Tuesday, which is the third day of the fifth week. Therefore, the lord of the second Sawan month is Mars. The first day of the third Sawan month is 31 days from Tuesday, which is the third day of the fifth week starting from Tuesday, which is Thursday. Therefore, the lord of the third Sawan month is Jupiter. Similarly, the lord of the fourth Sawan month is Saturn, the lord of the third day from Thursday, and the lord of the fifth Sawan month is the lord of Monday from Saturday, which is Som, and the lord of

the sixth Sawan month is Mercury and the lord of the seventh Sawan month is Venus. This sequence will continue from the eighth Sawan month. Therefore, the first day of the Sawan month that comes on the third day in the order of days and its lord is the lord of that Sawan month. Now if one looks closely, it will be known that the order of the Maspatis is as follows according to the order of the planets—Sun, Mars, Jupiter, Saturn, Moon, Mercury, and Venus; then Sun, Mars, Jupiter, Saturn, etc. If this cycle begins from the Moon, then their order will remain the same as the order in which they are considered sequentially distant from the Earth.

The lord of the year—the day from which the monsoon year begins is considered the lord of that year. If the beginning of the first monsoon year is on a Sunday, then the beginning of the second monsoon year will be on the 361st day from Sunday, which is the fourth day of the week after 51 weeks, which is Wednesday. Therefore, the lord of the second monsoon year will be Mercury. The beginning of the third monsoon year will be 361 days after the second monsoon year, so it will be the fourth day from Wednesday, which is Saturday, whose lord is Saturn. Therefore, the lord of the third monsoon year will be Saturn. Similarly, the lord of the fourth monsoon year will be Mars, the fourth day from Saturday. The lord of the fifth monsoon year will be Venus, the fourth day from Tuesday. The lord of the sixth monsoon year will be the Sun, the fourth day from Friday. The lord of the seventh monsoon year will be Jupiter, the fourth day from Monday. And the lord of the eighth monsoon year will be the Sun, the fourth day from Tuesday. In this way, the same sequence will begin from the eighth monsoon year. The sequence of these lords is as follows: Sun, Mercury, Saturn, Mars, Venus, Moon, Jupiter; then Sun, Mercury, Saturn, etc. Therefore, if the sequence is matched according to the days, the coming monsoon year's

SOURCE PAGE 66

The first day will be the fourth from the first day of the Sawan year. And if the order of the planets is combined, starting from Saturn, every third planet is the lord of the year. These things are written in the form of sutras in the 78th and 76th shlokas of Bhūgolādhhyāya:-

> मन्दादधःक्रमेण स्युश्चतुर्था दिवसाधिपाः ।
> वर्षाधिपतयस्तद्वत्तृतीयाश्च प्रकीर्तिताः ॥ ७८ ॥
> ऊर्ध्वक्रमेण शशिनो मासानामधिधपाः स्मृताः ।
> होरेशाः सूर्पतनयादधोऽधः क्रमशस्तथा ॥ ७६ ॥

Sūrya Siddhānta, Bhūgolādhhyāya

Both ways of knowing the year lord, month lord, day lord, and hora lord are shown by two pictures (Figure 5 and 6).

SOURCE PAGE 67

By knowing this rule of the names of the days and the relations of the astrologers and the month-makers, the derivation of the 51st and 52nd shlokas can be easily understood.

The number of days that have passed is the remainder when the day (Sawan day) that has come until the desired time is divided by seven. Since the creation began on Sunday, Sunday is the first day of the week and Saturday is the previous day, i.e., the seventh day. Therefore, if 5 remain, it should be understood that it is Jupiter's day, whose midnight completes that day because Jupiter is the fifth day of the week. As in the previous example, when the number of the day is 7, 14, 40, 41, 31, 603, when divided

by seven, the remainder is 1. Therefore, the day for which the day was calculated is the first day of the week, Sunday. However, this day is until the midnight before Basant-Panchami, so Basant-Panchami will be Monday.

To know the month-makers, the desired day should be divided by 30, and the remainder is the number of Sawan months from the beginning of creation. Multiply these Sawan months by two and add one and divide by seven, because the order of the month-makers changes on the third day according to the days, and the same order begins again after seven months. The remainder is the lord of that month that is running on the same day of the week. As in the above, when the day is divided by 30, there are 23, 81, 34, 71, 53 Sawan months and 13 Sawan days. When the number of these Sawan months is multiplied by two and one is added, it is 47,62,66,42,107. When divided by 7, the remainder is 3. Therefore, the first day of the running Sawan month was Tuesday. Therefore, the lord of this month is Mars.

To know the year lord, divide the Savana months which are taken from 360 or above by 12, which gives the number of Savana years passed. Multiply these by three and add one and divide by seven because the order of the year lords changes according to the day and the same order begins again after seven years. The remainder (of weeks) is the lord of the Savana year of that same day because the week begins with Sunday.

As in the example above, when the ahargana is divided by 360 or the Savana months are divided by 12, the number of past Savana years is 1,68,44,55,621.

SOURCE PAGE 68

Multiplying this by three and adding one gives 5,65,33,67,734. Dividing this by 7 leaves a remainder of 1. Therefore, the start of the Sawan year was on a Sunday, and the lord of this year is Ravi.

This is the method for finding the year lord according to the Sūrya Siddhānta. In all the calendars of today, the year lord (varshapati) is considered the lord of the day when Chaitra Shukla Pratipada occurs, and the minister of the year is considered the lord of the day when Mesha Sankranti occurs. Meghesh is the lord of the day when the Ardra constellation appears, etc., and the result for the entire year is calculated with this idea. The rule in the Makaranda Sārini is different from the Sūrya Siddhānta:--

On the day of the full moon of Chaitra, this day is the king. On the day of Aries conjunction, this day is the minister. On the day of Cancer conjunction, this day is the supreme ruler. On the day of Libra conjunction (this) day is the lord of rasa. On the day of Aries conjunction, this day is the lord of nira. Moisture entry

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SOURCE PAGE 69

Divase yo vāraḥ sa meghādhipah. Dhanuḥ sankranti divase yo vāraḥ sa pashchimādhyāhipah* ॥

The practice of Sawan year and Sawan month is not known anywhere nowadays. Therefore, I do not know what use the rules given in Sūrya Siddhānta for finding the rain-master and month-master are. If any learned know, please inform. From the thirteenth verse, as I have written in its commentary, it is implied that the actual year is the solar year. Then what is the need to find the year-master and month-master according to Sawan year and Sawan month?

> यथा स्वभगणाभ्यस्तो दिनराशिः कुवासरैः ।
> विभाजितो मध्यगतो भगणादिग्रहो भवेत् ॥५३॥
> एवं स्वशोघ्रमन्दोच्चा ये प्रोक्ताः पूर्वयायिनः ।
> विलोमगतयः पातास्तद्वच्चक्राद्विशोधिताः ॥५४॥

Translation—(53) Multiply the number of *ahargana* by the great revolution of any planet and divide the product by the great Sawan days. The revolutions that result are the ones that the planet has completed according to its mean motion (from the beginning of its creation). Multiply the remainder by 12 and then divide by (great Sawan days) to find the number of quantities that the planet has completed in the current revolution. Now, multiply the remainder by 30 and divide by the number of great Sawan days to find the number of degrees (parts) that the planet has completed in the current quantity, etc. (54) In this way, the positions of the quick and slow revolutions mentioned before can also be known. The motion of the nodes is backward (towards the west), so by subtracting the quantities of nodes that are in the angular positions of the angular art (arcminutes) from the whole cycle, i.e., from 12 quantities, the remaining ones are the positions of the nodes॥ 53-54॥

Scientific Commentary—These verses explain the method by which the mean positions of the planets at a desired time are known. In short, it means that when a planet moves the revolution in a great period (in great Sawan days), how many revolutions it will make in the desired time of Sawan days. Therefore, this rule can be expressed in the manner of the three problems as follows:-

Great Sawan days: *ahargana*:: Great revolution: desired revolution

Venkateshwar Press's printed Makarand table of 1960 page 476

SOURCE PAGE 70

If 's' is taken as the great divisional savana day, 'a' as the desired degree, 'b' as the great revolution, and 'ba' as the desired revolution, then in short, it will be written as:-

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ba = a x b / s
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This is a fraction, which when simplified, the integer that results is the number of revolutions of the planets that have been completed up to that time, and when multiplied by 12 and simplified, the integer that results is the elapsed amount, and then when multiplied by 30 and simplified, the numerator of the current amount will be obtained. If the art of the aspect is also to be known, then one must keep simplifying by multiplying by 60.

This rule applies to all preceding planets, epicycles, and apogees. If the position of the nodes of a planet is to be known, the amount, numerator, aspect, and declination that result from the above method should be subtracted from the 12 zodiacs because the motion of the node is opposite, so it will move on the zodiac in reverse order. If it is calculated from mathematics that a certain node is at 2 degrees 3 minutes 5 aspects of 'ba' revolution, then it should be counted in reverse from the beginning point of Aries, that is, from the last point of Pisces, Aquarius, and Capricorn, by 3 minutes 5 aspects, that is, at 26 degrees 55 aspects of Capricorn. Therefore, if 2 degrees 3 minutes 5 aspects are subtracted from the 12 zodiacs, then 6 degrees 26 degrees 55 aspects will result, which means that it is at the 26 degrees 55 aspects of the tenth zodiac after the 6 zodiacs of the zodiac.

> द्वादशघ्ना गुरोर्याता भगणा वर्तमानकैः ।
> राशिभिस्सहिताश्शुद्धाषष्ट्या स्युर्विजयादयः ॥५५॥

Translation—Multiply the elapsed revolutions of Jupiter by 12 and add the serial number of the zodiac in which the current revolution of Jupiter is, to the product, and when the sum is divided by 60, the remaining is the serial number starting from the victory of the century of that serial number, it should be understood. 55

Scientific Commentary—the time in which a body moves in one mean motion is called a century. Therefore, in one revolution of Jupiter (in 433203206 Sawan days) there are twelve centuries, and in one century there are 36102672 Sawan days. Therefore, it is clear that a century is smaller than the solar year by 4023203 Sawan days and larger than the Sawan year by 102672 Sawan days.

SOURCE PAGE 71

1 Victory 21 Pramādi 41 Śrimukha 2 Jai 22 Ānanda 42 Bhāva 3 Manmath 23 Rākṣasa 43 Yuvā 4 Durmukha 24 Anal(Nala) 44 Dhātā 5 Hemalamba 25 Pingala 45 Iśvara 6 Vilamba 26 Kālayukta 46 Bahudhānya 7 Vikārī 27 Siddhārthī 47 Pramāti 8 Śārvari 28 Raudra 48 Vikram 9 Plava 26 Durmati 49 vṛṣa 10 Śubhakṛta 30 Dundubhī 50 Citrabhānu 11 Śobhana 31 Rudhirodgārī 51 Subhānu 12 Krodhī 32 Raktākṣa 52 Tāraṇa 13 Viśvāvasu 33 Krodana 53 Pārthiva 14 Parābhav 34 Kṣaya 54 Vyaya 15 Plavanga 35 Prabhava 55 Sarvajita 16 Kīlaka 36 Vibhava 56 Sarvadharī 17 Saumya 37 Śukla 57 Virodhi 18 Sādhāraṇ 38 Pramoda 58 Vikṛti 19 Virodhakṛta 36 Prajāpati 56 Khara 20 Paridhāvī 40 Angira 60 Nandan

Varahamihira in the Brihatsaṁhitā considers the beginning of the cycle of years not from Victory but from the 35th year of Prabhava. This practice is still prevalent today. It is necessary to discover when this practice began. There are many smaller divisions in the cycle of years, the beginning of which also starts from Prabhava. Their discussion will be in the final chapter called Mānavādhyāya.

1. Published in 1884 AD by Naval Kishore Press and translated by Pt. Durgaprasad Ji, page 60—

SOURCE PAGE 72

When it is to know which era is running at a certain auspicious time, then it should first know according to the 53rd verse what is the mean position of Jupiter at that time. It is very difficult to know the mean planet by taking the difference from the beginning of creation; therefore, if the difference is taken from the beginning of Kali Yuga, it will be more auspicious; because the beginning of the victorious era has also started from this time. Those who know the method of decimal fractions will be even more fortunate in using solar years instead of the difference. In this way, the mean position of Jupiter at the time of Aries-transit will be found.

From the beginning of Kali Yuga to the Aries-transit of 1681 AD, 5025 solar years will pass. Knowing how many revolutions Jupiter has completed and in which zodiac sign it will be at that time will reveal the era. In a great yuga, which is 43,20,000 solar years, Jupiter has 3,64,220 revolutions; therefore, in 5025 solar years

= 3,64,220 × 502543,20,000 revolutions = 3,64,220 × 502543,20,000 × 12 eras = 5083 · 60416666 eras are

which means that at the time of Aries-transit of 1681 AD, the 5084th era is running, and 60416666 parts have passed, and 09583334 parts remain.

There is a cycle of 60 eras, so dividing 5084 by 60 gives a remainder of 84 and 44, which means that the cycle of 60 eras has been completed 84 times, and the 44th era of the 85th cycle is running, and the 095833...part of the Aries-transit is remaining to pass.

If one wants to know how many days this will remain in the year sixteen hundred eighty-one Vikram, then this remainder should be multiplied by the number of Sawan days of a year, that is 36102672. Doing this yields 3405684 Sawan days, therefore sixteen hundred eighty-one passed 3405684 Sawan days from the time of the mean Aries equinox.

SOURCE PAGE 73

The end of the earth and the beginning of the year named Ishvara will occur. This is stated earlier that the true Aries solstice occurs 21707 Sawan days before the mean Aries solstice; therefore, the entry of Ishvara will occur on the 3607661 Sawan days or at 36 days 46 hours 8 minutes 40 seconds at the equinoctial shadow.

> विस्तरेणैतदुदितं संक्षेपाद्यावहारिकम् ।
> मध्यमानयनं कार्यं ग्रहाणामिष्टतो युगात् ॥५६॥
> अस्मिन् कृतयुगस्यान्ते सर्वे मध्यगता ग्रहाः ।
> विना तु पातमन्दोच्चान्मेषादौ तुल्यतामिताः ॥५७॥
> मकरादौ शशाङ्कोच्चं तत्पातस्तु तुलादिगः ।
> निरंशत्वं यताश्चान्येनोक्तास्ते मन्द चारिणः ॥५८॥

Translation—(56) The method of knowing the mean positions of the planets has been stated with detail so far; but for practical purposes, this work should be done briefly from the auspicious age. (57) At the end of this Satya Yuga, except for the nodes and apogees, the mean positions of all planets were the same as the beginning of Aries, meaning they had reached the starting positions of the seven planets. (5) The Moon was at the beginning of Capricorn and its node (Rahu) was at the beginning of Libra. The nodes and apogees of the other planets were not at any full degree due to the slow motion; therefore, nothing is said about them. (56-58)॥

Scientific Commentary—The method for finding the mean positions of the planets described in the 45-50 shlokas of this chapter is not practical due to the expansion of mathematics, as is clear from the examples given. Therefore, there is no need to calculate the years from the beginning of creation to the end of the Satya Yuga. The mean positions of the planets should be known by knowing the calculation up to the Ishatkal from the time the current age began, according to the succeeding lines of the 48th shloka and the 46-50 shlokas.

At the time when Surya Siddhanta taught Maya to Purusha, according to the second verse of this chapter, it is the time of the end of Satya Yuga. Therefore, in the 57th verse, there is an indication of making Ahargana from the beginning of Treta, and it is also shown that at this time all the planets were in the same mean position at the beginning of Aries. According to this rule, it is convenient to derive Ahargana from the beginning of Kali Yuga itself in Kali Yuga, which is also prevalent nowadays. Just as all the planets were in the same mean position at the beginning of Aries at the beginning of Treta, so in the beginning of Kali Yuga also at the beginning of Aries.

SOURCE PAGE 74

It was so; because the half Mahayuga from the beginning of Treta to the beginning of Kaliyuga is the time in which all planets complete their revolutions. Yes, because the Moon, due to revolving an odd Mahayuga in one Mahayuga, was not in the beginning of Capricorn but in the beginning of Cancer, but it was in the beginning of Libra. This opinion is of Sūrya Siddhānta. According to Bhaskaracharya1, the mean positions of other planets besides the Sun and Moon in the beginning of Kaliyuga were:-

Mars Mercury Jupiter Venus Saturn Lunar Apogee Rahu* Sign 11 11 11 11 11 4 5 arc 26 27 26 28 28 5 3 Kala 3 24 27 42 46 26 12 Vikala 50 26 36 14 34 46 58-

Until here, the method is described by which the mean position of the planets is obtained at the time of midnight of Lanka or Ujjain. In the following verses, the method for knowing the mean position of the planets at the time of midnight in the countries to the east and west of Lanka will be described.

> योजनानि शतान्यष्टौ भूकर्णो द्विगुणानि तु ।
> तद्वर्गतो दशगुणात्पदं भूपरिधिर्भवेत् ॥ ५६ ॥

Translation—The diameter of the Earth is 1600 yojanas times 800; taking the square root of the square of ten times the product is the circumference of the Earth. 56॥

Scientific Commentary—If the diameter of the Earth is taken as 'v', then its circumference = $\sqrt{v^2 \times 10}$ = v x √10 = v x 301623, which proves that the circumference is 301631 times the diameter. Nowadays, this relation is understood to be correct to four decimal places, which is very different from 301623. But one should not understand that the author of Sūrya Siddhānta intended the exact relationship between the diameter and the circumference

1. Siddhanta Shiromani Ganitadhyaya page 32-33; printed in Calcutta in 1615 AD.

* This position of Rahu is written in the Siddhanta Shiromani printed in Calcutta and in the Siddhanta Shiromani edited by M. M. Panpādev Shastri; but according to my calculation, it should be at 6 degrees 26 minutes 47 seconds 204 arc minutes at the beginning of Kali Yuga, which is obtained by subtracting the position given above from the 12 zodiacs.

SOURCE PAGE 75

The exact relation was not known; because in the second chapter the ratio of semi-circumference and circumference is taken as 3438:21600, which makes the circumference 3014136 times the diameter. Therefore, in this verse, the circumference is taken as √(10) times the diameter, for convenience, to summarize the mathematical operations; just as nowadays when practical methods are used, some take it as 22/7 and some as 3014, and where very fine calculations are required, it has to be taken to five-five-seven-seven decimal places.

What value of this relation was taken in different theories will be known from the following quotations.

The value of the relation between diameter and circumference has been stated in two ways by Bhaskara and the Second Aryabhata, one subtle and the other gross and practical.

The relation between diameter and circumference will be indicated by the π sign as is the custom nowadays, that is, if the diameter is 1, then the circumference is π, when the value of π is practical

SOURCE PAGE 76

As according to 22, 3014, 30142, 301416 etc. as necessary.

The second word 'yojana' in this verse is of great importance. Nowadays people consider yojana to be four kos generally, but the value of kos is not fixed by itself. In some provinces, kos is very small and in some provinces, it is very large. Similarly, the magnitude of yojana is also not fixed. This is the reason why the values of the earth's circumference or meridian are given in different numbers in different theories. It will be revealed from the quotations written below what values of the meridian are given in the theories:-

From the five Siddhāntikas, according to the first view, the geodetic is 10186/10 yojanas, Aryabhata2 and Lalla, 1050, current Sūrya Siddhānta, 1600, Siddhānta Shiromani3, 15813/4, Second Āryasiddhānta (Mahāsiddhānta)4 2106, according to the modern European view5, latitude 7627 miles, latitude 7600,

From the above numbers it is apparent that the yojanas of Varahamihira, Aryabhata, and Lalla are approximately equal, and the yojanas of Sūrya Siddhānta and Siddhānta Shiromani are also approximately equal; but the yojanas of the first three Acharyas are approximately one and a half times the yojanas of these two. Therefore, these two types of yojanas will be compared with the current mile. In our treatises, the Earth is considered perfectly round, so this difference is not kept.

1. Doctor Thibo and Pt. Sudhakar Dwivedi have given the value of the circumference of the earth as 3200 yojanas in verse 18 of the Panchasiddhantika on page 34, which when divided by 3.1416, is 1018.6 yojanas for the diameter of the earth.

2. Aryabhatiya page 10, 5th verse of the first pada.

3. Goladhyaya page 20, Bhuvankosh verse 52.

4. Mahasiddanta page 161, Bhuvankosh verse 35.

5. Sir Robert Ball’s Spherical Astronomy pp, 44.

SOURCE PAGE 77

The equatorial circumference is different from the polar circumference. Therefore, it is appropriate to take the polar circumference for comparison because the Acharyas fixed the magnitude of the circumference by this measure. Therefore,

According to Aryabhata, 1050 yojanas = 7600 miles therefore 1 yojana = 7600 miles 1050 = 1752 miles If there are four kos in 1 yojana then according to Siddhanta Shiromani 1581 yojanas = 7600 miles therefore 1 yojana = 7600 1581 miles = 5 miles therefore 1 kos = 1/4 yojana = 1 1/4 miles

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1cos = 7.52 / 4mile = 1.88mile |
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Nowadays, 1 kos is understood as equal to 2 miles, so the yojana of today is very similar to the yojana of Aryabhata. The kos in Siddhāntashiromṇi is probably the same as the 'gau-kos' (go-kos) of today, which is still prevalent in some provinces.

Now the question remains how the Earth's circumference was measured. Nothing is written on this subject in the Sūrya Siddhānta. Bhaskaracharya says that measure the distance between two places situated on the north-south line in yojanas. Also find the difference between the latitudes of those two

places. Then Trajarah should know that when the distance between two places is such yojanas due to the difference in so many latitudes, what will be at 3600. The proof of this is:

In Figure 7, the planimetric difference of two places (S, Sa) located on the same North-South line should be measured. Then the difference of their latitudes and the angle between them should be known.

1. Sphere Chapter, Bhuvankosha, page 13, verse 14— The difference is that of the meridian, then the analysis is done. The ratio is connected with the circle, and the proof is by the circumferences. Or, Mathematics Chapter, page 56, verse 1— The difference of the time of the sun's path is known here. The proper arrangement is known.

SOURCE PAGE 78

u = North Pole to Mount Meru.s, same as North - South line (meridian)

Two positions.

latitude = < latitude.

lat, = < latitude.

Difference between the two latitudes = < latitude.

Sūrya Siddhānta: 360 °:: Sasa: Circumference

∴ Circumference = 360 ∘ × Sasa

The method of finding latitude will be explained in several ways in the third chapter called Tripraśnādhhyāya.

The circumference is measured in this manner even today; only due to subtle instruments is this work done more accurately now.

SOURCE PAGE 79

> लम्बज्याघ्नस्त्रिजीवाप्तस्स्फुटो भूपरिधिः स्वकः ।
> तेन देशान्तराभ्यस्ता ग्रहभुक्तिर्विभाजिता ॥६०॥
> कलादि तत्फलं प्राच्यां ग्रहेभ्यः प्रविशोधयेत् ।
> रेखाप्रतीची संस्थानां प्रक्षिपेत्तु स्वदेशजम् ॥६१॥

Translation—By multiplying the circumference of the place by the sine of the latitude and dividing by the radius, the true circumference of the place is obtained. The meridian plan of the place should be multiplied by the daily motion of the planet and the product should be divided by this true circumference. (61)(If the daily motion is taken in the arts, then the result is in the arts. If the place is to

the east of Lanka, then this result should be subtracted from the mean planet at the time of midnight of Lanka, and if the place is to the west of Lanka, then it should be added. By doing this, the mean planet (mean positions of the planets) at the time of midnight of the place are obtained.60-61)

Scientific Commentary—According to algebra these shlokas can be expressed as follows:-

Spute circumference = land perimeter X height / radius (7)

Longitude = daily motion of planet X (in degrees) circumference (2)

> mean planet of the time of half-night at one's place = mean planet of the time of half-night at Lanka bowtie difference in longitude. (3)

If the place is east of Lanka, then the negative sign should be taken, and if it is west, then the positive sign should be taken.

To understand its derivation, it should first be known what are latitude, meridian, etc.?

Sine—if the length of any side of a right-angled triangle is divided by the length of its hypotenuse, the resulting quantity is called the sine of the angle opposite to that side. Figure 8 has a right-angled triangle with sides a, b, c; therefore, the sine of the angle a = a/c and the sine of the angle b = b/c. Of the right-angled triangle

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SOURCE PAGE 80

The length of the chord is greater than the length of the side; therefore, the sine of the angle opposite to the side will be less than one, so it is written in decimal fraction. This is the practice of today. In ancient times when the promotion of decimal fractions did not exist, the sine of the angle was written in integers.

To know the sine of an angle, a circle is taken in our theories, whose radius (semi-diameter) is 3438 units and the circumference is 21600 units, so that each unit is equal to one art. Because the circumference is like a circle in which there are 360 degrees or 360 X 60 = 21600 arts. Then two radii are drawn from the center to the circumference such that the angle between them is the same as the angle whose sine is to be known, and a perpendicular is drawn from the point of intersection of the radius and circumference to the other radius. The number of units (arts) of this perpendicular is the sine of that angle. In Figure 6, there is a center; there are three radii, a, a u, and a u u that are drawn from a to the circumference. Perpendiculars from u or u u are dropped on a. The measurement of the radius is taken as 3438 units, and the measurement of u u or u u e in these units will be the angle u a u

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SOURCE PAGE 81

The sine of the angle of u is called sine. Those who are only familiar with the present practice may be confused; therefore, they should understand this difference well. The value of the radius was taken as 3438 because when the circumference is divided into arcs, the value of the radius is 34373/4 arcs, which is considered by the modern fine calculation, the nearest integer of which is 3438. As many arcs as there are in a modern radian, the magnitude of the radius is considered equal to the magnitude of the full arcs.

The meridian—the circle of the Earth that lies midway between the North and South poles at an equal distance and divides the Earth into two equal halves is called the equator; the half of the geography

north of the equator is called the northern hemisphere and the south is called the southern hemisphere. The celestial poles (the point in the sky that is just above the North or South Pole of the Earth, the North Star is usually about 10 degrees from the North Pole) appear on the horizon from this line. Here, the latitude is zero and the longitude is 60°. Therefore, the equator is also called the meridian. In Figure 10, V is the equator. If a circle (Parallel of latitude) is drawn on the Earth parallel to the meridian from a place 'S', its magnitude is called the meridian of 'S'. From the equator

* 1 Radian = 57°2658 = 34:7.748 arc

SOURCE PAGE 82

B = center of Earth. U = North Pole of Earth (Sumeru). D = South Pole of Earth (Kumeru). V = the point of the equator that is just south of S. S = desired place; the North-South line of that place. angle V B S = latitude of S. angle S B U = longitude of S. U D = axis of Earth. S B = distance of Earth's axis from S = sine of latitude of S, according to Siddhānta method.

As you go north or south, the meridian circumference decreases, and at the poles, the meridian circumference becomes zero. Similarly, the latitude increases and the longitude decreases, and at the poles, the latitude becomes 60^(°) and the longitude becomes zero. The figure also shows that the meridian circumference of place 'S' is S

SOURCE PAGE 83

The radius 's bh' is also called the sine of 's' latitude, because the length of s is s bh u, and the side opposite to it is s bh.

From geometry it is proven that the ratio in the perimeters of two circles is the same as in their radii or diameters, therefore,

v: s bh:: v va vi: s sa si

∴ s sa si = v va vi X s bh / v bh = earth circumference X sine of colatitude / radius, when the radius is 3438 and the value of the sine of colatitude is in the arts according to the Siddhānta method, whose table is given in the second chapter.

If the meridian circumference is to be calculated according to the present practice, then s sa si = earth circumference × sine of colatitude (Sine of Colatitude) while the sine of declination is given in decimal (because in this way sine of colatitude = s bh/s bh = s bh/v bh)

Latitude—Figure 11 is a picture of the surface of the half-sphere of the Earth in which the latitudes of the places C, A, S, and SA of the northern hemisphere are the same, therefore these four places'

SOURCE PAGE 84

The true circumference is the same. The north-south lines of these places are in the order of U SP D, U AL D, U SP D and U SA PA D. If there are the locations of Avanti (Ujjain) and Lanka on the AL D line, then it will be called the standard meridian of Bharatvarsha; just as today the north-south line from Greenwich is called the equator of Europe and America. The segment of the true circumference of a place that falls between the north-south line and the mean line of that place is called the longitude (in yojan), as the longitude of S is SA, the longitude of Sa is SA and the longitude of C is CA. Similarly, the longitude of P is PL, the longitude of Pa is PL and the longitude of P is PL. It is also clear from the

picture that although P and S are on the same north-south line, the longitude of P and S (in yojan) are not the same because the true circumference of S is smaller than the true circumference of P (circumference of the earth). If there is a place K on this line, its longitude K (in yojan) will be even smaller. The word longitude is used in the 60th verse according to this definition. However, this definition is not simple and practical. In the 64th verse, the discussion of longitude is further. This is also a simpler definition of longitude; therefore, it is appropriate to explain it here as well.

As many places on the same north-south line have the same noon or midnight at the same time, as in places k, s, p, and kh. But the places that are to the east of this line have noon or midnight earlier, and the places that are to the west have it later. As much earlier the noon is at s from a (the meridian), we call that the time difference of longitude of s. We can express this in units of time; if we write it in minutes, we will call it longitudinal degree or longitudinal-nadi, and if we write it in hours, we will call it longitudinal hour or minute. From this definition, we have the facility to easily reveal the time difference of longitude of k, s, p, and kh all from the same thing, whereas in sums, their longitude differences will have to be written differently.

In this way, the time that is behind the meridian by as much as on the line C will be called the western longitude of C.

The noon or midnight happens because the Earth rotates on its axis from west to east once every 24 hours or once every 60 minutes like a top, due to which celestial bodies like the sun, moon, stars, etc., revolve from east to west.

SOURCE PAGE 85

It appears to be applied. This direct motion of celestial bodies is called the motion caused by the flow of air in our theories. For further convenience, it will be written as if the sun sometimes revolves, because assuming this does not cause any hindrance in the calculation.

In figure 12, the center of the Earth and two places on the P, L, and the equator are shown; P is the semi-diameter of the Earth which is drawn very extended to clarify the figure, in reality, the distance of the sun is twenty thousand times the semi-diameter of the Earth. The sun revolves once in sixty minutes along the path of the Sun. R is a tangent line touching the equator, which is the horizon of P. When the sun is above it, the place P is visible. When the sun comes near the point 'R' above the horizon, it will be sunrise for the inhabitants. At the time P is noon, the sun will be on R. When it comes to R, the inhabitants will see the setting sun, and when it comes to R, there will be midnight in P. Similarly, the rise of the sun from the place L will be seen when it is at R, noon will be when it is at 'Ri', sunset will be...

SOURCE PAGE 86

The time will be when it is on 'Ru' and midnight will be at that time when it is on 'Ra'.

It will remain there. It is clear from the diagram that the time when the sun rises at 'Pa' will be as much later on 'La' than the time it takes for the sun to go from 'Ra' to 'Ra'. However, in going from Ra to Ra, it has to revolve the angle of R to Ra or Pa Bh La because the angle between the two tangents of the circle is equal to the angle between the radii drawn from the points of contact. This fact is even more easily understood from the positions of the sun at noon or midnight; because it has already been stated that when the sun comes to 'Ri', it is noon at 'Pa' and when it comes to 'Ri' it is noon at 'La', so the time it takes for the sun to go from Ri to Ri is the same as the time it will be noon at La being as much later

behind Pa. This time is called the longitudinal time between Pa and La. The longitudinal difference of Pa and La can also be expressed by the angle Pa Bh La and the longitudinal difference can be written in degrees, arc minutes, or arc minutes, whether the unit for expressing the longitudinal difference is in minutes, or in degrees, both are convenient, and where needed, they are written. It is clear that the sun moves one revolution, i.e., 3600 in 60 minutes or 24 hours, so it will move 60 and 150 in 1 hour; therefore, if the longitudinal difference of two places is one degree, there will be a difference of 10 minutes or 4 minutes in their times of noon or midnight. In short, it is written that the longitudinal difference of both is 10, 10 minutes or 4 minutes. Generally, the principle of measuring longitudinal difference is from the meridian. For those places that are to the east of the meridian, their longitudinal difference should be written as 'East' and for those that are to the west, their longitudinal difference should be written as 'West', otherwise there is a fear of confusion.

From the picture, it can also be easily seen that if the mean position of any planet is calculated at the time of midnight of Lanka (Lanka) then it will not be the mean position of the 'P' position at the time of midnight of 'P' because 'P' is to the east of Lanka, so there will be midnight there earlier, and since the planet is always in motion, it will remain some before the mean position of Lanka. How much earlier it will be, this should be determined by Trajarah, that when the planet moves so much in 60 hours, how much it will move in the longitude of 'P'. That which comes, should be subtracted from the mean position of Lanka at midnight. If the position is west of the meridian, then the midnight there will be behind the midnight of Lanka by the same amount as the longitude of that position, and the planet so

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SOURCE PAGE 87

It will move a little further in time. Therefore, what is obtained by Trairashi for the places in the west should be added.

If the difference in longitude is not written in yojanas but in hours or degrees, then the simple form of the rule of the 60th verse will be:-

60 hours: Longitude hour:: Planet's daily motion: Motion in longitude hour

A r th a t d e sh a n t r f l = d e sh a n t r gh a ḍ ī X g r h k ī d ai n i k g t i / 60 gh a ḍ ī (4)

This one equation will yield the result of the first two equations given below the 60th verse and will also be simpler; because for those equations, the longitude yojana has to be made according to the next 64-65 verses from the longitude hour. Therefore, why not do the direct action?

> आगे के श्लोक में यह बतलाया गया है कि मध्यरेखा पर कौन-कौन नगर
> पड़ते हैं ।

> राक्षसालयदेवौकश्शैलयोर्मध्यसूत्रगा ।
> रोहीताङ्कमवन्ती च यथा सन्निहितं सरः ॥६२॥

Translation—(62) Rakshasalaya means the countries between Lanka and Devlok, which are like Rohitaka, Avanti, Kurukshetra etc. (they are on the mean line)॥62॥

Scientific Commentary—In the scientific commentary of the previous verse, the mean line has been well discussed regarding the difference in countries. It is necessary to state here that the place where the North-South line from Ujjain meets the viṣuvat rekhā is called Lanka. This Lanka of astrology is the same Lanka where Ravana lived or some other imagined place, and it is assumed for the convenience of

mathematics, it cannot be said with certainty. Some people call the present Singhal Island (Ceylon) as Ravana's Lanka and Polan Narua as Ravana's capital and estimate that it is a corruption of Patalasya-nagar.

It is worth considering whether Rohitaka is currently Rohtak or if there was some other place with this name; because the country of present Rohtak is given as 'East' in Indian Chronology pg. 160 as 162 seconds, which suggests that Rohtak is on the mean line

1. See Shravana 1680 Vi Madhuri page 6-7.

SOURCE PAGE 88

There are 8 degrees east. There is no means here to know what the latitude of Kurukshetra is nowadays, so it cannot be proven that Kurukshetra is exactly on the meridian or is some east-west removed from it.

In the next three verses, it is stated how the difference in longitude is known from the lunar eclipse and how the distance in yojanas is derived from the difference in longitude:

> अतीत्योन्मीलनादिन्दोः पश्चात्तद गणितागतात् ।
> यदा भवेत्तदा प्राच्यां स्वस्थानं मध्यतो भवेत् ॥६३॥
> अप्राप्य वा भवेत्पश्चात् एवं वाऽपि निमीलनात् ।
> तयोरन्तरनाडीभिः हन्याद्भूषरिधि स्फुटम् ॥६४॥
> षष्ट्या विभज्य लब्धैस्तैर्योजनैः प्राक्तथापरे ।
> स्वदेशपरिधिज्ञेयः कुर्याद्देशान्तरं हि तैः ॥६५॥

The translation -(63) one knows by mathematics when the fully eclipsed moon will begin to emerge from the darkness (from the middle night, how many hours behind), and one should observe by sighting what time (how many hours behind the middle night of one's place) the fully eclipsed moon begins to emerge from the darkness at one's location. If the observed time (known by instrument sighting) is more than the mathematically determined time, then it should be understood that one's location is east of the meridian and (64) if the observed time is less than the mathematically determined time, then it should be understood that one's location is west of the meridian. Similarly, by observing that time, it can be known that at what time the full image of the moon goes into darkness. The difference between the mathematically determined and the observed times is called the longitude of one's place or the longitude in hours (because time is usually written in hours). Multiplying this longitude in hours by the circumference (65) and dividing the product by sixty, the resulting value is the east longitude yojanas (if the place is east) and west longitude yojanas (if the place is west). The longitude of the planets should be calculated from this longitude (according to the method stated in verses 60-61). 63-65

1. Perhaps that is why Bhaskaraacharya did not write Rohtak on the meridian— Yallankojjayiniputa Ruparikurukshetraadi desansprishat.

The sutra is Merugatavudhairnigadita samadhya rekha-bhuvam.

SOURCE PAGE 89

Scientific Commentary—Nimilan=(1) the closing of the eyes, (2) disappearance, (3) the complete image of the moon going into darkness. Therefore Nimilan period that

SOURCE PAGE 90

This is the time when the total or partial eclipse begins. This is also called the totality period.

Emergence = (1) opening of the eyes, (2) appearing, (3) the full moon emerges from the darkness. Therefore, the emergence period is the time when the full image of the moon begins to emerge from the darkness.

Touching period is the time when the moon's image begins to enter the darkness, that is, the time when the actual eclipse begins.

Salvation period is the time when the entire image of the moon comes out of the darkness.

Figure 13 shows the center of the Sun, the center of the Earth, and the tip of the shadow of the Earth. The moon revolves around the Earth in its orbit. When the center of the moon reaches the point in the orbit where 1 is written, the moon's image touches the Earth's shadow, so this position of the moon is the position of the touching period at the time of eclipse. At this time, the moon's image appears to enter the darkness. When the moon reaches the position indicated by the number 2, its entire image is in the darkness. This is the state of the totality period or the occultation period. When the moon reaches the position indicated by the number 3, the moon begins to emerge from the darkness. This is the state of the emergence period. And when the entire image of the moon comes out of the darkness, as indicated by the number 4, then it is the state of the salvation period.

Among these four events, the event that occurs in the sky is also seen on the ground at the same time. However, the sunrise or noon in all places on the ground, from which clocks can be accurately corrected, does not occur at the same time, as demonstrated in the scientific commentary of 60-61 shlokas which define the definition of longitude, therefore the time of seeing an event in clocks of different places is different. Clocks of places to the east of the meridian

2. Instruments for knowing time.

1. If the speed of light is also considered, it would be more accurate to say that any event of the moon is seen one and a quarter seconds behind on the ground.

SOURCE PAGE 91

it remains the same as the time of the country because the sunrise happens earlier here. Therefore, the eclipse seen at this time will be more than the time of the meridian and less in the western places. The time of the eclipse seen at the meridian is obtained even by calculation. Therefore, by knowing from calculation when the eclipse will be seen at the meridian and when it will be seen according to the time of the local clock, it can be easily known what is the difference in the local times of these two places. This difference is called the time difference of the place.

To know the time difference, the discussion on knowing the position of the obliquity period has been done earlier. The reason for this is that at that time the moon is about to come out of darkness, and there is darkness on the Earth's surface, so as soon as the lunar image starts coming into light, it is seen clearly and there is no much inaccuracy in knowing the time. There is some difficulty in knowing at what time the moon completely enters darkness at the time of the culmination period, so some

inaccuracy may occur in calculating the time difference. For the time of the touch period and the moksha period, it may not be known for many moments when the actual event happened, so these are not used to calculate the time difference.

How the time difference is known from the time difference yojana, it should be understood from the scientific commentary of 60-61 shlokas. It is clear that the sun completes the revolution of the Earth in 60 hours, so it revolves in 60 hours around the circumference of the meridian of any place, so the part of the circumference that is the difference from the meridian of that place will be complete in the time difference of that place. Trairashika expresses this as:-

60 hours: time difference hour:: Circumference: time difference yojana.

Some other methods of knowing the time difference need to be discussed briefly here.

Methods of knowing the difference in longitude:—One method has been written above. This is a very old method and when modern-like fine instruments were not built, there could be no other method than this. Most of the methods prevalent nowadays are transformations of this, if there is any difference, it is that nowadays the precise knowledge of the time of celestial events such as eclipses etc. can be known, by which the difference in longitude, as far as possible, can be known with great subtlety. Some inaccuracies remain in the method mentioned above.

SOURCE PAGE 92

Hey, its reason is that the commencement or culmination of the lunar eclipse cannot be seen exactly from the naked eye or a telescope instrument. If two nearby observers sit with their own clocks to see at what time the culmination begins and secretly write down that time when each of them observed the culmination, it is seen that there is a difference of two-three minutes in the times they observed. Perhaps this is the reason for the two-three minute difference in the longitudes of Ujjain, Kurukshetra, and Rohtak, although this is stated on the same north-south line, i.e., the equator, in the 62nd verse. Below, the longitudes of these places are compared with Greenwich:

Longitude (angular) from Greenwich to Ujjain (black) from Ujjain to Ujjain[2] 75°46‘6” East, time of shadow minute second Kurukshetra[3] 76°20’, 0°33‘54” East 0 5 36(2 15.6) Rohtak ‘76°35’, 0°48‘54”, 0 8 6(3 15.6) Kashi[4] 83°3‘4”, 7°16’58”, 12 50(26 8)

From these numbers it is proven that there can be a difference of two three minutes in the method of knowing the difference in longitude from a lunar eclipse.

Second method—just as the Moon revolves around the Earth and it experiences eclipses, similarly, 4, 5 bodies revolve around Jupiter and are seen by the sextant. These are called the moons of Jupiter. When they enter the shadow of Jupiter, they also experience eclipses. The moons of Jupiter are very small and their periods of revolution are also short, so they experience eclipses frequently. Due to eclipses, the time of their hiding and appearing is given in the nautical almanacs according to Greenwich time. Therefore, if the hiding or appearing of the moons of Jupiter in a place according to its local time

1. Godfray’s Treatise on Astronomy. Sixth edition, page 261. 2. Indian Chronology, page 60. 3. Imperial Gazetteer of India. 4. Bharat Bhraman (66 minutes of Kashi are given in the Makrand Sarini, which is 4 minutes less than the above value).

SOURCE PAGE 93

When the time is seen, the difference from the time given in the navigator's almanac is the longitude of that place from Greenwich. But this method is also as crude as the above method because the moment of the moon of Jupiter hiding or appearing cannot be definitely known even by naked-eye observation, but it does not have as much inaccuracy as in the first method.

Third method—by comparing the moments of the appearance and disappearance of the breaking stars with the local times of different places, the longitude can be known with subtlety if the time of the breaking of the stars can be determined beforehand and there is no error in their recognition.

Fourth method: Knowing the longitude by sending news by electricity—If there is such a relation between two places that news can be sent from one place to another by electricity, then the longitude of these two places can be easily known because the time taken for the electrical news to reach is so short that any inaccuracy that may occur is not like an error.

Suppose one wants to know the longitude of Lucknow from Kashi. The observers of both cities should keep the same type of clock, as if the clock of one tells the time of Sawan, then the clock of the other also tells the time of Sawan. Both clocks should be reconciled with their local time so that each clock can accurately tell the local time of its place. Since Kashi is to the east of Lucknow, the local time of Kashi will be ahead of the local time of Lucknow, and the difference between these two is the longitude of Kashi from Lucknow. At the time when the clock of Kashi shows 'S1' time, a signal should be sent from Kashi. At the time when this signal reaches Lucknow, the time on the clock of Lucknow should be seen. If the time in this clock is 'S2' and it is assumed that the signal reached Lucknow at the same moment it was sent from Kashi, then the longitude of Kashi from Lucknow will be proved by the equation written below:-

```
d1 = s1 - s2
```

But the value of the difference in longitude that will be obtained from this equation will be some less than the actual longitude because some time must necessarily pass for the electrical signal to reach from Kashi to Lucknow. If the value of this time is 'y' and the actual longitude from Kashi to Lucknow is 'd', then the form of the aforementioned equation will be:-

d = (s1 + y) - s2 = d1 + y (7)

SOURCE PAGE 94

Because at the time the news reaches Lucknow, the time in Kashi will be 'S1 + y'. To know the value of 'y', one should send a signal from Lucknow to Kashi and then know the local time of both. Suppose that at the time the signal was sent from Lucknow, it was 'Sa2' time in the Lucknow clock, and at the time the signal reached Kashi, it was 'Sa1' time in the Kashi clock, and if it is assumed that the signal does not take any time to reach, then the difference between these two will be the longitudinal difference of Lucknow, which is as follows:-

```
d2 = sa1 - sa2
```

But the value of d2 will be some more than reality because it must take some time to reach the signal, which will be like 'y', so the actual longitudinal difference

d = sa1 - (sa2 + y) = (sa1 - sa2 - y) = d2 - y (2)

(1) and (2) adding the same sides of the equations

```
2d = d1 + d2
```

(d1 + d2) / 2 (3)

which means that the time difference that arrives when sending the longitude from Kashi to Lucknow should be added to the longitude known when sending the reverse signal from Lucknow to Kashi. Then, adding both and dividing by two, the actual longitude will be found. There are also many other ways to know the longitude which are useful to sailors and in which they have to take help from the almanac or the clock obtained from Greenwich; therefore, they are not described at this place.

> वारप्रवृत्तिः प्राग्देशे क्षपार्धेऽभ्यधिके भवेत् ।
> स्वदेशान्तरनाडीभिः पश्चादूने विनिर्दिशेत् ॥६६॥

(66) The direction of the day where the direction is to the east of the meridian is that the beginning of the day is that much earlier than the midnight of that place as the longitude of that place. In the place to the west of the meridian, the direction of the day occurs that much earlier than the midnight of that place as the longitude of that place.

Scientific Commentary—According to this rule, the direction of the day in Kashi is 73 minutes before the time when

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SOURCE PAGE 95

According to the local time in Kashi, at night, it is 12 hours 73 minutes and 12 seconds, which is 12 hours 26 minutes 12 seconds, and in Bombay, the tendency of the day which is 26 minutes west of Ujjain from Ujjain occurs 26 minutes or 11 minutes 36 seconds before 12 o'clock at night. This means that the time tendency of other places in India should be understood at the same time when the midnight occurs on the meridian of India. Therefore, the position of the planets which is mathematically proven at the time of midnight of Lanka or Ujjain is the time at which the day tendency occurs in other places. That is why if one needs to find the planet at the time of midnight of a place, the difference or addition of the longitude should be done.

This opinion is of the Sūrya Siddhānta that the transit of the day occurs at the midnight of Ujjain in all places. Acharyas such as Brahmagupta,1 and Bhaskaracharya have considered the transit from the time when the sun rises in Lanka because according to them, the beginning of creation happened at the time when the first sun appeared in Lanka and the beginning of the first day also happened at this time. This rule is generally prevalent nowadays, yes, followers of the Vaishnava sect3 consider the transit of the day from the midnight and consider the same date during the day for religious acts as the time that remains in the previous half-night, hence their Ekadashi is usually on the day of Dwadashi. Most calendars also give the planet at the time of true midnight.

In these two views, the belief in the direction of the day from midnight is simpler and more comprehensive. All places located on the same north-south line have either midnight or noon always, but they do not occur together at sunrise except in two days of the year. Sunrise is the sine of the sun's declination and the latitude of the places.

> १. जगति तमोभूतेऽस्मिन् सृष्ट्यादौ भास्करादिभिः सृष्टैः ।
> मस्याद्दिनप्रवृत्तिर्दिनवारोऽर्कोदयात् तस्मात् ॥ ३३ ॥
> ब्राह्मस्फुट सिद्धान्त - मध्यमाधिकार ।

> २. लङ्कानगर्य्यामुदयाच्च भानोस्तस्यैव वारे प्रथमं बभूव ।
> मधोःसितादेर्दिन मास वर्ष युगादिकानां युगपत् प्रवृत्तिः ॥१५॥
> सिद्धान्त शिरोमणि गणिताध्याय पृष्ठ ७

3. Madhuri, Volume 2 Number 4 page 437.

SOURCE PAGE 96

According to this, there is some forward and backward movement which will be explained in the third chapter. Then in the countries of the east and west, there is also a great difference in the sunrise time due to the difference in longitude. Due to all these reasons, the day-direction sometimes becomes an hour or more behind or before the sunrise, which is very intricate. But if the day-direction is considered from midnight, then there can be no difference due to the revolution of the sun and the latitudes of the places. Yes, the difference in longitude still has to be made, but this will also happen during the night, so there can be no disturbance. In public practice as well, the morning, evening, or journey of any day is done before the sunrise, which shows that generally the beginning of the day is taken two or three hours before sunrise. This subject has been discussed a lot in texts like Dharma Sindhu' and Nirnaya Sindhu.

Nowadays in European countries, the date changes from midnight and the beginning of the day is considered, that is why English dates also change from midnight. Many people understand that considering the trend of the day from midnight is the English view, but this is a mistake. We also have the rule of considering the trend of the day from midnight.

Until here it has been stated that what is the mean position of a planet at the time of midnight of a place and how it is known. In the next verse it is being stated that if the mean planet is to be found at any other time of the day besides midnight, what should be done.

> इष्टनाडीगुणा भुक्तिषष्ठ्या भक्ता कलादिकः ।

> गते शोध्यो युते गम्ये ग्रहस्तात्कालिको भवेत् ॥६७॥

(If one wants to know the mean motion of a planet at any time other than midnight) multiply the time that is before or after the desired time from midnight by the daily mean motion of the planet (which is convenient to write in terms of arts) and divide the product by 60. If the remainder is not obtained, subtract it from the mean planet of midnight if the desired time has already passed before midnight.

1. From sunrise, before the hour, morning evening, after sunset, the hour. Dharma Sindhu, first section page 2 Decision Sagar Press (1826)

SOURCE PAGE 97

and add if the auspicious time comes behind midnight. By doing this, the instantaneous position of the planet will come out. ॥67॥

Scientific Commentary—it is clear that the mean position of the planet at midnight does not remain at any other time because the planets are constantly moving. Therefore, to know the mean position of a planet at any auspicious time before or after midnight, it is necessary to know how much the planet will shift at that time. This can be easily known from Trāiraśika—

60 clock: Auspicious time:: Daily motion: Motion in the auspicious time ∴ Motion in the auspicious time = Auspicious time × Daily motion/60 clock Therefore, the position of the planet at the desired time = Position of midnight ± Auspicious time + Daily motion/60 clock

If the auspicious time is before midnight, the minus sign should be kept, and if it is after, the plus sign.

This is so clear that there is no need to increase the size of the book by giving an example.

> भचक्रलिप्ताशीत्यंशैः परमं दक्षिणोत्तरम् ।
> विक्षिप्यते स्वपातेन स्वक्रान्त्यंशादनुष्णगुः ॥६८॥
> तन्नवांशं द्विगुणितं जीवस्त्रिगुणितं कुजः ।
> बुधशुक्रार्कजाः पातैर्विक्षिप्यन्ते चतुर्गुणै ॥६६॥
> एवं त्रिघनरन्ध्राकर्रसार्का दशाहताः ।
> चन्द्रादीनां क्रमादेषां मध्यविक्षेपलतिप्तिकाः ॥७०॥

Translation—(68) Due to its node, the Moon shifts more than 270 degrees north or south from the point of its nearest ecliptic. (66) Two-sixths of this shift are due to Jupiter, three-sixths to Mars, and one-third to Mercury and Venus and Saturn due to their respective nodes. (70) Thus, the mean displacements of the six planets (Moon, Mars, Mercury, Jupiter, Venus, and Saturn) are 270, 60, 120, 60, 120, 120 degrees in sequence.॥68-70॥

Scientific Commentary—the description of the node of the Moon is in the commentary of the previous 33rd verse. In Figure 4, the lunar orbit and the ecliptic intersect each other

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SOURCE PAGE 98

It is shown. At the time when the moon is at its node, at that time it appears on the ecliptic circle, and at other times it appears some distance north or south of the ecliptic circle. The time and the amount of distance it is shifted can be easily known by mathematics. At the time when the moon is 60° ahead or behind the node, it is at the extreme difference from the radiance circle. In Figure 4, this extreme difference is indicated by the arc or the chord. This is called the extreme displacement of the moon. Similarly, other planets also move north or south from the ecliptic circle, whose mean displacements are given in 66-70 shlokas. There is a very close relationship between the displacements and the nodes of the planets, which is why our ancient teachers thought that the nodes push the planets north or south.

Sūrya Siddhānta 1 Brahm-Sphuta Siddhānta 3 Maha Siddhānta 4 Siddhānta Darpaṇa 5 Tālmī 6 Ādhunik Chandra 4°30’ 4°30’ 4°30’ 5°6‘0” 5°0’ 5°8‘42” Mangala 10°30’ 10°50’ 10°46’ 10°51‘0” 10°0’ 10°51‘1” Budha 20°0’ 20°32’ 20°18’ 20°44‘0” 7°0’ 7°0‘10” Guru 10°0’ 10°16’ 10°14’ 10°18‘0” 10°30’ 10°18‘42” Shukra 20°0’ 20°16’ 20°10’ 20°18‘0” 3°030’ 3°23‘37” Shani 20°0’ 20°10’ 20°10’ 20°26‘0” 20°30’ 20°26‘36”

SOURCE PAGE 99

From the table above, it will be seen that there is a big difference between the modern values of the deviations of Mercury and Venus and the values given in the theories. The reason for this is that the modern deviation values are heliocentric, that is, they are those that appear from the center of the Sun and the values in our theories are geocentric, that is, they appear when viewed from the center of the Earth. Due to the variation in the observer's locations, there is no big difference in the deviations of those planets that are far from the Sun. However, a big difference does occur in the deviations of the planets near the Sun, Mercury and Venus, which will be clarified by the example below:-

In the given figure 14, R is the center of the Sun, P is the center of the Earth, and B is the ecliptic and A is the orbit of Mercury. From R, the orbit of Mercury forms an angle of B or bar P to the ecliptic, which is 70°10' according to modern opinion. But from P, the orbit of Mercury appears to form an angle of B, which is much less than the angle of B because P B (the distance between Mercury and Earth) if 1 is, then B (the distance between the Sun and Mercury) is only 3871. The cell value can be easily obtained from trigonometry because when the sine of a triangle is divided by the angle opposite to it, the result is the same. Therefore—

sine < bpar – – – – – – = – – – – – – barpab

Atajyā < bapar = bar / pab × jya < barpa = · 38171 × jya70 ° ' 10 ″ =.3871 × .1216 = 0.0472

∴ < Bipar = 2 ° 42 ′

SOURCE PAGE 100

This is the geocentric mean displacement of Mercury according to modern theory, which is 10' more than the mean displacement in Siddhānta Shiromani. The values in Siddhānta Darpaṇa are very close to modern theory.

Similarly, the mean deflection of Venus (around the Sun) is 3023^(prime)37^(primeprime) and the mean difference from the Sun is 0.7233 when the Earth's is 1, therefore if in figure 14, the places b, b are replaced by s, s, and s, s is taken as the orbit of Venus, then the previous relationship will be-

sine < super = .72331 × sine \ 30° 2' 37'' \ =.7233 × .0562 \ =.0428 \ therefore sine < super = 2° 27' \

The Sūrya Siddhānta is more than 11 and less than 1 from the Sūrya Siddhānta Darpaṇa.

From this it appears that according to our old teachers, Mercury and Venus have only 10 or 11 degrees less in their mean motions than the modern values, which are very subtle when looking at the position of that time.

SOURCE PAGE 101

Chapter on True Positions

(Brief Description)

[1-11 shlokas—How do the invisible forms of the time named śīghrocca, mandocca and pāta cause variation in the motion of the planets? 12-13 shlokas—Names of the eight types of motions of the

planets. 14 shloka—Need for explanation for the equality of the positions of the mathematically derived and directly observed planets. 15-16 shlokas—Method of knowing the sine of the twenty-four sections of a right angle. 17-21 shlokas—Table of what the sine of which section is. 22nd shloka's reverse—Method of knowing the versed sine. 23-27 shlokas—Table of what the versed sine of which section is. 28 shloka—The rule for knowing the sine of the inclination from the sine of the ultimate deviation. 29-30 shlokas—How to make the sine of the distance from the mean center and the sine of the distance from the meridian. 31-32 shlokas—Method of knowing the sine of other angles besides the sections given in the table, in proportion to the sine. 33 shloka—If the sine is known, how is the arc or angle to be known? 34-35 shlokas—What are the values of the mean and true perimeters of the positions of the seven planets in odd and even terms? 36-37 shlokas—What are the values of the mean and true perimeters of the five planets in odd and even terms? 38 shloka—What is the magnitude of the mean and true perimeters at a point between the two? 39 shloka—The rule for knowing the mean position. 40-41 purvadh—The rule for knowing the true position. 41 shloka's latter part-42 shloka—Method of knowing the true position. 43-44 shlokas—How to perform the calculation of the mean and true positions of the planets to know their true positions. 45 shloka—The mean position or true position of the center of meṣādi should be added and the position of tulādi should be subtracted. 46 shloka—Need for the calculation of the difference in distance. 47-46 shlokas—Method of knowing the true mean position from the mean true position of the planets. 50-51 shlokas—Method of knowing the true position from the mean true position; when do the curved motions occur? 52 shloka—The cause of the curved motion. 53-54 shlokas—How far do the five planets like bhaumādi deviate from the śīghracca and where do they reach and cease the curved motion. 55 shloka—Due to the different magnitudes of the true perimeters, the curved motions occur at different intervals. 56-57 shlokas—The rule for knowing the deviation of the planets. 58 shloka—Planets

SOURCE PAGE 102

The rule to know the true motion of the planets. 56 shlokas—The rule to know the value of the night of the planets. 60 shlokas—The method to know the zodiac. 61—The method to know the sine of the horizon and the sine of the variable. 62-63 shlokas—The rule to know the magnitude of day and night from the arc of the variable sine. 64 shlokas—The values of the constellations and dates and the method to know in which constellation the planet is. 65 shlokas—The method to know the yuga. 66 shlokas—The method to know the tithi. 67 shlokas—The names and times of the four fixed bodies and their times. 68 shlokas—How many revolutions the seven variable Karan months make. 69 shlokas—Half tithi is equal to one Karan.

In the first chapter named Chapter on Mean Motions, the method to know the positions of the planets according to the mean motion is explained. However, the position of the planet known by this method is very different from where the planet is directly seen. Some rituals need to be performed to eliminate this difference. This chapter explains how these rituals are performed. From these rituals, the position of the planets comes the same as seen in the clear sky by mathematics. Therefore, the name of this chapter is kept as Chapter on True Positions.

> अदृश्यरूपाः कालस्य मूर्तयो भगणाश्रिताः ।
> शीघ्रमन्दोच्चपाताख्या ग्रहाणां गतिहेतवः ॥१॥
> तद्वातरश्मिभिर्नद्धास्तैस्सव्येतरपाणिभिः ।
> प्राक्पश्चादपकृष्यन्ते यथासन्नं स्वदिङ्मुखम् ॥२॥
> प्रवहाख्यो मरुत्तांस्तु स्वोच्चाभिमुखमीरयेत् ।
> पूर्वापरापकृष्टास्ते गतीर्यान्ति पृथग्विधाः ॥३॥
> ग्रहात्प्राग्भगणार्धस्थः प्राङ्मुखं कर्षति ग्रहम् ।
> उच्चसंज्ञोऽपरार्धस्थस्तद्वत्पश्चान्मुखं ग्रहम् ॥४॥
> स्वोच्चापकृष्टा भगणात्प्राङ्मुखं यान्ति यद्ग्रहाः ।
> तत्तेषु धनमित्युक्तमृणं पश्चान्मुखेषु च ॥५॥

Translation—(1) The forms of time named epicycle apogee and node which cannot be seen by the eye and which revolve on the circle of declination due to the motion of the planets. (2) These forms, if (of the planets) are in the east with their right and left hands, pull the planets towards them according to the distance as they are tied to them by ropes of air to the east if they are in the east, and to the west if they are in the west. (3) The air named Pravaha also pushes these planets towards their poles. This is the reason why the planets pulled towards the east or west

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SOURCE PAGE 103

A variation occurs in the motions. (4) If the planet is to the east of the high point and not more than 6 degrees or 1800 from it, it pulls the planet towards the east from the mean place, but if it is more than 1800 away (due to being to the west of the planet), it pulls the planet towards the west. (5) The clear position is obtained by adding as much as the planet has moved towards the east from its respective high points in the mean place and by subtracting as much as it has receded towards the west from the mean place. The added quantities are called the cube quantity and the subtracted quantity is called the negative quantity. (1-5)

Scientific Commentary—in these five and the next 6—11 shlokas, our Acharyas have attractive imaginations from which it is proven that they worked with how subtle descriptions. They saw that a planet, while revolving, reaches a place where its true motion becomes extremely slow at some time. They determined the position of the planet's apogee by this very thing. The position of the apogee is also not fixed, but it moves with very slow motion, so it is considered to move on the rāśicakra (circle of the zodiac), which is based on the revolution. The ordinary motion of the planets on the rāśicakra is to the west of the east. When the planet reaches its apogee, due to its extremely slow motion, its mean motion is less than the mean motion. Therefore, when the planet moves ahead of the apogee, it does not reach the place where it should be with the mean motion during the day, but remains behind. In this way, a difference arises between the mean and true positions of the planet. This difference increases daily, and when the planet moves 60° ahead (towards the east) of the apogee, this difference is the most. After this, this difference begins to decrease, but the planet remains behind the mean position until it moves 180° ahead of the apogee. At 180° from the apogee, the mean and true positions of the planet become one. It is natural to imagine that when the planet is 180° less than the apogee and remains towards the west, the apogee pulls it towards the mean position. Therefore, by performing a negative calculation in the mean position, the true position of the planet is obtained. As the planet moves further away from the apogee, the true motion increases accordingly; therefore, it was understood that the attraction of the apogee increases and decreases according to proximity.

1. Some discussion of apogee, epicycle apogee, and node is in 'Vigyan' part 16 pages 187-191 or in the Vigyan commentary of the Chapter on Mean Motions, verses 26-33.

SOURCE PAGE 104

At the time when the planet reaches apogee from 1800, its motion is very high. This is the inferior position of the planet. When the planet moves further from this point, its daily true motion remains more than the mean motion, so it moves further than where it should reach with the mean motion and continues to move forward day by day. Therefore, by performing calculations in the mean position of the planet, the true position is found. When the planet goes beyond 1800 from apogee, it is to the east of the apogee planet within 1800. Therefore, here too, the planet seems to be pulled towards the apogee. For this reason, it is imagined that if the planet is towards the apogee, i.e., towards the east, it pulls towards the east, and if it is towards the west, it pulls towards the west.

In the given figure 15, the path of the Sun is shown. The center of the Earth is not at the center of the Sun's path.

For convenience, we will call a planet by two names, mean planet and true planet, the difference between which is—the mean planet is the hypothetical planet that is considered to revolve around the Earth on the zodiac with mean motion and the true planet is

SOURCE PAGE 105

The planet that is seen directly orbiting the Earth. The motion of the mean planet is always the same; but the motion of the true planet varies. The average of the daily true motions gives the mean motion. Therefore, it is clear that the true motion is sometimes less and sometimes more than the mean motion. When the planet is at its apogee, its true motion is very slow. At this place, the mean and true planets are together. But because the mean planet moves faster than the true planet, it moves ahead, and the true planet lags behind. In the figure, m and ma are the positions of the mean Sun and s and sa are the positions of the true Sun. Therefore, to know the position of s or sa, subtraction is required from the position of m or ma. When the mean Sun reaches the nadir, that is, moves 1800 ahead of the apogee, the true Sun also appears to be at the nadir. At this place, the motion of the true Sun is very high and it is much faster than the mean Sun, so the true Sun remains ahead of the mean Sun after moving beyond the nadir. Si and ci are the positions of the true Sun and mi and mi are the positions of the mean Sun. Here too, the true Sun appears to be shifted towards the high side and is ahead of the mean Sun, so to know its position, addition to the position of the mean Sun is required.

The reason for the difference between the mean and true positions of the Sun and the Moon can be explained by such imagination, but there is even more variation in the mean and true positions of the five planets—Mars, Mercury, Jupiter, Venus, and Saturn. Therefore, the imagination of the apogee was also made along with the imagination of the epicycle apogee. The way this was imagined is estimated by Bhaskaracharya as follows1:-

When the Sun is ahead of the Sun by Saturn, Jupiter and Mars, then the true planets are ahead of the mean planets, that is, they appear to be moving towards the Sun. But when the Sun is behind them, then the true planets are behind the mean planets, that is, they appear to be moving away from the Sun. Therefore, the scholars imagined that the epicycles of these three planets remain with the Sun. Therefore, it is natural to assume that these planets, along with their epicycles, also pull towards the

Sun. If this imagination were further extended, then Newton's theory would be known in taking the Sun itself as the epicycle or the attractive force of these planets.

Knowing that the planets are pulled towards the places of the apogee and epicycle, it is estimated that this place is some

1, Siddhanta Shiromani Ganitadhyaya page 20

SOURCE PAGE 106

They possess special power and are invisible; therefore, they are called invisible divine forms, which remain drawn towards the planets by an invisible rope of air and are also assisted by an air called Pravaha.

The matter of the node has already been written. What is written about the node of the Moon is also applicable to the nodes of other planets. When a planet comes to the northern node, it appears on the ecliptic. When it moves further from here, it becomes north from the ecliptic. Until it reaches 1800 degrees further from the southern node, i.e., 1800 degrees from the northern node, the ecliptic remains north. In such a situation, the northern node remains to the west of the planet. That is why it is stated in the next 7th verse that the node (northern node) located to the west of the planet up to 1800 degrees pulls the planet towards the north, and the position of the node to the east up to 1800 degrees pulls it towards the south. This is also invisible, and it appears that the planet is pulled towards the north or south by the ecliptic. Therefore, divine power is also considered in this. However, the real reason is that the orbits of the Sun and the planets are not in the same plane, due to which the orbit of each planet appears to intersect the orbit of the Sun at two points.

In the next 6-11 verses, it is stated that the planets whose size is larger pull less due to their apogees, epicycles, etc., and those that are lighter pull more. This estimation is the result of subtle depiction and is completely in accordance with the theory of attraction.

Based on these eight verses of Sūrya Siddhānta, some scholars say that the inventor of the law of attraction cannot be called Newton, but rather our own ancient astrologers. When considered impartially, it is proven that the revered Acharyas made the estimations with their imagination and reasoning, which are highly commendable considering the condition of that time. They certainly understood that there is a force causing the variation in the motion of the planets, but it was not known how this force works, whether it is only in the epicycles, apogees, and nodes of the Earth and the planets, or in all substances of the world, and by what operation of mathematics it can be proven. Newton is the inventor of this universal law of attraction. There is no doubt that if the study and teaching of astrology proceeded in India in the same way as during the time of Bhaskaraacharya, Ganesh Daivagya, etc., or as during the time of Copernicus, Tycho Brahe, Kepler, Newton, etc., in the 16th and 17th centuries in France, Germany, and England.

SOURCE PAGE 107

Then it is possible that the theory of attraction was already manifested to our teachers in the form that Newton established. Here, the imagination regarding attraction remained only in the form of a hypothesis, and Newton turned it into a theory.

Considering the different motions of the planets in this place, the hypothesis of attraction has been made, so it will not be inconsistent if the theories of Copernicus, Kepler, and Newton regarding the

motion of the planets are briefly stated.

Copernicus's Imagination

In the book that Copernicus wrote in 1587 (1530 AD), he showed that if the Earth and other planets are considered to be revolving around the Sun, then the apparent crooked and straight motions of the planets can be easily understood. This is called the method of Copernicus.

Kepler's Laws

(1) The orbit in which it revolves around the Sun is elliptical in shape, with the center of the Sun at one focus. (2) The line joining the centers of the Sun and any planet forms equal areas in equal time. (3) The relationship between the squares of the periods of two planets is the same as the relationship between the cubes of their mean distances from the Sun. Now, in short, it is stated that what calculations Kepler made to determine these constants.

This will be experienced that as an object moves farther away, it appears to become smaller because the angle formed by that object with the eye becomes smaller when it is far away. Suppose 'n' is the position of the eye and 'k x' is an object that is moving farther away. When it is at the position 'k x', an angle 'k n x' is formed with 'n', and when it reaches the position 'k a x', an angle 'n a x' is formed with 'n', which is smaller than the 'k n x' angle. For this reason, the same object will appear smaller at the position 'k a x', although there is no difference in its actual size. (See figure 16).

SOURCE PAGE 108

If the sun's image is seen every day by setting it, then it does not appear to be of the same size every day. When the sun is at 160 degrees of Sagittarius (on January 3rd), its image appears largest. On this day, the value of its image is 32'35.2''. On this day, its daily apparent motion is also the fastest, i.e., 61'6.6''. After this, the sun's image becomes smaller and its motion becomes slower gradually. When the sun is at 160 degrees of Gemini, i.e., it increases by 1800 from the first place, the image is smallest, i.e., 31'30.7'', and the daily apparent motion becomes the slowest, i.e., 57'11.5''. The reason for the apparent increase and decrease in the size of the image is not that the size of the sun itself actually changes, but that the distance of the sun keeps changing. This is also the opinion of our principles.1

If the value of the half of the sun's image is s and the nearest distance of the sun from the earth is k, then the sine of the angle that will be formed on the earth from the half-image of the sun is s/k

But on this day the image of the sun is 32^(prime)35.2^(primeprime), therefore the half-image will be 16^(prime)17.6^(primeprime).

Therefore sine 16'17.6'' = s/k

But when the angle is very small, there is no difference in the values of the angle and the sine of the angle, when the value of the angle is in Circular measure or the sine value is written according to the Indian custom!2

1. Sūrya Siddhānta Lunar Planet Chapter verse 1 - 3

2. See the Scientific Commentary of the 60-61 verses of the Chapter on Mean Motions.

SOURCE PAGE 109

∴ sum = 16' 17.6' yas = k × 16' 17.6"

Similarly, the image of the sun is 31'30.7" or half the image is 15'45.4" when the distance of the sun is 'of' is very large,

s / ka = 15'45.4" yaas = ka × 15'45.4' ∴ ka × 16'17.6" = ka × 15'45.4" (q)

The place where the sun appears largest is when it moves 180° ahead, it appears smallest. Therefore, the distances of the 'upper extracted k' and 'ka' are in a line. Therefore, if the position of the Earth is in figure 1 as the position of the sun, then s and sa will be the positions of the sun when the sun appears largest and smallest in sequence, i.e., when p s = ka and p sa = ka

If each side of equation (1) is subtracted from 1,

1 - k / ka = 1 - 15'45.4" ka 16'17.6" yaa ka - k / ka = 32.2" / 16'17.6' (2)

and if 1 is added to each side of the equation (1), then

ka + kka = 32'3" 16'17.6" (3)

now if equation (2) is divided by the sides of equation (3), then, ka-k / ka+k = 32.2" / 32'3" = 32.2" / 1623" = approximately 1/60

this relationship reveals that p is the focus of the ellipse whose major axis is sa, the midpoint of sa from the center is m, and the eccentricity is 1/6; because the distance from the center of an ellipse to its focus is the long

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SOURCE PAGE 110

When divided by half of the axis, the value of the chord comes out. There, the double of the distance from the center to the navel and the length of the long axis of k is.

In this way, if a distance s is drawn as the long axis, p as its navel, and d as the chord, then the distance of a radius vector p to si which makes an angle heta with the line s can be known from this—

psi = mas (1 − ch2) / 1 + ch X cojy

When ch = 60 = 0167 and m is the mean distance of the sun and earth.

Therefore the value of 1/psi changes according to 1+ch × cojy

which are written in brief as:-

1 / k ∝ 1 + chokojya

Where k is the difference of the Sun from the Earth (hypotenuse or Radius vector). This relation follows directly from the observation. Therefore it is proved that the Sun revolves in an ellipse

1. Ashutosh Mukhopadhyay's Geometry of Conics, Chapter, II. proposition III.

2. Loney’s Elements of Coordinate Geometry, pp. 307 and 229(1910 edition.)

SOURCE PAGE 111

It is located and the Earth is at the focus of this ellipse. It is more accurate to say that the Earth forms an elliptical orbit around the Sun, and the Sun remains at the focus of this orbit.

Its proof is given in 'Vigyan' part 12 pages 75-76, 188-186, 203 to 207. This is Kepler's first law.

It is stated above that the fastest speed of the Sun is 61' 10" and at this time its maximum image is 32' 35", and the slowest speed is 57' 12" and at this time the minimum image is 31' 31". Therefore, it is clear that the difference in the fastest and slowest troughs is equal to 3'58"/56'11" of the mean motion, or equal to 1/14 of the small difference, and the difference in the maximum and minimum sizes of the true images is equal to 1'4"/32"3 of the mean image, or equal to i. of the small difference. Therefore, the change in the true image is related to 1:1 + 1/14 and the change in the true motion is related to 1:1 + 1/14.

But $1 + 1/15 = (1 + 1/30)^2$ of the small difference.

∴ The change in motion is related to $1:(1+1/40)^2$.

Whatever is seen at any time will be found that the angular velocity of a planet changes according to the square of the true diameter. But the true diameter of the Sun changes according to the inverse of the distance to the Sun, as stated in the previous pages. Therefore, the angular velocity changes according to the square of the true diameter or the inverse of the hypotenuse squared. In short

$$\text{angular velocity} \propto (\text{true diameter})^2$$

$$y \propto (1 / \text{hypotenuse})^2$$

In figure 16, the position of the Earth is s, the position of the Sun is s, and p is the angle that the Sun moves in 1 day. Similarly, the second position of the Sun is s and p is the angle that the Sun moves in 1 day. s, p, or s, c are very close to each other, so there is such a small difference in the values of p s and p c that they can be considered equal. Similarly, p s and p c can be considered equal. In such a situation, the triangle p s c can be understood as a segment of the circle whose center is p and the radius is p s or p c. Therefore

SOURCE PAGE 112

$$\text{Area of this sector} = < \text{Suppasi} \times (\text{Pasa})\ ^{\wedge}22 = \text{Angular velocity of the station} \times (\text{Pasa})\ ^{\wedge}22$$

$$\text{AurasapseeKshetrphal} = < \text{latitude} \times (\text{latitude})\ ^{\wedge}(22) = \text{location's angular velocity} \times (\text{latitude})\ ^{\wedge}(22)$$

But it has been stated above that

The velocity propto 1 (cron)^2 = a × 1 (cron)^2 is the same.. Area of the arc = Area of the station × (pas)^2 = Area of the station × (pasa)^2 = Area of the station × (pas)^2 = Area of the station × (pas)^2

SOURCE PAGE 113

Sūrya Siddhānta

Scientific Commentary

Chapter on Mean Motions

Chapter on True Positions

Chapter on the Three Problems

The late

Sūrya-śukra-dūrī = (12.645)^(13) = (0.378)^(13) = (3781000)^(13) = 110 (3.78)^(13) = 110 × (73 + 35)^(13) = 110 × 7 (1 + 572)^(13) = 710 (1 + 1.02)^(13) = 710 \(1 + 13 × (.102) + 12 × 13 (13 - 1) (.102)^2 + |13 × 13 × (13 - 1) (13 - 2) (.102)^3 + cdots\) = 710 (1 + 0.034 -0.00116 +0.00066 — a very small number) = 710 × 1.0326 = 7.230310 = .723

SOURCE PAGE 114

Kepler made these three laws from the subtle representations of the planets in the years 1664-1674 (1606-1616 AD). He did not know what forces cause the motions of the planets according to these laws. No proof of these laws could be established for 75 years. After this, Newton proved that universal gravitation is the cause of all this. The three laws on which Newton proved this are called the laws of motion, are famous by that name, and are:-

First Law—If no external force is applied, every object either remains in its state of rest or moves in a straight line with uniform motion.

Second Law—The change in motion is according to the force applied and this change occurs in the direction of the force applied.

Third Law—To every action, there is an equal and opposite reaction, meaning that for every action, there is a reaction in the opposite direction, i.e., the same action but in the opposite direction.

These laws are self-evident. For more information, one should read any book of Dynamics.

From Kepler's first and second laws, Newton proved that every planet moves due to a force directed towards the Sun and whose magnitude is inversely proportional to the square of the distance between the Sun and the planet. From Kepler's third law, Newton proved what is the relation between the increase in the motion of one planet and the increase in the motion of another planet, and from this he derived the theory of universal gravitation, which is:-

Every particle of matter attracts every other particle by a force that varies according to the product of their masses and inversely proportional to the square of the distance between them.

Now it has to be proved that if a line is drawn from a fixed point to a moving particle and it covers the same area in the same time, then the direction of the force with which that particle is moving is towards that fixed point. This is briefly based on Differential Calculus and kinematics

SOURCE PAGE 115

it can be proved, which will be given later. This is proved with some expansion on the basis of simple mathematics.

Suppose that 's' is a fixed point and a particle of an object revolves around 's' forming a polygon area of k, kh, kg, khc and kx, xk, xg, xhc in the same time or 1 moment. Also suppose that the values of these sides are different and until the particle remains on one side, its motion remains uniform. The areas of the triangles s k, s k g, s g kh, s khc should also be considered equal.

Now it is evident that in the same time, the particle revolves around 's' and forms the same area. According to the first law of motion, until the particle polygon

1. This argument is based on the pages 173-175 of Heroes of Science: Astronomers.

SOURCE PAGE 116

Until no straight line is being formed by the field, no force is acting on it and it is going in a straight line with its obtained force, but as soon as it starts turning from one side to another side, some force must surely act on it for a moment, by which it changes its previous straight course to the second straight course.

Consider the condition at the time when the particle is at x. If no force is applied at this time, in the next moment it will go along the path x p in the same straight line as k x, and the line k x p will be a straight line, and x p and k x will be equal because there will be no difference in motion. Join p with g and s. The base of the triangle s k x p is x p, which is equal to k x and lies in the same straight line as k x, so according to geometry, both triangles s k x and s x p have equal areas. It is assumed from the beginning that the areas of triangles s k x, s x g, etc., are equal. Therefore, it is proven that the triangles s x p and s x g are also mutually equal, which are on the same base s x, so according to geometry, these two triangles are between the parallel lines s x and g p, which means the line g p is parallel to s x. Draw a line parallel to x p that intersects the s x line at point r. Then there will be a parallelogram area x p g r. At the time when the particle was at x, if no force had been applied, it would have reached point p; but due to the force, it reached g, so it is apparent that the first motion of the particle at x was x p, and due to the force, it changed to x g. Therefore, according to the 'parallelogram of velocities' of kinematics, due to the applied force, the motion of x p combined with the motion of x r in the particle, that is, in the motion of the particle at point x which was in the direction of x p, the second motion was obtained in the direction of x r, due to which the particle reached point g. Therefore, due to this combined force, the object turned towards s. Similarly, it can be proven that the force applied at the angle points g, h, c of the polygon area is also in the direction of s.

Now imagine that this polygonal area is made up of crores of extremely small polygons and the particle revolving around s completes each small polygon in one croreth part of a pal, then it is apparent that crores of forces will be exerted on that particle in the direction of s. Therefore, it is proven that the particle has taken an uninterrupted curved path leading towards s.

SOURCE PAGE 117

It is done by the force. If the imagination is further increased and the sides of the polygon become so small that they cannot be bounded and their number is infinite, this argument can still be applied. Therefore, it is proven that if a particle moves on such a path around a fixed point that it forms the same area in the same time, then the force continuously acting on this particle is in the direction of that fixed point, that is, that fixed point is continuously attracting that particle.

If S is taken as the center of the Sun and K, X, Y, etc., are the positions of a planet, then according to Kepler's second law, the force acting to make the planets revolving around the Sun is the attractive force of the Sun. Similarly, the planets also pull their satellites.

> दक्षिणोत्तरयोरेवं पातो राहु स्वरंहसा ।
> विक्षिपत्येष विक्षेपं चन्द्रादीनामपक्रमात् ॥६॥
> उत्तराभिमुखं पातो विक्षिपत्यपराद्धगः ।
> ग्रहं प्राग्भगणाद्धस्थो याम्यायामपकर्षति ॥७॥

Translation—(6) The planets like the Moon etc. displace north or south by their velocity due to the deviation from their node or Rahu ecliptic. (7) When the node is less than 6 degrees or 180° from the planet, it removes the north from the ecliptic, and when it is less than 6 degrees from the planet to the east, it removes the south from the ecliptic.

Sūrya Siddhānta—the simple meaning of these two shlokas is that by looking at the positions of the planet and its node, one should understand whether the planet is exactly on the ecliptic or is shifted some north or south from it. If both the planet and the node are at the 'same place', it should be understood that the planet is on the ecliptic. If the planet is ahead of the node, i.e., east, but not more than 180° away, then it will be shifted north from the ecliptic, and if the planet is behind the node, i.e., west, but not more than 180° away, then it will be shifted south from the ecliptic. The reason for this is not the attraction or repulsion of Rahu, but rather that the orbit of a planet is not in the plane of the ecliptic, so the planet does not always remain on the ecliptic. The two points where the planet's orbit and the ecliptic seem to meet are called the node. When the planet is at these two points of its orbit, then it appears on the ecliptic, otherwise the ecliptic

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SOURCE PAGE 118

It is according to what is said above as North or South. The distance of the planet from the meridian of the ecliptic is called the deviation of the planet. This occurs on the circle that makes a right angle with the meridian (ecliptic pole).

> बुधभार्गवयोः शीघ्रात्तद्वत्पातो यदास्थितः ।
>
> तच्छीघ्राकर्षणात्तौ तु विक्षेप्येते यथोक्तवत् ॥८॥

Translation—(8) When the nodes of Mercury and Venus are in accordance with the rules written in the above (6, 7 shlokas), they remove the planets from the meridian to the North or South by attraction in the epicycles.

Scientific Commentary—The rules stated in the 6, 7 shlokas apply only to the Sun, Moon, Mars, Jupiter, and Saturn. To know the positions of the two planets Mercury and Venus, one should see which direction and how far they are from their epicycle nodes. If the epicycle node is to the East but less than 180°, the planet will be North of the ecliptic, and if it is to the West but less than 180°, the planet will be South of the ecliptic.

> महत्वान्मण्डलस्याऽर्कः स्वल्पमेवाऽपकृष्यते ।
> मण्डलाल्पतया चन्द्रस्ततो बह्वपकृष्यते ॥६॥
> भौमादयोल्पमूर्तित्वाच्छीघ्रमन्दोच्चसंज्ञितैः ।
> दैवतैरपकृष्यन्ते सुदूरमतिवेगिताः ॥१०॥
> अतो धनर्णं सुमहत्तेषां गतिवशाद्भवतु ।
> आकृष्यमाणास्तैरेवं व्योम्नि यान्त्यनिलाहताः ॥११॥

Translation—(6) The Sun's sphere is very large, so it is pulled very little by its pole. The Moon's sphere is small, so it is pulled a lot. (10) The spheres of planets like Mars etc. are very small, so their epicycle apogees pull them very far at great speed. (11) Therefore, they have to undergo a lot of wealth and debt transformations due to their motion. In this way, these planets move in the sky while being pulled by their epicycle apogee deities and the push of the flowing air.

Scientific Commentary—Our teachers have seen that compared to the Sun, the Moon remains more to the East or West from its mean position, and Mars, Mercury, Jupiter, Venus, Saturn etc. see more to the East or West than their mean position first

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SOURCE PAGE 119

The names of the planets, when the Earth is taken as 1 unit of mass, when water is taken as 1 unit of mass, when the gravitational force of water is taken as 1, the Sun is 16^(°)1'18" 4,32,860 106.2 3,26,360 1.40 27.61, Mercury is 3".34 1,504 0.380 0.055? 5.14 0.38, Venus is 8".40 3,783 0.955 0.807 5.14 0.86, Earth is 5".80 3,963 1.000 5.56 1.00 1.00, Mars is 4".68 2,108 0.532 314.50 3.62 2.57, Jupiter is 1'37".36 43,850 11.06 344.50 1.37 2.57, Saturn is 1'24".75 38,170 64.07 0.64 1.01 1.01, Neptune* is 34".28 15,440 14.40 1.25 0.65 0.67, and the distance between them is mean. *This nomenclature is according to the calculation of light. ?The exact value of the number for which this sign is present has not yet been determined. The shapes of the X planets are also not perfectly round; they also have some flattening at the poles, like our Earth, so they are also + qualitative. The apparent angular value of the projection of the planet's semi-diameter is when the observer is at a distance from the planet that is the same as the distance from the Sun to the Earth.

SOURCE PAGE 120

Therefore they considered the orbits of these planets to be smaller than the Moon as they appear directly, and they determined that their orbits are very small, for this the attraction of their apogees and apogees has a great effect.

But there are real reasons for seeing some east or west from the mean positions of the planets, which are the shapes of the orbits of the planets. The orbits of the planets are elliptical in shape whose eccentricity is not one; that is why the difference is mainly in the sizes of the orbits of the planets, not in the shapes of their orbits. Knowledge of their sizes will be clear from the table on the previous page, which is taken from Robert Ball's 'Spherical Astronomy' page 462. The semi-diameter of the Moon is 1076 miles.

> वक्रानुवक्रा कुटिला मन्दा मन्दतरा समा ।
> तथा शीघ्रतरा शीघ्रा ग्रहाणामष्टधा गतिः ॥१२॥
> तत्रातिशीघ्रा शीघ्राख्या मन्दा मन्दतरा समा ।
> ऋज्वीति पञ्चधाज्ञेया या वक्रा सानुवक्रगा ॥१३॥

Translation—(12) There are eight types of motions in the planets called arc, arc of arc, crooked, slow circle, straight, quick, and quick. (13) Among these, the very quick, slow, slow, and straight motions are straight, that is, when the planet has these motions, it appears to go from west to east in the circle and the arc and crooked motions are called arc motion because when the planet has such motions, it appears to go from east to west in the circle in reverse. When the planet has straight motions, then it is called straight and when there are curved motions, then it is called crooked.

Scientific Commentary—the reason why these different motions occur in the planets is not explained anywhere in our theories, because the reason cannot be well explained until the Earth is considered stationary. Yes, if the Earth is also considered to revolve around the Sun like other planets, which has also been proven by many experiments, then it is easily understood how these eight motions appear in the planets; although in reality the planets are continuously moving from west to east revolving around the Sun. In this regard, I quote what I wrote in 'Vigyan' on page 264-266 of part 13.

SOURCE PAGE 121

If you plant a flag in the field and make a circle with a distance of 4, 7, 10, 15, 52, 65, 162, and 300 furlongs with a height of three feet for each circle; appoint one horseman near each circle; you stand near the flag and command the horsemen to stand near their respective circles in such a way that they all appear in a straight line, and then all the riders begin to circle the circle with this speed so that the nearest one completes one revolution in 88 seconds, the one some distance away in 225 seconds, the third in 365 seconds, the fourth in 687 seconds, the fifth in 4332 seconds, the sixth in 10756 seconds, the seventh in 510 minutes, and the eighth in 1003 minutes, when they complete the revolution, just as the horsemen will know that you are circling them in seconds, so the planets in the solar system appear to revolve around the sun in days. The only difference is that the riders will circle on a surface, but the planets also shift some north and south.

If you do not stand near the flag but ride on the third horse of the flag and begin to circle with the previously said speed, you will see the flag and the horsemen, the same scene that is visible to the inhabitants of Earth when they circle the sun of the planets. Sometimes it seems that the planets are moving forward, and sometimes it seems that some are moving backward, and sometimes they also appear to be stationary.

From the example of the riders, you must have known that if a rider circumambulates the flag while sitting on the third horse, the peculiarity of the motions of the horses on both the outside and inside is seen, like the peculiarity in the motions of other planets when riders on the horse of Earth see the motions of other planets. To explain this, we will have to take two examples—one of a planet that is between the Earth and the Sun, and another of a planet that is outside the Sun and Earth. For the former, Mercury and for the latter, Mars have been taken in pictures 21 and 23.

In Figure 21, the largest circle is the zodiac. The sun, revolving on it, seems to make one revolution in a year. Where it is written as vernal equinox, when the sun appears, the spring season begins, and on this day, day and night are equal. From here, the zodiac is divided into twelve parts. Therefore, whatever

part is written as Aries, Taurus, etc., should be understood as Aries, Taurus sine. The beginning of Aries sine is on March 21, 22.

SOURCE PAGE 122

The sidereal Aries month begins about 23 days after tropical Aries, on 13 or 14 April. The Sun appears to pass through the zodiac signs from Aries to Taurus, Taurus to Gemini, and so on.

The circle smaller than the zodiac is the ecliptic. The Earth moves around the Sun, which is in the center, one revolution in a year. The Sun appears to travel through the zodiac signs for the inhabitants of the Earth, from Aries to Taurus, from Taurus to Gemini, from Gemini to Cancer, from Cancer to Leo, etc. From this movement, we can know that the Sun is moving. In figure 21, its movement is shown starting from Aries and

SOURCE PAGE 123

The hands of the clock move in the direction opposite to which the Earth moves at points 2, 3, 4, etc. The direction of movement can be known from the direction of the arrow. Because the Earth's daily motion is uneven, the points of the latitude are shown with unequal differences.

The smallest circle is the orbit of the planet Mercury. Suppose that Mercury starts moving from point B1 and appears to go in the adverse direction of the clock and to the zodiac signs of Aries, Taurus, Gemini, etc., while the Sun resides in them. In the figure, B1 is written where Mercury is at the time when the Earth is at P1. When Mercury goes to point 2, then the Earth goes to point 2 in its orbit. When Mercury goes from point 2 to point 3 in its orbit, then the Earth goes from point 2 to point 3 in its orbit. Similarly, one should understand for all the points, as when Mercury is at point 7 in its orbit, then the Earth remains at point 7 in its orbit, etc. If one wants to see in which direction the zodiac will appear to the Earth from Mercury, then combine the points where Mercury and the Earth are at that time and take them to the zodiac. The place where this line reaches will be the position of Mercury. For the simplicity of the figure, the lines joining the Earth and Mercury are not shown, but from Mercury to the zodiac are revealed by these cut lines. For example, when the Earth is at points P1 and Mercury at B0, the line joining P2 and B1 reaches point 1 in the zodiac, that is, to the inhabitants of the Earth, Mercury will appear at point 1 of the zodiac or at the end of Aquarius. When the Earth reaches P2, then Mercury reaches B2 and appears at point 2 in the zodiac or in Pisces, and because Mercury is at B3 (that is, when the Earth is at point 3 in its orbit), Mercury will appear at point 3 at the end of Pisces in the zodiac. When Mercury is at 4 in its orbit, then the Earth will also be at 4 in its orbit and to the inhabitants of the Earth, Mercury will appear at point 4 in the zodiac, that is, a little ahead of 3. When Mercury comes from 1 to 2 in its orbit, then the Earth also came from 1 to 2 in its orbit and we see Mercury going from Aquarius to Pisces in the zodiac. When Mercury goes from 2 to 3 in its orbit, then the Earth also went from 2 to 3 in its orbit and the inhabitants of here see Mercury going ahead from 2 to 3 in Pisces in the zodiac. This time Mercury did not advance as much in the zodiac as before, that is, the speed of Mercury

SOURCE PAGE 124

The previously became slow. The Mercury zodiac-cycle moved very little forward in reaching from 3 to 4, so there is no exaggeration if it is said that the speed of Mercury is not the same. In such a situation, Mercury seems to be somewhat stationary. When Mercury goes to 4 to 5 in its orbit, the Earth also goes to 4 to 5 in its orbit, and it will appear to the inhabitants of Earth that the Mercury zodiac-cycle goes backward from 4 to 5, meaning Mercury has become retrograde, this will be felt. When Mercury goes to

5 to 6 in its orbit, the Earth also goes to 5 to 6 in its orbit, and it will appear to us that the Mercury zodiac-cycle goes backward from 5 to 6 from Pisces to Aquarius, but the speed will be very fast. Here too, Mercury will be called retrograde, although it is moving in the same order in its orbit. When Mercury goes to 6 to 7, it appears that the zodiac-cycle goes backward from 6 to 7. But while going from 8 to 6 in its orbit, it will appear to come straight in the zodiac-cycle from 8 to 6, meaning the speed of Mercury will be direct, but it will remain very slow. Now it will be clear that although Mercury and the Earth appear to go in the same direction according to the Sun, the inhabitants of Earth feel that the Mercury zodiac-cycle moves forward from 1 to 4 and backward from 4 to 7. When it moves forward, it is called direct, and when it moves backward, it is called retrograde. When it is direct, its speed does not seem the same, but sometimes it seems to increase very fast, sometimes it becomes slow, and sometimes it seems to stop. And when it is retrograde, the speed seems fast, very fast, slow, slower, and steady. One more thing is worth knowing. When the Earth is at P and

There is one more thing to know here. When the Earth is at P and Mercury is at B, then the line joining the Sun and Earth reaches the end of Capricorn, meaning the Sun appears in Capricorn, but Mercury in Aquarius. Therefore, Mercury remains to the east of the Sun and appears in the west after sunset. From P2, the Sun appears in the beginning of Aquarius and Mercury in the beginning of Pisces, P3 from which the Sun seems to have moved some further in Aquarius, but Mercury appears to have reached the end of Pisces, and the distance between the Sun and Mercury is the greatest in this case. In such a situation, if the line joining the Earth and Mercury is extended, it will touch the orbit of Mercury, and the angle it forms with the line joining the Earth and the Sun will be the largest. This is called the greatest elongation of the Sun and Mercury, and this difference is to the east of the Sun. A few days behind the greatest elongation, the motion of Mercury becomes retrograde and the difference

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SOURCE PAGE 125

It begins to shrink and as it shrinks, Mercury comes between the Earth and the Sun, meaning the distance becomes zero. In this situation, Mercury rises and sets with the Sun. This is also called the inferior conjunction of Mercury. When the distance between the Sun and Mercury becomes approximately 120°, due to the light of the Sun, Mercury is not visible to the naked eye. Therefore, it is said that Mercury sets in the west because Mercury disappears in the west while being seen in the west. Before the inferior conjunction, when Mercury becomes retrograde, it moves towards the right hand side and the Sun towards the left hand side, so the distance between the Sun and Mercury quickly decreases, meaning it comes to the west after setting and in a few days it comes to the west of the Sun and appears in the east before sunrise, then it is said that Mercury has risen in the east. When Mercury is 120° behind the Sun, it starts to be visible. Only then is it considered to have risen, and the motion also becomes regular in a few days from retrograde.

In this way, behind becoming regular, Mercury moves further away from the Sun, and when it moves between points 7 and 8 in its orbit, the distance between it and the Sun becomes maximum. Then Mercury moves closer to the Sun and in one and a half months it becomes so close to the Sun that it is not visible to the eye. According to Sūrya Siddhānta, when the distance remains 140°, it is considered to set. When the distance between Mercury and the Sun becomes zero, both come above the horizon simultaneously. In this situation, there is the superior conjunction of Mercury and the Sun. At the time of superior conjunction, Mercury remains regular.

At the time of the outer conjunction, both Mercury and the Sun appear to go to the left, therefore Mercury takes more days to move away from the Sun, that is, when Mercury sets in the east, it remains

set for a longer time than the time of setting in the west and rises later in the west.

It is written that in the west Mercury sets when the difference between the Sun and Mercury becomes less than 120, and sets in the east when the difference between them becomes less than 140. The reason for this is that at the time of the inner conjunction, Mercury remains very close to the Earth, so its image appears larger, and it is visible until it is 12 units away from the Sun. But at the time of the outer conjunction, Mercury also moves far from the Sun, so its image is small.

SOURCE PAGE 126

It appears and it disappears only when its distance is 140. Venus is the largest visible at the time of inner conjunction and the smallest at the time of outer conjunction.

This must have proven that the planets do not move somewhere behind the setting of the sun but move so close to the sun that they are not visible to the eye. Yes, with a telescope, they can be seen no matter how close they are to the sun.

The orbits of both Mercury and Venus are within the orbit of the Earth, so what is said for Mercury also applies to Venus. The only difference is that Venus's orbit is larger than Mercury's orbit, so at the time of inner conjunction, Venus becomes extremely close to the Earth. When viewed with a telescope, both Mercury and Venus show the same phases as in the Moon, when both planets appear full and round, because at that time the entire illuminated image remains in front of us. When the planet moves a little aside, the entire illuminated part is not visible to us, and day by day the image becomes fragmented. However, more light is received because the distance decreases, so even fragmented planets give more light due to being close. At the time of inner conjunction, the illuminated part of the planet is towards the sun, so we do not get any light from the planet and it appears in the telescope as a black spot. The phases of Venus are shown in Figure 22.

In Figure 23, only that part of the zodiac circle is shown where Mars appears retrograde and then direct. The Sun is in the center. The Earth and Mars revolve around the Sun in such a way that they appear in the zodiac as moving from Aries to Taurus, from Taurus to Gemini (from the Sun). A person in the Sun cannot see any planet in retrograde; they see all planets revolving in the same direction. Yes, the Earth dwellers see Mars as direct, fast, slow, fixed and retrograde, slow moving, then direct. Mars's

One revolution is completed in 686 days, so a 10° revolution is 686 X 10 360

It takes 16 days and in this time the Earth moves about 160, because the Earth's one revolution is completed in 365 days, which means in 1 day there are approximately 10 revolutions.

Suppose the Earth is at point 1 in its orbit and Mars is also at point 1 in its orbit, then to the Earth dwellers, Mars is at point 1 in the zodiac

SOURCE PAGE 127

It will be seen. When the Earth reaches point 2 of its orbit in 16 days, Mars will also travel 100 and reach point 2 in its orbit and we will see that it is at point 2 in the zodiac. When the Earth reaches point 3 in the next 16 days

SOURCE PAGE 128

It will reach, Mars will also reach point 3 in its orbit and it will appear that it has moved slightly ahead in the zodiac. Here Mars will seem relatively fixed for some days, because within 16 days, it has moved very little from 2 to 3 in the zodiac. When the Earth and Mars reach point 4 in their respective orbits, then Mars will appear at point 4 in the zodiac, meaning it will appear behind. This is called that Mars is retrograde although the motion of Mars in its orbit is the same straight. Between Earth, Mars, and the Sun, there is a position, meaning the Sun is to the right of Earth and Mars to the left. In this way, the distance of the Sun becomes 6 zodiacs or 1800. This situation is called that Mars is in opposition to the Sun. When the Sun sets, Mars rises in the east, and when the Sun rises, Mars sets in the west. In this situation, Mars is very close to the Earth, so its image appears very large, and when seen through a telescope, many things about the planet Mars are visible at the same time.

When the Earth and Mars are at point 5 in their respective orbits, then we see Mars at point 5 and behind in the zodiac. When both planets come to point 6 in their orbits, then Mars appears at point 6 slightly ahead in the zodiac, here too Mars seems fixed for some time, then it appears to move forward.

When the Sun is between the Earth and Mars, that is when the Earth is at 4 and Mars is at 'M', then the distance of Mars from Earth is very great. This situation is called 'Mars is in conjunction with the Sun'. In this condition, the image of Mars appears very small.

From these two pictures it is revealed that the planets within the geocentric orbit appear retrograde at the time when conjunction is about to occur and they remain retrograde at the time of conjunction. However, the planets outside the geocentric orbit appear retrograde when they are at a distance of 6 zodiac signs or approximately 1800 from the Sun and when they are in opposition, they remain retrograde at that time. The inferior planets are set at the time of both the inner and outer conjunctions of each revolution, that is, because they are near the Sun, they are not visible to the naked eye in the fierce light of the Sun, but the superior planets have only one conjunction in one revolution, only then they are said to be set.

SOURCE PAGE 129

Thus it is revealed that understanding the eccentric motions of the planets is very easy by taking the Earth as moving. If the Earth is considered stationary, it cannot be explained in any way why the planets have retrograde motion.

> तत्तद्गतिवशान्निर्यं यथा दृक्तुल्यतां ग्रहाः ।
> प्रयान्ति तत्प्रवक्ष्यामि स्फुटीकरणमादरात् ॥१४॥

Translation—(14) I respectfully state that the rules by which the planets attain the apparent equality in these motions, that is, they appear directly in the place of the node, are the rules by which they come to be seen, and I say these rules are known by mathematics.

Sūrya Siddhānta - this verse is of great importance. It proves that our ancient Acharyas used to determine the true positions of the planets so that there would be no difference in mathematics and direct observation. For this, subtle rules were made in the Chapter on True Positions. However, as I have explained in the Scientific Commentary of the 6th verse of the Chapter on Mean Motions, whether the instruments are gross or subtle, some experimental error remains in determining them, so a difference occurs in time, which requires periodic revision. This is called 'seed' correction. For example, suppose a clock slows down by one second per day, then in 60 days it will be 1 minute and in 1 year it will be 6 minutes behind. But in practice, it will be said that the clock is very pure; because the

difference of 1 minute in 60 days or 1 second per day is not like (the difference). If this difference always occurred and the clock was not revised, then in many years such a difference would occur that it would be impossible to consider it as not like (the difference), and revision would have to be done. Just as a difference of 1 second per day in a clock can take a very large form in some time, similarly, in the periods of revolution of the Sun, Moon, etc., the difference of even 1 moment becomes very large in hundreds of years. This is why the seed correction has to be done. The practice of making revisions in between was accepted by our ancient Acharyas, whose incarnations I will give below; but there has been a difference of opinion on this subject for some days. One side says that there is no right to make any commentary or revision on the ancient texts; whatever is in them should be accepted as is. The other side says that revision is entirely appropriate. The arguments of both sides, as I have found them, are given below:-

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Pandit Indranarayan Dwivedi Jyotish Bhushan, the resident of Prayag, gives this comment along with the translation of this verse—

Many people here interpret the meaning of 'dṛktulyata' as visual calculation; but this is their illusion. It is clearly known from the earlier viewing that the meaning of dṛktulata here is that the position appears in an invisible sight according to the calculation described here, otherwise no visible planets could have been proven according to this calculation, because all the signs extracted for the visible planets today should have always been so, this is known to those who know the science of the sphere.

The essence of this quotation is perhaps that according to the rules laid down in this text for finding the true positions of the planets, the position of the planets is not the same as what is seen by direct sight. Therefore, dṛktulyata does not mean direct sight, but rather the invisible sight which the sages knew by their power of yoga.

This astrologer, Shri Girijaprasad Ji Dwivedi, who is the principal teacher of the Naval Kishore Vidyalaya in Lucknow nowadays, writes very clearly in the 'Prabhā Bhāṣya' of the 'Prabhā Śaromaṇi' in the 'Goladhyāya' on pages 6, 7 of his Siddhanta Shiromani:

"...The twofold nature of mathematics is thus fourfold by the difference between the seen and the unseen. Therefore, perform the calculation as follows. Mathematics is always derived from what is seen by what is seen." And the unseen result is stated as follows." These are the Sanskrit texts of Kamalakara, who are followed by the teachings of Maharshi according to the true understanding. These are the modern Sanskrit texts used for the prediction of fruits by employing the true positions. From these three Rasas, the unseen calculations are derived. In many times, the fluctuations are heard in the results of the fruits by the following of the results of the calculations. Therefore, the wise and learned should examine the entire path based on these rules. The results obtained from these calculations are used only with the visible calculations such as the planetary rise. The wise and learned often debate the unseen calculations regarding the results of the calculations. The further details are in the other calculations, not in the above one.

"Mathematics is of two kinds, by the difference between the seen and the unseen. The seen is that which is seen by the eyes, such as eclipses, risings, conjunctions, and elongations, etc. And the unseen is that which is not seen, such as tithiyoga, etc. The result of the eclipse etc. is obtained from seeing it. And the result of vows and fasts etc. is obtained without seeing. The command of the fruit is only by the sages

1. Sūrya Siddhānta, page 35 published by the Hindi Literature Conference of Prayag.

2. Published in 1611 AD in the press of Lucknow.

SOURCE PAGE 131

It is obtained from the experiential sentences. Whatever fruit is available according to the positions of the planets, man will be able to know that. No one, except the sages, can imagine and know this fruit.

The method by which the planets are made clear in the ancient texts is appropriate for the clear planets in astrology. Because the 'bantes' (divisions) of the time etc. are based on those clear planets of the Shruta and Smarta actions. Therefore, it is appropriate to conduct the religious system and its conduct with the same mathematics from which the tithi etc. are proven.

In the present, the scholars of Europe have determined many new signs by a subtle instrument and by bringing them into use in the planets, proving the subtle-clear planets. Taking this clear method, English mathematical scholars nowadays prove the planets by showing them in several almanacs and use them to determine the date etc., and follow the religious system accordingly. But this is entirely improper and obstructs religion. Because according to ancient mathematics, when the fasts of Ekadashi etc. are proven, it will be impossible for it to be proven by this new subtle mathematics at that time. By obstructing the words of the sages, the religion will be overthrown. The sentences of the sages will be realized by following their custom. It is clear that the reliance on the mathematics spoken of by the sages is appropriate for the religious system.

The new astronomical treatises are written in the ancient texts as 'seed'. And the ancients also used it as a means. But this seed has been made suitable for knowledge at the exact time of planetary observations, etc. In unobserved mathematics, it is not as nowadays. Therefore, the new treatises of Europe today are only suitable in observational mathematics. There is no obstacle in using it in that. Because the arrangement of it was done by the Acharyas in this very way.

As — 'the unseen results are not revealed.'

Mathematics is that which is seen, and from that always arises.

That is, for the unobserved mathematics, one should rely only on the mathematics of Sūrya Siddhānta, Suryaokti, and Sūrya Siddhānta, and for the observed mathematics, one should always do mathematics with that which has a dialogue with the true sky and mathematics.

Thus, impartiality and religious wisdom prove that every thoughtful man should accept the observed and unobserved mathematics according to those rules.

SOURCE PAGE 132

I wish. To consider only the apparent is very unjust and a misstatement of the truth, like the Charvakas.

In this long excerpt, the Sanskrit verse given for proof is from Acharya Kamalakar's Siddhanta-TattvaViveka, which was written in 1580 and 1715. In this text, Acharya Kamalakar has made some unjust points in the Sun Siddhanta, the proof of which is in the words of M.M. Sudhakar Dwivedi as follows:-

“Here, the Sun Siddhanta is made, a book made by Bhaskar Muni and others. It is not suitable for the imagination of any highly skilled scholar to make the sphere of the calculations of the geometry of the sphere by the desire for great pleasure.”1

I do not know if there is any other ancient opinion in this regard. If any great person would be kind enough to tell me, I would be very grateful and accept it with thanks. I desire to know this that Acharya Kamalakar's these rules should be followed in the course of religious duties, as 'nirvijārkokta' planets. Whether anyone has accepted this or not, because hundreds of years ago, the Makrand Sarini and Grahalaghava etc. are practical in the means of astrology for making Panchangas etc., in which 'Beej Sanskara' has been done. Some of the proofs are given below:-

(1) There are many Karan texts of astrology; but as much as the propagation of planet abbreviation is in reading and teaching, not so much of others. Based on this, Panchangas are made in many countries and according to them, everyone performs the correct source-based actions. This is the solar-sided Karan text. Although the author of the text has also relied on some part of the solar and lunar sides. The predominance of the solar side has been running since a very ancient time, not just at this time. The mathematics of the solar and lunar sides of Aryabrahma were already messed up during the time of Acharya Barah Mihir (Shaka 427). Lunar side Panchangas are also prevalent in some places. For example, the 'Chandu' Panchangas run by the astrologer named Chand in Jodhpur, but they are not accepted because they are un-Anarshmulaka.[2]

1. Ganaka Tarangini page 68

2. Mentioned lines were written by Girijaprasad Dwivedi in the 'Madhuri' of Kartik 1680 on page 505.

SOURCE PAGE 133

This passage is evidence regarding the seed correction in the Makaranda table.

(2) "...the thousand parts of Kaligata 1000 Shadi 4.42. 46 Saturn seed and Dhanam|| It is necessary that Budh seed and Dhanam be included with 1.35.15..."6.17.1. The Sun seed is not present with the fraction of Shani-trijaya 3.8.31. The seed of Shani and Shukra is deficient 4.42. 46. The seed of Sanskrit and Budh..."1 The famous astrologer Shankar Balakrishna Dikshit writes in pages 184 and 257 of his Marathi Indian astrology text:— (3) "The seeds of the planets are the seed correction of the Sun Siddhanta in the Makarandagrathas"...; "The Makarandakas have the seed correction of the Sun Siddhanta, which has been written before regarding that" From these quotations it is proven that for hundreds of years, in all the Panchangs that are formed according to Makarandasarin or Grahalaghava, there is revision according to the seed correction. Therefore, I have the view that the statement of Kamalakar ji was never accepted in practice. Kamalakar ji has made the meaning of the verse of Acharya Vashistha as "It is this concise summary- the scripture is valid. The Sun and the Moon will change in each age" by changing the word 'visrasti' to 'vistruti', but this is entirely unscientific, misleading and against the ancient scientific method, and it is written only to support his side. Now I write what has been said for more than one and a half thousand years in support of the other side, which will prove how our ancient astrologers have viewed the ancient texts of astrology and what arguments they have written for the revision of them from time to time. At the time when Suryansh Purusha began to teach the Sun-Siddhanta to Mayasura, it was said, 'The scripture is that which the former sage Prakasha taught. The change of ages is only...' This is the 6th verse of the Chapter on Mean Motions; which has been explained. Then when the sages received the teaching of astrology from Mayasura, then the first Mayasura gave the teaching in 21 verses in the chapter of Bijop- Nayanadhyāya what he had learned from Suryansh Purusha, which explained the reason,

1. Makaranda Table page 36, printed in Bombay.

SOURCE PAGE 134

> "चक्रानुपातजोमध्यो मध्यवृतांशजः स्फुटः ।
> कालेन दृक्समो न स्यात् ततो बीजक्रियोच्यते ॥ ५ ॥
> बीजं निःशेष सिद्धान्त रहस्य परमं स्फुटं ।
> यात्रापाणिग्रहादीनां कार्याणां शुभसिद्धिदम् ॥ २१ ॥"

That is, because time is taken, equality is not established, therefore the method of seed action is explained. From the seed action, travel, marriage, and other auspicious events are fruitful only from the Sanskrit fine planets.

But it is regrettable that the pundits of the earlier side consider this chapter as abrogated. In my understanding, this matter is that what Suryansh Purush said is that this must be abrogated because it is the seed correction of Mayasur, not of Suryansh Purush. But if it is assumed that Mayasur has the same thing as what the sages said, then the Surya Siddhanta found by the sages is prevalent at this time, then there is no need to consider it abrogated. The point is also that the prevalent Surya Siddhanta is the one whose teaching Mayasur gave to the sages. In this, the chapter on the seed and the end states that it should be clear what teaching Mayasur received from Suryansh Purush and what Mayasur increased by his own experience.

Regarding the relationship of right ascension, Brahmagupta writes in Shak 550, Samvat 685, that-

> "प्रतिदिवस विसंवादाद् ग्रह तिथि करणक्ष दिवसमासानाम् । ग्रहणग्रह-
> योगादिषु पादं पादेन कः स्पृशति ॥ ५७ ॥

Knowing this, the efforts of the wise are daily.

In which there is only one observation of the eyes, it is always.

In these two shlokas, M. M. Sudhakar Dwivedi writes in the tilak:-

"In the conjunctions of planet-tithi-karaṇa-riksha-divasa and in the conjunctions of eclipse-planet and so on, the foot which touches the earth by the shadow of the east, the lower part of the foot, the part which touches the foot, is the one which spreads the shadow of the foot, 'the farthest touch of the foot is the shadow of the foot'"

"In the Tantra-bhransha, there is the Tantra-ganana, and in the opposition of the east, and this previously stated, the explanation of which is known daily, the Vedas, etc., in that Tantra, the visa etc., and the effort is as

SOURCE PAGE 135

It is the calculation of the eyes. In this system, where the calculation of the eyes is always, that is the system is esteemed-measure.1

In the above narration, along with planets, conjunctions, etc., the words Tithi, Karana, Raksha (constellation) also appear; from which it is revealed that there is a regulation of the calculation of the eyes even for what Pandit Girijaprasad ji did not see, and the need to perform seed correction is indicated. Therefore, for the calculation of the visible, Brahmagupta ji did not consider it against the scriptures or the ancient sayings. He has strongly refuted what he considered against the scriptures.

The renowned Bhaskaraacharya ji writes in the year 1207 of 1072 AD:-

“Yatra vivahotsavjatkādau khetaiḥ sphuṭairev phalasphuṭtvaṁ. Syāt prochyate tena nabhaścarāṇāṁ sphuṭakriyā drigganitayakṛdaya.2”

This means that the clarification of the planets for activities like journey, marriage festival, etc., yields more results than the clarification of the planets, and the method of clarifying the planets is the same which results in the calculation of the eyes.

Above is the evidence that how much the planet-shorthand is considered acceptable nowadays. In the course of this planet-shorthand, the father of Acharya Ganesh Daivagya, Acharya Keshav, in the year 1553 of 1418 AD, wrote in a book named Graha-Koutuk in ancient texts in favor of making amendments:-

In the eclipses of the Sun, the Moon, and Mercury, they are visible as numbers. In the slow sky, in the conjunction of constellations and planets, more than five parts are visible in the rising. ……And there is in the years and in the periods of enjoyment of these. And there will be many more in many times. Because the names of the revolutions and the conjunctions of the planets are visible in many ways, and there will be many more in many times. ……And in the future, by observing the present events of the conjunctions of constellations and planets with the knowledge of the good ones, the calculations of the revolutions of the planets are done. ……And the small calculations are done by projecting the years and periods of enjoyment. ……And by the opposite methods of the lunar eclipse dates, which I have established as the ultimate result, the lunar altitude is made in place of the center and the sphere. There, the result is increased by the utmost.

1. Tantrapariksha chapter pages 166-170, edited by M. M. Sudhakar Dwivedi, Brahma Sphut Siddhanta.

2, Siddhānta Śiromani Ganitādhayāy page 56.

SOURCE PAGE 136

There, the Moon is seen in the shadow of the Sun. It is dependent on the higher Brahman. The Sun is the center of all directions and holds the rays. Other planets, constellations, and the conjunctions of planets are seen as occurring in the events. By observing these, the planets are seen to be in the shadow of the Brahman. Brahma, Budha, Brahma, Surya, Shukra. The five parts of Saturn are seen as greater. By observing these current events, the mathematics of the planets are performed. "1"

From this long exposition it is clearly evident that by what means the revolutions of the Sun, Moon, and other planets should be calculated by observing the eclipses. This is also ordered for the future. This exposition also clearly mentions the Sūrya Siddhānta. Based on these rays and seeds of the father, Acharya Ganesh Daivagya created 'Grahalaghava', in which the 16th verse of the Chapter on Mean Motions states that in the year 1577, 1442, the year is written.

"So also, the great ones like the Sun, the Moon, and Rahu, are seen by the five parts of Saturn. The motion of Shukra, which moves towards the center, is seen by the true planets. The religious duties and other things are done by them. "2"

From this it is revealed that Ganesha orders all auspicious works like festivals etc. to be determined by the eyes alone, not by the 'nirbija' Sūrya Siddhānta.

In its commentary, Mallari ji writes in the year 1682, 1547, "……These planets are seen by the eyes alone, the true planets…… In this book, the true planets and the religious duties etc. are done. The eclipses, auspicious days like Ekadashi, etc. are done. No policy. Politics, punishment, etc. The auspicious work, vow, marriage etc. These days are derived from this book. This feeling is such that in which, in which time, the things that are seen by the eyes alone are considered as the shadow of the body."3

Then Mallari ji says, "That planet which is not seen by the yantra by observing the sky, yet is seen by the observer, is the mean planet. By knowing the motions of the planets through the yantra, the mean planet is understood to be among the true planets by the observer."4

SOURCE PAGE 137

Mallari ji writes in one place, "For the duration of the planet revolution, the planetary motion is the sum of the mean motions and the mean motions of the mean motions. Because of the small deviations in the minute quantities, the result is the gross motion. And so, the motion is the same. And in this time, the motions are seen as the same. . . . In these, the observations of the planets are taken by marking the arcs and the grahaṇāni (eclipses) which are completed by the treatises of the planets. This is mentioned by Acharya.

In this excerpt, to prove the argument that Mallari ji has taken, the treatises of Varaha-Mihir, Vashistha, Sūrya Siddhānta, and Brahmagupta are given, because I have already given them. For the purpose of finding equality, the examination can only be done by observation. Therefore, the rule can be called pure only when there is equality in mathematics and observation. Mallari ji's point is that the numbers obtained by observation have some grossness 'due to the small deviations', for which one should make amendments by observing current events from time to time.

Readers may have become bored with many long excerpts, so I will not give the excerpt from Acharya Ganesh Daivagya's book Vṛhatthichinṭāmaṇi. Although in this, it is explained that why there was a difference among Brahmacharya, Vashistha, Kashyap, Mayasura, Aryabhata, Durgasingha Mihir, Brahmagupta, Keshav, etc., and why the need arose to create new treatises at that time and how to make amendments when further need arises. Still, one cannot leave without the last verse which is:-

"How if the celestial bodies, the lunar nodes, and the planets are observed in the same time, then the conjunction is pure, as stated by the light of the intellect obtained from the knowledge of the great knowledge."2

After reading these quotations, who will not agree that our old Acharyas and scientists have shown by rational arguments that the trigonometrical

1. M. M. Sudhakar Dwivedi edited grahalaghav page 55.

2. M. M. Sudhakar Dwivedi edited Ganakatarangini page 63.

SOURCE PAGE 138

For the time being, the principles in the Siddhanta texts need to be revised from time to time, and should all worldly affairs be done knowing the date, yuga, karana, nakshatra etc with these revisions? Nowadays, no 'English mathematician' can understand more than what the old Acharyas said in support of their side.

> राशिलिप्ताष्टमो भागः प्रथमं ज्यार्धमुच्यते ।
> तत्तद्विभक्त लब्धोनमिश्रितं तद् द्वितीयकम् ॥१५॥
> आद्येनैवं क्रमात्पिण्डान्भक्त्वा लब्धोनितैर्युतैः ।
> खण्डकैस्स्युश्चतुर्विंशज्ज्यार्धपिण्डाः क्रमादमी ॥१६॥

Translation—(15) The eighth part of the number of arts in a quantity is called the first 'sine'. By dividing this and subtracting the remainder from the result obtained by subtracting the profit from it, the second sine is obtained. (16) Similarly, by dividing all the sines from etc. by the first sine and adding the products, subtracting the sum from the first sine and adding the remainder to the last sine, the resulting sum will be the next sine. In this way, there will be 24 bodies in sequence.

Scientific Commentary—What is sine and how is its value derived from trigonometry has been discussed in the Scientific Commentary of the Chapter on Mean Motions, verse 60. You should look at the second picture given below that verse. In the 15th verse above, the word 'jyartha' is used instead of 'jya', one should not be confused by this. Both are considered equal in meaning. 'Jyartha' is said for 'jya' because the simplest way to know the 'sine' of an angle u a aa is to make a sector u a a whose central angle is double the desired angle a, then draw the chord or sine u a a of this sector and halve it. Just half of this chord (jyartha) is the sine of the desired angle u i. That is why jyartha and jya are synonymous (Figure 24).

From this verse it also appears that the Acharya did not understand any difference in the arc or sine of one part of a degree which is 8th, i.e., 33 degrees or 225 arcminutes. After this, the method for how the sines of multiples of 33 degrees, three times, four times, etc., are found is explained. In short, the method can be written in the language of algebra as follows:

Sūrya Siddhānta

Scientific Commentary

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sin(n) - sin(n-1)p = sin(n+1)p - sin(n)p = sin(n+1)

and sin(n+1)p - sin(n)p = 2sin(n)p - sin^2p = 2sin(n)p - 2cos^2p = 2sin(n)p (1 - cos^2p) = 2sin(n)p cos^2p = 2sin(n) (1 - cos^2p) = 2sin(n)x sup (2) - sin(n+1)p - sin(n+4)p = 2sin(n+3)p - (sin(n+2)p + sin(n+4)p) = 2sin(n+3)p - 2sin(n+3)p cos^2p = 2sin(n+3) (1 - cos^2p) = 2sin(n+3)x sup (3) - sin(n+1)p - sin(n+1)p = 2sin(n) - sin(n-1)p - sin(n+1)p = 2sin(n) - sin(n-1)p + sin(n+1)p = 2sin(n) - 2sin(n)cos^2p = 2sin(n) (1 - cos^2p) = 2sin(n)x sup (n)

By adding the sums of the same sides of the equations (1),(2),(3)···(n)

t1 - t n+1 = 2 ujyā p(jā p + jā 2 p,
+ jā 3 p +...jā n p)

But $t_1 \rightarrow t_{n+1} = jā\ p + jā\ n\ p - jā(n+1)p$

$\therefore jā\ p + jā\ n\ p - jā(n+1)p = 2\ ujyā\ p$
$\times(jā\ p + jā\ 2\ p + \ldots jā\ n\ p)$

$\therefore jā(n+1)p = jā\ n\ p + jā\ p - ujyā\ p(jā\ p + jā\ 2\ p + \ldots jā\ n\ p)$

Here $p = 3045' = 225'$

$\therefore$, $2\ ujyā\ p = 2\ ujyā\ 225' = 2(1 - \cos(225'))$

$= 2(1 - .6678) = 2 \times .0022 = 44/10000 = 1/227$

$= 1/225$ with a decimal

$\therefore jā(n+1)p = jā\ n\ p + jā\ p - 1/225 \times$
$(jā\ p + jā\ 2p + \ldots jā\ n\ p)$

Tatvaashvinoṅkābdiveda rūpabhūdharartavaḥ .
Khānkāṣṭau pañcaśūnyeśā vāṇbhigunendaivaḥ ॥17॥
Śūnyalocanpañcaikāśchidrarūpmunīndavaḥ .
Biyaccandrātidhṛtyo guṇarandhramvarāśvinah ॥18॥
Muniṣaḍyamenātraṇi candrāgnikṛtadasrakāḥ .
Pañcāṣṭhaviyākṣiṇi kuñjarāśvinagāśvinah ॥19॥
Randhrpañcaṣṭakayamavasvadrayankayamastathā .
Kṛtaṣṭhaśūnyajvalanā nagādriśśivahnayaḥ ॥20॥
Ṣaṭpañcalocanguṇāścandranetrāgnivahnayaḥ .
Yamādravahnijvalanā randhrśūnyārnvāgnyah ॥21॥
Rūpāgnisāgarguṇā vasutrikṛtavahnayaḥ .
Projbhyotkramena vyāsārdhadutkramjārdhipiṇḍakāḥ ॥22॥

(17) 225, 446, 671, 860, 1105, 1315;(18) 1520, 1719, 1610, 2063;(19) 2267, 2431, 2585, 2728;(20) 2856,

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SOURCE PAGE 142

2678, 3084, 3177;(21) 3256, 3321, 3372, 3406;(22) 3431, 3438 are the sines of 24 bodies of a right angle in sequence of degrees, 23 degrees, 73 degrees, 114 degrees, 15 degrees etc. If these are subtracted in reverse order (from reverse) by a radius of art, i.e., subtracting 3438, then the sines of 24 bodies of a right angle in sequence of reverse are found. Their values are also given in the next five shlokas.

Scientific Commentary - there is no need to say more about this. After the next five shlokas, the values of these sines and reverse sines will be compared with the values obtained by the current method.

The rule written to know the values of the reverse sine is very simple and fundamental. If the last number in 3438 is subtracted from 3438, zero will remain, which is the reverse sine of zero degrees, and if 3431 is subtracted, 7 will remain, which is the reverse sine of 225 art. This can be known based on geometry as follows—if there is an angle of 225' art in figure 24, then the value of u a is 225', the value of

u i is 225'(with a comma), the value of a i is 3431', and the value of i a is 7'. This is the value of i a which is the reverse sine of the angle u a i. Similarly, the sines and reverse sines of other bodies can be known.

> मुनयो रन्ध्रयमला रसषट्का मुनीश्वराः ।
> द्व्यष्टैका रूपषड्दस्राः सागरेषुहुताशनाः ॥२३॥
> खर्तुवेदा नवाद्यर्था दिङ्नागास्त्यथकुञ्जराः ।
> नगाम्बरवियच्चन्द्रा रूपभूधरशङ्करः ॥२४॥
> शरार्णबहुताशैका भुजङ्गाक्षिशरेन्दवः ।
> नवरूपमहीध्रैका गजैकाङ्कनिशाकराः ॥२५॥
> गुणाश्विरूपनेत्राणि पावकाग्निगुणाश्विनः ।
> वस्वर्णवार्थयमलास्तुरगर्तुनगाश्विनः ॥२६॥
> रन्ध्राष्टनवनेत्राणि पावकैकयमाग्नयः ।
> अष्टाग्निसागरगुणा उत्क्रमज्यार्धपिण्डकाः ।।२७।।

Translation—(23) 7, 26, 66, 117, 182, 261, 354;(24) 460, 576, 710, 853, 1007, 1171;(25) 1345, 1528, 1716, 1618; (26) 2123, 2333, 2548, 2767; (27) 2686, 3213, and 3438 are the bodies of reverse sines in sequence.

A table is given below which is the summary of the above eleven shlokas:-

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SOURCE PAGE 143

The value of the arc when inverted according to the Indian method, the value of the arc according to the modern method, the value of the arc according to the modern method when the arc is inverted according to the Indian method, the value of the arc when the radius = 3438 radius = 3438 radius = 1 radius = 3438 when inverted the value of the arc when radius = 1 1 2 3 4 5 6 7 13045’ 225 224.85.9654 7.0022 2 7030’ 448 448.65.1305 26.0086 3 11015’ 671 670.72.1951 66.0162 4 150 860 886.82.2588 117.0341 5 18045’ 1105 1105.01.3214 182.0761 6 22030’ 1315 1315.05.3827 261.0761 7 26015’ 1520 1520.58.4423 354.1031 8 30015’ 1716 1716.00.5000 460.1340 9 33045’ 1610 1610.05.5555 576.1685 10 37030’ 2063 2063.05.6088 710.2066

See Figure 18 page 166 'Science' part 18 number 5

SOURCE PAGE 144

1 2 3 4 5 6 11 415/ 2267 2267.02 ·6564 853 ·2481 12 4500 2431 2431.01 ·7071 1007 ·2626 13 48045/ 2585 2584.708 ·7516 1171 ·3406 14 52030’ 2728 2727.55 ·7634 1345 ·3612 15 56015- 2856 2858.55 ·8315 1528 ·4445 16 6000 2978 2977.31 ·8660 1716 ·5000 17 63045/ 3084 3083.45 ·8666 1918 5577 18 67030’ 3177 3176.37 ·6236 2123 ·6173 19 70115’ 3256 3255.75 ·6466 2332 ·6786 20 7500 3321 3320.85 ·6656 2548 ·7412 21 7845’ 3372 3371.65 ·6808 2767 ·8046 22 8230’ 3408 3408.75 ·6614 2686 ·9346 23 86015’ 3431 3430.85 ·6678 3213 ·9346 24 9000 3438 3438.00 10000 3438 1.000

SOURCE PAGE 145

Scientific Commentary—in Sūrya Siddhānta, such relations (ratios) of trigonometry are given. To find the cosine of the coterminal angle, this rule has been applied in practice that if the sine of an angle is given, then the cosine of the angle obtained by subtracting that angle from 60° is the same, that is, the

sine of an angle is equal to the cosine of its complementary angle. The value of the tangent of an angle is not given nowadays, but its application is indirectly done by dividing the sine of the angle by its cosine.

If the value of the angle is such that it lies between the above-given bodies, the method of knowing its sine, cotangent, or versed sine from the proportional parts is explained in the next 31-34 shlokas. Similarly, if the value of the sine is known, the method of finding the arc (angle) from it is also in these shlokas.

Bhaskaracharya has explained the method of knowing sine and arc, and the subtle method.

> ज्या के पर्याय क्रमज्या, भुजज्या, बाहुज्या, अर्द्धज्या इत्यादि तथा कोटिज्या
> के लम्बज्या भी प्रयोग किये गये हैं ।

> परमापक्रमज्या तु सप्तरन्ध्रगुणेन्द्रवः ।
> तद्गुणा ज्या त्रिजीवाप्ता तच्चापं क्रान्तिरिष्यते ॥२८॥

(28) The value of the great revolution sine is 1367 art. Multiplying this by the sine of the (fractional part), and dividing the result by the radius, the sine of which is the angle, that is the value of the revolution.

SOURCE PAGE 146

Scientific Commentary—in this verse, it is shown how the 'sine' is treated. Along with this, the rule is also stated that what is the mutual relationship between the sides and angles of a right-angled spherical triangle is. The value of the ultimate revolution sine is stated as 1367 art; from which it is apparent that the value of the ultimate revolution is 24°; because the sine of 24° is 1367 art in the above manner; although by pure calculation it is the sine of 23°58‘31".

In the given figure 25, there is the section of the vernal equinox and the section of the ecliptic and the section of the equator. It is the section of the polar circle, that is, the section of that circle which passes through the pole and makes a right angle with the point on the equator. The angle SP is the angle between the ecliptic and the equator (obliquity of the ecliptic) which is 24° according to the above verse. The distance of S from S on the ecliptic and the portion of S of the section of the equator, while SP is a right angle, that is, SP, the point S is called the declination. This is also called the arc. According to the given rule,

```
sine (Vas) x 1367 = sine (Sap) 3438
```

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Athajya ( Vas ) x sine ( savpa ) / radius = sine ( sap)
```

If the radius is taken as 1 instead of 3438, as is the current practice, then 4067 must be kept instead of 1367 arts. This will bring some simplicity in multiplication and division, and then the form of this formula will be only this.

```
sine (vas) x sine (savap) = sine (sap)
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This is famous nowadays with one rule of Napier with some differences, which the mathematician named Napier published in his book 'Mirifici Logarithmorum Canonis Descriptio' in Edinburgh in 1614

AD or 1671 AD. This is the device to remember Napier's rule:

*See Taddhunter and Lehem's Spherical Trigonometry of 1611 page 50

SOURCE PAGE 147

The two sides that form the right angle, the complements of the hypotenuse, and the complements of the other two angles of a right spherical triangle, excluding the right angle, are called the circular parts of the triangle. In this way, a right spherical triangle has 5 circular parts. These five parts are arranged around a circle in the same order in which they are arranged. The order in which they are in the triangle. Suppose A, B, C are

It is a spherical triangle. A, I, U are the points where the sides of the triangle meet A, U A; A I, U I; and A U, I U. U A I, A I U and A U I angles are expressed in brief by the letters A, I, U. Similarly, the side opposite to angle A is U I as 'A', the side opposite to angle I is A U as 'I', and the side opposite to angle U is A I as 'U'. The simple rule is that the angles of the triangle are expressed by short vowels and the sides opposite to them by long vowels of the same type. The sides of the spherical triangle are also measured by angular values. If I is a right angle, then this triangle is called a right-angled spherical triangle. Then the side opposite to it, I, is called the hypotenuse. [See figure 26]

In Napier's rule, the two sides a, b, and c of the right-angled spherical triangle, excluding the right angle, are the complements of the angles a, b, and c by $\pi/2$ - a, $\pi/2$ - b, and $\pi/2$ - c, of the spherical segments, as shown in the figure [see figure 27].

SOURCE PAGE 148

Choose one of these five and keep its name as the mean section. The two sections adjacent to it are called the adjacent sections; the remaining two sections are called the opposite sections. Now Napier's rules can be written as follows:-

(1) Sine of the mean section = product of the tangents of the adjacent sections.

(2) Sine of the mean section = product of the sines of the opposite sections.

> यही दूसरा नियम उपर्युक्त श्लोक में नेपियर से कम से कम एक हजार वर्ष
> पहले प्रयोग किया गया है ।

> ग्रहं संशोध्य मन्दोच्चात् तथा शीघ्राद्विशोध्य च ।
> शेषं केन्द्रं पदैस्तस्माद् भुज ज्या कोटिरेव च ॥२६॥
> गताद्भुजज्या विषमे गम्यात्कोटिः पदे भवेत् ।
> समे तु गम्याद्बाहुज्या कोटिज्या तु गताद् भवेत् ॥३०॥

Translation—(26) The remaining parts obtained by subtracting the apogee and epicycle apogee of a planet from its mean position are called the mean center and the quick center respectively. Make terms from these and knowing the terms, calculate sine and

SOURCE PAGE 149

Make the arc. (30) The sine of the part that remains in the odd position is called the cosecant, and the sine of the part that is moved is called the secant, but in the even position, the sine of the moved part is

called the secant sine and the sine of the remaining part is called the cosecant sine.

Sūrya Siddhānta—it is stated in the fourth and fifth verses of this chapter that up to 1800, the apogee or epicycle apogee, which is situated behind its planet, pulls towards it, i.e., towards the east, according to the proximity, by which the position of the true planet can be known by performing a calculation in the mean planet, etc. In the above 26th verse, it is stated how the distance of the mean planet from the apogee or epicycle apogee should be calculated. The difference between two points of a circle can be shown in two ways. If the difference between the points m, ma, mi, and mi is not measured in the direction of the arrow in figure 15, then it will be um; uma, umi, and umi in sequence. But if the difference between these points is measured in the opposite direction from u, then the difference between u and m is 360°—um, the difference between ma and m is 360°—uma, the difference between mi and m is 360°—umi and the difference between mi and m is 360°—umi. The direction indicated by the arrow in the figure is called the opposite or apasvya direction in Sanskrit texts, nowadays it is called 'positive' or 'direction opposite to the clock'. People living to the north of the celestial equator see the sun, moon, and planets, etc., moving in this direction in their orbits. The opposite direction is called vilom, pratilom, savya, and nowadays 'negative' or 'clock direction' in Sanskrit. Due to the daily motion of the Earth, people living in the northern hemisphere see the sun, moon, stars, etc., moving in this direction. In Sūrya Siddhānta, the method of knowing the difference between the planets from the epicycle apogee or apogee, which are in sequence the fast kendra or slow kendra, is stated to be measured in the opposite or negative direction. Therefore, it is said that the mean planet should be subtracted from the epicycle apogee or apogee. However, many other scholars like Brahmagupta, Bhaskaraacharya have written to measure the difference between the mean planet and the apogee from the opposite direction and the difference between the epicycle apogee and the mean planet from the opposite direction. The reason for this is that the mean planet moves in the opposite direction because it is more rapidly moving from the apogee, and the epicycle apogee moves in the opposite direction because it is more rapidly moving from the mean planet; therefore, the mean planet moves in the opposite direction from the epicycle apogee. Whether the difference between the mean center or the fast kendra is measured, both have the same meaning. Bhaskaraacharya's method is natural and some of the Sūrya Siddhānta are misleading.

SOURCE PAGE 150

When the mean center and the epicycle center of the planet are known, then it is necessary to know their sine and cosine, because they are further required. Those who are familiar with modern trigonometry can know directly because they know how the cosines, sines, etc. from zero to 3600 can be known. But in ancient times, to find the sine of any angle from zero to 3600, they first looked at which position (quadrant) it is in. If the mean center or epicycle center is within zero and 60°, then in the odd position, above 60° but less than 180°, then in the even position, above 180° and less than 2700, then in the odd position, and above 2700, then in the even position. In short, the first and third positions are called odd positions and the second and fourth positions are called even positions.

To know in which position the planet is, the mean center or epicycle center should be divided by 60°. If the remainder is zero or 2, then it should be understood as an odd position, and if 1 or 3 is, then it should be understood as an even position. The remainder that is left is called the arc part. Subtracting this remainder from 60° gives the part that is to be subtracted. If it is an odd position

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SOURCE PAGE 151

Calculate the sine of the portion of the past and the mean portion of the present. This is called the versed sine. But if the portion is odd, it is called the quotient of the sine of the portion of the present and the sine of the past.

This matter can be understood easily from figure 28. The given circle is the orbit of a planet. 'U' is the position of the apogee or apogee. Mi, Mi, Ma, M are the mean positions of a planet. Therefore, moving in the opposite direction, Umi, Umi, Uma and Um are the mean centers of the planet in the first, second, third and fourth positions or in the odd, even, odd and even positions. In the first position, Umi is past and Mi is future; therefore, the sine of Umi, i.e., Mi, is called the versed sine of Mi, and the sine of Mi, i.e., Mi, is called the quotient of the sine. In the second position, Pm is past and Mi is future, therefore, the sine of Pm, i.e., Mi, is called the quotient of the sine of Mi, and the sine of Mi, i.e., Mi, is called the versed sine. In the third position, 'Ni Ma' is past and 'Ma Pu' is future, therefore, the sine of Ni Ma, i.e., Ma Ja, is called the versed sine, and the sine of Ma Pu, i.e., Maka, is called the quotient of the sine. Similarly, in the fourth position, Pu is past and Mu is future, therefore, the sine of Pu, i.e., Ma Ka, is called the quotient of the sine, and the sine of Mu, i.e., Ma Ja, is called the versed sine.

In short, it should be said that the difference between the mean planets is the versed sine of the line drawn from the line, i.e., the lower line, passing through the mean point. The difference of the mean planet from the line that forms a right angle with this line, passing through the mean point, is called the quotient of the sine. If the radius is equal to 3438 units, then the measurements of Mi, Ma Ja and M Ja in these units will be the versed sine, and the measurements of Mi Ki, Ma, and M Ka in these units will be the quotient of the sine.

> आगे के दो श्लोकों में यह बतलाया गया है कि किसी अंश की ज्या कैसे
> निकालनी चाहिये ।

> लिप्तास्तत्वयमैर्भक्ता लब्धा ज्यापिण्डकं गताः ।
> गतगम्यान्तराभ्यस्तं विभजेत्तत्वलोचनैः ॥३१॥
> तदवाप्तफलं योज्यं ज्यापिण्डे गतसंज्ञिते ।
> स्यात्क्रमज्याविधिरयं उत्क्रमज्यास्वपि स्मृतः ॥३२॥

Translation—(31) To know the sine of a certain fraction, divide it by 225; the remainder is the past sine; the remainder is the past sine.

SOURCE PAGE 152

And multiply the difference of the sines of the two celestial spheres by and divide the product by 225. (32) By adding the result to the sine of the geocentric sphere, the same angle will be the sine of the desired part. Similarly, the versed sine should also be calculated.

Scientific Commentary—the sines of 24 celestial spheres are stated in the 17-22 shlokas of this chapter. In addition, if the sine of any intermediate angle is to be known, it should be known from shlokas 31-32. Suppose we want to know the sine of 66°. First, it should be seen in which body 66° is. The bodies are placed at a difference of 225' art or 3045' or 34 degrees, so the art of 66° should be made and divided by 225 or 66° should be divided by 34. The method of making the art is explained in the shloka, therefore

```
66 ° = 66 X 60 ′ = 3660 ′
```

3660 ′ ÷ 225 = 1711 / 225

Therefore, the geocentric body is 17 and the mean body is 18.

18th circle sine = 3177 ′

17th arc sine = 3084 ′

, geetgamyanter = 63 ′

Now it should be known from Trairashi that when the difference between the geocentric and geocentric bodies is 225, then there is a difference of 63' in their sines, therefore when the geocentric body is 135' more than the desired part of the geocentric body, what will be the difference between the sine of the geocentric body and the sine of the desired part. That is

225: 135:: 63: Desired difference

∴ Desired difference = 135 × 63225 = 3 × 635 = 2765 = 56 ′ less than the difference.

Adding this to the sine of the geocentric motion, i.e., in 3084', results in 3140'. This is the sine of 66°.

If the value of the angle is in full degrees, the sine can be calculated without making art, as can be seen in the above example where the sines of 660 can be extracted:-

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SOURCE PAGE 153

66 ° ÷ 33 / 4 = 22 x 4 / 5 = 88 / 5 = 173 / 5

Difference in sines of the 17th and 18th bodies

= 63 ′

∴ 63 × 35 = 2765 = 56

17th sine = 3084

∴ 66 ° sin = 3140

> अगले श्लोक में यह बतलाया गया है कि यदि ज्या दी हुई हो तो कोण कैसे
> जाना जा सकता है ।

> ज्यांप्रोज्झ्यान्यत्तत्वयमैहत्वातद्विवरोद्धृतम्।सङ्ख्यातत्वाश्विसंवर्गेसंयोज्यधनुरुच्यते॥३३॥

Translation—(33) If one wants to know what fraction (arc) the given sine is, first see which among the sines of 24 bodies can be subtracted from the given sine. Multiply the remainder obtained by this subtraction by 225 and divide the product by the difference of the latitudes and longitudes, add the remainder to that product which results from multiplying that body by 225 whose sine was subtracted.

Scientific Commentary—In this verse, if the sine is known, the method of knowing the angle is explained. This will be clear by an example. Suppose the sine of an angle is 3140', now one wants to know what the angle is.

17—According to 22 verses, the sine of the 17th body is 3084' and the sine of the 18th body is 3177'. Therefore, subtracting 3084' from 3140' leaves a remainder of 56'. The difference between the sines of the latitudes and longitudes is 63'.

63′ : 56′:: 225′ : Ishtakala herefore Ishtakala = 56 imes 22563 = 420031 = 135′ The difference of the sine of the 17th body is 17 imes 225 = 3825′ herefore The sum of both is 3960′ herefore The sine of which is 5140′ have 3960′ or 660

SOURCE PAGE 154

रवेर्मन्दपरिध्यंशा मानवश्शीतगो रदाः ।
युग्मान्ते विषमान्ते च नखलिप्तोनितास्तयोः ॥३४॥
युग्मान्तेऽर्थाद्रयः खाग्निः सुरास्सूर्या नवार्णवाः ।
ओजे द्व्यगा वसुयमा रदा रुद्रा गजाब्धयः ॥३५॥
कुजादीनां ततश्शीघ्रा युग्मान्तेऽर्थाग्निदस्त्रकाः ।
गुणाग्निचन्द्राः खागाश्च द्विरसाक्षीणि गोऽग्नयः ॥३६॥
ओजान्ते द्वित्रिकयमाः द्विविश्वे यमपर्वताः ।
खर्तुदस्रा वियद्वे दाश्शीघ्रकर्मणि कीर्तिताः ॥३७॥

Translation—(34) At the end of even positions, the apogee of the Sun is 14° and that of the Moon is 320. At the end of odd positions, the apogee of each is 20 degrees less. (35) The apogees of Mars, Mercury, Jupiter, Venus, and Saturn at the end of even positions are 75°, 300, 330, 120, and 460 respectively, and at the end of odd positions are 720, 280, 320, 11°, and 480. (36) The epicycles of these five planets at the end of even positions are 235°, 133°, 700, 2620, and 360, and (37) at the end of odd positions are 232°, 1320, 720, 2600, and 400 which are mentioned for the swift motion.

SOURCE PAGE 155

Sūrya Siddhānta—the difference that occurs in the mean and true positions of a planet due to the apogee is called the mean motion, and the difference that occurs in the mean and true positions due to the epicycle apogee is called the quick motion. This changes according to the distance of the apogee or epicycle. The greatest difference between the mean and true positions of the planets is due to the apogee, and the greatest difference due to the epicycle is called the quickest motion. This is known by sighting, i.e., by the gnomon instrument. The circle drawn by taking the sine of the true mean motion as the semi-diameter is called the mean circumference. Similarly, the circle drawn by taking the sine of the quickest motion as the semi-diameter is called the quick circumference or the running circumference. If a circle is drawn and the Earth is taken in the center and the mean planet is considered to be moving on the circumference, then the circumference is called the planet's orbital circle or orbital circle. If 360

equal parts are made of this orbital circle, the extension of the mean circumference of the Sun will be at the end of the samapodas. Similarly, one should understand the magnitudes of the mean and quick circumferences of other planets. This can also be written that the mean circumference of the Sun is 360 of the orbital circle. In Figure 26, if the position of the Earth, the orbital circle of any planet, and its mean and true positions are shown, and if the value of the latter is the ultimate, then the sine of the mean motion, which is the linear distance between the lines in between, is called the ultimate mean motion. The smaller circumference drawn by taking the sine of the mean as the semi-diameter and the sine of the mean as the mean is called the mean circumference. If the extension of the orbital circle is considered as 360 parts, then the circumference of the mean circumference is the arc corresponding to the number of parts. Similarly, one should understand the length of the quick circumference. This magnitude also varies according to the opinions of different teachers. The reason for this is that the value of the ultimate mean motion is not always the same; it is changing gradually. The ultimate mean motion of the Sun decreases by 3 degrees in one thousand years. At this time, the ultimate mean motion of the Sun is 1055'. In Sūrya Siddhānta, the ultimate mean motion of the Sun is 2013'41". There is also an inaccuracy due to the imprecision of sighting.

> ओजयुग्मान्तरगुणा भुजज्या त्रिज्ययोद्धृता ।
>
> युग्मे वृत्ते धनर्णं स्यादोजादूनाधिके स्फुटम् ॥ ३८॥

(38) Multiply the difference between the end of odd and even periods by the sine of the difference of the slow center or fast center and divide by the radius.

SOURCE PAGE 156

If the mean center or the quick center is in the same position and the mean or quick circumference of the end of the odd term is less than the end of the same circumference of the mean or quick circumference, then add that difference to the prime vertical circumference, then the true mean or quick circumference of the center will be obtained. But if it is more than the circumference of the odd term, then subtracting that difference from the prime vertical circumference will yield the true circumference.

Scientific Commentary - The view of the Sūrya Siddhānta is that the value of the mean or quick circumference changes according to the hypotenuse of the mean center or quick center. The place where its magnitude is what should be extracted from trigonometry because it is given what are its values at the ends of the mean and odd terms. To know the value of the intermediate place, it should be reasoned that when the difference of the circumferences of the terms is given on the difference of the hypotenuses (the absolute value of the hypotenuse), then what will be the difference of the hypotenuses of the true center. This rule can be written in brief as—

True Mean Circumference

=Mean Circumference ± Circumferences of the ends of odd and even terms-

```
difference x true center's hypotenuse / radius
```

As the mean circumference of the Sun is 840', the difference between the mean circumferences of odd and even positions is 20', therefore if the desired mean center is 'A' then the true mean circumference will be

840' - 20' x akibhuja / 3438'

Because the mean circumference is greater.

Similarly, the true mean circumferences and the quick circumferences of other planets should be calculated.

> तद्गुणेभुजकोटिज्येभगणांशविभाजिते।
>
> तद्भुजज्याफलधनुः मान्दं लिप्तादिकं फलम् ॥ ३६॥

Translation—(36) The true mean circumference is obtained by multiplying the sine of the arc and the sine of the latitude by 360 (if the true mean circumference is in fractions) or: divide by two (if the true mean circumference is in degrees). The result of the multiplication and the quotient (if in degrees). The sine of the arc whose sine is the result is called the mean.

Scientific Commentary - This rule can be written in brief as:-

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SOURCE PAGE 157

Bhujphal = Sputamandperidhi X Bhujjya / 360

Quotifal = Sputamandperidhi X Quotijya / 360

The angle whose sine is the arc (degree) is called the mean angle. In the above equations, 360 will be the same when the mean circumference is in degrees. If the mean circumference is in arcs, then 21600 should be put instead of 360.

Its derivation is as follows: To know what the difference is between the mean and true positions of the planet, our teachers imagined that the mean planet always revolves around the Earth in the opposite direction with the same speed, and the true planet revolves around a smaller orbit, in the opposite direction, such that in the same amount of time that the mean planet completes one revolution in its orbit (in the orbital circle), the true planet completes its revolution on the smaller orbit. The point where the true planet appears in the orbital circle while revolving on the smaller orbit is called the position of the true planet. This will be clearly understood from Figure 30. In this, P is the center of the Earth. The larger circle drawn with a radius of P is called the orbital circle, taking P as the center. On this orbital circle, the mean planet is considered to move in the opposite direction with a mean motion. M, Ma, Mi, Mi, Mu, Mu, Me, Ma, the mean planet has eight positions, M is the position where the difference between the mean and true planets is zero. That is, the apogee of the planet is in this direction. The number 1 and s are written in this place in the orbital circle, which shows that the mean and true planets are together at this place, and starting from this place, the orbital circle is divided into four parts at differences of three quantities in the opposite direction. That is why the digits 4 at the end of the first part, 7 at the end of the second part, and 10 at the end of the third part were written. The small circles drawn according to the smaller orbit, taking the positions of the mean planet as mean, are the true smaller orbits. To clarify the figure, the extent of the true smaller orbits and the orbital circle are not shown in the same proportion as they are directly seen or given in the texts. The smaller orbit has been drawn with some addition. According to the Sūrya Siddhānta, the values of these true smaller

orbits are also not the same everywhere. The lines P M, P Ma, P Mi, etc., where they reach the point far from the smaller orbit, the digits 1 are also written on the smaller orbit. Starting from here, the smaller orbit

SOURCE PAGE 158

There are written the numbers 4, 7, 10 in the opposite direction on the difference of three three quantities or ninety ninety degrees. At the time when the mean planet is at M, the true planet is at the point on the apogee of the small circle where 1 is written. This is the position of the apogee of the planet; therefore, U is also written there. In the time it takes for the mean planet to go from M to Ma on the orbit, the true planet goes from 1 to Ga on the small circle (the small circle is also called the small circle); because the angular velocity of the orbit of the mean planet and the small circle of the true planet (the small circle is also called the small circle) is the same, the line joining Ma and Pa is parallel to Pm, which is called the inferior line. The line joining Ga and Pa is called the gnomon. This is the distance of the true planet from the center of the Earth. This gnomon intersects the orbit at point Sa, so the true planet appears at point Sa on the orbit. This point is called the position of the true planet. The angle Sa dhanu or Sa pa is called the small fruit. The angle Ma ghanu or Mp Ma is called the small center, and Ma Sa dhanu or Mp Sa is called the true center; therefore, the difference between the true center and the small center is called the small fruit. The inferior line from Ma

SOURCE PAGE 159

The perpendicular is the hypotenuse of the mean center. The perpendicular of the line from M to M is called the ratio of M to M. This is the perpendicular on the line that forms a right angle with P M and is drawn at point P. The part of the perpendicular drawn from G to P M is called the hypotenuse and the ratio of M.

Similarly, when the mean planet is at the points of the orbital circle, then the true planet is at the points of the mean circle, etc. In such a situation, the true planet is seen at the points of the orbital circle, etc.

It falls. For these points also, the sine of the arc, the crore sine, the result of the arc, the crore result, etc., should be understood as previously stated.

When the mean center is three signs or equal to 600, then the mean planet is on M. In this condition, the true planet is in the utmost difference from the mean planet on S. This is called the utmost mean result. When the mean center is six signs or equal to 1800, then the mean planet is on M and the true planet is on G; therefore, the true planet appears at the point of the orbit. At this place, the mean result is zero and on the mean hypotenuse G all

SOURCE PAGE 160

It is smaller. When the planet is at the apogee, it is very close to the Earth. This place is called the nadir of the planet.

When the mean center is equal to 6 signs or 270, then the mean planet and the true planet are at the apogee. The difference between the mean and true planets is also absolute at this place. In the figure, the apogee is the utmost apogee.

According to the Sūrya Siddhānta, the apogee is measured in the opposite direction; therefore, according to this method, the numbers 1, 4, 7, 10 on the orbital circle and the apogee circle should be

written as in 31 figures. From this, all matters of the relation of the fast-kendra can also be known. That is why both are explained by the same figure in the Sūrya Siddhānta. However, some difficulty arises in understanding this. Bhaskaracharya has used this figure only to know the relation of the fast-kendra and all its relations like śīghraphala, śīghrakarṇa, etc. Two figures do not cause confusion. With the help of these two figures, the derivation of verses 36, 40, 41, 42, and 45 is easily understood.

It is stated in the 36th verse that

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Bhujphal = Sputamandperidhi X Bhujjya / 360
```

```
andcotifal = sfutamandperidhi X kotijya / 360
```

because when the mean planet is at m then maja jya, m ka koti jya, gabha bhujphal and bhama cotifal are called.

and ghabama = majap, because both are right angles.

therefore riangle gha gha m and riangle maja p are similar.

, gabha: gamma:: maja: measure

```
ghabamajagamamaporga ghabagamamajamaap
```

But the radius of the circle of the periphery is ga and the measure is the radius of the orbit, and the two radii have the same relation as their perimeters, therefore

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SOURCE PAGE 161

Gama = Mean Motion (Surface) • Gamma = Surface Motion • The surface motion of the celestial body = Surface motion of the celestial body Body X Surface Motion (1)

If it is in the arc of the circumference, then the value of the orbit is 360 and if it is in the arts, then the value of the orbit is 21600.

Similarly, Gama:: Puja: Measure

Tama = Puja Gama M M Surface Motion Tama = Gama Surface Motion P M Surface Motion of the celestial body P M Surface Motion Q M = Surface Motion of the celestial body Q M X Surface Motion = Motion (2)

Thus, the rule of the 36th verse has been proven. Thus, the result obtained is the sine of the angle, that angle is called apogee. In figure 30, the angle of the center is gapabha, so the gapabha angle is the apogee. To know the value of this angle according to the Indian custom, one should first know the value of the semi-chord. riangle Pbhaga and riangle Pmasa are similar.

```
Therefore      sama /    sap =     gabha /   gap
```

a th va sa ma = saap imes gabha gāp = trijya imes bhujphal mand karṇa (3)

What is obtained from this equation is the sine of the mean motion, from which the mean motion can be known from the table of sines. But in the verse the arc of the gabha

SOURCE PAGE 162

It is taken as the mean result and the necessity of equation (3) is not stated; the reason for this is that the value of the mean orbit of a planet is so small that the difference between the mean hypotenuse and the radius is very small, due to which the arc of the true position is taken as the mean result. If one wishes to make a subtle calculation, then what is stated in equation (3) must also be considered; as there is a rule for the quick result in the next 40-42 verses; because the larger value of the quick orbit results in a much larger difference between the quick hypotenuse and the quick result. Therefore, according to the 36th verse, there remains a great inaccuracy in taking the quick position as the quick result.

> शैघ्रं कोटिफलं केन्द्रे मकरादौ धनं स्मृतम् ।
> संशोध्यं तु त्रिजीबातः कर्य्यादौ कोटिजं फलम् ॥४०॥
> तद्बाहुफलवर्गैक्यान्मूलं कर्णश्चलाभिधः ।
> त्रिज्याऽभ्यस्तं भुजफलं चलकर्णविभाजितम् ॥४१॥
> लब्धस्य चापं लिप्तादिफलं शैघ्रमिदं स्मृतम् ।
> एतदादौ कुजादीनां चतुर्थे चैव कर्मणि ॥४२॥

Translation—(40) If the quick center is above 6 degrees (2700) and within 3 degrees (600), add the quotient to the radius, but if the quick center is above 3 degrees and within 6 degrees, subtract the quotient from the radius; (41) Square what is obtained and add it to the square of the true position and take the square root of the sum, which is the quick hypotenuse or the moving hypotenuse. Multiply the radius by the moving hypotenuse and divide by the moving hypotenuse, (42) The sine of the arc (angle) is the quick result. This quick result is useful for the first and fourth calculations of the five planets like Mars etc.

Scientific Commentary—The equation (3) discussed at the end of the commentary of the 36th verse is very necessary for knowing the quick result. The form of this equation for the quick result will be—

sama = radius X true position / quick hypotenuse In this, the true position that has come is known from the 36th verse, the value of the radius is already fixed, only the value of the quick hypotenuse remains to be known, the rule for which is stated in the first half of the 40th and 41st verses. Figure 31 shows that gap, gip, gup, and gap are the moving hypotenuse. Among these

SOURCE PAGE 163

Thus it is revealed that if the center is within the first and fourth terms, i.e., within 3 amounts and above 6 amounts, then the cotangent should be added to the radius, but if the center is between the second and third terms, i.e., above 3 amounts and within 6 amounts, then the cotangent should be subtracted from the radius, then by adding the square of what comes, to the square of the result, the square root will be found. These four equations can be written in one equation as:--

Chapter on True Positions

Cancer is the fourth sign and Capricorn is the tenth, so 'Karkadau' means the sixth sign from the fourth sign and 'Makradau' means the third sign from the tenth sign. The words Makradi and Karkadi might cause the illusion that when the planets are in these signs, the appropriate sign of wealth or debt should be used. Therefore, I have used terms instead of signs in the translation, which is more clear to my understanding.

The late

sama = radius X epicycle result / declination

SOURCE PAGE 164

The sine of which angle is the śīghraphala.

In the latter part of the 42nd verse, it is stated that the need for śīghraphala arises only to know the true positions of the five planets Mars, Mercury, Jupiter, Venus, and Saturn, and not for the Sun and the Moon. The true positions of the Sun and the Moon come only from the correction of the maṃda phala, as stated in the next (43rd) verse.

If one wishes to express the 36-41 verses in a single equation according to algebra, its form will be:-

shadow hypotenuse = (3438 ± (epicycle circumference X latitude) ^{2} / 21600 + (epicycle circumference X colatitude) ^{2} / 216001 / 2

> इसमें शीघ्र केन्द्र की ज्या और कोटि ज्या भारतीय रीति से निकाल कर
> उपर्युक्त ग्रह के 'भुज ज्या' और 'कोटि ज्या' के लिए लिखना चाहिये । शीघ्र स्फुट
> परिधि ३८वें श्लोक के अनुसार जानना चाहिये और इसे कलाओं में लिखना चाहिये ।

> मान्दंकमैंकमर्केन्दोभौमादीनामथोच्यते।शैघ्रंमान्दंपुनर्मान्दंशैघ्रंचत्वार्यनुक्रमात्॥४३॥

Translation—(43) The Sun and the Moon are clarified by only one correction of the slow fruit; but for the five planets like Mars, an correction of the quick fruit requires two saṃskāroṃ of the slow fruit behind which a fourth time the correction of the quick fruit is done.

Scientific Commentary—our ancient teachers have only described the method of performing a certain rite to know the true position of the Moon. However, this does not actually reveal the true position of the Moon. The Moon is such a small body that it is affected by all the planets, due to which many variations arise in its motion. Therefore, nowadays, by performing small rites of 40, the true position of the Moon can be known with purity. Among these forty rites, five are very large which must be done. Their discussion will be further in the place where the method of knowing the true positions of the planets according to the present system will be described.

The method of the four rites for knowing the true positions of the five planets like Mars is described in the next 44th verse.

SOURCE PAGE 165

> मध्ये शीघ्रफलस्यार्धं मान्दमर्धफलं तथा ।
> मध्यंग्रहे मन्दफलं सकलं शैघ्रमेव च ॥४४॥

Translation—(44) Subtract the epicycle from the epicycle to find the fast kendra and the śīghraphala. Add half of this śīghraphala to the madhyama planet (if the fast kendra is less than 6 degrees) and subtract (if the fast kendra is more than 6 degrees); whatever results from adding or subtracting is called the first correction yukta madhyama planet. Subtract the mandocca from this first correction yukta madhyama planet, consider the remainder as the maṃda kendra, and make the maṃda phala. Half of this maṃda phala, whatever results from adding or subtracting to the first correction yukta madhyama planet, is the second correction yukta madhyama planet. Subtract again from the mandocca and consider the remainder as the dūsarā maṃda kendra, and make the dūsarā maṃda phala. Whatever results from adding or subtracting to the madhyama planet is called the maṃda true planet. Subtract the śīghraphala from the śīghrocca to find the fast kendra and śīghraphala, and whatever results from adding or subtracting this śīghraphala to the maṃda true planet is called the true planet.

Scientific Commentary—These rules can be written in the manner of algebra as follows:—

Epicycle—Mean-planet= fast kendra, the śīghraphala of which is called the first śīghraphala.

First correction yukta medium planet

```
= mean planet ± epicycle apogee ( first ) / 2
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Mandoch - first correction yukta medium planet = Sanskrit Manda center whose Manda- fruit is the first Sanskrit Mandaphal.

Second correction yukta mean planet

```
= first-order mean planet ± apogee ( first-order ) / 2 = mean planet ± ( first-order ) epicycle apogee ± ( first-order ) apogee / 2
```

Apogee - Second correction yukta mean planet = Second correction mean center whose mean motion is the second correction mean motion.

SOURCE PAGE 166

```
apogee = mean planet ± second ( Sanskrit ) slow motion.
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Shighrochcha - maṃda true planet = second quick center whose quick result is the second quick result.

```
True planet = apogee ± mean planet ± second mean motion ± second mean motion
```

This is the method of knowing the true planet in the words of Sūrya Siddhānta. However, in practice, much hassle has to be done with this, so simple rules can be made with the help of this.

You should remember the definitions written below:

1. apogee - mean planet = apogee center

2. epicycle apogee - mean planet = epicycle center

The first result obtained from the epicycle center is the first epicycle result. (1)

The sum of the first three mean planets = mean planet ± equinoctial shadow result 2 (2)

First Sanskrit apogee

= apogee - sum of the first three mean planets = apogee - (mean planet) ± equinoctial shadow result / 2) = (apogee - mean planet) ± equinoctial shadow result / 2) = apogee center ± equinoctial shadow result / 2 (3)

From this it appears that to know the mean center of the first Sanskrit, equation (2) is not required, but by inverting half the first result of the mean center and adding it (according to algebra*), the work will be done. The mean result obtained from this will be the first mean result or the first Sanskrit mean result.

*According to algebra, the meaning of adding is that if one number is positive and the other is negative, the result obtained by subtracting the negative number from the positive number is also called the sum of the positive and negative numbers, although in arithmetic, this sum is called the difference between them.

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the mean planet = equinoctial shadow ± mean planet shadow / 2 = mean planet ± epicycle apogee shadow / 2 (4)

second Sanskrit mean center = apogee - second Sanskrit mean planet = apogee (mean planet ± first epicycle apogee shadow / 2 ± first mean shadow / 2) = (mean center ± first epicycle apogee shadow / 2) ± first mean shadow / 2 (5)

which proves that to know the second Sanskrit mean center, add half the sign of the first mean shadow to the first Sanskrit mean center. Therefore, equation (4) is also not needed. The mean shadow obtained from the second Sanskrit mean center is the second mean shadow.

(y)

mean true planet = mean planet ± second mean shadow (6) second epicycle center = epicycle apogee - mean true planet = epicycle apogee - (mean planet ± second mean shadow) = (epicycle apogee - mean planet) ± second mean shadow = epicycle center ± second mean shadow (7)

it is proved that to know the second epicycle center, add the opposite sign of the second mean shadow to the epicycle center. Therefore, equation (6) is also not needed. The epicycle shadow obtained from the second epicycle center is the second epicycle shadow.

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Clear planet = slow clear planet + second quick result = medium planet + second slow result + second quick result (8)

From which it is proved that in the mean planet, the second mean fruit and the second quick fruit should be added according to bīja gaṇita arthāt add what is positive and subtract what is negative. The second mean fruit and the second quick fruit should be known from equations (5) and (7).

In short, the rule is as follows:—

(1) If the sign of the quick fruit is reversed and added to the mean centers (according to algebra), the first Sanskrit mean center will come. Its mean fruit is the first Sanskrit mean fruit or the first mean fruit. (2) If the sign of the second mean fruit is reversed and added to the first Sanskrit mean centers, the second Sanskrit mean center will come. Its mean fruit is the second Sanskrit mean fruit or the second mean fruit. (3) If the sign of the second mean fruit in the quick center is reversed and added, the Sanskrit quick center will come, whose quick fruit is the second quick fruit. (4) If the second mean fruit and the second quick fruit in the mean planet are added without reversing the sign, the true planet will come.

The rule for knowing the true planets in Sūrya Siddhānta is this. Other commentators have acted in some different ways, for which there is no need to compare. Such intricate rules were made only so that the position of the true planet might be known exactly. Therefore, all those new methods by which the position of the true planet could usually be known exactly were brought into use. This is why there is a difference in the opinions of the commentators. Not all real positions can be known by just these rules; anyone can test this nowadays; therefore, it is my view that it is extremely necessary to use those methods by which this can be correct. From this perspective, I will also write in the Scientific Commentary about those new methods from which similarity may arise in astronomy and mathematics. But first, some examples should be given, from which it can be easily known how the true planet can be known by these rules. For this, I will give examples of the three planets: Sun, Mercury, and Jupiter.

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Example 1—1676 AD, what were the true positions of the Sun, Mercury, and Jupiter in Ujjain on the half-night of Vasanta Panchami?

First, we need to know the position of their apogee—

From the beginning of creation to the Aries solstice of 1676 AD, 1,65,58,85,023 solar years have passed (see Chapter on Mean Motions); there are 387 revolutions of the Sun's apogee in one Kalpa; therefore, up to the Aries solstice of 1676 AD

1, 65, 58, 85, 023 × 3874, 3200, 00, 000 = 1, 65, 58, 85, 023 × 4348, 00, 00, 000 = 8410.305568648
=175revolution2degree17fraction17 ′ 31 ″.1703

That is, the position of the Sun's apogee on the day of the Aries solstice of 1976 AD was

2nd 17017′31”.1703. The motion of the apogee is so little (of the Sūrya Siddhānta

According) is considered correct for many years the apogee. In one example, the apogee of Buddha has 362 parts of 17; 6 Aries multiplied by 368432 × 107 revolutions. The part of the apogee of Buddha 432 × 10 7 = 4468.535552627 revolutions = 166b7ra10028'26. “ 54

∴ 1676 in the mean motion of Mercury is 7.10028' 26. " 54. At this time the mean motion of Jupiter = 1,65,58,85,023 X 600 / 432 X 107 revolution = 16558.85023 / 48 revolution = 407.21022' 36. " 21

:. In 1676 AD, the position of the apogee of Jupiter is 5rā 21022’ 36. “21

SOURCE PAGE 170

To know the mean positions of these planets, one should calculate by taking the arc from the beginning of Kali Yuga, as stated in verse 56 of the Chapter on Mean Motions.

The arc from the beginning of Kali Yuga to the half-night of Basant Panchami of 1976 was 18,34,977 (as calculated according to the Chapter on Mean Motions).

When there are 43,20,000 revolutions of the Sun in one Mahayugya Sawan day, i.e., in 157,76,17,828 Sawan days, then the revolutions in 18,34,977 Sawan days = 18,34,977 ×432 ×10^4/1577617828 = 5023 Bhira 8° 12’ 6”

∴ Surya Siddhanta = 68012.6 ″

Similarly the mean position of Jupiter

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The mean center is more than 3 and less than 6 rāśiyoṃ, therefore it is in the second position, and the last part is 66°5‘22” and the movable part (by subtracting the last part from 60°) is 20°54‘38” . Therefore, according to the 30th verse, the sine of the movable, which is 20°54‘38”, will be the hypotenuse of the sine of the sine and the sine of the latitude of 66°5’22” .

20054' 38 " = 20055' fractional part = 20 X 60 + 55art = 1255'

According to the 31st verse, when 1255’ is divided by 225, the last body came to 5 and the 6th body to 130’.

5th-degree sine = 1105'

Sixth " " = 1315' difference of both sines

Now 225: 130’:: 210: desired

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late

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This is considered the mean result. If further and finer calculations are to be done, then the action of 40-42 shlokas should also be done as shown in the equations (3) of the commentary on the science of the 36th shloka. However, doing so requires a lot of mathematics and the difference is very small, so there is no need to perform the action of 40-42 shlokas for the mean result.

This mean position of the Sun should be added because the mean is in the first two positions, as stated in the 5th and the following 45th shlokas. Therefore, the true position of the Sun in Ujjain will be the mean midnight of Vaspant Panchami, which will be 6th 80 12 6" + 47" arthāt 6th 80 56" 6".

To know the true position of the Sun—

Gurukamandocche − Gurukamadhyamasthan = 5th 21022' 36" − 6th 16052' 37' = 11th 1030'

Gurakashigrochcha - Gurukamadhyamasthan = Suryakamadhyamasthan - Gurukamadhyamasthan = 6° 12' 6" - 16° 52' 37" = 18° 19' 32" = 18° 20' s = 78° 20'

The quick center is less than 3 rāśiyoṃ, therefore it is in the odd term; therefore, the sine of the quick altitude is 780 20’ and the sine of the quick meridian is 11^ o 40’.

78 ° 28 ′ = 78 X 60 + 20art = 4700art

4700 / 225 = 20Pind + 200Kala

The sine of the 20th planet = 3321‘; The sine of the 21st planet = 3372‘; The difference between the two sines = 51’ 225: 200:: 51: desired difference

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225: 25:: 216: desired difference
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∴ fast koṭijyā = 671’ + 24’ = 665’. The Guru's fast paridhi is 720 in odd division and 700 in even division, so the difference between both is 20 and according to the 38th verse

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Sūrya Siddhānta =  = 70^° + 20 × 33663438  = 70^° + 20 difference  = 72^°
```

```
Therefore, shighrabhujphal = 72 × 3366360 [ verse36 ] = 673'
```

```
72 × 665360  = 136 '
```

The center is in the first line, so the radius should be added according to the 40th verse of Shikhara, therefore.

```
=  ( 3 × 38 + 136 ) 2 + 6732  [ Verse 41 Latter half ] =  35772 + 6732  =  1, 27, 94, 929 + 4, 52, 929  =  1, 32, 47, 858  =  3640 ’
```

SOURCE PAGE 174

```
Quick result = 3438 X 673 / 3640art [ verse41part, 42part ] = 636art
```

This is the first śīghraphala. It is positive, because the fast center is in the first position. If further actions are to be done according to Sūrya Siddhānta, one should do as written in the translation of the 44th verse. But this is a very long procedure, so I act according to the brief rules written at the end of the Scientific Commentary of this verse:-

According to Rule(1) + 636’ divided by two, reversing the sign gives - 318’ which is - 5018’. When added according to algebra to the mean center of the Guru at 11rā1030’ it resulted in 10rā26°12’. This is the first Sanskrit mean center. Its mean result is the first Sanskrit mean result.

10rā26012’ is more than the new sign, so it is in the fourth position, which has 1rā26012’ which is 56012’ past and 33048’ to go.

```
33° 48’ = 33 X 60 + 48 degrees = 2028 degrees = 6 Pind + 3’
```

sine of the 6th body = 1610'

```
10th " " = 2063'
```

∴ difference of both sines = 183'

225; 3':: 183' : desired difference

∴ desired difference = 3 X 183 / 225 = 2'

∴ apogee sine = 1610' + 2' = 1612'

The difference in the mean perimeters of Jupiter is 10, therefore according to the 38th verse,

Apogee = 330 - 1° * 16123438 = 330 - 33' = 1647'

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Sūrya Siddhānta = 1647 × 16122160 [Verse 36] = 172 '

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The result is also negative. Therefore, by reversing its sign and adding it to the epicycle center according to rule (3), the Sanskrit epicycle motion is obtained. Therefore, the Sanskrit epicycle = 78°20’ + 2046’ = 8106’

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This is in the first position, therefore its sine of the epicycle and sine of 8054' will be the sine of the epicycle

8106' = 81 X 60 + 6arcmin = 4866arcmin = 21pind141' sine of the 21st pind = 3372' sine of the 22nd pind = 3406' the difference between both = 37' 225:141:: 37: desired difference ∴ desired difference = 141 X 37225 = 23' ∴ sine of the epicycle = 3372' + 23' = 3395' 8° 54' = 8 X 60 + 54arcmin = 534arcmin = 2pind84' sine of the 2nd pind = 4463sine of the 2nd pind = 671difference = 222' ∴ 225:84:: 222:desired difference ∴ desired difference = 84 X 222225 = 83' ∴ sine of the equinoctial shadow = 446 + 83' = 532'

The circumference of the Sun is 720 in the prime vertical circle, therefore this time also the true circumference will be 720 like the first.

∴ sine of the epicycle, result = 72 X 3365360 arcmin = 676'

and the true equinoctial shadow = 72 X 532360 = 106'

This will be added in the short radius.

SOURCE PAGE 177

Therefore, shadow hypotenuse = (3438 + 106)^2 + 6762 = 35442 + 6762 = 3608'

therefore Busarashi Gharphal = 3438 × 6763608 [Shlokas 41, 42] = 647 kala = + 10° 47' (2)

∴ According to rule (4),

True Teacher = Mean Teacher + Second Result + Second Quick Result. = 6th 16052' 37'' - 2046' + 100047' = 6th 27053' 37''

From this example, it will be clear that to determine each planet, two times the quick result and two times the apogee result have to be calculated, and for each, the sine of colatitude, the meridian, the arc of the circumference, the sine of the quick result, and the quick result have to be calculated. If a table of sine and arc of latitude for each fraction or half-fraction from zero to six hundred is given, it can be known by looking at the tables of sine and arc of latitude. This table can be useful for all the planets. Similarly, a table of the apogee and quick result of each planet can also be made, which will make the long process of clarification very brief and eliminate the hassle of multiplication and division. This also happens in practice. Nowadays, the Makrand table is more useful.

Similarly, the true position of Mercury can also be known. The position of mean Mercury is the same as that of the Sun. The position of epicycle apogee has to be known. And the things have to be done in the same way as for the Sun. There seems no need to increase the size of the book by giving an example.

> अजादिकेन्द्रसर्वेषांमान्देशैघ्रेचकर्मणि।धनंग्रहाणांलिप्तादितुलादावृणमेवच॥४५॥

(45) When the fast center or slow center is less than 6 degrees, then the fast result or slow result is positive, so it is added to all operations, and when the fast center or slow center is more than 6 degrees, then it is subtracted.

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Scientific Commentary—Aja or Mesha is the name of the first sign, therefore Ajadi Kendra means the center from the first sign to 6 signs, and Tulādi Kendra means the center from the seventh sign to the twelfth sign, as explained for Karkādi and Makrādi in the 40th verse. The reason for addition and subtraction has been explained in the Scientific Commentary of the 5th verse and in many other places (see figure 15).

> अर्कबाहुफलाभ्यस्ता
> ग्रहभुक्तिविभाजिता ।
>
> भचक्रकलिकाभिस्तु लिप्ताः कार्या ग्रहेऽर्कवत् ॥४६॥

(46) Multiply the sun's meridian (mean position) by the daily true motion of the planet and divide the product by 12 quantities of arc, that is, 21600 arcs, and add or subtract this from the planet's true position (if the sun's mean position is positive) or subtract (if the sun's mean position is negative), so the planet will be visible in the true midnight.

Scientific Commentary—at the time when the mean sun comes on the meridian circle, that time is the mean noon and at the time when the true sun comes on the meridian circle, that time is the true noon. Similarly, at the time when the mean sun is in the nether (the part of the meridian circle that is below the horizon) there is the mean midnight and at the time when the true sun is in the nether there is the true midnight. From this it is apparent that the true midnight will be as much earlier or later as the true noon will be in the nether as the true sun will be in the nether as the mean sun. But the difference between the true and mean sun is called the mean motion; therefore, in the time it takes for a segment of the meridian circle similar to the mean motion to cross the meridian, the true midnight will be as much earlier or later.

As much as the planet will move in this time, by adding or subtracting the mean midnight time in the true midnight time, the true midnight time will be the true midnight time. To make a subtle calculation, it should first be known how long the arc of the ecliptic or the violation of the meridian will take for the mean motion, but even not doing so results in very little inaccuracy. Therefore, in short, all that is to be done is that in the time it takes to violate the meridian of the geocentric cycle, the planet moves forward at the same speed as its daily motion, so in the time it takes to violate the ecliptic of the mean motion, what will be the speed of the planet.

SOURCE PAGE 179

Some special discussions will be furthered in this regard when 'time equation' is written on.

> मध्ये शीघ्रफलस्यार्धं मान्दमर्धफलं तथा ।
> मध्यग्रहे पुनर्मान्दं सकलं शैघ्रमेव च ॥४७॥
> ग्रहभुक्तेः फलं कार्यं ग्रहवन्मन्दकर्मणि ।
> कर्क्यादौ तद्धनं तत्र मकरादावृणं स्मृतम् ॥४८॥
> दोर्ज्यान्तरगुणा भुक्तिस्तत्वनेत्रोद्धृता पुनः ।
> स्वमन्दपरिधिक्षुण्णा भगणांशोद्धृता कलाः ॥४६॥

Translation—(47) By subtracting the daily motion of the apogee of the moon from its mean daily motion, and taking the action told in (48 46 shlokas), the apparent daily motion of the moon is obtained by subtracting or adding from the mean daily motion. (48) The mean daily motion of other planets

should be known in the same way as the action of knowing the mean motion of a planet by the mean daily motion of other planets, which is—multiply the mean daily motion by the difference of the versed sines and divide the product by 225; (46) Multiply the latitude by the revolution angle and divide by (if the latitude is in fractions, by 260 and if it is in degrees, by 21600), the latitude will be in degrees. If the mean center is in the second and third positions (in the order of positions), then add, and if it is in the first or fourth positions (in the order of positions), then subtract. By doing this, the apparent daily motion of the sun and the moon and the mean daily motion of other planets are known.

Scientific Commentary—By subtracting the daily motion of the apogee from the mean daily motion of a planet, the daily motion of its mean center is found. This is called the center motion of the planet. However, the motion of the apogee of other planets, except for the moon, is so small that leaving it out will not cause any error, so the mean daily motion of other planets is considered the center motion, only the procedure for knowing the center motion for the moon is prescribed. Just as by giving the correction of the mean fruit in the mean planet, the mean true planet is obtained, similarly by giving the correction of the motion of the mean or the motion of the mean fruit in the mean daily motion, the mean true motion is found. For the sun and the moon, this becomes the apparent daily motion. For other planets, the action told in the next 50—52 shlokas should also be done. The origin of this rule is as follows:—

The distance the true planet travels in a day is the true daily motion of the planet. Therefore, to know the true daily motion, it is sufficient to know only that any

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SOURCE PAGE 180

What were the positions of the true planets at the beginning and end of the day; the difference between these is the true daily motion. However, to know the positions of the true planets at the beginning and end of the day, a lot of multiplication is required, so the above simple operation can also be done, the proof of which is:-

Dainikspashtgati = (Dinke) Antkaspashashtgraha — (Dinke) Aarambhakaspashashtgraha = (Antkamadhyamgraha ± Antkamandphal) — (Aarambhakamadhyamgraha ± Aarambkamandphal) = (Antkamadhyamgraha — Aarambhakamadhyamgraha) ± (Antkamandphal — Aarambkamandphal) = Madhyamadainikgati ± (Antkamandphal — Aarambkamandphal) (1)

But (Dinke) Antkamandphal = Mandparidhi X AntakekendrakibhujaJyaKadhnu [Shloka36] 360

And (Dinke) AarambkamandphalMandparidhi X AarambkekendrakibhujaJyaKadhnu360

Therefore the difference between these two (by gross method) = MandparidhiAntakekendrakibhujaJya – Aarambh360KekendrakibhujaJya (2) But (Dinke) AntakekendrakibhujaJya = (DinkeAarambhakendra + KendrakiDainikgati)KibhujaJya

= sine of the pole of the center of the beginning of the day + difference of the sines of the moving and revolving spheres X daily center motion225 [verse31 - 32]

By performing this in the equation (2) and leaving the equal positive and negative terms

a not the mean motion - the motion of the beginning = the mean position x the motion and movable bodies sine difference X the daily motion 360 225 (3)

SOURCE PAGE 181

This is the form of the latter part of the 48th verse and the earlier part of the 46th verse of this equation (3), in which the difference in the sines of the 'gone and to-go bodies' is mentioned in short as 'dojyaantar'.

By elevating equation (3) in equation (1)

Daily true motion = mean daily motion ± (manda circumference / 360) × (difference between the preceding and following sine increments × mean daily motion / 225). (4)

The reason for investing in the center of Karkādi and debt in Makrādi is that when the mean center is more than 3 and less than 6 signs, then the true daily motion is greater than the mean daily motion, otherwise it is less. (See figures 26 and 30). The time taken by the mean planet to reach from mi to mu or mu to me is the same time taken by the true planet to reach from si to su or su to 'se', that is, in the same time, the true planet moves more than the mean planet; therefore, the daily motion of the true planet will also be greater than the daily motion of the mean planet. Etc.

> मन्दस्फुटीकृतां भुक्तिं प्रोज्झ्य शीघ्रोच्चभुक्तितः ।
> तच्छेषं विवरेणाऽथ हन्यात्त्रिज्यान्त्यकर्णयोः ॥५०॥
> चलकर्णहृतं भुक्तौ कर्णे त्रिज्याऽधिके धनम् ।
> ऋणमूनेऽधिकात्प्रोज्झ्यं भुक्तिं वक्रगतिर्भवेत् ॥५१॥

-(50-51) Subtract the mean true daily motion from the daily motion of the epicycle, multiply the remainder by the difference between the radius and the shadow hypotenuse, divide the product by the shadow hypotenuse, and add the resulting quantity to the mean true motion if the radius is greater than the hypotenuse, and subtract it if it is less. If the result is negative and greater than the mean true motion, the remainder will also be negative. This will be the daily curvilinear motion.

Scientific Commentary—this rule can be written according to algebra as follows:—

True daily motion = apogee ± (epicycle apogee - mean motion) (shadow hypotenuse rightarrow radius) shadow hypotenuse (5)

This sign is the sign of revealing the difference. Among the two numbers between which this sign is, the smaller number should be subtracted from the larger one.

SOURCE PAGE 182

Example—to find the clear daily motion of the Sun and the Earth.

The mean daily motion of the sun is 56^(°)8''. Therefore, according to equation (4) [see udā0 1] the true daily motion of the sun

= 56 ′ 8 ″ ± 83311600 × 210225 × 56 ′ 8 ″ (here in the second degree of the mean center therefore do not take the plus sign) = 56 ′ 8 ″ + 83321600 × 210225 × 56 ′ 8 ″ = 56 ′ 8 ″ + 2 ′ 7 ″.7 = 61 ′ 16 ″ after the

small difference

Guru's mean daily motion = 4' 56'' Guru's mean true motion = 4' 56'' ± 164821600 × 161225 × 4' 56'' Here the fourth position of the mean center is, so the minus sign should be taken. therefore Guru's mean true motion = 4' 56'' - 164821600 × 161225 × 5' = 4' 56'' - 25'' = 4' 34''

The motion of the epicycle of the Guru = the mean motion of the Sun = 56'8" . The epicycle hypotenuse = 3608 Therefore, by raising all these values in equation (5) and taking the positive sign because the epicycle hypotenuse is greater than the radius of the epicycle,

True motion of the Sun

Mean motion = 4' 34" + (56' 8" - 4' 34") (3608 - 3438) 3608 = 4' 34" + 54' 34" * 1703608 = 4' 34" + 2' 34" = 7' 8"

SOURCE PAGE 183

> दूरस्थिताच्च शोघ्रोच्चाद् ग्रहश्शितिलरश्मिभिः ।
> सव्येतराकृष्टतनुर्भवेद्वक्रगतिस्तदा ।।५२।।
> कृतत्तृचन्द्र वैदेन्द्रैः शून्यत्येकैर्गुणाष्टिभिः ।
> शररुद्रैश्चतुर्थांश केन्द्रांशैभू सुतादयः ॥५३॥
> भवन्ति वक्रिणस्तैस्तैः स्वैस्वैश्चक्राद्विशोधितैः ।
> अवशिष्टांशतुल्यैः स्वैः केन्द्र रुज्झन्ति वक्रताम् ।।५४।।
> महत्वाच्छीघ्रपरिधेः सप्तमे भृगुभूसुतौ ।
> अष्टमे जीवशशिजौ नवमे तु शनैश्चरः ॥५५॥

(52) When the planet is far from its epicycle (more than three zodiac signs difference) then the ropes from which it is drawn become slack. Due to this, the planet is pulled in the opposite direction and the motion becomes a curve, that is, it is reversed. (53) When the final epicycles (from which the second epicycle is extracted) of Mangala, Budha, Guru, Shukra, and Shani are in succession 164, 144, 130, 163, and 115 degrees (54) then their circular motion begins, and when the epicycles are in succession which come from subtracting 3600 from the above epicycles (that is, 196, 216, 230, 167, and 245 degrees) then the circular motion ends, that is, the planet becomes rectilinear again. (55) Due to the increase of the larger circumference of the epicycle, the circular motion of Shukra and Mangala stops at the same time when the epicycle is in the seventh sign, Budha and Guru when the epicycle is in the eighth sign, and Shani when the epicycle is in the sixth sign.

Scientific Commentary—the true cause of the circular motion of the planets is explained in detail in the science commentary of 12-13 shlokas. Here it is further explained when the circular motion begins and ends and how it can be known by mathematics. The numbers given above for the fast center are according to the mean value, so some variation is seen in reality. In the 55th shloka it is stated that according to the extension of the fast perimeter, when the fast center is in the seventh, eighth, or ninth sign of the fast perimeter, the circular motion ends. This is clear from seeing picture 30. The true planet while moving on the fast perimeter reaches two places where the fast shadow hypotenuse touches the fast perimeter. In such a situation, the fast shadow hypotenuse, the radius of the fast circle, and the radius of the orbit circle form a right-angled triangle with these three lines. The angle of this triangle formed at the midpoint of the fast circle is fast

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It is smaller than big according to the radius of the circumference. Similarly, the value of the epicycle also decreases and increases for the beginning and end of the arc motion.

Up to here, the method of knowing the true positions of the planets according to Indian tradition has been described. Bhaskaraacharya and many other Acharyas have also described another method by which the true positions of the planets are obtained in the same way, but it is not written out due to fear of expansion. Now, in brief, it will be explained how the astrologers of Western countries find the true positions of the planets according to the new method.

According to Kepler's law, to know the true position of a planet, it first has to be seen where the planet, which is elliptical in shape in its orbit and whose focus is the Sun itself, is. Then it has to be known where that planet will appear from the Earth at that time.

Suppose the orbit of a planet is elliptical and R is the position of the attractive Sun, which is at the focus of the ellipse. The beginning of the revolution of the planet is considered from the time when the planet is at the nearest distance from the Sun, i.e., at A, and if the planet appears at point G at a distance of 'D' from that time, then A-G is called the apogee of the planet, which will be denoted by the letter S hereafter.

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The circle drawn with radius a or a from k taken as the center k touches the ellipse at points a, a. The auxiliary circle of such an ellipse is called. If g is dropped on the long axis of the ellipse from g, then when extended, it will cut the auxiliary circle at point g. If g is joined to the center k, then the angle ak g is called the eccentric anomaly of g. The eccentric anomaly will be denoted by the letter u. These are measured in s and u thermal values. If the planet's daily mean motion is in the thermal value 'b', then the revolution period of the planet will be 2 ↟rac(π)(b) days because one b cycle is 3600 in thermal value and 2π in angular value. If the planet's daily motion were always the same as 'b', then the difference of the planet from d in the day would be d imes b. This d imes b is called the mean minor center, which will be denoted by 'm' hereafter. If the planet's angular velocity is constant, the mean minor center is also the true center.

The relationship between the mean minor center and the eccentric anomaly based on the properties of the ellipse and the true minor center and the eccentric anomaly can be known as follows:-

According to Kepler's second law,

Area a_g = rac(a b)(2 π) rac(X b)(2 π) But rac(Area a_g)(Area a_g) = rac(t)(t) = const. rac(t)(π t). t = rac(Area of ellipse)(Area of auxiliary circle)

The long and short axes of the ellipse are in this order. • Area a_g = Area a_g imes • Area of ellipse / Area of auxiliary circle

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= area perigon^2

but area of the lune = area of the circle - area of the lune × = t^2u - bga × k^2 = t.2u^2 - t.sinu × t.cos^2 = t^2 (u - csinu)

Here c is the eccentricity of the long circle.

It has been proved that

area = (d)/(2) area of ellipse^2 π therefore (d)/(2) π = t^2 / 2 (u - csinu) / gata^2

yabh. d = u - czjau (1)

This equation reveals the relation between the mean center and the eccentricity. To establish the relation between the true center and the eccentricity: The polar equation of the ellipse is,

r = t(1 - c^2) / 1 + c cosec where r is the distance between the planet and the center of the planet. But according to the definition of the ellipse,

r = c X n y m k re kh a (directrix) se g ka antar = c X b se n y m k re kh a ka antar = c X (kendra se n y m k re kh a ka antar - kendras se b ka antar) = c X (t / c - b c) = c X (t / c - t cosec u) = t - c X t cosec u ∴ krn = t (1 - c cosec u) (2)

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∴ t (1 − c^2) 1 + ccot¨ = t (1 − ccot¨u) which means 1 − c^21 + ccot¨ = 1 − ccot¨u or 1 + ccot¨ = 1 − c^21 − ccot¨u ccot¨ = 1 − c^21 − ccot¨u − 1 = c (cot¨u − c)1 − ccot¨u ∴ cot¨ = cot¨u − c1 − ccot¨u ∴ 1 − cot¨ = 1 − cot¨u − c1 − ccot¨u = 1 − ccot¨u + cot¨u + c1 − ccot¨u and 1 + cot¨ = 1 − ccot¨u + cot¨u − c1 − ccot¨u ∴ 1 − cot¨1 + cot¨ = 1 − ccot¨u − cot¨u + c1 − ccot¨u + cot¨u − c = (1 − ccot¨u) (1 + c) (1 + ccot¨u) (1 − c) = 1 + c1 − c × 1 − ccot¨u1 + ccot¨u ∴ * spre2s = 1 + c1 − c × spre2u or spre s2 = 1 + c1 − c × spre u2 this equation shows the relationship between the center and the apogee of the eccentricity.

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From equations (1), (2), and (3), one can know the values of the true center, true chord, and d. However, this is not simpler in practice. It is more useful if the value of the mean center can be known to find the value of the true center and chord. For this, equation (3) has to be expanded using trigonometry, which is done as follows:-

It is clear from Loni's Trigonometry Part 2 or Taddhunter's Trigonometry or M. M. Sudhakar Dwivedi's Chalan Kaln Page 42 that

s / 2 √ − 1 − s / 2 √ − 1

spere s2 = i − is − 1 − s − 1 × 1 − 1, \

Here i is the basis of the Napierian logarithm, whose value is according to algebra 1+1+(1)/(2)+(1)/(3)+(1)/(4)+...i etc. when |4 means 4× 3× 2 × 1, |3 means 3× 2× 1, |11 means 11× 10× 6×...2× 1.

3 2 − 1 − 3 2 − 1

Similarly u2 = i - iu2 - 1 - u2 - 1 imes 1 - 1

Therefore, the equation (3) will be

si2 − 1 − − si2 − 1si2 − 1 + − si2 − 1 = 1 + cx − cx

ui2 − 1 − ui2 − 1ui2 − 1 + ui2 − 1

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or

s - 1 u - 1 i - 1 i - 1 - = 1 + c 1 - c × s - 1 1 - c u - 1 i + 1 i + 1 (k)

If 1 is added to each side of equation (a)

OCR equation (fraction layout unclear): sa − 1 2 i − − − − − − − = sa − 1 i + 1 1 + ca (i − 1 − 1 − 1) + 1 − ca (i − 1 + 1 + 1) 1 − ca (i − 1 + 1) (kha)

And if each side of the equation (k) is subtracted from 1

2 - 1 + 1 \ \ \ \ = 1 - c (i - 1 + 1) - 1 + c (i - 1 - 1) 1 - c (i - 1 + 1) (g)

Dividing the left side of equation (kha) by the left side of equation (ga) and the right side of (kha) by the right side of (ga)

begin(aligned) & i√(-1) = frac(√(1+overline(c)(i√(-1)-1)+√(-1)))(√(1-overline(c))(i√(-1)+1)-√(-1)+(i√(-1)-√(-1))) \\ & quad i√(-1)(√(1-overline(c))+√(1+overline(c)) + √(1-overline(c))-√(1+overline(c)) \\ & = frac(-i√(-1)(√(1+overline(c))-√(1-overline(c)))+√(1-overline(c))+√(1+overline(c)))(-i√(-1)(√(1+overline(c))-√(1-overline(c))) + √(1-overline(c))+√(1+overline(c))) \\ & = frac(-√(1+overline(c))-√(1-overline(c)))(√(1+overline(c))+√(1-overline(c))) \\ & quad 1 - i√(-1) × frac(√(1+overline(c))-√(1-overline(c)))(√(1+overline(c))+√(1-overline(c))) end(aligned)

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if √(1+c)-√(1-c)/√(1+c)+√(1-c) is taken instead of

then the form of this equation will be

this -1 & = iu -1 - p & = 1 - iu - 11 - iu - 1 × p & = iu -11 - p × iu -1 × iu -1 & = iu -1 × 1 - pi - u - 11 - piu -1

taking the logarithm of each side,

log(iu√(1-p)) = log(iu√(1-p)) + log(1 - p iu√(1-p)) - log(1 - p iu√(1-p)) (h)

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(3 - 1u - 3 - 1u' - i) +..... therefore equation (h) will be, so s - 1 = u - 1 + p (u - 1 - u - 1i - i) + p22 (2 - 1u - 2 - 1u' - i) + p33 (3 - 1u - 3 - 1u' - i) +..... thus s = u + p × i - 1 - i - u - 1 - 1 + p22 × i^2 - 1 - i - 2u - 1 - 1 + p33 × i^3 - i - 3u - 1 - 1 +..... but i^3 - 1 - i - u - 12 - 1 = sin u, i^2u - 1 - i - 2u - 12 - 1 = sin 2u, etc. so s = u + p × 2sin u + p22 × 2sin^2 u + p33 × 2sin^3 u +..... or s = u + 2 (sin u + sin^2 u + sin^3 u + p44sin^4 u +..... (c)

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If in the equation (c), instead of u, sin u, sin^2 u, etc., their values are kept in such a way that u is not present but v is, which is possible from equation (1), then such an equation will be found in which only v, v and v will remain and which will be very useful for practice. However, the values of u, sin u, sin^2 u, etc., can only be found in the form of v and v when u, sin u, sin^2 u, etc., are expanded according to Lagrange's Theorem. Therefore, in short, it should first state what Lagrange's Theorem is. This theorem is given in M.M. Sudhakar Dwivedi's Chalan Kalan pages 107, 110 and in William Sun's 'Differential Calculus' pages 151-153. The form of this theorem is:—

If r = h + y. If f(r) is such an equation in which h and y are independent quantities and f(r) is a function that depends on the value of r, then r has another function

f(r) = f(h) + y. f(h). f'(h) + y^2 / 2. ta / tah [f(h)]^2f'(h) + y1 / 3. ta^2 / tah^2 [f(h)]^1f'(h) +..... + yn / n. tan - 1 / ta(h) n - 1 [f(h)] n f'(h) +...

Here f'(h), f(h) have the first instantaneous relation dy/dx, and ta/tah', ta^2 ta^3 tah^2, tah^3 etc., have the first, second, third instantaneous relations of the terms written in parentheses.

Equation (1) is of the form

B. D = u - czjau

w = b. d + czjau = m + czjau, where m = b. d.

in the form in which

r = h + y. f (r)

where r, h and y are equal to u, m and c respectively.

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If phi(u) = u then phi(m) = m and phi'(m) = 1

∴ u = m + c. sin. 1 + c ^ 2 / 2ta / tam [sin].1 + c ^ 3 / 3. ta ^ 2 / tam ^ 2 (sin) ^ 1 + c ^ 4. / 4ta ^ 3 / tam ^ 3 [sin] 4.1 + c ^ 5. / 5. ta ^ 4. / ta ^ 4 [sin]. 5 - c ^ 6. / 6. ta5 / tam5 [sin] 6.1 +... etc.

According to part 2 of Loni's trigonometry, if the expansion of any power (sin m) of sin m is not even, then this will be:-

sin m = 1m - 1m - 1sin m - nsin (n - 2) m2 (- 1) ^ 2 + n (n - 1) ^ 2sin (n - 4) m - n (n - 1) (n - 2) ^ 3 | 2 | 3sin (n - 6) m +..... etc

If not odd,

sin m = 1m - 1m - 1sin m - nsin (n - 2) m2 (- 1) ^ 2 + n (n - 1) ^ 2sin (n - 4) m - n (n - 1) (n - 2) ^ 3 | 2 | 3sin (n - 6) m +..... etc ∴ · taatam ([sin] 2) = ta (tam) (1 - cos^2m2) = sin^2m, ta^2tam^2 (sin^3m) = ta^2 (tam)^2 (6sinm - sin^4m) = 6sin^3m - 3sin^4m = 38 (3sin^3m - sinm ^4)

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ta^3 (sin^4m) = ta^3 tam^123 (-1)^2 * (cos^4m - 4cos^2m + 2 * 3 / 2 * 1 = 3 / 2 (4^2 sin^4m - 4 * 2 sin^2m) = 4 (2sin^4m - sin^2m) ta^4 tam^4 (sin^55) = ta^4 tam^4 124 (-1)^2 * (sin^5m - 5sin^2m + 5 * 4 / 2sin^3m) = 1 / 24 (5^4 sin^5m - 5 * 3^4 sin^3m + 10sin^4m) ta^4 tam^4 (sin^65) = ta^4 tam^4 124 (-1)^3 * (cos^6m - 6cos^4m + 6 * 5 / 2cos^2m - 6 * 5 * 4 / 3 * 1 / 2) = 1 / 25 (6^4 sin^6m - 6 * 4^5 sin^4m + 15 * 25sin^2m) ∴ u = m + chjam + c^2 / 2sin^2m + c^3 / 3 * 3 / 4 (3sin^3m - sin^m + c^4 / 4 * 4 (2sin^4m - sin^2m) + c^5 / 5 * 1 / 24 * (5^4 sin^5m - 5 * 3^4 sin^3m + 10sin^4m) + c^6 / 6 * 1 / 24 (6^5 sin^6m - 6 * 4^5 sin^4m + 15 * 25sin^2m) +..... if chjam, sin^2m etc. are taken separately, then u = m + (c - 1 / 8c^1 + 1 / 162c^5) sin^m + (c^2 / 2 - c^4 / 6 + c^3 / 48) sin^2m + (3c^3 / 8 - 27c^5 / 128) sin^3m ext(

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Sūrya Siddhānta + (ch43 - 4ch615) sin4m + 125ch5384sin5m +.....

The sine of 6m in this equation and the multipliers of the sines following it and the terms in which the number after the sixth power of ch is present are omitted because their values are not equal.

Equation (1) can also be written as:-

ch sin u = 6 - m

Then sin u = u - m / ch

which means that if m is subtracted from the expansion of u and the remainder is divided by ch, then the expansion of sin u will be.

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Tatam (sine2m X 2cos2m) = ta / tam (1 - cos2m / 2 X 2cos2m) = ta / tam (cos2m - cos22m) = ta / tam (cos2m - 1 + cos4m / 2) = 2sin4m - 2sin2m

ta2tam2 (sine3m X 2cos2m) = ta2tam2 (3sinm - sin4m X 2cos2m) = ta2tam2 (3sinmcos2m - sin3mcos22m) = ta2tam23 / 4 (sin3m - sinm) - 1 / 4 (sin5m + sinm) = ta2tam23 / 4 (3sin3m - 4sinm - sin5m) = 1 / 4 (-32sin3m + 4sinm + 52sin5m), ta3tam3 (sine4m X 2cos2m) = ta3tam31 / 8 (cos4m - 4cos2m + 3) 2cos2m = ta3tam31 / 8 (2cos4mcos2m - 4.2cos22m + 6cos2m) = ta3tam31 / 8 (cos6m + cos2m) - 4 / 8 (1 + cos4m) + 6 / 8cos2m = ta3tam31 / 8 (cos6m - 4cos4m + 7cos2m - 4)

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= 16 (6 3 sin6m - 4 4 sin4m + 7 × 2 3 sin2m): therefore sin2u = sin2m + c (sin3m - sinm) + c22 (2sin4m - 2sin2m) + c3 | 3 - × 14 (25sin5m - 27sin3m + 4sinm) + c6 | 4 - × 18 (216sin6m - 256sin4m + 56sin2m) +..... = (- c + c36) sinm + (1 - c 2 + 7c424) sin2m + (c - 6c38) sin3m + (c 2 - 4c43) sin4m + 25c324 sin5m +.....

If phi = sin 3 theta then phi = sin 3 theta and phi' = 3 cos 3 theta, therefore according to Lagrange's theory,

Sūrya Siddhānta = sine^3m + ch jya^3m + ch^2 / 2ta / m jya^2m + ch^3 / 3ta^2 / m^2 jya^3m + dots = sine^3m + 3/2ch (sine^4m - sine^2m) + ch^2 / 2 × -3/4 (5sine^5m - 6sine^3m + sinem) + ch^3 / 6 × 3/8 (36sine^6m - 48sine^4m + 12sine^2m) + dots = 3ch^2 / 8jyam - (3ch / 2 - 3ch^3 / 4) sine^2m + (1 - 6ch^2 / 4) sine^3m

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Similarly, sin4u = sin4m + sinjam X sin4m + sin2ta / 2tamsin2m X sin4m = sin4m + 2ch (sin5m - sin3m) + sin2 / 2 (6sin6m - 8sin4m + 2sin2m) = sin2sin2m - 2chsin3m + (1 - 4ch2) sin4m + 2chsin5m +.....

and sin5u = sin5m + 5 / 2ch (sin6m - sin4m) +..... = - 5ch / 2sin4m + sin5m + 5ch / 2sin6m +...

In this way, the values of u, sin u, sin2 u etc. of the equation (ch) have been obtained, but the values of p, p2, p3 etc. are yet to be known. Here p

is kept for 1 + ch - 1 - ch1 + ch + 1 - ch .

The extension of any power of this can be known according to Lagrange's theorem. However, extensions up to five or six powers which do not require more than seven digits can also be known by the Binomial Theorem, which are shown here:-

p = 1 + c - 1 - c1 + c + 1 - c = 1 - 1 - c2c = 1c (1 - 1 - c2) = 1c (1 - (1 - c2)12)

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Now, if the detailed values of p, p2, p3 etc. and u, sin u, sin2 u etc. are calculated in the equation (c), then its form will be:-

S = m + (c - c^3 / 8 + c^5 / 162) sinm + (c^2 / 2 - c^5 / 6 + c^6 / 48) sin2m + (3c^3 / 8 - 27c^5 / 128) sin3m + (c^4 / 3 - 4c^6 / 15) sin4m + 125c^5 / 384sin5m + + 2 (c / 2 + c^5 / 16) [(1 - c^2 / 8 + c^4 / 162) sinm + (c / 2 - c^2 / 6 + c^5 / 48) sin2m

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+ ((3c^2)/(8) - (27c^4)/(118)) sin3m + ((c^3)/(3) - (4c^5)/(15)) × sin4m + (125c^5)/(384) sin5m +.....] + (1)/(2) ((c^2)/(4) + (c^4)/(8) + (5c^6)/(64))[(-c + (c^3)/(6)) sinm + (1 - c^2 + (7c^4)/(24)) sin2m + (c - (6c^3)/(8)) sin3m + (c^2 - (4c^4)/(3)) × sin4m + (25c^3)/(24) sin5m +.....] + (1)/(2)((c^3)/(8) + (3c^5)/(32))[(3c^2)/(8) sinm - ((3c)/(2) - (3c^3)/(4)) × sin2m + (1 - (6c^2)/(4)) sin3m + ((3c)/(2) - 3ch^3) × sin4m + (15c^2)/(8) sin5m + (6c^3)/(4) sin6m +.....] + (1)/(4)((c^4)/(16) + (c^6)/(16))[c^2 sin2m - 2c sin3m + (1 - 4c^2) sin4m + 2c sin5m] + (1)/(2) × (c^5)/(32) [-(5c)/(2) sin4m + sin5m + (5c)/(2) sin6m]

In this value, the terms following 6 of m and all the terms whose multipliers are 6 or more have been omitted because this cannot cause any particular inaccuracy. When this value is simplified, such terms will also be found whose multipliers are more than 6. Leaving these also and collecting the multipliers of sin m, sin^2 m, etc.

s = m + (2c - 1 / $4c^3$ + 5 / $66c^5$) sinm + (5 / $4c^2$ - 11 / $24c^4$ + 17 / $162c^6$) $\sin^2 m$

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sūrya - siddhānta + (1323 - 4364 + 5) sin3 + (10366 + 4 - 451480 + 6) sin4 + 1097660 + 5 sin5 (6)

This is the principal equation to reveal the relationship of the mean and true positions of the mean and true planet. From this it is known that if the observer is in the mean of the Sun, then at what time the

mean and true positions of any planet are in its own orbit. The mean and true positions of the planet that is placed in the place of its center of revolution can be known from the equation (6). For practical convenience, the multiples of sin, sin2, etc., can be expressed in a single number by taking the actual value of sin, sin2, etc.

c = 0.048254 c^2 = 0.0023284 c^3 = 0.0001124 c^4 = 0.0000054

No need to know the values of c5, c6 because if the digit 5 is in the sixth decimal place and it is left, there can be an error of 1 arc minute. Therefore,

s = m + (.066508 - .0000281) sin + (.0026106 - .0000025) sin2 + .0001218 sin3 + .0000058 sin4 (6)

This equation is sufficient to know the true position of the Sun-centered Earth. If the sines of m, 2m, 3m etc. are used in the Indian manner in art or declination, then what is in addition to m on the right side of the equation (j)

*The center does not remain fixed for several reasons but changes very slowly, so its value is somewhat different at different times. This center is from the end of the year 1656 AD.

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In Kala or Vikala there will be and from the middle of the sun this will be the mean motion of the earth. If the sines are expressed in a way different from the current custom, then in order to simplify, the number that comes in decimal fraction besides m will be in radians, for which it will have to be multiplied by 3437.75 or 206265 to make the art or vikala. The result will be the same from both methods.

As the equation (j) was obtained for the Guru, a simple formula can be obtained by elevating the center of the orbit of each planet into the equation (ch). The center of the orbit and other necessary things of each planet will be given in a table further.

There is an equation to know what the distance of a planet from the Sun is at what time. According to Lagrange's theory, this is obtained by knowing the value of 5 - cos u from the equation (2) which is hypotenuse = t(1 - cos u). According to Lagrange's theory,

1 - ch cos(ch) = (1 - ch cos(m)) + ch sin(m)

(1 - chokajam) + ch2 | 2 tatam sin 2 m X chajam + ch3 | 3 tatam2 sin 3 m X chajam = 1 - chokajam + ch22 - ch22 cojyam + 38 ch 3 cojyam - 38 ch 3 cojyam3m - ch43 X cojyam4m + ch43 cojyam2m +..... = (1 + ch22) - ch (1 - 38 ch 2) cojyam - ch22 (1 - 23 ch 2) cojyam2m - 38 ch 3 cojyam3m +.....

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∴ karṇa = t((1 + c^2/2) - c(1 - 3/8 c^2) sin m - c^2/2(1 - 2/3 c^2) sin 2m - 3/8 c^3 sin 3m)(jha) will be the form of the equation(jha) for the chord of the guru for the karṇa, 5202.8((1+.0011642) -(.048254 -.0000421) sin m -(.0011642 -.000018) sin 2m - .0000421 sin 3m) or 5202.8(1.0011642 -.0482116 sin m -.0011624 sin 2m -.0000421 sin 3m) or 5208.86 - 251.06 sin m - 6.05

sin 2m - sin 3m

5202.8 is the mean chord of the Jupiter from the sun when the mean chord of the Earth is taken as 1000. Similarly, the formula for knowing the chords of other planets can be simplified.

The true center of the planet that comes from the equation is the distance of the planet in its orbit from its perihelion. If the orbit of the planet lies on the same plane as the orbit of the Earth, i.e., the ecliptic, then this is also the distance of the planet in the ecliptic. However, the plane of the orbit of each planet makes some angle with the plane of the ecliptic, which is called the true perigee of the planet and is discussed in the last three or four verses of the first chapter, so by performing some operations on the true center of the orbit, the ecliptic true center is obtained.

Suppose the orbit of the planet is p and the ecliptic is pa, the north node of the planet is pa, the center of the sun is r, and the perihelion of the planet is n. The true

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The line and g of the meridian is that g circle goes to the polar of the meridian. Then the distance of the true center and g of the orbit and the distance of g are the true hypotenuses of the planet according to the (six) and (seven) equations. n also goes to the polar of the meridian. The distance between n and g in the meridian is the true center of g. According to Napier's rules, one can easily know the distances of n and g. Then by knowing the difference between both, the distance of n and g (true center of the meridian) can be known. But in practice, simplicity is at the time when it is known that what is subtracted or added in p n or p g to get the value of p n and p g. The distance of the planet from the meridian obtained by subtracting or adding as much as is obtained from the node is determined. Therefore, it should be known how to calculate the result. According to the definition

```
pan - pana
```

```
Planet motion = arc - arc
```

The instantaneous shadow of the planet, <g> p <g> planet's ultimate shadow, the distance or inclination of the planet from the node is called the planet. <g> p is a right angle, therefore <g> p is a spherical right-angled triangle, and according to Napier's rules,

```
( 1 ) sine ( 60 ° − degree ) = versed sine ( arc ) X versed sine
```

```
(2) sine (gaga) = cosec(60° - gapa) x cosec(60° - pag)
```

```
(3) spere(gaga) = sine(paga) x spere(gapa)
```

(4) spere(paga) = cosec(gpaga) spere(pag) sin(pag - paga) = sin(pag) cosec(paga) — cosec(pag) sin(paga) (t)

From the first four formulas, the values of cosec(paga) and sine(paga) should be established in the ultimate, instantaneous, and mean planet. From formula(3)

```
sine(paga) = spere(gaga) / spere(gpaga)
```

```
From formula(4), sine(paga) / cosec(paga) = cosec(gpaga) spere(pag) ∴ cosec(paga) = sine(paga) / cosec(gpaga) x
```

spere(pag)

SOURCE PAGE 205

= sphere(gaga) sphere(gapga) × 1 × cosec(pag) cosec(gapga) × sin(pag) = sphere(gaga) sin(gapga) × cosec(pag) sin(pag)

By elevating the values of sine(p g) and kojyā(p g) in the equation(t),

sine (p - pa) = sine (p) × spere(gga) sine(gpga) × cojya(pa) sine(pa) - cojya (pa) \ × spere(gga) spere(gpga) \ \ = spere(gga) × cojya(pa) sine(gpga) - cojya(pa) × spere(gga) spere(gpga) \ \ = spere(gga) cojya(pa) sine(gpga) \(1 - cojya(gpga) \) \ \ = sine(gga) cojya(gga) × cojya(pa) sine(gpga) × utkramjya(gpga) \ but sutra (2), sine(gga) sine(gpga) = sine (p)

∴ sine (p - pga) = sine (p) cos(p) / cos(gga) × versed sine (gpga)

Except for Mercury, the ultimate arc of all planets is not more than 30.4; therefore, their instantaneous arc will be even smaller; from which there is no error in assuming that the zodiac (g g) is equal to one. In such a situation,

sine (p - pa) = sine (p) cot(p) = versed sine (pa) = 3 / 2$\sin^2$(p) versed sine (pa) sin(peripl) = 1 / 2perigee versed sine X sine^2(apogee)

(ठ)

By knowing the resultant of the planet and its nadir from this equation, the true center of the ecliptic and the declination can be known.

Now it remained to know in which direction and how far the planet appears from the center of the Earth. It is self-evident that to know the direction and distance of any planet from the Earth, it is necessary to know where the Earth itself is.

SOURCE PAGE 206

This equation is known from (cha) because the motion of other planets has to be done in the surface of its orbit. When the position of the Earth is fixed, the position of the Sun can be easily known; because the direction from which the Sun appears to the Earth is the same as the direction from which the Earth will see the Sun at 1800. Therefore, the position of the Sun is obtained by adding or subtracting 1800 from the true center of the Earth. By subtracting the position of the Sun from the true center of the planet's ecliptic, the epicycle center can be found. In figure 34, ra and gu are the positions of the Sun, Earth, and Jupiter respectively. ra is the point where the Sun appears from the center of the Earth; therefore, the angle rā and gu is the epicycle center of Jupiter. The angle pra and gu is the same as the angle 1800 - rā - gu. Therefore, the two sides of the triangle pra and gu are pra and gu, because they are the true hypotenuses of the distances from the Sun to the Earth and Jupiter, and the angle between them is also known. Therefore, gu, <ra pa gu and <pa gura can be known, because trigonometry of Loni part 1 page 104 or Hall and Knight's trigonometry page 171 is clear that

Sūrya Siddhānta - rgup = rgu - rp / rgu + rpsprerpgu + rgup / 2

SOURCE PAGE 207

Sūrya Siddhānta

Scientific Commentary

Chapter on Mean Motions

Chapter on True Positions

Chapter on the Three Problems

These are the interplanetary and swift arcal spherical surfaces, that is, in that condition where the planet appears in the arc but in reality the planet remains some north or south. Therefore, if the swift arc is divided by the angular measure of the planet's instantaneous shadow, then the actual swift arc will be known. Similarly, by performing the same in the spherical interplanetary, the actual interplanetary is known. All these things are clarified together from figure 35 as per Gadfly's 'Astronomy' page 274—

The surface is a revolution plane, in which the orbit of the Earth, i.e., the ecliptic, is P C. The orbital circle P A G P intersects the revolution plane at points P P. P is north and P is south. R, P, and G are the actual positions of the Sun, Earth, and the planet in order; G is the perpendicular dropped on the revolution plane from G; V is the vernal equinox point; G R G and V R G are the longitudes of the Sun and the portion (Longitude) of the planet, respectively. G P G and V P G are the geocentric longitudes and portions of the planet, respectively. P V R are parallel. V R P is the portion of the Sun's geocentric longitude of the Earth; therefore, V R P+180 is the portion of the Sun's geocentric longitude. The triangle R P G of the P A G P figure is obtained from the triangle R P G by P G. The ecliptic shadow and V R P of the planet P G are the revolution plane. P G is the actual shadow and V R P is the actual inclination.

SOURCE PAGE 208

pg = pga / cojya < pgga

According to modern astronomy, the method by which the true positions of the planets are known is merely an indication. In the places known as true positions by this action, some subtle differences are seen in the places reached by direct measurement. The reason for this is that a planet is not only attracted by the sun but also by other planets and satellites, due to which it deviates from that position

SOURCE PAGE 209

It is seen that it is known by the above method. Therefore, if one wants to know the true position with subtlety, one should also consider the effect of the attraction of other planets which causes the change. But this subject is very serious. For its complete information, knowledge of higher mathematics such as Physical astronomy, Dynamics, Chalan Kalan, Chalrashikalan etc. is also necessary. Therefore, the consideration of it is not done here for fear of expansion.

If the true position of the Moon is found according to the method mentioned above, it is seen that the position known by the arc sometimes moves three-three degrees forward and backward from it. The reason for this is that as the Moon revolves around the Earth, it also completes its revolution around the Sun in a year; therefore, the effect of the Sun's attraction along with the Earth's attraction also has a very large effect on the Moon, causing the deviation of the Moon to take a very large form. Therefore, some main considerations regarding the Moon need to be made, which are discussed briefly.

The simple form of the mean motion that should be considered according to Kepler's law is necessary to state.

```
Therefore = 0.0548442c^2 = 0.003076c^3 = 0.00016466c^4 = 0.00000605
```

There is no need to know the values of c or the numbers following it because they are very small. Elevating the values of the powers of c in equation (6) will give the form of the mean motion of the Moon as:-

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Sūrya Siddhānta
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If the sines of m, 2m, 3m are written in decimal fractions according to the modern method, the multiples of sine m and sine 2m in radians are the angles or

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SOURCE PAGE 210

It will be even simpler to multiply by 3437.75 or 206265 to write in the variations. This is because one radian is approximately 3437.75 arcminutes or 206265 arcminutes. Doing this will give its form as:-

```
s = m + 376' 56.4sin + 12' 54.7sin2m + 36'' .6sin3m + 2'.0sin4m
```

Here it should be remembered that the mean center is understood from the pedigree nowadays as low. If the mean center is understood from the old tradition as high then

```
s = m - 376' 56.4sin + 12' 54.7sin2m - 36'.9sin3m + 2'.0sin4m
```

Similarly, the multipliers of sin m, sin2m, etc., of the principal equations of other planets can be written in terms of arcminutes or variations.

In the right side of this equation is the portion of the moon from high to mean center, which is the mean result, and by performing the correction in the mean center, the true moon is obtained. It is clear that the first term in this mean result, which is 376^(56.4) sin m, is very large, and there is the second term 12^(54.7) sin 2 m. However, when the value of m is 60^(°), the value of sin m is 1 and the value of sin 2 m is zero, so the value of the true mean result depends on the first term and is often equal to 377^() or 6017^() . However, our astrologers have taken the value of the moon's true mean result to be approximately 50, so this question arises: why is there such a difference?

When the true mean result is understood as 376^(56.4) sin m, when the true position of the moon is calculated by removing this, it is seen that there is sometimes a difference of up to 10^(20^()) between the actual position and the mathematically determined positions. Upon continuous removal for many years, it becomes apparent that when the moon is about 60^(°) away from the apogee on the day of

Amavasya or Purnima, the moon determined by the mean result is 1020^() ahead of the true moon, and when the moon is 600 ahead of the apogee or nadir, the moon determined by the mean result is 1020^() behind the true moon. In the first case, the mean result is 376^(56.4) or 6016^(56.4), which means that subtracting 6016^(56.4) from the mean result of the mean planet gives the true planet. However, the determined planet is 1020^() ahead of this.

SOURCE PAGE 211

Therefore, the mean motion should be added to the true planet in the Sūrya Siddhānta by 1020'. Therefore, if instead of subtracting 6016'56".4 and adding 1020', the difference between these two, which is 4056'56".4, is subtracted, the result will be the same. Therefore, if we assume the mean motion is 4056'56".4 instead of 6016'56".4, no loss is understood. In the second case, 6016'56".4 has to be added and 1020' has to be subtracted, and instead of that, if the difference between these two, which is the same 4056'56".4, is added, no change will occur. When the moon is high even on the day of the full moon, the mean motion is zero because the mean center is zero. At that time, there is no difference in the positions of the mean and true moon. This proves that by eclipsing the true mean motion on the day of the full moon or new moon, one can know the value of the true mean motion to about 50, which is given in our ancient texts. Therefore, it can be said that the value of the true mean motion is indeed 50, not 6016. However, it has also been found by eclipsing that when the moon is at 60° from the apogee on the eighth day of each side, even giving the mean motion of 50, the moon is still 30 behind, that is, the eclipsed moon is 80 behind the mean moon. And if on the eighth day the moon is at 600 from the nadir, the eclipsed moon is not 50 but about 80 ahead of the mean moon. Therefore, it must be assumed that taking the true mean motion as 50 does not cause any loss on the day of the full moon or new moon, but a difference of about 30 is seen on the eighth day. Our ancient astrologers did not know this because, in my understanding, they calculated the difference between the mean and true moon from the time of eclipse and found the mean motion as such, as revealed by the quotation of Keshavacharya, which is given in the commentary of the 14th verse of this chapter. From this quotation, it is also revealed that Keshavacharya saw the moon by eclipse as 5' less according to the Sūrya Siddhānta, as previously shown that the mean motion and the chuti correction together are 4057', which is the case on the full moon or new moon. Therefore, Keshavacharya's eclipse is very subtle.

Therefore, it is necessary that the reason for this difference be seen elsewhere. It is clear that this difference has a special relation to the moon's height, and it has also been seen that it does not always remain the same, but decreases and decreases until it sometimes becomes zero and negative and sometimes positive, therefore this is fixed-

SOURCE PAGE 212

It also has a time (periodical). This was determined by the Greek astronomer Talmi in the second century of Vikram, but no one was able to find the cause of this. Newton proved by the law of attraction that the attraction on the Moon is not only from the Earth but also from other planets, and the cause of the above great difference is the attraction of the Sun. Physical astrophysics has proven by mathematics that this difference is due to the attraction of the Sun, and the main form of this phenomenon is+1026.5 sin[2(Moon- Sun)] Moon mean center. The subsequent terms which are very subtle have been left out.

Talmi named this phenomenon evection, which is now prevalent. Swargiya Bankatesh Bapu Ji Ketkar called it chuti sanskara in his astronomical calculations. In this term, Moon - Sun means the difference between the Sun and the Moon which is manifested by the name of Tithi in our place. At the time of Amavasya or Purnima, the value of Moon - Sun is zero or 1800, so the form of this term is 1026.5 sin(-

Moon mean center) or -1026.5 sin(Moon mean center), which is in the form of the mean-result sanskara, and when the mean-result is added, it is subtracted, and when the mean-result is subtracted, it is added, the result of which is that if the mean-result is considered equal to the difference between these two, there is no loss.

Since the mean-result depends on the value of chuti, the decrease of the mean-result indicates that the chuti of the lunar orbit has decreased, and the increase indicates that the chuti is increasing. That is, from this decrease and increase, it is firmly estimated that the size of the lunar orbit does not always remain the same. This also completely matches the law of attraction, as will be shown further.

But when the Moon—Sun is 600 or 2700, that is the eighth, then its form is 10 20' 26.5" sine[2 X 60°—Moon mean center] or 10 20' 26".5 sine m, which is in the form of the mean result, but if the mean result is positive, then this is also positive, and if the mean result is negative, then this is also negative. Therefore, assuming the mean result as 50 sometimes results in a difference of 30 ahead or behind. For this reason, the times written in the Panchangs made according to Indian custom in the seventh, eighth, and ninth are modern

SOURCE PAGE 213

The times passed according to the tradition sometimes remain 14, 15 hours ahead or behind. This may also be revealed in the calendars made from the Panchangs of Bapudev Shastri and the Makaranda sāriṇī of Kashi.

The way Newton has explained the reason for this in brief is as follows: The angle between the orbits of the Moon and the Earth is only about 50, so assuming both on the same plane will not cause any special loss, but it will be simpler.

Suppose the orbits of the Sun, Earth, and C_1, C_2, C_3, C_4 are the orbits of the Moon. It must not be forgotten here that the distance between the Sun and Earth, i.e., P, is 400 times the distance between the Earth and the Moon, i.e., P_1. It is also necessary to understand that the deviation of the Moon occurs because the Sun is attracting the Earth and the Moon unequally, and this deviation is due to the difference in their attractions. If there were no such difference, i.e., if the Sun's attraction on the Moon and Earth was the same, then there would be no deviation because then both would move back and forth together and there would be no variation in the relative motion of the Moon.

Figure 36 makes it clear that as long as the Moon moves from C4 to C1 to C2, it remains nearer to the Sun than to the Earth. That is, the Moon remains nearer to the Sun than to the Earth from the eighth day of the new moon phase to the eighth day of the full moon phase. Therefore, according to the law of attraction, in the first state, the attraction of the Sun on the Moon is greater, meaning the Moon is kept some distance away from the Earth due to the greater pull towards the Sun. This shows that the perturbing force is pulling the Moon away from the Earth. Since R is very far, this force acts in the direction of R, which is approximately parallel to the direction of R. In the second state, the Earth is more pulled, so the Moon lags behind, which shows that the perturbing force is pushing the Moon away from the Earth in the direction opposite to the Sun. It has already been said that the Sun is very far, so the perturbing force is being pulled away from the Moon in the direction parallel to R, away from R. Therefore, it is proven that this force always keeps the Moon and the Earth apart, acting in a direction approximately parallel to R, and its effect is that

Heroes of Science: Based on Astronomers

SOURCE PAGE 214

Time is zero when the moon is near C_2 or C_4 because at that time the moon and the Earth are at the same distance from the Sun.

Suppose one wants to know how the declination force is working when the moon is at A between C_4 and C_1 and lies on the line of Earth and Sun. The declination force can be expressed from the A line and according to the 'parallelogram rule of motion', this force can be divided and expressed in the A and B and C lines when the A line is the tangent to A arthāt in the direction of the moon's motion and the C line is in the direction of the radius vector and acts against the Earth outwards. The part (resolved part) of the declination force acting in the C direction is the moon

SOURCE PAGE 215

SOURCE PAGE 216

Where it keeps moving away from the Earth, the result is that the lunar orbit becomes longer, as is apparent from figure 37.

From the full line the true lunar orbit is revealed and from the cut line the new form of the lunar orbit is revealed due to the force of deviation. This makes it clear that the deviation of the lunar orbit has increased because the deviation of the long period is more, which makes its shape longer and when it is less, it becomes somewhat spherical. This is also seen in direct observation, as indicated before.

The equation 'ch' also proves that the main term of the apogee is dependent on the value of the deviation. If the deviation is more, then the apogee is also more, and if the deviation is less, then the apogee is also less. Therefore, it is proven that due to this force of deviation, if the deviation of the lunar orbit is in the direction of the nadir of the Moon, it will become more, which will also increase the apogee. The apogee is called the deviation (evection) as much as the reality increases.

Conversely, if it is 60° ahead or behind the nadir of the Sun (see figure 38), the shape of the lunar orbit will become somewhat round and the deviation will be less, which will make the apogee less than the reality by the same amount that it increased in the first situation. In such a condition, the deviation will be negative.

This proves that the part of the force of deviation which is acting outwards of the lunar orbit in the direction of the nadir of the Moon causes such a deviation in the Moon that the need for the deviation is created.

Now attention should be paid to that part which is working in the direction of the tangent of the lunar orbit. This results that as long as the moon (see figure 36) remains between C4 and C1 or between C2 and C3, the declination power also works in the state of the moon's ordinary motion and intensifies its ordinary motion (which occurs due to the attraction of the Earth). But when the moon is between C1 and C2 or between C3 and C4, then the declination power works against the moon's ordinary motion and slows it down somewhat. This fact is very evident in the quarter and ekadashi of every lunar month, so it causes some change in the times of these dates. Due to this inequality, one more correction also has to be done in the moon, which is the fortnightly-

SOURCE PAGE 217

They are called correction (variation). In Jyotiganyat, it is called Tithi-Sanskara. It also has many terms in which the form of the main term is this.

35’ 41“•6 sin 2(Moon-Sun)

When it has become certain that due to the revolution of the Earth the distance of the Moon from the Sun sometimes becomes less and sometimes more, due to which there is deviation in the Moon

SOURCE PAGE 218

It becomes not difficult to understand that the distance of the sun from the earth, which changes throughout the year, causes some difference in the position of the moon, and this is not fully explained by the above two saṃskāroṃ. Therefore, the need for another correction also arises, which is called the annual correction; its main point is this.

11’ 41”.67 sine(Sun-mean-center)

Thus, the four main saṃskāroṃ of the moon have been briefly discussed, and it has also been shown in three ways what their causes are. In addition to these, there are many minor saṃskāroṃ that cannot be understood without good knowledge of higher mathematics, and which were invented in the last hundred years when there was a subtlety in mathematics and scientific instruments.

It is stated that the inventor of the Chuti-sanskar is Tali, who lived in Greece in the second century of Vikram. However, its reason was not known before Newton. Tycho Brahe invented the Pākṣika-sanskar and the Annual-sanskar with his extraordinary representational power (born 14th Oct 1546, died 24th Oct 1601 AD). The reason was not known to him either because he did not have good knowledge of higher mathematics and the theory of attraction at that time. Some indication of the Tithi correction was also made by a Muslim astrologer named Abdul Wafa.

Ancient Indian astrologers did not pay attention to any other correction besides the Mandaphal correction. Munjal1 has discussed one correction and one Pākṣika-sanskar, similar to the Chuti correction

SOURCE PAGE 219

and Nityananda ji1 (in 1666 of śaka 1561) discussed the fortnightly and lunar nodes rituals; but they were not propagated. From Siddhanta-darpan it appears that M. M. Chandra Shekhar Samant also discussed the rituals.

The form of the principal equation of the Moon with these four rituals will be:-

```
s = m + 376' 56" 4jyam + 12' 54".7' jya2m apogee + 36".6jya3m + 2.0jyam + 1020' 26".5jya [ 2Chandra - Ravi ) - m ] kutyasanskar + 35' 41".6jya2 ( Chandra - Ravi ) fortnightlysanskar + 11' 11.67jya ( Suryamandakendra ) annualsanskar
```

Here 's' is the true center of the Moon and 'm' is the mean center of the Moon, when the mean center is calculated from perigee. In astronomy, other terms for the chuti and fortnightly rituals are also given which are not given here. Due to the slow change of chuti, a difference of a few degrees in the numbers occurs, which needs to be taken into account.

> आधुनिक ज्योतिष का इतना परिचय देना मेरी समझ में पर्याप्त है।
> उदाहरण देने से विस्तार बहुत हो जायगा; इसलिए उदाहरण नहीं दिये जाते ।

> कुजाकिगुरुपातानां ग्रहवच्छीघ्रजं फलम् ।
> वामं तृतीयकं मान्दं बुधमार्गवयोः फलम् ॥५६॥
> स्वपातोनाद्भहाज्जीवा शीघ्रात्तु बुधशुक्रयोः ।
> विक्षेपघ्नाऽन्त्यकर्णाप्ता विक्षेपस्त्रिज्यया विधोः ॥५७॥

Translation—(56) For the positions of the nodes of Mars, Saturn and Jupiter, perform the ritual of the second quick-result of each in the same way as done with the planets, that is, if it is positive then add it and if it is negative then subtract it. Thus

1. Here, the fortnightly ritual of mandaphalatritah and the lunar distance of the mean of the Sun and Moon and the rest will be established...(Ganaka Tarangini, page 101)

SOURCE PAGE 220

By doing this, the true positions of these three planets' nodes will be known. However, in the positions of the nodes of Mercury and Venus, subtract the mean motion of each from the other's mean motion, which is used in the third calculation to clarify the planet. Do this to find the true nodes of Mercury and Venus. (57) Subtract the true position of each of Mars, Saturn, and Jupiter from its own node, and find the sine of the remainder. Multiply this sine by the mean motion of the planet and divide by the final epicycle to find the true deviation or arc. However, in the positions of the epicycle of Mercury and Venus, one should subtract their true nodes and find the sine of the remainder, and multiply this sine by the mean motions of Mercury and Venus and divide by the final epicycle. To know the true arc (deviation) of the Moon, it is sufficient to subtract the position of the node (Rahu) from the true position of the Moon, multiply the remainder by the mean motion of the Moon, and divide by the radius.

Scientific Commentary—For example, the method for knowing the true arc of Jupiter is written. The true position of Jupiter in Ujjain on the half-night of Bais Panchami of 1676 AD, calculated, is 6 degrees 17 minutes 58.2 seconds.

Therefore, it will be easier to find the true arc of Jupiter at this time. In one kalpa, the node of Jupiter moves 174 revolutions, so on the day of Bais Panchami of 1676 AD, when 1,65,58,85,023 solar years have passed since the beginning of creation. The position of the node of Jupiter

= 1, 65, 58, 85, 023 × 1744, 32, 00, 00, 000 = 1, 65, 58, 85, 023 × 267', 00, 00, 000 = 78b6ra10°16'58.2 “

or 610 1658-2” because there is no need to write the entire revolution. However, the motion of the nodes is in the opposite direction or westward direction.

†This is from the time of the Aries solstice of 1676, but since the motion of the node is very slow, there is no loss in considering it the day of Vaspant Panchami also.

SOURCE PAGE 221

Therefore, the position that has come for the node is negative. Subtracting this from 12 signs will give the position of the node of Jupiter according to the ordinary method. Therefore, the position of the node of Jupiter = 2nd 16° 40’ 20”†

According to the above 56th verse, the second quick result should be processed like the planet. On the day of Vaspant Panchami, the second quick result of Jupiter + 10° 47’ and the final quick hypotenuse 3608 is.

Therefore, the second quick result of the inscribed node = 2^(16)040 ′ 2 ″ + 1047 ′ = 3^(0027 ′ 2 ″

But the true position of Jupiter = 6^(27) ° 53 ′ 37 ″ ∴ According to the 57th verse, the difference between the node and Jupiter = 6^(27) ° 53 ′ 37 ″ − 3^(0) ° 27 ′ 2 ″ = 3^(27) ° 26 ′ 35 ″

This is the center of the deviation of Jupiter on the day of Vaspant Panchami.

Multiplying the sine of this deviation center by 10 or 60’ according to the 66-70 verses of the Chapter on Mean Motions, and dividing by the final quick hypotenuse, the true deviation or arc of Jupiter will come out.

The sine of 3rd 270 26' 35" is equal to the sine of the part of the second foot's gnomon.

∴ sin 3rd270 26' 35" = sin 2nd2033' 25" = sin62033' 25" = 3050 arc ∴ true position of Jupiter = 3050 × 603608 arc = 50' 43"

The deflection center is less than 1800, so Jupiter is north of the ecliptic and the true position of Jupiter is 50' 43".

SOURCE PAGE 222

In the same way, Mars and Saturn can also be seen. Some variation has to be made for Mercury and Venus, that is, to know their point of deviation, the second mean fruit which is useful in the third action in their nodes has to be inverted and subtracted from the locations of the epicycles. What needs to be done after this is the same as the above method.

To know the true shadow of the Moon, there is no need to do all this hassle; because in this, no correction of the quick action has to be done. Therefore, the same rule applies for this as is stated for the Sun in the 28th verse, that is, multiply the sine of the Moon's point of deviation (the difference between the true Moon and Rahu) by the radius which is 3438 from the Moon's ultimate deviation, which is 4030', and divide by the value of the sine, then the true shadow of the Moon will be found. If the point of deviation is less than 1800, the northern shadow will be, otherwise the southern shadow (see verse 7 and its Scientific Commentary and figure on page 22 (4)). If we assume that in figure 25 on page 122, 'va' is the position of Rahu, 'va' is the ecliptic, and 'va' is the Moon's orbit, then 'sa' will be the northern shadow of the Moon.

> विक्षेपापक्रमैकत्वे क्रान्तिर्विक्षेप संयुता ।
> दिग्भेदे वियुता स्पष्टाभास्करस्य यथागता ॥५८॥

Translation—(58) To know the true declination of a planet, add the true shadow (deviation) of that planet to its (mean) declination if the shadow and declination are of the same type, that is, if both are

northern or both are southern. But if there is a difference in their directions, the difference between the two will be the true declination. The rule stated earlier (in the 28th verse) is sufficient to know the true declination of the Sun (because the Sun moves only on the ecliptic).

The difference between the circle of the true planet and the circle of the ecliptic that lies on the circle of the latitude is called the true displacement of that planet (see page 22, chapters 7, 8) and the difference between the circle of the tropics and the circle of the equator that lies on the circle of the pole is called the true motion of that planet.

In figure 36, there are the circle of the ecliptic, the circle of the equator, the latitude (ecliptic pole), and the pole. The position of a planet in its orbit is not shown in the figure for simplicity. The planet at this time

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SOURCE PAGE 223

The uttara is shown of the ecliptic. If the planet is at point G, then it will be to the south of the ecliptic. C P Kadamba makes a right angle on the ecliptic and D G makes a right angle on the meridian circle. P is also a part of D G and makes a right angle on the meridian circle. G P, the deflection of G, P is the middle ecliptic and C, the deflection of G, is the true ecliptic. Similarly, the deflection of G, the south deflection of the planet G, P is the middle ecliptic of the planet G, and C, the deflection of G, is the true ecliptic. In the first situation, the middle ecliptic and the deflection are both north, so by connecting these two, the true northern ecliptic will come according to the rule. But in the second situation, the deflection is south and the middle ecliptic is north, so the true northern ecliptic will be found from the difference between these two.

Here is a point for consideration. It is a section of the G kadamba-protavritta and the P chadha-protavritta; therefore, the sum of these two will not be the same as G ch. But the Acharya has written this. From this, it should be understood that in the Acharya's view, this difference is understood to be so little that the resulting crudeness is understood as not being. Bhaskaraacharya ji therefore

---

SOURCE PAGE 224

It has been ordered to perform the motion by not using this method as unapplied1, which is written in a simpler manner based on Napier's rules without writing the expansion in fear here. For convenience, the simpler form of figure 36 is taken from figure 40.

In this figure, the true position of planet G, the true deflection of planet P, the true declination of planet C, the sine portion of planet V, the latitude of planet C, and B G

The arc of the ultimate circle is the bow. Therefore, it is clear that triangle G P V and triangle G C V are right-angled spherical triangles and the angle P V C is the angle between the circle of declination and the circle of latitude, which is called the ultimate in the 28th verse.

If the sine portion of the planet V and the true deflection of planet P and the true deflection of planet G are known, then according to Napier's rule (2) in the right-angled triangle G V P,

Cotangent(V G) = Cotangent(G P) X Cotangent(V P)(1) according to rule (1)

sine(v p) = tangent(g p) x cotangent(g v p) or cotangent(g v p) = sine(v p) x cotangent(g p)(2)

1 deflection: it becomes towards Kadamba. How the difference in direction of the pole is understood with the revolution in conjunction with the pole. The difference in direction then depends on the motion..... Siddhanta Shiromani Ganitadhyaya, page 215

SOURCE PAGE 225

From these two equations, the angles g v and g p can be known. Then in the right-angled triangle triangle g v c, according to rule (2)

jya (g c) = jya (w g) × jya (< g w p + < p w c) and n m (1) k e a n u s a r s p r s h e k h a (w c) = s p r s h e k h a (w g) × k o j y a (< g w p + < p w c) (3) (4)

If the deflection and sine portion of any planet or star are known from equations (1), (2), and (3), then its declination can be known, and with the help of equation (4), its latitude can be known.

Similarly, if the obliquity and declination are known, then the sine portion and the deviation can also be known.

> प्रहोदयप्राणहता स्वखाष्टैकोद्धृता गतिः ।
>
> चक्रासवो लब्धयुताः स्वाहोरात्रासवस्स्मृताः ॥५६॥

(56) The quantity that arises when the daily motion of the planet is multiplied by the number of breaths in which it rises in the zodiac sign, and then divided by eighteen hundred, and the result is added to the rest of the cycle, is the magnitude of the planet's ahoratra.

Scientific Commentary—in the 11-13 verses of Chapter on Mean Motions, nakṣatra-ahorātra, ghaḍī, pala, prāṇa, sāvana day etc. have been discussed in detail. It is stated there that one nakṣatra-ahorātra is equal to 21600 of asu or pranas, and the sāvana day is usually 4 minutes or 10 pals or 56 pranas more than the nakṣatra-ahorātra because the sun moves approximately 1 degree towards the east per day while the constellations or stars remain in the same place. Just as the time between one sunrise and the next sunrise is called sāvana day, similarly, the time from the time a planet rises in the east to the time it rises again on the next day is called that planet's ahorātra. If the planet is in the direct motion, its ahorātra will be more than the nakṣatra-ahorātra, and if it is retrograde, it will be less. The method of knowing how much more or less the planet's ahorātra is than the nakṣatra-ahorātra is explained in this verse. According to how much the planet moves forward or backward during the day, the planet's ahorātra will be more or less than the nakṣatra-ahorātra. In the third chapter named Chapter on the Three Problems, in verses 41-43, this is elaborated.

SOURCE PAGE 226

It will be shown which sign rises in which place in how much time. The time in which a sign comes above the horizon, that is, rises (according to the Nakshatra period), is called the rising-life or udayasu of that sign in that place. Therefore, it can be easily known from the three problems that when the rising of a sign happens in the same time as the rising-life, the rising of the planet in that sign in a day will be what fraction of the rising-life. Just as the Ahoratra of Nakshatra is not much, by adding the same number of lives more than has passed, the planet will come into the horizon again on the second day. A sign is equal to 30 degrees or 30 X 60 or 1800 degrees. The daily motion of the planet is also generally expressed in degrees, so this ratio is -

1800 degrees: Planet's daily motion:: Sign's rising-life: Ist Antar

∴ IstAntar = RashikaUdayPran X PlanetkiGati / 1800Kala

Only by adding (if the planet is direct) and subtracting (if the planet is retrograde) the Ahoratra of the planet is known in 21600 lives.

There is a slight inaccuracy in this rule. If the planet moves on the ecliptic where the signs are, then this rule would be perfectly correct, but it moves in its orbit, due to which it is either north or south of the ecliptic. If it is south, it will rise a little earlier, and if it is north, it will rise a little later. If the difference in the daily rising time of the planet is known according to the equation (4) of the Scientific Commentary of the 58th verse and the difference in latitudes is converted into lives, then adding it to 21600 lives will give the exact rising-lives of the planet.

In the following many verses, the method of knowing is explained that for how much time the planet will remain above the horizon and for how much time below the horizon, that is, how much time the planet's day and night are. For this, it is necessary first to know what is the *char prana* of the planet, which is known according to the following verses:-

> क्रान्तेः क्रमोत्क्रमज्ये द्वे कृत्वा तत्रोत्क्रमज्यया ।
>
> होना त्रिज्या दिनव्यासदलं तद्दक्षिणोत्तरम् ॥६०॥

SOURCE PAGE 227

> क्रान्तिज्या विषुवद्भाघ्ना क्षितिज्या द्वादशोद्धृता ।
> त्रिज्यागुणा होरात्रार्धकर्णाप्ता चरजासवः ॥६१॥

(60) Knowing the sine of the true revolution and the versed sine of the planet, subtract the versed sine from the radius, which is 3438 art, then the semi-diameter of the circumcircle is obtained. This is also called dyuja. If the revolution is south, then the semi-diameter of the circumcircle is south, and if the revolution is north, then it is north. (61) When the sine of the revolution is multiplied by the equinoctial shadow and divided by 12, the subtended arc is obtained, and by multiplying this by the radius and dividing by the semi-diameter of the circumcircle, the resulting quantity is called charjya. The arc of the bow of charjya is called char pran.

Scientific Commentary—the essence of the third chapter named Chapter on the Three Problems is contained in these two shlokas, therefore the detailed discussion of the technical terms that appear in them will be found in that chapter. However, to understand the meaning of these two shlokas, it is necessary that the technical terms and some other things be discussed in brief.

Equinoctial shadow—the day when the sun is on the ecliptic, that is, the day when the sun comes on the straight land on the latitudes of Aries or Libra.

The shadow of the gnomon of the finger is as large as the meridian time, that is called the equinoctial shadow. In figure 41, on a flat land, the gnomon kha is erected straight on point kha and kha has a measure of 12 fingers, so if the shadow of kha on ga is kha then the measure of kha ga will be called the equinoctial shadow, obliquity, meridian, etc. The value of this equinoctial shadow is not the same

everywhere but increases and decreases according to the latitude. Where the latitude is zero on the equator, the value of kha ga is zero on the day of the vernal equinox. From the equator as in figure 41

SOURCE PAGE 228

As you go north or south, the value of the shadow will increase accordingly. In the northern hemisphere, the shadow will be in the north direction from the gnomon, and in the southern hemisphere, it will be in the south direction; therefore, the latitude of a place can be easily known from the shadow. In our country, for this reason, instead of expressing the latitude in degrees, the practice is to express it in the measurement of the shadow in fingers. The angle of the angle kha is called the latitude of place kha, therefore,

A q sha n shi spar shekha = kh ga / kh ka = palabha / shanku = palabha / 12 (7)

It is clear from this how the value of latitude can be known from the knowledge of the equinoctial shadow.

In our texts, besides sine, versed sine, and versed sine, no other trigonometric ratios are discussed; but the work and method of other ratios are taken from elsewhere; for example, the work of the tangent of the latitude is taken from the equinoctial shadow. To understand the declination sine, versed sine, and sine of declination, see the picture written below-

Assuming this, a slight inaccuracy remains due to the inclination, which will be discussed in the third chapter. However, this inaccuracy cannot cause any harm.

SOURCE PAGE 229

Sh is the place where one observes how long a planet remains above the horizon. The line of the horizon of that place and the line of the horizon of the meridian of the earth is the meridian line. The celestial north pole and the celestial south pole are the North and South celestial poles. The meridian circle and the cross of the meridian of the earth and the cross of the meridian of the earth are the meridian circles. Due to the daily motion of the earth, the planets, constellations, sun, etc., which appear to revolve on a circle and complete one orbit in a day, that circle is called the diurnal circle of that planet, constellation, or sun. These diurnal circles are parallel to the ecliptic. The diameters of three diurnal circles are shown in Figure 42 with the lines of Va, Vi, and Vu. The diameter of the Vi diurnal circle coincides with the diameter of the ecliptic. In this way, the same stars or planets appear to move as if they were on the ecliptic. On the day of the vernal equinox, the sun also (if its motion of declination is assumed to be constant for a short time) appears to move on this diurnal circle. If a planet is like the north declination and the Sagittarius, then the diameter of that planet's diurnal circle will be Va. Similarly, if the planet's south declination is like the Aquarius, then the diameter of its diurnal circle will be Vu.

From the figure it is revealed that the lines of the horizon in the meridian country are cut into two equal parts by the diameters of all the great circles. In the meridian country, as long as the sun, star, or planet remains above the line of longitude, it is seen or rises, and as long as it remains below this line, it is not seen or sets. Therefore, in the meridian country, where this line forms the horizon, the sun, the moon, the stars, all—no matter the declination—remain rising for 12 hours and setting for twelve hours. In this twelve hours, for 6 hours, they rise from the eastern horizon and reach the meridian circle, and for 6 hours, they descend from the meridian circle and go to the western horizon. (The slight difference caused by the refraction of light is not considered here for convenience). In places to the north or south

of the meridian country, only those planets or stars that lie on the equator remain visible for half a day and not visible for half a day, which means the diameter of their great circle is similar to the equator.

1. The great circle of the stars is exactly parallel to the equatorial circle. There is a slight difference in the direction of the great circles of the sun, moon, and planets because their declination always changes.

SOURCE PAGE 230

But the planet whose revolution is north of this star remains above the horizon for more than half a day in the northern circle, and the planet whose revolution is south remains above the horizon for less than half a day. In the southern circle, it is exactly the opposite. How much more or less time the planet remains above the horizon for half a day depends on the value of its revolution. If the revolution is more, then the difference is more, and if it is less, then less. The planet whose revolution is north of V and W in the figure will remain above the horizon until the time it rises from P to W, and then it will descend the same amount and go below the western horizon. It is stated above that it will take 6 hours to go from K to W; therefore, the time it will take to go from P to K will be 6 hours more than that, because the planet will come to the horizon at the time it reaches the point P due to the northward revolution. If the diameter of the orbit of the planet with a southern revolution is VU due to the southern revolution, then the time it will take to go from K to PA will be 6 hours less than that, and it will reach the horizon point PA behind the horizon. The part of the diameter of the orbit, VU, or the segment VU, or KU, is called dyujya because dy means day, revolution, or light. The segment VU is called charjyā of the planet with a northward revolution and the segment VU is called charjyā of the planet with a southward revolution. The arc of charjyā and the arc of this arc are called charakha because in one circle there are 360 × 60 arcs or 21600 arcs and in one celestial revolution there are the same number of revolutions. Here it should be remembered that DH and PA are drawn perpendicular to the circle from the circle.

Now we have to see how the rule of 60, 61 shlokas is proved with the help of figure 42.

The radius is, and the versed sine, and the angle, and the arc of the planet's revolution; therefore, the sine of revolution = the sum.

The versed sine of revolution

SOURCE PAGE 231

Jyā = vaka = lsha = shvi - vi l = trija - kratnika ki utkarmajyā (2)

This is the meaning of this 60th verse. The value of the right side is equal to the sine of declination.

∴ yajya = krantijya - kotijya error bhajak shapmen, kash = val = krantijya < kashpa = sthanakanksha ∴ akanksha sparshrekha = kap / kash = stitijya / krantijya but above equation (1) and it has been stated that (3)

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Latitude = equinoctial shadow / 12 ∴ equinoctial shadow / 12 = meridian arc / sine of declination
that meridian arc = sine of declination X equinoctial shadow / 12
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arcsin = sine of declination X equinoctial shadow / 12 ( 4 )
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But since both dh and ka are perpendicular to va and vi respectively, the mutual relationship between ka and va is the same as that of cha and sha also, that is—

vk: cup:: vis: scha

yashcha = cup X vis / vk = xitijya X trijya / dyujyachaashcocharjyabhikhtehain, isliye charjya = xitijya X trijya / dyujya

charjya = xitijya X trijya (5)

The rule of the 61st verse is proven by equations (4) and (5). The artistic arc of charjya is called char pran.

SOURCE PAGE 232

If the values of the meridian and versed sine in equation (5) are derived based on equations (2) and (4), then the simplified form of equation (5) will be:-

c y = krāṃtijyā * palabhā /12 * 1 / krāṃti koṭijyā * trijijyā = krāṃtijyā / krāṃti koṭijyā * palabhā /12 * trijijyā = krāṃti sparśarekhā * akṣāṃśa sparśarekhā * trijijyā (6)

That is, to get the sine, multiply the latitude of a place by the tangent of the planet's revolution and then multiply by the radius. If the sine value is in decimal form according to the current method, then no need to multiply by the radius on the right side of equation (6), and the simplified form of the sine will be—

c jaya = kranti sprasherkha × akshanshsprasherkha (7)

This proves that if the revolution and latitude are known, the sine can be easily known, and the hassle of knowing the meridian, etc., is not necessary.

The proof of equation (7) is as follows based on Napier's first rule:-

In Figure 43, there is a place in the northern hemisphere whose latitude is less than U degrees or U degrees arc. U is the horizon circle of S, and the points north, east, south, and west are in order U, P, D, and P points. Kh is the arc of the great circle of that planet or star that lies above the horizon when the planet's declination is equal to V degrees.

The rising point of the planet G is on the great circle which is perpendicular to the celestial pole D at the equator. Therefore, the time it takes for planet G to rise from point G or to reach the meridian circle at point V is the same time it takes for point V to move ahead of C in the same amount of time. But until C reaches the east point P, it will remain below the horizon. When it reaches the east point P, it will be 6 hours behind V. The time it takes for C to go from C to P is the same time the rising of the planet G has already occurred. Therefore, the sine of the planet's arc is

SOURCE PAGE 233

sine will be. To know its magnitude, Napier's first rule is very suitable because G is a right-angled spherical triangle whose G is a right angle, the inclination of the planet's revolution is known, and the

angle G is equal to the angle of the arc or the arc of the latitude, which is the complementary angle to the arc or the latitude. Therefore—

> ज्या(पू च) = स्पर्शरिखा(ग च) X कोटि स्पर्शरिखा < ग पू च, अथवा
> चर ज्या = क्रान्ति स्पर्श रेखा X अक्षांश स्पर्शरिखा ।

> तन्नाडिका उदक्क्रान्तौ धनं हानिः पृथक्स्थिते ।
> स्वाहोरात्रचतुर्भागे दिनरात्रिदले स्मृते ॥६२॥
> याम्यक्रान्तौ वियर्यस्तं द्विगुणे ते दिनक्षपे ।
> विक्षेपयुक्तोनितया क्रान्त्या भानामपि स्वके ॥६३॥

Translation—(62) If the arc of the variable sine obtained in the above manner is North, then adding it to the fourth part of the declination of the planet will give half the day and subtracting it will give half the night. (63) If the declination is South, it will be the opposite, that is, subtracting the variable art of the fourth part will give half the day and adding it will give half the night. Doubling half of the day or night will give the day-

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SOURCE PAGE 234

The mean and night time will be known. Similarly, to know the day time or night time of a constellation, i.e., a star, one should find the true position by adding or subtracting the deflection in its revolution, and then know the day time or night time by knowing the arc-time from the true position.

Scientific Commentary—these verses do not require special explanation because the explanation given to the preceding verses and the figures provided for it can easily prove this rule. The same rule is given in the last line to know the arc-time and day time and night time of the constellations, which has already appeared in the Scientific Commentary. Yes, the same mistake will occur in adding or subtracting the deflection to know the true position. Therefore, the true position of a star should also be determined according to the method shown in figure 40.

Example—Suppose the north true position of a star is 29°30'. What will be the day time and night time of it in Prayag?

(1) Sūrya Siddhānta kī rīti se — Prayag kī palbh * = 5 angula 41 angula = 5.7 angula (sthūl rīti se) tārē kī krānti = 22030 ′ ∴ krāntijyā = 1315 aur krāntikī utkrāmajyā = 261 ′ ″ ∴ dyujyā = 3438 ′ − 261 ′ = 3177 ′
kṣitijyā = 1315 × 5.712 carjyā = kṣitijyā × rījā dyujyā = 1315 × 5.712 × 34383177 = 2576652638124 = 676 ′

*Taken according to the equinoctial shadow and latitude of Prayag on page 76.

SOURCE PAGE 235

∴ variable art = 681' ∴ variable time = 681 life = 113 shadow 3 life = 1 hour 53 shadow 3 life add this to 15 Nakshatra hours because the inclination is north so half of the dayman = 16 hour 53 shadow 3 life full dayman = 33 hour 47 shadow full nightman = 26 hour 13 shadow

If one wants to know the dayman or nightman of the sun, then by adding double the variable life to the fourth part of the asu of the sun's ahuratra and subtracting and doubling to find the asu of the dayman and finding the asu of the nightman, if the inclination is north. One should know according to the 56th verse of the asu of the sun's ahuratra.

In the same way, by adding double the variable life to the fourth part of the asu of the planet's ahuratra, the asu of the dayman and by subtracting and doubling, the asu of the nightman will be found if the inclination is north.

It should be kept that the dayman or nightman that are obtained in this way will be in the units of Nakshatra time. A separate action will be required to convert to the units of Sawan day. A Nakshatra-ahuratra is equal to 21600 lives while a mean Sawan day is equal to 21656.14 lives.

(2) In a new manner—

Latitude of Prayag 25° 25'

Revolution of stars 22030'

∴ latitude = meridian line X declination line = line25 ° 25 ′ X line22030 ′ =.4752 X.4142 =.1968

∴ variable = 11 ° 21 ′ = 113shadow3

The Earth moves in 10 motions in 4 minutes, 10 *pal* or 60 *prana*. Therefore, the variable 'char' is kept as equal to 113 *pal* and 3 *prana*.

It is clear that according to the new method, there is no need for *dyujya*, *kshitijya*, etc. Yes, the table of the tangent is necessary, which can be made like the tables of sine and cotangent.

SOURCE PAGE 236

Method of knowing constellations

> भभोगोऽष्टशतीलिप्ताः खाश्विशैलास्तथा तिथेः ।
> ग्रहलिप्ता भभोगाप्ता भानि भुक्त्या दिनानि च ॥६४॥

Translation—(64) One constellation enjoys 800 zodiacs and one tithi enjoys 720 zodiacs. The number of constellations that have passed, which is the enjoyment of a constellation, is obtained by dividing the enjoyment of 800 zodiacs by the enjoyment of one constellation, and the remainder is the motion of the next constellation. If one wants to know when the planet came into the current constellation, dividing the passed zodiacs by the planet's daily motion yields the number of days, hours, etc. Subtracting the passed zodiacs from 800 and dividing the remainder by the daily motion will show when the planet will remain in the current constellation.

Scientific Commentary—It has been stated that the 27th part of the ecliptic is also called a constellation. The ecliptic circle is equal to 360 degrees or 360 X 60, i.e., 21600 zodiacs. Therefore, one constellation is equal to 21600 ÷ 27 = 800 zodiacs. For convenience, the names of each constellation have been kept —

In addition to these 27 constellations, the constellation of Abhijit is also needed in some places. It falls between Uttarashadha and Shravana. Uttarashadha

SOURCE PAGE 237

The last fourth part and the first fifteen parts of śravaṇa are understood as the enjoyment of abhihita. Thus, the enjoyment of abhijita was 2533 arts.

In ancient times, there were different values for 28 constellations instead of 27, regarding Abhijit. Bhaskaracharya says that the astronomers such as Pulish, Vashishtha, Garg, etc., determined the subtle values of the constellations for the fulfillment of the results of marriage journeys, etc.:

The mean daily motion of the Moon is considered to be 760^(°)35''. Its declination is 1185^(°)52''.5 and half 365^(°)17''.5.

The enjoyment of each of the remaining 15 constellations is 760^(°)35''. Subtracting the sum of all these enjoyments from 21,600 arts, what remains is the enjoyment of the auspicious one. In this way, the enjoyments of twenty-seven constellations together are equal to 6 × 3/2 + 6 × 3/2 + 15 × 1 or 27 mean daily motions, which is = 27 × 760^(°)35'' = 27 × (800 - 6'25'') = 21600'' - 254'15''.

Thus it is proven that the enjoyment of the auspicious time is 254’ 15” which is 45” more than the value of the Muhurta concern-gem.

From all these things it is understood that the values of the constellations were determined in ancient times according to the mean motion of the moon and by observing the bright stars that indicate the constellations. But as astrology developed from behind, it must have become known

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SOURCE PAGE 238

That division does not match; therefore, for convenience, the division of the ecliptic was made in only 27 constellations and each part was considered to be 800 degrees.

Example—Suppose you want to know in which constellation Jupiter was on the day of Vaspant Panchami (1676 AD).

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Gurukāspasthabhoga = 6th27° 53' 37'' = 20° 054' sthūlroopse = 12474'
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Dividing this by 800 gives a remainder of 15 and 404′.

Therefore, Jupiter has passed the 15th constellation and is in the 16th constellation Vishakha, has consumed 474′ of Vishakha, and has 326 degrees remaining. To know how long Jupiter will remain in Vishakha. According to the rule, 326 degrees should be divided by the daily motion of Jupiter. However, this rule is not subtle for Jupiter and other slow-moving planets because for 326 degrees to move, Jupiter needs a long time during which its motion will not be constant. Therefore, more subtle consideration is required.

What has been written earlier regarding the tithi is sufficient. Special discussion will be made in the next 66th verse.

Method of knowing the yuga

> रवीन्दुयोगलिप्ताभ्यो योगाभ भोगभाजिताः ।
> गतं गम्यं च षष्टिघ्नं भुक्तियोगाप्त नाडिका ॥६५॥

Translation—By adding the true positions (nirayana bhogashas) of the Sun and the Moon and finding their art, and dividing by 800, the number of past yugas is obtained. From the remainder, it is known

how many arts of the current yuga have passed. If this remainder is subtracted from 800 arts, it will be known how many arts of the current yuga remain. If this past navigable art is multiplied by 60 and divided by the sum of the true daily motions of the Sun and the Moon, then it will be known how many hours before the current yuga began and how many hours after it will end.

Scientific Commentary—When the Sun and the Moon both advance by 800 arts from the beginning of Ashvini, 1 yuga passes, when they advance by 1600 arts, the second yuga passes, etc. Similarly, when they together advance by 3600

SOURCE PAGE 239

When 21600 kala Ashvini are passed, the 27th Yuga is completed. Then the first Yuga begins. The names of the 27 Yugas are:-

1 Viskambha 10 Gandha 16 Paridha 2 Preeti 11 Vriddhi 20 Shiva 3 Ayushman 12 Dhruva 21 Siddha 4 Saubhagya 13 Vyaghat 22 Sadhya 5 Shobhan 14 Harshan 23 Shubha 6 Atigandha 15 Vajra 24 Shukra 7 Sukarma 16 Siddhi 25 Brahma 8 Dhriti 17 Vyatipata 26 Indra ya Ain 9 Shula 18 Variyān 27 Vaidhriti

An example will be sufficient to understand the rule. Suppose you want to know which Yuga was current at the time of the Aries solstice of 1681 A.D., and when it began and when it ended?

The positions of the Sun and Moon and the daily motions of the first Aries solstice will have to be clarified.

There are 5025 solar years and 18,35,423.080625 mean Sawan days from the beginning of Kali Yuga until the time of the Aries solstice of 1681 A.D. The beginning of Kali Yuga occurred at midnight of the Guru in Ujjain, so 1681 A.D. Aries solstice occurred 0.080625 days after the midnight of Saturday in Ujjain. It will be good to clarify the Sun and Moon at the time of midnight for convenience.

The way in which the true position of the Sun has been calculated, similarly, the positions of both the Sun and the Moon should be clarified. The essence of the calculation is:-

The position of the Sun is the position of the Moon, the sum of both is Saturday (midnight) 356°55'16" 65°38'53" 65°34'6" Sunday (midnight) 0°53'56" 108°16'56" 106°10'55" Daily motion 58'43" 120°38'3" 130°36'46"

The first sum becomes more than 360°, so 360° was left and 65034'6" was taken

Now 65° 34' 6" = 5734' 6"

SOURCE PAGE 240

When divided by 800, 7 leavels and 134' 6" remain. Therefore, the eighth conjunction of the remaining ecliptic is present at the half-night of the remainder and it has passed 134'6" and 665'51" remains.

In 60 hours, the motion of the Sun and the Moon together is 13°36'46" or 13.6128.

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134' 6" = 2014' 6" = 20.2358
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665' 51" = 11° 5' 51" = 11° .0675
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13.6128: 2.2358:: 60 hours: auspicious time

Sūrya Siddhānta

Scientific Commentary

Chapter on Mean Motions

Chapter on True Positions

Chapter on the Three Problems

The following was calculated according to the mean period. To do it according to the true period, the time equation and the longitude from Ujjain for the other place will also be necessary. The discussion of the time equation will be specially made in the third chapter. If time calculation is to be done from sunrise, then the latitude will also be necessary.

> अर्कोनचन्द्रलिप्ताभ्यस्तिथयोभोगभाजिताः।गतंगम्यंचषष्टिघ्नंनाड्योभुक्त्यन्तरोद्धृताः॥६६॥

(66) By subtracting the apparent position of the sun from the apparent position of the moon in the true position of the moon and making the resulting art, divide it by the duration of a tithi, i.e., 720 arts, the elapsed tithi will be obtained; the remainder will be the elapsed arts of the current tithi. When this is subtracted from 720 arts, the elapsed arts of the current tithi will be obtained. By multiplying the elapsed and elapsed arts of the current tithi by 60 and dividing by the difference in the daily true motions of the sun and the moon, it will be known when the beginning and end of the current tithi occurred.

SOURCE PAGE 241

Scientific Commentary—for this work also the sun and the moon have to be clarified. To see which date was current on the midnight of the night of Shani near the Aries solstice of 1681.

Subtracting the sun's true position from the moon's true position at the midnight of Saturday does not decrease, so subtracting the sun's position from the sum by adding 360° to the moon's position gives 65° 43’ 37”. Similarly, the difference in the positions at the midnight of Sunday is 107° 22’ 57”. The difference in their daily motions is 110° 36’ 20”.

There is a date of 720 art or 120. Therefore, when 65°43’ 37” is divided by 120, the remainder is 7 and 11° 43’ 37”. From this it appears that the eighth date, which is Ashtami, is currently at the time of midnight, and 11° 43’ 37” has passed, with 16’ 23” remaining. Multiplying this 16’ 23” by 60 and dividing by 11° 36’ 20” yields 1 hour 24 minutes. Therefore, the end of Ashtami has occurred one hour 24 minutes after the midnight of Saturday.

To know the end time of the tithi from the sunrise in another place, the same rituals have to be performed as mentioned in the Yoga.

The rules stated for knowing things like Tithi Yoga are very difficult, so tables are used for practical purposes, which makes it very easy to know the beginning or end time of Tithi Yoga, etc. The principle

of making tables for expansion is not stated here out of fear. If understanding is required, it will be stated in the appendix at the end.

> ध्रुवाणि शकुनिर्नागः तृतीयं तु चतुष्पदम् ।
> किस्तुघ्नं च चतुर्दश्याः कृष्णाया अपरार्धतः ॥६७॥
> बवादीनि तथा सप्त चराख्यकरणानि तु ।
> मासेऽष्टकृत्व एकैकं करणं परिवर्तते ॥६८॥
> तिथ्यर्धभोगं सर्वेषां करणानां प्रचक्षते ।
> इत्थं स्पष्टगतिः प्रोक्ता सूर्यादीनां खचारिणाम् ॥ ६९॥

(67) Shakuni, Naga, Chatushpada, and Kistughna four fixed positions remain in sequence from the latter half of the full moon of each fortnight until the half-moon. (68) After that, the seven movable positions (Bava, Balava, Kaulava, Taitila, Garaj, Vanija, Vishti) revolve in sequence eight times in a month. (69) Each

SOURCE PAGE 242

The enjoyment of the Sun should be considered as half the tithi. Here the method of explaining the solar and lunar planets is mentioned.

Scientific Commentary—the sequence of fixed points described here differs from some pracalita paṃcāṃgoṃ in the pracalita paṃcāṃgoṃ. In these, after Shakuni, Nag and Kistughna are found written. What is the reason for this and when did this sequence begin is worth considering. The second name of Visti is Bhadra, which is considered inauspicious in auspicious deeds. Which tithi and which karana is experienced in each lunar month will be revealed by the table below.

Date Date of Shukla Paksha Purvaddh tithi of Shukla Paksha Uttarardh tithi of Krishna Paksha Purvardh Krishna Paksha Uttarardh 1 Kistughna Bav Walav Kolav 2 Balav Kolav Taitil Garaj 3 Taitil Garaj Vanij Vishti 4 Vanij Vishti Bav Walav 5 Bav Balav Kolav Taitil 6 Kolav Taitil Garaj Vanij 7 Garaj Vanij Vishti Bav 8 Vishti Bav Balav Kolav 9 Walav Kolav Taitil Garaj 10 Taitil Garaj Vanij Vishti 11 Vanin Vishti Bav Balav 12 Bav Balav Kolav Taitil 13 Kolav Taitil Garaj Vanij 14 Garaj Vanij Vishti Shakuni 15 Vishti Bav Nag Chaturpad

The full moon of the 15th day of the waxing phase and the new moon of the 15th day of the waning phase are called. The full moon indicates 15 and the new moon 30.

The scientific commentary of the second chapter, called *Spashtādhikāra*, of the Sūrya Siddhānta has ended.

SOURCE PAGE 243

Chapter on the Three Problems

(Brief description)

[1-4 verses—drawing lines indicating directions by placing a gnomon on a flat land. 5 verses-knowing the direction of the shadow by knowing the difference between the shadow of the gnomon and the line from its tip to the east-west. 6 verses—definitions of the prime vertical circle, the meridian circle, and the viṣuvanmaṇḍala. 7 verses—definition of the agra. 8 verses—knowing the shadow hypotenuse by

knowing the magnitude of the gnomon and its shadow. 9-10 verses—knowing the revolutions, shadows, and variations of the planets by knowing the precession angle. 11 verses—checking the precession angle by eclipsing the sun for the day of agra or viṣuvat. 12 verses—definition of the equinoctial shadow. 13 verses—knowing the arc and latitude from the equinoctial shadow. 14-15 verses—knowing the latitude by knowing the arc and revolution of the meridian sun. 16 verses—knowing the equinoctial shadow from the latitude. 17-16 verses—knowing the revolution of the sun from the latitude and the arc of the meridian sun, knowing the true sine of the sun from the sun's revolution, and knowing the true sine of the meridian from the apogee. 20-21—knowing the meridian shadow and shadow hypotenuse by knowing the latitude and the revolution of the sun. 22 verses—knowing the agra of the time of rising of the sun by knowing the agra of the sun. 23-24 verses - knowing the length of the shadow by agra and equinoctial shadow. 25 verses—the method of knowing the shadow hypotenuse when the sun is in the prime vertical circle. 26 verses—knowing the shadow of the sun when the sun's revolution is less than the latitude. 27 verses—the second method of knowing the agra. 28-31 verses—knowing the latitude of the sun from the knowledge of the ratio and the result when the sun is in the east or west. 32 verses—knowing the arc by knowing the latitude. 33 verses—knowing the shadow and shadow hypotenuse from the latitude and the arc. 34-35 verses—knowing the sine of declination and the shadow and shadow hypotenuse by knowing the difference from the variation. 36-38 verses—knowing the time of the shadow by the shadow and shadow hypotenuse. 36 and 40 the first half—the second method of knowing the portion of the sun by knowing the revolution from the agra. 40 the second half and 41 the first half—drawing the ecliptic. 41 the second half and 42 verses—the method of knowing the rising times of the zodiac signs in Lanka. 43-44 verses—the method of knowing the rising times of the zodiac signs of Aries, Taurus, and Gemini in Lanka and the rising times of the zodiac signs in other places. 45-47 verses—knowing at what time the sign is in the east horizon. 48 verses—knowing the meridian! 49-50 verses—knowing the time by knowing the sign.

SOURCE PAGE 244

In this chapter, some hold that the number of shlokas is 50 and some hold that it is 51. Those who consider the number of shlokas to be 50 say that in the 11th and 20th shlokas, there are 6 steps instead of 4 in each. Those who consider 51 believe that each shloka has four steps. Therefore, both opinions are identical in my understanding. At this time, I have four reminiscences of Sūrya Siddhānta, but regretfully, the order of the numbers of shlokas is not the same in two of them. The order of the numbers is also messed up in the book edited by Indra Narayan Dwivedi, so for convenience, I have considered the 11th and 35th shlokas as three lines or six steps each. Considering the 20th shloka as having 6 steps causes the problem that the rules are not complete in any subsequent shloka, but the latter half of one shloka and the first half of another shloka have to be combined. Considering the 35th shloka as having 6 steps, this inconvenience remains only in shlokas 36-42.

In this chapter, many methods of obtaining information about direction, country (place), and time from the sun's zenith are described. Only one instrument is used for the zenith, which is called gnomon. A straight pointed rod of a hard metal or ivory was placed vertically on level ground, and all work was done from its shadow. This is called the gnomon. The Yantradhyaya describes other instruments, but in this place, only the gnomon is discussed. It is clear that the image of the sun appears very large, and the tip of the shadow of the gnomon cannot be determined with much subtlety, so the things that can be known from the gnomon are some gross ones. Nowadays, more subtlety can be achieved by sighting with a telescope, but in ancient times, the gnomon was very useful. Our astrologers have shown great skill in obtaining the subtlety that could be achieved by sighting with it. There is no doubt in this at all.

Some such things have also been invented with the help of the gnomon instrument, without which knowledge of direction, country, and time remains coarse, so there is a need to discuss them as well. Therefore, in the Scientific Commentary, the parallax of rays (refraction of light), the cause of the motion of the ecliptic, the deviation of the axis (nutation), the aberration of light, and the equation of time will be described in detail.

SOURCE PAGE 245

Method of Determining Directions

> शिलातलेऽम्बुसंसिद्धे वज्रलेपेऽपि वा समे ।
>
> तत्रशङ्कङ्गुलेरिष्टैः समं मण्डलमालिखेत् ॥१॥
>
> तन्मध्ये स्थापयेच्छङ्कुं कल्पितद्वादशाङ्गुलम् ।
>
> तच्छायाग्रं स्पृशेद्यत्र वृत्तं पूर्वापराह्णयोः ॥२॥
>
> तत्र विन्दू विधातव्यौ वृत्ते पूर्वापराभिधौ ।
> तन्मध्ये तिमिना रेखा कर्तव्या दक्षिणोत्तरा ॥३॥
>
> याम्योत्तरदिशोमध्ये तिमिना पूर्वपश्चिमे ।
>
> दिङ्मध्यमत्स्यैः संसाध्या विदिशस्तद्वदेव हि ॥४॥

(1) Draw a circle with the diameter of one and a half angulas according to the gnomon on the surface of the stone leveled by water, or on a level platform made of vajralepa (sarkhi lime etc.). (2) Establish a cone of twelve angulas in the vertical form at the center of this circle. Mark the points where the shadow of this cone touches the circle before and behind the meridian. (3) Make two points on the circle at those locations. These two points are called the forenoon and afternoon points. Draw a north-south line between these two points with a plumb line. (4) Draw east-west lines between the north-south directions with a plumb line. In this way, draw lines of directions such as northeast between the two directions with a plumb line.

Sūrya Siddhānta—the easiest way to know whether a plane is level is to fill water one or a one and a half inches deep by hiding it with wet soil around the edges of the plane and see with a straight plumb line whether the water depth is the same or different everywhere. If the water depth is the same everywhere, then it should be understood that the plane is level. Nowadays this work is done with a Spirit level.

Vajralepa – in the beginning, many types of spices were mixed in lime to make such a mortar that became as hard as thunder. Such mortar is called Vajralepa. There is one method for making Vajralepa in the Barahi Samhita:-

Chapter 71 verse 1-3

SOURCE PAGE 246

Make a decoction by cooking the raw fruits of tendu, raw fruits of katha, semal flowers, sallki seeds, bark of bondage, and these all in water, when the eighth part of water remains, take it out and mix in

the juice of shrivas (gum of a simple tree), gūgala, bhilawa, kundru, ral, linseed, and bel giri ground, then vajralep is prepared.

Timi - If two circles are drawn intersecting each other, the part between them becomes fish-shaped. This is called timi.

In Figure 44, the place of the gnomon is in the center of the circle. When the shadow of the gnomon before noon is the same as the shadow of 'k', its tip reaches the point on the circumference of 'k'. When the shadow behind noon is the same as the shadow of 'kh', its tip reaches the point on the circumference of 'kh'. Just these points 'k' and 'kh' are taken as centers, and two circles of equal diameter are drawn from them, which form an area in the shape of timi. The line joining their general points is the North-South line. The direction of North should be indicated by 'u' and the direction of South by 'd' on the points where this line cuts the first circle. Then, taking 'u' and 'd' as centers, two more arcs of equal diameter are drawn, and their general points are joined by a straight line. This will be called the East-West line. 'p' should be written for the West direction and 'pu' for the East direction. Then, taking 'u' and 'p' as centers, two arcs of equal diameter are drawn, and the point where the circumference will be cut between their general points is the direction of the NW angle and the point where it will be cut between 'pu' is the direction of the fire angle. Similarly, the direction of the North-East and South-West angles can also be known.

Proof—At the time of rising, the point on the eastern horizon where the sun appears, rises towards the south, and the shadow of a vertical stick or gnomon moves towards the north, becoming smaller. At noon, the sun comes to the meridian. At that time, the shadow is the smallest and in the exact north direction. After this, the sun begins to descend slightly towards the north, and the shadow moves from the north towards the east, becoming larger. The time before noon when the gnomon's shadow makes a larger angle towards the west from the north is the same time behind noon when the shadow makes the same larger angle towards the east from the north. From noon the same

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SOURCE PAGE 247

Forward and backward in time, the length of the shadow is also equal*. Therefore, when the length of the shadow is equal to half the diameter of the stretched circle, the line that divides the angle formed between them into two equal parts will be the North-South line. To draw this bisecting line, it is ordered to draw arcs of equal diameter and make a shadow, which is according to the method of geometry. According to this rule, lines indicating other directions can be drawn. The line joining the points fixed by the noose of the shadow at the forenoon and afternoon on the circle is also the East-West line, but in order to work with the tilted gnomon in the future, it is necessary that all the lines drawn indicating directions pass through the center of the gnomon. Therefore, it is ordered to make a shadow from the North-South points of the circle and draw the East-West line.

Due to the constantly changing nature of the ecliptic, the slight imprecision that arises for its correction has been regulated by Bhaskaraacharya dagger and other astronomers, but there seems no need to describe them. These corrections maintain the simplicity of the above method. If a difficult rule is required for accuracy, why not use the method of knowing Digansha, which is discussed in this chapter?

> चतुरश्रं वहिः कुर्यात्सूत्रैर्मध्याद्विनिर्गतैः ।
>
> भुजसूत्राङ्गुलेस्तत्र दंतौरिष्टप्रभाग्रतः ॥५॥

(5) Draw lines touching the points reached by the perimeters drawn with the same breadth as the shadow from the center to the north-south and east-west lines. Make a rhombus area by drawing lines that form a right angle with the meridian.

The revolution of the sun always changes, therefore there is some difference in the revolutions before and after noon, which introduces some crudeness in the above statement, but it should not be understood as an equality. At the time when the motion of the revolution is very slow, that is, at the time when the sun is near the points of uttarāyana or dākṣiṇāyana, this matter will be purer.

> तित्कालापमजीवयोस्तु विवराद्भाकर्णमित्याहताल्लम्बज्याप्तमिताङ्गुलै-
> रयनदिश्यैन्द्री स्फुटा चालिता ॥८॥

Mathematics Chapter, Privileges of Inquiry page 104-105

SOURCE PAGE 248

Draw a straight line as straight as the rays of the sun, which reaches the circumference. Join the point where the tip of the ray reaches the circumference with the center of the gnomon, then the direction of the shadow will be known.

Scientific Commentary—in Figure 45, śa is the center of the gnomon and śa is the shadow of some time. Drawing the circumference from the radius of śa to śa is done. pa is the meridian or east-west line and u is the north-south line. The difference between the tip of the shadow from the east-west line is equal to cha and the difference from the north-south line is equal to ka. cha is called the side of the shadow and ka is called the extent of the shadow. The meaning of this verse is that if the length of the shadow and the side are known, then the direction of the shadow can be known. According to the current practice, this can be said as follows: if the coordinates of the side of the shadow are known, then the shadow can be drawn. The difference from the tip of the shadow from the east-west line and the difference from the tip of the shadow from the north-south line

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SOURCE PAGE 249

They call the extent of the shadow. The relationship between the shadow at a certain time and the hypotenuse of it is explained in 23-24 shlokas.

If the line is extended by 6, it will cut the circumference at point S. The sun will be in the direction where the shadow of the gnomon is S at this time. The distance from the east point to the sun in this time is the angle that can be taken from the east-south angle. This angle is the azimuth of the sun at this time. In this picture, the sun is south of the east line. If the sun is north of the east line, then the shadow, azimuth, hypotenuse, etc., will be as per the 46 pictures.

The day when the sun is on the equinoctial line, that day, i.e., on the day of the vernal or autumnal equinox, the shadow of the gnomon being as large as the subsolar point is called the subsolar point, equinoctial shadow, or latitude. If the equinoctial shadow of a place is equal to that of V, then the line drawn parallel to the east-west line from V is called the subsolar line. The difference between the tip of the shadow and the subsolar line is called the sine of colatitude. In the 45-46 pictures, this is the sine of colatitude.

> ★सममण्डल, उन्मण्डल और विषुवन्मण्डल ★
> प्राक्पश्चिमाश्रिता रेखा प्रोच्यते सममण्डलम् ।
> उन्मण्डलं च विषुवन्मण्डलं परिकीर्त्यते ॥६॥

SOURCE PAGE 250

Translation—(6) The prime vertical, meridian, and equatorial circles are on the east and west points.

Scientific Commentary—The word meaning of this verse is—the line going from the east and west points is called the prime vertical, and the same is also called the meridian and the equatorial. However, in reality, these three words have different meanings, which is why in the translation I have written the meaning that is appropriate against many other commentators. It seems that the pure form of this verse is not this, but it has been done due to confusion. Ranganatha ji in his profound meaning commentary tried to prove the uniformity of these three words by considering this to be pure, but it does not seem logical because these three words have had different meanings since very ancient times, and the only similarity in them is that these three circles pass through the east and west points.

The prime vertical is the vertical circle that passes through the cross and the east and west points.

SOURCE PAGE 251

The meridian (six o'clock line) is the circle that passes through the east-west points and the north-south celestial poles. This is the horizon on the meridian.

The celestial equator is the circle that passes through the east-west points and is at the same distance from the north and south celestial poles.

Description of Figure 47—

Description of Figure 47— Sh...the position of the observer is ...the southern celestial pole, ...the north point, ...the cross, ...the full point, ...the horizon circle, ...the southern point, ...the pole, ...the meridian, ...the western point, ...the prime vertical circle, ...the north celestial pole, ...the equatorial circle, ...the east-west line and the general point of the equatorial circle

SOURCE PAGE 252

In Figure 42, only one straight line is drawn for each circle or sphere. The east-west line of the meridian is drawn in both. The line parallel to the line passing from the front of the equinoctial shadow is called the subsolar line. The difference between the tip of the shadow and the subsolar line is called agra.

Chapter on True Positions

> अग्राज्या
> रेखा प्राच्यपरा साध्या विषुवद्भाग्रगा तथा ।
> इष्टच्छायाविषुवतोः मध्यमग्राऽभिधीयते ॥७॥

sine of declination

shadow hypotenuse

gnomon

> शङ्कुच्छायाकृतियुतेर्मूलं कर्णोऽस्य वर्गतः ।
> प्रोज्झ्य शङ्ककृतिमूलं छाया शङ्कुविपर्ययात् ॥८॥

(8) Gnomon and Shadow
By adding the squares of each, taking the square root gives the shadow hypotenuse. By subtracting the square of the gnomon from the square of the shadow hypotenuse and taking the square root, and by subtracting the square of the shadow from the square of the shadow hypotenuse and taking the square root, the gnomon is obtained.
Scientific Commentary—If two sides of a right-angled triangle are known, the way to know the third side is the same here as gnomon, shadow, and shadow hypotenuse.

It also applies in this regard. The essence of this verse is:-

$$\text{shadow hypotenuse} = \sqrt{\text{gnomon}^2 + \text{shadow}^2};\ \text{shadow} = \sqrt{\text{hypotenuse}^2 - \text{gnomon}^2};$$

$$\text{gnomon} = \sqrt{\text{hypotenuse}^2 - \text{shadow}^2}$$

Method of knowing precession angle

> त्रिंशत्कृत्या युगे भांशैश्चक्रं प्राक्परिलम्बते ।
> तद्गुणाद्भूदिनैर्भक्ताद् च्युगणाद्यदवाप्यते ॥६॥
> तद्दोस्त्रिघ्ना दशाप्तांशा विज्ञेया अयनाभिधाः ।
> तत्संयुक्ताद् ग्रहात्क्रान्तिच्छायाचरदलादिकम् ॥१०॥

SOURCE PAGE 253

Translation—(6) In one yuga, the constellation-circle oscillates like a pendulum 600 times towards the east. Multiplying this 600 by the number of days of the great yuga (10) and dividing by the number of days of the season, multiply the result by 3 and divide by 10. Whatever remains is called the precession angle. By imprinting this on the positions of the planets, one should know the revolution, shadow, declination, etc., of the planets.

Scientific Commentary—The common point on the ecliptic where the sun comes from the north of the ecliptic and the tropics is called the vernal equinox. When the sun reaches 60 degrees beyond the vernal equinox, its northward movement stops and it begins to return towards the south. At this time, the onset of the southward motion begins. Therefore, the point where the sun turns to the south is called the summer solstice. The time when the sun continues to move towards the south from the beginning of the southward motion, which is similar to 6 months, is called the southward motion. After 3 months from the beginning of the southward motion, the sun comes again on the vernal equinox. This point is called the autumnal equinox because the autumn season begins at this time. The sun continues to move towards the south for 600 further, then turns back to the north. The point where the sun begins to return to the north is called the winter solstice. The onset of the northward motion begins from this time. The southward and northward points are called the apogees. In Figure 36, the order of V, D, SH, and U are vernal equinox, southward motion point, autumnal equinox, and winter solstice. The circle

that occurs on the apogees, celestial poles, and latitudes is called the solstitial colure. In Figure 36, the circle of V, D, SH, U is the solstitial colure.

The declination point does not always remain in the same place in the sky but is moving towards the west, therefore the constellation or star group that has Uttarayana or Dakshinayana nowadays does not have it in the ancient times. Vedanga1

SOURCE PAGE 254

In astrology it is written that when the sun was at the beginning of the Nakshatras Śravishtha or Dhanishtha, the beginning of Uttarayana started, and when the sun reached the half of the Ashlesha Nakshatra, the beginning of Dakshinayana started.

In Barahmihira's Barahi Samhita, while discussing this, he writes that in ancient times, Dakshinayana was at the half of Ashlesha and Uttarayana at the beginning of Śravishtha, but now as soon as the sun enters the sign of Ka, Uttarayana happens when the sun enters the sign of Makara. If this is not the case, one should ascertain by observation.

Nowadays, the beginning of Dakshinayana starts at the beginning of the Ardra Nakshatra and the beginning of Uttarayana starts at the half of the Moola.

Thus it is proven that the Uttarayana point was at the beginning of Dhanishtha in the time of Vedanga Jyotisha and now at the half of the Moola. Therefore, it is clear that the motion is towards the west. Because of this, the Spring Equinox point or Autumn Equinox point is also moving towards the west. The westward movement of the Spring Equinox point is called precession of equinoxes by European astronomers, so the phenomenon that our astrologers have written as Ayana-Chalan is called precession of equinoxes by Western astronomers. Bhaskaracharya Ji2 has equated Ayana Chalan and Vishuvatkranti-Valay-Patachalan.

There is a great difference in the opinions of our ancient astrologers regarding Ayana Chalan. The view of Sūrya Siddhānta is that the beginning point of the Nakshatra cycle moves or oscillates by up to 27 degrees on both sides of the Spring Equinox like a pendulum (Figure 46). A is also called the beginning point of Ashvini. This movement means that at the beginning of the Yuga, the Spring Equinox and the beginning point of Ashvini were together. After that, the beginning point of Ashvini started moving towards the east, and when it moved 27 degrees further from the Spring Equinox, then it again moved from the Spring Equinox

> १. आश्लेषाद्धदक्षिणमुत्तरमयनं रवेर्धनिष्ठाद्यम् । नूनंकदाचिदासीद्येनोक्तं पूर्वं शास्त्रेषु ॥१॥ साम्प्रतमयनं सवितुः ककंटकाद्यं मृगादितश्चान्यत् । उक्ताभावे विकृतिः प्रत्यक्ष परीक्षणैर्व्यक्तिः ॥२॥

Barahi Samhita, Adityachar pages 16, 17.

2. There are also the motions of (the motion of the ecliptic) and these are the parts of the nodal motion that are famous.

SOURCE PAGE 255

It began to return and slowly became with the vernal equinox. After this, it started to go towards the west from the vernal equinox and went back towards the vernal equinox after going 27 degrees. This

sequence is called a complete oscillation. Such 600 oscillations occur in a great age, that is in 43,20,000 solar years. Therefore, one oscillation takes 7200 solar years and a quarter oscillation or a motion of 27° takes 1800 solar years.

To know how much the initial point of Ashvini is from the vernal equinox, that is, the precession angle, one should use the rules mentioned in 6-10 shlokas which will be clear with an example.

Suppose one wants to know the precession angle of 1682 AD.

From the beginning of creation to the beginning of the current Kali Yuga, 1,65,58,80,000 solar years have passed* in which 2,71,650 oscillations have been completed according to one oscillation in 7200 years, so a new oscillation began at the beginning of the Kali Yuga. Therefore, it is convenient to use the calculations of Kali Yuga etc. to know the precession angle.

* See Chapter on Mean Motions

SOURCE PAGE 256

Will be. The rule in the text says to work with ahargana, but there can be no inaccuracy in working with solar years to know the precession angle of Aries. From the beginning of Kali Yuga to the Aries solstice of 1682 AD, there are 5026 solar years. Therefore,

7200 : 5026 :: 1 oscillation : desired oscillation.

That is, desired motion = 5026 / 7200 = 5026 X 360 / 7200 fraction = 251018 ′

This is in the third pada. Therefore, according to the 30th verse of the Chapter on True Positions, the preceding part of the third pada has been completed. Multiplying this by 3 and dividing by 10 gives the precession angle of 21023'24". To find the precession angle of the auspicious time, one can calculate the motion for the number of days behind the Aries equinox that needs to be known, in a rate of 54" per year, and add it to the precession angle of the Aries equinox.

It is clear that the supreme value of Bhuj is 60°, so if it is multiplied by 3 and divided by 10, then 27° comes, which is the supreme value of the precession angle according to the Sūrya Siddhānta.

There is one thing to note here. It is said in the 6th verse that the constellation-circle reflects towards the east, meaning that upon starting the motion, it first moves towards the east. Therefore, as long as it moves 27 degrees towards the west of the vernal equinox, it is in the first quadrant, when it moves back towards the east of the vernal equinox, it remains in the second quadrant, until it moves 27 degrees towards the west of the vernal equinox, it remains in the third quadrant, and when it moves back towards the east of the vernal equinox, it remains in the fourth quadrant. Therefore, it is proven by the above calculation that the initial point of Ashvini is 21023’ 24” west of the vernal equinox. However, in reality, the initial point of Ashvini is east of the vernal equinox at this time, as is clear from the next 11th verse as well. Therefore, it must be assumed that the initial point of Ashvini moves towards the west upon starting the motion, which is contrary to the verse. Therefore, it seems that the Acharya

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SOURCE PAGE 257

It is considered that the vernal equinox moves by 27° east and west of the beginning point of Ashvini, and there is some error in the text. Because other scholars consider the precession or the sine of

declination to be moving. When the precession moves 27 degrees in 1800 years, then the motion in 1 year is 54 arc minutes. Therefore, according to the view of Sūrya Siddhānta, two things are proven—(1) the vernal equinox may move 27 degrees forward or backward from the beginning of Ashvini, and (2) its annual motion is 54 arc minutes.

What is the cause of the precession is not found anywhere in Indian astrology. Rangnathji considered the will of God as the cause in his profound commentary Prakashika ṭīkā.

The opinion that is of Sūrya Siddhānta is also of Soma Siddhānta*, Romash Siddhānta*, Shakalya Brahma Siddhānta*, and Laghuvashishtha Siddhānta*. Second Aryabhata* and Parashara* also do not consider the full revolution of the precession, but according to their opinion, the vernal equinox moves only 24 degrees before or after the original point, not 27°. The method of knowing the precession angle laid down by the Second Aryabhata suggests that the annual motion of the precession is not always the same. Yes, the mean annual motion is considered to be 46.3 arc minutes. Parashara considers the annual motion to be 46.5 arc minutes.†

Conversely, the view of Munjal[1] is that the precession or the vernal equinox travels in the opposite direction and completes a full circle in one Kalpa, moving 1,66,666 revolutions. Bhaskaraacharya[2] also considered this. According to this, the annual motion of the precession is 56.6007 arc minutes, which is approximately 1 art. Therefore, in practice, Munjal, Bhaskaraacharya, Ganesh Daivagya, etc., consider the annual motion of the precession to be 1 art.

Varahamihira, Brahmagupta, etc., did not write about the calculation of the precession angle. It seems that the magnitude of the precession angle was very small during their time

SOURCE PAGE 258

And the knowledge of the course of the obliquity was not known to them either. Varahamihira has only written that in the beginning of Dhanishtha, the northward motion was, and at their time, in the beginning of Makara, that is, on the first Pada of Uttarashadha. He has also said that if there is a variation, one should work from the direct shadow. Besides this, no rule for knowing the obliquity is written. Brahmagupta has not indicated anything either. The reason for this is that Bhaskaracharya[1] writes that in the time of Brahmagupta[2], the precession angle was very small, so he could not find out this.

The knowledge of the movement of the vernal equinox was known to the Greek astronomer Hipparchus seventy years before the Vikram era. While making a list of the stars, he found that the difference between each star in this list and the list made one and a half hundred years earlier was an additional 2 degrees. From this, he concluded that the vernal equinox is receding. The annual motion of the vernal equinox that he calculated was less than 3 less than 36 degrees. Ptolemy determined this motion of the vernal equinox in the beginning of the third century of Vikram. After this, there was no further advancement in Greek astrology regarding the movement of the vernal equinox and other matters.

But a prince named Albatauni of Arabia, who was a skilled astronomer, determined the annual motion of the vernal equinox with some accuracy of about 637 AD. Shankar Balakrishna Dikshit writes that Albatauni had determined the annual motion of the equinox to be 55.5 degrees. After this, Nasiruddin established a observatory near the northern western border of present-day Iran and determined the annual motion of the vernal equinox to be 51 degrees at the beginning of the 14th century Vikram.

SOURCE PAGE 259

The formula for the annual motion of the vernal equinox in the spring equinox nowadays is according to Newcomb.

50”.2453+0”,0002225 v

Where v is the number of years passed after 1850 AD or 1607 BC. According to this, the annual motion of the vernal equinox at the beginning of 1682 BC is 50”.2453+0”.0002225 X 75 =50”.262

The discussion of the ideas of Indian, Greek, Arabic, and European scholars regarding the motion of the ecliptic is briefly made so that it may be revealed how independent the views of our astrologers are in this regard. Now the question arises that why our astrologers considered the annual motion of the ecliptic as 1 art when the pure motion is approximately 50.262 degrees. The reason for this is that our astrologers call the precession angle the difference that is from the beginning point of Aries to the vernal equinox. And the beginning point of Aries is not determined by the arc but by calculation. For calculation, according to the Sūrya Siddhānta, a year is considered to be 365 days 15 hours 31 minutes 31.4 arc, whereas according to modern discovery, the value of the pure year is 365 days 15 hours 22 minutes 56.87 arc (see table on page 20 of the Chapter on Mean Motions). In this way, the value of our year is 8 minutes 34.53 arc more than the pure year. In this time, the sun moves 8.361 degrees per day at a rate of 58°42".46. Therefore, if the motion of the vernal equinox is 50" .262 according to the pure year, then according to our year, the vernal equinox becomes 50".262+8".361=58".653 degrees westward. That is, if the value of the solar year is kept as per the Sūrya Siddhānta, then by considering the motion of the vernal equinox as 58".653 every year, accuracy is achieved. This proves that Munjal, Bhaskaraacharya, Ganesh, etc., considered the motion of the ecliptic as 1 art or 60 degrees, which is only 1.347 degrees more than the truth at this time. At the time when Munjal determined the motion of the ecliptic as 56.6007 degrees by direct observation, the inaccuracy must have been a little less because in 686 B.C. the pure motion of the ecliptic

९. Ball’s Spherical Astronomy pp. 187.

This is the true daily motion of the sun on the day of the Aries solstice and is derived according to the Sūrya Siddhānta.

SOURCE PAGE 260

50 .2853 — 0.0002225 X (1607 — 686) = 50 .2453 —.2043 = 50 .041 will be.

From this it is apparent that our Acharyas had calculated the motion of the ecliptic in their independent and unique way so minute that it was only 1.568 degrees more than the truth, which is very commendable considering the instruments of that time.

Now in brief, it will be considered that the motion of the ecliptic is like the motion of the pendulum as in Sūrya Siddhānta, Soma Siddhānta, Parashara Siddhānta and

the view of Mahasiddhānta or complete revolution as the view of Munjal or Bhaskaraacharya etc. is.

It is not written in the texts like Sūrya Siddhānta etc. why the motion of the ecliptic is like the motion of the pendulum. Brendand1 and perhaps on the basis of these, Vignanananda2 Swami has tried to explain this in this way:-

7. Brennand’s History of Hindu Astronomy, London 1896.

2. Shri Sūrya Siddhānta Bāngānuvāda and Commentary, Calcutta 1606 AD

SOURCE PAGE 261

Manlo U V S Krantivritt and K is its projection, that is Kadamba, the pole of the viṣuvavṛtta (equator) and the projection of the dha da ayanānta- vṛtta (solstitial colure). Consider the projection of that path along which the northern pole is moving in the opposite motion due to the ayan chalan. The two points of the Krantivritt are those where the southyana point, due to the ayan chalan, is reflected, so it is at an angle of 270 difference from 2 and 2. According to the theory, the distance of Kadamba from the northern pole is 240 from dha. Draw the tangents from the two and two points to the four and b points which intersect each other at the point h. The time taken for dha to move 360 degrees on the vṛtta is the same time taken for the point da to move from two to two, then return from two to two and then come back from two to two. Therefore, while the pole is revolving around Kadamba, the celestial circle appears to move like a pendulum on both sides of the point h.

But this does not provide any satisfaction because ancient texts prove that the vernal equinox point has been more than 27 degrees ahead of the beginning place of Ashvini. It is written in Shatpatha Brahman that the rising of the Kritikas occurs exactly in the east direction and other stars rise from the east direction, which shows that at that time the Kritikas were exactly on the equator. Nowadays, there are 270 degrees of north rise in Prayag. From this, it can be calculated that at the time when the Kritikas were on the equator, where the vernal equinox point was. The sum of the star Kritika (n – Tauri) is 36°6’ and 402’ from the direct meridian. If the angle between the ecliptic and the equator (paramepagram) is assumed to be 240, then it is easily known what the Kritikas' position was with respect to the vernal equinox in the time of Shatpatha Brahman. In figure 51, the present vernal equinox point and the p and pa equatorial circles are. And the vernal equinox point and the ch in the time of Shatpatha-Brahman.

> १. एकं द्वे त्रीणि चत्वारीति वा अन्यानि नक्षत्राण्यथैता एव भूयिष्ठा यत्कृत्ति
> कास्तद्भूमानमेवैतदुपैति तस्मात्कृत्तिका स्वादधीत ॥२॥ ‘एता ह वै प्राच्यै दिशो
> न च्यवंते सर्वाणि ह वा अन्यानि नक्षत्राणि प्राच्यै दिशश्च्यवंते तत्प्राच्यामेवास्यैतद्दिश्या
> हितौ भवतस्तस्मात् कृत्तिका स्वादधीत ॥ ३ ॥ शतपथ ब्राह्मण २. १.२.[भारतीय
> ज्योतिष शास्त्र पृष्ठ १२७ में उद्धृत]

2. Indian astrology text page 455.

SOURCE PAGE 262

It is a circle. The circle is a cluster of stars shown on the ecliptic. The distance of the star, which is facing the southward, is considered to be 402'. The negative consequence of the circle's śatapatha brāhmaṇa time is the beginning point of Ashwini and the consequence of the circle is considered to be 3606'. The fraction is approximately 22041' on the day of the Aries solstice of 1682 AD.

The negative bhoga of the Shatpatha Brahmana time of the constellation is the beginning point of Ashvini and the bhoga of the constellation is considered to be 3606'.

The precession angle is approximately 22041' on the day of the Aries solstice of 1682 AD.

In a spherical right triangle in A according to Napier's rule,

In a spherical right triangle in A an(B) = an(C) imes an(60^ ° - A) = an 402' imes an(60^ ° - 24^ °) = .0705 imes 2.2460 = .1583

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therefore va = va + ad + da = 220 ° 41 ′ + 36 ° 6 ′ + 6 ° 6 ′ = 67 ° 56 ′
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SOURCE PAGE 263

Therefore, it is proven that the vernal equinox point, where it was in the time of the Shatapatha Brahmana, is 67° 56' west at this time. However, according to the Sūrya Siddhānta, the maximum difference of the vernal equinox points should not be more than 54°. Therefore, it is proven that the Sūrya Siddhānta holds that the vernal equinox points do not move 27° east and west from the mean position.

In this regard, K.I. Mahashay says that one should not understand the Pleiades as the Pleiades star cluster but should understand the Spring Equinox, just as modern European scholars understand the Aries (First point of Aries). Believing this, the Shatapatha Brahmana does not appear as ancient as it does according to the aforementioned calculation. The aforementioned calculation proves that the time of the Shatapatha Brahmana is 4867 years or 2885 years before Vikram from today, which seems impossible to K.I. Mahashay. However, in my understanding, K.I. Mahashay is in error. Where the word Krittika appears in the aforementioned passage, it is used in the plural, which indicates that Krittika means the Pleiades star cluster in which 6 stars are visible to the naked eye. If it were understood as the Spring Equinox, it would never be used in the plural. Besides this, it is worth considering whether knowledge of the east direction can be obtained by seeing this point if it is not understood as the same star cluster but as the Spring Equinox? Can knowledge of the east direction be obtained by looking at the Aries today, or by the first star (δ Orionis) of the Apogee cluster, which is currently on the ecliptic? Without overstating this, it will be briefly stated now that according to modern physical astronomy, what is the reason behind the precession or the deviation of the Spring Equinox, which will also prove that it is a full revolution and not a motion like that of the sphere.

What result emerges from direct observation?

If the obliquity and revolution of a star at some time are compared with the obliquity and revolution of that star at another time, then it is seen that these are very

1. Memoirs of Archaeological Survey of India No. 18. Hindu Astronomy by G. R. Kaye pp. 23-24.

SOURCE PAGE 264

It changes. For example, the obliquity 1 and the declination are:-

From the first January of 1850, obliquity 1 hour 5 min 23 sec declination + 88° 30’ 46”

From the first January of 1600, obliquity 1 hour 23 minutes 0 seconds declination + 88° 46’ 53”

It can be said that the reason for the change in obliquity and declination is that the star itself is moving. The second assumption may be that the change occurs in the axes of coordinates with which the obliquity and declination are determined.

In 50 years, the revolution of the pole star increased by 16’ 4” which shows that the difference between the pole and the pole star is decreasing by approximately 16’ per year, meaning either the pole star is

moving towards the pole or the pole is moving towards the pole star. When the differences between the distances of other stars from the pole star are compared, it is seen that they are not changing as much as the difference between the pole and the pole star. The proper motion of the pole star is so subtle that it cannot cause a difference of 16’ per year. It has also been observed that in 50 years, the difference between the pole and other stars has also become very small, but their mutual difference remains as it is. From all these things, it follows that the difference between the pole and the pole star is not decreasing due to the motion of the pole star, but is decreasing due to the motion of the celestial pole.

If the pole constantly changes its position, then it is also necessary that the equator, which always remains 60 degrees away from the pole, changes its position continuously. But even while remaining due to the equator, the mean inclination of it with respect to the ecliptic always remains almost the same. This inclination moves only a few aspects in and out from the mean value. Since the sun's ultimate revolution was as in 1850 AD, it was also as in 1600 AD, therefore between the equator and the ecliptic

1. Balls’ Spherical Astronomy pp. 171. Generally people understand that the pole star appears to be in the same place and other stars revolve around it, but this is not correct. The pole star also revolves around the celestial pole, which is invisible, and is very close to it, so no difference is known.

2. See figure 40 and its description.

SOURCE PAGE 265

The angle generally remains constant. This proves that the equator moves in such a way that it always intersects the ecliptic at the same angle, and the subsolar point (spring or autumn subsolar point) moves in the opposite direction to the motion of the Earth. This proves that the kadamba (ecliptic pole) is fixed and the celestial pole (equatorial pole) revolves around it at the same distance. This motion is called the precession of equinoxes or the motion of the node. This motion occurs due to the attraction of the Sun and the Moon, so it is called luni-solar precession.

Cause of luni-solar precession

It has been proven that the change in direction along the axis on which the Earth rotates once every 24 hours is due to the movement of the ecliptic point. The reason for the deviation in the direction of the Earth's axis is that the Earth is not a perfect sphere, but is somewhat flattened at the poles and bulged at the equator. Therefore, the resultant attraction of the Sun and the Moon does not pass through its center. In Figure 52, let s be the Sun, k be the center of the Earth, dh be the axis of the Earth extended to the sky, va be the equator, a be the point where the Sun's attraction is acting, and aa be the direction of the Sun's attraction. If the Earth were a perfect sphere, a and k would be at the same point, so the plane of the equator would not tilt towards the Sun. The figure also reveals that half of the meridian, which is towards va, is near the Sun, and the other half, which is towards v, is far from the Sun. Therefore, the Sun's attraction will be weaker towards v, which results in the meridian plane tilting towards the Sun, causing the 'axis of the Earth dh to wobble'. It also appears that the plane of the equator, as it tilts, will eventually meet the plane of the ecliptic, where the Sun resides, and the axis of the Earth will start to form a right angle with the ecliptic, and the pole and the south will become one. However, this is not the case. Because the Earth is rotating on its axis at a very high speed, the angle between the planes of the equator and the ecliptic will always remain approximately the same, and the pole and the south will continue to revolve in a circle around each other.

Exactly the same thing happens in the rotation of Lattu or Furki as well. At the time

SOURCE PAGE 266

SOURCE PAGE 267

It is working in a straight line, the line of the axis of which one end is attached to the Earth's surface and the other end is revolving some distance from the vertical line, revolving around it. As long as the motion of the top is fast, it continues to revolve around a slightly inclined vertical line towards the Earth's surface in the same manner. In the time taken for one revolution around the line, it is not known how many times the top revolves on the axis of the Earth. Similarly, the Earth revolves once every 24 hours on its equatorial axis, which can be compared with the vertical line, one revolution every 25000 years.

SOURCE PAGE 268

It happens because of which the vernal equinox moves in the opposite direction of the sign of the year approximately 5003 BC. Several letters are one in figures 52 and 53 to show similarity. In figure 52, the Earth is like a top, its center of attraction is A, D is the axis of the top, and if a line is drawn parallel to the line of A and C, then it will be the same as D and C.

That which is said about the Sun also applies to the Moon. The influence of the Moon is somewhat more than that of the Sun because the Moon is very close to the Earth.

The influence of either the Sun or the Moon is greatest at the time when their north or south revolution is greatest. At the time when they are on the equinoctial motion, their influence is zero. However, the retrograde influence of the planets also affects the motion of the vernal equinox. The magnitude of the deviation of the planets is 0.11 degrees towards the west per year. The deviation does not occur in the vernal equinox due to the planets, but rather the Earth's orbit also deviates, which causes the plane of the ecliptic to waver, and the angle between the ecliptic and the equinoctial motion (ultimate inclination) is decreasing by approximately half a degree per year. However, this is within a limit, that is, it is 13 degrees less or more than the mean position.

It has been stated until now that the celestial pole of Kadamba is revolving on a circle due to the solar-lunar motion. But this is some grossness. The reason for this is that the Moon does not always remain on the ecliptic, but becomes about 5 degrees north or south of it, and its node (Rahu) occurs approximately once every 16 years.

The time taken for the sun to return to the vernal equinox point from the spring equinox point is called the tropical year. This is the same as the mean savana day, which is 365.242216.

The time taken for the Sun to return to the same point by moving from one point of the ecliptic is called the sidereal solar year. This is the same as the mean savana day of 365.256374. This is also called the true revolution of the Sun or Earth. (See Chapter on Mean Motions page 20)

According to the Sūrya Siddhānta, the revolution of the Sun (365.258756 mean savana days) is considered the solar year here. (See Chapter on Mean Motions page 20)

SOURCE PAGE 269

It completes the orbit. Due to the motion of Ayan, the way the celestial pole orbits the ecliptic circle with a diameter of 23°27' is the same as the way the pole of the lunar orbit also orbits the ecliptic circle with a diameter of 50 due to the opposite motion of Rahu, and the orbit of the pole of this lunar orbit is done by the celestial pole due to that part of Ayan motion which is due to the influence of the Moon.

Due to these two reasons, the celestial pole sometimes remains a little ahead of the mean position and sometimes behind, and its distance from Kadamba sometimes becomes a little less and sometimes a little more. Therefore, the actual path of the celestial pole is elliptical. This

The cycle of 456 changes is usually 16 years, in which Rahu completes one cycle. The slight change in the position of the celestial pole due to the Moon is called nutation. The result of nutation is that the vernal equinox point, which is determined by the solar lunar and planetary Ayan motion from its mean position, sometimes remains ahead and sometimes behind. Due to this, the inclination (peripline) between the ecliptic and the tropics also sometimes becomes a little less and sometimes a little more than its mean value. The nutation was discovered by the astronomer Bradley during the period from 1784 to 1768 A.D. (1727-1741 AD) by the continuous observation of the 'G' star (ext(extgreek(Draconis))). The clear knowledge of nutation is found in Figure 54.

SOURCE PAGE 270

Suppose that Dhi is a small circle which revolves around the mean pole of Kadamba. Taking Dhi as the center, draw an ellipse whose major axis is in the line of Kadamba and is 18.5 larger, and whose minor axis is 13.7 larger on the same small circle. As the mean pole takes the long circle with it in the same motion in the opposite direction, the true pole revolves around the perimeter of the long circle in 6768 days (in the period of Rahu's revolution).

The mean difference and the true difference of Dhi from Kadamba are. At the time when Dhi is on the long axis, the mean and true positions of the vernal equinox coincide; otherwise, the true position of the vernal equinox is some ahead or behind the mean position. Similarly, when Dhi is on the minor axis, the mean and true difference or inclination (the angle between the ecliptic and the tropics) of Kadamba and the pole coincide, otherwise, some less or more.

The annual motion of the obliquity and the order of precession is not constant, but there are some subtle changes in them. The values of these in one year are revealed by the following formulas, established on the basis of physical astronomy and higher mathematics:-

1) After 'v' years from 1600 AD, the annual mean motion (vernal equinox motion) =50.2564+0.0002225 v*(1)

and the angle between the tropics and the ecliptic (if other smaller terms are omitted)

= 23 ° 27 ′ 8 “.26 - 0 “.468 v * (2)

The formula for the correction that has to be made in the mean position of the vernal equinox due to the axial deviation is --

- 17”.235 sin(Rahu) - 1”.27 sin(2 sin Sun)*(3)

The formula for the correction that has to be made in the mean value of the angle between the ecliptic and the tropics is-

+ 6”.21 cot(Rahu) + 0”.55 cot(2 sin Sun)*

(4)

R. S. Ball's Spherical Astronomy pp. 177, 186-187.

SOURCE PAGE 271

The portion of Rahu's sine that is obtained in the auspicious time, that is, the distance of Rahu on the ecliptic from the point of the equinox, is called the sine of Rahu. The portion of sine of Rahu and the sun is called the sine of the sun.

Due to this motion, the difference of each star from the point of the vernal equinox is always increasing, due to which the sine portion of the star increases. If the portion of the star at the beginning of the year is given, the portion at any other time is known by this formula--

ta = t + 50 .26v - 17 .235sin (rahu) - 1 .27sin (2sun) (5)

where

t = the mean sine portion of the star at the beginning of the year,

v = the difference between the auspicious time and the beginning of the year (in decimal form of the year)

The second term on the right side of this formula is 50.26 because the larger it is, the greater the portion of the star will be.

Sine Rahu is in the third term, which if it is zero or 1800, then the sine of Rahu will be zero. At this time, the third term will completely disappear, that is, when the portion of sine of Rahu is zero or 1800, the third term will fly away. And when sine of Rahu is 60°, the value of the third term is -17.235 and when sine of Rahu is 270°, the value of the third term will be +17.235. Because of this, the true position of the vernal equinox point in one period of Rahu's revolution is about 6 years before and 6 years after the mean position.

Even due to the slight deviation of the axis due to the sun, it is indicated by the fourth term. Its cycle changes in 6 months because when the sine portion of the sun is zero, 600°, 1800°, 2700° and 3600°, the value of the fourth term is zero, and when the portion of the sun is 45°, 135°, 225°, 315°, then its values will be in succession—1", + 1", - 1" and + 1".

Similarly, changes continue in the order of the meridian due to the deviation of the axis. Now an example of the method of knowing the precession angle is given from these formulas:-

† This formula was determined at a conference of astrologers which took place in May of 1866 AD in Paris (see R. S. Balls' Spherical Astronomy pp. 186.)

SOURCE PAGE 272

The formula for the annual motion of the vernal equinox is given in equation (1).

But this year is the sine and our year, as considered according to Sūrya Siddhānta, is larger than it. Therefore, it should be known from the first Trāirāśika what is the motion of the precession in one year,

that is, when the precession motion is 50”.2564+0”.0002225 in 365.242216 days, what will it be in 365.2587565 days. Upon simplification, the formula in the beginning of the year 1600 AD is as follows.

50”.258676+.00022251 *

And in the beginning of the year 1622 AD, the formula for the annual precession motion comes from simplifying the above formula by replacing 'v' with 22—

50”.263571+.00022251 ¨(7)

From the beginning of January to the Aries solstice period, there are approximately 102 days or 2.763 years, so if the value of v in formula (7) is replaced by 2.763 and simplified, then the annual motion of Aries in the 1676 years is

50.263633+.000 2251 v(8)

This will be the case if the value of v is taken according to our Siddhantic year. This formula shows that the vernal equinox point moves how much behind a point on the ecliptic per year (according to our theory). However, our Siddhantic year is 0.002382067 days larger than the pure sidereal solar year† Therefore, in such time, the point of our Aries solstice moves a little forward per year. To know the magnitude of this, the difference above should be multiplied by the true daily motion of the sun during the Aries solstice period.

SOURCE PAGE 273

58'46.2 is the difference between the portions of the sun before and after the Aries solstice of 1622 AD in 'Nautical Almanac'. Therefore, the true daily motion of the sun on this day is 58'46.2. Multiplying this by the above difference gives 8'3666 of the sun's motion. Therefore, one should understand that due to the lengthening of our year, there is an increase of 8'3666 in the precession angle per year. Including this in the formula (8) will give the formula for the annual precession motion at the solstice period—

50.263633+.00022251 + 8'3666 or 58.663233+.00022251 and (6)

If one wants to know what will be the increase in the precession angle in a year, then this must be used —

58.663233 + .00022251(and(v+1)/2) † or 58.663233 + .000111255 + .000111255 +² or in short 58.663344 + .000111255 +²(10)

The increase in the precession angle that arises from this should be done according to the formula (3) of the axial deviation.

Now it is to see what was the precession angle in the Aries solstice period of 1676 AD. To do this, it is sufficient to know what is the portion of the sun's declination in the Aries solstice period, the knowledge of which is ordered in the next 11th and 17-18th verses.

In this chapter, it will be explained further that how the sine of the sun can be known by measuring the shadow of the gnomon and how the precession angle can be known from it. Nowadays, this work can be done very precisely with theodolites.

By inserting 1, 2, 3,...v in place of w in the second term of the sutra(6), and adding them all, we get .00022251 x(v(v+1) / 2). This is like adding the numbers of an arithmetical progression.

SOURCE PAGE 274

is. If the arc of the sun is known at the same time as the Aries solstice, then this is the precession angle. However, in the absence of the theodolites, this fact can also be known from the nautical almanac. Therefore, from this, the arc of the sun at the time of the Aries solstice of 1676 Vikram or 1622 AD is calculated. According to the Sūrya Siddhānta, the Aries solstice of 1676 Vikram occurred on Thursday, April 13th, after 6 o'clock in the morning in Ujjain, at 18 degrees 47 minutes 12 arcminutes of the ecliptic. Kashi is 72 arcminutes 50 arcminutes before Ujjain. Therefore, the Aries solstice in Kashi will be at 20 arcminutes 0 arcminutes and 2 arcminutes after 6 o'clock in Kashi. However, Kashi is 83°3‘4” or 13 arcminutes 50 arcminutes 31 arcminutes before Greenwich. Therefore, at the time the Aries solstice occurred in Kashi, it must have been 6 arcminutes 6 arcminutes 31 arcminutes after 6 o'clock in Greenwich.

The portion of the sun at the mean meridian on April 12 this year*

Therefore, in 60 hours, the true motion of the sun was 58^(°)46".

On April 13, the Greenwich meridian time was 6 o'clock plus 6 hours or 15 minutes, and the Aries equinox occurred at 6 o'clock 6 minutes 31 seconds PM, so the time from the equinox by 15 minutes—6 minutes 6 seconds 31 seconds PM = 8 hours 50 minutes 26 seconds PM, the meridian of April 13 occurred at which the portion of the sun was 22046°17”.7. It is revealed from this that the portion of the sun in the equinox period will be less than this. But the true motion of the sun is 58’46”.2, so in 8 hours 50 minutes 26 seconds PM the sun

```
58' 46".2 X 8h50p26viple60h
```

or 8'36". 606 would have run.

† The difference in longitude of Kashi (from Greenwich) is 83°3'4" East and Ujjain is 75°46'6" East. The difference between these two is 7016'58", which is similar to 72 lunar months. Therefore, this is the difference in longitude from Ujjain to Kashi.

* Nautical almanac for 1922 pp. 40.

SOURCE PAGE 275

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Sūrya Siddhānta
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By adding whatever increase comes according to the formula (10) to this, the mean fraction of the ecliptic period of another month will be obtained. If the angular deviation is done with the help of formula (3), then the clear fraction of the ecliptic period will be obtained. If you want to know the fraction of any other time besides the ecliptic period, then if you add the number of days that have passed since the ecliptic to the mean fraction of the ecliptic, in which days have passed (when the speed of motion is approximately 58.66* times per year), and do the angular deviation, then the clear fraction of that time will be known.

Chapter on Mean Motions

Chapter on True Positions

Sūrya Siddhānta

On Monday, the solstice of Aries on May 1982, 6 hours and 6 minutes 34 seconds 36 minutes after the mean time in Kashi, 34 minutes 36 seconds of the equinoctial shadow were cast. There are 165 Sawan days from May 5 to Kartik Shukla 8, which is the same as 05336 solar years. In this time, at a rate of 58”.66 per year, the mean precession angle 58”.66 X.5336 = 31”.32 will increase. Therefore, on Kartik Shukla 8, the mean precession angle will be 220.41’ 5”.41.

To know the inclination for the correction, it is necessary to know the portion of Rahu and the portion of the Sun of the sign of Saturn on Kartik Shukla 8. On this day, the declination of Rahu

* If you want to know more accurately, you should determine the precession motion of that year from Sutra(6).

SOURCE PAGE 276

The portion is 3rd 7013'8". Add 22041'5" to know the portion of sine. Then it is 3rd 26°54' 13" or in gross form 3rd 26°54' or 116°54". This is the sine portion of Rahu.

Similarly, the sine portion of the sun should be known. The sine portion of the sun at midnight of the full moon of Kartik Shukla 8 will be 186°8'10", so the sine portion of the sun at 6 o'clock in the morning will be 188°23" in gross form. Adding 22041'5" to this, its sine portion is 21104" in gross form. Therefore, according to Sutra (3) on this day, the axial precession occurred

```
= -17''.235  sin 116^  54' - 1'.27  sin 2  imes 211^  4' = -17''.235  sin 6006' - 1''.27  sin (360^
+ 6208') = -17''.235  imes.8666 - 1''.27  imes.8841 = -14''.64 - 1''.12 = -16''.06
```

When added to the mean precession 22041'5".41, the true precession of the full moon of Kartik Shukla 8 morning is 22040'46".35.

Ketkarji has given a table in his astronomical calculations from which the appropriate precession is 6' or 7' less. The first reason for this is that Ketkarji considered the beginning of the Aries transit at the time when the star Chitra remains at 180 on the sun, whereas according to Sūrya Siddhānta, the Aries transit happens 7 hours earlier. The second reason is that Ketkarji used the pure sidereal year and the calculation was done considering the year according to Sūrya Siddhānta.

Examining the precession angle by sighting

> स्फुटदृक्तुल्यतां गच्छेदयने विषुवद्वये ।
> प्राक्चक्रं चलितं हीने छायार्कात्करणागते ॥११॥
> अन्तरांशैरथोंद्धृत्य पश्चाच्छेषैस्तथाऽधिके ।

Translation—(11) On the days of Uttarayana and Dakshinayana or the day of the equinox, it can be easily seen which direction the constellation has moved. If the portion of the sun from the shadow of the sun (the method of which is explained in the following 17-19 verses) is less than the portion of the true sun, then it should be understood that the difference between these two is the distance the constellation circle or the beginning point of Ashvini has moved to the east

SOURCE PAGE 277

That is, it is before the vernal equinox point. But if it is more, then it should be considered that the constellation-circle has moved westward by the same amount.

The Scientific Commentary—the portion of the sun's shadow that comes from the shadow is the portion of the sun from the vernal equinox point (the sidereal portion), and the portion that comes from mathematics, the method of which is explained in the Chapter on True Positions, is from the beginning of Ashvini. Therefore, the difference between these two is the actual precession angle. This proves that the precession angle should also be tested by the quadrature. The opinion of the Sūrya Siddhānta author is that the beginning of Ashvini, which is also considered the beginning of the ecliptic, can move up to 27^ o before or 27^ o west of the vernal equinox point. No more than this. Many ancient teachers also have this opinion. But some teachers hold a different opinion, which has been discussed earlier. Ancient sentences also prove that the vernal equinox point has been more than 27^ o before the beginning of Ashvini of today. What is proven by physical optics has already been stated above. But the direct proof of its truth will be found when the vernal equinox point actually moves more than 27^ o west of the beginning of Ashvini. According to the Sūrya Siddhānta, the precession angle of 27^ o will be completed 374 years from now in the Kali Yuga of 5400 or in the Vikram Samvat of 2356. But its proof will be found by quadrature even earlier because in the period of the Aries solstice of 1682, the mean precession angle will be 22040^ o 34^ q .06, which is only about 4.16^ q 26^ q less than 27^ o. By setting up an equation by making the formula (10) similar to this and finding the value of 'v', the number of years will come out in which the precession angle will increase by this much. Now

Sūrya Siddhānta

See the quotation from the commentary of the Śatpatra Brāhmaṇa in the previous shlokas and the related calculations.

SOURCE PAGE 278

Therefore it is apparent that it will only be proven after two or four years from 1682+266 that the vernal equinox is a full revolution or motion.

If it is direct that the vernal equinox may lag due to the full revolution, then there will be a great need for amendment in the method of Indian calendar-making and in determining dates and festivals.

The First Method of Knowing the Equinoctial Shadow

> एवं विषुवति छाया स्वदेशे या दिनार्धजा ।
> दक्षिणोत्तरयोरेव सा तत्र विषुवत्प्रभा ॥१२॥

Translation—(12) In this way, on the day the Sun is on the equinoctial circle, the shadow cast by the gnomon on the north-south line of a place at noon is the equinoctial influence or equinoctial shadow of that place.

Scientific Commentary—Much has been written regarding the equinoctial shadow on pages 204-205 and in the commentary of the seventh verse of this chapter. On the 205th page, it is stated that when the tangent of a place's latitude is divided by the gnomon, it gives the equinoctial shadow of that place. Therefore, it is necessary to state here that the value of the equinoctial shadow according to this verse is

gross. Because on the day of the vernal equinox of Aries or Libra, when the shadow of the gnomon is measured at noon and the equinoctial shadow is known, the Sun is not exactly on the vernal equinox at noon but is some ahead or behind. The conjunction of the Sun coming exactly on the vernal equinox at noon happens after many years. On this day, the precession of the Sun changes by approximately one zodiac sign per hour. Therefore, one should know the precession of the Sun at noon by knowing the pure time calculation of the vernal equinox of Aries or Libra and after performing the correction, know the pure equinoctial shadow. The method of performing the correction will be shown in the next 14-15 verses.

The First Method of Knowing Latitude

> शंकुच्छायाहते त्रिज्ये विषुवत्कर्णभाजिते ।
> लम्बाक्षज्ये तयोश्चापे लम्बाक्षौ दक्षिणौ सदा ॥१३॥

SOURCE PAGE 279

Translation -(13) By multiplying the gnomon and its shadow (here equinoctial shadow) separately by the radius, i.e., 3438, and then dividing each product by the obliquity, the latitudes and latitudes will come, which will be the arc and latitude in order. (In the meridian) These are always south.

Scientific Commentary—The essence of this verse is:-

```
sine of colatitude = gnomon x radius / obliquity
```

```
sine of latitude = equinoctial shadow X radius / obliquity
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At noon on the day of the transit of Aries or Libra, the shadow hypotenuse of the 12-inch gnomon is the obliquity, equinoctial shadow or latitude. In picture 41 of page 204, 'k' is the obliquity. Therefore

```
obliquity = sqrt(equinoctial shadow^2 + gnomon^2)
```

Page 55, Figure 7 and Page 56, Figure 10 state that what are the latitude and longitude of a place. From these figures it also appears that the latitude and longitude of a place together are equal to 600. In Page 204, it is also said regarding Figure 41 that the angle between the meridian and the gnomon is the latitude. Therefore, it is proven that the angle between the meridian and the equinoctial shadow is the longitude, because both <kha ka ga and <ka ga kha are complementary angles.

```
Therefore sin of colatitude × sin of latitude = gnomon
according to our principles (in arts) = gnomon × radius
```

The sine of the longitude is also called the co-latitude of the latitude because the sum of longitude and latitude is 60°. Similarly, the sine of the latitude or co-latitude

```
= equinoctial shadow ( in decimal fractions )   or  equinoctial shadow × radius of the circle ( in degrees )
```

SOURCE PAGE 280

It is also stated in pictures 42-43 that the height of the pole from the uttara point of the horizon is equal to the latitude. This may cause readers to doubt which definition of latitude is correct. Therefore, it should be decided here that these three definitions of latitude are the same.

From pictures 7 and 10 it is clear that the angular difference of any place from the equator on the north-south line is its latitude, and the angular difference of the place from the pole on the north-south line is its longitude. If the plane of the equator is extended towards the sky, this is called the prime vertical circle, and if the plane of the north-south line is extended in the sky, it is called the meridian circle. Similarly, when the line joining any place from the center of the Earth is raised, the point it reaches in the sky is called the astrolabe of that place. Therefore, it is proven that the difference between the astrolabe of a place and the prime vertical circle is also a latitude, and the difference between the astrolabe and the celestial pole on the meridian is the longitude. Therefore, the latitudes of the kha dhanu sthāna and the longitudes of the kha dha dhanu sthāna in pictures 40 and 43 were obtained. However, the dha vi dhanu or u kha dhanu is equal to 60 degrees. Therefore, if the common arc is removed from each, then the remaining kha vi and u dha will be equal, that is, the difference between the prime vertical circle and the astrolabe is the same as the difference between the uttara point of the horizon and the northern celestial pole. Similarly, it can also be proven that the dha kha dhanu is equal to the vi dhanu, that is, the height of the prime vertical circle from the south point of the horizon is also equal to the longitude. It is also clear that the prime vertical circle of any place in the northern hemisphere is always to the south of the prime vertical circle, so the kha vi remains the latitude and the vi da remains the longitude in the northern hemisphere.

Now it remains to prove that the difference between the gnomon and the subsolar point is equal to the latitude.

It should be noted that in figure 55, the gnomon is shown much larger than the Earth.

Sūrya Siddhānta + Scientific Commentary + Chapter on Mean Motions + Chapter on True Positions + Chapter on the Three Problems

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SOURCE PAGE 281

u v d = latitude of the Earth north-south line d = southern pole c = point of the equator r = on the equator the position of the sun sh = shadow hypotenuse g = gnomon x = position of the place cross x = equinoctial shadow bh = center of the Earth

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SOURCE PAGE 282

But the distance between the Sun and the Earth's center is approximately 6 crore 26 lakh miles and the Earth's semi-diameter is b_s or b_h (because s h = 12 inches) 4000 miles. Therefore <s r b is such a small angle that it can be considered zero (in reality, this angle is about 6 arcminutes). Therefore

<s s = < r b = < v b = latitude

That is, the angle between the gnomon and the subsolar point is equal to the latitude. Therefore, the angle between the equinoctial shadow and the subsolar point, which is the complement of the former angle, is equal to the meridian.

Example—the equinoctial shadow of Prayag is 5 angulas 41 vyangulas or 5.68 angulas, then tell the latitude of Prayag.

Subsolar point of Prayag

Sūrya Siddhānta

Scientific Commentary

Chapter on Mean Motions

Chapter on True Positions

> मध्यच्छाया भुजस्तेन गुणिता त्रिभमौर्विका ।
> तत्कर्णाप्तधनुर्लिप्ता नतास्ता दक्षिणे भुजे ॥१४॥
> उत्तराश्चोत्तरे याम्यास्तत्सूर्यक्रान्तिलिप्तिका ।
> दिग्भेदे मिश्रितास्साम्ये विश्लिष्टाश्चाक्षलिप्तिकाः ॥१५॥

SOURCE PAGE 283

(14) To find the meridian altitude of the sun, multiply the shadow of a day's noon, which is also called the radius, by the shadow hypotenuse of the meridian and divide, and make an arc of the quotient, then the meridian altitude of the sun will be found. If the shadow is to the south, then (15) it will be the northern altitude, and if the shadow is to the north, then the southern altitude. If the altitudes and the directions of the sun's declination are different, and their sum is equal to one, then it will be the difference in latitude.

Sūrya Siddhānta - The circle drawn perpendicular to the horizon from a point (star, center of the sun, etc.) to which the science extends is called the Vertical Circle. The difference between that point and the x-coordinate from the x-coordinate is called the zenith distance of that point, and the difference from the horizon is called the altitude of that point. The prime vertical circle is also a vertical circle. But its special feature is that it is at the points east and west of the horizon. The meridian circle is also a vertical circle at the north and south points. Therefore, when the sun is on the meridian circle, the difference of the x-coordinate from it is its meridian altitude. If the sun is also on the equator circle, then this will be the same as the latitude. If the sun is not on the equator circle, then it will be either north or south of the equator circle. The difference of the sun from the equator circle at the meridian time is the sun's declination, which is called the north or south declination according to whether the sun is north or south of the sun's equator. Similarly, the sun can also be north or south of the x-coordinate. If the sun is north of the x-coordinate, the shadow will be towards the south, and the difference between the sun and the x-coordinate will be called the northern altitude. But if the sun is south of the x-coordinate, the shadow will be towards the north, and the altitude of the sun will be south. Figure 56 clearly shows how the meridian altitude and declination of the sun can be known:

The meridian circle, the northern celestial pole, the cross, the equinoctial circle, and the meridian circle have a common point, and r, ra, and ri are three different positions of the sun. r is south of the sun's equinoctial circle, so at this time the sun's south transit is r, but the sun's north transits are ra and ri. Similarly, r and ra are the southern parts and ri is the northern part of the sun. The latitude of the location of x is. It is apparent from the figure that

SOURCE PAGE 284

vx = xr — vr (1) = khara + vra (2) = vari — khari (3)

In equation (1), the declination and the sine of declination are both south, and in equation (3), the declination and the sine of declination are both north. However, in equation (3), the declination is south and the sine of declination is north. It is revealed that when the directions of both the declination and the sine of declination are the same, their difference and if they are different, by adding them, the latitude can be known. If the declination, the sine of declination, and the latitude are considered, their relationship will be revealed by this equation—

n ± k = a

Here the sign of plus will be written when the directions of both n and k are different, and the sign of minus will be written when their directions are the same.

In the two above shlokas it is stated that how the declination is known by measuring the meridian shadow of the gnomon. It is clear that the angle between the gnomon and the shadow hypotenuse is the declination. Therefore

SOURCE PAGE 285

Nataanshja = shadow X radius / shadow hypotenuse [according to Siddhāntīrikē]

From the sine of Nataansh, by adding or subtracting as needed, the latitude can be obtained according to verse 28 of the Chapter on True Positions of the sine of Nataansh in the revolution of the Sun.

Example—If the noon shadow of the gnomon in Prayag is 2.12 angulas on a certain day, and the latitude of Prayag is to be determined. The noon shadow hypotenuse of Prayag

= gnomon 2 + shadow 2 =122 + (2.12) 2 =148.46 =12.16 angulas

∴ Nataanshja = shadow × radiusshadow hypotenuse = 2.12 × 343812.19 = 568 ′

Therefore nataansh = 60° 51 ′

This latitude is towards the south and the revolution of the sun is north. Therefore, the sum of both will be the latitude of Prayag.

Therefore this way Prayag's latitude = 6 ° 51 ′ + 15 ° 25 ′ = 25 ° 16 ′

> दोनों रीतियों से निकाले गये अक्षांशों में कुछ अन्तर है । इसका कारण प्रत्यक्ष है । छाया की नाप स्थूल होती है जिसका कारण पहले बतलाया जा चुका है । यदि छाया छोटी हो तो अशुद्धि और भी बढ़ जाती है ।

> पलभा जानने की दूसरी रीति यदि अक्षांश ज्ञात हो—
> तज्ज्याऽक्षज्याऽथ तद्वर्गं प्रोज्झ्य त्रिज्याकृतेः पदम् ।
> लम्बज्याऽक्षगुणोऽकंधनः पलभाप्तोऽवलम्बकः ॥१६॥

Translation—(16) Knowing the latitude mentioned above, make the sine of latitude and find the square root of the remainder after subtracting the square of the sine of latitude from the square of the radius

SOURCE PAGE 286

Then the colatitude will be obtained. Dividing the sine of latitude by multiplying the sine of colatitude by 12 gives the equinoctial shadow.

Scientific Commentary—The first method to know the equinoctial shadow is used on the day of the transit of Aries or Libra. By the second method, the equinoctial shadow can be calculated by knowing the latitude by knowing the transit and declination of the sun for a given day.

radius^2 - sine of latitude^2 = sine of colatitude and sine of latitude × 12 sine of colatitude = equinoctial shadow

Proof—In Figure 10 page 56, the sum of colatitude, latitude, radius, and angle is a right angle, so if the sine of latitude is known, then

$Sabhā^2 = Sabh^2 - Bhabhā^2$ therefore sin of colatitude = $\sqrt{}$ $radius^2 - sin\ of\ latitude^2$

In Figure 41 on page 204, kh shadow, kx g gnomon, kh latitude and kx longitude are, therefore according to the current method

sine of latitude = g / g

sine of colatitude = kx / g

- sine of latitude = g imes g imes kx imes kg imes kx imes kg = equinoctial shadow imes gnomon = equinoctial shadow imes 12 arcminutes

∴ equinoctial shadow = sine of latitude imes 12 / sine of colatitude (1)

SOURCE PAGE 287

In this manner, a lot of multiplication and division is required to find the value of the equinoctial shadow. Therefore, if a table of tangents is made, this work can be done easily because latitude and longitude are complementary angles, therefore

sine of latitude - - - - - - - - - - - - - - - - - - - = latitude, sine of colatitude

Equinoctial shadow = 12 X latitude tangent this matter is also shown in page 205. (2)

Example—if the latitude of Prayag is 25^°25', what will be the equinoctial shadow of Prayag?

(1) According to the method of Sūrya Siddhānta, sin of latitude = 25° 25’ the sin = 1 × 74’ therefore sin of colatitude = √(r^2 - latitude^2) = √(34:8^2 - 1474^2) = √((3438 + 1474)(3438 - 1474)) = √(4612 × 1664) = 3106’ equinoctial shadow = latitude × 12 / sin of colatitude = 1474 × 12 / 3106 = 5.66 inches (2) According to the new method— equinoctial shadow = 12 × latitude tangent = 12 × spread × 25° 25’ = 12 × 0.4752 inches = 5.7024 inches = 5.7 inches

Instead of tangent, sine is written for simplicity, just as cosine is written for cotangent.

SOURCE PAGE 288

To measure the revolution of the Sun, know the sine portion—

> स्वाक्षार्कनतभागानां दिक्साम्येऽन्तरमन्यथा ।
> दिग्भेदेऽपक्रमश्शेषः तस्य ज्या त्रिज्यया हता ॥१७॥
> परमापक्रमज्याप्तचापं मेषादिगे रविः ।
> कवर्यादौ प्रोज्झय चक्रार्धात्तुलादौ भार्धसंयुतात् ॥१८॥
> मृगादौ प्रोज्झय भगणात् मध्याह्नार्कस्फुटो भवेत् ।
> तन्मान्दमसकृद्वामं फलं मध्यो दिवाकरः ।।१९।।

(17) If the latitude and the declination of the mean solar time of its place are in the same direction, then find their difference and if they are in different directions, then add them. Whatever comes is the mean solar revolution of the sun. Multiply its sine by the radius (18) and divide by the sine of the ultimate revolution of the sun and make an arc of the arc of the latitude. If the sun is in three zodiac signs like Aries, then this (arc) will be the true sine portion of the mean solar time. If the sun is in three zodiac signs like Taurus, then subtracting this arc from 6 signs will yield the true sine portion of the mean solar time. If the sun is in three zodiac signs like Libra, then adding this arc to 6 signs will yield the true sine portion of the mean solar time. And if the sun is in three zodiac signs like Capricorn, then subtracting this arc from 12 signs will yield the true sine portion of the sun. By performing the inverse operation several times in this true sine portion, the mean sine portion will be obtained.

Scientific Commentary - in 14-15 shlokas, the method of knowing the latitude by adding or subtracting the meridianal portion and the declination of the sun is explained. In the 17th shloka, the method of finding the revolution by knowing the latitude and the declination is given. Therefore, this is the second form of the first method, and just as addition or subtraction is required there, so is it here too. The reason for this can also be understood from the three equations in relation to figure 56.

The rule of addition and subtraction will be easily understood from this equation -

a ± n = k

In which 'a', 'n', and 'k' indicate latitude, declination, and revolution respectively, the sign of the number will be written at the time when the directions of latitude and declination are different, otherwise the sign of minus will be used. One thing must be kept in mind here. People generally understand this, and nowadays this practice is also that the north

SOURCE PAGE 289

The direction of latitude in a circle is understood as North, but in this rule, its direction is understood as South because in the northern circle, the direction of the equator from the cross is South.

When the ecliptic is known, the portion of the sun can be known from the 28th verse of the Chapter on True Positions; because it is stated (see page 122 figure 25) that

Jya (vs) × 13673438 = Jya (sp) Therefore Jya (vs) = Jya (sp) × 34381367 (7)

Where sin (portion of the sun) is the sine of the portion of the sun, ecliptic of the sun is the ecliptic of the sun, and 1367 is the ultimate ecliptic of the sun. This is the form of the last part of the 17th verse and the first half of the 18th verse.

If the value of sine is to be used in decimal fraction according to the current custom, then an even simpler form will be—

ja (sp) ja (vs) = ja (svp) (2)

Now it has been well proven that the revolution of the sun is not 240 but its mean value is approximately 23°27’ at this time and it is decreasing by about half a degree every year. Therefore, if one wants to know the portion of the sun's revolution nowadays, one should calculate by considering it as equal to 23°27’.

Example—on one day, the revolution of the sun at noon is 16017’ south and this is in the sign of Makara, then tell the true sine portion.

```
sin (portion) = sin (revolution) / sin (great revolution) = sin16° 17' / sin23° 27' = .2804 / .3676 = .7047
```

```
∴ portion = 440, 48 ′
```

SOURCE PAGE 290

Since the Sun is in the latitude of Makara, this portion should be the sine portion of the Sun after subtracting 12 degrees or 3600. Therefore, this is the sine portion of the Sun for this day

```
= 360 ° - 44 ° 48 ′ = 315 ° 12 ′
```

It is also clear from the picture 36 of page 200 and experience that the time it takes for the Sun to go from the vernal equinox to the south point, that is, to go from the sine Aries to three signs, the corresponding time in which its northern revolution increases from zero to 23°27’. When the Sun goes from the south point (from the beginning of the sine Cancer) to the autumn equinox, its northern revolution decreases and becomes zero from 23°27’. The solar declination increases from zero to 23°27’ when the Sun goes from the autumn equinox, that is, the sine Libra, to the post meridian (until the beginning of the sine Makara), and then it decreases and becomes zero from the beginning of the sine Makara to the vernal equinox.

The rule given above for finding the portion only tells that how far the sun is from the vernal or autumnal equinox. If the sun is between three zodiac signs, i.e., between Aries and Pisces, then the obtained portion is the distance of the sun from the vernal equinox, so this is the sine portion. If the sun is within three signs from the beginning of Cancer, then the obtained portion is the distance of the sun in the opposite direction from the autumnal equinox. But the autumnal equinox is 6 signs away from Aries, so the obtained portion has to be subtracted from 6 signs, then the sine portion of the sun from the vernal equinox is obtained. If the sun is between three signs from Libra, then the obtained portion is

the distance of the sun in the opposite direction from the autumnal equinox, so 6 signs have to be added, then the sine portion of the sun from the vernal equinox is obtained. And if the sun is between three signs from Capricorn, then the obtained portion is the distance of the sun in the opposite direction from the vernal equinox. Therefore, by subtracting this portion from 12 signs, the distance of the sun in the opposite direction from the vernal equinox (portion) is obtained.

It is stated above that the sun's apogee decreases by about half in the year. It would be good to give the formula so that the apogee can be easily known at any time.

SOURCE PAGE 291

Sūrya Siddhānta

Scientific Commentary

Chapter on Mean Motions

Chapter on True Positions

Chapter on the Three Problems

the late

As in Chapter on True Positions, when the correction of the apogee is done, the true Sun appears from the mean Sun; similarly, in this manner, when the inverse correction of the apogee is done in the true Sun that has come, the mean Sun comes. However, it is stated in the Scientific Commentary of the Chapter on True Positions that the true Sun does not appear by merely giving the correction of the apogee in the mean Sun by scientific method, therefore it is proven that the mean Sun cannot come by doing the inverse correction of the apogee in the true Sun that appears according to the method of the 17-16 shlokas of this chapter. Therefore, it is said to do the unperformed action, that is, consider the mean Sun that comes by giving the inverse correction of the apogee once and then do the correction of the apogee. Do the correction of the apogee in that mean Sun. By doing this many times, the mean Sun will come.

Knowing the shadow and shadow hypotenuse of the midday (from the Sun's declination and latitude)—

> स्वाक्षार्कापक्रमयुतिः दिवसाम्येऽन्तरमन्यथा ।
>
> शेषं नतांशस्सूर्यस्य तद्बाहुज्याऽथ कोटिजा ॥२०॥

SOURCE PAGE 292

> शंकुमानांगुलाभ्यस्ते भुजत्रिज्ये यथाक्रमम् ।
> कोटिज्यया विभज्याऽऽप्ते छायाकर्णावहर्दले ॥२१॥

Translation—(20) If the latitude and the declination of the sun are in the same direction, add them and if they are in different directions, subtract them. Whatever is obtained is the apparent meridianal portion of the sun. Make its sine of latitude and sine of colatitude. (21) The angular value of the gnomon, i.e., 12, when multiplied by the sine of latitude and divided by the sine of colatitude, will give the shadow at noon, and when the gnomon is multiplied by the radius and divided by the sine of colatitude, the shadow hypotenuse at noon will be found.

will be. Scientific Commentary—This is the opposite of the 14th verse. The simple form of these two verses is— (1)

```
a ± k = n
```

shadow = sine (n) x 12 / sine (n) shadow hypotenuse = sine X 12 / sine (n) (2) (3)

Where a is the latitude, k is the mean motion of the sun, and n is the apparent meridianal portion of the sun. In equation (1), the sign of plus should be written when the directions of latitude and motion are the same, and the sign of minus when their directions are different. The direction of latitude is always considered south in the Northern Hemisphere, which has been explained before.

whose explanation has already been done. It has been stated in the commentary of the 15th verse that the angle between the gnomon and the shadow hypotenuse is called the natajyā. Therefore

not nataṁjya = chāyā / chāyā karan vartamana parthana sura (k)

not nataṁkoṭjya = shanku / chāyā karan (kha)

```
∴ chāyā = nataṁjya imes chāyākaran = nataṁjya imes shankuntaṁṭja : = jya ( n) imes 12 = kojya ( n)
```

SOURCE PAGE 293

If the table of the tangent is taken, then its simple form will be-

```
Shadow = 12 X sphere ( n ) ( 4)
```

It is proved by the above equation (b) that

```
Shadow hypotenuse = gnomon / ntaanshkotijyacurrentas
```

If the value of the ntaanshkotijya is written according to the Indian system, then

```
Shadow hypotenuse = gnomon X radius / ntaanshkotijya
```

```
or shadow hypotenuse = 12 X radius / sine ( n )
```

Example—If the sun's north declination is 15°25’ and Prayag's latitude is 25°25’ on a certain day, what will be the shadow and shadow hypotenuse in Prayag at noon on this day? [See example of 14-15 shlokas]

Prayag is in the northern hemisphere, so the direction of its latitude is south according to the rules of the shlokas and the direction of declination is north, so the difference between these two will be the portion of the sun.

```
∴ n = 25 ° 25 ′ - 15 ° 25 ′ = 10 °
```

Shadow = sine (n) × 12 / cosecant (n) = sine10° × 12 / cosecant10° = 567 × 12 / 3384 = 2.12 inches

shadow hypotenuse = 12 × arcsecond = 12 × 34383384 = 12.16 arcminutes

SOURCE PAGE 294

(2) In a new manner: —

Shadow = 12 x spre100 = 12 x.1763 inches = 2.1156 = 2 - 12 inches

Shadow hypotenuse = 12 imes sine100 = 12.6848 = 12.16 inches

Knowing the declination and the shadow of some auspicious time of the Sun to know the direction—

> क्रान्तिज्या विषुवत्कर्णहताऽऽप्ता शंकुजीवया ।
>
> अकाग्रा सेऽष्टकर्णघ्ना मध्यकर्णोद्धृता स्वका ॥२२॥
>
> विषुवद्भायुताऽर्काग्रा याम्ये स्यादुत्तरो भुजः ।
>
> विषुवद्भा विशोध्योदग्गोले स्याद्बाहुरुत्तरः ॥२३॥
>
> विपर्ययाद् भुजो याम्यो भवेत्पूर्वापरान्तरे ।
>
> माध्याह्निके भुजो नित्यं छाया माध्याह्निकी मता ॥२४॥

Translation—(22) The rising time of the Sun is obtained by multiplying the sine of the declination of the Sun by the obliquity of the ecliptic and dividing by the gnomon, which is 12. It is obtained by multiplying this by the shadow hypotenuse of the auspicious time and dividing by the mean motion, which is the radius. (23) If the Sun is in the southern hemisphere, that is, if the declination of the Sun is south, adding the equinoctial shadow to the hypotenuse and subtracting the equinoctial shadow from the hypotenuse gives the southern side. (24) If the Sun is in the northern hemisphere and the equinoctial shadow is smaller than the hypotenuse, performing the opposite action, that is, subtracting the equinoctial shadow from the hypotenuse, gives the northern side. The shadow at noon is always the meridian side.

Scientific Commentary—in the commentary of this authority, the discussion of agra and agrajya has been made in the 5th and 7th shlokas. In the 7th shloka, the definition of agra is given, "The difference between the line of the subsolar point and the line of the equinoctial line from the tip of the desired shadow is called agra." In the description of Figure 85, 46, six agrajya and six bhuj are described. However, there is a fear that calling six as agra or agrajya will cause much confusion, so six which in the 7th shloka is called agra and in the scientific commentary by me

SOURCE PAGE 295

Sūrya Siddhānta is written, or as Bhaskaraacharya writes, it would be more appropriate to say Karnavrittāgra. Agra should only be understood as the angle by which the Sun, planet, or star differs

from the east or west point on the horizon circle. In Figure 43, the position of the rising planet is at point G of the horizon circle, and the east point is P, so the rising agra of the planet is P dhanu. Similarly, the setting agra of the planet is P dhanu because P and G are the positions of the planet at the time of setting and the west point. If a perpendicular is dropped from points G or P to the east-west line, its value will be treated as the rising agra. In Figure 42, P is a straight line, the agra of the rising planet. Except for the time of rising, in any other time, the difference between the point where the Sun will fall on the vertical circle and the east or west point will be called the istakalik agra.

In the 22nd verse, the apragjya of the rising sun, the shadow hypotenuse of the time of the ecliptic, and the radius of the mean motion have been used, so they should be taken into account. Some translators have considered the mean motion to be the shadow hypotenuse of noon, but this is an illusion. Rangnath ji has proven the radius in this way, "The mean motion is the half of the diameter of the chord."* The term chord is also used in Chapter on Mean Motions, verse 56. This same rule is repeated in the next 27th verse of this authority, where the radius is used instead of the mean motion. Therefore, the meaning of the mean motion is nothing other than the radius. The essence of this verse is:-

agrajya = kartanja × vishrutakkrana / 12 (7)

Gravitational constant = G × r^2 = G × r × R^2 where R is the radius of the shadow and r is the radius of the shadow. Then R = r × cos latitude × meridian angle [show 13 and latitude] so gravitational constant = G × r × cos latitude (2) (3)

SOURCE PAGE 296

The value of the sine of the meridian of the rising or setting of the sun can be obtained from the equation (3), therefore this is good. In this way

The meridian arc = sine / sine of the shadow x gnomon (4)

It is clear from figures 45, 46, how the hypotenuse of the shadow is found by adding or subtracting the equinoctial shadow in the meridian arc. When the sun is in the south latitude, that is when the sun's declination is south, the equinoctial shadow will always be added to the meridian arc and the sum will be the north side because in such a situation, the tip of the shadow is always north of the viṣuvadbhāgragā line (see figure 45), if the sun is in the north latitude, that is when the declination is north, then as long as the sun remains north of the prime vertical circle, the shadow will remain south of the east-west line, so the hypotenuse of the meridian arc will be greater than the equinoctial shadow. In this situation, subtracting the equinoctial shadow from the meridian arc will give the hypotenuse (see figure 46). But if the sun is south of the prime vertical circle, then the shadow will be between the east-west line and the viṣuvadbhāgragā line. In this situation, the equinoctial shadow will be greater than the meridian arc and the second will have to be subtracted from the first.

Meridian arc leftrightarrow equinoctial shadow = hypotenuse (y)

The positive sign will be taken when the sun's declination is south, that is when the sun is in the six zodiac signs like Libra, and the negative sign will be taken when the sun's declination is north, that is when the sun is in the six zodiac signs like Aries, etc. In the previous situation, if the shadow is south of the east-west line, then the hypotenuse will be in the south and if the shadow is north of the east-west line, then the hypotenuse will be north and the meridian arc will have to be subtracted from the equinoctial shadow.

Nowadays, according to the rules of spherical trigonometry, the equation (5) is determined as follows:-

A triangle dh kh r is a spherical triangle, therefore*

kojya dh kh r = kojya (dh r) - kojya (kh r) imes kojya (dh kh) jya (kh r) imes jya (dh kh) (k)

or kojya (digansh) = kojya (dhruvantar) - kojya (nataansh) imes kojyalambansh / jya (nataansh) imes jya (lambansh) (kh)

*See Taddhunter and Lehme's spherical trigonometry page 21.

SOURCE PAGE 297

Introduction to the Figure

The meridian circle is the circle of the horizon. The north circle is the circle of the tropics. The north-south circle is the circle of the meridian. The east point is the eastern point. The north point is the northern point, the south point is the southern point. The cross is the meridian circle and the circle of the tropics. The general point of the circle is the sun's meridian circle when the revolution is in the north, east, and south, which are the three positions of the sun. The three vertical circles are the ascensions of the sun, the setting of the sun, and the transit of the sun. The directions of the sun are the east, east, and west. The zenith of the sun is the meridian, the east, and the west.

SOURCE PAGE 298

But the difference of latitude, the difference of declination and the difference of latitude are complementary, therefore

sine (difference of latitude) = sine (difference of declination) - cosine (latitude) imes sine (latitude) / sine (declination) imes cosine (latitude) = sine (declination) / sine (latitude) imes cosine (latitude) - cosine (latitude) imes sine (latitude) (g) (h)

But the line of position (latitude) = shadow / 12

The arc of the circle (latitude) = 12 / sine

sine (declination) = sine / sine herefore sine (difference of latitude) = sine (difference of declination) / cosine (latitude) imes sine / sine - 12 / sine imes equinoctial shadow / 12 = 1 / sine (declination) sine / sine imes equinoctial shadow / latitude - cosine (latitude) imes equinoctial shadow herefore shadow imes latitude = circle - equinoctial shadow [Table of Trigonometry (4)] (6)

but shadow X = true shadow hypotenuse, because in figure 46 angle Sshuya angle Chashbhishthkalakraghai whose sine = chash / chash therefore shadow X = chash X chash / chash therefore hypotenuse = hypotenuse - equinoctial shadow

in this figure the sun is north of the prime vertical circle, therefore the hypotenuse will be south. If the sun is on a latitude like the south of the prime vertical circle, then in the spherical triangle

cot angle Dhkhra = cot (Dhara) - cot (Khara) × cot (Dkh) / sin (Khara) × sin (Dkh)

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∴ sine (60 ° + apogee) = sine (60 ° + declination) − sine (node) / sine (node) × sine (longitude)

∴ - sine (latitude) = - sine (declination) sine (node) * sine (latitude) * sine (longitude) - cosine (node) * cosine (latitude)

: like before

- sine (latitude) = 1 / shadow - sine of declination X shadow hypotenuse / latitude sine - equinoctial shadow = 1 / shadow - chord circle - equinoctial shadow -

or - sine (latitude) X shadow = - chord circle - equinoctial shadow, - altitude = - chord circle - equinoctial shadow

SOURCE PAGE 300

Therefore, Bhuj = Karnavrittagra + Palabha

Here, Bhuj is obtained by adding the equinoctial shadow to the arc of the circle.

, when the declination of the sun is south, the equinoctial shadow must always be added to the arc of the circle.

Figure Introduction

u h k v p d = north circle u = north point d = south point k = khastak and d = south point p = east point k p = samamandal v p = equator circle a p = sun's ahoratra circle when rotation is south a = udaya point r = sun's desired place k r aa = sun's upward circle u a = sun's diurnal direction u aa = sun's instantaneous direction a p = diurnal angle aa p = instantaneous angle k d a = sun's diurnal night time < k d r = sun's instantaneous night time k r = sun's night part; a r = sun's zenith part

SOURCE PAGE 301

If the sun is in the prime vertical circle, the first way to know the shadow hypotenuse is

> लम्बाक्षजीवे विषुवच्छाया द्वादशसङ्गुणे ।
> क्रान्तिज्याप्ते तु तौ कर्णौ सममण्डलगे रवौ ॥२५॥

Translation—(25) If the sun is in the prime vertical circle, the shadow hypotenuse is obtained by multiplying the sine of colatitude by the equinoctial shadow or the sine of latitude by 12 and dividing each by the sine of declination.

Scientific Commentary—The essence of this verse is:— when the sun is in the prime vertical circle,

Shadow hypotenuse = sine of colatitude X equinoctial shadow / sine of declination = sine of latitude X 12 / sine of declination

At the time when the sun is in the prime vertical circle, the shadow of the gnomon remains on the east-west line, in figure 57 the sun will be at the point 'S' where the prime vertical circle and the prime vertical circle intersect at the point 'P' in this position. In such a case, the angle D is equal to 600 and the instantaneous meridian will be zero. Therefore, the left side of the equation (6) of the previous verse will be the valid zero; therefore, the right side of this equation will also be the valid zero. Therefore

sine of declination X shadow hypotenuse latitude = equinoctial shadow = equinoctial shadow X latitude of declination / sine of declination but latitude of declination = sine of colatitude

∴ shadow hypotenuse = equinoctial shadow X sine of colatitude / sine of declination (7)

It is stated in the commentary of the 13th verse that

sine of colatitude = gnomon / meridian

SOURCE PAGE 302

sine of latitude = equinoctial shadow / declination

∴ solar noon = gnomon sine of colatitude = equinoctial shadow sine of latitude

∴ equinoctial shadow X sine of colatitude = gnomon X sine of latitude

If the place of shadow hypotenuse is replaced by gnomon X sine of latitude in equation(7), its form will be—

shadow hypotenuse = gnomon X sine of latitude / sine of declination

This can also be proven by Napier's second rule because at the time the sun is in the prime vertical circle, the azimuth will be 60° and the altitude will be zero, therefore the angle angle dha kha sa in figure 57 will be a right angle. Therefore, the triangle angle dha kha sa will be a spherical triangle, and the relationship between its sides and angles will be as per Napier's second rule:-

cotitude (thas) = cotitude (thak) X cotitude (khas)

Here, theta is the polar distance of the sun, phi is the nodal distance of the sun, and theta is the length.

cotangent (polar distance) = cotangent (nodal distance) x cotangent (length)

But the polar distance is the supplement of the revolution, therefore

sine of declination = cotangent (nodal distance) x sine (latitude)

butantaanshkojya = 12 / shadowhypotenuse × [see shloka21equation (x)]

∴ jyakranti = 12 / chhayakarn × jya (akshansh)

arthatchhayakarn = 12 X akshansh jya / krantijya

from this the other form can also be known like before.

the method to know the shadow hypotenuse when the sun is in the prime vertical circle

> सौम्याक्षोना यदाक्रान्तिः स्यातदाद्युदलश्रवः ।
>
> विषुवच्छाययाभ्यस्तः कर्णोमध्याग्रयोद्धृतः ॥२६॥

SOURCE PAGE 303

Translation—If the North declination is less than the North latitude, the true solar shadow is obtained by multiplying the meridian shadow hypotenuse by the equinoctial shadow and dividing by the meridian arc (7).

Scientific Commentary - The essence of this verse is:-

When the Sun is in the prime vertical circle,

Shadow hypotenuse = meridian shadow hypotenuse x equinoctial shadow / meridian arc (7)

Proof - According to the 22nd verse,

Midday arc × Midday shadow hypotenuse

∴ by raising this value of the meridian arc in equation (1)

shadow hypotenuse = meridian shadow hypotenuse X equinoctial shadow X radius of rising X sine of declination X meridian shadow = prime vertical circle X radius of rising (2)

In equation (2), the value of the rising sine is obtained by raising it according to the first line of the 22nd verse or according to equation (1) there,

shadow hypotenuse = equinoctial shadow X radius / sine of declination X latitude / 12 = equinoctial shadow X 12 X radius / sine of declination X latitude

But13th verse according to12 X radius / meridian = sine of colatitude

∴ shadow hypotenuse = equinoctial shadow X sine of colatitude / sine of declination

It is one form of the rule of the 25th verse. Therefore it is proven that when the Sun is in the prime vertical circle

shadow hypotenuse = meridian shadow hypotenuse x equinoctial shadow / meridian hypotenuse

the second method of hypotenuse —

precession angle / latitude =

> सेष्टकर्णहताभक्तात्रिज्ययाऽग्राङ्गुलात्मिका॥२७॥

SOURCE PAGE 304

Translation—(27) When the sine of the sun's revolution is multiplied by the radius and divided by the sine of latitude, the rising declination is obtained, which when multiplied by the shadow hypotenuse and divided by the radius, the declination of the sun is obtained.

Scientific Commentary—In the 22nd verse, the method of knowing the declination is explained. The only difference is that there, the sine of declination is multiplied by the obliquity of the ecliptic and divided by 12, and here the sine of declination is multiplied by the radius and divided by the sine of latitude, which is the same (see verses 13 and 22).

When the sun is in the four angles of east, agni, etc., the method of knowing the altitude or declination -

> त्रिज्यावर्गार्धतोऽग्रज्यावर्गोनाद् द्वादशाहतात् ।
> पुनर्द्वदशनिघ्नाच्च लभ्यते यत्फलं बुधैः ॥२८॥
> शङ्कुवर्गार्धसंयुक्त विषुवद्वर्गभाजितात् ।
> लब्धं तु करणी नाम तां पृथक्स्थापयेत्ततः ॥२६॥
> विषुवच्छायाऽर्कं वधादग्रज्यासङ्गुणात्तथा ।
> भक्तात्फलाख्यं तद्वर्गसंयुक्तकरणीपदम् ॥३०॥
> फलेन हीनं संयुक्तं दक्षिणोत्तरगोलयोः ।
> याम्ययोविदिशोश्शङ्कुरेवं यामोत्तरे रवौ ॥३१॥
> परिभ्रमति शङ्कोशच शङ्कुरुत्तरयोश्च सः ।
> तल्लिज्यावर्ग विश्लेषान्मूलं दृग्ज्याऽभिधीयते ॥३२॥

Translation—(28) By taking half the square of the radius, subtracting the square of the rising declination from it, multiplying the remainder by 12, and then multiplying the product by 12, the result obtained by the scholars (26) is divided by the sum of the square of the gnomon and the square of the equinoctial shadow, which is the gain, which is called karani. The scholars keep this separate. (20) Multiplying 12 by the product and also by the rising declination, the result obtained is also divided by the sum of half the square of the gnomon and the square of the equinoctial shadow, and the gain is called phal. Adding the square of phal to karani and taking the square root of the sum, (31) if the sun is in the south celestial sphere, that is, if the sun is in the south declination, subtract phal from the square root, and if the sun is in the north celestial sphere, add phal to the square root. By doing this, whatever

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Now the gnomon of the agneyadi angles is called the angle gnomon. (32) When the sun is in the south, the angle gnomon is in the first agnikona and in the nairutya angle after noon. But when the sun is in the north, the angle gnomon is in the first ishan kona and in the vaivy kona after noon. Taking the square root of the difference of the squares of angshanku and radius gives the sine.

Scientific Commentary - The essence of these 5 shlokas is:-

```
Karani = (radius^2 / 2 - ajaya^2) * 144 / (12^2 / 2 + palabha)
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```
Phal = 12 X palabha X ajaya / (12^2 / 2 + palabha^2)
```

```
Angshanku = sqrt(Karani + phal^2 + phal)
```

```
Dṛjya = sqrt(radius^2 - angshanku^2)
```

At the time when the sun is in the northeast, fire, southwest, or northwest corners, the sine of its altitude is called the sine of the angle cone and the sine of its declination is called the sine of the sight. At any other time, the sine of the declination is also called the sight and the sine of the altitude is called the cone. Therefore, the difference between this cone and the cone with 12 fingers should be carefully noted. Therefore, when the azimuth of the sun is 45°, the sine of the angle of the portion it remains above the horizon is the sine of the cone and the sine of the portion it remains below the cross is the sine of the sight. Therefore, the value of the sine of the cone or the sine of the sight can be easily obtained with the help of the equations (a), (b), or (c) of figure 57. Equation (c) is as follows—

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Scientific Commentary
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sine ( latitude ) = equinoctial shadow / declination [verse13]
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sine (declination) = 12 X sine (obliquity) / declination [verse22]

cosine (latitude) ′ = 12 / obliquity [verse13]

It is proven by the equation that

sine (obliquity) X sine (portion) x cosine (latitude) = sine (declination) — cosine (portion) X sine (latitude)

By raising the values of sine (agra), sine (latitude), etc.

12 × sin (antevers) × 12equant = 12 × prime vertical circle - sin (antevers) × equinoctial shadow

In the other side of this equation, the ante-sine is the sine of the sunrise meridian of the sun. Since there is the obliquity in the denominator of each term of this equation, removing this general number will make no difference. If the equinoctial shadow, the netaansh, and the ante-sine are denoted by the letters p, n, and a respectively, and the obliquity is removed, then

1 / √ 2 × sine (n) × 12 = 12 × a - cosine (n) × p

By squaring both sides,

12 × sin^2 (n) × 122 = 122 × a^2 × p^2 × cos^2 (n) - 2 × 12 × a × p × cos (n)

But based on the 16th verse, sine^2(n) = radius^2 - cosine^2(n)

Therefore, the form of the appropriate equation will be

12 imes 122 imes [radius^2 - cosine^2(n)] = 122 a^2 + p^2 imes cosine^2(n) - 2 imes 12 imes a imes p imes cosine(n)

SOURCE PAGE 307

or simplifying

122 [r^2 - a^2] = cos(n) [1222 + p^2] - 2 × 12 × a × p × cos(n)

sine 2 (n) − 2 × 12 × a × p122 + p2 sine (n) − 122 [radius22 − a2] 1222 + p2 = 0

Instead of the third part, use the formula and the second part is 12 × a × p/12^2 + p^2

Instead of writing the fruit, it will have this form

cos^2(theta) - 2 cos(theta) sin(theta) = arccot(result) or cos^2(theta) - 2 cos(theta) sin(theta) = result Adding (result)^2 to each term to make the first part a perfect square: cos^2(theta) - 2 cos(theta) sin(theta) + sin^2(theta) = result + sin^2(theta). The first part of this equation is [cos(theta) -

sin(theta)]^2. Therefore, cos(theta) - sin(theta) = √(result) + sin^2(theta) or cos(theta) = sin(theta) + √(result) + sin^2(theta). But sin(theta) = sin(precession angle) = sin(latitude) = angle of gnomon. Therefore, angle of gnomon = √(result) + sin^2(theta) + sin(theta). Thus, it is proven that when the revolution of the sun is north, then by adding the square of the result to the result and taking the square root and adding it to the result, the angle of gnomon is obtained. If the revolution of the sun is south, then according to figure 58-

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- sin (agra) X sin (nataansh) X cos (akshansh) = - sin (kranti) - cos (nataansh) X sin (akshansh)

by raising and simplifying the values of sine (apr), sine (cranti), etc.

– 1 / √ 2 x sine (n) x 12 = – 12a – p X cos(n)

or $1/\sqrt{2} \times 12 \times \text{sine}(n) = 12a + p \times \cos(n)$

by squaring both sides

1222 × sin2 (n) = 122a2 + p2 × cos2 (n) + 2 × 12 × a × p × cosyan

or 1222 [radius2 - cos2 (n)] =122a2 + p2 × cos2 (n) + 2 × 12 × a × p × cosya (n) ∴ 122 [radius22 - a2] = cos2 (n) [1222 + p2] + 2 × 12 × a × p × cosyan ∴ cos2 (n) + 2 × 12 × a × p122 + p2 cosya (n) = 122 [radius22 - a2] 1222 + p2

or cos2(n) + 2 X phal X cosya(n) = karani

. [cosya (n) + phal] 2 = karani + phal2

∴ cosya (n) + phal = √ karani + phal2

∴ cosya (n) = √ karani + phal2 - phal

Therefore, it is proven that when the revolution of the sun is south, subtraction has to be done.

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> स्वशंकुना विभज्याप्ते दक्षिज्ये द्वादशाहते ।
> छायाकर्णौ तु कोणेषु यथास्वं देशकालयोः ॥३३॥

Translation—(33) Multiply the sine and the radius by 12 and divide by the gnomon. The quotient will be the shadow and hypotenuse in the order of time.

Scientific Commentary—It is stated in 28-32 shlokas that how the latitude sine (sine of gnomon) and declination sine (sine of sight) are calculated when the sun is in the zodiacs. In the 33rd shloka, it is stated how the shadow or shadow hypotenuse is calculated from the sine and the gnomon.

The essence of this rule is:—

(Angle) Shadow = dṛgjyā imes 12 / koṇaśaṃku

(angle) shadow hypotenuse = radius x 12 / gnomon

This is similar to the 21st shloka. Therefore, its derivation is also the same.

Example—When the declination of the sun is 150 north or south, what will be the gnomon and sine when the latitude is 250 25' in Prayag?

The viṣuvatkarṇa of Prayag = 13.28 angulas

, equinoctial shadow = 5.68 angulas

Therefore, the udayakālika sine of that day

= sine15° × 13.28 / 12 [see shloka22] = 860 × 13.28 / 12 = 685 '

Result = (radius22 - sine22) × 14472 + equinoctial shadow2 = (5606622 - 670225) 14472 + 32.26 = 4636667 × 144104.26 = 6822524

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fruit = 12 × equinoctial shadow × sine of declination72 + equinoctial shadow^2 = 12 × 5068 × 685104026 = 644

Angle of cone = root + product 2 ± product = 6822524 + 414736 ± 644 = 7237260 ± 644 = 2660 ± 644 = 3334' or 2046'

Therefore, when the declination is north, the angle of the gnomon is 3334' and when the declination is south, the angle of the gnomon is 2046'.

It is stated that the angle of the gnomon is the arc of the meridian or the sine of the altitude, therefore if the meridian or altitude is to be known, the arc of the gnomon must be made.

Here, angle of the gnomon = sine of altitude = 3334'

altitude = 75° 57′

declination = 60° - 75° 57′ = 14° 3′

When declination is south

Uttanshjyā = 2046 ′

∴ altitude = 36° 32′

and declination = 60° - 36° 32′ = 53° 28′

When the precession of the sun will be north,

hgya = r^2 - h^2 = 34382 - 33342 = (3438 + 3334) (3438 - 3334) = 6772 * 104 = 836 art

purtudjya = ntaanshjiya = 836

ntaansh = 13° 56′

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There is a difference of 4 arts in both answers because the decimal places were left when calculating the square root.

If you want to know what the shadow or shadow hypotenuse of the gnomon will be on the angles (directions), you will have to use the 33rd verse. When the sun's declination will be north, then

Shadow = sine imes 12 / gnomon = 836 imes 12 / 3334 = 302 inches

Shadow hypotenuse = radius imes 12 / gnomon = 3438 imes 12 / 3334 = 12.37 inches

There is no special facility in the new method to know the value of the angle of the gnomon; yet, it would be good to give an example. It has been proven before that when the sun is in the east or northeast angle, then the sine of Agra + 1/ sqrt2 and when it is in the south or southwest angle, then the sine of Agra - 1/ sqrt2 (see figures 57, 58). Therefore, according to the equations (g) of verses 22-24,

± 12 = ± sine150^ circ - cosine (n) imes sine250^ circ 25' sine (n) imes cosine250^ circ 25' +

ya ± · 7071 = ± · 2588 - cos(n) × · 4262sin(n) × · 6032 therefore ± · 7071 × · 6032 × sin(n) = ± · 2588 - · 4262 × cos(n)

†150 sine of declination will be positive when the declination is north, that is when the sun is in the northern hemisphere. But when the declination is south, its sine will be negative.

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When squaring both sides,

$.4076\sin^2(n) = .0670 \pm .2222\cos(n) + 1.842\cos^2(n)$

$or.4076 (1 - \cos^2 n) = .0670 \pm .2222\cos n + 1.842\sin^3 n$

∴ 5621 $\cos^2 n$ ± 1222 cosn - 3406 = 0

∴ cosn = ±.2222 ± (.2222)^2 + 4 ×.5621 ×.34062 ×.5621 = ±.2222 ±.62561.842 = 1.1478sin.70341.8421.842

Therefore, the sine of unnatāṃśa is = 6663 or 5640

latitude = 75° 46' N 36° 27' N

From this, the sine, shadow, etc., can also be known.

The method of knowing the altitude by knowing the right ascension, latitude, and declination—

> त्रिज्वोदक्क्षरजायुक्ता याम्यायां तु विवर्जिता ।
> अन्त्या नतोत्क्रमज्योना स्वाहोरात्रार्धताडिता ॥३४॥
> त्रिज्याभक्ता भवेच्छेदो लम्बज्याघ्नोऽथ भाजितः ।।
> त्रिभज्यया भवेच्छङ्कुस्तद्वर्गं प्रविशोधयेत् ।
> त्रिज्यावर्गात्पदं दृग्ज्या छायाकर्णौ तु पूर्ववत् ॥ ३५।।

Translation—(34) If the sun is in the northern hemisphere, the result is obtained by adding the right ascension to the radius and if the sun is in the southern hemisphere, by subtracting it. Then, by subtracting the versed sine of the local time from the remainder and multiplying by the sine of the meridian (35) and dividing by the radius, the chord is obtained. Multiplying this by the sine of colatitude and dividing by the radius gives the gnomon (sine of the altitude of the local time). Subtracting the square of the gnomon from the square of the radius and taking the square root of the remainder gives the sine of the altitude (sine of the right ascension of the local time) from which the shadow and shadow hypotenuse should be known previously.

Scientific Commentary — The essence of these two verses is:-

(1) Antya = radius ± versed sine

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(2) chheda = (antyā - natoktramjya) × dyujya / trijya (3) shanku = chheda × lambajya / trijya (4) dṛjya = √(trijya^2 - shanku^2)

By raising the values of equation (2) and (1) in equation (3),

Gnomon = (sin - rectitude motion sine) × diurnal sine × latitude sine = (rectitude sine ± meridian motion sine) × latitude sine = (rectitude sine - rectitude motion sine ± meridian motion sine) latitude sine^2 diurnal sine × latitude sine = (meridian sine ± meridian motion sine) latitude^2 diurnal sine × latitude dagger = (meridian sine ± meridian motion sine) latitude^2 × declination sine

ya..... (k) *

This matter can be easily proven by spherical trigonometry. Some new words have come here, so it is necessary to explain them first:-

End—in Figure 42, there are chajya and chajya and vish radius. Therefore, cha vi and chvi are end.

Nata kala—the time taken for a planet or star to come on the meridian from some time is called the west sidereal time of that planet or star, and the time that remains after passing the meridian of that star or planet is called the east sidereal time of that star or planet. If the sidereal time (hour angle) and declination of a planet or star are given, its position can be easily determined. The magnitude of the sidereal time is known by the angle between the polar circle of the planet or star and the meridian. The polar circle is perpendicular to the equator, so the sidereal time is also known by the arc from the equator to the pole of the star or planet.

See figure 24 and pages 116, 118-119

SOURCE PAGE 314

It is located between the ecliptic and the meridian circle, which can be known. In figure 57, kha dha ra, kha dha sa and kha dha rā angles are the hours of the sun while the sun is at the points ra, sa and rā in sequence. This place is east of the meridian circle, so it is the hours of the sun east. When a planet or star is on the meridian circle, its hour is zero. The hour is generally written in degrees. If the east hour of a star or planet is 15^ 0, it should be understood that it will arrive on the meridian circle in 15 imes 4 minutes or 1 hour (sidereal).

The advanced time is called the advanced time when the hour is subtracted from half the day time. The east advanced time is the time from the rising time of the planet or star to the auspicious time, and the west advanced time is the time from the auspicious time to the setting time. The west advanced time is the time when the planet or star is west of the meridian circle. The advanced time or its abbreviation advanced should be understood as different from the declination as the hour is different from the minute.

In chapters 34-35, it is stated how the latitude, declination, etc., can be known from the hour of a planet or star, if its hour, latitude and declination are known. Its proof is based on spherical trigonometry. (See figure 57)

Suppose the sun is at r. The cosine of kha dha ra in the spherical triangle dha kha ra

= cojya (xr) - cojya (dhkh) × cojya (dhr) jya (dhkh) × jya (dhr) cojya (ntkal)
= cojya (ntansh) - cojya (lmbandsh) × cojya (dhruvantr) jya (lmbandsh) × jya (dhruvantr)
= cojya (ntansh) - jya (akshansh) × jya (krantti) cojya (akshansh) × cojya (krantti)
= cojya (ntansh) cojya akshansh × cojya krantti - jya (akshansh) × jya (kranti) cojya (akshansh) × cojya krantti
= cojya (ntansh) super (akshansh) (9)

X sphere (declination) (2)

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And variable * = tangents of the Circle of the Meridian equations (2) and (3) are added, angle (time) + variable = angle (arcsecond) / angle (latitude) × angle (revolution) (3)

atvanatcotijyā + charjya = kojya (ntansh) / akshanshkotijyā X krantikotijyā

Nataanshcotijya = (Natcotijya + charjya) X akshcotijya X krantikotijya (kh)

Nantaśa arc is also called a cone. From this formula, the value of the cone will be in decimal fraction according to the current method. If it is to be written according to the Indian method, then it will have to be divided by the square of the radius(3438).

This formula will be useful when the sun is in the northern hemisphere. If the sun is in the southern hemisphere, then the declination will be negative (see explanation of figure 42). In this situation, the polar distance will be more than 600 degrees, which will make the cotangent (declination) negative. Therefore, the sine (latitude) X sine (longitude) of the right-hand side of equation (2) will also be positive, which will keep the cotangent negative in equation (3) but will not make any other difference. Therefore, the general form of equation (3) will be—

Natansh Kotijya

= (n t k o t i j y a ± c r j y a) X a k s k o t i j y a X k r a n t i k o t i j y a (g)

In which the plus sign will be taken when the revolution of the sun or planet is north and the minus sign when the revolution is south.

By knowing the value of the ntaansh cotijya or the value of the gnomon, the sijnya, shadow, shadow hypotenuse, etc., can be known as before, so there is no need for expansion. Example—if the revolution of the sun is 150 north or south, what will be its ntaansh at the time when the forenoon of the sun in Prayag is 3 hours 30 minutes?

According to the method of Sūrya Siddhānta

charjya = krantijya X palbha X trijya / 12 X krantikotijya [See page 208]

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= sine15° × 5.7 × 343812 × cosec15° = 860 × 5.7 × 343812 × 3321 = 438

butanta = radius + versed sine ∴ antya = 3438 ± 438 = 3876yā3000night = 3hours30minutes = 52030

versed sine = 52030' = 1345

therefore cheda = (3876 − 1345) × 33213438 yā (3000 − 1345) × 33213428 = 2531 × 33213438 athavā 1655 × 33213438
therefore gnomon = cheda × lambajyātrijyā = 2531 × 3321 × 31063438 × 3438 athavā 1655 × 3321 × 31063438 × 3438 = 2206 athavā1444parantuyahā̃śaṃkuunnatāṃśakījyākelieprayuktahai l isaliejabasūryauttaragolameṃhogātabaiṣṭakālameṃunnatāṃśakīja06kalāaurajabasūryadakṣiṇagolameṃhogātaba unnatāṃśakījyā1444kī lisaliepahalīdaśāmeṃ —

Unnatansh = 40° and Natansh = 60° - 40° = 50° and in the second daśā unnatansh = 24°52’ and natansh = 15°8’

firsttensine = radius - 2 gnomon^2 = 11816844 - 2206^2 = 11816844 - 4876681 = 664163 = 2634

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∴ In the first daśā, shadow = sine × 12 / gnomon = 2634 × 12 / 2206 = 14031 inches In the second daśā, sine = √(trijyā² - 1444²) = √(18819841 - 2085136) = 3120 ∴ In the second daśā, shadow = 3120 × 12 / 1444 = 25‘63 inches Newly— based on equation (kha), the node co-latitude = (node co-latitude ±-verse) × latitude co-latitude × declination co-latitude but verse = spread declination × spread latitude[see page 206 = spread 15° × spread 25°25’ =.2676 ×.4752 =.1273 ∴ node co-latitude =(co-latitude 52°30’ ±.1273) × co-latitude 25°25’ co-latitude 15° =(.6088 ±.1273) ×.6032 ×.6656 =.7361 ×.6032 ×.6656 or.4815 ×.6032 ×.6656 =.6422 or.4201 ∴ When declination is north, then node will be 50°3’ and when declination is south, then node will be 65°9’ In the first daśā, 12 inches of gnomon shadow = 12 spread 5003’ = 12 × 1.1640 = 14.328 inches In the second daśā, shadow = 12 × spread 65°6’ = 12 × 2.1564 = 25.613 inches

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Knowing the time of day by measuring the shadow —

> सी समय को छाया नापकर नतकाल जानना —
> अभीष्टच्छाययाऽभ्यस्ता त्रिज्या तत्कर्णभाजिता ।
> दृग्ज्या तद्वर्गसंशुद्धात् त्रिज्यावर्गाच्च यत्पदम् ॥३६॥
> शङ्कुस्य त्रिभजीवाघ्नः स्वलम्बज्याविभाजितः ।
> फलं त्रिभज्ययाऽभ्यस्तं स्वाहोरात्राधंभाजितम् ॥३७॥
> उन्नतज्या तया हीना स्वान्त्या शेषस्य कार्मुकम् ।
> उत्क्रमज्याभिरेवं स्यात्प्राक्पश्चाच्च नतासवः ॥३८॥

Translation—When the shadow of the desired time is multiplied by the hypotenuse and divided by the shadow hypotenuse, sine is obtained. Subtracting the square of sine from the square of the radius and taking the square root yields the gnomon. Multiplying the gnomon by the radius and dividing by the sine of colatitude gives the hole. Multiplying the hole by the sine of declination and dividing by the versed sine gives the sine of inclination. Subtracting this from the node gives the remainder, which is understood as the versed sine, and forming an arc from the versed sine body, the east or west time of day is known.

Scientific Commentary—The summary of these three shlokas is:—

(1) shadow x radius / shadow hypotenuse = sine (2) √(radius^2 - sine^2) = gnomon (3) gnomon x radius / sine of colatitude = hole (4) hole x radius / sine of declination = sine of inclination (5) node - sine of inclination = nto-versed sine

These are the opposites of the rules written in shlokas 34-35, so their derivation is the same. Yes, here, the value of the sine of the shadow, i.e., the sine of the ntaansh, should be known as in the rule mentioned in the 13th shloka. It has already been stated that the angle between the gnomon and the shadow hypotenuse is the ntaansh, so when the shadow is divided by the shadow hypotenuse, a decimal fraction * and when this result is multiplied by the radius, the value of the sine of the ntaansh of the arts is obtained.

Pandit Indranarayan Ji Dwivedi writes in relation to this, "Although in the calculation contrary to 34-35 shlokas, to make the time of night in the upper shlokas"

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The method is stated, but in this way, some difference arises in the local time. This is what Bhaskaraacharya corrected (see Siddhanta Shiromani).

But in my understanding, this difference does not arise because the rule is incorrect, but rather because of the crudeness of the measurement of the shadow. If the shadow is measured exactly up to two or three decimal places and no crudeness arises even in multiplication and division, then there can be no error in knowing the local time by this method.

Example 1—If the shadow in Prayag is 14.33 angulas at a certain time and the declination of the sun is 150, tell the east or west local time and also tell what time is on the clock?

Mathematical procedure—

Shadow = 14.33 angulas

∴ Shadow hypotenuse = √ 12^2 + $(14 - 33)^2$ = 18 * 66 angulas

∴ sine = 14.33 X 3438 / 18.66 = 1636art

∴ gnomon = √ 34682 - 26 · 362 = 2207art

∴ intersection = 2207 × 34383106

and versed sine = 2207 X 3438 / 3106 x 3438 / 3321 = 2526art

end = 3876 (like before)

, versed sine = 3876 - 2526 = 1347art

natkal = 1347kalaaka (utkramjyaakeanusar) dhanu = 52032 ′ [see page120 - 121] = 3hours30minutes8seconds

If the local time is before, then by subtracting from 12 hours and if it is west, by adding to 12 hours, the time of the sundial will be found.

∴ If the local time is before, then in the sundial

12 hours - 3 hours 30 min 8 seconds = 8 hours 26 min 52 seconds.

And if the local time is west, then 3 hours 30 minutes 8 seconds have passed after noon in the sundial, that is 3 o'clock 30 minutes and 8 seconds.

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It should be noted that this time of the clock is the true local time. To reconcile it with railway time, the time equation and the longitude equation will have to be done, which will be discussed at the end of this chapter.

New method—

Spere (Nataansh) = Shadow / 12 = 14.33 / 12 = 1.1641

∴ Nataansh = 5203 ′

Gnomon = Nataansh koṭi sine = Cojya 50°3’ = proved in equation (b) that

It is proved in equation (b) that the arc of the meridian

= (arc of meridian + versed sine) X arc of latitude X sine of declination,.6421 = (arc of meridian +.1273) X.6032 X.6656 = (arc of meridian +.1273) X.8724

∴ arc of meridian + 1273 = 6421 / 8724 = 7360

arc of meridian = 7360 - 1273 = 6087 ∴ time = 52030 ′ = 3hours30minutes

Therefore, if the meridian is west, the time will be 8:30 and if the meridian is east, it will remain 3:30.

It will be easier to find the time in a new way if equation (1) is used directly. An example is given below:- Example 2—shadow = 14.33 inches and precession = 150 uttara, what is the time in Prayag?

Spere (Nataansh) = shadow / 12 = 14 · 33 / 12 = 1 · 1641 ∴ Nataansh = 50 ° 3 ′ ∴ Cojya (Natkal) = Cojya50 ° 3 ′ − sine25 ° 25 ′ × sine15 ° / Cojya25 ° 25 ′ × Cojya15 °

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Sūrya Siddhānta

Scientific Commentary

Chapter on Mean Motions

Chapter on True Positions

Chapter on the Three Problems

The late

Kajya (latitude) or jya (declination) negative therefore cojya (local time) = cojyanatansh + jya (latitude) + jya (declination) cojya (latitude) X cojya (declination) = cojya65°6' + jya25°25' X jya15° cojya25°25' X cojya15° =.4202 + .4262 X.2588.6032 X.6656 =.4202 + 1111 =.5313.8724 =.8724 =.6060 ∴ local time = 52°26' = 3hours26minutes5

To find the portion of the sun knowing the longitude at a certain time-

> इष्टाग्राघ्ना स्वलम्बज्या स्वकर्णाङ्गुलभाजिता ।
> क्रान्तिज्या सात्रिजीवाघ्नी परमापक्रमोद्धृता ॥४०॥
> तच्चापं भादिकं क्षेत्रं पदै र्भध्याह्निको रविः ।

Translation—(40) Multiply the longitude of the desired time by the sine of colatitude and divide by the shadow hypotenuse of the desired time, then the quotient will be the sine of the sun's declination. Multiply this by the radius and divide by the sine, the quotient's arc

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SOURCE PAGE 322

Make. Then according to the 17-19 shlokas, determine the position of the Sun by making the position in which it is, according to the sine of the bhoga.

Scientific Commentary—Its summary is:-

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Karnagra X sine of colatitude shadow hypotenuse = sine of declination
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Revolutionary X radius perimeter sine = Sun's sine fraction sine

The first rule explains the method of knowing the declination of the Sun from the meridian (or the meridian circle) in the time of the ecliptic, which is the opposite of the 27th and 22nd shlokas [See the equation of the 2nd shloka(4)]

The second rule by which the sine portion of the sun is obtained knowing the declination is found in the 17-16 shlokas of this chapter and in the 28th shloka of the Chapter on True Positions. Therefore, there is no need to repeat here.

The line on which the tip of the shadow moves—

> इष्टे हि मध्ये प्राक्पश्चाद् वृत्ते बाहुत्रयान्तरे ॥४१॥
> मत्स्यद्वयान्तरयुते स्त्रिस्पृक्सूत्रेण भाभ्रमः ।

Translation—On the day when the line of the tip of the shadow of the gnomon is to be drawn, determine three points of the shadow tip before and behind the meridian of that day. Make a triangle from the first and second points and the second and third points. Extend the lines passing through the general points of each triangle so that they meet. Taking the point where they meet as the center, draw a circle passing through the three points of the shadow tip. This circle will be the apparent line of the shadow tip on that day.

Scientific Commentary - In reality, the path of the tip of the shadow is not circular but is in the shape of a hyperbola. Therefore, this rule is impure, which has been accepted by Bhaskaraacharya, Ranganatha ji, etc. Therefore, there is no need to think much about it.

The method of knowing the rising time of the signs of sine, etc., in Lanka and Ishta Sthan

> त्रिभद्युकर्णार्धगुणाः स्वाहोरात्रार्धभाजिताः ॥४१॥
> क्रमादेकद्वित्रिभज्यास्तच्चापानि पृथक् पृथक् ।

SOURCE PAGE 323

> स्वाधोऽधः प्रविशोध्याथ मेषाल्लङ्कोदयासवः ॥४२॥
> स्वागाष्टयोऽर्थगोऽगैकाश्शरत्र्यङ्कःहिमांशवः ।
> स्वदेशचरखण्डोना भवन्तीष्टोदयासवः ॥४३॥
> व्यस्ता व्यस्तैयुतास्तैस्तैः कर्कटाद्यसवस्स्मृताः ।
> व्युत्क्रमेण षडैवैते भवन्तीष्टास्तुलादय ॥४४॥

Multiply the sines of one, two, and three zodiacs in sequence by the mean motions of three zodiacs in sequence, and divide the products by the arc of the mean motions (mean motions) of one, two, and three zodiacs in sequence, and keep the resulting products separate by forming the arcs of the bows of the results. The first is the rising of Aries in Lanka, and the remainder obtained by subtracting the first from the second is the rising of Taurus in Lanka, and the remainder obtained by subtracting the second from the third is the rising of Gemini in Lanka. (43) Their values are 1670, 1765, and 1635 as Asu or Prana respectively in sequence. By subtracting their respective sections from the auspicious place, the rising of Aries, Taurus, and Gemini zodiacs of the auspicious place are known. (44) These are the rising of the three zodiacs in Lanka in reverse order. By adding their respective sections in reverse order, the rising of Cancer, Leo, and Virgo of the auspicious place will be. These six risings are the risings of Libra, Scorpio, Sagittarius, Capricorn, Aquarius, and Pisces in reverse order.

Scientific Commentary—Sine Aries means the moment when the sine Aries zodiac begins to rise from the point before the horizon, and until the moment when the 30 degrees of the circle of declination does not come above the horizon from the vernal equinox, the sine Aries zodiac continues to rise. The time taken for the 30 degrees of the circle of declination to rise from the vernal equinox is called the rising time of the sine Aries zodiac. This subtlety is expressed in Asu. That is why this time is called the rising of the sine Aries zodiac. After this, the time taken for the next 30 degrees of the circle of declination to rise is called the rising time of the sine Taurus zodiac or UdayaSu. Similarly, one should understand the risings of the other sine zodiacs.

To know how much time it takes for a certain quantity to rise in a place, it is easier to first know how much time that quantity rises on the equator. Once this is known, some calculations need to be done on the time of rise on the equator to know the rising time of one's own place.

SOURCE PAGE 324

In our astrological principles, the viṣuvat rekhā and Ujjain are considered to be at the general point of the meridian line passing through Lanka. Therefore, the rising point of Nirakshades is called the rising point of Lanka. The rule for knowing the rising points of the Aries, Taurus, and Gemini signs in Lanka is given in the latter part of verse 41 and the 42nd verse, which is understood from figure 56.

U, P, D, Po—the points to the north, west, south, and east of the horizon of Lanka.

P and Po—the great circle which is also the prime vertical circle in Lanka.

K and Ka—the circle of declination.

V—the beginning point of the Vernal Equinox or the first point of the Aries sign of the sign of Pisces.

It is also the location of the North Pole.

Due to the daily motion of the Earth, the time taken for the northern part of the equator to come above the horizon is the same time taken for the part of the ecliptic to come above the horizon. Therefore, the time taken for the rise of va is the same as the time taken for ba

SOURCE PAGE 325

It seems to occur when P rises. But the time when the entire viṣuvadvṛtta(3600) rises is called one sidereal day which is equal to 21600 lunations (page 6); therefore, when the time of 21600 lunations passes in the rising of 3600 or 21600 phases of the viṣuvavṛtta, it will take 1 lunation time for 1 phase to rise. Therefore, if the value of P is known in terms of phases, the rising time of P will come out in the same number of lunations.

Now we have to see what is the mutual relationship between V and V P. V P is a right-angled spherical triangle whose angle V P is a right angle, and the angle P V is the angle between the viṣuvadvṛtta and the declination circle, which is the true revolution of the sun. The sides of this spherical triangle are the revolution of P to the point of the declination circle, the revolution of V, the sine portion of the point of V, and the sine portion of the point of V P (see page 21). Therefore, based on Napier's first rule, the relationship between V and V P can be known because cot sine ↟rac(angle)(V P) = tangent(V P) imes tangent(V)

sine(latitude) tangent = true meridian radius / sine portion imes cotangent(latitude) = cos^2 23^ o 27^ '/ cos(sine portion)

But our Acharyas did not deal with the behavior of the tangent or cotangent, so they derived the relationship between them from the spherical triangle of V by this way:-

sine(latitude) sine(longitude) sine(revolution) sine(revolution)

It is stated on page 7 that the time from the rising of a star until its next rising is called Nakshatra Ahuratra or Nakshatra Din. Therefore, the time from the rising of the Vernal Equinox point until its next rising should not be considered Nakshatra Din because in such time, due to the motion of precession, this point will be approximately 0.14 degrees westward and will rise 0.002 degrees earlier. However, this difference is so subtle that there is no loss in understanding both definitions as the same in practice. Nowadays, Western astrologers use the definition of Nakshatra Din as given above.

SOURCE PAGE 326

But angle u ka = theta v po

∴ sin(u ka) sin(u v ka) = sin(v ka) sin(v po)

∴ sin(v po) = sin(v ka) × sin(u v ka) sin(u ka)

But v ka = sine of 'ka' fraction

∠ u v ka = ∠ u v po − ∠ ka v po = 60 ° — solar revolution

∴ sine (uva) = Sūrya Siddhānta = 24 ° of latitude (according to Siddhantic opinion) = three celestial coordinates (page 208)

sine (arc) = sine (subtraction - kapu) = sine (60° - ' ka' kīkranti) = ' ka' kīkrāntijyā = ' ka' kīdyujyā

Therefore, sine (wpu) = ' of the geogony X of the geogony of the revolution

The value of 'ka's portion in the arts will be the same as in the sciences. In this simple equation, the time of rise of the portion of 'ka' will be the same as the arc placed in place of the portion of 'ka'. If 300, 60° and 60° are placed in its place, then the rises of the portions of 30, 60 and 60 degrees, i.e., the rises of the signs of Aries, Aries and Taurus, and the rises of the signs of Aries, Taurus and Gemini in order, will come. If the rises of the signs of Aries and Taurus are subtracted from the rises of the signs of Aries, Taurus and Gemini, and if the rises of the signs of Aries and Taurus are subtracted from the rises of the signs of Aries, Taurus and Gemini, then the rises of the signs of Gemini will be obtained.

If the portion of 'ka' in equation (1) is 600, then the revolution of 'ka' of the sun will be the supreme revolution of the sun. In such a case, the value of the sine of the portion of 'ka' will be 3438 in the Siddhantic method and 1 in the modern method. Therefore, since the revolution of 'ka' is equal to the supreme revolution of the meridian, the right-hand side of the equation will be equal to 3438 or 1, and the value of vpu will also be equal to 600. This means

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SOURCE PAGE 327

It happened that when va is 60^(°), then va is also 60^(°). Therefore, the ascensions of the three zodiac signs like Aries will be 60 × 60 = 5400, which are equal to 15 Nakshatra hours or 6 Nakshatra hours.

In the first half of the 42nd verse, the ascensions in Lanka of the zodiac signs Aries, Taurus, and Gemini are given as 1670, 1765, and 1935 respectively, which are obtained according to the rule above from equation (1) and will be clarified by the examples written below.

Example—What are the ascensions in Lanka of the Taurus sign?

First, the ascensions of the Aries sign should be known. For this, the fraction of 'ka' in equation (1) must be kept as 30. At this time, 'of' sine is the final point of Aries and the initial point of Taurus sine, which can be known from the 28th verse of the Chapter on True Positions.

' sine of declination = sin30° × 13673438 = 1716 × 13673438 = 668 art

∴ Surya Siddhanta = 703 degrees = 11° 43′

Sūrya Siddhānta

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That is, the rising of the Aries sign is 1670.

Now one should know the combined risings of the Aries and Taurus signs.

SOURCE PAGE 328

The portion of 'ka' at this time is 600 and the declination of the last point of Taurus will be its declination.

Taurus's declination = sine60° X 1367 / 3438 = 2678 X 1367 / 2438 = 1210art

∴ sine of revolution of Aries = 20 ° 38 ′ but 20 ° 38 ′ of versed sine = 222 ′ ∴ the chord of 20 ° 38 ′ = 3438 - 222 = 3216 ′ ∴ from equation (1),

sine (wpu) = sine60 ° X 3140 / 3216 = 2678 X 3140 / 3216 = 2608art

∴ Vapu = 57 ° 48 ′ = 3468 ′

∴ The sum of the rising of Aries and Taurus = 3468 but the rising of Aries = 1670 ∴ the rising of Taurus = 1768

In the verse, 1765 is written instead of

It has already been stated above that the sum of the sines of Aries, Taurus, and Gemini is 5400 and it has been proven that the sum of the sines of Aries and Taurus is 3468, so the rising of Gemini is the difference between these two, i.e., equal to 1632. 1635 is given in the verse. This difference is due to the crudeness of the calculation.

Now it has been proven that the true precession of the Sun is not 240 but 1680.23°26'57.35 and it decreases by approximately 0.468 every year [see page 268]. In this way, a deficiency of 1 degree in the precession usually occurs in a hundred and one years. Therefore, the first 50 years of the 21st century of Vikram

SOURCE PAGE 329

To know the rising of Aries etc. by considering the supreme revolution as 23027', there will be sufficient subtlety.

With the new method, the sine of 23027' = 0.3676 which should be understood as the supreme sine of the sun by the new method.

According to the 28th verse of the Chapter on True Positions, the sine of the final point of Aries.

= sine30 ° X.3676 / 1 [new method radius = 1] =.5 X.3676 =.1660

, the revolution of the final point of Aries = 11°26'

Similarly, the sine of the final point of Taurus = sine60 ° X.36761 =.866 X.3676 =.3446

, the precession of the last point of the sine of Taurus = 2706'.7

To know the rising for the equinoxes from these values of the precessions, by making the appropriate correction in equation (1), for the sine of Aries

sin (equator) = sin30° x cos23° 27' / cos110° 26' = .5 x .6175 /.6768 = .4682 ∴ equator = 27° 55' ∴ rising of sine Aries = 1675 sine Aries and Taurus combined rising for sine (equator) = sin60° x cos23° 27' / cos10° 6.7 = .866 x .6175 /.6387 = .8464 ∴ equator = 57° 46' sine Aries and Taurus rising = 3466

∴ sine Aries and Taurus rising = 3466

SOURCE PAGE 330

∴ sin(Aries) = 3466 - 1675 = 1764

and sin(Gemini) = 5400 - 3466 = 1631

Based on Napier's first rule, the sine of Aries can also be found from this equation.

tangent of the equator = cos(23027') sin(300) = 0.61751.7321 cos(5267)

∴ latitude = 27° 54'.5 = 1674.5

Sūrya Siddhānta

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The following are the rising times of the three zodiac signs of Aries, Taurus, and Gemini in Lanka according to the ancient method: Aries 1670 278 111 1675 276 111.7 Taurus 1765 266 120 1764 266 119.6 Gemini 1635 323 126 1631 322 128.7

SOURCE PAGE 331

Now it is to see what are the rising points in any place other than the equator, whose northern latitude is a, of the three zodiac signs Mesha etc.

The arc of the horizon of that place whose northern latitude is a is the North Celestial Pole, and the arc of the vernal equinox, and the arc of the ecliptic which is below the horizon.

At the time when the vernal equinox point keeps rising, at that time it is at the due east point. Therefore at this time both the ecliptic and the equator are at the due east point. The time in which the arc of the ecliptic comes above the horizon is the same time in which the arc of the equator comes above the horizon, so the rising points of the arc of the ecliptic are the same as the rising points of the arc of the equator. A meridian is drawn from the point of the ecliptic which is exactly on the horizon, which is the pole of the equator below the horizon.

SOURCE PAGE 332

It is found at the point. Therefore, the point of the east part of the equinoctial circle is the meridian of the solstice. The point of the declination circle in Lanka and the point of the equinoctial circle together come to the horizon as stated above. However, the east part of the latitude remains below the horizon when the point of the latitude comes to the horizon in the latitude. Therefore, it is the same as the ascensions of the east parts of the latitudes, which is obtained by subtracting the east parts from the east parts. It is stated in pages 206-210 that this is the true time of the east part of the point. Therefore, it is proven that by subtracting the true time from the ascensions of Lanka, the ascensions of the auspicious places will be obtained. It is stated on page 210 that the true sine = declination line X latitude line.

(1) When latitude = 30°, then the declination of latitude = 110°26'. Therefore, for Prayag whose latitude is 25°25', the true sine of the point of latitude = 110°26' X 25°25' = 2032 X 4752 = 09666 ∴ the arc = 5°33' ∴ the arc of the point of latitude = 333 ∴ the ascensions of the east parts of latitude in Prayag = 1675 - 333 = 1342, that is, the ascensions of the signs of Aries in Prayag = 1342 (2) When latitude = 600°, then the declination of latitude = 200°7 = 20010'. Therefore, then the true sine of latitude in Prayag = 200°10' X 25°25' = 3671 X 4752 = 1745 ∴ the arc of latitude = 10°3' ∴ the arc of latitude = 603 ∴ the ascensions of the east parts of latitude in Prayag = 3466 - 603 = 2866, that is, the ascensions of the signs of Aries and Taurus in Prayag = 2866.

That is, the ascensions of the signs of Aries and Taurus in Prayag will be in 2866 ascensions. But the sign of Aries ascends in 1342 ascensions. Therefore, the sign of Taurus of Aries will ascend in 2866 - 1:342 = 1524 ascensions.

SOURCE PAGE 333

(3) When ka = 60° then the revolution of ka is 23° 27' therefore the declination of ka in Prayag is spere 23° 27' × spere 25° 25' = 4337 × .4752 = .2061 therefore the arc of ka is 11054' therefore the arc of ka is 714 therefore the rising of ka in Prayag is 5400 - 714 = 4686

That is to say, the signs of Aries, Taurus, and Gemini will rise in 4686 degrees in Prayag. But the signs of Aries and Taurus rise in 2866 degrees, so the sign of Aries Gemini will rise in 4686 - 2866 = 1820 degrees.

Thus it is apparent that the arc of the last point of Aries is 333, the arc of the last point of Taurus is 603, and the arc of the last point of Gemini is 714. The difference between the first and the second is 270, and

the difference between the second and the third is 111. These are mentioned in the latter part of the 43rd verse of Taurus and Gemini, which will be clarified by the bracket below:-

The zodiac signs rise in Lanka, the ascent in Assu, the ascent in Prayag, Aries 1675 - 333, 1342, Taurus 1764 - 270, 1524, Gemini 1631 - 111, 1820

In the first half of the 44th verse, it is stated how the rising signs of Cancer, Leo, and Virgo will be known. In Lanka, the rising signs of Cancer will be the same as those of Gemini, the rising signs of Leo will be those of Taurus, and the rising signs of Virgo will be those of Aries. Adding their respective arcs will yield the rising signs of the auspicious places, which will be clear from the following brackets:-

SOURCE PAGE 334

The rising of the zodiac signs in Lanka in the ascensions in Assu and the rising in Prayag in the ascensions of Cancer1631 + 111 2042 Leo1764 + 270 2064 Virgo1675 + 333 2008

Its derivation is as follows:-

To know the declination of any point of the meridian, the equation (1) is used, which is

Kakebhoganshija X Paramkrantikotijyā (vp) = KakiKrantikotijyā

It is proven by both ancient and modern methods that the sine of an angle is equal to the sine of its supplementary angle [see pages 126-128] i.e., sin(k) = sin(180° - k) where k is the measure of an angle. Therefore it is proven that

sine (v p u) = sine (180° - v p u) and sine (kaka bhogansh) = sine (180° - kaka bhogansh) is lie sine (180° - v p u) = sine (180° - kaka bhogansh) paramount fraction sine kaka ki paramount fraction (2)

It is stated above that when the fraction of the year, which is v ka bhogansh, is 600, then the fraction of the year, which is v po, is 57°46', therefore according to the equation (2), when the fraction of the year is 180° - 600 = 120°, its fraction of the year will be 180° - 57°46' = 122°11'. This means that in the same time, as 120 degrees of the ecliptic rises in Lanka from the vernal equinox, the ecliptic rises by 122°11'. However, the first three signs of the ecliptic rise in the same time as the ecliptic also rises by 600. Therefore, the fourth sign rises in the same time as the ecliptic by 122°11' - 600 = 32011'. However, the ecliptic's

SOURCE PAGE 335

32011' = 1611', therefore the time of its rising is 1631 asu. Therefore the sine of the Cancer sign's rising is 1631 which is also the sine of Gemini.

In this way it can be proven that the ascensions of the Aries ascensions of the sign of Sagittarius and the ascensions of the Taurus ascensions and the ascensions of the Virgo ascensions and the ascensions of the Aries ascensions of the sign of Gemini are the same.

Now it is to know what will be the rising of the sun in another place, suppose in Prayag, than the rising in Cancer.

This picture is the same as the 60th picture, the only difference is that the 'v' in it is less than 600 and here 'v' is like 1200.

From the picture it is revealed that the time when V is equal to 1200 will rise in Prayag in the same time as V rises in the east. But the declination of V is the same as the east and west, in which the east and west portion is below the horizon. Therefore

SOURCE PAGE 336

V po = V po ch — po ch

But the revolution of the point of Aries is the same as the revolution of the last point of Taurus, that is, 2010, because the revolution which increases from the point of Spring to 600 parts, in the same order, decreases from 600 to 180 degrees, that is, the revolution of the last point of Taurus is the same as the revolution of the last point of Cancer, and the revolution of the last point of Aries is the same as the revolution of the last point of Leo, etc.

Therefore po ch = 1003'

But po ch = 122011'

Because this is the arc of the equinoctial shadow of 1200.

Therefore v = 122011' - 1003' = 11208' = 6728'

∴ 120 Bhogash ke Udayasu = 6728

But the Udayasus of the first three zodiacs = 4686

, the Udayasu of Cancer = 6728 - 4686 = 2042

which comes by adding 111 to the Udayasus of Cancer in Lanka.

Similarly, it can also be proven that the Udayasus of 150 Bhogash, i.e., from Aries to Leo, will be what in Prayag. Then subtracting the Udayasus of the first four zodiacs will yield the Udayasu of Leo, which can also be obtained by adding 270 to the Udayasus of Leo in Lanka.

The final point of the sign Virgo whose portion is 180 is found again from the equinoctial line, that is, this is the place of autumn equinox; therefore, it rises just before it like the spring equinox and its declination is also 180°.

Similarly, it can be known what the rising positions of each sign from Aries to Virgo in Lanka and other places in the Northern Hemisphere are. Now it is to show what the rising positions of each sign from Libra to Pisces are. In the latter part of the 44th verse, a very simple rule is given for this that the rising positions from Aries to Virgo are the rising positions from Libra to Pisces in reverse order, that is, the rising positions of Virgo are the same as the rising positions of Libra, the rising positions of Leo are the same as the rising positions of Scorpio, etc.

SOURCE PAGE 337

The zodiac signs in Lanka are in the ascending arc in Prayag, the zodiac signs are 1 Aries 1675 - 333 1342 12 Pisces 2 Taurus 1764 - 270 1524 11 Aquarius 3 Gemini 1631 - 111 1820 10 Capricorn 4 Cancer 1631 + 111 2042 6 Sagittarius 5 Leo 1764 + 270 2064 8 Scorpio 6 Virgo 1675 + 333 2008 7 Libra

To explain its origin, it will be sufficient to state only what the ascensions of the tulā are.

From Figure 62 it is revealed that the time taken for the part of the meridian of the ecliptic to appear on the horizon of Prayag from the autumn equinox is the same time taken for the part of the equator of the tropics to appear on the horizon.

but = s + q

If the point of ka is taken as the final point of the tulā, then sha will be equal to 30 degrees of sha. sha and cha form a right-angled spherical triangle because ka is the polar circle of ka, which is at a right angle to the equator. Therefore, according to Napier's rules in this right-angled spherical triangle

sine (arc) = versed sine (arc) X co-verse (arc)

Sūrya Siddhānta (chash) = cojya23°27' / cospare30° =.5267 [See page 307]

∴ chash = 27°54.5 = 1674.5 jola in the sky is the rising of Virgo.

SOURCE PAGE 338

This figure is similar to 56, 60 figures, the only difference is that here is the position of the solstice where the meridian becomes south of the tropics. The point of the meridian of the meridian of the meridian.

To know the value of the arc of the quadrant, a right-angled spherical triangle should be used in which the point of the quadrant is the south of the meridian. This is the same as 11026' when the declination is 30°. The angle of the quadrant is the angle between the equator and the horizon, which is the same as the longitude of Prayag (see page 257).

Therefore according to Napier's rule

sine (qp) = sin(qa) x cos(qp) = sin11°26' x cos(60° -25°25') = sin11°26' x sin25°25' = .0666. qp = 5°33' = 333' [see page 306] Therefore qp = 1674.5 + 333 = 2008 arc

SOURCE PAGE 339

Therefore, the sines of the ascensions of Libra (at the conjunction) are the same as the sines of the ascensions of Virgo.

Similarly, it can also be proven that the ascensions of Scorpio, Sagittarius, etc., are in the same order as the ascensions of Leo, Cancer, etc.

Bhogansh Vishuvansh Kranti Uttar Ansh Kala Ansh Kala Ansh Kala 0 0 0 0 0 0 30 0 27 55 11 26 60 0 57 46 20 10 90 0 90 0 23 27 120 0 122 11 20 10 150 0 152 51 26 180 0 180 0 0 0 Kranti Dakshin 210 0 207 55 11 26 240 0 237 46 20 10 270 0 270 0 23 27 300 0 302 11 20 10 330 0 332 51 26 360 0 360 0 0 0

SOURCE PAGE 340

From the table above it will be revealed that the portions of the 12 principal points of the ecliptic, the obliquity, and the ecliptic are.

It is revealed that by simply adding the rising points of the zodiac signs like Aries, Taurus, etc., in Lanka, which are the ascensions, the obliquity comes, but keep in mind that if you want to know the obliquity of a point of the ecliptic that is not among these 12 principal points, the ratio method will not work, because some inaccuracy occurs. The best way for this is to use the method of Figure 56 and Equation (1).

From what has been written so far, the ascensions of the zodiac signs can be known, but there is also a promotion of the Nakshatras whose beginning is from the first points of Aries and Ashwini. This point was the place of the Spring Equinox in Vikram's 6th century (see the commentary of 6-1 shlokas). Therefore, it is necessary that the ascensions of the Nakshatras also be mentioned briefly.

It is stated that if the precession angle is subtracted from the sine portion, the precession portion comes, but the precession angle increases by about 58.66 per year (see page 50) and at the time of the Aries solstice of 1682 AD, it was about 22041'. (see page 252). For convenience, the calculations of the eccentrics have been omitted, which makes very little difference in practice. Assuming the precession angle is 22041' means that when the sine portion is 22041', the Nakshatra portion is zero, that is, when the Nakshatra of Aries begins, and when the sine portion is 52041', the Nakshatra of Aries ends and the Nakshatra of Taurus begins. Similarly, the Nakshatras of Gemini, Cancer, etc., should be determined.

To know the rising of Aries for the Navamsha, it is necessary to see how much time it takes for the part of the ecliptic that is between 2041' and 52041' sine portions to rise. Therefore, it is first necessary to know how much time the part between the Spring Equinox and the initial point of Navamsha Aries rises. Then it is necessary to know how much time the part between the Spring Equinox and the final point of Navamsha Aries rises. The difference between the two is the rising of Navamsha Aries. For this, the inclinations of the initial and final points of Navamsha Aries need to be known.

SOURCE PAGE 341

sine of declination = sine 22'41" X sine 23°27' [page 305] = .3856 X .3676 = .1534 ∴ The declination of the initial point of Aries = 8° 46' The declination of the initial point of Taurus = sine 520 41' X sine 23°27' = .3165 ∴ The declination of the initial point of Taurus = 18° 27' This is also the declination of the final point of Aries. The sine of the obliquity of the initial point of Aries = sine 2204' X cotangent 23°27' cotangent 8°46' = .3856 x .6175 = .6882 = .3580 ∴ Obliquity = 20°56' = 1256' ∴ The rising of the Ayan part in Lanka = 1256 The chord of the initial point of Aries of the initial point of Aries in Prayag = spread 8°46' X spread 25°25' = .1551 X .4752 = .0737 ∴ Chord = 4°14 ∴ The chord of the initial point of Aries = 254 ∴ The rising of the Ayan part in Prayag = 1256 - 254 = 1005 Similarly the sine of the chord of the final point of Aries or the chord of the initial point of Taurus = sine 52041' X cotangent 23°27' = cotangent 18°27' = .7653 X .6175 = .6486 = .7662

sine of the obliquity

SOURCE PAGE 342

∴ viṣuvāṃśa = 50°17’ = 3017’ Therefore the rising of the Ayan bhag and Nirayana Mesh in Lanka = 3017 But the sine of the final point of Nirayana Mesh in Prayag = Spere 18°27’ X Spere 25°25’ = ‘3337 X 4752 = .1586 Charaansh = 6°7’ = 54°7’ ∴ The rising of the Ayan bhag and Nirayana Mesh in Prayag = 3017 - 547 = 2470

> but in Prayag the rising of the declination = 1005 ∴,, the rising of Aries = 1465 ∴,,,,, the rising time = 244 minutes

= 4 hours 4 minutes

If the method of knowing the rising of every Nirayana rashi is written with the above description, there will be no special advantage except for the increase in the size of the book. The following table has been made in the same manner so that it will be known how the rising of the Nirayana rashi or the time of rising can be calculated in a place:

The sign of the first part of the revolution in the second column indicates that the revolution is in the North direction and the sign indicates that the revolution is in the South direction. The order of the revolutions of the seven signs from Libra is like the order of the first 6 signs only with a difference in direction.

The Charaansh of the initial point of each sign in the third column is given, and the number obtained when written in arts is the Charaasu or Charpran of that point. When the revolution is North, the Charaasu has to be subtracted in the Northern Hemisphere and when the revolution is South, the Charaasu has to be added in the Northern Hemisphere (see pages 206-210). That is why Charaansh is written as negative in the first 6 signs and positive in the last 6 signs. This is the Charaansh of Prayag. To know the Charaansh of other places, one should write the latitude of the desired place in the formula of Charajya = Revolution tangent X Latitude tangent and calculate.

SOURCE PAGE 343

Nirayan Rashiyas etc. point's radiance in Prayag, etc. point's circumscription art fraction art section section etc. point's visuvansh Lanka in Udayansh Prayag in Udayansh Prayag in Udayansh (Nakshatra) ghadi pal Prayag in the time of rise from Aries etc. pal fraction art fraction art fraction art fraction art art ghadi pal Aries + 8 46 - 4 14 - 4 53 20 56 26 18 24 25 4 4 Taurus + 18 27 - 6 7 - 2 40 50 17 31 43 29 4 51 8 Gemini + 23 15.3 - 11 47 + 0 58 80 0 32 30 32 28 5 35 Cancer + 21 32.7 - 10 46 + 4 2 114 30 30 33 34 35 5 46 Leo + 13 58 - 6 47 + 5 24 145 3 28 14 32 38 5 36 Virgo + 2 54 - 1 23 + 5 37 17 27 42 31 16 5 33 25 Libra - 8 46 + 4 14 + 4 53 200 26 18 31 5 42 31 25 Scorpio - 18 27 + 6 7 + 2 40 230 17 31 43 23 5 40 42 Sagittarius - 23 15.3 + 11 47 - 0 58 0 32 30 33 28 5 15 48 Capricorn - 21 32.7 + 10 46 - 4 2 264 30 30 33 26 31 52 31 Aquarius - 13 58 + 6 47 - 5 24 24 325 3 28 14 22 50 56 Pisces - 2 54 + 1 23 - 5 37 17 27 42 24 5 3 41 60 Aries + 8 46 - 4 14 380 56

SOURCE PAGE 344

The arc given in the fourth column is the difference between the arcs of the initial points of the two adjacent signs, by which it is known what difference occurs in the arc from the initial point of the first sign to the final point. For example—

Aries Arc Section = Taurus Arc Section = − 607 ′ − (4 ° 14 ′) = − 6 ° 7 ′ + 4 ° 14 ′ = − 4 ° 53 ′

The arc of the Leo sign

= Virgo-Ascendant - Leo-Ascendant = - 1° 23' - (- 6° 47') = - 1° 23' + 6° 47' = + 5° 24'

It is apparent from observation that the arc of the first 6 zodiac signs is in the same proportion as the arcs of the other 6 zodiac signs in succession. There is only a difference in the plus or minus signs.

In the 5th column, the latitude of the initial point of each zodiac sign is given. If this is written in terms of arts, then in so many degrees, the initial point of that zodiac sign will rise in Lanka with the vernal equinox. If the difference between the latitudes of the two adjacent rāśiyoṃ is calculated, then these will be the rising latitudes of the upper zodiac sign in Lanka, which is given in the 6th column. If this is written in terms of arts, then this number will be the rising degrees of that zodiac sign in Lanka. In the rising degree of the zodiac sign in Lanka, if the arc of the same zodiac sign is positive, adding it and if negative, subtracting it, the rising degree of that zodiac sign in the auspicious place is obtained, by writing it in terms of arts, the number of rising degrees of that zodiac sign will also be obtained. In the 8th column, the rising time of each zodiac sign is not written in rising degrees but in hours and minutes, which is more practical but some is crude because 6 arts equal 1 minute, and when the whole minutes do not come when dividing by 6, more than half are taken as 1 and those which come less than half are left out.

SOURCE PAGE 345

In the 6th column, it is shown what time it takes for the entire zodiac to rise from the beginning of Aries. For example, if one wants to know the time it takes for the entire Leo to rise from the beginning of Aries, the answer in the 6th column in front of Leo is 25 ghati 52 pal, which means that the signs of Aries, Taurus, Gemini, Cancer, and Leo rise in Prayag in 25 ghati 52 pal. This column will be of great help in knowing the ascendant. Therefore, this is also given here.

It is necessary to note that this time is according to the sidereal value which is somewhat different from the sidereal value. (See pages 7, 8).

This table proves that the necessary corrections to know the rising of the zodiac signs in a place can be made easily by knowing only the arc. But this table cannot always be useful because due to the motion of the ecliptic, the portions of the beginning point of each declination sign and the obliquity keep increasing. This causes some difference in the declination, arc, and arc segments. But this difference is very subtle because due to the motion of the ecliptic, only about 1 art increases in the portion every year, so it is necessary to make a correction in the table once in at least 25 years later.

To know at what time which point of the ecliptic is in the east horizon as the ascendant

> गतगम्यासवः कार्या भास्करादिष्टकालिकात् ।
> स्वोदयासुहता भुक्तभोग्या भक्ता खवह्निभिः ॥४५॥
> अभीष्टघटिकासुभ्यो भोगयासून् प्रविशोधयेत् ।
> तद्वदैष्यभलग्नासून् एवं यातांस्तथोत्क्रमात् ॥४६॥
> शेषं त्रिंशत्क्रमाभ्यस्तमभुक्तोदयभाजितम् ।
> भागादिहीनं युक्तं च तल्लग्नं क्षितिजे तथा । ४७॥

(45) At the time when the ascendant is to be known, the apparent Sun and the actual Sun should be known. The portion of the Sun that is on the zodiac is called gataansh, and the portion of the zodiac that remains for the Sun to enjoy is called bhogyansh. By multiplying the rising signs of the zodiac by gataansh and dividing by 30, and by multiplying gataansh and bhogyansh by 30 and dividing by 30, bhogyasu are known. (46) From the time the sunrise has passed until the auspicious time, the bhogyasus should be subtracted. The rising signs of the subsequent zodiac are to be subtracted from the remainder.

SOURCE PAGE 346

We should subtract the rising signs of the amounts beyond this from the rest. Similarly, by subtracting the rising signs of the following signs, when the remainder is such that the rising signs of the further amounts do not decrease, this is called the impure amount (the amount that does not decrease). But if one wants to know the ascendant from the setting signs, then the rising signs of the signs that have already risen before sunrise should be subtracted in the time of sunrise, in reverse order, that is, first subtract the setting signs, then subtract the rising signs of the amount that is behind the sign of the sun, then subtract the rising signs of the sign behind that, and so on, (47). Finally, if some remains, it should be multiplied by 30 and divided by the rising signs of the impure amount. If the operation is done from the setting signs, subtracting the quotient from the impure amount, and if this operation is done from the setting signs, adding the quotient, it is known that which point of the ecliptic is the ascendant in the horizon at that time.

Scientific Commentary — The commentators have discussed the addition and subtraction of the precession angle in the latter part of the 47th verse, which is beyond my understanding because when the ascendant is known from the true position of the sun and the true positions of all the planets are done in the zodiac signs, what is the need to know the ascendant from the sun. The confusion arises in this sense that the two ways of finding the ascendant in these three verses are given, which are usually the same. If the time from sunrise to the auspicious time is less than 30 degrees, it will be easier to use the rising signs, and if the auspicious time is close to the next sunrise, it will be convenient to use the setting signs of the next sunrise. That is why the need for adding and subtracting the final result arises. This will be clarified by the two examples below.

Example 1.—What are the ascensions when the sine of the sun's declination is 34516 and the true daily motion of the sun is 57'21" at 16 hours 15 minutes and 52 minutes 10 seconds from sunrise.

Because multiplication and division will be very much needed in calculating the variations, the sine of the sun's declination will only be taken up to the arts later.

1st method—

First, it should be known what will be the sine of the sun's declination at 16 hours 15 minutes from sunrise.

SOURCE PAGE 347

In one hour the sun moves 57’ 21” forward therefore. In 15 hours, 14’20” 15”, and in 1 hour, 57’21”, and in 15 minutes, 14’20”, therefore the sun moves 16 hours 15 minutes 15’32” forward. This

∴ In 16 hours 15 minutes 15' 32", the Sun moves forward. Therefore the apparent portion of the Sun from sunrise = 3°50'16" 38" = 3°50'17"

In the auspicious time, the portion of the sun in the Cancer zodiac is 5017' and the portion of enjoyment is 300 - 5017 = 24043 = 1483'.

But the rising of Cancer is 2075^dagger (in Prayag). Therefore, when 300 degrees or 1800 arts of Cancer rise in 2075 asu, then how much time will 24^°43' or 1483' rise, that is

Bhogyasu = 1483 × 20751800 =1710 therefore Bhogyakaal = 285pal = 4hours45pal

That is to say, the Cancer zodiac will rise for 4 hours 45 minutes until sunrise. Then the rise of the Leo zodiac will begin.

The auspicious time is 16 hours 15 minutes, the time when the shadow of the North is in the zodiac of Cancer is 4, 45, the difference is 11, 30, the time of the rise of the Lion is 5, 36, the difference is 5, 54, the time of the rise of Virgo is 5, 33, the difference is 21 minutes. This is the elapsed time of the scale.

Just as the 60th part of the art is called arcseconds, so the 60th part of arcseconds should be considered prativikalā for which three signs ('') are used.

† If the time of rise is written in minutes instead of the time of rise, the calculation will be simpler, but some crudeness will arise.

SOURCE PAGE 348

Therefore, in the auspicious time, the sign of Libra has risen by 21 minutes and will rise by 5 minutes and 21 minutes because the rising time of Libra in Prayag is 5 hours 42 minutes. Therefore, in the auspicious time, the sign of Libra is in the eastern horizon, which means the ascendant. To know which point of Libra is the ascendant, one will have to use the ratio. Because when 30 degrees of Libra rise in 5 hours 42 minutes, i.e., 342 minutes, then how many will have risen in 21 minutes.

342: 21:: 30: Portion of Libra

∴ Libra's portion = 21 × 30342 = 1050' 31'' = 1051'

.. The 6th ascendant is at 16 hours 15 minutes from sunrise. Here 1051' has been added to the rising signs.

Rari custom—

If one wants to know what is the ascendant at 52 hours 10 minutes from sunrise, it will be more convenient to use the portion of the next day's sunrise.

Ishkalaseglesuryodayakāsamay = 60 hours - 52 hours 10 minutes = 7 hours 50 minutes

Angle of sunrise in the time of the sun's declination = 3501' 6 " + 57' 21 " = 35058' 27 "

60timeinSūrya'smotion = 57'21".73,,,, = 7'10"8'30minutes,, = 16"7'30minutes = 7'26"

Ishkālamēsūryakānirayanbhogāṁśa = 35058' 27'' - 7' 26'' = 35051'

Therefore, the portion of the sun in the Cancer sign at the ishtakal is 5051’ = 351’

Therefore, like before, gatasu = 351 x 2075 / 1800

SOURCE PAGE 349

326 Sūrya Siddhānta = 405 ∴ Elapsed time = 67 minutes = 1 hour 7 minutes. Sunrise is 7 hours 50 minutes before sunrise. The beginning of karka will be 6 hours 43 minutes before the rising of mithuna.

The difference is 1 hour 8 minutes ∴ In the auspicious time, there is 1 hour 8 minutes remaining for the rising of the entire Taurus. However, the rising of Taurus by 300 or 1800' takes 261 minutes. ∴ 261: 68:: 1800: Ascendant's portion 68 X 1800 261 = 421' = 7°1 and Taurus's portion (past portion) = 30° - 7°1' = 220°56' ∴ The ascendant of the auspicious time = 1ra22056'. Here the final remainder is reduced. In this regard, it should be kept in mind that the rising or rising time of the signs are shown in the naked-time and the auspicious time is known according to the sundial, so this happens in the Sawan period (see p. 7, 8 and 212). 1 Sawan day = 60 Sawan hours = 21656.14 asu = 3610 minutes (naked-time) physically = 60 hours 10 minutes (naked-time) ∴ 6 Sawan hours = 6 naked-time hours + 1 naked-time minute, which proves that to convert the Sawan period to the naked-time, by adding 1 minute for every 6 Sawan hours, the naked-time is obtained. But in calculating the ascendant by taking the true Sun, the auspicious time

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SOURCE PAGE 350

When the clear sun of the auspicious time is extracted, the question remains that what is the difference between the place where the sun is in the auspicious time and the point on the horizon up to the rising point of the ecliptic? The part of the ecliptic that is between the instantaneous or auspicious time sun and the rising point of the ecliptic is the same as the number of Nakshatra periods in which the sun travels from the rising time to the auspicious time from the place of sunrise. Yes, if one wants to know how many Nakshatra periods have passed from the rising time to the auspicious time, then one will have to calculate the distance from the point where the ecliptic was rising at the time of sunrise to the auspicious rising point. The rising point of the ecliptic is some west of the sun in the auspicious time because the sun moves some east until the auspicious time. This consideration must be made at the time when the ascendant of the auspicious time is to be found from the declination of the rising sun. An example of how to find the ascendant is also given below:

In this way -

What is the ascendant 16 ghaṭīs and 15 palas after sunrise?

The solar time point of the ecliptic becomes some west of the Sun at the auspicious time because the Sun moves some east until the auspicious time.

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Therefore, Karkakabhogyansh = 30° - 501' = 24056' = 1466'
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1800: 1466:: 2075: Bhogyasu

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∴ Bhogyasu = 1466 × 20751900 = 1728 = 288pa
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16 ghadi 15 pal is according to the sundial, therefore it is in the unit of Sawan period.

Sawan Nakshatra

6 ghadi = 6 ghadi 1 pal

∴ 16 ghadi 15 pal = 16 ghadi 18 pal(gross form)

Now in this, one should subtract the time of the bhogyāsu of Cancer and Lion, and the time of rising of Virgo respectively from the east.

16 ghadi 18 pal Cancer's bhogyakāla 4 ghadi 48 pal difference 11 ghadi 30 pal Lion's udayakāla 5 “36 pal

SOURCE PAGE 351

Difference 5 hour 54 minutes

Rising time of Virgo 5,, 33 minutes

Setting time of Libra 21 minutes

The calculation after this is the same as before.

It is proven that whether the sidereal time is taken by knowing the portion of the sun's declination and considering the sidereal time as the nakṣatrakāla, or the true time is taken by knowing the portion of the sun's declination and converting the true time to the nakṣatrakāla, there is no difference between the two. Yes, there is some ease in working by making the sidereal time from the sidereal time.

In this method, the rising time of each zodiac sign must be subtracted from the true time. If the 320th page is taken from the 6th column of the table, there may be even more convenience.

Time of the bhogyakāla of Cancer in sunrise = 4 hours 48 minutes

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Parantukarkkaudayakaal = 5hours46minutes
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difference between both = 0hours58minutes
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∴ Time of the past of Cancer in sunrise = 0 hours 58 minutes

Time of rise from the beginning of Aries to the end of Gemini in Nirayana = 14 hours 30 minutes

∴ Time of rise of the rising part of the ecliptic in sunrise = 15 hours 28 minutes auspicious time 16 hours 18 minutes

The rising time of the ascending part of the ecliptic in the auspicious time = 31 ghadi 46 pal

From which the rising time to Virgo may decrease because it is 31 ghadi 25 pal

∴ The motion of Libra = 21 pal

That is, in the auspicious time, the sign of Libra is the ascendant. To know which point of the sign of Libra is the ascendant, one should also perform the action further, as before.

Bhaskaraacharya has described the method of finding the ascendant from the Sun, which is purer because it has been stated that the 30 degrees of any sign do not rise at the same time, so it is appropriate that the rising time of each degrees

SOURCE PAGE 352

They should go separately. But this work is troublesome, so if the rising time of each part is taken as equal and the ratio is used as everyone does, then there will be no difference in the sign of the ascendant. Yes, there will be a slight difference in determining the rising point of the sign. Therefore, if the purity of the ascendant or the dodecimal portion is to be known, one should work as before with the

sine of the Sun. In such a situation, performing the correction of the precession angle will give the knowledge of the Nirayana ascendant.

Method of knowing the mean ascendant—

> प्राक्पश्चान्ततनाडोभिः तस्माल्लंकोदयासुभिः ।
> भानौक्षयधने कृत्वा मध्यलग्नं तथाभवेत् ॥४८॥

U, P, D, P = North, West, South, and East points of the horizon of Prayag. K V ka = Ecliptic. U Dh Kh M Da = Meridian. Ka = Rising Ascendant. K = Setting Ascendant. M = Mean or Tenth Ascendant. V = Vasant Sampat. Vi = Vitribha Ascendant. Kh = Khastikas.

SOURCE PAGE 353

Translation—(88) Know the segment between the arc of light between the instantaneous Sun and the meridian circle from the rising of the Sun and Lanka in the east or west. If it is the eastern meridian, subtract it from the instantaneous Sun, otherwise add it, then the mean ascendant can be found.

Scientific Commentary—The point of the declination circle that lies on the meridian is called the mean or the decimal point. The point of the declination circle that is very close to the cross is called the tritibha. The tritibha is found by subtracting 3 signs in the rising sign or adding three signs in the setting sign.

The time it takes for the rise of signs in Lanka is the same time it takes for them to cross the meridian. This is the same for all places. For example, the rising portion of the Aries sign in the Nirayana in Lanka is 26°18'. Therefore, the rising of Aries in Lanka was 1758. In the same time, the Aries sign crosses the meridian in all places. Similarly, one should understand about the other signs.

The reason for this is that the rising portion of a sign in Lanka is the segment of the ecliptic for which it takes the same time to rise as it takes to come above the horizon. It also takes the same time to cross the meridian for this segment of the ecliptic. Therefore, this sign will also take the same time to cross the meridian.

Example 2.—What are the mean ascendant after 16 ghadi 15 pal and 52 ghadi 10 pal from sunrise when the portion of the Sun in the Nirayana is 34516" and the true daily motion of the Sun is 57'21"?

First Section—

First, one must know what the time difference between sunrise and the local time is, i.e., how much before or after this time is exactly noon. Therefore, it is necessary to know how many hours and minutes the noon will be from sunrise. For this, the calculation of the variable is necessary. However, the variable depends on the revolution of the sun and the latitude of the place. Therefore, one must first know what is the revolution of the sun at the time of sunrise.

SOURCE PAGE 354

The portion of the sun's declination = 3 Ra 501'6" precession angle = 22041'

The sine portion of the Sun = 3 R 50°27'42' = 117°42' sine of declination = sine 117°42' X sine 23 27' = sine(180° - 117°42') X sine 23 27' = sine 62018' X sine 23°27' = .8854 X .3676 = .3523 declination = 20°38' north declination = spar 20°38' X spar 25°25' = .3766 X .4752 = .1760 arc = 10°18' = 618' time =

618 asu = 103 shadow = 1 hour 43 shadow half-day = 15 hours + 1 hour 43 shadow = 16 hours 43 shadow which means the meridian will be exactly at 16 hours 43 shadow from sunrise. But the auspicious time is 16 hours 15 shadow when the sine portion of the sun is 50°16'38" or 50°17' (see the scientific commentary of verses 45-47). Therefore the preceding time = 28 shadow = 168 asu the sun is in the Cancer sign which has an Udayasus of 1833 in Lanka (writing the value of the 6th column of Sarini in terms of arts gives the number of asus). Now it is to see that when 1833 asus violates the entire Cancer sign, i.e., 1800 artitudes, then which part of the Cancer sign will be violated in 168 asus.

1883: 168:: 1800: Mean Part.

∴ desired part & = 168 × 18001833 & = 165 ′ & = 2045 ′

This is the section between the meridian circle and the circle of declination of the sun. But the sun is at 3^(5017)'. Therefore 3^(5017)' - 2045' = 32032' is the meridian ascendant.

SOURCE PAGE 355

The second section— when the time is 52 ghadi 10 pal then western sidereal time = 52 ghadi 10 pal - 16 ghadi 43 pal, that is, after noon, 35 ghadi 27 pal will remain after sunrise by 52 ghadi 10 pal. At this time the sun's declination = 3 ra 5051’ therefore the bhogyāṃśa of Cancer sign = 24°6’. When the entire Cancer sign violates the meridian circle in 1833 asu, how many asus will it violate its 24°6’? 1800: 1446:: 1833: time of violation of the bhogyāṃśa. Time of violation of the bhogyāṃśa = 1446 x 1833 1800 = 1476 asu = 246 pal = 4 ghadi 6 pal. Now western sidereal time = 35 ghadi 27 pal. The difference in the violation of Cancer is 4 ghadi 6 pal after noon. The difference in the violation of Leo is 26 ghadi 36 pal. The difference in the violation of Virgo is 4 ghadi 37 pal. The difference in the violation of Libra is 4 ghadi 53 pal. The difference in the violation of Scorpio is 5 ghadi 17 pal. The difference in the violation of Sagittarius is 5 ghadi 25 pal. The difference in the violation of Capricorn is 5 ghadi 5 pal. The difference is 1 ghadi 21 pal. Therefore, the ascendant is on the meridian circle of Aquarius. Because the last difference does not decrease the time of violation of Aquarius, so this is the impure sign. Now

SOURCE PAGE 356

Time of violation of the meridian of Kumbha = 4 hours 42 minutes shadow = 1664 asu

1 hour 21॥ shadow = 81॥ shadow = 486 asu

Therefore, when 1800 degrees of violation occurs in 1664 asus, what will be in 486 asus.

1694: 486:: 1800: gtaansh

∴ gtaansh = 486 × 18001694 = 520 ′ = 8 ∘ 40 ′

∴ Aquarius 8°40’ has crossed the meridian

∴ Chapter on Mean Motions = 108040'

Knowing the time from the true Sun and ascendant-

> भोग्याशेनूनकस्याथ भुक्तासूनधिकस्य च ।
> संपिण्ड्यान्तरलग्नासूनवं स्यात्कालसाधनम् ॥४६॥
> सूर्यार्द्वने निषाशेषे लग्नेऽर्काधिके दिवा ।
> भचक्रार्धयुताद्भानोः अधिकेऽस्तमयात्परम् ॥५०॥

(46) Add the ascensions of the zodiacs that are less than in the signs of the ascendant and true Sun, and the ascensions of the signs that are more than in them, and the full zodiacs between both. The ascensions of these are known as the auspicious times. (50) When some night remains, that is, behind midnight and before sunrise, the zodiac of the Sun is less than the zodiac of the ascendant, the zodiac of the Sun during the day is less than the zodiac of the ascendant, and even by adding 6 zodiacs to the zodiac of the Sun behind sunset, the zodiac of the ascendant is more.

The translation of the third chapter named Vitprashnādhikāra has been completed.

Scientific Commentary—the method by which the rising ascendant is known from the zodiac of the sun, which is given in verses 45-47, the opposite (reverse) action of which is explained in verses 46-50. Therefore, there does not seem to be a need to explain its derivation. In the daytime, the rising ascendant is ahead of the zodiac of the sun, so the zodiac of the sun is less than the ascendant. After midnight, the sun remains ahead of the rising ascendant, so at that time the ascendant is less than the sun. At the time of sunset, the rising ascendant remains exactly 6 zodiacs ahead of the sun, so after this, the rising ascendant is greater than the true sun (ṣaḍbhā sun) which is 6 zodiacs advanced.

SOURCE PAGE 357

The time that comes behind the midnight is the time from that time till sunrise according to this rule, and the time that occurs in the day or after sunset is the time from sunrise till that time. This rule will be clarified with an example:-

Example—The clear sunrise time of the Sun is 3501'6

The clear daily motion is 57'21'

First section—

Here the bhogyāsu of the clear Sun is more than the ascendant, so the bhogyāsu and bhuktāsu of the sign of the Sun and the sign of the ascendant should be added. These bhogyāsu and bhuktāsu should be known like before.

Second section—

Here the rising ascendant is less than the sign of the Sun. Therefore, the bhogyāsuoṃ of the rising ascendant should be added to the bhuktāsuoṃ of the sign of the Sun. Their values need to be known as before. The time when the ascendant will be 1^(22056)' is the time after midnight, so the bhuktāsuoṃ of the true Sun of the next sunrise should be taken.

The true Sun = 3^(501)'6' of the same day's sunrise time

True daily motion

```
Sūrya Siddhānta = 3451' 6' = 57° 21'
```

∴ The true Sun of the next day's sunrise time = 3^(5058)'27” = 3^(5058)' in substance

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∴ ! Sūrya Siddhānta = 5 ° 58 ′ = 358 ′ Lagnabhogyaansh = 30 ° − 220 ° 56 ′ = 7 ° 1 ′ = 421 ′ Suryakebhaktasu = 358 × 20751800 = 413 = 66pal

Lagnakabhogyaakaal = 421 X 261 / 1800pal = 68pal

∴ Sūrya Siddhānta and Lagna bhogyakāla = 66 + 68pala = 2ghaḍī17pala

Sūrya and Lagna between Gemini zodiac rising time = 5 hours 35 minutes both sum = 7 hours 52 minutes

Therefore, there is 7 hours 52 minutes remaining for the sunrise.

The method of knowing the time from the ascendant can only be practically used when the identification of the signs and constellations is good. Therefore, it is necessary that there is complete information about the signs, constellations, and other famous stars. In the eighth chapter of Sūrya Siddhānta, there is a discussion of some constellations and stars in the chapter named Yutyaadhikar, so there it will also be stated which are the famous stars from which the time in the sign can be easily known.

Here, it is sufficient to state that knowing the time from the mean ascendant is more convenient. If it is known what was the declination of the sun at noon and which star, whose declination is known, is on the meridian circle at night, then it can be easily known how much time has passed since noon because the difference between the declination of the star and the declination of the sun is the same in the degrees, in which that star comes after noon on the meridian circle.

If the declination of a star is not known, the gross knowledge of time can be obtained by just identifying the star groups of the ecliptic. For this, it is also necessary to know which constellation the sun is in. It has already been said that the 27th part of the ecliptic is called a constellation and the entire ecliptic is a

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The Earth seems to revolve in the constellation during the day, therefore a constellation violates the meridian in 60 ghaḍī = 23 ghaḍī or two and a quarter ghaḍī or the meridian, or 6 constellations violate the meridian in 20 ghaḍī or 8 hours. From this calculation, the time known will not differ from the actual time by more than half an hour.

Example—if the Sun is in the constellation of Punarvasu, at what time will it be on the meridian of the śravaṇa constellation?

In the Nakshatra table of 213 pages, Punarvasu is the 7th Nakshatra and Shravana is the 22nd Nakshatra. Therefore, there is a difference of 15 Nakshatras between these two.

The difference between 6 constellations is 20 degrees or 8 hours.

Sūrya Siddhānta, Scientific Commentary, Chapter on Mean Motions, Chapter on True Positions, Chapter on the Three Problems

It will be on the meridian 133 hours behind or at 1:15 in the night from midnight the śravaṇa nakṣatra meridian.

Nowadays, time is not known from the sundial but from clocks that run on the force of a chain, whose time is somewhat different from the sundial, so those who calculate the ascendant from the clocks prevalent nowadays and tell the results of astrology do not have exact knowledge of the ascendant. Besides the Panchang of Mahamohopaladhyaya Bapudevji Shastri of Kashi, no other Panchangs have come where this point is well discussed. Therefore, it is very necessary here to state what the relationship is between the sundial and the chain clocks.

True time, mean time, and time-equation—

In Chapter on Mean Motions pages 7, 8, it is stated that the time from the time a star rises until it rises again is called Nakshatra day and the time from one sunrise of the sun to another sunrise is called Sawan day. However, knowing the exact time of rising is very difficult because for this work, there should be a horizon where there are no trees, etc., which is generally impossible for everywhere because such a plain is found very rarely where there are no trees in the east or west direction for many kosa. Even if such a horizon is found, the rising of the sun or star happens some time before the proper time due to the bending of the rays of light, the magnitude of which varies according to the different conditions of the atmosphere.

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is. Therefore, for very fine calculations, the time is not tested from the time of sunrise but from the time of noon. Therefore, the definition of Sawan or Nakshatra day is made as follows today:-

The time from the moment the center of the Sun comes to the meridian circle until the moment its center comes to the meridian circle is called 'Spatha Sawan day'.

The time from the moment the Spring Equinoctial point comes to the meridian circle until the moment this point comes to the meridian circle is called 'Nakshatra day'.

The motion of the Spring Equinoctial point is generally the same. Therefore, the Nakshatra day is always the same*. However, a lot of difference occurs in the magnitude of the Sawan day because the daily motion of the Sun constantly changes. This can be easily found by clocks that indicate the Nakshatra period. If a clock is made such that when the Spring Equinoctial point comes to the meridian circle, it shows exactly 12 o'clock, such a clock is called Nakshatra clock (Ghatika Yantra). With such a Ghatika Yantra, it is easily known how much difference occurs in the magnitudes of the Sawan days. For example, four Sawan day magnitudes are given for 1606 AD:-

1606 AD(Nakshatra period) hour minute second 1st January's true noon to 2nd January's true noon 24 44 24.6 2nd April's true noon to 3rd April's true noon 24 3 38.5 3rd July's true noon to 4th July's true noon 24 4 7.5 2nd October's true noon to 3rd October's true noon 24 3 37.6

Due to the deviation of the axis, the position of the vernal equinox becomes mean or western before the true position (see page 247). However, this does not cause such a small difference in the magnitude of the sidereal day that it is not a loss to consider it as such.

†Ball's Spherical Astronomy page 215

SOURCE PAGE 361

From this it appears that the value of the Sawan day is not the same. The value of the Sawan day from the noon of January 1st to the noon of January 2nd is 46 4 seconds larger than the value of the Sawan days of the second and third April, etc. It is impossible to make a clock that changes its pace according to the motion of the sun because a clock running by mechanical force will always run at the same pace. Therefore, the time known by such clocks is different from the time of a sundial because the value of the Sawan day is known from the sundial, which changes every day. If a clock that tells the sidereal time is used, it is not convenient in practical behavior because the sidereal time of 24 hours is about 4 minutes shorter than the 24 hours of the solar day. Therefore, if the solstice of the sun today is exactly 12 o'clock in the sidereal clock, then the solstice of the sun tomorrow will be at 12:04 in the sidereal clock. In this way, the solstice of the sun falls four minutes behind each day, and in 15 days, the solstice of the sun will be at 1 o'clock in the sidereal clock, in 1 month, the solstice of the sun will be at 2 o'clock in the sidereal clock, and in two months, it will be at 4 o'clock in the sidereal clock, etc. Thus, it is evident that the rising and setting of the sun may occur at any time of the day according to the sidereal clock, which may not be useful for practical behavior because time is generally determined by the rising and setting and solstice of the sun.

To eliminate this dilemma, astrologers have decided that for the work of astrology, such clocks should be used that indicate the sidereal time, but the clocks should also have worldly behavior so that they indicate the hours, minutes, seconds, or clock of the mean Sawan day. Doing this, the time of these clocks will necessarily differ from the sundials that indicate the true Sawan day, but this difference does not exceed 1: minute. The value of the mean Sawan day is the mean (average) of the true Sawan days of several years. The mean value of the four days written above 1606 AD is 24 hours 3 minutes 57.1 seconds, which is very close to the mean value of a Sawan day. If the mean value of the values of several years of true Sawan days is calculated, then a mean Sawan day is similar to the sidereal time of 24 hours 3 minutes 56.555 seconds. The examples below will clarify how the mean Sawan day value is known by observation. Determining the value of the mean Sawan day

On the fourth day of July of 1836, at the time when the center of the true Sun was in the meridian

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On the circle at that time its true right ascension was 6 hours 54 minutes 703 seconds. Similarly, on the day of July 4th of 1860 AD or in the time of Yamottaro-Llanghankal, the true right ascension of the Sun was 6 hours 53 minutes 54.61 seconds. Determine the value of the mean Savan day from this.

First it is necessary to know how much time passed in the sidereal time from July 4th of 1860 AD at 6 hours 53 minutes 54.61 seconds (sidereal time).

It is clear that in one year, that is, in the time from one sidereal Aries solstice to the next sidereal Aries solstice, the Vernal Equinox passes one time less than the Sun. This is because due to the motion of the Earth, the Sun moves one degree towards the east every day, causing it to pass one less time than the Vernal Equinox by approximately 4 minutes each day. Thus, in one year, the Sun falls behind by 1 day, meaning the Sun's passage of the Aries solstice is one less than the Vernal Equinox's passage of the Aries solstice.

From July 4th of 1836 AD to July 4th of 1860 AD, there are 54 years in which there are 13 more years (leap years) such as 1840, 1844, 1848, and the remaining 41 years are ordinary years. Therefore, this period is equal to 41 X 365 + 13 X 36 = that is, equal to 19723 Savan days. It has been proven above that in one year, the passage of the Vernal Equinox is one time more than the passage of the Sun, so in 54 years, the passage of the Vernal Equinox will be 54 times more. Thus, in the above period, there were 19723+54=1977 sidereal days. Therefore, from the time of the Sun's passage of the Aries solstice of July 4th of 1836 AD to the time of the Sun's passage of the Aries solstice of July 4th of 1860 AD, there were 19777 days 6 hours 53 minutes 54.61 seconds - 6 hours 54 minutes 7.03 seconds, that is, 19776 days 23 hours 56 minutes 47.58 seconds in sidereal time.

Therefore, this sidereal period was equal to sixteen thousand seven hundred twenty-three clear Sawan days. Now if the above sidereal period is divided by sixteen thousand seven hundred twenty-three, the value of one mean Sawan day in the sidereal period is 24 hours 3 minutes 56.555 seconds. Therefore

The part of the meridian that the polar circle reaches at the point where it meets the tropics is called the meridian part of the center of the sun. This fraction is revealed in degrees, arts, arcseconds, and also in hours, minutes, and seconds. One degree is equal to 4 minutes or 10 minutes of arc.

SOURCE PAGE 363

1 medium Sawan day = 24 hours 3 minutes 560555 seconds (Nakshatra)

The difference between mean and true Sawan days is explained by astrologers by imagining a sun that is considered to move at the same speed on the ecliptic and comes behind the sidereal time of 24 hours 3 minutes 54.555 seconds every day on the meridian. The moment this imagined sun crosses the meridian is the mean meridian, and it is exactly 12 o'clock in the clocks indicating the mean period. It is shown that the mean value of the true Sawan days throughout the year is the same as the mean Sawan day, so it is apparent that the time taken by the aforementioned imagined sun to complete one revolution on the ecliptic is the same time taken by the true sun to complete one revolution on the ecliptic. Therefore, the mean daily motion of the true sun on the ecliptic is the motion of this imagined sun on the ecliptic. It is proven that in the same time, the portion of the imagined sun on the ecliptic increases as much as the portion of the true sun increases.

It is revealed from the table of 320 pages that as long as the portion of the true Sun is less than 60 degrees, its declination is less than the portion of the mean. However, the declination of the hypothetical Sun is always equal to the mean portion of the true Sun. Therefore, it is proven that as long as the declination of the hypothetical Sun or the mean portion of the Sun is less than 60 degrees, the polar transit of the hypothetical Sun is to the east of the polar transit of the true Sun. Therefore, the polar transit of the true Sun comes before the hypothetical Sun on the meridian circle and the true meridian comes before the mean meridian. Therefore, at the time when the sundial indicates 12 o'clock according to the true Sun, the clocks indicating the mean time will also show 12 o'clock. That is, the sundial will be faster than the mean clock. The faster it is, the more the time of the mean clock will be found by subtracting from the sundial time.

Similarly, as long as the mean portion of the true Sun is more than 60 degrees and less than 180 degrees, that is, when the Sun's sine will be between the sine of Cancer and the sine of Virgo.

It becomes clear from this table that if the declination of the imagined Sun is taken as the solstice and the declination of the true Sun is taken as the solstice, then at what time the true Sun's pole-transit is ahead or behind the imagined Sun's pole-transit.

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Until then, the polar transit of the true Sun remains to the east of the hypothetical Sun. Because the latitude of the true Sun will be greater than the latitude of the hypothetical Sun, which is equal to the mean portion of the true Sun (see Table 320). In this situation, the true Sun will pass behind the hypothetical Sun, meaning the sundial will remain behind the mean hour (slow or sluggish). Therefore, by adding the difference of the latitudes of both in the time of the sundial, the time of the mechanical clock (mean clock) will be found. Similarly, when the true Sun is in three zodiac signs, the sundial will be ahead of the mean hour, and by subtracting the difference of the latitudes of the true Sun and the hypothetical Sun from the time of the sundial, the time of the mean clock will be found, and when the true Sun is in three zodiac signs, the difference of the latitudes of both in the time of the sundial will be added to find the mean time.

But the true Sun does not move with the same speed on the true Sun circle always. Sometimes its speed becomes fast and sometimes slow. Therefore, for this reason also the true Sun will not come to the true Sun meridian at the time when the mean Sun comes, as stated above. The true Sun and the mean Sun stay together only at the apogee and nadir of the true Sun circle (see pages 80-82). When the Sun moves ahead of the apogee, the daily motion of the true Sun is less than the daily motion of the mean Sun, so the true Sun falls behind the mean Sun, that is, the true Sun remains ahead of the mean Sun towards the east. Therefore, the true noon comes before the mean noon. For this reason also the time of the sun-dial remains ahead of the mean time. This condition remains until the Sun reaches the nadir. From here on, the motion of the true Sun is greater than the mean motion, so the true Sun remains ahead of the mean Sun towards the east. Therefore, the true noon is behind the mean noon, that is, the sundial remains slow compared to the mean time.

Due to these two reasons, that is, the movement on the true Sun circle and the difference in daily motion, there is some difference between the true time and the mean time. The time taken to subtract or add from the true time to find the mean time is called the 'Equation of Time'. It is also written as:-

Mean Time = True Time + Equation of Time*

* Venkatesh Vapu Ketkar has named it Udayantara in his astronomy (Jyotg. Ga. page 75)

SOURCE PAGE 365

When the time equation is positive, it is added and when it is negative, it is subtracted.

Determining the equation of time-

Now it is proven that the difference between the hypothetical Sun's obliquity and the true Sun's obliquity is called the time equation. Therefore, the time equation can be easily derived according to the method written below:-

It is shown in pages 302-303 that

```
Declination line = ultimate meridian / sine of declination line
```

If the obliquity of the ecliptic is taken as v, for the ultimate revolution as k, and for the true sine portion as b, then

s p r e v = k o t i j y a k X 1 kosvare bh = k o t i j y a k X s p r e v (9)

This equation is in the form in which the equation relating the true center and the eccentric is indicated [see page 164 equation(3)]. Therefore, this equation may also be in the manner written in pages 164-168. In this way, it can be proven that

```
2v = 2b + 2 (- spere2k / 2sin2b + 1 / 2spere4k / 2sin4b - 1 / 2spere6k / 2sin6b +..... )
```

or

```
v - b = - spere2k / 2sin2b + 1 / 2spere4k / 2sin4b - 1 / 3spere6k / 2sin6 +... ( 2 )
```

```
here - spread2k / 2 = - 1 - cosjac / 1 + cosjac = cosjac - 1 / cosjac + 1
```

Therefore, just as the value of p was determined in page 167, similarly here the value of spre2 k/2 has come.

Each term of the equation (2) is in angular values (radian) (see page 162). Therefore, if we multiply its right-hand side by 3437.75 then

---

SOURCE PAGE 366

The value of V-B will be known in the arts and in the arcs. From this formula we can easily know the declination of the sun's any fraction of sine.

It is stated (see page 306) that there is no loss in taking the value of 3027' up to the first half of the 21st century of the supreme revolution of the sun, therefore

```
SPRE2k / 3 = SPRE23° 27' / 2 = SPRE11° 43' .5 = ( 2075 ) 2 = .04305
```

```
SPRE4  k2 = (.04305 ) 2 = .00185
```

```
SPRE6k / 2.00185 X.04305 = .00008
```

Therefore, by multiplying the right-hand side of the equation (2) by 3437.75 and raising the values of spore 2 k/2, spore 4 k/2, etc., its form will be—

```
v - bh = 147' .663sin2b + 3' .18sin4b - 0' .06sin6b +... ( 3)
```

The terms on the right-hand side of this equation are becoming smaller so quickly that omitting the terms following the third term cannot cause any loss. Even if the third term is omitted, there is no particular loss. In this way, the difference between the true portion and its declination can be easily known in degrees or minutes, by which the table of the declination and portion can be made as easily as a table of 320 pages.

Now with the help of this formula, it is also necessary to know the relation between the hypothetical 'mean declination of the Sun' and the declination of the true Sun because the time equation is the difference between the declination of the true Sun and the hypothetical Sun. However, the declination

of the hypothetical Sun is the same as the mean portion of the Sun. Therefore, the work can be easily derived from equation (2) and equation (6) of page 178.

The brief form written below equation (6) of page 178 will be sufficient— s = m + 2 c sin s + 5/4 sin2m

Here s = true mean center, m = mean center, and c = the center of the Earth which is the same as 1670.

SOURCE PAGE 367

The difference between the planet and the apogee is called the mean center (see page 162). Therefore, if the portion of perigee is added to the mean center, the portion of the planet will be obtained. If the portion of the perigee of the Earth is assumed to be zero, then

```
True portion = s + ni
```

and mean portion = m + ni

Since the true portion is considered as 'bha', it is appropriate to consider the mean portion as 'bha'. Therefore

```
bha = s + ni or bha = m - ni
```

```
bha = m + ni or bha = m - ni
```

By raising these values of s and m in the brief form of equation (cha)

```
b − n = m − n + 2 sin ( m − n ) + 5 / 2 sin^2 ( m − n ) or
```

bh = pha + 2 c jya (pha - ni) + 5 / 2 c² jya² (pha - ni) (4)

Using equations (2) and (4), an equation in which bh is not present can be found. From such an equation, the relation between the apparent latitude and the mean latitude or the latitude of the hypothetical Sun can be easily known. It is a fact that the sum of the two appropriate equations will also yield terms whose multipliers are very small and whose inclusion will greatly increase the form of the equation, but will not yield more benefit. Therefore, those terms whose multipliers are less than 0.0001 will be omitted. By substituting the right-hand side of equation (4) for bh in equation (2) and by omitting those terms whose multipliers are less than 0.0001, we obtain the equation written below.

```
v = pha + 2sin(phi - ni) + 5 / 4sin²(pha - ni) - spere² k / 2sin² [ pha + 2sin(phi - ni) + 5 / 7sin²(pha - ni) ] + 1 / 3sware⁴ k / 2sin⁴ [ pha + 2sin(phi - ni) + 5 / 8sin²(pha - ni) ] here 5 / 8sin² [ pha - ni ) also very small so if we leave this in the fourth and fifth positions, it will get the following form in order.
```

```
− spre² / 2sin [ 2tan + 4sin (tan − sin) ] and
```

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+ 1 / 2spere4k / 2jya [4bha + 8cjya (bha - ni)] in this secoutapad = - spre2k / 2jyarba X kojya [4cjya (bha - ni)] + kojya2bha X jya [4cjya (bha - ni)] = - spre2k / 2jyarba2bha + kojya2bha X 4cjya (bha - ni)

Because 4 sin(bha-ni) is a very small angle, its cotangent will be equal to one and its sine will be the same. (See Hall and Knight's Trigonometry page 262)

Similarly the fifth term

= 1 / 2spere4k / 2jya4bha X kojya [8cjya (bha - ni)] + kojya4bha X jya [8cjya (bha - ni)] = 1 / 2spere4k / 2jya4bha + kojya4bha X 8cjya (bha - ni) because its second term's multiplier is less than 0.0001

Because the multiplier of its second term is less than 0.0001, therefore

v = h + 2sin(h - n) + 4 / 8v^2 sin^2(h - n) - spe^2 / 2sin^2(h) + 4sin(h - n) cos^2(h) + 1 / 2spe^4 / 2sin^4(h)... (5)

but2sin(h) - n) cos^2(h) = sin(h - n + 2h) + sin(h - n - 2h) = sin(3h - n) - sin(h + n)

Therefore, if the equation (5) is simplified and its terms are written in order according to the difference of the terms,

v = h + 2sin(h - n) - spe^2 / [sin^2(h)]

SOURCE PAGE 369

+ $2c^2k$ / 2sin(b + n) + 5 / $4c^2\sin^2(b - n)$ - $2c^2k$ / 2sin(3b - n) + 1 / $2\sin^4k$ / $2\sin^4b$... (6)

From this equation, the relationship between the declination of the hypothetical Sun and the mean sine of the Sun and the declination of the true Sun and k can be known. All the terms that follow 'b' on the right side together constitute the equation of time. Among all these terms, the first two terms 2c sin(b - n) - $k^2/2\sin^2b$ are of greater importance because the multipliers of the other terms are so small that they can be omitted. Therefore

w = b + equation of time

where time - equation = $2c\sin(b - n) - k^2/2\sin^2b$

This is expressed in radians. If it is to be expressed in arcminutes, it should be multiplied by 3437.75 because 1 radian = 3437.75' and the motion of one arc of the meridian circle is in arcminutes. Therefore, the form of the equation of time in arcminutes will be

3437.75 $\sin(b - t) - \sin^2b$ (7)

If the place of k is kept as 0.01675 and the place of ni as 281036'14" which was the sine portion of the sun's declination* in the epoch of 1676 AD,

* This sine portion of the sun below is known from this formula of page 626 of the Nautical Almanac of 1625 AD—

Mean longitude of solar perigee

= 281 ∘ 13 ′ 15 ″.0 + 6189 ″ · 03 r + 1 ″ · 63 au 2 + 0 ″ · 012 au 3

When the sine portion of the declination of the first day of January of 1600 AD at noon is and au is the difference of the Julian century from that time to the instant. For the meridian transit of the 1676 AD ecliptic au = 8137 / 36525 when the number of days from the first noon of January 1600 AD to the meridian transit of the 1676 AD ecliptic is 8137 and the number of Julian century days is 36525.

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2sin(sin - cos) = 2(sin cos^281°36'14'' - cos^281°36'14'') = 2sin cos(360° - 78°23'46'') - cos(360° - 78°23'46'') = 2sin cos(360° - 78°23'46'') - cos(360° - 78°23'46'') = 2sin cos(78°23'46'' + cos(78°23'46'')) = 2 X 01675 (.2011sin + .6766cos) = 00674sin + 03282cos ∤ ∴ Time equation (7) will be 3437.75 (.00674sin + .03282cos - 043.5sin^2) = 23' .17sin + 112' .83cos - 148' .0sin^2 (8) By establishing the value of the mean portion of the sun's arc and other values, the time of the sun can be known, from which the time of the year can be known. This will happen in these ways.

It is apparent that this value of the time equation is not pure for all time because this form came at the time when the southern declination of the sun was understood as 281036'14". The sine portion of the sun's declination increases by about 1 degree per year, so this understanding will not be more inaccurate for 10 or 15 years. Even with the decrease of the sun's ultimate revolution, some difference occurs, but its speed is very slow, so it cannot be very different for 100 years because of this.

It is stated that the time elapsed in the sidereal clock after the transgression of the latitude point of the vernal equinox is called sidereal time. If there is no sidereal time for a certain time and the declination of the true Sun is the same at that time, then the time elapsed after the transgression of the latitude of the true Sun is not. Therefore, the hour angle of the true Sun at that time is

† The time elapsed from the time of the transgression of the latitude of the Sun or star to the right ascension is called the hour angle of the Sun or star. Nowadays, the difference between the east hour angle and the west hour angle is not considered as in page 260

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− 148 ′ sin (2 X 300 ° 50 ′ 33 ″) =23 ′ 17 (− sin56 ° 6 ′ 27 ″) + 112 ′ 83 (cos56 ° 6 ′ 27 ″) − 148 ′ sin601 ° 41 ′ 6 ‴ = − 23 ′ 17 X.8586 + 112 ′ 83 X.5126 − 148 ′ (− ′ 8804) = − 16 ′ 86 + 57 ′ 84 + 130 ′ 30 = + 168 ′ 25 = + 168 ′ 25asu = + 28equinoctial shadow

If more accuracy is required, the value of the time equation should be known with the help of equation (6). It should also be noted that the mean position of the sun taken above is known according to the Sūrya Siddhānta. If the mean sine portion of the sun is extracted from the true declination, it is 30105'2". Therefore, if the time equation is derived from equation (6) and the true declination mean sine portion, it will be the same as the time equation given in the Nautical Almanac. The curve for revealing the time equation

The form of the time equation in the asus is in formula (7)

3437.752sin (phi - ni) - spore2k / 2sin

601° 41' 6" = 360° + 241° 41' 6' = 360° + 180° + 61° 41' 6" ∴ sin601° 41' 6" = sin (180° + 61° 41' 6") = - sin61° 41' 6"

† The procedure is as follows:

In the time of the Aries solstice in 1676, according to the mathematical almanac of 1622, the mean sine portion of the sun was 20^ 044'58". There are 284.4167 mean days from the Aries solstice to the half-tithi of Vaspant Panchami, and the motion of the mean sine portion of the sun is 0^ 068564733536 per day. Multiplying these gives 280^ 03345625 or 280^ 020'4". Adding this to the sine portion of the Aries solstice gives 301^ 05'2"4.

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If Ni is taken as 281° 36'

Bh - Ni = Bh - 281° 36' = Bh - (360° - 78° 24') = Bh + 78° 25' - 360° = Bh + 78° 24' †

Therefore the form of the above formula will be

3437.752sin (bh + 78° 24') - sphere^2sin^2bh

In this, by raising and simplifying the values of c and spore2 k/i, this will be the form.

115.165.4 sine(bhā + 7 024') - 147.665 sine 2 bhā(ka) from which two curves can be drawn whose equations are as follows r = 115.565 sine(bhā + 78°24')(kha) ra = - 147.665 sine 2 bhā(khā)

If the values 0, 30, 60, 60, etc. are assumed in place of 'bh', then upon simplification, the values of r, ra, and the time equation will be according to the table in page 351.

This table also gives the solar and English dates. The values of the time equation in these dates are correct. To know the time equation of the dates, one should use figure 64. The dates of the solar month written are according to the ecliptic. For example, 6 Pisces means the 6th day from the Vernal Equinox of Pisces, 6 Leo means the 6th day from the Vernal Equinox of Leo.

It has been stated before that the time equation is dependent on the mean sine portion of the sun's declination and the sun's ultimate revolution, which is not fixed, so the values of the time equation given in the table or figure 64 are not pure forever. There is a subtle difference of one art per year in these. The mean sine portion of the sun increases by approximately 1 art per year. Thus, in 60 years, there is a difference of one part.

† Adding or subtracting 360 degrees from an angle does not cause any difference in the values of its sine, and co-latitude, etc., nor does it cause any difference in the value of that angle itself.

SOURCE PAGE 374

It goes. Therefore, the values of the equation for 60 years should be the same as those known from table figure 64, so there will be no big mistake.

The solar month English date fraction of the time calculation: 0 + 18.8 0 + 18.8 6 Pisces 23 March 30 + 18. 2 - 21.4 - 3 26 Aries 22 April 60 + 12.4 - 21.4 - 6 6 Taurus 23 May 90 + 3.6 0 + 3.6 8 Gemini 22 June 120 - 6 + 21.4 + 15.4 7 Cancer 23 July 150 - 14 4 + 21.4 + 7 6 Leo 22 August 180 - 18.8 0 - 18.8 6 Virgo 22 September 210 - 18.2 - 21.4 - 36.6 5 Libra 22 October 240 - 12.4 - 21.4 - 33.8 5 Scorpio 21 November 270 - 3.6 0 - 3 6 7 Sagittarius 22 December 300 + 6.1 + 21 4 + 27.5 8 Capricorn 21 January 330 + 14.4 + 21.4 + 35.8 8 Aquarius 20 February 360 + 18.8 0 + 18.8 6 Pisces 23 March

Now the method of drawing the curve is briefly explained (see figure 64).

The abscissa is taken as the abscissa and the ordinate as the ordinate for the mean sine portion of the sun.

SOURCE PAGE 375

Two arcs have been formed. The difference between two passing vertical lines is considered to be 3 degrees and the difference between two passing horizontal lines is one minute. When bhā is zero, then r + 18.6 minutes and rā

is zero, therefore a point for r is made on the 16th horizontal line in the positive direction in front of the zero portion. When bhā is 30 degrees, then r + 18.2 and rā - 2.14 are. Therefore, in front of the 30 portion, the 18th in the positive direction

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SOURCE PAGE 376

A point for a point on a horizontal line and a point for the vertical lines 21st and 22nd in the negative direction are fixed. Similarly, other points are also fixed. By joining all the points that reveal r, a curve of k is formed. Similarly, by joining the points that reveal a, a curve of kha is formed. With the help of these two curves, the curve of k is drawn as follows—

On zero degree, r = +18.8 and a = 0. Their sum will also be +18.8, so the point of the curve of k will also be at +18.8, meaning the points of the curves of khkhkh and kkkkk will be normal in this situation. At 30 degree, r = +18.2 and a = -21.4. Their sum is -3.2. Therefore, the point of the curve of kkkkk will be on the third horizontal line in the negative direction. At 60 degree, r = +12.4 and a = -21.4. Their sum is -6. Therefore, the point of the curve of kkkkk will be on the sixth horizontal line in the negative direction. At 60 degree, r = +3.6 and a = zero, so the point of the curve of kkkkk will be on the fourth horizontal line in the positive direction. This will be on both curves khkhkh and kkkkk, etc.

Where the curve of the axis of x intersects four points, it is revealed that the time equation is zero. In these times, the mean portion of the sun is approximately 23, 82, 160, and 273.5 in sequence. This portion corresponds to the 2nd of Aries, the 1st of Gemini, the 16th of Leo, and the 10th of Sagittarius. Therefore, the time equation is zero in these dates. This means that in these dates, the true time and the mean time are the same. The time equation for the 17th Libra is -41 minutes. This means that on this date, the mean time (time of a mechanical clock) will be found by subtracting 41 minutes from the true time or the time of the sundial. Similarly, for the 26th of Capricorn, the time equation is +36 minutes, that is, the mean time will be found by adding 36 minutes to the time of the sundial on this day.

These vertical and horizontal lines are not shown in the figure. They can be estimated by the numbers of the *bhogansh* and the positive and negative numbers of 10, 20, 30 given next to them.

SOURCE PAGE 377

Now it has been proven that the time of the sun-dial is not pure because those who take the ascendant based on the time of a clock connected to the railway or telegraph do not consider the time of the sun-dial, as there is sometimes a difference of 41 minutes or 16 minutes between the sun and mean clocks. Besides this, a difference occurs due to the difference in longitude because the time of the clocks of the railway or telegraph of India is advanced by 53 hours or 13 hours 45 minutes from the mean time of Greenwich. Therefore, this time is correct only for those places whose mean time is 53 hours or 82030' before Greenwich. Mirzapur is 82038'10" before Greenwich. Therefore, Mirzapur's mean time is 8'10" or 8 and a quarter minutes more than the Indian mean time. If the consideration of a quarter minute is not taken, it can be said that the Indian mean time used in railways and telegraphs is the same as the mean time of Mirzapur. Therefore, to know the mean time of the previous places, the factor of longitude should be added, and to know the mean time of the western places, the factor of longitude should be subtracted. This will be clear from the examples below:-

Example 1—In Prayag, what will be the time on the railway clock when the portion of the sun's declination is 3501'6" in the time that 16 hours 15 minutes have passed according to the sun-dial after sunrise?

On this day the sun is at the 6th degree of the Cancer zodiac, therefore it is the 6th Tithi of Cancer. In the table, the 7th Tithi of Cancer is time equation + 15.4 minutes. Therefore, only from the table it can be found that on this day the time equation is approximately time equation + 15 minutes. From figure 64 where the curve of the time equation is given, it can be found that the 6th Tithi of Cancer was more or less than time equation + 11.4 minutes. It is clear from seeing that the ordinate of the point of the

curve of 7 Cancer is less than the days before 7 Cancer, so it is determined that 6 Cancer is some less than time equation + 15.4 minutes and 4 Cancer is exactly + 15 minutes. Therefore, the desired time equation is approximately + 15.3 minutes. This is positive, so it should be added in 16 hours 15 minutes. Therefore, when the sun-dial in Prayag

Some people understand that the time of Madras remains in the observatory clock, but this is a delusion. There is indeed an observatory in Madras and previously that same time was kept in all clocks, but now the rule has been changed.

SOURCE PAGE 378

According to this, if 6 karka is 16 ghadi 15 pal, then the mean time of Prayag will be 16 ghadi 30.3 pal.

But the mean time of Prayag is less than the Indian mean time because the longitude of Prayag is 81055^(°)15'' west of Greenwich and the Indian mean time is 82030' west of Greenwich. Therefore, the longitude of the Indian mean time is 34'45'' west of the longitude of Prayag. When there is a difference of 10 in longitude, there is a difference of 4 minutes or 10 *pala* in time, and when there is a difference of 1 degree, there is a difference of 1 *asu* in time. Therefore, when the difference is 34'45'', the difference in time will be approximately 35 *asu* or 6 *pala*. Since the longitude of the Indian mean time is to the west of Prayag, the Indian mean time is ahead of the mean time of Prayag. Therefore, in the desired time, the Indian mean time will be approximately 16 hours 30.3 *pala* + 6 *pala* = 16 hours 36 *pala*. This is similar to 6 hours 38 minutes 24 seconds.

The clear time from sunrise to noon on this day is 16 hours 43 minutes 43 seconds (see page 331), which is 6 hours 41 minutes 12 seconds. But noon is exactly 12 o'clock, so 12 hours - 6 hours 41 minutes 12 seconds = 5 hours 18 minutes 48 seconds the sun must have risen.

Sūrya Siddhānta = 5h 18m 48s

Sūrya Siddhānta time until the auspicious time is in the mean time = 6h38m24s therefore Railway clock time = 11h57m12s

That is, at this time the railway clock will be 11 hours 57 minutes and 12 seconds.

Example 2—If the time elapsed in the clock after noon is 5 hours 24 minutes from the railway clock, what will be the time in the sundials of Kashi and Prayag? In the sunrise period of this day, the portion of the sun is 6th 15°23'34''.

The sun is at the 16th degree of the tulā sign, therefore this day is the 16th date of the solar month named tulā. Figure 64 reveals that for the 5th date of tulā, the time equation is -37.5 minutes and for the 20th date, it is -41 minutes. This proves that the time equation has increased by approximately -3.5 minutes in 15 days. Therefore in 11 days

SOURCE PAGE 379

That is, on the 16th day of Tula, the time equation will increase by -40.2 minutes, which is approximately -16 minutes. It is stated that

Mean Time = True Declination + Time Equation therefore 5 hours 24 minutes - True Declination + (-16 minutes) therefore True Declination = 5 hours 24 minutes + 16 minutes = 5 hours 40 minutes

This will be in the sundials of places located on the longitude line 823 degrees west of Greenwich because the clocks of the telegraph offices and railway stations of the entire India remain in the mean time of this longitude line.

Kashi is 83°3'4" or 83°3' east of Greenwich, which is 33' more than 82°30', so the true sidereal time of Kashi will be 33' less or 53 minutes more than the above sidereal time, which is equal to 2 minutes 12 seconds or 2 minutes. Therefore, at that time, the sundial of Kashi will show 5 hours 42 minutes.

The longitude of Prayag is 81055'15" west, so it is 34' 45” behind 82030' by 34’ and 45”. Therefore, the sundial of here will be 34° 30' or 2 minutes 16 seconds behind. Therefore, the sundial of Prayag at this time will be 5 hours 37 minutes 41 seconds.

Example 3—What will be the Lagn in Prayag at the time given in the second example?

It is necessary to know the exact time of the first sunrise. For this, the ecliptic time of Prayag should be known.

Sūrya Siddhānta = 615° 23' 34'' 1682.616Tula ko precession angle = 220° 41' 6'' [De.Pg.253] ∴ Sūryakāsayanbhogāng = 78° 4' 40'' = 78° 5' = 6Rashi + 38° 5'

∴ Sūrya Siddhānta = sin38° 5' x sin23° 27' [DevaPṛcāda] =.6168 X.3676 =.2454

SOURCE PAGE 380

∴ declination = 14°12’ The portion of the sun's transit of the sun is more than 6 from the zodiac, therefore it is the southern transit. True sine = spread latitude × spread declination = 25°25’ × 14°12’ = 4752 × 2530 = 1202 [•] spread = 6°54’ [•] spread shadow = 66 shadow = 27 minutes 36 seconds

Since the transit is south, the sun will rise in Prayag at 6 o'clock 27 minutes 36* seconds in the sundial. However, the time equation for this day is -16 minutes. Therefore, the mean time of Prayag in the sunrise time = 6 hours 27 minutes 36 seconds - 16 minutes = 6 hours 11 minutes 36 seconds

To know what will be the mean time of Bharatvarsha at the sunrise time of Prayag, 2 minutes 16 seconds must be added because Prayag is 2 minutes 16 seconds west, so its mean or true time will be that much behind the mean time of Bharatvarsha, therefore at the time of sunrise in Prayag, the railway clock will show 6 hours 13 minutes 35 seconds.

The mean meridian time from sunrise is 12 hours - 6 hours 13 minutes 35 seconds or 5 hours 46 minutes 25 seconds, and it is 11 hours 10 minutes 25 seconds until the evening of 5 o'clock 24 minutes. This is the same as 27 clocks 56 shadows. Therefore, in the auspicious time, there are 27 clocks 56 shadows after sunrise. This is the mean Sawan period. It will be easier to know the ascendant by converting this to the Nakshatra period.

6 Sawan clocks = 6 Nakshatra clocks + 1 Nakshatra shadow (page 3) ∴ 28 Sawan clocks = 28 Nakshatra clocks + 5 Nakshatra shadows ∴ 27 clocks 56 shadows (Sawan) = 27 clocks 56 shadows + 5 shadows (Nakshatra) = 28 clocks 1 shadow (Nakshatra)

Due to the refraction of light, sunrise occurs some before this, which will be discussed further.

SOURCE PAGE 381

The portion of the sun's declination in the rising time = 6 degrees 15 minutes 23 seconds 34 seconds = 6 degrees 15 minutes 24 seconds. Therefore, in the rising time, the ascendant of the Libra sign is 15 degrees 24 minutes. The rising time of the Libra sign in Prayag is 5 hours 42 minutes. When the rising is 5 hours 42 minutes, then it will be 15 degrees 2 minutes 51 minutes and 30 seconds, 5 minutes 42 seconds, 6 minutes 1 second, 1 minute 8 seconds. Therefore, the rising of 15 degrees 24 minutes will be 2 hours 55 minutes 34 seconds, which means the time of enjoyment of Libra = approximately 2 hours 56 minutes. The rising time of Scorpio = 5 degrees 44 minutes, Sagittarius = 5 degrees 15 minutes, Capricorn = 4 degrees 25 minutes, Aquarius = 3 degrees 48 minutes, Pisces = 3 degrees 41 minutes. The sum is 25 degrees 36 minutes, meaning the rise of the Pisces sign has already occurred in the time of 36 minutes from sunrise. Therefore, in the auspicious time, the Aries sign is rising, so this is the rising ascendant. This is commonly called the ascendant. To know which point of Aries is the ascendant, one should use the ratio. Auspicious time = the rising time of the end of Pisces = 25 hours 36 minutes. The time of enjoyment of Aries = 2 hours 22 minutes = 142 minutes. The rising time of Aries = 4 hours 4 minutes = 244 minutes. 244 minutes : 142 minutes :: 30 degrees : portion. Therefore, the portion = 142 * 30 / 244 = 17 degrees 29.5. Therefore, the ascendant of Aries is 17 degrees 27.5.

SOURCE PAGE 382

Example 4—If the declination of the true Sun in Prayag at sunrise is 6th 15023'34” in Prayag, then what will be the Indian mean time in Ujjain at the time the Sun will be at the meridian there?

Ujjain is at a longitude of 75°46’ East and a latitude of 23°6’ North from Greenwich. The longitude of Prayag is 81055'15” and the latitude is 25025’.

The difference between Ujjain and Prayag is 81055'15” - 75046’ = 609'15” West. Therefore, the true noon of Ujjain will be 24 minutes 37 seconds behind the true noon of Prayag. It is stated in the third example that the sunrise in Prayag will be at 6 hours 27 minutes 36 seconds according to the sundial. Therefore, at the time of sunrise

```
Natkal = 12 hours - 6h27m36s = 5h32m24s
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That is, the true noon will be 5 hours 32 minutes 24 seconds after sunrise. But at the time of sunrise, the Indian mean time is 6 hours 13 minutes 35 seconds, so the Indian mean time at the time of true noon in Prayag = 6 hours 13 min 35 sec + 5 hours 32 min 24 sec

= 11 hour 45 minutes 46 seconds

Ujjain is 24 minutes 37 seconds west of Prayag, so here it will be 24 minutes 37 seconds behind the true noon of Prayag. However, at the time of the true noon of Prayag, the Indian mean time is 11 hours 45 minutes 46 seconds, so at the time of the true noon of Ujjain, the railway clock will show 12 hours 10 minutes 26 seconds.

The comment in page 80 states that due to the bending of rays, sunrise occurs some before the time of calculation. Therefore, it is necessary to explain what the bending of rays is and how the difference in the magnitude of the day caused by it should be corrected.

Refraction of Light

Air, water, glass, amber are substances in which light enters and travels to the other side. Therefore, they are called transparent. When light passes from one homogeneous transparent substance to another homogeneous transparent

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When it goes into matter, its direction is not the same as it was in transparent matter before. This phenomenon is called the deflection of the ray or simply deflection. Some writers have named it refraction of the ray, but for the sake of convenience of many things, it seems better to call it deflection. From this, a big difference in the reality and apparent positions of an object is seen. Sometimes objects take on a divided form. But this place is not appropriate for discussing all these events. Only what is related to astrology will be stated here. Giving a small example for experience will be sufficient:-

Fill a glass with water and keep it in the sun and see how much of the glass is illuminated by the sun. Pour the water and keep the glass in the same place. This time, some less part of the glass will be illuminated. In figure 65, there is a glass 'ga'. If water is filled and it is kept in the sun, it is seen that from 'ga' to 'bha' is illuminated, that is, it is seen that the sun reaches even to the edge of the glass. But when the water is poured and the glass is kept in the same place, it is seen that now only the part 'ga' to 'kha' of the glass remains illuminated, the sun does not reach the edge. This shows that if the ray is in the direction 'gā kha' in the air, then it becomes in the direction 'gā bha' as soon as it dissolves in the water.

SOURCE PAGE 384

If a ray of light enters from one homogeneous transparent substance to another homogeneous transparent substance in the direction of the direction, then the two rays are parallel. The first ray is called the incident ray, the second ray is called the refracted ray, and the third ray is called the emergent ray. If the line is normal at the point, then the first angle is called the angle of incidence and the second angle is called the angle of refraction. The normal to the incident and refracted rays and the point of entry lie in the same plane.

The relationship between the incident and refracted angles is revealed by the formula below—

The sine of the angle of incidence equals the sine of the angle of refraction

If the angle of incidence is denoted by a, the angle of refraction by b, and the constant by d, then the simplified form of the above formula will be

sin a = d; or sin a = d X sin b sin b

From the change of A, V will also change in such a way that the relation of their sines will always be the same as that of D. The result of D is the property of two transparent substances.

SOURCE PAGE 385

it changes according to. This is called the index of refraction of one transparent substance to another transparent substance. When light goes from a thin transparent substance to a dense transparent substance, the refracted ray bends towards the perpendicular, meaning the refracted angle is smaller than the angle of incidence. But when light goes from a dense substance to a thin substance, it moves

away from the perpendicular. In figure 66, since substance H is denser than substance H, the refracted ray in H has bent towards the perpendicular, and when it comes out from H and enters H, it has moved away from the perpendicular. If the direction of light is reversed, i.e., the direction in H is not P, then the direction in H will become S. The light that passes through many transparent substances along which it travels along a curve or a broken line, if the direction is reversed, it also returns along the same line.

Suppose H, I, three transparent substances are at the same level which are mutually parallel.

If the index of refraction of HA from H is D and the refractive index of HI from H is D, then the index of refraction of HA from HI can be known easily. It can be experienced from the experiment that if light enters H from HA and HI, then the direction it has in H is the same at the end, meaning the angle of node formed at the entry point of H and HA is also formed at the exit point when HI exits H. According to the law of refraction

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sin a = sin b x sin c; sin a = sin d x sin c

or sin b / sin c = sin d / sin a

From this it is proved that if the refraction of a substance from one substance to another is d and the refraction from the first substance to the third substance is d, and if the angle of aberration and the angle of refraction of the ray going from b to c is alpha and beta respectively, then the refraction of the ray going from b to c in the second substance is d / a.

Refraction related to astrology—when the rays of light coming from celestial bodies that come to the earth enter the atmosphere, they undergo refraction. This refraction is called astronomical refraction. The density of the atmosphere is not the same from top to bottom. As the distance from the earth increases, the atmosphere becomes thinner, so the total atmosphere is not homogeneous. When a ray coming from a celestial body enters the atmosphere, it first goes into a very thin layer, and as it approaches the earth, due to entering layers of less dense to more dense, it bends slightly towards the vertical and reaches the earth. Therefore, the path of it in the atmosphere is curved. The direction of the ray when it reaches the earth is in which the celestial body is seen.

A ray from a star enters the atmosphere at a point in the direction of k. From this point, its path is not straight. The ray has to enter the different levels of the atmosphere from k to the observer's position d, which become progressively denser. Therefore, the ray also becomes curved and finally reaches d. At the point d of this curve, there is the tangent d to d. The observer realizes that the star is in the direction d. If a line parallel to k is drawn from d to t, the star appears in the direction t at that time when there is no atmosphere that bends the ray. Therefore, the effect of the atmosphere is that

The direction is absolutely pure, but the star is so far that the angle of the star is equal to zero.

SOURCE PAGE 387

The apparent position of the star is due to the fact that the star appears to be slightly raised towards X. Therefore, due to the rotation, the portion of the celestial body is slightly less and the altitude is slightly more. In the figure, the magnitude of this rotation is equal to the angle of T and D. The actual portion of

T is D X and the true portion is T D X. At the time when the celestial body is in the horizon, its rotation is about 35' maximum.

Now it will have become apparent that due to the atmosphere, the apparent position of a celestial body is not the same as it should be in reality. Therefore, if the rotation is not taken into account, some error remains in the calculation. Below is a table*

* R. S. Ball's Spherical Astronomy page 120.

SOURCE PAGE 388

It is known from this that how much the portion of a star is reduced due to the motion. This table is for the time when the atmospheric pressure is equal to the pressure of 10 inches of mercury and the temperature is equal to 500 Fahrenheit. Some difference occurs in a different state.

True portion motion true portion motion true portion motion 0 0" 35° 41" 70° 2'36" 50 5" 40° 46" 75° 3'34" 100 10" 45° 58" 80° 5'19" 15° 16" 50° 1'6" 85° 6'51" 20° 21" 55° 1'23" 87° 14'23" 25° 27" 60° 1'41" 88° 18'16" 30° 34" 65° 2'4" 88°40' 22'23" 90° 35' approximately

From this table, the true portion of any star can be easily known. For example, if the true portion of a star is 60°, then its true portion will be 6001'41". It is also worth noting that the star which is exactly above the head (on the cross) has the same true and apparent position, and if the apparent portion is less than 45°, its motion is not more than 1". If the apparent portion is more than 20°, there is 1" motion for every 10 portions.

Simple exposition of atmospheric motion-

For simplicity, it would be good to understand that the Earth is a perfect sphere and there are thin layers in the atmosphere from bottom to top whose centers are the same as the Earth

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SOURCE PAGE 389

It is the center. It should also be assumed that the declination of each latitude is constant everywhere in that latitude, but the declination of one latitude is different from the declination of another latitude.

In Figure 66, the relationship between such two latitudes ha ha and hi hi is shown. Suppose that when light comes from zero (aether) to ha ha, its declination is dha and when light comes from zero to hi hi, its declination is dhi. Suppose that the direction of the ray in ha ha is ch and in hi hi, the same ray's direction becomes ch j.

If B is the center of the Earth and ch = ka, ch = ki, angle Bch = at angle Bch = i and angle sch = u, then according to page 363, sin u / sin i = dhi / dha. therefore sin u = sin i × dhi / dha but in the triangle Bch, sin u / ka = sin a / ki therefore sin u = sin a × ka / ki

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SOURCE PAGE 390

$$\therefore \sin \times dh = \sin a \times ka$$

or what X dh X sin = ka X tha X sin a

This rule is correct for two adjacent levels. In this way, this simple rule is derived:

If the atmosphere is assumed to be made up of spherical homogeneous layers whose center is the same as the center of the Earth but whose densities vary from one another, then when a ray of light passes from one level to another, the product of the sines of the meridian, radius, and angle of inclination of a level remains constant.

This rule can also be revealed by the formula below-

ka X dhya X sine = k X dhya n (7)

Where a, b and c are respectively the radii, refractive indices and angles of refraction, and a, b, c are respectively the radius, refractive index and angle of refraction of the Earth. The angle of refraction in the layer touching the ground is usually the same as the nadir, therefore there is no true nadir of any celestial body.

If these levels are considered very thin, then the path of the ray will be a curved line instead of a straight line. Suppose this curve is the one on which the ray enters the Earth's surface at point J in these thin, thin levels respectively. There is a tangent to this curve at the six points. The refractive index and radius of the level into which the ray enters at the six points are dhā and radius. This tangent becomes the same as the ray that forms the curve for some distance from the ray, so the angle formed at this point is the same as the bha angle. Suppose this is like ā, when the ray enters the uppermost level of the atmosphere, its direction is the same as its actual direction in zero. At the time the ray reaches the point J on the Earth's surface, the tangent of the curve at this point J indicates the direction in which the ray reaches the observer's eye. If the cross-section of point J is kha, then this kha angle is the true portion of the star. The angle between the tangent of the curve at the uppermost level and the tangent of the curve at the point on the Earth's surface is called the motion related to astrology. From knowing this, information about the true and actual position of a star can be obtained. This is commonly called motion. If

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SOURCE PAGE 391

If magnitude and assumed then that (or) means the difference between the motions of two levels of the nearest [or] and that (or) means the difference between the latitudes of those two levels. If alpha and beta are the angular positions of these two levels and delta, phi are their latitudes then

```
alpha - beta =        ta (v) and delta - phi = ta (phi)
```

```
therefore beta = alpha - ta (v) and delta = phi + ta (phi)
```

```
But if X delta X sinalpha = ki X phi X sinbeta
```

If the two thin levels of the atmosphere are very close to each other, then their radii are usually equal, therefore in such a case

```
delta X sinalpha = phi X sinbeta = [ phi + ta (phi) ] sin [ alpha - ta (v) ] = [ phi + ta (phi) ] sin [ sinalpha (v) - sinphi (v) ]
```

But it is very small and in terms of temperature, so sine of ta(v) = ta(v) and cosecant of ta(v) = 1

SOURCE PAGE 392

∴ dhā X sin a = [dhā + tā(dhā)][sin a - tā(va) X cos a] = dhā X sin a - dhā X tā(va) X cos a + tā(dhā) X sin a because the product of tā(dhā) and tā(va) in the fourth term is very small, so it is omitted.

Or therefore 0 = ta(dha) × sin a - dha × ta(va) × cos a sin a × cos a = ta(va) ta(dha) ta(va) ta(dha) = sin a ta(dha) (2)

Such an equation can be known from equation(1) and(2).

From equation(1), sin a = k x dh X sin n / k x dh

This is revealed by trigonometry that

> Sūrya Siddhānta

This is the simple relation (differential equation) of this astrological behavior. If the index of the uppermost level is taken as 1 and the index of the lowest level is taken as theta, and if the integration of the appropriate differential equation is done between these limits, then the entire knowledge of astrological behavior can be known. But there is difficulty in doing this that the differential equation has two variables whose mutual relationship cannot be known because we do not know exactly that 24

SOURCE PAGE 393

At what height of the Earth is the altitude? But even without knowing this, the differential calculus of the above equations can be derived by one trick, from which the true motion can often be known exactly.

In this device, k/k is taken as 1+epsilon when the magnitude of epsilon is very small because k is the radius of some level of the atmosphere and k is the radius of the Earth. It is also known that the height of that atmosphere in which the property of bending (refraction) of the rays occurs is at most 50 miles. The radius of the Earth, that is k, is 4000 miles, so k/k = 4050/4000 = 1 + 1/80. It is clear that epsilon = 1/80 and its square, cube, etc., are so small that they can be omitted. Imagining this

> $v = \int 1d(theta)\ 1d(theta^2 - \theta^2 + 2\theta^2)^2 = \int 1d(theta)\ 1d(theta^2 - \theta^2)\ (1 + 2\theta^2 1d^2 - \theta^2)^{12} = \int 1d(theta)\ 1d(theta^2 - \theta^2) - \int 1\theta.d(theta)\ (1d^2 - \theta^2)$

> To simplify this form, it will be — v = psupren + fspren3n

> v = apsere + fspere3n (1)

Where p and f are fixed quantities and no true position is known. The values of p and f can be known by direct observation.

Cassini's Formula—

Cassini, the astronomer, imagined that the atmosphere is homogeneous from top to bottom, meaning it has the same density. The formula for the motion obtained from this imagination is similar to the

formula mentioned above. The magnitude of the motion known from this is satisfactory for latitudes up to 80°. If the latitude is more than 80°, there is a coarseness in the magnitude of the motion.

SOURCE PAGE 394

In this imagination, it has to be assumed that the ray coming from zero bends once upon entering the atmosphere and then that same direction remains up to the Earth's surface.

Suppose there is a ray that is bent at point C.

Radius of Earth = R = r

r = R(1+r) while r is very small as previously stated.

```
A = precession angle = ∠ true = ∠ true + ∠ apparent = revolution + apparent angle = v +        heta
```

8 x ch c sprashta nataansh

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According to the law of motion

```
sin(v + va) = sinv * cosva + cosva * sinv = v * cosva + sinv
```

Where d is the refractive index of the atmosphere.

It is clear that v is very small. Therefore

```
sin(v + va) = sinv * cosva + cosva * sinv = v * cosva + sinv
```

```
∴ sin(a) + b sin(a) = d sin(a)
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∴ b = (d - 1) sin(a)
```

In the triangle a b c,

```
sin(a) - b - c = k sin(a) (1 + b)11 + b
```

```
sin(a) = sin(b) / 1 + b
```

```
r(a) = sin(a) / cos(a) = sin(a) / √ (1 - sin(a)^2) = sin(a) / 1 + b / √1 - sin(a)^2 / (1 + b)^2 sin(a) = √ (1 + b)^2 - sin(a)^2 k sin(a) ∴ b = (d - 1) √ (1 + b)^2 - sin(a)^2 k sin(a) = (d - 1) √1 + 2b + b^2 - sin(a)^2 k
```

It is very small, so it can be left. In such a situation

```
jyanav = ( d - 1 ) ( cos2n + 2ch ) = ( d - 1 ) sin(1 + 2chcos2n) = ( d - 1 ) sin(1 + 2chsin2n) *
```

* The secant line is called the line of intersection which is the complement of Cosine, i.e., cotangent. The abbreviation of the secant line is taken as che. Similarly, the abbreviation of Cosecant is taken as koche.

SOURCE PAGE 396

Viprashnadhikar373 = (d - 1) shren (1 - x * x2) = (d - 1) shren1 - x (1 + shre2n) = (d - 1) shren1 - x - xspre2n = (d - 1) (1 - x) shren - x (d - 1) spre3n = pspren + fspre3n (2) this is the first form. here p = (d - 1) (1 - x) and f = - x (d - 1)

This formula can be used in practice at the same time when the values of p and f are known. Their knowledge can be directly observed, which will be discussed later. Suppose* it has been found that at a temperature of 500 Fahrenheit while the air pressure is equal to the pressure of 30 inches of mercury, the motions of 540 and 740 degrees are respectively 80°06 and 200°46.

According to the above formula, the two equations are these

80" · 06 = p (Spere54°) + f (Spere54°) 3200" · 46 = p (Spere74°) + f (Spere74°) 3

From these equations, the values of p and f are 58'0264 and 0'06682 respectively. Therefore, the simple formula for the motion at 50 fa0 and 30 inches of pressure is

v = 58.263 sphere - 0.06682 sphere 3 (3)

It is also apparent that f/p = 1/873, so as long as the sphere³n is not very large, that is, if the sun, or star, is not near the horizon, the second term can also be omitted without any loss.

If the declination is not more than 70° and there is no much difference in temperature, the use of the simple term written below will be appropriate to know the value of the variation.

Cuspren

Where taking 58"2 for k would be more accurate. This k is called the coefficient of refraction.

The refractive index of the atmosphere is 1.000264 at 0°C and 760 mmHg (see Ball's Spherical Astronomy page 117) and Cassini

Ball's Spherical Astronomy page 127

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According to this formula, the refraction is 1000283 at a temperature of 50° and a pressure of 30 inches.

Astrologers named Simpson and Bradley have also created formulas for refraction, but their analysis is not necessary here. It will be sufficient to give only Bradley's formula—

v = 58 " 361 s p r e (n – 406 v) (4)

From this formula, the magnitude of refraction up to 80° of declination is satisfactory. The positions of stars near the horizon can be determined exactly from this formula because even when the declination is close to 60 degrees, the magnitude of (n–4.06 v) will not be very large.

Differences in the magnitude of refraction occur due to the decrease and increase in temperature and atmospheric pressure. However, the discussion of all this is left out for fear of length. To know the magnitude of refraction of the atmosphere from the observation,

Several methods are used to know the values of the multipliers of the formula that has been established for refraction. Three of these methods will be discussed here. The first and second methods can be used from the observations of the same observatory if its latitude is not too low or high. The third method requires two observatories.

First rule—such a star should be chosen that it remains above the horizon at both times. In Figure 72, t and ta are the true positions of such a star. The true position of the star should be known at both times. Suppose the true position of the star is not at place t and not at place ta.

According to the sutras, its actual portion was

n + paspren + faspren3n

urna + susparena + susparena1

If the actual positions of the stars are taken, then th = th and kh + kh = 2 kh = 2 longitudes = 2(60° - a), that is, the sum of both actual latitudes is 2(60° - a), where a is the latitude of the place s.

∴ n + paspren + paspare ^{3} n + na + pasparena

+ cusparena = 180 ° − 2a

SOURCE PAGE 398

If the values of 'n' and 'a' which are known from the arc are raised in this equation, then a linear equation of the three unknown numbers p, f and a is obtained. Similarly, if the true positions of three stars are known from the arc, then three equations will be obtained from which the values of p, f and a can be easily known.

u s d = the uttara dakṣiṇa line of the place u, d = the uttara dakṣiṇa point of the horizon u dha kha v d = the meridian circle dha = the North Celestial Pole kha = the cross mark of the place va = the tropics and the meridian circle, t = the highest position of the star at the time of the meridian passage, ta = the lowest position of the same star at the time of the meridian passage, g = the summer position of the sun, sh = the winter position of the sun

SOURCE PAGE 399

Second method—when the sun is near the nodal points, the constant multipliers p, f can be known from their nodal parts.

In Figure 72, if xg and xh are the true nodal parts of the sun, then xg + xh = 2x

Suppose n has no true nodal part and sh has no true nodal part, then n + p sphere n + f sphere3 n + na + p sphere na + f sphere2 na = 2a

b = Earth's center = Earth's meridian = meridian + d, dh = North and South celestial poles, g = observatory of North Pole, u = observatory of South Pole, x = cross of g-place, xa = cross of u-place, v = circle of the equator, t = star, bhdha = Earth's axis

If a is known and p sphere3 n or p sphere3 na are left as very small due to the reason, then the value of p can be easily found. But this method takes 6 months.

SOURCE PAGE 400

The third method—in this method, the properties of motion are fixed by knowing the true position of the same star on the meridian by knowing the south position of the two observatories of uttara and dakṣiṇa.

The south position of the star from the first observatory is x and the north position of the star from the second observatory is y.

x = xv - vt = a - k

xa = xv + vt = a + k

when a, a are the latitudes of both observatories and the declination of the star. If there are no true positions of t from g and u

khat = n + pasparen + fspare ^{3} norkhat = na + pasparena + fspare ^{3} na ∴ n + pasparen + fspare ^{3} na + na + pasparena + fspare ^{3} na = khat + khat = a + aa

If the values of f spre^3 n and f spre^3 na are left very small

n + pasprena + na + pasprena = a + aa

The value of p can be known by determining n, na, a, and aa by observation. If the value of f also needs to be known, then knowledge of the true part of another star will be required.

For example, the parts of the star kha (□eta Andromeda) from the observatories of Greenwich and Uttarashah (Cape of Good Hope), whose latitudes are 510^ 028'38'' North and 33^ 056'4'' South respectively. The true southern part of the star from the first observatory was 16^ 020'3'' when it was on the meridian, and the true northern part of the same star from the second observatory was 66^ 01'50''. Therefore

16^ o20'3'' + p imes 16^ o20'3'' + 66^ o1'50'' + p imes 66^ o1'50'' = 5^ o28'38'' + 33^ o56'4'' = 85^ o24'42'' herefore p imes (spre16^ o20'3'' + spre66^ o1'50'') = 85^ o24'42'' - 85^ o21'53'' = 2'46''

yap (.2630 + 2.6063) = 2' 46'' = 166''

SOURCE PAGE 401

∴ p =1682.6023 = 58 " . 23

Due to the motion of celestial bodies, their rising and setting appear a little earlier and later respectively, so the day length increases.

The rising of celestial bodies is understood at the time when their center comes to the eastern horizon. At that time, their true position is 60°. However, it has been proven that the true position is greater than the apparent position due to motion. It has also been stated that when the apparent position is 60°, the motion is approximately 35'. Therefore, at the time of rising, the true position of the celestial body is approximately 60035'. This means that when the center of the celestial body appears to be on the horizon, it is approximately 35' below the horizon. Therefore, it is evident that when the center of the sun is 35' below the horizon, then it appears to have risen, and when its center does not reach 35' below the horizon, it appears to be setting. Because of this, the apparent day length is about 5 minutes or 12, 13 minutes more than the true day length. To know the exact magnitude of this increase, we should calculate the instantaneous motion of the sun at the time of rising.

Sūrya Siddhānta equation is [see page 261 equation (1)] sin (time) X sin (latitude) X sin (declination) = sin (latitude) - sin (latitude) X sin (declination)

If the values for the time, latitude, inclination, and apparent position are taken in the order of n, a, k, and n, a, k are shifted to the right or left side, then

sineA X sineK + versed sineX versed sine (not) X sineK.

If the instantaneous relationship between latitude and declination is to be found by keeping them as constant, then

If jya X (n) = cojya X cojyak X jya (nat) X ta (nat) but the rising and setting of the sun are 60^ o degrees, then jya = jya 60^ o = 1; cojya = 0 therefore

cojya (nat) = - spre X spre k (1) ta (nat) = cojya X cojyak X jya (nat) X ta (nat)

SOURCE PAGE 402

or

ta (n) = ta (n) / kojyā a X kojyā k X jyā (n) (2)

If the instantaneous motion of the latitude is raised by 35' instead of the place of ta(n), then the rotation of the true latitude of 60° is approximately, and if the right side of equation (2) is simplified, then it will be known how much the rise in rise time is due to the rotation.

Example 1—What are the times from the clear sunrise to the clear sunset on the day of the solstice of Cancer and the solstice of Capricorn in Kashi?

Nowadays, the sun's northward transit on the day of the solstice of Cancer is 23°27' and the sun's southward transit on the day of the solstice of Capricorn is 23°27'. The latitude of Kashi is assumed to be 25°18'.

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the late

Sūrya Siddhānta = 78 ° 10 ′ = 60 ° − 11 ° 50 ′ = 6hours − 47min020sec0 = 5hours12min040sec0 = 13hours2pa

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If there were no motion, this would be the time from sunrise to noon. But the magnitude of motion is approximately 35', so in equation (2), by raising the place of ta by 35', on the day of the Tropic of Cancer

Ta (Nat) = cojya25° 18' cojya23° 27' sin101050' = 35' = .6041 X.6175 X.6787 = 35' = .81184 = 43' .12 = 43.12asu = 7palaya2minutes52seconds

Even on the day of Makar Sankranti, there will be such an increase due to motion because the value of sine 78° 10’ = sin 101° 50’ and cosine 23° 27’ will not change, whether 23° 27’ is North or South transit because cos 23° 27’ = cos(-23°: 7’).

Therefore in Kashi the time of the rising of the node of the ecliptic and the true position = 16 hours 58 minutes + 7 minutes = 17 hours 5 minutes and the true position of the day = 34 hours 10 minutes similarly in Kashi the time of the rising of the node of the ecliptic and the true position

Natakal = 13hours2minutes + 7minutes = 13hours6minutesandtrueposition = 26hours18minutes

Example 2—What will be the true sidereal time in Kashi on the day of the conjunction of Aries and Libra?

If the precession is zero at the time of sunrise of the sun on the day of Aries or Libra, then the sidereal time of mathematics will be exactly 60° or an hour or 15 minutes. The magnitude of the increase due to motion will be found.

ta (nat) = 35 ′ / $\cos^2$25 ° 18 ′ $\cos_o$ ° sin60 °

SOURCE PAGE 404

= 35' sin250° 18' = 35'.6041 = 38'.7 = 38.7as = 6pal

∴ On the days of the true rising and setting of the ecliptic for Aries or Libra, the true local time = 15 ghadi 6 pal

∴ In these days, the true or true day length = 30 ghadi 12 pal

Thus it is proven that due to the motion of Aries and Libra on the days of the conjunction of Aries and Libra, the day length is 12 pal more than the night length. It is a well-known fact that the day length and night length are the same in all places on these days. Therefore, if someone wants to know the day of the conjunction of Aries and Libra by measuring the time from sunrise to sunset with the ecliptic and in the reverse manner, they will find that the conjunction of Aries will fall 3 days before the true conjunction and the conjunction of Libra will fall 3 days after the true conjunction.

In page 13 of the Makaranda table, the magnitude of the maximum day length for Kashi is given as 34 ghadi 5 pal and the minimum day length as 25 ghadi 55 pal. This proves that in this table, the magnitude of the local time on the day of the conjunction of Cancer was determined to be 17 ghadi 2.5 pal. Now it is to see whether Makaranda-kar determined this day length by mathematics or by the ecliptic.

Sūrya Siddhānta has considered the maximum revolution of the sun to be 24^ o. Therefore, it is estimated that Makarandakar must have used this revolution to know the sidereal time through mathematics. It is not known what latitude he considered for Kashi. Nowadays, it is determined to be approximately 25018^ o. Therefore, there seems no loss in assuming that Makarandakar considered the latitude of Kashi to be 25^ o. If 25^ o latitude is assumed, then in Kashi on the day of the solstice of Cancer

```
Sūrya-sine = 24° x 25° =.4452 X.4663 =.2076
```

```
therefore arc = 11056 ′ = 716arcus = 120arc = 2hours
```

SOURCE PAGE 405

Therefore, sunrise time = 15 + 2 = 17 hours

∴ Maximum day length in Kashi on the day of the Cancer solstice = 34 hours

From this it appears that the latitude of Kashi was considered to be a little more than 250 because only then the sidereal time could be 2 hours 2.5 minutes.

It is also estimated from this that all Acharyas from the time of Brahmagupta to the time of Ganesh Daivagya considered the maximum day length to be the same as the 24° of the supreme transit of the Sun because they did not know that due to the atmosphere, the true day length increases by about 14, 15 minutes compared to the apparent day length.

Considering the variation, the maximum day length nowadays is 34 hours 10 minutes. This is only 5 minutes more than 34 hours 5 minutes. This inaccuracy cannot be considered more than for the arc of rising and setting time.

The difference in the shape of the Sun due to variation—as the Sun rises and sets, the shape of the Sun appears larger and somewhat oval. The reason for this is that the increase in variation near the horizon is very intense. The image of the Sun is about 32 degrees. Therefore, at the time when the lowest point

of the Sun's image remains in the horizon, its true position is 60°, and the true position of the highest point of the image is about 32 degrees less. Due to this difference, the lower point remains higher and the upper point remains less than it. This causes the apparent vertical to appear as if the image is 5 degrees less, making the Sun appear oval.

Further investigation of the motion will increase greatly. If one wants to know when the upper image of the sun appears on the horizon, then the amount that comes from adding 35' + the semi-diameter of the sun or 35' + 16' elevation in place of ta(n) in the equation (2) on page 376 should be added to the time calculated mathematically.

To know the rising time of the moon, a calculation called Lamba Sanskar has to be done. Therefore, the explanation of parallax will be given further.

Parallax*

The positions of the sun, moon, and planets, which are known from the new methods explained in Spṣṭāadhikar, can be seen exactly as from the Earth's center. But the surface of the Earth

*A lot of help from Loomis' Practical Astronomy has been taken in writing this section.

SOURCE PAGE 406

When viewing from a place, some difference is seen in those places. If two observers see the Moon at the same moment from two places on the Earth's surface, they do not see in the same direction. Therefore, it is necessary to know how much difference the celestial bodies appear from the actual positions when viewed from a certain place. The difference in the directions of the celestial bodies seen when viewing from the Earth's center and any place on the Earth's surface is called latitude.

In Figure 74, the Earth is the center or geocenter, a place on the Earth's surface where the observer is seeing the Moon. The line D K is the meridian line that goes to the X of the place. From the place, the observer will see the Moon in the D J direction, and from the geocenter B, the Moon will be seen in the C H direction. The difference in these directions is the same as the angle B C. This is the altitude of the Moon from the place.

The angle of the moon's portion is equal to that of kha and ca, which is called the true portion of the moon. The angle of the moon's portion is equal to that of kha and bha, which is called the actual portion of the moon. The diagram proves that the true portion of the moon is the actual portion of the moon + latitude.

It is clear that due to the inclination, the true position of the Moon's declination becomes more than the actual declination, therefore the Moon's latitude becomes as much less. This

SOURCE PAGE 407

Because the Moon appears to hang some distance from its true position. That is why this distance is called length. This length also affects the portions of the Moon and other planets, the shadow, the viṣavāṃśa, the declination, etc., the explanation of which will be given later.

Suppose that r is the radius of the Earth; c is the distance of the Moon from the center of the Earth; n is angle h c, the true position of the Moon; na is angle h d c, the true position of the Moon; ja is the length of the Moon's position;

in the triangle d h c

```
bhd - - - = - - - jyabhchadjya)bhchad
```

But the sum of angle h d c and angle k d c is 18^(°) therefore jyabhchad = jyak d c. It is proved by raising the letters of the symbols written above that

```
vak - - - = - - - jyalajyaana
```

```
Athvajyala = 3 / k x sine
```

```
Its meaning is this, the ratio of the radius of the earth / distance of the moon x true portion
```

From this it is proven that if the distance of the moon or any planet for a given place is given, then its length increases or decreases according to the sine of its true position. That is, if its true position is less, then the length will be less, and if it is more, then the length will be more. If the true position is 60°, meaning the moon or planet is rising or setting, then the value of its sine is 1, which is the maximum. In such a case, the length related to the position is also maximum, that is, the greatest. The maximum length is called the ultimate length or horizon length because such a large length occurs at the time when the celestial body is rising or setting and is on the horizon. It is also clear that when the body is exactly on the cross, its position is zero.

SOURCE PAGE 408

Then the sine of the declination will also be zero and the value of latitude will become zero. That is, when the celestial body is exactly above the cross on the head, there is no latitude related to the declination in it.

If the horizon latitude is expressed as l then

```
sine = v / k
```

If in the first equation the place of tra/ka is taken as sine l then

y = y X y (2)

This means that if the sine of the horizon's declination is multiplied by the sine of the true latitude, then the sine of the latitude will be obtained.

Knowledge of the declination can only be obtained from this formula if the true latitude of the body is known. If the actual latitude is given, then the other type of formula will work, which is proven as follows—

It is clear from Figure 74 that

```
na = n + la
```

Therefore, from formula (1)

```
jala = jala X ja ( n + la ) = jala ( jana*jala + koja*jala ) = jala*jana*koja + jala*koja*jana
```

Dividing both sides by cos(l)

```
Sperula = sin(l) + sin(l)cos(l) when done in one side
```

```
Sperula = sin(l) / 1 - cos(l)
```

From this formula, the value of latitude can be known at the time when the true position is given. However, it is not convenient to know the latitude in this way because a lot of multiplication and division has to be done. Therefore, the second form should be proven to simplify it.

If the numerator of the right side is divided by the denominator

sper la=sin l sin n+sin2 l sin n cos(l)+sin3 l cos(l) cos(l)4+sin5 l sin n cos(l)3...etc

---

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In this series, the subsequent terms become so small that there is no loss in taking only the first three terms, if the value of l is not more than 10.

Terms like spirala can also be kept in place where only la is because* la = spirala - 3 spira3 la

This is often the pure formula to express the arc in its tangent if the arc is very small. If only sin3 and sin3n are taken for the second term of this formula, there is no loss. In this case 3

```
la = sinl + sin2l + sin3l (sin2l - sin3l / 3) +..... = sinl + sin2l + sin3 (sin2l - sin3l / 3) +.....
```

But sin n cos n = sin2n / 2

```
and sin2n - sin3n / 3 = 3sin2n - sin3n / 3 = 3sin - 3sin3n - sin3n / 3 = 3sin - 4sin3n / 3 = sin3n * * / 3
```

Therefore la = sin l sin n + sin2l sin2n + sin3l sin3n / 2 +.....

* See Sudhakar Dwivedi's Chalan page 40

** See Hall and Knight's Trigonometry page 105(published in 1910)

SOURCE PAGE 410

The value that will come from this formula will be in radians. To express this in degrees, each term on the right side should be divided by sin 1" or multiplied by cos 1" because

```
1radian = 206265 " ∴ 1 " =.00000485radian ∴ sine1 " =.00000485
```

From this it is proven that to make a declination from the radian, divide the radian by 0.0000485. But 0.0000485 = sine 1", therefore to make a declination from the radian, the radian should also be divided by sine 1".

$\sin^2 + \sin^{22} + \sin^{32} / \sin^1 + 2\sin^1 + 3\sin^1 + (3) = \sin^2 + \text{versed sine}^2 + \text{sine}^{32} / \sin^1 = \text{arc of declination}^1$ which is written as the short arc^1

$\sin^2 = \sin^2 \ldots 1 / 2\sin^2 = 1 / \sin^2 = \text{chord}^2$, etc.

From this formula, the latitude of a celestial body can be calculated when its true position is given.

To know the latitude of the Moon, the three terms of this series are required, but for the Sun and planets, only the first term is sufficient because their latitudes are very small, so the values of the second and third terms are not equal.

Therefore, this formula will be sufficient for the latitudes of the Sun and planets:-

la = sin(l) / sin(1)

But when l is very small, then sin(l) / sin(1) = l

∴ la = l sin(l) (4)

SOURCE PAGE 411

Example 1—If the horizon of Venus's latitude is 30

la = ljyan = 30 " X sin60° = 30 " X.866 = 25".68

Example 2—If the horizon of the Sun's latitude is 8.6

Sūrya Siddhānta = 80 ° − 16 ° = 74 ° la = lajyāna = 8 ″.6 X sine74 ° = 8 ″.6613 = 8 ″.27

Example 3—If the horizon of the Moon's latitude is 60' 41".5, what is its latitude when its true position is 80° 16' 16"

Here the true position of the Moon is given, so the first formula must be used. Therefore

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Here the actual latitude is given, so one has to use formula (3). The radius 60' 41".5=8.246833 The radius 79° 16' 26”.33=6.662418 The chord 1”=5.314425 Sum =3.553676 But the chord

3578’.26=3.553676 ∴The first term of formula (3)=3578".26 =56’ 38".26 The sine² 60’ 41".5=6.4637 The radius 2X76° 19’ 26”.33=6.5612 The chord 2”=5.0134 Sum=1.0683 But the chord 11”.70=1.0683 ∴The second term of (formula 3)=+11”.70 The sine³ 60’ 41".5=4.740 The sine 3 X 76° 19’ 26”.33=6.628 Negative the chord 3” =4.837 Sum=6.505 ∴The value of the third term=-0”.32 Collecting the three terms gives latitude=56’ 38".26+11”.70-0”.32=56’ 46".67 It is stated that the horizon latitude = t where t is the distance from the geocenter to the place of the observer and the distance from the geocenter to the celestial body. But the Earth is not a perfect sphere, so the value of v is not the same everywhere. In such a situation, the value of the horizon latitude cannot be the same for all places. Therefore, before mathematics, the horizon latitude is known that occurs at a place on the meridian (equator). Then with its help, the horizon latitude and the true apparent latitude of other places and the apparent latitude of the ecliptic are known.

Suppose v is the meridian of the Earth, d is the North and South poles of the Earth, and v and a are two points of the equator. From the geocenter to the equator v

---

SOURCE PAGE 413

The distance of the point is v and the distance of the pole is d. The position of the observer is s and s is the tangent to the line of sight which is in the plane of the horizon of d. The line x is at a right angle to the tangent s of d, so this is the meridian of the position. Therefore, the true north of the position is x. If this meridian is extended within the Earth, it will reach the point v on the line v without reaching the center of the Earth. If a line is drawn from the center of the Earth to d and it is extended towards the sky, it will reach point x. Therefore, it is proven that the geocentric meridian of the position is x. x is called the geographical meridian of the position. It is stated in Chapter on Mean Motions on page 55 that the angle v is the geocentric latitude of the position, so the angle v is called the true or geographical latitude of the position. The meridian of the position, the angle v and the angle d from the radius of the Earth, which forms the angle v is called the meridian angle (angle of the vertical) of the position. The geographical latitude of a place and the geocentric latitude are expressed by the letters a and b.

It is proven from Mukhopadhyay's Geometry of Conics pages 63, 64 that

```
pf = pb X th2 / t2
```

In English, the geographical latitude is expressed as phi and the geocentric latitude as phi'.

SOURCE PAGE 414

J h a n t, th k r m s e d i r g v r t s e d i r g au r l g h u a k s k e a d e h a i n. P r n t u p d = p b h X s p r e ∠ p b h d = p f X s p r e ∠ p f d ∴ p b h X s p r e a = p f X s p r e a = p b h X t 2 / t 2 s p r e a ∴ s p r e a = t 2 / t 2 s p r e a (y)

This means that if the tangent of the geographical latitude of a place is multiplied by t2 t2, then the tangent of the geocentric latitude of that place will be obtained. By the inverse operation, if the geocentric latitude is given, the geographical latitude can also be known.

It is stated that t and t are half of the long and short axes of the Earth, whose values are as per Colonel Clark*

t = 2, 06, 26, 202fít = 3663.3mile (with remainder) t = 2, 08, 54, 865foot = 3646.8mile (with remainder) • t^2 20854865 ^{2} • t^2 = 20626202 ^{2} =.6631665 =.6632

If the geographical latitude of Dehradun is 30°18’51".8, then what is the geocentric latitude and the angle of the meridian?

According to the above formula,

Rupreā =.6632 imes Spre30018'51.8 herefore Lrispreā = Lri.6632 + Lrispre30016 = 1.6670 + 6.7670 = 6.764 a = 3008'40

This is the geocentric latitude of Dehradun. If this is subtracted from the geographical latitude, the angle of the meridian will be the same as 10'12”. The meridian's

*Hall’s Spherical Astronomy pp. 44.

SOURCE PAGE 415

The angle is called Reduction to Geocentric latitude in the Nautical Almanac. In the Nautical Almanac of 1627 AD, its value is written as 10^(prime)5^(primeprime)3^(primeprime). The reason for the difference is that in this calculation, the accuracy of the small quantities has been taken only to four decimal places.

The distance of any place from the center can be known in this way:-

In Figure 76 there is a section of an ellipse which starts from the point of the equator and goes to the other side of the equator to Va from the North Pole D. If a semicircle is drawn on Va and Da, then this will be the auxiliary circle of Va and Da. Ph is the angle of the geocentric latitude which will be indicated by A. The distance of the place from the geocenter will be indicated by the letter B. In the triangle P, Ph, D

pb = $\sin^2 a - \sin^2 b$

p = v, dzyaa = vzyaa

It is proven from page 65 of Mukhopadhyay's Geometry of Conics that

Padpada = bh d h bhav = that

∴ pada = tath × pad = tath × tjyaa

Parantupbh2 + pada2 = bhada2 = t2

∴ (vakojaa) 2 + (tath × tjyaa) 2 = t2

or t2 kojyaa2a + t2 / th2 X t2 tjyaa2a = t2

SOURCE PAGE 416

But it is proven on page 361 that t^2/t^2 = sine^2 ↟rac(a)(a) therefore sine^2 a ↟rac(a)(a) = t^2 here sine^2 a/a = sine^2 a ↟rac(cosine a/ sine a)(sine a) = sine a imes cosine a and sine a = sine a/ cosine a herefore t^2 cosine^2 a + sine a/ cosine a sine a imes cosine a imes t^2 = t^2

By multiplying each side by cosine a and collecting the common parts of each term

t^2 cosine a (cosine a imes cosine a + sine a imes sine a) = t^2 cosine a herefore t^2 cosine a cosine (a - a) = t^2 cosine a *

t^2 = t^2 cosine a / cosine a cosine (a - a) (6) = cosine a / cosine a cosine (a - a) when the radius of the meridian is 1

Take it.

This proves that if the geographical latitude of a place, the angle of its meridian, and the distance of the geocenter from the equator are known, then the distance of that place from the geocenter can be known.

Knowing the horizon distance of a place

Suppose the horizon distance of the Moon is l on the equator and li at another place. If the distance of the geocenter from the equator is t and the distance of that place is t, then it is clear from page 385 that —

sine = t / k

SOURCE PAGE 417

and the radius = t / kis jyala = t / tjyala yadit ko 1 manlya jyata jyala = t jyala (7)

This means that if the radius of the Earth is taken as 1 for the polar radius, then by multiplying the sine of the lunar polar horizon by the radius of a place, the horizon of that place will be found.

Example 2—If the lunar polar horizon is 53’ for the Moon, what will be the horizon in Dehradun?

The angle of the meridian is known in Example (1). Therefore, the radius of Dehradun should first be known from formula (6):-

Sūrya Siddhānta = Sūrya Siddhānta (a - a) sine30018' 52'' = sine3008' 40'' sine10' 12'' therefore 2tritra = sine30018' 52'' - sine3008' 40'' - sine10' 12'' = 6.6361 - 0.6.6366 - 10 = 1.6662 therefore tritra = 1.6666 therefore tr = 0.666 therefore Dehradun latitude sine = 0.666 × sine53' = 0.666 × 0154 = 0.01538 therefore Dehradun latitude = 52' 56'' *

* For such fine calculations, the table of the small ratios should be at least of seven decimal places, otherwise there will be much imprecision.

SOURCE PAGE 418

What difference does the inclination of the celestial body have in the true and actual latitudes of the celestial body?

From what has been said so far regarding inclination, it is clear that due to it, there is a difference in the coordinates of the celestial body, which has some effect on the latitude, declination, latitude, and the south.

Knowing the inclination of the latitude

Assume that the meridian of a place is the meridian of that place, the east, west, and north points of that place, the terrestrial cross and the northern celestial pole are the earth-centered. Assume that the true position of the moon as seen from the center of the earth is 'ch' and its apparent position as seen from the surface is 'cha'. The difference in the position of the moon is the inclination, for which 'la' is written on page 383. The angles 'kh', 'dh', 'ch', and 'kha' are the hour angles. Therefore, it is clear that due to the inclination, this apparent hour angle of the moon is greater than the true hour angle by the same angle as 'ch' and 'kha'. This angle is the moon's

SOURCE PAGE 419

The declination is the latitude. It is also clear that the true polar distance of the Moon is greater than its actual polar distance. Therefore, the true declination will be less than the actual declination. Therefore, the declination of the Moon will be equal to the difference of theta - theta.

Suppose the latitude of the Moon's horizon is l, the latitude of the equator is lambda, the true time is t, and the true declination and the observer's geocentric latitude are a. Then it is clear that the true time of the Moon is x theta = theta + l = theta

In the spherical triangle of theta and theta

sin(theta) = sin(theta) sin(theta) sin(theta + theta) sin(theta - theta) (k)

And in the spherical triangle of theta and theta

```
sin(theta) = sin(theta + theta) sin(theta - theta)
```

∴ sin (x c a d h) = sin (x d h) × sin (x d h c a) sin (x c a) (x)

But angle xcd = angle cdh

sin (A) sin (B) × sin (B + C) sin (A + C) But according to the formula (1) of page 385, sin (A) = sin A × sin (B + C) By obtaining this in the composite triangle (G) sin B = sin A / sin (B + C) × sin (B) × sin (B + C) C = A B Polar Distance = 60° - C therefore sin (B + C) = Opposite sin B = Adjacent = 60° - A therefore sin (B) = Opposite A angle B + C = angle B + angle C = D + L sin A × Opposite A × sin (D + L) therefore sin B = (G) (K)

SOURCE PAGE 420

```
manlokip = sin(x) / cos(a)
```

t by j y a l i = p x j y a (h + l i) = p j y a h cos(y) + p cos(y) j y a l i (7)

if each side is divided by cos(y)

sparle = p sin(h) + p cos(h) sparle

, sparle = p sin(h) / 1 - p cos(h) (2)

The extension of this formula can be made in the same series as the formula(2) is brought into the form of formula(3) in page 385—388. In this way

li = pjya gh / jya.1 " + p2 jya2 dh / jya2 " + p3 jya3 dh / jya3 " + (3)

Formula(1) can be used to know the latitude when the true time and the apparent time are known, and formula(2) or (3) is used when the true time is known.

Example 1—Tell the latitude of the Moon when the latitude of the observer's position is 36°57'7", the longitudinal distance of the Moon is 56'36.".8, the true time of the Moon is 24°5'11".6 and the true time of the Moon is 61010'47.".4.

The geocentric latitude of this place is 36°45'47".5 according to the formula(5) of page 361.

lari jya li=lari jya 56'36".8=8.236048 lari coja a=lari coja 310'45'47".5=6.885754 *lari chere k=lari chere 24°5'11'".6=0.036563 ∴ lari p=8.164365 lari jya gh=lari jya 61010'47".4=6.942572 lari cochere 1'="5.314425

* In page 367 it is assumed that p = sin(l) x sin(a) / sin(c) but

sin(c) = division line = chord, therefore p = sin(l) x sin(a) chord

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SOURCE PAGE 421

Therefore, the length of the first term of the formula (3) = 3.421362.. First term = 2638”.53 lri p^2 = 2 lri p = 6.3287 sine^2 = sine^2 imes 61010.47”.4 = 6.9267 lri cochere^2 = 5.0134 herefore length of the second term = 1.2688 herefore second term = +18”.57 lri p^3 = 3 lri p = 4.463 lri sine^3 gh = lri sine^3 imes 61010.47”.4 = 8.761 negative lri cochere^3 = 4.837 herefore length of the third term = 8.121 negative herefore third term = -0”.01 herefore li = 2638”.53 + 18”.57 - 0”.01 = 2650”.06 = 44‘17”·06 = 44‘17”·1 herefore apparent time of the Moon = 61010.47”·4 + 44‘17”·1 = 61055‘4”·5 If the apparent time is given, then the declination of the equator is known from formula (1) as follows:- lri sine li = lri p + lri sine 61055‘4”·5 = 8.164365 imes 6.645604 = 8.10666 herefore li = 44‘17”·06 In this way, a table of the declination of any place can be prepared. The parallax in declination of the Moon can also be known for this work by figure 77. Suppose the actual

This work will also provide figure 77. Suppose the actual parallax in declination, actual time, and actual apparent time are k and gh respectively, and due to the declination, the true parallax, true apparent time, and true apparent time are in the order of na, na, and gh respectively.

SOURCE PAGE 422

Let the Moon's declination parallax be lu. The passage compares spherical triangles Cha-Dha-Kha and Chā-Dha-Kha. It states that their angles at Kha are the same and that Cha-Dha is the Moon's true polar distance, 60° minus ka. The following OCR equation sequence retains source order; sine and cosine labels are rendered in English, but fraction layout requires checking against the scan: cosine(Cha-Dha)

= kojyā ca kha - kojyā ca kha kojyā dha kha sine ca kha sine dha kha kojyā cā dha = kojyā cā dha - kojyā cā kha kojyā dha kha kojyā cā kha dha = sine cā kha sine dha kha parantu ca kha dha and cā kha dha koṇa eka hī are and ca dha = caṃdramā of yathārtha dhruvāntara = 60° - ka cā dha =,, true,, = 60° - ka,, kojyā ca dha = kojyā(60° - ka) = sine ka and kojyā cā dha = sine of sine ka - kojyā na X sine ā sine of - kojyā nā X sine ā sine na = sine nā arthāt sine ka sine nā - sine ā sine nā kojyā na = sine of sine na - sine ā sine na kojyā nā yā sine ka sine nā - sine ā(sine nā kojyā na - kojyā nā sine na) = sine of sine na ∴ sine ka sine nā - sine ā sine(nā - na) = sine of sine na parantu nā - na caṃdramā of natāṃśa lambana is isalie sine(nā - na) = sine lā = sine li sine nā(dekho pṛṣṭha 385) yahā̃ li kṣitija lambana mānā gayā hai. ∴ sine ka sine nā - sine ā sine li sine nā = sine of sine na yā sine of sine na = sine nā(sine ka - sine li sine ā)(ka) isa samīkaraṇa from sine nā and haṭāne of lie golīya tribhuja ca dha kha and cā dha kha from isa prakāra kāma lenā hogā— sine ca kha dha = sine ca dha kha sine ca dha = sine ca kha and sine cā dha = sine cā dha kha parantu ca kha dha and cā kha dha eka hī are, isalie sine ca dha sine ca dha kha = sine cā dha sine cā dha kha

Sine of latitude = sine of latitude - sine of colatitude

SOURCE PAGE 423

Sūrya Siddhānta

Scientific Commentary

Chapter on Mean Motions

sa pa ra e ka = ja ya ā ka – ja ya ā la i ja ya ā ā / ja ya ā gha
ka o ja ya ā ka x ja ya ā gha = ja ya ā ka + ja ya ā la i ja ya ā ā / ka o ja ya ā ka x ja ya ā gha
ja ya ā dha = (ja ya ā ka / ka o ja ya ā ka – ja ya ā la i ja ya ā ā / ka o ja ya ā ka) x ja ya ā gha
= sa pa ra e ka (1 – ja ya ā la i ja ya ā ā / ja ya ā ka) / ja ya ā dha (9)

Chapter on the Three Problems

Sūrya Siddhānta

SOURCE PAGE 424

sine (c - ca) = --- --- --- cos(c)cos(a)

- sine(c-ca) imes spereka imes (1 - sine(declination)) = sin(longitude)

or

sine (c - ca) / cos(c)cos(a) = sin(a) - spereka / cos(a) (sine(declination) - sine(latitude))

but according to page 113 of Hall and Knight's Trigonometry

sinegha — sinegha = 2sinegha — h / 2cosinegha + h / 2

Where gha is apparent time and h is actual time.

Therefore gha - h = latitude = l

gha + h = gha - h + 2h = l + 2h

∴ sinegha - sinegha = 2sinele / 2cosinegha (l + 2h) / 2

and k - ka = difference of true and actual declinations = latitude = l (see page 366)

Sūrya Siddhānta

Scientific Commentary

Chapter on Mean Motions

Chapter on True Positions

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Therefore

jalu / kojyakkojyaka = jalijyaa - spreka / kojyakjyaghakojyakjyaali2 x jyalikojyaajyagha / kojyakkojyali - 2 x kojya (h + li / 2)

If both sides are multiplied by the sine of the cosine and simplified then

jyalu = jyalijyaakojyaka - jyalikoja a kojya (h + li2) jya ka kojya li2 (k)

Suppose that

kospref = kojya (h + li / 2) kospera / jyalii / 2

Then sine of the sine of the sine of the cosine - sine of the sine of the cosine is

lu = sin(r) (cos(a) - sin(a)cos(a)) = sin(a) (cos(a) - cos(a)) = sin(a) (sin(a))

For sin(a) the mean value is if sin(a) = sin(a) (r - a) - a = sin(a) = c - lu. ∴ sin(a) = sin(a) (r - a + lu) = sin(a) (r - a) cos(lu) + cos(a) sin(lu)

(2)

SOURCE PAGE 426

When both sides are divided by cosecant, the result is equal to (cos(phi) + cosec(phi)) and cosec(phi) = cos(phi)/(1 - cosec(phi)). In this way, the total length is = (cos(phi) / sin1" + cos(2) / sin2" + cos(3) / sin3" +..... (3) (4))

If the true declination is known, the declination can be found using formula (2) and if the actual declination is known, it can be found using formulas (3) and (4).

To know what difference occurs per hour in the declination of the equator:- The formula for the declination of the equator is given in page 367:-

```
If sinX cosecX singh = coseck and lidthnu are very small, then sinX = l and sinX = l
```

The difference seen between true time and apparent time due to declination is also very small, so there is no loss in considering singh as equal to singh for the convenience of practice. Therefore, the form of the above formula is as follows:-

```
li = li X cos(a) X sin(h) / cos(a)
```

In this formula, h is such that its difference keeps increasing very much every moment, the variation that arises in li, li, a, and k is so slow that these quantities can be considered constant for some time. Therefore, if h is considered a moving quantity and the instantaneous motion of li is calculated, then

```
t (li) = li X cos(a) / cos(a)cos(h)
```

Here ta(h) must be written in radians. If it is to know what difference arises in per hour in the declination of the equator, then ta(h) should be expressed in 150 radians. It is known that

SOURCE PAGE 427

=3.14156 radians ∴15° = 0261766 radians and 6° = 01047167 radians Example—what difference does the variation in the latitude of the Moon per hour have when the horizon of a place is 57', the geocentric latitude is 4211'21", and the Moon's declination is 25° and the lunar distance is 50°? Lari Li = Lari 57' = Lari 3420" = 3.53 026 Lari cojya a = Lari cojya 4211"21" = 6.866778 Lari cojya g = Lari cojya 50° = 6.808067 Lari ta(gh) = Lari ·261766 = 1.417666 Lari chere k = Lari chere 25° = 0.042724 ∴ Lari ta(li) = 2.672564 ∴ ta(li) = 4700"5 = 7'50.5 Knowing what difference does the variation in the declination per hour have:— it is proven in page 402 that the sine of the sine of a is the sine of the sine of a(gh + li/2) sine of li/2 If instead of the sine of lu and sine of li are taken as lu and li, and if gh + li/2 is taken as gh and sine of li/2 is taken as 1 and sine of ka is taken as— lu = li sine of a — sine of li sine of a gh now if the instantaneous motion of this equation is calculated by considering gh as the moving body, then ta(lu) = li sine of a sine of gh ta(gh) Example—what difference does the variation in the declination per hour have in the Moon's declination, when the horizon of a place is 57', the geocentric latitude is 4211'21", and the Moon's

Example—what difference does the variation in the declination per hour have in the Moon's declination, when the horizon of a place is 57', the geocentric latitude is 4211'21", and the Moon's declination is 25° and the lunar distance is 50°?

SOURCE PAGE 428

Lari57' = Lari3420 " = 3 · 534026LarikozyaA = Larikozya42011' 21'' = 6 · 866778LarijyaK = Larija25° = 6 · 625648LarijyaGh = Larija50° = 6 · 884254Laritā (h) = Lari261766 = 1 · 417666 ∴ Laritā (l) = 2 · 331675 ∴ Ta (l) = 214'' · 8 = 3' 34'' · 8

The effect of the inclination on the arc and declination

Just as formulas were established to know the latitude and declination, similarly such formulas can also be established to know the arc and declination. For this work, the point D in figure 77 must be understood as Kadamba (declination circle). In this case, the difference between Kadamba and Khswastik, which goes from X to U, is the circle on which there is the trigon position or Vitribha position (see figure 63 and page 330), D, C, and CA are the true and apparent declination of the planet on the Kadamba-Protavrittas. Therefore, D and CA will be the same as the true and apparent positions of the planet. The difference between D and CA is the part of the trigon position which comes from Khswastik, which when subtracted from 600 will give the part of the trigon position. This is the true difference of the trigon position from Khswastik, so it is the geocentric declination of Khswastik.

Suppose the geocentric declination of Khswastik or the part of the trigon position is Ta, the true declination of the planet is Sha and the apparent declination is Sha. The true difference between the arc and the part of the trigon position of the planet is D and CA, which can be summarized as the true analysis or simply V. If the latitude inclination and the arc inclination are also present, then like the equation (k) of page 367

Jya bhi = Jya likojya ta jya (V + bhi) / kojya sh (k)

It can also be written according to the method shown in page 367.

Sprebhiṃ = tjyāva /1 - tkojyāva (2)

SOURCE PAGE 429

whent = sin(latitude) / cos(latitude)

It is clear that the formulas (a) and b are also very small, so there will be no loss in writing their sines instead of arcs, but it will be simpler. Therefore

b = cos(latitude) / sin(latitude) (b)

Or if the shadow of the planet is very small, like the shadow of the moon during a solar eclipse, then the value of the arc will be approximately equal to 1. Therefore

b = cos(latitude)

This form is stated in the shlokas 7-8 of the Sūrya Siddhānta Chapter on True Positions. The declination or latitude—if the shadow is the latitude, then the equation of page 402

(a) like

sin theta = sin lambda cos delta cos alpha + sin lambda sin delta cos alpha cos lambda + sin lambda sin delta sin alpha cos lambda + cos lambda sin delta sin alpha

It is clear that bh / 2, that is, the portion of the arc is very small, so cojya bh / 2 = 1. In such a situation, if v + bh / 2 is replaced by v and sha is replaced by sh, then there will be no much difference, and the formula (gh) will become simple as follows: jya bh = jya li jya va cojya sh - jya li cojya ta jya sh cojya v(ng)

If sine of the ecliptic and sine of latitude are taken in place of them because they are very small then

bhu = li sine(ta) cosine(sha) - li cosine(va) sine(sha) cosine(va) (c). The OCR does not reliably preserve the equation layout.

To understand the similarity of the declination of the bhogansh with the declination of the vishuvansh and the similarity of the revolution of the kranti with the declination of the shar, one must remember that

BhoganshambankesutrammevishuvanshambankeSutrammeLi = horizonLi = horizonbhi = BhoganshambanLi = Vishuvanshamban

SOURCE PAGE 430

Tr = the part of the three-fold ascendant, a = geocentric latitude, v = meridian, g = true time, sh = true hour, ka = true declination, sa = true declination, bu = hour length, ya = declination length, lu = revolution length

Sūrya Siddhānta has named the part of the arc and the name of the hour length. In other treatises, the part of the arc is only called arc or true arc length, and the name of the hour length is called nati.

Now in brief, it will be explained what our Acharyas have written about the arc:-

Bhaskaraacharya has written* that when the daily motion of a planet is divided by 15, the absolute arc length (horizon arc length) of that planet is obtained. The reason for this is explained:-

The circle of the sky that is touched by a horizontal (plane) touching any place on the earth is called the horizon circle of that place. This horizon circle divides the sphere of the sky into two parts. This horizon circle is called the sensible horizon circle. If a plane of sky is extended parallel to this sensible horizon circle towards another plane of sky passing through the center of the Earth, the circle that this plane cuts in the sky is called the true horizon circle of that place. In Figure 78, the position of the observer and the center of the Earth on the earth surface are shown. The line drawn from D touching the earth surface intersects the orbit of the Moon at point C and the orbit of the Sun at point S. Therefore, C, S are on the sensible horizon of the place. If a plane of sky is extended parallel to this, passing through another plane of sky which cuts the orbits of the Moon and the Sun at points C and S respectively, then this plane is called the true horizon of the place of D. It is apparent that when the Moon and the Sun or any other planet is on the true horizon of the place of D, at that time it is in the plane B C S which is below the sensible horizon of D, so it will not be seen by the observer. In such a situation, the magnitude with which the planet remains below the sensible horizon is C or S (which is equal to B D, i.e., the semi-diameter of the Earth).

SOURCE PAGE 431

Therefore, there is nothing wrong in saying that when a planet is on the true horizon of a place, it is below the true horizon of that place by the same amount as the semi-diameter of the Earth, that is, its

latitude is the same as the semi-diameter of the Earth. If you understand 'cha' as 'cha' and 'sa' as 'sa', there will be no much difference because 'cha' or 'sa' is much smaller than the whole orbit. Therefore, it can be said that when a planet is on the true horizon, it is below the horizon of the observer by the same amount as the semi-diameter of the Earth in its orbit.

This has been stated before (see pages 16-17) that our teachers assumed that the planetary motion of each planet is the same. According to the verses 81-82 of the upcoming Geography chapter, the daily motion of each planet is 11858.72 yojanas. The semi-diameter of the Earth is 800 yojanas according to Sūrya Siddhānta and 760.5 yojanas according to Siddhānta Shiromani (see Chapter on Mean Motions page- 53). What was written in the previous text is exactly one-tenth of the planet's daily motion. According to the earlier text as well, the daily motion of the planet is the semi-diameter of the Earth

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SOURCE PAGE 432

It is often equal to fifteen times. Therefore, it can be said that at the time when the planet is on the actual horizon, at that time it is below the true horizon by the fifteenth part of its daily motion. That is, the true declination of the planet is equal to the fifteenth part of its daily motion. One day is equal to 60 *ghadi* (hours), so the fifteenth part of the motion in 60 *ghadi* is equal to the motion of four *ghadi*. This means that as much the planet moves in four *ghadi*, so much is its true declination (in the arts). In units of time, the true declination of the planet is equal to four *ghadi*.

If the daily qualitative motion of the planet, and its daily planometrical motion, and the utmost declination, and the distance of the planet's orbit from the Earth and the semi-diameter of the Earth can be expressed in this way, then the above statements can also be revealed as follows:-

```
L = g / 15 = y / 15c = t / k
```

Because if the planet is very far, its daily rotational motion, that is the arc the planet travels in one day, when divided by the semi-diameter of the orbit, its daily rotational motion is found, so g = y/k. But when y is divided by 15, what comes is equal to the semi-diameter of the Earth, so y/15 = tr.

From this it is proven that the rules that our Acharyas made to know the magnitude of the ultimate latitude are very similar to the rules made today (see p. 383). But the mistake in this was that the planetary motion is not the same as in the present, as proven by the modern observations. Therefore, the values of the ultimate latitude given by our Acharyas are very different from the ultimate latitudes obtained by the modern observations. This will be made clear by the comparative table on page 410.

Now it is necessary to state that our Acharya knew how to know the apparent portion of the planet by knowing its ultimate latitude and its true portion of the latitude and the declination or the latitude. Bhaskaracharya writes that (1) at the time when the planet is on Khastak, there is no inclination in it because from the center of the Earth and the observer

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SOURCE PAGE 433

According to the current arcs of the planets according to Bhaskaraacharya's Sūrya Siddhānta, the mean obtained, the true image obtained, the utmost extent, the least extent, the utmost extent, the least extent, the sun is deviated, deviated, deviated, deviated, the moon 236.5 8.7 6.0 1860 1656 mangala 3162.3 3186 3720 1740 2028 budha 125.7 3.5 16.9 4.4 21.2 guru 682.1 6.4 14.4 4.8 10.6 shukra 20.0 1.4 2.1 31.6 46.7 shani 384.5 5.0 31.4 6.6 60.0 8.0 0.8 1.0 15.8 16.5

The lines drawn to the planet are the same. (2) When the planet is in the Tribhona sign, that is, when the planet is at the point on the ecliptic that is three signs less than the rising sign, then the planet does not have the arc of inclination, only the arc of transit. (3) When the ecliptic makes the meridian from Swastika and forms the zenith, and the planet is on the ecliptic, then it does not have the arc of inclination, only the arc of arc. The rules for knowing what inclination and transit are in other conditions have been stated.

It is stated in page 406 that to know the length of the portion of time, it is first necessary to know the nataṃśa of the three signs of time or

It is known from this that Bhaskaraacharya considered the Earth to be a perfect sphere because only then this statement is correct.

SOURCE PAGE 434

What is the altitude because it is a part of the sine formula (x) of the altitude of the three-fold ascendant. The sine of the nadim of the three-fold ascendant is called the altitude and the sine of the altitude is called the arc of the nadim. As shown in Figure 76, when the rising of the ecliptic is south of the point east of the horizon, then the three-fold ascendant is west of the meridian because the three-fold ascendant is 3 signs or 10 degrees less than the rising ascendant. The meridian passing through the zenith, going through the Khastak and Kadamba points, is at right angles to the ecliptic and intersects the horizon at point F.

u p d f = the horizon circle of the place u d h k m v d = the meridian circle of the place d = the North Celestial Pole k = Khastak m = the mean ascendant v p = the tropics t m t a = the ecliptic t = the autumn equinox (Libra) k = Kadamba k h t f = the meridian passing through the three-fold ascendant u p = the arc of the rising ascendant

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SOURCE PAGE 435

Therefore, if the arc is theta and phi = 60^(°) = theta. If the common segment is removed from both arcs, then theta = phi

Spherical triangle in theta

```
sin (theta) = sin(theta) sin(theta)
```

But the angle between the equatorial and ecliptic is the true transit of the sun, and the theta angle is the same as the arc of the place.

```
Sūrya Siddhānta X sine ( t a ) ∴ sine ( a p u ) = sine of colatitude
```

This is the meaning of the third verse of the solar eclipse chapter. This sin(apu) is named uday or uday jya. But apu = phi, which is the same as the m k tna angle.

Now if the knowledge of the arc of kha in a spherical right-angled triangle ma kha tra is obtained, then the value of kha tra can be easily known because the angle kha tra is a right angle. It is clear that ma kha is the part of the mean ascendant which is the difference between the north revolution of the mean ascendant and the part of kha and iṣṭa sthāna. If the revolution were south, addition would be needed.

The sine of kha is named mean sine, the method of knowing this is told in the 4th and 5th verses of that authority. Therefore in the right-angled spherical triangle kha tra

sine (tkh) = sine (khmat) X sine (mkh)

If the spherical triangle kha ta is taken as a plane triangle, then sin(kha ta) = cos(ma kha ta) because kha ta and ma kha ta will have the same sum of 60°.

Therefore

sine (tkh) = cosecant (mkhat) sine (mkh) = cosecant (daf) X sine (mkh) = sine (mkh) √ 1 − sine² (daf) = √ sine² (mkh) − sine² (mkh) sine² (daf)

∴ short = √ mean sine² − mean sine² X rising²longitude = √ 1 − short²

SOURCE PAGE 436

This means the 5-6 shlokas of the Chapter on Solar Eclipses, where the radius is considered 1.

(sin300) ^ 2 / sight = (1 / 2) ^ 2 / sight = 1 / 4sight_latitude = sin_analysis_angle * / section = sin_analysis_angle / 14sight = 4 x sight x sin_analysis_angle

The magnitude of the latitude that will be found from this will be in hours. This formula is found in the formula (g) of page 406 where li = 4 hours = the ultimate latitude of the planet, "dṛggati = sine of the latitude of the three signs", and v = the angle.

For the declination or latitude, it is only given that multiplying the declination by the ultimate latitude gives the latitude. This method is very crude.

The need for latitude and declination arose in calculating solar eclipses. Therefore, it has been discussed in our texts in the Chapter on Solar Eclipses. However, nowadays the distance of planets and stars from the horizon is also determined by latitude. It is stated that the sine of the horizon latitude is sin(latitude). Therefore, if the sine of the horizon latitude and the distance of the observer from the Earth's center are known, the latitude can be easily found. Now in short, it will be stated how the latitude of a planet is measured.

Measuring the latitude of a planet— suppose there are two places on the Earth's surface in Figure 80 that are on the same longitude line and whose latitudes are also known with accuracy. At the time the planet comes on the meridian, the true latitude of it is D or not, and the true latitude of it is X or not. The sum of the latitudes of these two places is D, therefore

∠ D = 360^ circ - (∠ C + D + ∠ D) = 360^ circ - (180^ circ - N + 180^ circ - NA + ∠ D) = N + NA - ∠ D

* The difference between the portion of the planet and the three-fold ascendant is the analysis (see page 405)

SOURCE PAGE 437

But we do not need to know the angle of the d and da. We need to know the length of c from d and da, that is, we need to know the angles of c from d and da, which are the lengths of c from d and da. Suppose c from d=l and da from d=la and c from da=c. Now

```
jya = sin x / y
```

```
jyal = sin x / y = sin x / y
```

Because both d and da are the radii of the Earth, they can be considered equal. Therefore

```
jyaljyala - - - = - - - jyanja
```

or jya la = jya la x jya na

but l = c - la

```
∴ sin ( c - la ) = sinla X sina / sina
```

SOURCE PAGE 438

∴ sine and cosecant are - cosecant and sine are = sine la X sine not sine if each side of this equation is divided by sine and sine

```
Kosprela - Kosprech = sine / sine of sine
```

Or cospare la = cospare c + sin / sin c

Thus it is proven that if the netaṃśa of a planet is known by sighting it from two places, then the information about the angle ch can be obtained from the knowledge of the latitudes of those places. Then the information about la can be obtained from the above equation.

It is clear that the latitudes of the planets other than the Moon are very small, so if their arcs are kept instead of their sines of latitude, no loss can be incurred. In such a situation

```
sine (c - la) = sin(l) * sin(n) / sin(n - la) = la * sin(n) / sin(n - la).
```

```
∴ c = la + la * sin * n = la (1 + sin(n)) = la * sin(n) + sin(n).
```

∴ la = c * sin(n) + sin(n) * sin(n - la) (k).

From this formula, the simple length or horizon length of a planet can be known because if the horizon length is l then

SOURCE PAGE 439

```
sine = sine
```

or sine (x)

By collecting equations (a) and (b)

Li = c / 2sin + n / 2cos - n / 2 = c / sin + sin

Example--If the north latitude of the place is 56^°20'30" and the south latitude of the east is 33^°55'5" and the meridian declination of Mars from east and west is 68^°14'6" and 25^°2', what is the horizon length of Mars?

Dbhada = 56 ° 20 ′ 30 ″ + 33 ° 55 ′ 5 ″ = 63 ° 15 ′ 35 ″

n + na = 68° 14' 6" + 25° 2' = 93° 16' 6"

∴ c = dchada = n + na - dbhada = 63° 16' 6" - 63° 15' 35" = 31"

jyan = sin68° 14' 6" = .6287

jyana = sin25° 2' = .4231

∴ jyan + jyana = 1.3518

∴ kshitijlambanli = 31" / 1.3518 = 12" 63

It is apparent that to know the value of c in this way, we need to know the latitudes of two places. But if we see at what time the planet is on the meridian, how much it appears above or below from both places, then the value of c can be easily known. Suppose that in figure 81 is the position of the planet c and the position of some nearby star. From d, the difference between t and c is equal to the angle t d c, and from d, the difference between these two is equal to the angle d c.

Therefore d c d = t d c + t d c + d t d

But the star is so far that the angle t d is equal to zero. Therefore

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SOURCE PAGE 440

In this figure, from the place, it appears to be below T and from the place, it appears to be above T. Therefore, the sum of the differences of C and T is taken. If from both places, C appears on the same side of T, then the difference of the angles C and T is equal to the difference of the angles C and T.

In practice, it is difficult to observe a planet or star from two places with exactly the same longitudinal line. However, if the two places have a slight difference in their longitudes, the above rule may still apply because the inaccuracy caused by this will be the same as that of T.

This method can be used only to know the declination of the Moon and Mars. This method can only be pure for Mars if it is very close to the Earth, i.e., about 6 signs away from the Sun. This method is not useful for other distant planets because when the declination is 10 or 12 degrees

SOURCE PAGE 441

When taking action in this manner, some errors in sighting remain from which the result becomes very impure. The Moon is so close that even if the Earth is considered a perfect sphere, as understood for the above rule, and if bh and bha are considered equal, some inaccuracy remains. Therefore, to know the latitude of the Moon, bh and bha must not be considered equal, and their actual magnitudes must be taken. If l is placed in the place of jval and la is placed in the place of ja, according to page 414.

L = bhad / bch x sine

la = bhada / bch x sine

∴ c = l + la = bhad × sine + bhada × sin

butkṣitijalimbanali = 3 / bhachajhantra = Earth's radius

∴ v = v / l

∴ C X sin / L = B X sin + B X sin

∴ li = ch × trbhada × sine + bhada × sine

Here there are no actual parts of the Moon and the Sun. If one wants to know the value of li according to geographical or true parts, one should know the actual parts from the geographical parts as explained in page 383. From the above formula, it is proved that there is a difference in the apparent position of the horizon due to the variation in the observer's position because bh and bha will keep changing. This has also been seen by observation that the apparent position of the Moon's horizon appears different in different places. This is proof that the Earth is not a perfect sphere but an ellipsoid.

The declination of the Sun cannot be known by the above method. For this, several methods are used, two of which are written below:-

First method — two places D and DA are chosen on the surface that are near the equator and are very far from each other. For simplicity, also assume that the orbit of Venus is SS and the Sun is also on the plane of the equator on which D, DA

SOURCE PAGE 442

SOURCE PAGE 443

In the time of the arcs, the difference that exists in that time is the same time in which Venus moves from one point to another in its orbit. That is, in the same time, Venus appears to move as if it is revolving around the Sun at an angle equal to the angle of the Sun. Here it should be noted that this motion of Venus revolving around the Sun is equal to the difference in the motions of Venus and Earth. But we know that Venus and Earth come into a line after a certain time when they both revolve around the Sun, so the magnitude of the angle of śu sa śū or da sa dā can be known. When it is known how much angle is formed from two places on the Earth's surface at a point of the solar image, then it can be easily known from the method shown in figures 80 and 81 what is the length of the horizon of the Sun.

In practice, this method is not as convenient as it appears because the orbits of Venus and Earth are not in the same plane; the longitudes of the two places need to be known with great accuracy. This method was followed by Delisle.

The second method—in this method, there is no need to know the longitudes of the observer's places. Here, it is only seen how long Venus appears to move from one edge of the solar image to another from two different places. For this work, the places of the two observers should be very far apart but in the north-south direction.

Suppose that the two positions of the observer on the Earth are in the same plane, that is, on the plane of the paper, and the circle that shows the image of the sun, whose center is S, is at a right angle to this plane. The observer at the D position will see that the sun image moves in the direction of an arrow in its orbit, cutting the line K-X. But the observer at the D position will see that the sun image is cut by the circle G-H. The time taken for Venus to move from one edge to the other in front of the sun should be carefully observed and written down from each position. The speed with which Venus passes the sun image can be easily calculated. This is approximately 4 arcminutes in 1 minute. Therefore, when it is known how much time it takes for Venus to pass the lines K-X or G-H, the angular values of these lines can be easily and very accurately known. Therefore, the arcs of K-X and G-H

SOURCE PAGE 444

The magnitudes of half parts can also be known. However, the angular value of the solar image remains in the deviations. Therefore, the deviational values of s ch and s sh can also be known according to the geometry-

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SOURCE PAGE 445

```
s2 = sgh2 - ghch2
```

and sch2 = skh2 - khch1

Once the information of sch and sch is known, finding the difference between these two gives us the knowledge of chch. From this, the distance of chch can also be known in miles because if both triangles are considered similar to the dshu and chshu

```
chch = shchdadu
```

but the relation between shch and shd is known to us by Kepler's laws because shch is the distance from Venus to the Sun and shd is the distance from Venus to the Earth. Therefore, if shch is 723 and shd is 277 then

chch = 723dadu277

The Earth has two places on the surface, therefore their mutual distance can be easily known. In this way, the magnitude of ch can also be known in miles. But in the same way, the magnitude of ch can also be known in degrees. Therefore, when its magnitude is known in both degrees and miles, it is easily known that how far the sun is from the earth because

Chachakavikalatmakman = Chachakamanmilonmein26265PrithviSeSūryakīdūrī

∴ Earth-Sun-distance = 206265 X shadow-distance in miles in shadow-angles

The declination of the sun can be easily known from this. Hally invented this method in 1773 AD. The defect in both these methods is that it is very difficult to precisely ascertain the time of the inner touch of the images of Venus and the sun. The motion of Venus is so slow and the edge of the sun's image is so indistinct that several errors in the time of contact may occur.

Knowing the arc of the horizon, knowing the distance between the sun and the moon—it is stated that the sine of the arc of the horizon is (t)/(k), where t is the radius of the Earth and k is the distance of the planet from the geocenter.

SOURCE PAGE 446

∴ r = Earth's radius / sine of zenith distance

It is convenient to express the sine of the horizon inclination in terms of latitudes and declinations, so the radius of the Earth should also be written in terms of latitudes and declinations. It is stated that

radius X 2 X 3.14159 = circumference = 360 °

therefore radius = 360 2 × 3.14156 = 57 .2657765 = 3437' .74677 = 206264'' .8062 = 206265''

Sūrya Chapter on Mean Motions = 8 ".80

∴ Sūrya Siddhānta = 206265 / 8.8 = 23436 in Earth's radii

This distance is in Earth's radii whose latitude value is 3663.3 miles. Therefore the distance of the Sun = 22436 X 3663.3 miles = 62865786 miles.

Lunar mean horizon latitude = 57' 1.8 = 3422

∴ Lunar mean distance = 206265 / 3422 = 60.3 in Earth's radii = 60.3 X 3663.3 = 238687 miles

The extension of the Sun and Moon—if the angular semi-diameter of a celestial body is known by sighting and its latitude is also known, then its extension can also be known because

sine of angular semi-diameter = body's radius / body's distance sine of latitude = Earth's radius / body's distance * body's radius / Earth's radius = sine of angular semi-diameter of body / sine of latitude of body

SOURCE PAGE 447

Sūrya Siddhānta

Scientific Commentary

The declination of the moon is 12° 26' 16" and the relationship is 57' 1" .8, therefore the radius of the moon = radius of the earth × 15' 36".6 / 57' 1".8 = radius of the earth × 15' 36".6 / 57' 1".8 = 3663.3 × .275 = 1086.6 = 1060 miles figure 84 in angle p b c = angular declination of the moon angle b c d = elongation of the moon p c = radius of the moon b d = radius of the earth annual elongation

Annual parallax—it is stated that the stars are so far from us that their distance cannot be determined by looking at them from any two places on the Earth. But if a star is observed throughout the entire year, due to the annual motion of the Earth, a great difference occurs in the places from which the observer sees, which shows that there is some distance in the star as well. It has been proven that the distance from the Earth to the Sun is approximately 6,30,00,000 miles. It is known that the Earth revolves around the Sun in one year. For this, the Earth completes half a revolution in 6 months. Now if the observation of a star is made on one day and then the observation of the same star is made after 6 months, the difference in the places of the observer becomes 18,60,00,000 miles, due to which some change in the direction of the star is seen. This change is due to parallax. When the observer's

SOURCE PAGE 448

Even if the distance between two places is eighteen crore sixty lakh miles, the inclination of all stars is not visible because many stars are so far from us that the diameter of the Earth's orbit is equal to zero in front of them. Therefore, not even a few stars' inclinations could be measured by very subtle instruments.

Annual inclination—the annual inclination of a star is the angle that the Earth

SOURCE PAGE 449

It is formed on the star opposite to the semi-diameter of the orbit. In figure 85, if the Earth, Sun, and any star are in the positions of P, S, and T respectively, then the annual declination of T is the same as that of B or L.

Just as formulas have been established to know the declination of the Moon or a planet, a formula can also be established to know the declination of a star.

SOURCE PAGE 450

jyalu / jyasabhat = sabhat / sat

∴ jyalu = lu = sabsat × jyasabhat

That is, the annual declination of a star is in the ratio of the sine of the angle formed between the star and the Earth's center and the sine of the angle between that star and the Sun. It is clear that when the angle is equal to s heta 60°, that is, when the portion of the star is 60° ahead or behind the portion of the Sun, the magnitude of the declination is maximum. Therefore, the maximum declination of a star is seen twice a year. The formula for this is:-

tarekamhattamlamban = sabhat / sat

If the maximum declination is taken as lu then the declination of a star at any time

lu = lu X sin

Generally, the maximum length of a star is called the length of the star.

In the above formulas, lu and lu are in the decimal fraction. If these are written in degrees then

lu / 206265 = sab / sat

This proves that if lu is known, then sat, that is the distance of the star, can be known because sab is already known.

Example—if the annual declination of a star is 0.8, tell the distance of that star from the sun?

0.8 / 206265 = sabh / sat

∴ sat = 206265.8 × sabh = 2,57,831sabh

That is, the star is that far from the sun as the sun is from the earth multiplied by 2,57,831.

In miles this distance = 2,57,831 X 6,30,00,000 = 2,36,78,28,30,00,000

From this it is proven that if the distances of the stars are written in miles, then a behavior of a very large number which does not have subhita is necessary. Therefore

SOURCE PAGE 451

Astrologers have fixed another unit to reveal such a large distance, which is called light-year. The distance light travels in one year is called a light-year. This has been proven by many experiments that the sunlight reaches the Earth in 8 minutes 18 seconds, which means the speed of light is 1,86,000 miles per second. Therefore, in one year, light travels 365.2422 × 24 × 60 × 60 × 186000 miles or 58,66,58,82,50,880 miles. Therefore, this distance is called one light-year.

It should also be remembered that where calculations of such large distances are made, it is ordinary to have an error of millions of miles because if

Star names (English and Hindi), annual parallax, distance in Earth-Sun units, and distance in light-years. α Centauri... 0".750±0".01 2,80,000 4.4 61 Cygni... 0.37±0.02 5,60,000 8.8 Sirius Lubdhaka 0.37±0.01 5,60,000 8.8 Procyon Prashva 0.31 6,60,000 11 Altair Shravana 0.28±0.02 7,40,000 12 Aldebaran Rohini 0.17±0.02 14,00,000 22 Capella Brahma Hriday 0.12±0.02 17,00,000 27 Vega Abhijit 0.12±0.02 17,00,000 27 Polaris Dhruva Tara 0.07±0.02 30,00,000 47 Arcturus Swati 0.024 87,00,000 140 alpha Gruis... 0.015 1,40,00,000 220

SOURCE PAGE 452

In determining the declination of a star. If an error of .001 arc is allowed, which is not impossible, then there can be a great difference in its distance.

A table is given behind which one will know what are the declinations and distances of some stars. This table is taken from page 328 of R. S. Ball's Spherical Astronomy.

The distance of a light-year can be imagined in this way. When it is said that the brightest star in the celestial sphere, Lubdhaka, is 8.8 light-years away from us, it also means that the ray of light that is reaching our eyes at this time and introducing the introduction of Lubdhaka came from there 8.8 years earlier, that is, this ray tells the condition of Lubdhaka 8.8 years ago. Now, the knowledge of the condition of Lubdhaka can be obtained 8.8 years after today, not before it, just as the news of a distant friend that is received by a letter is the news of the time when the letter is written, not the time of its arrival.

Recently, such stars that are millions of light-years away from here have also been found by telescopes.

SOURCE PAGE 453

Due to the annual declination, the star appears to move on a small ellipse in a year.

In Figure 86, the Sun is S, P, Pi, Pu, Pe, and the Earth is shown moving in its annual orbit on four points. There is no place for the star. If two lines are drawn from P and S and further, reaching the stars located very far from T, then when viewed from S, the star will appear at T, and when viewed from the Earth, it will appear at P. Similarly, when the Earth is at the points Pi, Pu, and Pe, the star will appear at the points P, Pu, and Pa in sequence. The result will be that the star will appear to revolve on an orbit formed by P, Pu, and Pa. This small orbit will be in a plane parallel to the ecliptic and will appear to have the shape of an ellipse.

The orbit of a star that is at the polar circles, i.e., the parallels, is seen to be only circular because in such a condition, the plane of this small orbit will be at a right angle to our line of sight. However, the star that is on the ecliptic will appear to move east for six months and west for six months from the mean position, as if a body revolving on a circle only seems to move forward or backward when the plane of the circle is in the same direction as the observer's line of sight. The method for knowing the annual declination of the star is also generally the same as shown in Figure 81. But for this work, very fine observation is required, which does not seem necessary to discuss here.

Seismic Sign (Aberration)

It is stated above that light travels 1,86,000 miles in one second. The Earth also revolves around the Sun throughout the year, causing it to move 183 miles per second in its orbit because the magnitude of the Earth's orbit is 2 × 6,30,00,000 miles and there are 365.2422 × 24 × 60 × 60 seconds in a year, so

dividing the Earth's orbit by the seconds of a year results in approximately 183 miles. Due to these two motions, the positions of celestial bodies seen in a telescope are some ahead or behind the actual positions. The difference seen in the true and apparent positions of a body due to these two motions is called aberration. Before examining this, it is necessary to briefly state how the speed of light was measured and what difference is seen in the true and apparent positions of bodies due to the combination of these two motions.

SOURCE PAGE 454

The speed of light—there are many ways to measure the speed of light. The discussion here will be about the first method:

The speed of light was determined by a astrologer named Roman in the year 1732 Vikram Samvat. Before this, no one had imagined that light also requires some time to travel from one place to another. It is also astonishing how Roman found this out. You all must have seen lunar and solar eclipses many times. During a lunar eclipse, the Earth comes between the Sun and the Moon, so the Moon falls into the Earth's shadow. When the entire Moon falls into the shadow, a total eclipse occurs, and when only a part falls into the shadow, a partial eclipse occurs. Just as the Moon revolves around the Earth and sometimes appears half, sometimes quarter, sometimes three-fourths, similarly, four or five moons revolve around Jupiter. Jupiter's moons are so small that they cannot be seen without a telescope. These moons move very quickly into Jupiter's shadow as they revolve, so they are not visible for a while. Therefore, it can be said that when Jupiter's moons fall into its shadow, they are eclipsed. Even during these eclipses, they are known in the same way as during solar and lunar eclipses by calculating many years before. Sailors use these eclipses to tell time. Just by thinking about this, Roman found out the speed of light.

Imagine that in figure 87, there is the sun, and p, pa, pi are three places of the Earth on its orbit, and g, ga, gi are three places of Jupiter on Jupiter's orbit. The Earth and Jupiter revolve around the sun in succession in 1 and 12 years, respectively. At the time when the sun is in the positions s, p, and g of the Earth and Jupiter in succession, the Earth is very close to Jupiter, and at the time when the Earth is at pa, Jupiter is at ga, and the sun is in the middle, the Earth becomes very far from Jupiter. It takes the Earth 6॥ months to reach from p to pa. In 13 months, the Earth, while being from p to pa, reaches pi and comes between the sun and Jupiter. As the Earth moves from p and gets closer to pa, the apparent time of the eclipse of Jupiter's moon falls behind the time known by mathematics, and when the Earth reaches pa and Jupiter reaches ga, that is, the Earth becomes very far from Jupiter, then the apparent time is the most

SOURCE PAGE 455

75 hēn

It falls further behind. Romans had determined the method of knowing the time of eclipse by many observations, when the Earth was far and near the Sun in each situation. Therefore, the time of eclipse that came from this method was correct according to the mean value. From this time, the amount by which the actual eclipse fell began from the time when the Earth was extremely near the Sun, that is, when it was in the state of the point, and similarly when the Earth moved further ahead to the vicinity of Jupiter, the difference between the actual eclipse time and the mathematical eclipse time decreased. When the Earth was at the perihelion and Jupiter at the aphelion, the difference between the actual and mathematical times became zero, that is, actual

SOURCE PAGE 456

The time of eclipse was the same as that which came directly from mathematics. From Figure 87, it will be known that the greatest difference between the time of eclipse calculated from mathematics and the time of actual eclipse is equal to the difference in those times in which the light of the Moon of the Sun travels from g to p and from a to pa, that is, this difference will be the same as the time in which the light travels a distance equal to the diameter of the Earth's orbit.

From experience, it has been known that the difference in the time of eclipse when viewing from pa and p is the greatest and is equal to 16 minutes 36 seconds. The semi-diameter of the Earth's orbit is approximately 6,30,00,000 miles, so its diameter is 18,60,00,000 miles. Therefore, when the light takes 16 minutes 36 seconds to travel such a distance, its speed in one second is approximately 1,86,000 miles/996=1,86,000 miles.

Many other scientists conducted experiments to measure the speed of light. The results from all these experiments are mostly one. From these experiments, it has been proven that the speed of light is 1,86,350 miles.

When it is established that light is in motion, it is not difficult to understand that if moving light enters another moving object, its direction will change. For example, suppose a train is moving at a speed of 60 miles per hour. If a gun is aimed at the train in such a way that the bullet enters one side at a right angle to the direction of the train and passes through to the other side, will the bullet also form a right angle to the direction of the train within the train compartment? In the time it takes for the bullet to reach the rear wall of the train, the train will move some further, and the bullet will not hit the hole in the rear wall but will lag behind. Imagine that there is a compartment of the train moving towards R at a speed of 60 miles per hour or 88 feet per second, and B is such that the gun is aimed so that the bullet enters the compartment in the direction of B and enters from G. At the time the bullet reaches G, the compartment is in the R position. If the train were stationary, the bullet would pass through the hole at G and exit. But this is not the case because while the bullet enters from G and continues towards H, the train is also moving forward. Therefore, at the time

SOURCE PAGE 457

When the bullet reached the back wall, the box was in the position of Ra, and the point of Ga appeared in place of Ha. Therefore, seeing the bullet piercing Ga, one will see. The traveler sitting in the box will say that the bullet came from the direction of Ga, so the gunner must have been in the line of the place.

When it is raining and the drops are falling upright, even if a person holds an umbrella directly above, they are saved from getting wet, but if they move forward holding the umbrella in exactly the same way, they cannot be saved from getting wet because the upright falling drops also come to them obliquely due to their movement. The more the human's pace is, the more obliquely the drops will fall on them. This is also an example of this.

Similarly, when light enters the far-viewing instrument, its direction changes. Imagine that the actual position of a star is t and the eye of the observer is at d. If the observer is stationary and the motion (refraction) also

SOURCE PAGE 458

Then the star will always be seen in the direction of the star, whether the light from the star reaches the eye of the observer at the same moment the star moves or it takes some time to arrive.

But if it is assumed that the observer is moving in the direction of the observer, then the star will be seen in the direction of the star only when the light reaches the observer's eye at the same moment the star moves, but if it takes some time for the light to travel from the star to the observer, then the star will never be seen in the direction of the star.

Suppose that the axis of the tube is the line whose sides are parallel to m and d. When the light is descending in the tube from m to d, if at the same time the tube is moving in the same direction as d and the time it takes for the light to travel m to d is equal to the time the tube moves to d, then as in figure 8 it is apparent that the light does not reach d but d.

will reach. From this it will be known that the light is coming from the direction of d and the star is somewhere on the line of d m. Therefore, if the tube is movable and the star is at t, then it will not be seen in the direction of the axis of the tube but in the direction of d m. That is, the true position of the star will be at t which is in the same direction as the actual position.

SOURCE PAGE 459

It is extended in the condition in which the tube is going. In this way, due to these two arcs, there is a difference of the angle of t m ta and the true positions of the stars, which is called aberration.

It is easy to know what is the magnitude of t m ta or d m da angle because in the d m da triangle,

```
jādamadājyāmadadā
```

But the d da is the speed of the Earth in the same time as the time light travels in the same amount, so the ratio of the distances of d da and m da is the same as in the motions of the Earth and light. But the Earth moves 183 miles per second and light moves 5,86,000 miles, so

```
sine of Earth's motion / sine of light's motion = 18.5 / 186000 approximately = 1 / 10000 approximately
```

If the aberration is considered as the Earth, then sine da ma dā = sine bhū = bhū when the value of the Earth is in radians. In such a condition

```
Bhū = 1 / 10000 x sine_madada
```

If Bhū is written in aberrations, then

```
Bhū " = 1 / 10000 sine_madada = 1 / 10000 jyatadta206265
```

```
or Bhū " = 20 " .63 jyatadsa
```

20".63 is called the constant of the coefficient of aberration of the motion of the Earth. Its more accurate value is 20".47. If the angle of the aberration is the same as 60°, then it is clear that the maximum value of the coefficient of aberration of the Earth's motion will be 20".47.

It is clear that the motion of the Earth is present in the coefficient of aberration as a multiple. However, the motion of the Earth is not always the same; at the time the Earth is at its lowest, its motion is very fast, and at the time it is at its highest, its motion is very slow. Therefore, in the first case, the constant of the coefficient of aberration is 20".80 and in the second case is 20".13.

SOURCE PAGE 460

The effects of the portion of the Sun, stars, and distant planets due to the correction of terrestrial motion, the shadow, the solstice, and the declination are left unexplained due to fear of expansion. Here, it is discussed in a simple manner:-

Just as a star appears to revolve in a small orbit around its true position due to the annual declination, similarly, due to the correction of terrestrial motion, it also appears to revolve in a small orbit around its true position. This orbit is also parallel to the plane of the ecliptic. The size of its orbit also changes in the same way that the orbit of the star changes due to declination. At the time when its size is elliptical, its major axis is equal to 20”.47 and its minor axis is equal to 20”.47 X ja when ja is the shadow or deflection of the star.

It is clear that the correction of terrestrial motion of the star is in the same direction as the motion of the Earth, but the Sun is 60° ahead in the direction of the Earth's motion because the Earth's motion is in the direction of the tangent of the geodetic, which makes an angle of 60° with the semi-diameter of the geodetic, and the Sun is at the center of the geodetic. Therefore, it is proven that the correction of terrestrial motion of the star is towards the point of the ecliptic which is 60° behind the Sun, i.e., whose portion is 60° less than the portion of the Sun.

The star that is on the ecliptic pole, i.e., Kadamba, revolves around its true position in a circle throughout the year, the angular value of whose semi-diameter is 20‘46.

The star that is on the ecliptic oscillates forward and backward by 20”.46 from its true position on the ecliptic like a pendulum. Therefore, the total difference throughout the year is equal to 40”.6.

If a star remains in another place from which its shadow is like that of the sun, it appears to revolve on an ellipse throughout the year whose center is the true position of the star, whose semi-major axis is 20046/2 and whose semi-minor axis is 20046 sine and whose plane is parallel to the plane of the ecliptic.

Many readers will ask what is the difference between the annual declination and the annual motion. The answer is that the star, due to the annual latitude, which

SOURCE PAGE 461

Seeing it revolving in the class, its extension depends on the distance of the star, that is, the farther the star is, the smaller its colatitude will be, due to which the size of the orbit will also be smaller. The long axis of the orbit of the nearest star, which is seen due to the colatitude, is not more than 0^(°)76. However, due to the correction of terrestrial motion, the long axis of the orbit of the star seen is always 20^(°)40, and this is the same for all stars. The second point is that if the star is in the same direction as the sun or at exactly 180^(°) from the sun, the magnitude of the colatitude is zero, but the magnitude of the correction of terrestrial motion is maximum, i.e., 20^(°)47. The third point is that due to the colatitude, the star appears to be slightly shifted towards the sun, but due to the correction of terrestrial motion, the star appears to be shifted towards the point that is 600 behind the sun.

The effect of the motion of the planets on the geodetic signs happens in two ways, one due to the motion of the Earth; the other due to the motion of the planet. If the motion of the planet were the same as the motion of the Earth and in the same direction, then there would be no geodetic signs. In other conditions, the calculation of what the geodetic signs will be can be done easily separately.

The motion of the Moon is much less than the speed of light, therefore its cause can be considered as equal to zero in terms of terrestrial motion. Due to the motion of the Earth also, the motion of the Moon is not like terrestrial motion because the Moon also revolves around the Sun throughout the year along with the Earth. Therefore, the effect of terrestrial motion in the Moon is equal to zero.

Daily geodetic measurement—because of the daily motion of the Earth, the equator moves approximately 25,000 miles in a day because the circumference of the Earth at the equator is approximately 25,000 miles. The circumference of other places is shorter than this. Therefore, when there is a motion of 25,000 miles in a day, a motion of 3/10 miles per second is proven. But the annual motion of the Earth is 183/2 miles per second. Therefore, the one-twelfth part of the geodetic measurement due to the annual motion will be on the equator due to the daily motion. It will be even less at other places, so it is not unreasonable to understand it as such.

The story of the invention of this is very interesting. But there is no need to write in detail. Bradley named

SOURCE PAGE 462

The astrologer was checking the annual declination of the third star (gamma Draconis) of the constellation named ajagara at the time when he realized that this star does not remain fixed but changes in sequence over the year, a fact that was not understood from the known facts up to that time. Finally, he reached the principle that this annual change happens in all stars due to the annual motion of the Earth and the speed of light—Brandelay experienced this phenomenon in 1786 A.D.

The science commentary of the Chapter on the Three Problems, including the analysis of time-equations, motion, declination, and terrestrial motion-signs, has ended.

SOURCE PAGE 463

Sūrya Siddhānta

of

Scientific Commentary

Second Section

[Lunar Eclipse Chapter, Solar Eclipse Chapter, Position Chapter, Planet Chapter, Constellation-Planet Chapter, Rising Chapter, Shadow Chapter, Node Chapter, Earth Chapter, Astrology Upanishad Chapter, Mean Chapter]

Commentator: The late Mahāvīra Prasāda Śrīvāstava

Dr. Ratnakumari Swadhyaya Institute, Allahabad

SOURCE PAGE 464

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Introduction

Many texts of ancient literature have been published on behalf of Dr. Ratnakumari Swadhyay Sansthan —the texts of Swami Satyaprakash and Dr. Usha Jyotishmati, such as Bakhashali-Manuscript, Shulba-Sutras (in Sanskrit and English). In this tradition, we are publishing the Scientific Commentary of Sūrya Siddhānta by Swargiya Shri Mahabir Prasad Srivastava. This honor has been received from the generosity of the Vigyan Parishad, Prayag, for which we are indebted to the officers of the Parishad.

It would have been difficult for us to complete the arduous task of editing this text if we had not received financial assistance from the respected Raisahab Chaudhary Pratap Singh Ji and the trust established by him from Karnal. We are extremely grateful to Chaudhary Sahab. We have already published the first volume in March 1982, which includes Chapter on Mean Motions, Chapter on True Positions, and Chapter on the Three Problems. This second volume has 11 chapters. Thus, the entire Sūrya Siddhānta is completed in a total of 14 chapters.

May 10, 1983. S. Rangnaki, M. S-C., D. Phil, D. S-C. Director

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Chapter on Lunar Eclipse (Brief Description)

[1 verse—Value of the mean distance between the Sun and the Moon. 2-3 verses—Method of knowing the true diameters of each and the true diameter of the Sun in the Moon's orbit (in conjunctions and phases). 4-5 verses—Method of knowing the value of the diameter of the Earth's shadow in the Moon's orbit. 6 verses—Where the node of the Moon can be so that an eclipse may occur. 7 verses—In which date an eclipse may occur. 8 verses—Method of clarifying the Sun and Moon at the end of Amavasya and Purnima. 9 verses—Why an eclipse occurs. 10 verses—Method of knowing the magnitude of the occulted part. 11 verses—Method of determining whether a total eclipse or partial eclipse or no eclipse will occur. 12-15 verses—Method of knowing how long the eclipse and total eclipse will last. 16 verses—Method of knowing the beginning and end time of the eclipse. 17 verses—Method of knowing the beginning and end time of the total eclipse. 18-21 verses—Method of knowing how much part will be occulted at what time. 22-23 verses—Method of knowing the auspicious time knowing the magnitude of

the eclipse. 24-25 verses—Need for knowing the declination to draw the picture of the eclipse. 26 verses —Method of knowing the angular value of the image at the auspicious time.

To know how an eclipse occurs between the sun and the moon, some properties of light are first necessary. Therefore, these will be considered briefly first. It is everyone's experience that in the light of a lamp at night, the shadow cast on the wall of an object is somewhere light and somewhere deep. The deep shadow is in the middle and the light shadow surrounds the deep shadow. If the object is near the wall, the deep shadow is large and the light shadow is small. As the object moves further away from the wall, the extension of the shadow increases, but the deep shadow decreases and the light shadow increases. If the object is smaller than the lamp, there will also be a situation where the deep shadow does not fall at all, only the light shadow will be seen on the wall.

SOURCE PAGE 467

Yes, if the object is larger than the lamp, a deep shadow will always fall on the wall.

The part of the wall where the deep shadow falls, no part of the lamp's light reaches that part, but in the light shadow, some part of the lamp's light must reach. If an insect is in the deep shadow on the wall, it will not see the lamp at all, but in the light shadow, it will surely see some part of the lamp. This can be tested as follows:-

Keep a lamp or candle lit. Place a pencil, bullet, or any object smaller than the lamp a short distance away, standing or hanging. Hold a thin paper in your hand in such a way that both deep and light shadows of the pencil fall on it. Make a hole with a needle in the deep shadow and see if the lamp can see or not. The lamp will not see. Make a hole by piercing in the light shadow and see. Some part of the lamp will be seen.

From geometry, it can be known where the deep shadow will fall and where the light shadow will fall. It is also possible to determine the extent of these. To know how eclipses happen in the Sun and Moon, the mathematics of the deep and light shadows has to be done, so it is necessary to think about this well. Further, the deep shadow will be called shadow and the light shadow will be called umbra.

Suppose R is a luminous body and C is an opaque body. Both bodies are spherical. The rays of light from R spread in all directions, but the rays that fall on body C cannot proceed further than it. If lines are drawn that touch both these bodies, they will move forward, intersecting each other at point T. It will be of the shape of a cone. This will be the boundary of the shadow formed by body C. Shadows will not fall above, below, to the sides. (See Figure 60).

The lines that touch both these bodies and meet at point T form the boundary of the umbra.

If a sheet (screen) is kept in shadow in such a way that it is at a right angle to the line joining the centers of the r and c bodies, then the diameter of the circle of shadow that will be formed on this sheet will be ch cha, and the diameter of the circle of the umbra will be u u, which includes the diameter of the shadow (see figure 61). If a hole is made at a point p in the segment u ch and the light-emitting body is seen through this hole, then the upper part of the body which is above the point p will be seen. This point p a

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SOURCE PAGE 468

0 3 4 Figure 62

SOURCE PAGE 469

By extending the line, the luminous body is determined. If a hole is made somewhere in the shadow section, the part of the luminous body below will be seen. But if a hole is made in the shadow section, no part of the luminous body will be seen. The summary is that if the observer remains somewhere above the line of sight but below the line of the shadow, he will definitely see the upper part of the body but not the lower part. Similarly, if the observer remains below the line of sight and above the line of the shadow, he will definitely see the lower part of the luminous body but not the upper part.

If brought to the point, there will be no shadow here. The luminous body will not be seen, but its brightness will be seen all around. If it is moved further away from the point, another view will be seen. In the sections of the shadow of K, Kh, and Gg, the same thing will be seen as before, but in the space between the lines that form the boundary of the shadow in Kh and Gg, the entire edge of the luminous body will be seen, but there will be darkness in between (see figure 62). The figure shows that the lines drawn from any point between K and Gg that touch the body will reach both above and below the body. Since the body is round, it will be seen like a bead due to the darkness in between.

All these views can be seen by experiment. A round lamp, a ball, and wooden pegs are sufficient for these three things. The round lamp should be understood in place of the body and the ball in place of the peg. This experiment can be easily done in any place during a dark night.

All the things about a solar eclipse can be understood from this experiment. R should be considered the Sun and C the Moon. The place of Pat should be considered the Earth. Just as this body remains in both shadow and penumbra when it is near the body but in the lines forming the boundary of only penumbra or shadow when it is far, similarly the Earth remains in both the shadow and penumbra when it is near the Moon and only in the shadow when it is far. Since the Earth is much larger than the Moon, the entire Earth cannot fall in shadow or penumbra. The inhabitants of the part of the Earth that falls in shadow do not see the Sun at all. Therefore, a total eclipse of the sun or total eclipse occurs. The inhabitants of the part of the Earth that falls in penumbra

SOURCE PAGE 470

The inhabitants see a partial eclipse of the sun. If the shadow does not reach the Earth, it will remain in the same position as seen by k kh g gh pat. In such a situation, the part of the Earth that will be between the line forming the boundary of the shadow will see an annular eclipse.

Just as the Earth comes into the shadow or umbra of the moon, a total eclipse or annular eclipse of the sun is seen, similarly when the moon comes into the shadow of the Earth, the light becomes without light. This is called lunar eclipse. If the entire moon comes into the shadow, a total eclipse of the moon occurs, and if an incomplete part comes into the shadow, a partial eclipse of the moon occurs. In this situation, the moon will appear to be in eclipse of the sun, but it may not have the good fortune to see an annular eclipse because the size of the Earth is larger than the moon, so the moon can never come out of the shadow.

From the picture it is also clear that it is necessary to enter the umbra before reaching the umbra. It should be remembered that a lunar eclipse is seen only when the moon enters the Earth's shadow. If the moon only goes into the umbra, no eclipse will be seen, yes some impurity will also occur.

What is stated above is clear that according to the mutual distances of the Sun, Moon, and Earth, the magnitude of the shadow and umbra can also be less or more. It has already been stated that the

distance between the Sun and Earth and the distance between the Moon and Earth constantly increase and decrease. Due to the decrease and increase in distance, the angular sizes of these bodies appear to decrease and increase (see pages 84-85). Therefore, to know the magnitude of the angular sizes, it is necessary to know their true distances. However, in the Chapter on the Three Problems, it is stated what the relationship is between the size, length, and true distance of a body. Therefore, by knowing either the latitude or the distance, it can be known at what time the magnitude of the shadow is how much. From Figure 93, it is known how large the diameter of the Earth's shadow in the Moon's orbit is at the time of lunar eclipse.

Suppose that in figure 63 there is the moon entering into the shadow of the earth, therefore it is touching the line of sight because the sun and

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SOURCE PAGE 471

SOURCE PAGE 472

The shadows of the Earth are formed by the parallels of the Earth's general tangents pa and ri p only, which have no tip. The radii of the Sun and Earth are at a right angle to the tangent pa of the tangent pa. The circumference is parallel to para.

It is necessary to know first that the angle s is equal to ch because this angle is formed by the radius of the shadow at the center of the Earth which is in the Moon's orbit, therefore it will show the shape of the shadow in the lunar orbit.

```
∠ riper = rir - rariper = rar - papper = rar - papper = surya trijya - surya kalamban = t - l
```

The radius and colatitude of the Sun are indicated by tr and l.

```
∠ sapch = ∠ paspa - ∠ panpa = ∠ paspa - ∠ riper
```

Because P and N are parallel and neither intersects them.

```
Here angle Pspa = Ppa / Ps = Lunar declination = La
```

Taking La as the true declination or horizon declination will not make much difference. Therefore

```
angle Sapcha = La - (Tr - L) = L + La - Tr
```

This proved that if the angular value of the radius of the Earth's shadow in the lunar circle is subtracted from the sum of the horizon declinations of the Sun and the Moon, the angular value of the radius of the Earth's shadow in the lunar circle is the same. This is also called Bhūbhārddha.

It is known from experience that due to the Earth's atmosphere, its shadow is 10 times larger than the mathematically derived shadow because the above mathematics only consider the solid body of the Earth, not its atmosphere.

Example—If the latitude of the sun is 6", the latitude of the moon is 58'1", and the radius of the sun is 16'13", then tell the radius of the shadow of the Earth in the lunar orbit?

Bhūbhādha = l + la − t = 6 " + 58 ′ 1 ″ − 16 ′ 13 ″

SOURCE PAGE 473

Sūrya Siddhānta - = 58' 10" - 16' 13" = 41' 57"

This is the radius of the shadow proved by mathematics. Due to the atmosphere, the shadow increases by one and a half times. Therefore the total shadow

= 41' 57" + 41' 57" / 50 = 41' 57" + 50" = 42' 46"

It is revealed that the magnitude of the geodiameter (radius of the Earth's shadow) in the lunar orbit is not always the same because it depends on the declination and the angular radius of the Sun and Moon and these three things depend on the distances of the Earth from the Sun and Moon which always increase and decrease.

Now it will be explained what the opinion of Sūrya Siddhānta is on this subject.

The mean diameter of the images of the Sun and Moon and the true diameter of the Sun in the lunar orbit—

> सार्धानि षट् सहस्राणि योजनानि विवस्वतः ।
> विष्कम्भो मण्डलस्येन्दोः साशीतिस्तु चतुश्शती ॥१॥
> स्वध्यासौ त्रिज्ययाऽभ्यस्तौ स्वमन्दश्रवणोद्धृतौ ।
> स्पष्टौ स्वकौ स्वकौ भूमेस्तथा सूची शशाङ्कवत् ॥२॥
> स्फुटस्वभुक्तिगुणितो मध्यभुक्त्या हृतो स्वकौ ।
> रवेस्स्वभगणाभ्यस्तः शशाङ्कभगणोद्धृतः ॥३॥

(1) The mean diameter of the Sun's sphere is 6500 yojanas and the mean diameter of the Moon's sphere is 480 yojanas. (2) When the true diameter of someone is to be known, multiply his mean diameter by his true motion and divide the product by his mean motion. By multiplying the true diameter of the Sun by the great revolution of the Sun and dividing the product by the great revolution of the Moon, (3) or by multiplying the true diameter of the Sun by the lunar orbit and dividing the product by the Sun's orbit, what is obtained is the magnitude of the true diameter of the Sun in the lunar orbit. Dividing the diameters of the Sun and the Moon in the lunar orbit by 15, the phases of the Sun and the Moon are known.

Sūrya Siddhānta—Scientific Commentary—Chapter on Mean Motions, Chapter on True Positions, Chapter on the Three Problems.

The mean diameter of the Sun's image = 6500 yojanas

The mean diameter of the lunar image = 480 yojanas

SOURCE PAGE 474

Sputvyas = Meanvyas X Sputgati / MadhyamgatiChandrakakshamSineofSunSineofSun X MeanhaayugyBhagan / MoonhaayugyBhagan

Or = SineofSunSineofMoon / SineofSun

Here may arise the doubt whether the planimetric shape of the sun also increases and decreases because it is stated above that the apparent (true) diameter of the sun is dependent on its apparent motion, which constantly increases and decreases. But this is not the case. The planimetric shape of the sun does not increase or decrease according to the apparent motion; yes, the artistic or angular shape certainly changes, which has been well done in the Chapter on True Positions on pages 84-86. Here, although the magnitude of the mean diameter and the apparent diameter is stated in the calculations, it should be understood as angular because this is the meaning derived from what Bhaskaraacharya ji1 has written about the method of knowing this. Bhaskaraacharya ji says that on the day when the apparent or true motion of the sun is similar to the mean motion, on that day, in the rising time, take two sticks of the same length and join their two ends, placing eyes at two original places, and extend the image of the sun in such a way that the forward ends of these sticks touch the north and south edges of the image. In this situation, fasten the ends of the sticks in such a way that there is no difference in the distance of the forward ends. Now measure the distance of these ends with the same unit with which the length of the sticks was measured. This difference, if it is equal to some number of units, will be the diameter of the sun's image in the artistic sense. According to Bhaskaraacharya ji, this diameter is 32'31"33". If this is taken as 32'30" or 32'5" and the value of the sun's orbit is taken as 3, and the value of the sun is taken as 43,31,500 yojanas, then the planimetric value of the sun's image will be obtained as follows:-

1. Mathematics chapter page 171-172

2. Nowadays this work can be done with the tips of the dividers. The eye should be at the point where both arms of the compass meet.

3. Geography Chapter verse 86.

SOURCE PAGE 475

Figure 64

some angle = 3600 = 21,600 ′

∴ The Sun's orbit is also like 21,600 degrees, but in terms of degrees it is like 43,31,500, therefore

21600 ' = 43, 31, 500 yojana

∴ 3205' = 32.5 × 433150021600 yojan = 6517 yojan

Sūrya Siddhānta has considered the mean diameter of the sun to be 6500 yojanas. It appears from this that the rising image of the sun in this treatise has been taken less than 32°30', which is correct because due to motion, the rising image appears somewhat larger than reality.

Thus it is proven that the planimetric value of the images of the Sun or Moon was known from the artistic values.

The rule given for knowing the true diameter from the mean diameter is somewhat coarse because the change in the true motion of the Sun or Moon does not occur in the same proportion.

SOURCE PAGE 476

The change in their angular images from the ratio.

Two rules have been stated to know the true diameter of the Sun in the lunar orbit. (See Chapter on True Positions page 88).

```
Chacha = Bhacha = Lunar orbit radius half Surya orbit radius = Lunar orbit Surya orbit
```

Because the same ratio exists in the perimeters of two circles as in their diameters.

Therefore

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Sūrya Siddhānta
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Scientific Commentary

It has already been stated that the opinion of our teachers is that the daily planning motion of every planet is the same. Therefore, it is proven that the distance traveled by each planet in a great yuga or kalpa is the same for all planets. If the distance traveled by a planet in a great yuga is divided by the great yuga of that planet, then the value of the planet's orbit will come out in yojanas, this can also be written as:-

```
Chapter on True Positions
```

Chapter on the Three Problems

SOURCE PAGE 477

```
M / R = Sun's orbit
```

or m/c = chandrakaksha

If each side of the second equation is divided by the corresponding sides of the first equation

```
Mach ÷ Mar = Lunar orbit Solar orbit
```

```
Tuesday / c = lunar orbit / solar orbit
```

This way, the first rule of the apparent diameter of the sun is also proven.

It is necessary to state here that the apparent diameter of the sun and the extent of the sun's orbit are not as real as stated in our theoretical texts. Many doubts have proven that the latitude of the sun is not

more than 6 degrees, so its distance from the Earth is 9 crore 26 lakh miles according to the formula of latitude (see Chapter on the Three Problems page 384) and the semi-diameter of its body is 4,32,860 miles (see page 66). If the magnitude of a yojan is considered as equal to 5 miles (see page 54), then

Solar circumference = 432860 / 5 = 86578 yojanas, and

The medium distance of the sun = 6,26,00,000/5 = 1,85,80,000 yojan

This magnitude will be clarified by the table below, where all magnitudes are given in yojanas:-

Sūrya Siddhānta R.S.Ball's Spherical Astronomy Sun diameter 6500 6522 173156 Sun mean distance 686378 686377 1858000 Moon diameter 480 480 430 Moon mean distance 51566 51566 4750

SOURCE PAGE 478

In the latter part of the third verse, it is also stated that the magnitude of the diameter of the sun's image in the moon's orbit, when divided by 15, will come in the form of degrees. The reason for this is that the extension of the lunar orbit is considered to be 324000 yojanas, which is also equal to 360 degrees or 21600 arts. Therefore, when 21600 arts = 324000 yojanas, then 1 art = 324000 / 21600 = 15 yojanas, which means that 15 yojanas of the lunar orbit is equal to one art.

Magnitude of the diameter of the umbra in the lunar orbit —

> शशाङ्ककक्ष्यागुणितो भाजितो वाऽर्ककक्षया ।
>
> विष्कम्भश्चन्द्रकक्ष्यायां तिथ्याप्तो मानलिप्तिकाः ॥४॥
>
> स्फुटेन्दुभुक्तिर्भूव्यासगुणिता मध्ययोद्धृता ।
>
> लब्धं सूची महीध्यासस्फुटार्कश्रवणान्तरम् ॥५॥
>
> मध्येन्दुव्यासगुणितं मध्याकंव्यासमाजितम् ।
>
> विशोध्य लब्धं सूच्यास्तु तमो लिप्ताश्च पूर्ववत् ॥६॥

Translation—The quantity obtained by multiplying the true motion of the moon by the diameter of the Earth and dividing the product by the mean motion of the moon is called the list. Subtracting the diameter of the Earth from the diameter of the sun (5), multiplying the remainder by the mean diameter of the moon, and then dividing the product by the mean diameter of the sun. The remainder obtained by subtracting the list from it will be the diameter of the Earth's shadow in yojanas in the lunar orbit. When this is divided by 15, the diameter of the umbra will be known in arts.

Scientific Commentary—Here, the true motions of the moon and the sun will be denoted by the letters ka and ra. If the great cycles of revolution of the moon and the sun are divided by the great cycles of Sawan days and the arts of the list are made, then the mean daily motions of the moon and the sun are respectively 760' 56 and 56' 1362. The diameter of the Earth is considered to be 1600 yojanas (see Chapter on Mean Motions verse 56). Based on these values, the above two verses can be summarized as —

List = 1600 X cha / 760.56

Sūrya-sphutvyāsa = 6500 X rā / 56.1362 (dekhośloka2)

SOURCE PAGE 479

Sine of the umbra in the lunar orbit: =1600 imes 760.56 - (6500 imes 56.1362 - 1600) imes 480650

If this is divided by 15, then the lunar orbit umbra has a lunar shadow

The university = 10623 imes 960.56 - 32 imes 56.1362 + 7.88 At the time when the true motions of the moon and the sun are equal to their mean motions, the values 760.56 and 56.1362 are equal to each other. In such a situation, the lunar umbra's artful diameter is 106.67 - 32 + 7.88 = 82.55. As stated with the help of Figure 63 at the beginning, the arc of the Earth's shadow in the lunar orbit is 41'57", from which the diameter of the Earth's shadow is approximately 84'. Therefore, it is clear that the diameter of the Earth's shadow as obtained from the rules of Sūrya Siddhānta is approximately the same as the diameter obtained by a new method, although its instruments are crude and impure. The magnitude of the umbra's diameter in the Indian way is proven by the sum and difference of the three terms 106.67, 32, and 7.88.

SOURCE PAGE 480

Similarly, the magnitude of the earth's orbit can also be expressed by the difference of three terms 58^(°)1", 16'13" and 6" (see example on page 438). The double order of these three terms is 116", 32'26" and 18". This is very close to the second term of the Indian rule, 32'26". The first term here is 116' and there is 107 degrees, and the third term here is 18" and there is approximately 8', so the sum of the first and third terms becomes approximately 116'. This shows that the form of the earth's orbit that is proven by our theory is newly different only in the fact that the size and distance of the sun are considered very small here.

Proof: Imagine that the center of the Earth is P, the diameter of the Earth is Ph, the center of the Sun is R, the diameter of the Sun is Dh, the center of the mean Moon is Ch, the center of the true Moon is Ca, and the common tangents of P and H, and the lines of P and D, are parallel to the lines of B. It is clear that H is the diameter of the Earth's shadow in the plane of the true Moon, which, when viewed from Bhve, will be the same as the orbit of the mean Moon in the direction of Kg. If H and Hg are increased, then they will meet at points K and H in the orbit of the mean Moon.

From Figure 66 it is clear that

Bhuchayakavya = qagha - (kagha + gagha) = pf - (kagha + gagha)

In the similar triangle of P, in the x and y,

pka / path = kakha / tath

in the same way fa/fd = ga gha/d dh

but pka = fa ga and p th = fa dh

therefore ka kha th = ga gha dh = ka kha + ga gha th + dh = ka kha + ga gha th - th dh therefore pka p th = ka kha + ga gha th - pf ..... (7)

in the similar triangle of bhgha and k bh gh

bhchakagha - - - = - - - bhchakgh

SOURCE PAGE 481

But there are clear and mean distances of the Moon from the Earth and it is stated that the angular velocity changes according to the reciprocal of the hypotenuse (see Chapter on True Positions page 88) Therefore it can be assumed in a gross sense that the angular velocity also changes according to the reciprocal of the hypotenuse as revealed by the rules of Sūrya Siddhānta.

Therefore

Bhacha / Bhach = Moon's mean motion / Moon's true motion = m / s

∴ msa = pagh = pphakgh

∴ q = p X m = e X true motion mean motion

The names of these kgh are listed in verse 4.

It can be proven that the similar triangle of B can be established from B A and B C etc.

Bh cha = ka kha = ga gha = ka kha + ga gha bha cha kha ga gha + gha... (2)

Since p and m are equal in equations (1) and (2) therefore

. pdh X kakha + gagha / tdh - pf = bch X kakha + gagh / kakh + gagh

ya pathbhach = tgh - pfakkh + ggh

But the mean distance of the Earth from the Sun and the mean distance of the Earth from the Moon is such that their ratio = 4331500/324000 = 13.37 because according to Sūrya Siddhānta, 4331500 yojanas is the extent of the Sun's orbit and 324000 yojanas is the extent of the Moon's orbit, and 6500/480 = 13.54, which is the ratio of the mean diameters of the Sun and the Moon, which is approximately equal to 13.37. Therefore, there is no objection to take this value

path6500suryakamdhyavyas - - - = - - - = chandramakamdhyavyasbhach4806500tadh - pf - - - = - - - 480kkh + gh

SOURCE PAGE 482

K X + G H = (T D - P F) × 480/6500 But K X + G H = K H - X G = Listi - Lunar orbit in umbra and T H - P F = apparent diameter of Sun - diameter of Earth Listi - Lunar orbit in umbra = (apparent diameter of Sun - diameter of Earth) × 480/6500 ∴ Lunar eclipse in umbra = Listi -(apparent diameter of Sun - diameter of Earth) × 480/6500 This implies 4, 5 shlokas. When is eclipse possible - Bhanobhardhe mahochchaya tattulye'arkasme'thva. Shashankapate grahana kiyadbhagahikonke||6|| Tulyau rashyadihbhyaata mamavasyantkalikau. Suryendu purnamasyante bhardhe bhagadihbhissamo||7||

Translation - (6) There is a shadow of the Earth on the difference of 6 zodiac signs from the Sun. If the Moon's node is at the same distance from the Sun or the same angular portion of the Sun or some less or more distance from them, then an eclipse occurs between the Sun and the Moon. (7) The angular portions of the Sun and Moon, etc., are the same at the end of the new moon and at the end of the full moon at a difference of exactly 6 zodiac signs.

Scientific Commentary - Figure 63 reveals that the center of the Earth's shadow always remains on the line joining the centers of the Sun and Earth, so the Earth's shadow always remains 180 or 6 zodiac signs ahead of the Sun. Therefore, when the Moon is about 180 ahead of the Sun, it can enter the Earth's shadow, otherwise not. But when the Moon is 180 ahead of the Sun, then the end of the full moon occurs, so a lunar eclipse may occur around the end of the full moon. Similarly, when the Moon comes in front of the Sun and covers it, a solar eclipse occurs, but this is only possible when the portions of the Sun and the Moon are approximately equal, that is, when it is the new moon. Therefore, it is apparent that the Moon-

SOURCE PAGE 483

Eclipse occurs at the end of the full moon and at the end of the new moon. (See Figure 67)

Now this question may be that why lunar or solar eclipse does not occur at the end of every full moon and new moon? The reason is that the orbit of the moon

This is because the orbit of the moon is different from the orbit of the ecliptic. The absolute difference between these two is approximately 50, which is called the absolute deviation or absolute inclination of the moon (see Chapter on Mean Motions pg. 74-75). However, the apparent diameter of the sun and moon is approximately 16' and the diameter or arc of the Earth's shadow is approximately 42' (see the previous example). Therefore, an eclipse can only happen when the moon is in a position in its orbit that is near the ecliptic and the difference between the ecliptic and it is approximately 16' + 42' = 58' or less. This situation is possible at the same time when the moon is near the nodes, i.e., the points of intersection of the orbit and the ecliptic, of the new moon or full moon. But the nodes of the moon are always at an angle of 180° from each other, so this

SOURCE PAGE 484

It is clear that an eclipse can only occur if the portion of the sun is the same as or approximately the portion of Rahu or Ketu of someone at the time of amāvasyā or pūrṇamāsī, otherwise not. This will be well explained by Figure 68:-

Figure 68 Ch ch or sh = Moon's deviation at the time of pūrṇimānta time:- ch = Center of Earth's shadow ch = Center of the Moon ch or sha = Earth's shadow or Sun's shadow portion motion m = Earth's shadow and the sum of the diameters of the Moon p = Moon's node ch k or cha = Moon's shadow portion motion k ch or sha = "Motion of the shadow"

SOURCE PAGE 485

On the line of ka, cut it as if it were cha, and draw it parallel to of as if it were ka kha, and draw it parallel to khā. By combining khā, draw it to the point of the revolution of ca khā. This will be the relative path of ca khā and the moon, if it is assumed that the terrestrial shadow is fixed at two points.

If a perpendicular is drawn from cha to ca ba, then this will be the shortest distance between the centers of the moon and the terrestrial shadow. If a circle with a radius equal to ma, passing through cha as the center, is drawn which intersects ca ba at two points, sā, then these two points will be the positions of the touch and the time of salvation of the moon on the relative path. If lines parallel to ca ka are drawn from these points, the points where they will reach will be the actual positions of the touch and the time of salvation of the moon.

If the angle ka ca kha is taken as i and the angle ca khā is taken as ī, then

Spray = kakh / kach = sha / cha

sprei = kakha / kach = kakh / kach = sha / kach − kaka = sha / cha − ra

The sum of angles A B C and B C D is equal to a right angle because a perpendicular B Q is drawn on B C, but the sum of angles C D and D C is also equal to a right angle because C D is perpendicular to D C and C D is parallel to D C. Therefore, angle A B C = angle B C D = E.

∴ chha = chachkojyai = shkojyai

If the value of Sh cos i is greater than the sum of the diameters of the Earth's shadow and the Moon, then an eclipse will not occur.

But if Sh cos i is smaller than m, an eclipse will definitely occur. If it is to know whether a partial eclipse or a total eclipse will occur, the difference between both should be calculated. If the value of m - Sh cos i is smaller than the diameter of the Moon, it should be understood that a partial eclipse will occur, and if its value is greater than the diameter of the Moon, a total eclipse will occur because picture 66 shows that m - Sh cos i is the part of the Moon affected. Therefore, if the affected part is less than the diameter of the Moon, it is clear that a total eclipse cannot happen. But if the affected part is more than the diameter of the Moon, a total eclipse will definitely occur.

SOURCE PAGE 486

Figure 66 is a picture of the moon and the umbra at the time when the moon is at the nearest distance from the umbra, that is when the moon is at point b of Figure 68 and the umbra at c. It is clear that c is half the diameter of the umbra and c is half the diameter of the moon. The part of the moon affected is the same as j. Now we have to see what is the magnitude of j.

j = c - c = c - (c - j) = c + j - c = m - shkojyai

If m - shkojyai is equal to zero, that is, if m = shkojyai, then the eclipse will not occur because the moon will pass by touching the umbra. In such a case, the difference between the node and the center of the umbra will be found to be the magnitude of c-p—

It is apparent that angle c p c = i,

, sprei = cch / cp

∴ chap = chachasparei = shaparei

but it is assumed above that

m = shokayai

SOURCE PAGE 487

Sūrya Siddhānta

Scientific Commentary

Chapter on Mean Motions

Chapter on True Positions

Chapter on the Three Problems

The result is that the value of chap is also unstable. Here i is the angle between the ecliptic and the lunar circle, so it is known but i is unknown, so i should be known first. From the above relation it is clear that

Sūrya Siddhānta ÷ Sūrya = Chanda - Ra

therefore sphere = ca / ca - ra × sphere

∴ i, cha and ra of the mean values respectively 506', 760'35" and 59'8"

Thus

760' 35" / 760' 25' - 56' 8" = 760' 35" / 731' 27" = 1.08084
sprei = 1.08084 X
sprei5° 6' = 1.08084 X.0601 = .0674 ∴ e = 5° 34' ∴ chap = michereiiko
spreiim = cojyaisprei

SOURCE PAGE 488

= m / cos534' sin506'

But m==shadow and the sum of the diameters of the Earth and the Moon ∴ mean value of m = mean diameter of shadow + mean diameter of Moon shadow mean diameter of shadow = mean latitude of

Moon + mean latitude of Sun - mean diameter of Sun = 57'11" + 8'"5 - 16'1" = 41'18"5. Increasing this by 9/27 the mean diameter of shadow = 41'18"5 + 46"6 = 42'8" = 42'135 and the mean diameter of the Moon = 15'35" = 15'583 ∴ m = 42'135 + 15'583 = 57'72 ∴ cp = 57'72 / cos 5°34’ sin 5°6’ (cp) = sin 57'72 - sin cos 5°34’ - sin sin 5°6’ = 1.7614 - 6.6676 - 8.6546 = 2.8086 ∴ cp = 644' = 10°44"

This is the mean limit of the lunar eclipse. Similarly, it can be known that the maximum value of cp is 12036' and the minimum value is 60. This means that if cp is more than 12036', the lunar eclipse is impossible, and if it is less than 60, the eclipse will definitely happen, but if cp is more than 60 and less than 12036', the eclipse may be possible, which should be determined by the latitudes of the Sun and Moon and their true positions.

It has already been stated that cp is the distance of the node from the center of the shadow, but the center of the shadow is 180° ahead of the center of the Sun and the difference between the two nodes of the Moon is also 180°; therefore, if the difference between any node of the Moon and the Sun's node is more than 12036', the eclipse is impossible, less than 60

SOURCE PAGE 489

Then an eclipse must occur, and if it is between these two, then an eclipse is possible. Therefore, the maximum and minimum limits of lunar eclipse are 12036' and 60.

The maximum and minimum limits of solar eclipse—just as figure 63 proves that when the Moon comes into the shadow of the Earth, a lunar eclipse occurs, similarly it is also proven by the same figure that when the Moon comes between the tangents that form the shadow of the Earth at the end of the new moon, a solar eclipse must be seen somewhere on Earth because in such a situation, the shadow of the Moon must fall somewhere on Earth. Just as the limit of lunar eclipse can be known from the magnitude of the semi-diameter of the arc, similarly the limit of solar eclipse can be known from the magnitude of the semi-diameter of the arc.

∠ latitude = ∠ prime vertical circle + < meridian + < latitude = Sūrya radius − solar declination + lunar declination

therefore < Sūrya Siddhānta = 16'1" - 8" · 5 + 57'11" = 73'3.5 = 73'058

Sūrya Siddhānta - relation to eclipse formula = fish's isosceles, can be done here. Here m = angle shadow + Moon's diameter/2 = 73' .058 + 15' .583 = 88' .64 therefore formula = 88' .64 / sin5° 34' sin5° 6' therefore true formula = 88' .64 - sin5° 34' - sin5° 6' = 1.6476 - 6.6676 - 8.6546 = 2.6651 therefore formula = 688' .6 = 16° 28' .6

This is the mean limit of the solar eclipse. Similarly, it can be known that the maximum value and minimum value of the 6th in relation to the solar eclipse is 180.5 and 150.3 respectively. That is, if the difference of any node of the moon from the sun at the end of the new moon is less than 150.3, then it should be understood that a solar eclipse will definitely occur, and if this difference is more than 180.5, then a solar eclipse is not possible. But if this difference is between these two

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SOURCE PAGE 490

This means if it is more than 150.3 and less than 180.5, then an eclipse is possible, which should be determined by the sun, the latitude, and their true motions at the end of the new moon.

Lunar eclipse is visible in all places where the moon is in the rising. However, a solar eclipse is not possible in all places where the sun rises because it can only be seen from a few places due to the latitude of the moon and the very thinness of its shadow, which is not easy to determine.

> पर्वान्त काल में सूर्य, चन्द्रमा और पात को स्पष्ट करने की रीति-
> गतैष्यपर्वनाडीनां स्वफलेनोन संयुतौ ।
> समलिप्तौ भवेतां तौ पातस्तात्कालिकोऽन्यथा ॥८॥

Translation—(8) The time difference from the time when the sun and moon were made apparent to the time of the end of the lunar month or new moon, by knowing the true motions of the sun and moon for that time, by subtracting or adding from their true portions of the sun and moon, the ones that result are to be considered the apparent sun and apparent moon at the end of the lunar month. If the above time is behind the end of the lunar month, then subtraction should be done, and if it is before, addition should be done. But to know the true position of the node, its opposite action should be done, that is, if the above time is behind the end of the lunar month, the motion of the node for that time should be increased and if it is before, it should be decreased because the motion of the node is opposite.

Scientific Commentary—as explained in Chapter on Mean Motions 67th verse, it is stated that if the mean planet of a certain time is given, how to know the mean planet of another time? Similarly, here it is stated that if the true positions of the Sun, Moon, and Rahu of a certain time are known, how to know the true positions of the Sun, Moon, and Rahu of the parvānta time? This is necessary because for calculating eclipses, the true positions and motions of the Sun, Moon, and Rahu of the full moon and new moon intervals are used, as is revealed by the methods mentioned above.

Cause of eclipse—

> छादको भास्करस्येन्दुरधस्ताद् घनवद् भवेत् ।
> भूच्छायां प्राङ्मुखश्चन्द्रो विशत्यस्य भवेदसौ ॥६॥

SOURCE PAGE 491

Translation—(6) Because the Moon is below the Sun, it is covered by the clouds like a shadow. The Moon, moving towards the east, enters the terrestrial shadow, so the Moon is covered by the terrestrial shadow. Therefore, there is a shadow of the Moon in a solar eclipse.

Scientific Commentary—This has already been stated, so there is no need to repeat it here.

The magnitude of the grass—

> तात्कालिकेन्दुविक्षेपं छाद्यच्छादकमानयोः ।
> योगार्धोत्प्रोज्झ्य यच्छेश तमश्छन्नं तदुच्यते ॥१०॥

Translation—(10) Subtract from the sum of the diameters of the obscuring and the occulting shadow of the parvāntakālika Moon, the remainder will be the magnitude of the grass.

Scientific Commentary—This is clear from the explanation of figure 66. This figure applies equally to both the Sun and the Moon. In a lunar eclipse, there is an occulting shadow and an occulted shadow,

and in a solar eclipse, if the Sun's image is taken, there will be an occulted shadow and an occulting shadow.

State of total eclipse and partial eclipse—

> ग्राह्यमानाधिके तस्मिन्सकलं न्यूनमन्यथा ।
>
> योगार्धादधिके न स्याद्विक्षेपे ग्रास संभवः ॥११॥

Translation—(11) If the magnitude of the eclipse is more than the image of the moon, then the total eclipse, i.e., total eclipse and less, occurs. But if the deviation of the moon is more than the sum of the diameters of the obscuring and the obscured, then the eclipse cannot happen.

Scientific Commentary—This has also been explained in Figure 66. What is said about the moon there can also be applied to the sun.

Method of knowing the position and declination

> ग्राह्यग्राहकसंयोगवियोगौ दलितौ पृथक् ।
> विक्षेपवर्गहीनाभ्यां तद्वर्गाभ्यामुभे पदे ॥१२॥
> षष्ठ्या संगुण्य सूर्येन्द्वोः भुक्त्यन्तरविभाजिते ।
> स्यातां स्थितिविमर्दार्धे नाडिकादिफले तयोः ॥१३॥

(12) Keep the images of Chādya and Chādaka separate by adding and subtracting them and then taking half of each; subtract the square of the moon's deflection from the square of each. (13) Multiply each square root by 60 and divide the product by the difference in the true motions of the Sun and the Moon. The remainder obtained by division will be the latitude and the colatitude.

Scientific Commentary—the half time that the eclipse remains is called sthityardha and the half time that the total eclipse remains is called vimardadha. Or the time from the time of contact to the middle of the eclipse is called sthityardha and the time from the time of submersion to the middle is called vimardadha. Doubling sthityardha gives the total eclipse period and doubling vimardadha gives the time of the total eclipse.

In figure 10 there are six revolution circles, the lunar orbit, the node of the moon, the center of the terrestrial shadow, the center of the moon at the time of the solstice, the center of the moon at the time of the conjunction, the center of the moon at the time of the declination, and the center of the moon between the time of salvation and the eclipse. Here, for convenience, the terrestrial shadow is taken as constant, so the motion of the moon is relative to its orbit as shown in line b in figure 68. Therefore, it is proven that the motion with which the moon is shown to move along the line is the difference between the true motions of the moon and the sun. If it is assumed that the six are a plane triangle, there will be no objection. In such a case, when the six angles are right angles and the six is the lunar shadow, then

```
Chapter =  chha^2 - chha^2  =  Lunar Image + Terrestrial Image )^2 - (Lunar Shadow) ^2
```

If there is a difference in the clear daily motions of the Moon and the Sun, then the time in which the Moon will travel the same path with this motion will be thus known-

When the moon moves 60 hours in a day, then it will move 60 X a day.

If the value of a is taken as the value given above, then the sidereal time will be—

SOURCE PAGE 494

If Lunar Illumination + Earth Illumination^2 is written and

If this formula is simplified, then this will be the form—

sthityardha = 60ghaḍī X √ (mānaikyakhaṃḍa + śara) (mānaikyakhaṃḍa − śara) difference of the true daily motions of the Moon and Sun

Similarly

cha = √(cha^2 - chha^2) = √([(bhubhābim - chandrabimb / 2)^2 - (chandrashar)^2]) therefore vimardhakal = cha - ra / 60 x √([(bhubhābim - chandrabimb / 2)^2 - (chandrashar)^2]) if bhubhābim - chandrabimb ko manantarakhand likha jaay aur is sutrakko

It should be simplified

60degrees X √ (mean motion + arc) (mean motion - arc) / difference of the true daily motions of the Moon and Sun

The time that will be obtained by simplifying these will be in hours. But this will be crude because in its calculation the difference between the apparent daily motions of the sun and the moon and the lunar-solar shadow are taken, which will be very different from the apparent motions and the shadow from the time of touch or conjunction. Therefore, it is necessary that the difference in the apparent motions of each hour or hour before and after the lunar-solar shadow is calculated. If the apparent positions of the sun, moon, Rahu, and lunar shadow some time before and after the lunar-solar shadow are calculated from the above rule, then the apparent time and the mean time will be obtained. These will be purer than the previous ones. Similarly, if the apparent time and the mean time are calculated many times, then at the end a result will be obtained which will not be different. The action by which such pure apparent time and mean time are obtained is called as akshakarm. The procedure for this is explained in the next two shlokas.

SOURCE PAGE 495

Know the situated and unsituated periods from inaction—

> स्थित्यर्धनाडिकाभ्यस्ता गतयष्षष्टिभाजिताः ।
> लिप्तादि प्रग्रहे शोध्यं मोक्षे देयं पुनः पुनः ॥१४॥
> तद्विक्षेपैः स्थितिदलं विमर्द्धार्धं तथाऽसकृत् ।
> संसाध्यमन्यथा पाते तल्लिप्तादिफलं स्वकम् ॥१५॥

Translation—(14) By multiplying the daily motions of the Sun, Moon, and Node by the situated period (which is in clocks) and dividing by sixty, it is known how much the Sun, Moon, and Node move in the situated period. By subtracting these magnitudes from the portions of the equinoctial Sun and Moon, the tangential portions of the Sun and Moon are obtained, and by adding them, the nodal portions are obtained. (15) But to know the portion of the tangential Node, the amount the Node moves in the situated period should be added to the portion of the equinoctial Node, and it should be subtracted from the portion of the equinoctial Node to know the nodal portion because the motion of the Node is opposite. In this way, the situated and unsituated periods are calculated by knowing the portions of the tangential Sun, Moon, and Node, the shadow of the Moon, and the true motions of the Sun and Moon. Similarly, the knowledge of the tangential and unsituated periods can be known with subtlety many times from inaction. Similarly, the purity of the conjunction and the motion periods should also be known.

Scientific Commentary—Its derivation has been stated in the previous page, so there is no need to write more.

Method of knowing the tangential and unsituated periods and the conjunction and motion—

> स्फुटतिथ्यवसाने तु मध्यग्रहणमादिशेत् ।
> स्थित्यर्धनाडिकाहीने ग्रासो मोक्षस्तुसंयुते ॥१६॥
> तद्वदेव विमर्द्धार्ध नाडिकाहोन संयुते ।
> निमीलनोन्मीलनाख्ये भवेतां सकल ग्रहे ॥१७॥

Translation—(16) At the end of the true date, that is, at the end of the full moon and new moon, there is the middle time of the eclipse. Subtracting the situated period from this time gives the time of the tangential period, and adding it gives the time of the unsituated period. (17) Similarly, by subtracting the unsituated period from the middle time of the eclipse, the beginning of the conjunction period, i.e., the conjunction period, is found, and by adding it, the end of the motion period, i.e., the end of the total eclipse period, is found.

SOURCE PAGE 496

Scientific Commentary—it is so clear that there is no need to write more.

It is also necessary to state how the middle time, touch time, moksha time, sammilan time, and unmilan time of the eclipse are known according to other practices?

Figure 68 shows that the portions of the moon and the umbra of the Earth are equal at the end of the full moon. Therefore, the difference between the portions of the moon and the umbra will be equal to X(cha - ra) when cha and ra are the portions of the clock of the moon and the sun or the umbra. If the motion of the moon's shadow is the clock, then after X hours, its shadow will be equal to X sha. If the moon's shadow is sha at the time of the full moon, then after X hours, its shadow will be sha - X sha. Therefore, the difference between the centers of the moon and the umbra after X hours will be:-

```
Ma =    sqrt [ X (  cha -    ra) ]^2 + (    sha - X   sha)^2
```

Because the difference between the portions of the moon and the umbra will be equal to the hypotenuse of the moon's shadow and the difference between their centers, as shown in Figure 100 at the time of touch time and sammilan time. Squaring both sides of this equation

Ma2 = [h (ca -ra)]2 + (sh -h.sha)2 = (ca -ra)2h2 + sh2 - 2sh.sha.h + h2sha2 = [(ca -ra)2 + sha2]h2 - 2sh.sha.h + sh2 - ma2 = 0

This is the quadratic equation of h from which two values of h are found. It is proved that the moon comes at the same distance from the lunar eclipse before and after the full moon. If the value of the lunar phase is substituted in place of ma and two values of h are obtained, then these will be the times of eclipse and salvation. If these two values are imaginary, then it should be understood that there will be no eclipse, if they are equal, then it should be understood that the beginning and end of the eclipse will be together, that is, the moon will just touch the shadow but there will be no eclipse.

If the value of the lunar segment is substituted in place of ma and two values of h are obtained, then the beginning and end times of the total eclipse or conjunction and

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SOURCE PAGE 497

They will be known at the time of the declination. If these two values are imaginary, then it should be understood that the total eclipse will not occur, and if both values are equal, then it should be understood that the beginning and end of the total eclipse will happen together, that is, as soon as the full image of the moon comes into shadow, it will also begin to be outside the shadow.

From this equation, both values of h will be as below according to the following formula:-

h = 2sh. sha. ± 4sha^2 - 4 [(cha - ra)^2 + sha^2] [sha^2 - ma^2] 2 [(cha - ra)^2 + sha^2] = sha. sha. ± sha^2. sha^2 [(cha - ra)^2 + sha^2] [sha^2 - ma^2] (cha - ra)^2 + sha^2

Half the sum of these two values of G

Sh. Sha (cha - ra) ^{2} + Sha ^{2}

This is the middle time of the eclipse, that is, the eclipse is in the middle after the same time as the full moon.

The method of knowing the time of eclipse and the time of the touch of eclipse from the declination and the transit of the Sun and Moon, etc., which is very similar to the above method but is not written here in detail due to fear.

It is necessary to understand here that the rule given in Sūrya Siddhānta for knowing the time of eclipse and the time of moksha, etc., proves that the middle time of the eclipse occurs at the end of the full moon, but in the other method mentioned above, the middle time of the eclipse comes some time after the full moon. The second method is absolutely pure and the first is somewhat crude. The reason for this is clarified by Figure 68.

To know the magnitude of the grass from which the time of touch is how much time it is—

> इष्टनाडीविहीनेन स्थित्यर्धेनाकंचन्द्रयोः ।
> भुक्त्यन्तरं समाहन्यात् षष्ठ्याप्ताः कोटिलिप्तिकाः ॥१८॥
> भानोग्रहं कोटिलिप्ता मध्यास्थित्यर्धं सङ्गुणाः ।
> स्फुटस्थित्यर्धसम्भक्ताः स्फुटाः कोटिकलाः स्मृताः ॥१९॥

SOURCE PAGE 498

> क्षेपो भुजस्तयोर्वर्गयुतेमूलं श्रवः स्मृतः ।
>
> मानयोगार्धतः प्रोज्झ्य ग्रासस्तात्कालिको भवेत् ॥२०॥

Translation—To know what the magnitude of the grain is, some time behind the beginning of the eclipse but before the middle eclipse, subtract the ecliptic time from the mean position and multiply the remainder by the difference between the daily true motions of the moon and the sun and divide the product by 60. This quotient is called the sidereal time when the daily motions are expressed in degrees. (16) To know the magnitude of the solar eclipse grain, multiply the sidereal time obtained by the above method by the mean position and divide the product by the true position, the result of which is called the true sidereal time. (20) Taking the length of the moon at that time as the hypotenuse and adding its square to the square of the sidereal time to find the square root of the sum, and then subtracting the result from the arc, the remainder is the instantaneous grain.

Scientific Commentary—the derivation of this rule is well explained in figures 68, 66, and 100. The formula m = √([h(cha - ra)]^2 + (s - h × sha)^2) established in the Scientific Commentary of 16-17 shlokas is another form of this rule. In this formula, h is the time before or after the epoch, but in the rule, the time after the touch time is considered the desired time, so the order is to subtract the desired time from the actual time. By doing this, when the formula is simplified by replacing h with h in place of h, the magnitude of m that results can be found by subtracting it from the segment. Here, h(cha - ra) is already given because cha - ra is the difference in the motions of the Sun and the Moon per hour. If the difference in daily true motions is given, it has to be divided by 60, as stated in the rule. s - h × sha is the instantaneous arc and the instantaneous difference between the centers of the Earth and the Moon. The smaller this difference is than the segment, the smaller the magnitude of the arc, which is well explained in relation to figure 66. The only difference from that figure is that the nearest distance between the centers of the Moon and the Earth is taken, which is equal to s × jya e, and the distance taken there is that which occurs after the touch time.

If the sum of the diameters of the Earth and the Moon is indicated by the letter 'm' as before, then the magnitude of the arc after the time of touch will be:-

SOURCE PAGE 499

$$\text{Grass} = m - \sqrt{[(st - h) \times ra / 60]^2 + sh^2}$$

Where 'st' is the meridian, 'sh' is the instantaneous shadow of the moon, and 'ra' is the difference in the daily motions of the sun and the moon. If 'ra' is the difference per hour, then no division by 60 is required.

The relation to solar eclipse will be explained in the subsequent chapter on solar eclipses.

The magnitude of the grass after the middle eclipse but some time before the time of salvation—

> मध्यग्रहणतश्चोर्ध्वम् इष्टनाडीविशोधयेत् ।
> स्थित्यर्धानमौक्तिकाच्छेषात् प्राग्वच्छेदस्तु मौक्षिके ॥ २१॥

Translation—(21) When it is to know the magnitude of the grass some hours before the time of salvation after the middle eclipse, subtract the desired hour from the position of the time of salvation and calculate the grass according to the method described in the eighteen-twenty shlokas. This will show how much of the moon is remaining eclipsed before the desired hour of salvation.

Scientific Commentary—This rule is the same as the rule described in eighteen-twenty shlokas. From this it is known how much is eclipsed at the desired time and it is also known how much eclipsed part remains at the desired time because the increase in the eclipsed part in the order before the middle eclipse is the opposite order of the decrease in the eclipsed part after the middle eclipse.

If the magnitude of the arc is known, then know the auspicious time—

> ग्राह्यग्राहकयोगार्धात् प्रोज्झ्य संछन्नलिप्तिकाः ।
> तद्वर्गात्प्रोज्झ्य तत्कालविक्षेपस्य कृतिं पदम् ॥२२॥
> कोटिलिप्ता रवेः स्पष्टस्थित्यर्धेनाहता हृताः ।
> मध्येन कोटिलिप्ताभिस्स्थितिबद्वासनाडिकाः ॥२३॥

Translation—(22) Subtract the art of the part afflicted by the part from the remainder and square it, and subtract the square of the instantaneous shadow of the moon from this square and find the square root of the remainder, then (23) the value of Kotilipta will be found. Multiplying this Kotilipta by the true position in a solar eclipse and dividing the product by the mean position gives the Kotikala. The method to know the time of position from this Kotikala

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SOURCE PAGE 500

To make the clock, multiply the sidereal time by 60 and divide by the difference in the daily motions of the sun and the moon. Subtract the resulting quotient from the sidereal time to find out how many hours have passed after the occultation. This is named Grasanadika. This is the same.

Scientific Commentary—This is the opposite of the rule stated in 18-20 shlokas. There it is stated what is the magnitude of the gras after the desired time from the beginning, and here it is stated how the desired time is known if the magnitude of the gras is known. Therefore, its derivation is the same. Its form is as follows:-

$$\sqrt{(m - grass)^2 - s^2}$$

$$G = st - 60 / ca - ra \sqrt{(m - grass)^2 — sh^2}$$

Know the arc

> नतज्याक्षज्योर्घातः त्रिज्याप्तस्तस्य कार्मुकम् ।
> तदंशास्सौम्ययाम्यास्ते पूर्वापरकलापयोः ॥२४॥
> राशित्रययुताद्ग्राह्यात्कान्त्यंशे दिक्समे युतात् ।
> भेदेऽन्तराज्ज्यावलनसप्तत्यंशोऽङ्गुलात्मिका ॥२५॥

Multiply the sine of the netaṃśa of the circumposition of the chhaya graha by the sine of the latitude of the desired place, and divide the product by the radius of the ahoratravritta of the planet, and make the

quotient an arc. This is called the axis of the planet. If the planet is in the pāla, that is, before the yatrotarvitta, then the axis is towards the north, and if the planet is in the kapāla, that is, behind the yatrotarvitta, then the axis is towards the south. (25) Adding 60 degrees to the sine portion of the planet, the resulting portion is obtained by taking the inclination (according to chapter on true positions verse 28) by multiplying it by the sine of the ultimate revolution (of the ahoratravritta) and dividing by the radius. This quotient is called the inclination. This will be to the north or south. If the directions of the axis and the inclination are the same, add them, and if they are different, subtract them. The result obtained by doing this is called the sphutvalan. Dividing its sine by 70 gives the angular value of the arc.

Scientific Commentary—To know from which point the eclipse, moksha, etc., of the chhaya graha begins or ends, the inclination is required.

SOURCE PAGE 501

The angle at which the meridian is north or south from the east or west point of the chhaya graha is called sphutavalan. The east and west points of the planet's image are known in this way—they draw a great circle passing through the north-south points of the horizon from the center of the planet to the meridian, which is called the circle of position of that planet. This circle of position forms a right angle with the prime vertical circle. The vertical circle drawn from the center of the planet's image parallel to the prime vertical circle is also at a right angle to the circle of position. Among the points where this vertical circle intersects the edges of the planet's image, the one towards the east is called the east point of the planet and the one towards the west is called the west point of the planet. All these things are clarified by figure 101.

From Figure 101 it is proven that the ecliptic is at the difference between the eastern point of the image of the Sun and the angle of the ecliptic. This difference is called the declination. To know

U dh kh da = meridian circle of the auspicious place

U pūda = the horizon circle of the desired place (first half)

SOURCE PAGE 502

U, D, P = North, South, and East points of the horizon; X = Prime Vertical Circle; X P = Prime Vertical Circle; CR = Circle of Declination; G = Center of the image of the planet; U G D = Circle of position of G and D; DH = North Celestial Pole; K = Altitude (Circumference of Declination); K of ki = Position of Kadamba when sine of the Tropic of Cancer is on the circle of right ascension; K of sthāna jaba sāyana makara yāmottaravṛtta on hotā is = Position of Kadamba when sine of the meridian of Capricorn is on; G DH = Angle of declination of the planet; G K = Angle of latitude of the planet; DH G K = Angle of obliquity of the planet; U G K = Angle of right ascension of the planet; CR G PU = Parallel circle of the prime vertical circle passing through the center of the planet; PR G = Eastern point of the planet's image; PT G = Western point of the planet's image

To know the angle between the circle of position UG and the circle of Kadamba position K, it is necessary because UG and K are at right angles with XG and K in sequence, so the angle between the first two will be the same as the angle between the previous two. Therefore, the angle of the circle becomes the same as the angle of the circle of declination, which can be divided into two parts named UG DH and DH GK. The angle UG DH is called obliquity and the angle DH GK is called inclination. It is clear from the figure that to know the magnitude of the angle UG K, obliquity and inclination have to be added. However, if KG is between the circle of declination UG and DHG, then the angle of the circle

comes from subtracting inclination from obliquity. In the figure, the planet is in the pūrva kapāla (in the meridian circle) i.e., on the meridian circle.

SOURCE PAGE 503

The above is shown. In such a condition, the declination is to the north from the east point. If another figure is drawn in the same way and the planet is shown in the western skull, then it will be clear that the declination is to the south from the east point of the planet. In this way, the latter part of verse 24 is proven. The magnitude of the latitude is known by the spherical triangle g g d as follows:-

Spherical triangle in UG DH,

```
sine (rising) = sine (setting) = sine (east) sine (west) sine (north) sine (south) sine (meridian difference)
```

Because u x and u g are equal to 60°, the angle between them is equal to u x or g u or g u x, which is the part of the prime vertical circle. sine (polar distance) = sine of the planet's revolution = sine of the planet's obliquity; sine(d u) = sine of latitude = sine of the axis

jya (ugad) = aksjya × jya (khga) / graha ke ahoratvarttaki tirjya = aksjya × jya (khga) / krantika kotija jya... (9)

In this way, the first half of verse 24 is also proven. Here, the radius does not mean 3438 but the radius of the circumcircle which is equal to the sine of the ecliptic of the planet and the sine of the node of the planet, nor is the sine of the node the sine of the node of the planet or the sine of the node of the planet. Bhaskaraacharya has stated this rule2 to know its magnitude: multiply the nodal period of the occulted planet by 60 and if there is a solar eclipse, divide by the diurnal arc and if there is a lunar eclipse, divide by the nocturnal arc. It seems that when the occulted planet rises by 60° in the diurnal arc or nocturnal arc, the amount it rises in the nodal period is its node. However, this rule is gross. It seems that Sūrya Siddhānta has written the sine of the node of the occulted planet not for the sine of the node of the planet, but for the sine of the node of the prime vertical circle because no rule has been given to know the magnitude of the sine of the node of the planet. It is also possible that the sine of the node of the node is written for the sine of the node of the node. But these two meanings are not pure. Therefore, I have taken the sine of the node of the circumcircle in the translation.

1. See Chapter on True Positions page 207. 2. Mathematics chapter page 122

SOURCE PAGE 504

The part of the circumscription is known in this way with accuracy based on the spherical trigonometry as follows:

One should know the portion of the first planet from its sidereal time by the formulas proven in page 262 of (kha). Then, using the portion, one should know by the formulas proven in page 276 of koṇa dha kha ga. When the portion kha ga and koṇa dha kha ga or u kha ga is known, then the spherical triangle of kha kha ga, i.e., the three sides of kha kha ga, which are two sides u kha and kha ga and the angle between them, can be easily known by knowing the angle kha u ga because

```
Kospara (Sky) x sine (earth) = cosine (earth) cosine (sky) + Kospara (sky) 1 sine (earth)
```

But here x = 60°, therefore sin(x) = 1 and cos(x) = 0

∴ Coaspra (Khaga) = Coaspra (Khuga) sine (Ukha)

Cosper (Khug) = Cosper (Khag) / sine (Ukhag)

or Cosper (Khug) = sine (Ukhag) / Cosper (Khag) = sine (Ukhag) Spere (Khag)

but AngleUkhag = 60° + AnglePukhag = 60° + Agra2

∴ sine (Ukhag) = sine (60° + Agra) = cosecant (Agra)

∴ Spere (Khug) = Agrakotijya X Natanshsparshrekha

Therefore it has been proven that to know the tangent-sine of the planet's mean position, one should multiply the planet's agra by the tangent line of the planet.

By knowing the value of the sine of the angle or the sine of the arc of the arc of the bow, by substituting it into the formula (1) of page 478, the value of the latitude will be found.

1. See Todhunter and Leather’s Spherical Trigonometry, PP. 26.

2. See page 276

SOURCE PAGE 505

The value of the inclination can be known as follows:— in the triangle of the spherical triangle (a, b, c)

sine angle c = sine (c) * sine angle c

Here, the difference between the pole and the meridian is the same as the sun's ultimate revolution. The difference between the pole and the planet is equal to the angular distance of the planet's revolution, and the angle a c, between the meridian of a, the meridian of c, and the inclination of a is between a and c. As is clear from Figure 36 of page 200 that the declination point is 600 ahead of the vernal equinox and the transit of the northward point is 2700 ahead of the vernal equinox, that is, the inclination intersects the ecliptic at a right angle at the difference of 600 and 2700 from the vernal equinox, from the declination and the northward points. From the same figure, it is also revealed that the angle between the meridian of c, the meridian of a, and the inclination of c is the same as the angle between a and c, which is also the same as the angle between a and the bow. The portion of the planet and the bow is. Therefore, the sum of b and c is equal to 600, that is, the sine of c is equal to the angular distance of b. Thus, it is proved that the sine of the angle between the meridian of the planet and the inclination of the circle is equal to the angular distance of the portion of the planet. Therefore

sine of the portion of the planet and the bow = sine angle c = ... (2)

Thus the value of the angle k d or the declination is proven. In the first half of the 25th verse, it is briefly stated that by adding 60° to the portion of the planet, what comes forth is the sine of its declination, i.e., by multiplying its sine by the true sine and dividing by the radius of the planet's orbit. But the

radius of the orbit = true sine (see page 207) and the sine of what comes forth by adding 60° to the portion of the planet is equal to the sine of the portion of the planet because if the portion is b,

```
sine ( bha + 60 ° ) = versed sine ( dekhopṛṣṭha126 − 128 )
```

When the portion is less than 6 signs, the revolution is north, and when it is more than 6 signs, the revolution is south. Similarly, when bh + 600 is more than 6 signs, angle k g d should be considered south, and when it is less than 6 signs, it should be considered north.

SOURCE PAGE 506

If the directions of both declination and latitude are the same, adding them or taking their difference gives the magnitude of the true position. This is clear from figure 101.

From the formulas of declination and latitude, it can also be determined at what time their values can be maximum and at what time they can be zero. For example, take the formulas of latitude. This equation has 3 multipliers on the right side, in which the sine of the true motion is constant, but the sine of the planet's portion and the sine of the motion are variable. At the time when the portion is zero, the planet will be at the vernal equinox, so its motion will also be zero. In such a case, the values of these sines will be 2. Therefore, the latitude will be equal to the true motion, i.e., 23°27'. This will also be true at the autumnal equinox. This is what Bhaskaraacharya wrote in the 30th verse of the Grahadhyāya on eclipses. Similarly, when the portion is 600 or 2700, the sine of the portion will be zero, but the sine of the motion will not be zero because it will be approximately 240, so the latitude will also be zero, etc.

Here it has been stated how the magnitude of the true motion is known in parts or degrees. If one wants to know how to work with the true motion by measuring with an inch while drawing the figure, the magnitude of the true motion divided by 70 gives the magnitude of the motion in inches. This is stated in the 25th verse. Its proof is that the procedure was to draw a circle by taking the semi-diameter of 46 inches for drawing the image of the celestial body. It is approximately four times the gnomon of 12 inches, and in this way, one inch is approximately 70 degrees because the value of the radius is generally understood as 3438 degrees, and 46 x 70 = 3430, which is very close to 3438.

Knowing the value of the image in inches—

> सोन्नतं दिनमानार्धं दिनार्धाप्तफलेन तु ।
> छिन्द्याद्विक्षेपमानानि तान्येषामङ्गलानि तु ॥२६॥

(26) In the auspicious time, add the advanced period of the visible planet to the sum of the diurnal and diurnal arc. Divide this quotient by the diurnal arc. By dividing the artistic images of the visible and visible and visible planets by this quotient, the angular values of their images will be found.

SOURCE PAGE 507

The science commentary—in page 382 it is stated that due to the motion of rising and setting, the sun appears larger. The same condition is also true for the moon. Our teachers also knew this matter1. It was certain that there is no difference in the real bodies of the sun and the moon while rising or setting, therefore our teachers imagined that to express the apparent size of the sun or moon during the rising or setting time in fingers, three arts of the finger should be considered, and when this body is in the x-cross shape, four arts of the finger should be considered2. By doing this, the difference in shapes that is

seen will also be in their finger values. This is the matter of the apparent sizes of the images while rising or setting and the images located in the x-cross. If the planet is between the x-cross and the horizon, then to know the finger value of its image, they used the ratio in this way. When going from the horizon to the arc between x and the meridian, the value of the finger becomes 4 arts from 3 arts. What will be the value of the finger in the advanced time, which is from the rising or setting time to the auspicious time? But the time taken to go from the rising time to the meridian is called the diurnal arc. Therefore, when there is a difference of one art in the value of the finger in the diurnal arc, then what difference will there be in the advanced time. This difference = advanced time x 1 art / diurnal arc. Therefore, it is apparent that the image in the auspicious time

To know the angular value, one should take the value of 1 angula as 3 + advanced time X 1 art / half day. Therefore in the desired time

```
1angula = 3 + advanced time X 1art / diurnal mean = 3 X diurnal mean + advanced time X 1 / diurnal meanart = 3diurnal mean + advanced time / diurnal mean - art
```

1. See Gṇitādhya page 124

2. Bhaskaraacharya has considered the arts as the digits of art. See Gṇitādhayāy page 124.

SOURCE PAGE 508

```
= ( 2 + 1 ) daymean + advancedtimemean - art = daymean + daymean + advancedtimemean - art
```

Thus the derivation of the 26th verse is proven. However, it is necessary to state here that the size of a celestial body does not decrease in a sequence from sunrise to the meridian circle, therefore knowing the size of the planet by ratio is not pure as revealed by the table of motion given in page 365.

Example—Calculation of the lunar eclipse of Shravani Purnima of the year 1681 AD—

According to Sūrya Siddhānta—

First, the sun, the moon, and Rahu of this day should be clarified. Therefore, it is necessary to know what is the elapsed time from Kaliyuga to this day.

From the beginning of Kali Yuga to the beginning of Vikram Samvat, 3044 years from the beginning of Vikram to the Aries solstice of 1681, 1681,, from Kali Yuga and so on,,,,,, 5025,, 1 solar year = 365.258756 Sawan days

```
∴ 5025solar year = 5025 x 365.258756sawan day = 1835425.2486,
```

This is the time from the beginning of Kali Yuga to the mean Aries solstice of 1681 AD. The true Aries solstice occurs 2.1707 days earlier. Therefore, when subtracted, the time is 1835423.0782 Sawan days up to the Aries solstice of 1681 AD.

Now it is to see which tithi was on the day of Aries solstice.

```
Ichanmas = 26.530588SawanDin
```

Dividing the above Sawan days by this will be the number of lunar months passed and the remaining 8.442236 Sawan days will be the time from the mean new moon of Chaitra to the Aries solstice. Therefore, after 8.442236 days from the mean new moon of Chaitra, the solstice of Aries occurred. This proves that the Malas will not be observed this year, because when the Aries solstice occurs after the Krishna Paksha 4 of Vaishakh, then the year

SOURCE PAGE 509

There is a month called Malmas. In this way, the new moon of Chaitra is until Shravani Purnima by 4. Chandramasa occur. 1 Chandramasa = 26·530588 days 4,, = 118·122352 days half,, = 14·765264,, ∴ 4 ॥,, = 132·887646,,

1 lunar month = 26.530588 days; 4 months = 118.122352 days; half a month = 14.765264 days; therefore 4½ months = 132.887646 days. Values follow the supplied OCR.

From Kaliyuga to the Aries transit of 1681, there are 18354230782 days. From the Aries transit to Shravani Purnima, there are 124.4454,, From Kaliyuga to Shravani Purnima, there are 1835547.5236 days.

From the start of the Kali Yuga to the Aries ingress of 1681: 18354230782 days. From Aries ingress to the Śrāvaṇī full moon: 124.4454 days. From the start of the Kali Yuga to the Śrāvaṇī full moon: 1835547.5236 days. Values follow the supplied OCR.

From this calculation, the positions of the Sun, Moon, apogee, Rahu, etc., at the twilight of Shravani Purnima should be known from the Trairashika. It is stated in the Chapter on Mean Motions that there are 1577617828 Sawan days in one Mahayuga, in which there are 432000 revolutions of the Sun, 57753336 revolutions of the Moon, 488203 revolutions of the apogee, and 232238 revolutions of Rahu, so in 1835547 days

Chapter on Mean Motions states that in one Mahayuga there are 1577617828 Sawan days in which there are 432000 revolutions of the Sun, 57753336 revolutions of the Moon, 488203 revolutions of the apogee and 232238 revolutions of Rahu, so in 1835547 days

Revolution zodiac sign degrees arcminutes Sun of 5025 3 26 56·75 hue caṃdramā of 67182 6 23 36·21 ” caṃdrocca of 567 10 28 34·02 ” rāhu of 270 1 26 6·8 ”

SOURCE PAGE 510

The remaining parts after leaving the full revolutions of the Sun and Moon are their mean positions at midnight of Shravani Purnima. However, some modifications need to be made after leaving the full revolutions of the Moon and Rahu because in the beginning of Kali Yuga 1, the Moon's position was at the beginning point of Cancer and Rahu's position was at the beginning point of Libra, and Rahu's motion is opposite, therefore at the time of midnight of Shravani Purnima

The position of the Moon is 3 Ra + 10 Ra28°34'02 = 1Ra28°34'02

```
Rahu's position6Ra - 1Ra26°6'8 = 4Ra3053'2
```

The motion of the Sun's apogee is so slow that its position should be taken as 2Ra17°17'52 as per 2.

Sun's apogee = 2Ra17°17'52 - 3Ra26°56'75 = 10Ra17°17'77 = 3pada + 1Ra17°17'8

∴ the part of the fourth foot to be traversed = 1st 12°42'2 = 42042'02

The clear mean circumference of the Sun

840 - 20 × sine420° 42' 3438 = 840 - 20' × 23303438 = 840' - 14' = 826'

Bhujbal = 826 × 233021600 = 86' · 1 = 1026' · 1

This is the mean position of the Sun because its arc will be the same as this. It is negative because the mean center is in the Libra.

Therefore mean motion true sun = 3265' 75 - 1026' 1

1. Chapter on Mean Motions verse 57,58 and page 50

SOURCE PAGE 511

Sūrya Siddhānta = 118° 30' · 65

Darmakamandakendra = 1st 28° 34' 02 - 6th 23° 36' 21 = 4th 4054' 81 = 1 foot + 34° 54' 81

∴ second arc part = 55° 5' · 2

Lunar fine mean circumference

Lunar arc circumference = 320 - 20' × sin 55.5' 23.438'' = 320 - 20' × 28.83438'' = 320 - 16' = 31° 44' = 1604'

∴ lunar arc result =1604 × 281821600 =248' · 4 = 4° 8' · 4

This bow will also be this much.

Therefore lunar apogee result = 4° 8'.4

This is positive because the mean center of the moon is in six zodiacs. Therefore the true moon at midnight

= 23° 36' · 2 + 8' · 4 = 27° 47' · 7 = 27° 48'

From the true positions of the sun and the moon it is known that the moon is not 1800 degrees ahead of the sun but some less, so the full moon will be some behind midnight when the distance between the moon and the sun is exactly 1800. To know when the full moon will be, we need to know the true motions of the sun and the moon. We know that

The mean daily motion of the sun is 56’ 8,

Moon's, 760'35"

Moon's altitude, 6'41", = 6.7

and the daily mean motion of the Moon is 783'54".

SOURCE PAGE 512

Therefore, the apparent daily motion of the Sun (see Chapter on True Positions page 158)

> = 56' 8'' - 82621600 * 164 * 56' 8'' 225 = 56' 8'' - 1' 36'' = 57' 26'' = 57'.5

The daily motion of the Moon cannot be derived by the above method because the motion of the Moon is very fast. Therefore, to know the daily motion of the Moon, it is good to observe the Moon at midnight on a Friday after the full moon.

> Mean Moon = 623° 36'.21; Daily Mean Motion of the Moon = 13° 10'.58 therefore Mean Night Moon of Pratipada = 10° 6046'.76 = 10° 6046'.8 = 10° 6050' " " Lunar Altitude = 128° 34' + 6'.7 = 128° 40'.7 therefore Mean Night Moon's Center = 128° 40'.7 - 10° 6046'.8 = 3210° 50'.6 = 1foot + 21° 51' Subtly therefore Second Part of the Path = 68° 6' Lunar Circumference = 320 - 20' × sin68° 6' 3438 = 320 - 20' × 31613438 = 320 - 16' = 1601' therefore Lunar Shadow = 1601 × 311621600 = 281' This will be the same. Therefore Mean Shadow = 281' = 4041' therefore Apparent Moon = 10° 6050' + 4041' = 10° 11' 31'

SOURCE PAGE 513

and the difference between the full moon "" = 6 arc 27°48' is = 13°43'

In this way, the true daily motion of the moon is 13°43' = 823'

> Sūrya Siddhānta, 575

Therefore, the moon moves 765'5 more in a Saavan day than the sun.

The moon at the middle night of the full moon is 6 arc 27°47'06

> Sūrya Siddhānta

This is 43.1 less than 6. Therefore, when the moon moves so much further from the sun, it will be the full moon. But in 60 hours, the moon moves 765.5 degrees further from the sun. Therefore, 43.1 will complete it in 43.1 × 60/765.5 hours which is 3 hours 23 minutes. Therefore, the full moon ended 3 hours 23 minutes after the midnight of Thursday.

Now the Sun and Moon of the full moon should be clarified. The clear daily motion of the Sun = 57'5 ∴ The motion of 3 ghadi 23 pal = 3'14"5 = 3'24. Midnight time Sun = 3 ra 28°30'7 ∴ Full moon time Sun = 3 ra 28°33'6 = 118°34' The clear daily motion of the Moon = 823' ∴ The motion of 3 ghadi 23 pal = 46'24" = 46'4 Midnight time Moon = 6 ra 27°47.6 ∴ Full moon time Moon = 6 ra 28°34' = 268°34' Rahu's daily motion = 3'11" ∴ The motion of 3 ghadi 3 pal = 11" = 2' Midnight time Rahu = 4 ra 3053'2 Rahu's motion is opposite, so from this 3 ghadi 3 pal

SOURCE PAGE 514

Subtracting the Full Moon time Rahu = 4 arc 3053' = 123053'

The Moon from Rahu = 268°34’ - 123°53’ = 174°41’ ahead.

therefore lunar declination = sine174° 41' × sine4° 30' * 3438 = sine5° 16' × sine4° 30' 3438 = 316 × 2703438 = 25'

∴ Full Moon Moon Arc = 25’

This arc is after the revolution because the Moon from Rahu is less than 6 signs. (Spashta Shloka 7,)

Prime Meridian Rahu = 123 ° 53 ′

Sūrya = 118° 34' difference = 5016'

If the minimum limit of the lunar eclipse is less than 60. Therefore, the lunar eclipse will definitely occur. (Lunar Eclipse Chapter page 463)

According to verse 1, 3 of Lunar Eclipse Chapter

Sūrya Siddhānta = 6500 X 57 ′ 26 ″ / 56 ′ 8 ″ yojan

Moon ″ ″ ″ = 480 × 823' 760' 35'' yojan = 480 × 823760.583 × 115 kala = 32 × 823760.583 kala = 33.31 kala

Planar diameter of the terrestrial shadow in the lunar orbit according to shlokas 4, 5

= 1600 × 823790 · 583 − (6500 × 57' 26" 56' 8" − 1600) × 4806500 yojan

The shadow of the moon is calculated in the same way as the true shadow according to the 28th shloka of the Sūrya Siddhānta. The ultimate shadow of the moon according to Sūrya Siddhānta is 4030′.

SOURCE PAGE 515

Simplifying by dividing this by 15 gives the artistic shadow in the lunar orbit

Vyasorbhuvabimb = 1062 / 3 x 823 / 760.583 - 32 x 57' 26" / 59' 8" + 7.877 = 111.17 - 31.11 + 7.88 = 87.64

In a lunar eclipse, the umbra is the shadow. Therefore, the semi-diameter of the umbra = 87’ 64 ÷ 2 = 43’ 67

The true diameter of the Moon = 33‘·31

∴ Chādyakāvyāsardha = 33 ′.31 ÷ 2 = 16 ′ 66

∴ accepted part = 43 ′.67 + 16 ′.66 = 60 ′.63

and mānāntara khaṃḍa = 43.67 - 16.66 = 27.31

grascamān = 60'.63 - 25' (verse 10) = 3563

this is larger than the diameter of the lunar eclipse. Therefore, the solar eclipse is occurring (see page 460 and verse 11).

according to page 466,

sthityardha = 60 × \((60.63 + 25) (60.63 - 25) 765.5 \) = 60 × \(85.63 × 35.63 \) 765.5ghaḍī = 60 × 55.236765.5ghaḍī = 4ghaḍī20pala

60 hours × \((27.31 + 25) (27.31 - 25) \) 765.5 \ = 60 × \(52.31 × 2.31 \) 765.5 \ hours = 60 × 10.662765.5 \ hours = 51.7 minutes = 52 minutes

It has already been stated that the full moon ended 3 hours 23 minutes after the midnight of Thursday. By subtracting the time of the present, the time of eclipse will be, and by subtracting the time of the half-eclipse, the time of conjunction will come.

SOURCE PAGE 516

∴ time of touch = 3 hours 23 minutes - 4 hours 20 minutes = - 57 minutes

The eclipse will touch 57 minutes before the eclipse.

Summilation Time = 3 hours 23 minutes — 52 minutes = 2 hours 31 minutes

That is, the eclipse will begin after 2 hours 31 minutes of midnight, or the entire lunar disc will enter the shadow.

In this way, the time of ascent = 3 hours 23 minutes + 52 minutes = 4 hours 15 minutes, Chapter on Mean Motions

and mokshakal = 3 hours 23 minutes + 4 hours 20 minutes = 7 hours 43 minutes, the above

This time is the middle period of Ujjain, meaning the eclipse touch, conjunction, motion, and liberation will occur 57 *pal* before the middle night in Ujjain, followed by 2 *ghadi* 31 *pal*, 4 *ghadi* 15 *pal*, and 7 *ghadi* 43 *pal*. To know what time the touch, conjunction, etc., will occur in any other place, one should add the longitude based on verses 63, 64 of the Chapter on Mean Motions if the place is east of Ujjain and subtract if the place is west of Ujjain. Doing so will determine the touch time, conjunction time, etc., according to the middle period of that place. If it is to be known how many *ghadi* and *pal* after sunrise the touch, conjunction, etc., will occur in that place, then the true time must be extracted by performing the calculation of the time equation in the mean period and also performing the calculation of the variable time by taking the declination of the sun for that day.

The time-equation of this day—

Sūrya-mean portion = approximately 2240' ∴ Sūrya-true portion = 14240'

Therefore, the time equation can be easily derived according to the formula (8) of page 347 of the Chapter on the Three Problems or the formula (a) of page 350. According to formula (a),

Time - equation = 115.165sin (1420' 40'' + 78° 24') - 147.665sin (2 X 1420' 40'') = 115.2sin (1800' + 410' 4'') - 148sin 185° 20'

SOURCE PAGE 517

= − 115.2 sin41° 4' — 148 sin (360° — 74° 40') = − 115.2 X.657 + 148 X sin74° 40' = − 115.2 X.657 + 148 X.6644 = − 75.7 + 142.7 = 67asu = + 11pala

The sign of wealth reveals that adding 11 minutes in any true time of this day gives the mean time. Therefore, when the night will be 12 o'clock according to the sun, it will be 12 o'clock plus 11 minutes in the mean clock.

Sunrise of the Sun -

The true portion of the Sun at midnight is 3 ra 28°30‘7 or 3 ra 28°31’. But the mean morning time is considered to be 6 o'clock, so the midnight is 45 clocks after the mean morning time. The true daily motion of the Sun is 57‘5. Therefore, in 45 clocks, its motion is approximately 43’. Thus, the portion of the Sun at sunrise is 3 ra 28°31 - 43’ = 3 ra 27°48’. Adding the precession angle of 220°40’ to this gives 4 ra 20°28’. This is the true sine portion of the Sun. Therefore, according to page 306

sin of sunrise of the Sun = sin 4 ra 20°28’ X.3676 = sin 36°32’ X.3676 =.6336 X.3676 =.2521

∴ True sunrise of the Sun = 14°36’

The latitude of Kashi is 25^°20'. Therefore, the sine of the rising sun in Kashi on this day = 14^°36' × 25^°20' = .2605 × .4734 = .1233 therefore the declination = 7^°5' therefore the time of rising = 71 minutes, which is positive because the declination is north.

SOURCE PAGE 518

The difference in longitude from Ujjain to Kashi is 1 hour 13 minutes 50 seconds (see page 66) or 1 hour 13 minutes.

Adding the longitude of Kashi to the time of touch (Sparsha Kaal) of Ujjain at the time of touch of Kashi will show the eclipse at 16 seconds from the middle night of Kashi, which is -57 seconds + 1 hour 13 minutes, i.e., 16 seconds in Kashi.

However, the true middle night is 11 seconds above the middle night. Therefore, from the true middle night, 16 seconds - 11 seconds = 5 seconds, the eclipse will be seen in Kashi after 5 seconds.

The variable time of this day + 71 seconds = + 1 hour 11 minutes. Therefore, from sunrise, it will be 6 o'clock in the sun clock at 1 hour 11 minutes.

Therefore, until 6 o'clock in the morning from sunrise

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Although the time-samīkaraṇa is positive, it has been subtracted here for the reason that when the simple tense is known, adding the positive tense equation results in the middle tense, but when the middle tense is known and the simple tense is to be known, the positive tense equation has to be subtracted from the middle tense because the simple tense is less than the middle tense.

SOURCE PAGE 519

That is, the commencement of the total eclipse will be at 46 hours 44 minutes and 44 equinoctial shadows from sunrise in Kashi and the full lunar eclipse will become dark.

By adding double the meridian time in the time of touch, and double the meridian in the time of conjunction, the time of salvation and the time of conjunction will be known.

Meridian = 4 hours 20 minutes

∴ Eclipse position = 8 hours 40 minutes in the time of touch 46 hours 16 minutes, from sunrise ∴ Time of salvation 54 hours 56 minutes "

Time of conjunction = 54 minutes

∴ Eclipse position = 1 hour 44 minutes conjunction - time from sunrise 4644 on ∴ Time of revolution from sunrise 5128

Essence of all

According to the above calculation according to the letters of Bapudev Shastri

Gpa pal gpa pal sparsha 46 16 46 8 sammilan 66 44 48 36 madhya 50 36 50 40 unmilan 51 28 52 43 moksha 54 56 55 11

The times given in the letters of Bapudev Shastri in regard to the eclipse period are very much in agreement with the Nautical almanac. Therefore, these times are absolutely pure. The times calculated according to Sūrya Siddhānta are very different from these. Therefore, now it is to see what is the reason for this difference?

When the fundamental angles of the eclipse according to Sūrya Siddhānta are compared with the angles calculated from Jyot-Ganita, it appears that the portions of the sun and the moon according to both methods are almost the same, but there is a big difference in the portions of Rahu, due to which there is a great difference in the shadow of the moon. Therefore, it is necessary to know what happens if the true portion of Rahu is known by a new method and the shadow of the moon is known from this shadow and the calculation of the lunar eclipse is done.

SOURCE PAGE 520

According to Jyotirganita, the portion of Rahu is 12004.5. According to this text, the precession angle for this year is 22047. The Aries solstice of 1676 years according to Sūrya Siddhānta, at which time the precession angle was 22037.381 (see page 252). Increase in precession in two years

= 58 . 663344 X 2 + 000111 X 32 = 117 . 3267 + 001 = 117 . 3277 = 1' 57 . 33

124 days after the Aries solstice of 1681 years, Shravan became a full moon, so the increase in precession in 124 days will be 16.63. Therefore, Shravani Purnima

kedinayanaansh = 220 ° 37 ′ 38 ″ · 1 + 1 ′ 57 ″ · 3 + 16 ″ · 6 = 220 ° 36 ′ 55 ″ = 220 ° 40 ′ difference

It is revealed that the Aries point of Jyotirganita was 7’ ahead of the Aries point of Sūrya Siddhānta this year. Therefore, when the portion of Rahu according to Jyotirganita is 12004.5, the portion of true Rahu from the Aries point of Sūrya Siddhānta will be 120011.5. By knowing the lunar eclipse time from the shadow of the moon from this portion of Rahu, it will be known how much the eclipse time calculated according to Sūrya Siddhānta differs from reality by only modifying the motion of Rahu.

Sūrya Siddhānta according to

According to the observation or astronomical calculation, the true position of the moon is 506' not 4030' as written in the Sūrya Siddhānta. Therefore, the true position of the moon is

sin506' X sin178° 22'5 = 3438 sin506' X sin1037'5 = 3438 = 306 X 67.53438 = 8.76

SOURCE PAGE 521

Therefore, the lunar instantaneous position was also 8.76. By keeping this value of the lunar position along with the segments of the meridian and the meridian arc, the calculation is done before, as before.

Garkaparamana = 60' · 63 - 8' · 76 = 51' · 87

Position = 60 hours × ((60.63 + 8.76) (60.63 - 8.76)) 765.5 = 60 × \(66.36 × 51.87 \) 765.5 hours = 60 × 56.66765.5 hours = 4 hours 42 minutes
Time of eclipse position = 6 hours 24 minutes

Ymardardha = 60 × \((27.31 + 8.76) (27.31 - 8.76) \) 765.5 = 60 × \(36.07 × 18.55 \) 765.5 ghadi = 60 × 25.867765.5 ghadi = 2 ghadi1.65pal

Therefore the time of the inundation or full eclipse = approximately 4 hours 3 minutes.

h. full moon time = 323mean night position - 442touch time - 116

The minus sign indicates that 1 hour 16 minutes is before midnight. The previously written rituals are summarized as follows:-

Touch time Kashi country latitude -16 +13

SOURCE PAGE 522

Time equation - 011 variable time + 11 medium time from medium midnight till + 45 04554 will be. Full moon time after 323 midnight - 21.7 conjunction time 121.3 after midnight + 13 degrees of latitude of Kashi - 011 variable time + 11 morning time till midnight) will be at 45 04834.3. Commencement of the zenith of the sunrise of Kashi) 43.4

At Kāśī, unmīlana will occur at 52 ghaṭīs 33 palas 07 vipalas after sunrise (OCR notation: 52 33 07).

The eclipse will be at 52 3307 from the sunrise of Kashi

In this way, it is proven that using the actual share of Rahu instead of the Rahu's share of the Surya-siddhanta brings the beginning and end of the sarvagrāsa grahaṇa very close to reality. From all these things, it is apparent that the Bhagan period of Rahu in the Surya-siddhanta is very impure. Compared to this, according to Brahmagupta and Bhaskaracharya, the calculation of Rahu is very pure and very close to reality.

Thus it is proven that using the actual share of Rahu instead of the share of Rahu in Sūrya Siddhānta brings the commencement and end of the total eclipse very close to reality. From all these things it is apparent that the period of revolution of Rahu in Sūrya Siddhānta is very inaccurate. According to Brahmagupta and Bhaskaraacharya, the calculation of Rahu is much purer and very close to reality.

SOURCE PAGE 523

To know this, modifications should be included in your Siddhanta texts which are proven by trigonometry. To provide complete information about such modifications, it is necessary that we have an observatory through which the study of modern astrology can be done easily. In my understanding, it is unrighteous that we depend on the nautical calendars made in Western countries for calculating eclipses, lunar occultations, planetary risings, planetary settings, etc., in our Panchangs, but do not manage the independent study of their principles.

Now in brief, it will be shown what are the fundamental measures according to the astronomical calculations for this eclipse.

Full-term Fundamental Measures

The true Sun 118°56'·23 true Moon 268°56'·23 Sun's daily motion 57'·6 Moon's daily motion 14°8'·1 Rahu 120°4'·5 Sun's length 0'·15 Moon's ultimate length 56'·3 Moon's image 32'·4 Earth's image 87'·0 Lunar segment 56'·6 Inter-segment 27'·2 Precession angle 220°47' Moon's north-east 6'·6 Sun's daily motion in the morning 14°31'·3

This is stated before that why there is a difference in the precession angles. Due to this difference, there will be a difference of 7' in the portions of the Sun, Rahu, and Moon. If the calculation of an eclipse is made from these fundamental measures, there will be a difference of 2 or 3 minutes from the time given in the nautical Panchang. The reason for this is that only the daily motions of the Sun and Moon have been taken, whereas for minute calculations, one should know the motion of each hour, the touch, conjunction, and declination times.

SOURCE PAGE 524

In the same way, the calculation of the shadow of the moon should also be done as explained in Figure 68. Doing this will make the calculation very extensive, so it is not shown.

Calculation of the fine divisions of the time of touch and the time of salvation-

To find the fine divisions, it is necessary to know the declination and the arc of motion. For the declination, the instantaneous revolution and the sidereal time of the moon should be known.

Calculation of the declination —

As stated above, the position is 4 hours 42 minutes in which the moon moves 104°28” or 104.5’ because the daily motion of the moon is 823’.

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SOURCE PAGE 525

> TouchtimeLunarFraction = 267° 26' 5precession angle = 22040' TouchtimeLunarPositionFraction = 32006' 5SimilarlyLunarPositionFraction = 322018' 5TouchtimeLunarMeanMotionSine = sin23° 27' sin32006' 5 =.3676 X.6405 =.2546 ∴ MeanMotion = 14° 46' SouthLunarShadow = 0° 14' 6North

Spashtikalachandrasāyanabhogaṃśa = 32006' 5

SOURCE:
, clear revolution of the moon of the touch-time = 14° 31’ 4 South

In this way, moksha time lunar pleasure portion = 322018' 5

This is according to shloka 58. If pure calculation is to be done, one should follow the method mentioned in page 202, which is left here out of fear of expansion.

> Sūrya Siddhānta = 14 ° 5 ′ Southward = 3 ′ North ∴, Chandra Spashta Kranti = 14 ° 2 ′ South

Moon of the touch-time clear revolution = 14° 31’ 4 South

Calculation of the night-time—

To know the time of the moon, one must first know the time of the sun. In the meridian time, the sun remains on the meridian circle, and in the midnight time as well, it remains on the meridian circle below the horizon, therefore the center of the Earth's shadow remains on the meridian circle in the midnight time because the center of the Earth's shadow remains at exactly 180 degrees from the sun.

Therefore, the time of the center of the Earth's shadow in the midnight time is zero. If it is known how much before or after midnight the time of touch is, then it is easily known how far east or west the time of the center of the Earth's shadow is in the time of touch. During a lunar eclipse, the moon is very close to the Earth's shadow, and it is known that the moon is at what distance from the Earth's shadow, therefore the time of the moon can be easily known.

Time of sunrise of the sun = 111
Time from mean morning to mean night = 450

SOURCE PAGE 526

From sunrise to midnight = 46 11 time from sunrise to touch = 45 54 time before midnight = 017

Therefore, according to what has been said above, the center of the Earth's shadow at the time of touch will be at the meridian circle after 17 minutes = 102 arc minutes, that is, after the beginning of touch, the center of the Earth's shadow will come onto the meridian circle after 17 minutes.

The time of salvation from sunrise to the hour of the sun = 55 18,, time of midnight = 46 11 time after midnight to salvation = 6 7 Therefore, the arc of the center of the Earth's shadow at the time of salvation is 6 hours 7 minutes or 547 minutes or 3282 arc minutes.

Now it is to see what is the portion of the Earth's center at the time of touch? The portion of the sun at the time of full moon = 118°34’ in the position of sthiryarth 4 hours 42 minutes the motion of the sun = 4’.5 ∴ The portion of the sun at the time of touch = 118°26’.5 The difference between the Earth's center and the sun = 180°00 ∴ The portion of the Earth's center at the time of touch = 268°26’.5 The portion of the moon = 267°26’.5 ∴ The difference between the Earth's center and the moon at the time of touch = 10

To convert this to sidereal time use shlokas 56 of spashtadhikar. The moon is in the Aquarius sign, from which the rise of Lanka is 1764 1 (see page 314). Therefore 10 rise of 1764 ÷ 30 = 56.8 = 60 The anterior arc of the center of the Earth at the time of touch = 102 arc minutes The difference between the Earth's center and the moon towards the west = 60 arc minutes The arc of the moon at the time of touch = 42 arc minutes = 42’ The portion of the sun at the full moon = 118°34’.

1. To know the sidereal time, one should use the rising signs of Lanka because the consideration of the mean ascendant is done from the rising signs of Lanka, see pages 326-330.

SOURCE PAGE 527

Sūrya Siddhānta, Scientific Commentary, Chapter on Mean Motions, Chapter on True Positions, Chapter on the Three Problems.
Position (1)/(4) hour 42 minutes of the sun's motion = 4^(°)05' therefore Portion of the immortal sun = 118^(°)38'5" Difference between the center of the Earth and the center of the sun = 180^(°) therefore Portion of the immortal center of the Earth = 268^(°)38'5" Portion of the moon = 266^(°)38'5" "Portion of the moon" 10 "Difference between the center of the Earth and the moon to the east" Immortal center of the Earth's west time = 3282 asu Difference between the center of the Earth and the moon to the east = 60 asu therefore Immortal time of the moon = 3222 asu = 3222' = 53^(°)42' Sine of the tangent of the moon = 25^(°)20' X Sine 14^(°)31'4 = .4734 X .2560 = .1226.. Tangential arc = 7^(°)2' According to the equations (g) of page 262, the arc of the meridian = (cosine 42' - sine 7^(°)2') cosine 25^(°)20' cosine 14^(°)31'4 = (.6666 -.1226) X .6038 X .6806 = .8773 X .6038 X .668 = .7675

therefore Tangential arc = 36^(°)53' According to page 276, sine agra = sine 14^(°)31'4 sine 36^(°)53' X cosine 25^(°)20' + cosine 36^(°)53' X sine 25^(°)20' = .2508 /.6412 X .6038 + 1.1667 X .4734 = .4328 +.5665

SOURCE PAGE 528

= 06663 ∴ agra = 87 ° 54 ′ ∴ purvavindusechandramakadigansh = 87 ° 54 ′ dakshinauruttarvinduse,, = 60 ° + 87 ° 54 ′ = 177 ° 54 ′ dakshin ∴ spre (khug) = agrakotijya X natanshsparshrekha = cojya87 ° 54 ′ X spre36 ° 53 ′ = 0366 X 08356 = 0306samprotvrittakantaansh = 1 ° 45 ′ jya (akshavalan) = jya25 ° 20 ′ jya1045 ′ / cojya14 ° 31 ′ = 0131 / 0668 = 0135 ∴ akshavalan = 0 ° 46 ′ uttarsparshkalinchandramakabhogansh267 ° 26 ′ 5aynansh220 ° 40 ′ sparshkalinchandramakasaynbhogansh320 ° 6 ′ 5

Adding 600 to this results in the sine of the moon's declination being 5006.5, whose transit will be north. Therefore, the precession will also be north.

∴ According to the formula (2) of page 480

jya (akshavalan) = sin230 ◦ 27 ′ × cos320 ◦ 6 ′.5cos315 ◦ 31 ′.5 =.3676 ×.7676cos315 ◦ 31 ′.5 =.3157 ∴ akshavalan =18 ◦24 ′ north

∴ sparshkalinsphutvalan = 18 ◦24 ′ + 0 ° 46 ′ = 16 ° 10 ′ north

SOURCE PAGE 529

Lunar_liberation_arc_sine = 25° 20' 14° 2' =.4734 X.2466 =.1183 ∴ Liberation_arc = 6° 48' ∴ Liberation_tangent_sine = (cos53° 42' - sin6° 48') cos25° 20' cos142' = (.5620 - .1183) X.6038 X.6702 =.4737 X.6038 X.6702 =.4154 ∴ Liberation_tangent = 65° 27' sinagra = sin14° 2' / sin65° 27' cos25° 20' + cos65° 27' X 25° 22' =.2425 /.6066 X.6038 +.4568 X.4734 =.2650 +.2163 =.5113 ∴ Agrad = 30° 45' ∴ Western_point_Moon_liberation_direction = 300° 45' South ∴ Sphere (Khug) = Agrad_tangent X Tangent_line = cos30° 45' sin65° 27' =.8565 X 2.1863 = 1.8817 ∴ Prime_vertical_circle_tangent = 620' 1' ∴ sin (inclination) = sin25° 20' sin620' 1' / cos14.2' =.4276 X.8830 /.6702 =.3778 /.6702 =.3864 ∴ Inclination = 220° 55' South

SOURCE PAGE 530

Fraction of the Moon during Moksha time = 266° 38’ 5

Precession angle = 22040 ′

Sine fraction of the Moon = 322018‘5

Adding 60 to this will result in a sine fraction of 520-18’ 5 which is the true north.

Therefore the declination will be north.

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Chapter on Mean Motions

Chapter on True Positions

Chapter on the Three Problems

The study of the fourth right called Lunar Eclipse

SOURCE PAGE 531

Chapter on Solar Eclipse Authority (Brief Description)

Verse 1—When are the perpendicular and declination of the sun zero. Verse 2-8—Rules for knowing the perpendicular. Verse 6—Determining the time of Amavasya by giving the correction of the perpendicular and by inaction. Verse 18-11—Rules for knowing the declination. Verse 12—The declination-season is known by the sum or difference of the declination and the moon's season. Verse 13—From the declination-season, the position, etc., should be known. Verse 14-17—Knowing the perpendicular of the touch and the time of moksha, the calculation of touch and time of moksha should be done by inaction.

When is there absence of perpendicular and declination—

> मध्यलग्नसमे भानौ हरिजस्य न संभवः ।
> अक्षोदङ् मध्यभक्रान्तिसाम्ये नावनतेरपि ॥१॥

Translation—(1) When the sun is on the tribhone ascendant, there is absence of the portion of the perpendicular. When the north of a place is equal to the latitude and the north declination, that is, when the sun is on the x-cross, there is absence of declination, i.e., season-perpendicular.

Scientific Commentary—The translation of the chapter on mean motions in this verse is the tribhone ascendant, although it is stated on page 326 that the mean ascendant is different from the vitrim ascendant or tribhone ascendant. However, here the Acharya has written tribhone as mean because it is between the rising and setting signs, although the same word is ambiguous in two meanings. If the meaning of mean ascendant taken is the one considered in verse 48 of the Chapter on the Three Problems, then the feeling is impure, so here the meaning of mean-ascendant is tribhone ascendant. If the sun or any planet is on the tribhone ascendant, the portion of the perpendicular is zero. Its proof is explained in detail in the Chapter on the Three Problems (see p. 410). It has also been stated regarding the season-perpendicular.

> देशकालविशेषेण यथाऽवनतिसंभवः ।
> लम्बनस्यापि पूर्वान्यदिग्वशाच्च तथोच्यते ॥२॥

SOURCE PAGE 532

(2) the state of the country which is different from the condition mentioned in the first verse according to the time and when the sun is before or to the west of the zodiac sign. The discussion here is about the portion of the arc and the colatitude that is enjoyed in it.

लग्नं पर्वान्तनाडीभिः कुर्यात्स्वैदयासुभिः ।
तज्ज्याऽन्त्यापक्रमज्याघ्ना लम्बज्याप्तोदयाभिधा ॥३॥
तथा लङ्कोदयैर्लग्नं मध्यसंज्ञं यथोदितम् ।
तत्कान्त्यक्षांशसंयोगो दिवसाम्येऽन्तरमन्यथा ॥४॥
शेषं नतांशास्तन्मौर्वी मध्यज्या साऽभिधीयते ।
मध्योदयज्याऽभ्यस्ता त्रिज्याप्ता वर्गितं फलम् ॥५॥
मध्यज्यावर्गतः प्रोज्ज्व्य दक्क्षेपः शेषतः पदम् ।
तल्लिज्यावर्गविश्लेषान्मूलं शङ्कुस्स दग्गतिः ॥६॥

(3) The portion of the time after the new moon is obtained by knowing the ascensions of the auspicious places (of the zodiac) and multiplying its sine by the sine of the ultimate revolution, and then dividing the product by the sine of the latitude or the latitude-longitude of the auspicious place, the resulting quantity is called the rising or rising sine. (4) In the time after the new moon, according to what was said before from the ascensions of the zodiac, if the mean ascendant or tenth ascendant is known, add its declination to the latitude of the auspicious place if their directions are the same. If the directions are different, find the difference between the declination and the latitude. (5) Whatever is obtained by adding or subtracting is the ntasha of the mean ascendant. The sine of this is called the mean sine. Multiply the mean sine by the rising sine and divide the product by the radius and square the result. (6) Subtract the year of the mean sine from the square and find the square root of the remainder. Whatever is known to find the square root is called the arc. Subtract the square of the arc from the square of the radius and find the square root of the result, whatever is obtained is the gnomon or the arc.

Scientific Commentary—the derivation of the action shown in these four shlokas is well explained in Chapter on the Three Problems pages 411-413. From the picture 76 of the same place it appears when to add and when to subtract the revolution of the mean ascendant and the latitude of the auspicious place. The method of knowing their directions is the same as explained in pages 265-266.

यहाँ लग्न का अर्थ सायन लग्न समझना चाहिये ।
नतांशबाहुकोटिज्ये स्फुटे दक्क्षेपदग्गति ।
एकज्यावर्गतश्छेदो लब्धं दग्गति जीवया ॥७॥

SOURCE PAGE 533

मध्यलग्नाकविलेषज्या छेदेन विभाजिता ।
रवोन्द्वोलंम्बनं ज्ञेयं प्राक्पश्चाद्घटिकादिकम् ॥ ८॥

Translation—(7) The sine of the nataṃśa of the tenth sign in a gross form can be called the sight and the co-latitude. What results when the square of the sine of a quantity is divided by the figures of the sight is called the shadow. (8) The distance of the sun from the three-quarters sign is called the analysis or analysis part. Dividing its sine by the shadow gives the east or west altitude of the sun or the moon in clocks. If the sun is east of the three-quarters sign, then the east altitude is east and if it is west, then the west altitude.

Scientific Commentary—The pure forms of sight and co-latitude are the same as stated in the 6th verse. However, the way to know them is long, so a shorter method is described in the first half of the 7th verse, which is gross. In this shorter method, the nataṃśa of the mean sign is taken as the nataṃśa of

the three-quarters sign because there is very little difference between these two. But this inevitably brings grossness.

> छेद और विश्लेषांश से सूर्य और चन्द्रमा का भोगांश लंबन आनन की जो रीति यहाँ बतलायी गयी है उसकी उपपत्ति त्रिप्रश्नाधिकार पृष्ठ ४१३ और पृष्ठ ४०५-४०६ में बतलायी गई है ।

> मध्यलग्नाधिके भानौ तिथ्यन्तात्प्रविशोधयेत् ।
> धनमूनेऽसकृत्कर्मं यावत्सर्वं स्थिरीभवेत् ॥६॥

Translation—(6) If the portion of the sun from the three-quarters sign is more, then the sun is east of the three-quarters sign, so the difference between the portions of the sun and the moon altitudes should be subtracted from the end of the new moon. But if the portion of the sun from the three-quarters sign is less, then the difference between the portions of the sun and the moon altitudes should be added at the end of the new moon. The time that comes when the difference of the portions of the altitudes is processed in this way with the end of the new moon is the time of the portion altitude- Sanskrit-end of the new moon. In this time, the difference between the altitudes of the sun and the moon is calculated again by the aforementioned method and the above altitudes are added, subtracted. The time that comes from this is then calculated as an altitude and it is also processed. Such an action should be done until the time becomes stable, that is, when the end of the new moon comes again after processing the altitudes repeatedly, then this work should be stopped. By doing this, the middle time of the eclipse is known.

SOURCE PAGE 534

Scientific Commentary—the calculation becomes purer from non-action as previously stated (see page 266). At the time when the moon, while moving towards the east, becomes equal to the portion of the sun, at that time the end of the new moon occurs. This is called the mathematical proven new moon end. When the sun is to the east of the three-fold ascendant, the moon, due to declination, hangs to the east and becomes in front of the sun before the new moon. Therefore, as much as the relative declination of both, the end of the true new moon occurs earlier. For this reason, in the first half of the shloka, it is said to subtract the declination from the mathematically proven time. Conversely, when the sun is to the west of the three-fold ascendant, the moon does not appear in front of the sun even in the mathematically proven new moon time because, due to declination, it hangs a little lower towards the west. Therefore, as much declination occurs at this time, the true new moon time is behind. That is why adding this declination results in the true new moon.

> दृक्क्षेपश्शीततिग्मांश्वोर्मध्यभुक्त्यन्तराहतः ।
> तिथिघ्नत्रिज्यया भक्तो लब्धं साऽवनतिर्भवेत् ।।१०।।
> दृक्क्षेपात्सप्ततिहृताद् भवेद्वाऽवनतिः फलम् ।
> अथवा त्रिज्यया भक्ता सप्तसप्तकसङ्गुणा ॥११॥
> मध्यज्यादिग्वशात्तस्या दिग्ज्ञेया दक्षिणोत्तरा ।
> दिक्साम्ये सेन्दुविक्षेपयुक्ता विश्लेषितान्यथा ॥१२॥

(10) Multiply the difference in the mean daily motions of the Moon and the Sun by the radius fifteen and divide the product by the radius. Doing so, the resulting quantity is the latitude or declination or

latitude. (11) Or, if the declination is divided by the north, the resulting quantity is the declination, or if the declination is multiplied by 46 and the product is divided by the radius, the resulting quantity is the declination. (12) The declination is in the north or south direction according to the direction of the mean sine. That is, if the direction of the mean sine is south of the cross, then the direction of the declination will also be south, and if the direction of the mean sine is north of the cross, then the direction of the declination will be north. If the directions of the Moon's arc and the declination are the same, they should be added, and if they are different, they should be subtracted. Doing so, the resulting quantity is the declination of the Moon, the arc, or the deviation.

The essence of the ten and eleven shlokas of Vigyanbhashya is this.

```
n = ( lunar daily motion - solar daily motion ) x arc / 15 x radius
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SOURCE PAGE 535

```
Dṛkṣe = dṛkṣe X 46 / radius
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Here it should be noted that the radius is equal to 3438 art. If the versed sine, that is the netaansh of the three-fold inclination, is written in the Indian manner, then division by the radius will be necessary, but if the value of the versed sine is in decimal fraction according to present practice, then the value of the radius will become 1 instead of 3438.

The value of nti in page 407 of Chapter on the Three Problems has been proven—

If bh=li sine tva cojya sh—li cojya tva jya sh cojya va

Here bh, li, tva are taken for the altitude for shralamban, the ultimate altitude for the planet, the netaansh of the three-fold inclination, and sh and va for the shadow of the planet and the precession angle respectively.

At the time of solar eclipse, the shadow or sh is very small, if it is taken as very small, then sine sh and cojya sh must be taken as 1 according to present practice. In such a case

```
bu = latitude
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It will be. That is, what comes from multiplying the sine of the nataṃśa of the three-fold ascensions by the sine of the declination is the nati. This is explained in verse 10. Knowledge of the natis of both the sun and the moon at the time of solar eclipse is necessary because the relative nati of the moon is the same as the difference between these natis, so it is said to multiply the declination by the difference in the motions of the sun and the moon. It is stated on page 406 that the ultimate latitude of a planet is one-twentieth of its daily motion, so the relative ultimate latitude of the moon is considered to be one-twentieth of the motions of the sun and the moon. Thus, the derivation of the tenth verse is proven. Now it is clear that the nati that arises is gross. To know the value of the nati with purity, one should use the formula proven on page 407.

In the 11th verse, the other ways of knowing the motion are two forms of the first way. The difference between the mean daily motions of the Moon and the Sun = 760.6 - 56’. 1 = 731.5. Therefore, its 15th part = 48.77 = 46 in gross form. If this value is raised in the first formula then

SOURCE PAGE 536

n = 46 × arcsec = 46 × arcsec3438 = arcsec3438 = 46arcsec = arcsec70468 = arcsec70 and position measurement proof

> तथा स्थितिविमर्दर्धग्रासाद्यं च यथोदितम् ।
> प्रमाणं वलनाभीष्टग्रासादि हिमरश्मिवत् ॥१३॥

Translation—(13) According to the method explained in the chapter on Sun eclipse regarding the position, the apparent position, the arc, the proof, the arc of the desired, etc., that is, the combination, the inclination, the time of eclipse, etc., should be known. The positions and apparent positions that will arise from this are called the mean positions, apparent positions.

Scientific Commentary—After the action of inclination and nti, the calculation of solar eclipse is done in the same way as the calculation of lunar eclipse has been explained. Because just as in lunar eclipse the Earth's shadow is the obscurer and the Moon is the obscured, so in solar eclipse the Moon is the obscurer and the Sun is the obscured. The relationship between obscurer and obscured is the same in lunar eclipse too.

> स्थित्यर्थोनाधिकात्प्राग्वत्तिथ्यन्ताल्लम्बनं पुनः ।

> स्थित्यधोनाधिकात्प्राग्वत्तिथ्यन्ताल्लम्बिन पुनः ।
> ग्रासमोक्षोद्भवं साध्यं तन्मध्यहरिजान्तरम् ॥१४॥
> प्राक्कपालेऽधिकं मध्याद् भवेत् प्राग्ग्रहणं यदि ।
> मौक्षिकं लम्बनं हीन पश्चार्धे तु विपर्ययात् ॥१५॥
> तदा मोक्षस्थितिदले देयं प्राग्ग्रहणे तथा ।
> हरिजान्तरजं शोध्यं यत्रैतत्स्याद्विपर्ययात् ॥१६॥
> एतदुक्तं कपालैक्ये दिग्भेदे लम्बनैकता;
> स्वे स्वे स्थितिदले योज्या विमर्दर्धेषि चोक्तवत् ॥१७॥

Translation—(14) According to verse 6, subtract the position that comes in the time of Amavasya due to inaction and add the time of eclipse according to the 13th verse. Then, knowing the portions of the time of eclipse and the time of release, find the difference from the portion of inclination of the middle time of eclipse.

SOURCE PAGE 537

(15) If the eclipse is in the anterior skull, that is, if the portion of the sun during the eclipse time is more than the portion of the three-fold ascendant, then the declination of the touch time will be more than the declination of the mean time and the declination of the liberation time will be less than the declination of the mean time. But if the eclipse is in the posterior skull, that is, if the portion of the sun during the eclipse time is less than the portion of the three-fold ascendant, then the magnitude of the declination will be in reverse order, that is, the declination of the touch time will be less than the declination of the mean time and the declination of the liberation time will be more than the declination of the mean time.

(16) In both situations, whether the touch and moksha are in the anterior skull or in the posterior skull, according to the 14th verse, the true position should be known by adding the difference of the latitudes obtained by the touch-position and the moksha-position to the mean position. However, if there is a condition contrary to the rule stated in the 15th verse, that is, if the touch-position latitude in the anterior skull is less than the mean latitude and the moksha-position latitude is greater than the mean latitude, or if the touch-position latitude in the posterior skull is greater than the mean latitude and the moksha-position latitude is less than the mean latitude, then the difference obtained according to the 14th verse should be subtracted from the touch or moksha position latitude to obtain the true position latitude.

(17) When the tangents, mean, and motion are all in the same skull, then the difference of the above latitudes should be taken and the above action should be performed. If the tangent is in one skull and the mean in another skull, or the mean is in one skull and the motion in another skull, then the latitudes of the tangent and the mean, or the latitudes of the mean and the motion, should be joined with their respective positions. Similarly, the true position and the apparent position should also be known.

Scientific Commentary—it is clear that according to the 6th verse, the inclinations of the Sun and the Moon that occur in the time of Amavasya-antakaal or the middle of the eclipse are different from the inclinations of the time of touch and the time of liberation because the parts of the Sun, Earth, and Moon's continuous motion at the time of touch and liberation are different, and it is shown in the Chapter on the Three Problems that the inclination depends on the nataansh, that is, if the nataansh is more, the inclination is also more, and if the nataansh is less, the inclination is also less (page 383). Therefore, in the 14th verse, the need to know the inclinations of the time of touch and the time of liberation is indicated, and the difference in inclinations from the middle time is shown. If both touch and liberation are in the same protection, that is, the eclipse is the touch and liberation before the rising of the trilobhan lagna, then

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SOURCE PAGE 538

It is clear that at the time of touch, the portion of the Sun or Moon will be greater than the portion of the Sun or Moon in the mean time and less in the time of salvation because the ecliptic is the highest point of the declination above the horizon, and the Sun and Moon rise respectively, meaning their portion decreases, so the portion of the time of touch is greater than the portion of the mean time, and the portion of the time of salvation is less than the portion of the mean time. But if both touch and salvation are in the western hemisphere, then at the time of touch, the portion of the Sun will be less than the mean time portion, and the portion in the time of salvation will be greater than the mean time portion because the Sun or Moon descend in the western hemisphere, so their portion increases.

Adding the difference between the time of touch and the mean time of the latitudes to the mean position of the anterior hemisphere gives the true position of touch because the position calculated first is according to the latitude of the mean time, but the latitude in the time of touch is some more, so due to this, the Moon hangs a little lower, i.e., towards the east, so the touch is seen a little earlier, meaning the value of the position increases. But in the time of salvation (being in the anterior hemisphere), the latitude remains less compared to the mean time, so adding the difference between these two also gives the true position of salvation because when the latitude in the time of salvation is less, the Moon does not hang as much towards the east as in the mean time, so it remains for a longer time in front of the Sun and the position of salvation also increases.

Due to the obliquity in the western skull, the moon hangs towards the west, which causes a some delay in it coming in front of the sun because the motion of the moon is always towards the east and east, and due to the obliquity, it seems as if it is also going towards the west. For this reason, the time of eclipse is some behind the time of the mathematical amāvasyānta time. However, the lunar portion of the touch time is less than the mean time because the eclipse occurs some time after the touch time, and in this time, due to the daily motion of the earth or the flow of air due to the ancient view, the sun and moon all appear below. Therefore, the touch time obliquity in the western skull is less than the mean obliquity due to the lesser portion of the lunar portion, which has the effect that the eclipse does not take as much delay in touching as it does in the mean time, that is, due to the lesser obliquity at the time of touch.

SOURCE PAGE 539

The true position of the apparent touch increases. In the same way, because the portion of the moon at the time of salvation is more than the mean portion, the salvation-time declination is more than the mean declination. This has the effect that the moon remains in front of the sun for a longer time because at the time of salvation the moon appears above the sun, but due to the greater declination, it does not go up but remains hanging down, which causes some delay even in the time of salvation, that is, the true salvation position also increases.

Thus it is proven that whether the touch and moksha are both in the anterior skull or both in the posterior skull, in every case the time of eclipse increases, that is, the time of touch is some earlier and the time of moksha is some later. Therefore, by adding the difference in the perpendiculars of the time of touch and the mean time and the time of mean and moksha to the mean, the true position is found. By subtracting the time of the touch-time true position from the mean time of the eclipse and by adding the time of the moksha-time true position to the mean time of the eclipse, the true time of moksha is obtained.

Up to this, the first half of the 16th verse has been explained. The latter part does not make sense because the condition given in it is against nature. In the anterior skull, the tangential inclination will always be greater than the mean inclination because the tangential part is always greater than the mean part, and by the same logic, the rectilinear inclination will always be less than the mean inclination. Similarly, in the posterior skull, the tangential inclination will always be greater than the mean inclination. Yes, there is another matter if there is a total solar eclipse or lunar eclipse. But in such a situation, calculations will have to be done in a special manner, and it will be necessary to know only for a place when the eclipse occurs in the total eclipse. In the first case, one will have to consider how much of the sun is eclipsed at the time of sunrise and how long this eclipsed part will take to pass out. For this action, the verses of the lunar eclipse chapter 18, 16, and 20 will have to be used. In the second case, that is, in a lunar eclipse, one will have to consider how much of the sun is eclipsed at the time of sunset and how many eclipses have touched it before. For this action as well, the same verses will have to be used.

What has been said so far is for the state where touch and liberation are in one art. If touch is in one skull and the mean in another skull or the mean in one skull and liberation in another skull then the signs of touch and mean time

SOURCE PAGE 540

Or by joining the latitudes of the middle and the time of salvation, what is obtained should be added to the mean position because in such a situation the eclipse period will increase greatly. Suppose the eclipse occurs in the anterior skull and the middle in the posterior skull. It is clear that in such a situation the middle period of the eclipse will be in the posterior latitude in a short time, that is, because the moon will move in front of the sun in a short time due to the moon moving away. But at the time of touch, the latitude of the moon will be towards the east, so the touch will happen a little earlier. From the first reason, the middle period of the eclipse will shift a little back, and from the second reason, the time of touch will be a little earlier, so the time from the touch to the middle period will increase due to both reasons. In such a situation, to find the true position of the time of touch, the sum of the latitudes of the touch and the middle period should be added to the mean position, but for the true position of the time of salvation, the difference of the two latitudes should be added to the mean position because both the middle period and the time of salvation will be in the posterior skull, only the time of touch will be in the anterior skull. But if the touch and the middle are in the anterior skull and the salvation is in the posterior skull, then the difference of the latitudes should be added to the mean position for the true position of the time of touch and the sum of the latitudes should be added to the mean position to know the true position of the time of salvation.

That which is explained here to know the time of touch and liberation should also be known in that same manner the times of combination and motion.

Example - Calculation of the solar eclipse of Magha Krishna Amavasya of the year 1682 of Kashi—

According to Sūrya Siddhānta -

First, the sun, moon, lunar altitude, and Rahu of this day should be clarified. Therefore, it is necessary to know the sidereal time from the beginning of Kali Yuga until this day.

From the beginning of Kali Yuga to Shravani Purnima of 1681, there are 183554705236 days. From the time of Shravani Purnima of 1681 to the time of Amavasya of Magha of 1682, there are 17 lunar months because there is no Malas in between. One lunar month is equal to 26.530588 Sawan days. Therefore

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1681KisharavaniPūrṇimāntaka183554705236days17LunarMonth = 502002daysHalfLunarMonth = 14.7653days
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∴ From Kali Yuga to Amavasya of Magha of 1682, there are 1836064·3086

∴ The sidereal time until midnight of this day = 1836064

SOURCE PAGE 541

The remainder is 6 when divided by 7. Therefore, the time of the observation is the midnight of Wednesday, as the beginning of Kali Yuga occurred at the midnight of Thursday. However, the true Maghi Amavasya of 1682 was on Thursday, so the above observation is the midnight of the Chaturdashi of Magh by the Purnimanta calculation and the midnight of the Chaturdashi of Pausha by the Amanta calculation. The end of this Amavasya occurred around the midday of Thursday. Therefore, the Moon, Sun, etc., of both the Chaturdashi and Amavasya should be clarified in the midnight period. Just as the positions of these planets are known in page 485, doing so here will also yield the mean position at the half-night of Magha Krishna 14 (if the full revolution is not written)—

```
Sun = 8R26A33.144Moon = 8 " 25 " 49.653Moon Altitude = 0 " 26 " 6.116Rahu = 2 " 23 " 30.386
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Three signs should be added to the position of the Moon and the position of Rahu should be subtracted by 6 signs (see page 486).

Therefore, in Ujjain on the midnight of Wednesday of Magha Krishna 14 of 1682 AD

SunMeanPosition = 8R26033 ′ 144Moon = 82546 ′.653MoonAltitude = 3266 ′ 116Rahu = 3626 ′ 611

SunApogee = SunAltitude - SunMeanPosition = 2R17° 17' · 52 - 8R26° 33' · 14 = 5R17° 44' · 38 = 1Pad + 2R17° 44' · 38

∴ secondpadakamyabhag = 12015'62 = 735'32

Sūrya Siddhānta - circumference = 840' - 20' × 735'.63438 = 840' - 20' × 7303438 = 840' - 4' = 836'

SOURCE PAGE 542

∴ Bhujphal = 836 × 73021600 = 28 ′ 254

This is the Sun-mean result. This is the center of the positive axis etc. (see page 154). Therefore, the true Sun at the mean night of Wednesday = 8^(°)26'33'' · 144 + 28' · 254 = 6^(°)1'368

= 826 ° 33 ′ · 144 + 28 ′ · 254 = 60 ° 1 ′ · 368Sūrya Siddhānta * = 56 ′ · 8 ″ + 836 × 216 × 56 ′ · 821600 × 225 = 56 ′ · 8 ″ + 2 ′ 13 ″ · 6 = 61 ′ 21 ″ · 6 = 61 ′ · 36Lunar apogee = Lunar apogee — Mean Moon = 26 ° 6 ′ · 116 − 25 ° 46 ′ · 653 = 70 ° 16 ′ · 463 = 2paad + 10 ° 16 ′ · 463 = 2paad + 30 ° 16 ′ · 463 ∴ Third paad past part = 30 ° 16 ′ · 46 = 1816 ′ · 5Lunar meridian apogee − circumference = 320 − 20 ′ X Bhujya 1816 ′ · 53438 = 320 − 20 ′ X 1735.553438 = 320 − 10 ′ 310 ′ 50 ′ = 1610 ′ ∴ Bhujphal = 1610 X 1735.5521600 = 153 ′ · 468 = 20 ° 33 ′ · 468

The true Moon is the mean Moon. It is negative because the center of the Moon is in Libra. Therefore, the true Moon at midnight on Wednesday

SOURCE PAGE 543

Sūrya Siddhānta - = 825046' 653 - 2033' 468' = 823016' 185

It is apparent from the true Sun and Moon that the Moon is about 7 degrees west of the Sun at midnight of Wednesday, therefore the new moon will be the next day. To know when the new moon will be, one should know the true motion of the Moon.

The mean Moon at the midnight of the fourteenth = 825046' 653

SOURCE PAGE 544

Sun's true daily motion = 61' 036difference between the daily motions of the Moon and the Sun = 13° 12' 0343 = 762' 0343

True Sun at midnight = 6ra01'36d

difference between the two = 8ra23°16'185=6° 45' 213=405' 213

The difference between the Sun and the Moon is 762' 343 in 60 hours, so the difference of 405' 213 will be in 405.213 X 60 hours, which is similar to 30 hours 41.1 minutes. Therefore, the end of Maghi Amavasya in Ujjain occurred 30 hours 41.1 minutes after the middle night of Wednesday or 15 hours 41.1 minutes after 6 AM on Thursday.

Kashi is 1 hour 12.8 minutes before Ujjain (see page 251). Therefore, the end of Amavasya in Kashi occurred at 16 hours 53.6 minutes after 6 AM on Thursday.

Now the apparent Sun, Moon, and Rahu should be clarified.

Sun's apparent motion in 60 hours = 61' 36

∴ 30,,,, = 30' 68

Sun's motion in 30 minutes = 30 '.68 =.5113

Sun's motion in 10 minutes =.17041,,,, =.0170.1,,,, =.0017

∴ Sun at the time of New Moon = 6.032' 778

60 in the moon's true motion = 14°13'70330,,,, = 7°6'851530 in the shadow = 7'114210,,,, = 2.37141,,,, = .23711,,,, = .023730 in 41.1 in the shadow = 7°16'568

SOURCE PAGE 545

Wednesday's mean night true moon = 8^(°)23'16''085
Instantaneous true moon = 6^(°)00'32''78360
Motion of Rahu in the hour = 3'11'' = 3'183330
In a minute = 026510,,,, = 000881,,,, = 0006
∴ 30 hours 41 minutes motion of Rahu = 1' · 628
Wednesday's mean night Rahu = 3^(°)6'26'' · 611
∴ Instantaneous Rahu = 3^(°)6'28''0
Sun's image angular diameter = 6500 × 61.3656.133 yojan ^1
Moon's image angular diameter = 480 × 853.703760.583 yojan
Moon's orbit angular diameter of Sun's image = 6500 × 61.3656.133 × 43200057753336 yojan ^1 = 6500 × 61.36 × 43200056.133 × 5775336 × 15
= 33' · 634
Moon's angular diameter in the arts = 480 × 853.503760.583 × 115
= 34' · 555

Time of sunrise in Kashi—

First it is necessary to know what was the revolution of the sun during the sunrise period in Kashi. It is apparent that the portion of the sun during the sunrise period is approximately 6°015' and thus the revolution is about 15' before the initial point of the sun in the Capricorn sign according to table 470, 21032.7 and some

SOURCE PAGE 546

It will be less and south. In Kashi, its arc will be approximately 10°47’ and the time will be 1 hour 4708 minutes. Therefore, the true sunrise in Kashi will be at 6 o'clock 1 hour 4708 minutes.

Time equation—to know the mean sine portion of this day, adding 15’ to the mean sun of midnight gives the morning mean declination portion of approximately 8°26’48’. If the precession angle 220°41’ is added to this, the mean sine portion is 8°22026=262026’.

herefore Time - equation = 23' · 17 sin 262026' + 112' · 83 cos 262026' - 148' sin 2 × 262026' = 23' · 17 (- sin 67031') + 112'.83 cos 67031' - 148 sin 584° 58' = - 23' · 17 ×.6240 + 112' · 83 ×.3824 - 148' (-0.7067) = 21.4 + 43.2 + 104.6 = 126.4 sin = + 21.1 minutes

Time of true sunrise in Kashi = 6 o'clock 1 hour 47.8 minutes. Time equation = + 21.1 minutes

herefore Mean time of sunrise in Kashi = 6 o'clock 2 hours 8.6 minutes but the mean time at the end of the new moon = 6 o'clock 16 hours 53.6 minutes. herefore Time from sunrise to the end of the new moon = 14 hours 45 minutes, meaning the end of the new moon in Kashi occurred at 14 hours 45 minutes from sunrise.

Now if the motion of the sun is decreased by 14 hours 45 minutes of the instantaneous sun of the new moon, then the true sunrise time of the sun will be known, from which the solar rising motion, etc., can be known accurately.

1. 584° 58’ = 360° + 180° + 44°58’ ∴ sin 584°58’ = sin(180° + 44°58’) = - sin 44°58’ = - 7067

SOURCE PAGE 547

The motion of the sun in 60 hours = 61'36 15 " = 15'34 and 15 shadow = 0'256 14 hours 45 shadow = 15'084 Amavasya-time true sun = 6°32'778 = 6°017'664 = approximately 6°017'7. The sine portion of the sun's rising time of Kashi = 6°22058'7 = 6°22056'. Sine 6°22056' X.3676 (see page 306). The sine of the sun's declination = ———————— 1 = — sine 67°1' X.3676 = —.6206 X.3676 = —.3663 ∴ South transit = 21°26' Kashi's rising sine = Spere 21°26' Spere 25°20' = .3636 X.4734 = 1863 arc = 10°44' transit time = 644 asu = 107'3 shadow = 1 hour 47'3 shadow Therefore, the true sunrise in Kashi occurred at 6 o'clock 1 hour 47'3 shadow. Time equation = 21'1" ∴ The mean time of sunrise in Kashi = 6 o'clock 2 hours 8'4 shadow But Amavasya-time = 6 o'clock 16 hours 53'6 shadow ∴ Time from sunrise to Amavasya-time = 14 hours 45.5 shadow Time from sunrise to noon = 15 hours — transit time = 13 hours 12.7 shadow ∴ The transit time (west) of Amavasya-time = 1 hour 32.8 shadow The rising ascendant and mean ascendant of Amavasya-time, according to Sūrya Siddhānta:

The rising of the signs of the zodiac in Kashi (like page 314)

SOURCE PAGE 548

Aries 1345 minus Mean

Taurus 1525 "Aquarius Gemini 1821 "Capricorn Cancer 2041 "Sagittarius Leo 2063 "Scorpio Virgo 2005 "Libra New Moon True Sun = 6° 32'8 = 22041' › Therefore, True Sun at New Moon = 6230 13'8 = 62314' The portion of Capricorn sign = 6°46' = 406' In Kashi, the rising of Capricorn sign = 1821 1800: 406:: 1821: The portion of Capricorn = 406 × 1821/1800 = 411 Time from sunrise to New Moon = 14 hours 45.5 minutes. Shadow = 885.5 Shadow = 5313 minus the portion of Capricorn = 411 The rising of Aquarius sign = 1525 Pisces "1345 Aries "1345 Sum 4626 Subtracting this sum by 5313 portions leaves 687 portions. These are the portions of Taurus ascendant. But the rising of Taurus is 1525. 1525: 687:: 1800: The portion of Taurus › True position of Taurus = 687 × 1800/1525 = 811 degrees = 13°31' Therefore, True position of Ascendant = 30° + 13°31' = 43°31' Therefore, True position of Ascendant in Threefold Ascendant = 43°31' - 60° = 360° + 43°31' - 60

SOURCE PAGE 549

The sine of the Sun at the time of new moon is in the sign of Capricorn, whose ascent is 1631 (see page 314). Therefore, the sine of Capricorn in 1631 asus violates the meridian of any place (see three shlokas 48 and page 330). In the time of new moon, the western motion of the Sun is 1 hour 32.8 minutes = 62.8 minutes = 557 asus.

When the 30 degrees or 1800 art of the sign of Capricorn in 1631 asus violates the meridian, then in 557 asus, 557 ×1800/1631 art = 519 art = 8°36’ will occur. Therefore, the mean ascendant is 8°36’ west of the Sun, and knowing the mean ascendant will be achieved by adding this to the Sun's portion. However, adding this makes the Aquarius sign the mean ascendant, so it is best to first see how long the Capricorn sign violates and how much Aquarius moves in the remaining time.

The sine of the Sun at the time of new moon is 6 ra 230 14’ therefore the portion of Capricorn is 6046’ which is equal to 406’.

1800: 406:: 1631: Capricorn's portion

∴ Capricorn's portion = 406 X 1621 / 1800 = 435.5 asus

But the time is 557 asus, therefore Aquarius's portion = 121.5 asus. Aquarius's ascent is 1764 asus, therefore

1764:1215::1800:Gemini's portion

∴ Gemini's portion = 121.5 x 1800 / 1764 = 122arc = 20'2"

∴ The mean position of Aquarius in the time of Amavasyaanta is the mean position. That is, the sine portion of the mean position = 10 arc 2"

rising sine = latitude sine X true declination sine / latitude sine = sine43° 31' X sine23° 27' / sine (60 - 25° 20') =.6886 X.3676 /.6038 = 3032

∴ The agra of the rising sign = 17°36'

SOURCE PAGE 550

The sine of the mean ascendant = 10° 20' = 302', herefore sine of the mean ascendant = sine 302' imes sine 23°27' = - sine 57°58' imes sine 23°27' = -.8477 imes .3676 = -.3373 herefore South node = 16°43' Latitude of Kashi = 25° 20' herefore sine of the mean ascendant = 45° 3' Page 413 according to the first equation of the page, sine of the latitude of the three-fold ascendant = sine(90° - 42024') = sine 47°36' = sine.7385

Here, the calculation of the arc of sine and co-latitude is done according to the decimal table in which the radius is taken as 1. If this table is not available, one should use the method described on page 413. It will be even more convenient to use the short blank table. A table can also be made to know the arc of the three-fold ascendant so that the calculation can be done easily. It is stated on page 328 and in other places that since the degrees of any quantity do not rise at the same time, if the ratio is used, the knowledge of the rising point of the quantity will remain coarse. In such a situation, there will also be coarseness in the ascendant of the three-fold ascendant in the method mentioned above because the three-fold ascendant comes when 60 degrees are subtracted from the rising point of the ecliptic. Therefore, it is necessary to know the rising point of the ecliptic or the rising ascendant with purity for the calculation of the solar eclipse. From this idea, the following method is written.

Local time—the time from the moment the vernal equinox point or sine Aries comes on the horizon of a place until some auspicious time is called local time. Page 318 - 16 states that in Prayag in the part of Ayana

SOURCE PAGE 551

How the rising points are known. In that, the rising points of the Ayan part are stated as 1005. This can also be said in this way that the time when the initial point of Aries comes on the horizon circle in Prayag is equal to the sidereal time of 1005 asu. Similarly, the time when the initial point of Taurus comes on the eastern horizon in Prayag is equal to the sidereal time of 2470 asu. From this it is apparent that if one wants to know what the sidereal time is at a certain place, one should first know the rising sign at that time, then know the difference between the rise sign's sidereal portion and arc, and find the difference between both. This difference is the sidereal time of that time.

Similarly, if the time of the solstice is known at some time, then the rising ascendant of that time can also be known. However, this task is not that easy in the reverse manner. Therefore, the methods for directly knowing the rising ascendant from the solstice and the solstice from the rising ascendant are written here:-

In the new style of the rising time:

From Figure 61 it is clear that the spherical triangle in the south,

```
sinepuka / sine8kavpu = javkav / sine8vpuka
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Here, the rise of the ascendant is the agra of the sun, the arc of angle is the supreme revolution, the sine portion of the rise of the ascendant is the sine portion of the sun and angle and the sun = 180° - angle and the sun = 180° - ideal position's longitude.

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, sine ∠ apogee = sine ( 180 ° − latitude) = sine latitude
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∴ sinevapuka = precession angle X sineoboganshlatitude

This is also a formula to know the rising angle of the sun which can also be obtained from the formula on page 267 and the formula on page 272(3). The rising angle of the sun can be known in this way.

Sūrya Siddhānta = 622056' = 262° 56'

Latitude of Kashi = 25°20′

SOURCE PAGE 552

sine = sine23°27' x sine262056' / cos(25°20')

but sine262056' = - sine (360° - 262056') = - sine6701'

The negative sign indicates that the rising point is south of the pūrva bindu.

sine = sine23°27' X sine67°1' / cos(25°20°)

arc length = arc length23°27' + arc length67°1' - arc cos(25°20') = 6.56666 + 6.6641 - 6.6561 = 6.6076 ∴ arc = 23°55'

The sine of this is also called the sine of the sunrise time or the sine of the meridian. With this, the sunrise time should be known.

The time of sunrise—if the angles of the spherical triangle are revealed by the letters i, u, a, and their opposite sides by the letters e, i, u, respectively, then by spherical trigonometry it is revealed that

Spreu / 2 = sin3 (a + i) / sin3 (a - i) x spre (a - e)

The sine portion of the Sun is 262056' or 1800+112056, which means that the autumnal equinox is 112056' before the point of the solstice. There are 30 degrees from the rising time of the vernal equinox to the rising time of the autumnal equinox. Therefore, the vernal time of the autumnal equinox is 30 degrees or 1800. Therefore, if it is known how long it takes for 112056' from the autumnal equinox to rise in Kashi, then the vernal time of this point can also be known. In such a case, the hypotenuse of the spherical triangle S in Figure 62 is S's side S's 112056', the angle P is 23055', angle S P = Kashi's arc = 600 - 25° 20' = 60040', S P = S's vernal time, therefore according to the formula given above in spherical trigonometry.

See Todhunter and Leathem's Spherical Trigonometry page 75

SOURCE PAGE 553

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The equinoctial time of Amavasya:—the moment when the autumnal equinox point will come on the eastern horizon, then 20 hours 55.7 minutes later the sun will come on the horizon when its sidereal fraction of the autumnal equinox will be 112056'[1]. But the equinoctial time of autumn equinox from spring equinox is 30 hours, so at the time of sunrise of Maghi Amavasya the equinoctial time is 50 hours 55.7 minutes. This is in sidereal measure. But the time of Amavasya from sunrise is 14 hours 45 minutes. This is in sidereal measure which is approximately 14 hours 47.5 minutes of sidereal measure. (See page 326). Therefore,

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1. This can also be known in the manner stated in pages 314-15.

SOURCE PAGE 554

Rise and setting of the ascendant and declination from the viṣuvakāla—now one needs to know what is the sine portion of the rise of the ascendant when the viṣuvakāla is 34°16'? This can be easily known with the help of figure 60 where the sine of angle is 23^(°)27' and the sine of 8^(°) is 180^(°) — angle is 180^(°) — the latitude is 180^(°) - 64^(°)40' = 115^(°)20'

If the three angles of a spherical triangle are revealed by the letters a, b, c and the opposite sides by the letters a, b, c respectively, then the two formulas of spherical trigonometry1 can be revealed as follows:-

Sphere 12 (a + e) = cosine12 (a - i) cosine12 (a + i) × sphere2

(1)/(2) (a - e) = (sin)/(2) (a - i) / sin / 2 (a + i) × (e)/(2)

With the help of these two formulas, the values of both a and b can be known. In this way, from the spherical triangle and setting of figure 60

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! vaka + kapu = 62046 ′..... (1)

Similarly, from the second formula,

spere 3(va - kapu) = sin 45° 56’. 5 / sin 66° 23. 5’ x spere 17° 6’. 5

therefore Lispere vaka - kapu2 = lirijya 45° 56'5'' + lispere17^° 6'0'' - lirijya66^° 23'0'' = 6'8565 + 6'4866 - 6'6713 = 6'3748

∴ vaka - kapu / 2 = 13 ° 20 ′

!: va - kapu = 26°40’..... (2)

Adding equation(1) and (2),

2vaka = 86 ° 26 ′

∴ vaka = 44 ° 43 ′

And subtracting equation(2) from equation(1),

2kapu = 36 ° 6 ′

∴ Kapu = 18 ° 3 ′

Thus it was proved that if the time of the solstice is known with the help of the two above formulas, the rising ascendant and the declination can be obtained at any time. Therefore,

amavasyaantkalkasayanudaylagna = 44° 43′

andudayanklakkiuttaragra = 18° 3′

On page 524, the sine of the ascendant is 43°31′ and on page 524, the declination of the rising ascendant is 17°36′ which are very different from the numbers obtained by the new method. The reason for this is that there is the ratio of the rising ascendant known there which is gross.

whensayanlagna=44° 43′ then tribhonlagna = 44° 43′ - 60° = 360° + 44° 43′ - 60°

∴ amantkaliktribhonlagna = 314° 43′

Knowing the mean meridian of the unpunctual time—the mean meridian of that time which has come from the Bisuvaka in the unpunctual time, fifteen hours or 60° less at the same time.

SOURCE PAGE 556

This will be because the point of the obliquity circle on the meridian circle is the same as the obliquity time of the mean ascendant and the point of the obliquity circle on the eastern horizon is the same as the obliquity time of the rising ascendant. However, the difference between these two points of the obliquity circle is equal to 15 minutes or 60°.

In Figure 63, if we assume that v is 34^ 16', w is 44^ 43', and the common point of the meridian circle and the obliquity circle is c, then the value of w in the spherical triangle can be easily known because

c = c_(pu) - v_(pu) = 60^ - 34^ 16^ ' = 55^ 41^ ' angle c = 23^ 27^ '

and angle w c = 60^ . Because this is the angle between the equator and the meridian, according to Napier's first rule (see page 125),

cos23^ 27^ ' = sin55^ 41^ ' x cosw = sin55^ 41^ ' cosw herefore cosw = sin55^ 41^ ' - cos23^ 27^ ' = 10 imes 1658 - 6 imes 6625 = 10 imes 2033 herefore w = 57^ 57^ ' herefore sin mean ascendant = 360^ - 57^ 57^ ' = 302^ 3^ '

It is only 1' larger than the median zodiac in these 524 pages. This means that according to Sūrya Siddhānta, the median zodiac that has come is absolutely correct. The reason for this is that the median zodiac and the sun are very close; if the sun were far from the median zodiac, there would also be a difference. Knowing the nataṃśa of the three zodiacs—

The nataṃśa of the median zodiac is 45°3' according to the method of Sūrya Siddhānta (see page 525). This method is absolutely pure. According to the method of knowing the nataṃśa of the three zodiacs or the epicycle, which is explained on page 413, the nataṃśa of the three zodiacs is 42018' if the rising zodiac is taken as 1803' according to the new method of udaya ascendant. But this is very crude. Therefore, spherical triangle m x v (figure 63)

SOURCE PAGE 557

It would be appropriate to directly calculate the value of x y. Here, x y is the nataṃśa of the tribhija or tribhone lagna, m is the nataṃśa of the madhya lagna, and m y is the difference between the madhya lagna and the tribhone lagna, which is similar to 314° 43’ - 3020’3’ or 120° 40’. Since angle m y x = 60°, according to Napier's second rule,

Cojyamakh = Cojyakvi + Cojyamvi

∴ sine of declination = sine of latitude = sine of latitude 4503′ sine of latitude 12°40′

∴ arcsec = arcsec45° 3' - arcsec12° 40' = 6.8461 - 6.6863 = 6.1598

∴ xv = 43° 36′

∴ tribhonlagnakants = 43° 36′

There is another way to know which is based on Napier's second rule with the help of angle M and kha and Mkha of the same spherical triangle. Both methods are abhinna of the netaṃśa of the three-fold ascendant. Therefore, this should be valid rather than the method mentioned in page 413 of Sūrya Siddhānta.

Projection = sine43° 36' = 6866

Dṛgagati = Tribhonlagnakīunnatāṁśjya = cojya43° 36' = 7242

1 / 4 x 7242

Immemorial Triangulation = 314 ° 43 ′ (page 530)

AmantakalikaSāyanaSūrya = 263 ° 14 ′ (page524)

∴ Instantaneous Analysis Angle = 21026 ′

therefore Surya-chandrama-lamban = sine-analysis-section = 4 × 0.7242 × sin 21° 26' = 4 × 0.7242 × 0.3662 = 1.0608 hour = 1 hour 3.65 pal

This is the western declination because the sun is west from the three-gon ascendant. Therefore, when added to the new moon period, the arc-declination-circumference-new moon period will occur.

Sūrya Siddhānta time of sunrise to new moon = 14 hours 45 minutes

Western Hemisphere Altitude = 1hour3.6minutes

SOURCE PAGE 558

∴ Time from sunrise to the time of the nadir-transit-new moon = 15 hours 48.6 minutes

That is, due to the nadir, the moon will be at the position of sunrise by 15 hours 48.6 minutes. This is not entirely pure either, so it is necessary to do an uncalculated action, meaning one should see what nadir occurs at 15 hours 48.6 minutes from sunrise. For this time, the rising ascendant, tribhone ascendant, mean ascendant, etc., should be known, for which the same action will have to be done as seen from page 528 till now.

15 hours 48.6 minutes (Sawan) = 15 hours 51.1 minutes (Nakshatra) Sunrise of the Sun = 50 hours 55.7 minutes (Page 528) ∴ Latitude - Sanskrit - Amavasya time of the time = 6 hours 46.8 minutes = 40° 41' approximately

Simplifying the equations in page 526 by replacing 34°16 with 40°41′ will result in the rising ascendant and agra, as the multipliers are general. Therefore

Sine 3(theta + phi) = sin 45° 56' 5" + sin 20° 21' 5" - sin 66° 23' 5" = 6'8565 - 6'5661 - 6'6713 = 6'4543 therefore theta + phi = 36° 14' 2 theta + phi = 72° 28'.....(3) Sine 3(theta - phi) = sin 45° 56' 5" + sin 20° 21' 5" - sin 66° 23' 5" = 6'8565 - 6'5661 - 6'6713 = 6'4543 therefore theta - phi = 15° 53' 2 therefore theta - phi = 31° 46'.....(4) From equations (3) and (4), theta = 52° 7' phi = 20° 21'

SOURCE PAGE 559

∴ The rising ascendant is 52^°7’ and 20^°21’ at 15 hours 48.6 minutes from sunrise. ∴ At this time, the three-fold ascendant is 52^°7’ - 60° = 322^°7’ and the solstice is 40^°41’ ∴ Like page 530, c = 60° - 40^°41’ = 46^°16’ ∴ s v m = spere 46^°16’ / kojya 23^°27’ ∴ l r s v m = spere 46^°16’ - lri kojya 23^°22’ = 10.0657 - 6.6625 = 10.1032 ∴ v m = 51^°45’ ∴ sain mean ascendant = 36^°0 - 51’45’ = 308^°15’ The sine of the mean ascendant is 308^°15’ × sine 23^°27’ = -sine 51^°45’ × sine 23^°27’ ∴ lri sine = 6.8650 + 6.5666 = 6.4646 ∴ The south transit of the mean ascendant is 18^°13’ The north latitude of Kashi is 25^°20’ ∴ The latitude of the mean ascendant is 43^°33’ The difference between the mean ascendant and the three-fold ascendant is 322^°7’ - 308^°15’ = 13^°52’ ∴ The co-latitude of the latitude of the three-fold ascendant is kojya 43^°33” / kojya 13^°52’ ∴ lri latitude = lri kojya 43^°33’ - lri kojya 13^°52’ = 6.8602 - 6.6872 = 6.8730 The latitude of the three-fold ascendant is 41^°43’ The true daily motion of the sun is 61’.37 ∴ The motion of the sun in one hour is 1’.023 ∴ The motion of the sun in 3 minutes is 0.051 ∴ The motion of the sun in one hour and 3 minutes is 1’.07 The apparent sine of the sun is 263^°14’ ∴ Longitude-latitude-apparent-sun is 263^°15’ The three-fold ascendant is 322^°7’ The sine of the sun is 263^°15’

SOURCE PAGE 560

The calculation is as follows: Analemma =28^°52’ sight = prime vertical circle of the ascendant = latitude 41^°43’ therefore chord = 1/4 sight = 1/4 latitude 41^°43’ therefore sun's altitude = sine Analemma/ chord = 4 latitude 41^°43’ sine 28^°52’ = 4 × 0.7464 × 0.4828 time = 1.442 time = 1 hour 26.5 time from amāvasyānta to sunrise = 14 hours 45 time of sun's altitude = 1 hour 26.5 time therefore second altitude-Sanskrit-new moon time = 16 hours 11.5 time knowing the prime vertical circle of this time, one should know the altitude:- 16 hours 11.5 time (Sawan) = 16 hours 14.2 time (Nakshatra) time of sunrise of the solstice = 50 hours 55.7 time therefore time of solstice of the second altitude-Sanskrit-unmanifest-time = 7 hours 6.6 time = 43° approximately Lirspere 3/2(v of + of of) = Liricord 45^°56’.5+ Lirspere 21^°30’ - Liricord 66^°23’.5 = 6.8422+6.5654 - 6.5464 = 6.8612 therefore v of + of = 37^°54’ therefore v of + of = 75^°48’ Lirspere 3/2(v of - of) = Lir sine 45^(°)56’.5+ Lirspere 21^°30’ - Lir sine 66^°23’.5 = 6.8565 + 6.5654 - 6.6713 = 6.4806

SOURCE PAGE 561

∴ Vaka - Kapu2 = 16046' 5

∴ 16^°46' · 5/2 ∴ v_k - v_p = 33^°36' ∴ v_k = 54^°43' · 5 and v_p = 210^°5 ∴ Rise of ascendant at 11.5 minutes from sunrise is 54^°43' · 5 and 10^°4' · 5 ∴ At this time, the tribhona ascendant is 54^°43' · 5 - 60° = 324^°43.5 and the viṣuvakāla is 43° ∴ v = 60° - 43° = 47° Spere 47° ∴ Spere v_m = cojya 23^°27' ∴ Lari Spere v_m = Lari Spere 47° - Lari cojya 23^°27' = 10.0303 - 6.6625 = 10.0678 ∴ v_m = 46^°27' ∴ Saiyan madhyalagna = 36^°0 - 46^°27' = 310^°33' ∴ The angular velocity of madhyalagna is jya 310^°33' × jya 23^°27' = -jya 46^°27' × jya 23^°27' Lari angular velocity = 6.8807 + 6.5666 = 6.4806 ∴ Period = 17^°66' Latitude of South Kashi = 25^°20' Nataṃśa of madhyalagna = 420^°56'

Difference between madhya ascendant and tribhona ascendant = 324^°43' · 5 - 310^°33' = 14^°10' · 5 ∴ Nataṃśa of tribhona ascendant = cojya 420^°56'/cojya 14^°10' · 5 ∴ Lari Nataṃśa angular velocity = Lari cojya 420^°56' - Lari cojya 14^°10' · 5 = 6.8646 - 6.6665 = 6.8781 ∴ Nataṃśa of tribhona ascendant = 400^°57' Speed of 1 hour of the sun = 1.023 Speed of 20 minutes of the sun = 0.341

SOURCE PAGE 562

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In this way, it occurs at the time of Amavasyaant fifteen hours forty-eight point six minutes from the first lamba, sixteen hours eleven point five minutes from the second lamba, and sixteen hours sixteen point six minutes from the third lamba. From this it is apparent that there is only an eight minute difference in the previous Amavasyaant times. If the ritual is performed two or three more times, the difference will become zero. In that situation, the Amavasyaant time that comes will be the pure Amavasyaant. It is estimated that the Amavasyaant time that has come for the third time will be only two or three minutes more than the pure Amavasyaant. Therefore, there is no special benefit in performing the ritual two or three more times for two or three minutes, except for the hassle. Therefore, it is assumed that the lamba-sanskrta-pure-amavasyaant time is sixteen hours twenty minutes from sunrise. This should be understood as the middle time of the solar eclipse. The action of the sixth verse has ended here. Nati-

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SOURCE PAGE 563

The difference should be multiplied by the small angle. However, in my understanding, if multiplied by the difference of true motions, there will be more accuracy.

Difference of the daily motions of the Sun and Moon = 762' · 343 sin = sin 40° 57' therefore result 762' · 343 × sin 40° 57' 15 = 762' · 43 × · 655415 = 34' · 62

No need to divide by the radius here because the value of sine is taken as a decimal fraction. This is south because the position of the mean is south.

The motion of the Moon in 60 hours = 14°13"·7

Sūrya Siddhānta = 14' 13" .7 30p = 7' 6" .6 5" = 9' 11" .1

Therefore, the motion of the Moon in 1 hour 35 minutes = 22'31.7” = 22’5

Mathematics-derived amāvasyāntakālika caṃdramā = 6 RA 32' 8

∴ Lunar-Sanskrit-New-Moon-instantaneous Moon = 6RA°55'3

,,,, Rahu = 36028' 0RahusechandraMakaAntar = 524° 27' 3 = 174° 27' 3

therefore lunar sine = sin174° 27' · 3 × sin4° 30' 3438 = sin5° 32' · 7 × sin4° 30' 3438 = 332° 7 × 2703438 = 26'.13

This answer is correct because the Moon is ahead of Rahu but less than 6 signs.

∴ NatisanskrtaChandrashar = − 34 ′.62 + 26 ′.13 = − 8 ′.46
ArthatansanskrtaDakshinChandrashar = 8 ′.46
ChandrakakamaMansauryaBimbakasphutvyas = 33 ′.634
Chandramakasputvyas = 34 ′.555

SOURCE PAGE 564

Chadyathavasūryakavyāsaradh = 16' 0817chadakaathvāchandramākāvyāsaradh = 17' 278 ∴ Mānaiyakhanda = 34' 10Mānāntarakhanda = 0' 46

GrasakaParimaan = Manaikyakhand - NatisanskrutachandraShar = 34' · 1 - 8' · 46 = 25' · 61

It is smaller than the diameter of the lunar image, therefore a partial eclipse will occur, not a total eclipse. (See page 460 and verse 11 of Ch. Gr.)

According to page 466

Position = 60 hours × \((34.1 + 8.46) (34.1 - 8.46) \) 762.343 = 60 × \(42.56 × 25.61 \) 762.343 = 60 × 1060.73762.343 = 60 × 33.026762.343 = 198156076234 hours = 2 hours3000 minutes = 2 hours30 minutes
Time difference = = 16 hours20 minutes - 2 hours30 minutes = 13 hours50 minutes

That is, the touch of the eclipse will occur at 13 hours 50 minutes and equinoctial shadow in Kashi. But this is gross. To make finer calculations, the latitude and declination of this time should also be calculated, as stated in shlokas 14-17.

13 hours 50 minutes (Sawan) = 13 hours 52.3 minutes (Nakshatra)

Sūrya Siddhānta = 50hours55.7minutes

∴ time of touch of the time of the equator = 4hours48minutes = 28 ° 48 ′

SOURCE PAGE 565

∴ Latitude 3(Lat + Lon) = Longitude 45° 56'5 + Latitude 14° 24' - Longitude 66° 23'5 = 6.8422 + 6.4065 - 6.5464 = 6.7053 ∴ Latitude 3(Lat + Lon) = 26° 54' ∴ (Lat + Lon) = 53° 48' Longitude 3(Lat - Lon) = 6.8565 + 6.4065 - 6.6713 = 6.2647 because 3(Lat - Lon) = 110'6' ∴ Lat - Lon = 220'18' ∴ Lat = 38^°3' and Lon = 15^°45' ∴ The rising of the Ascendant at 13 hours 50 minutes from sunrise is 38^°3' and 15^°45'. ∴ At this time, the Tribhona Ascendant = 38^°3' - 60° = 30^°03' and "Sidereal Time = 28^°48' Page 532 like Lat = 60° - 28^°48' = 61^°12' ∴ Lat Lon = Lat 61^°18' / Lon 23^°27' = 10^°2568 - 6.6625 = 10.2673 ∴ Lon = 63^°14' ∴ Sidereal Midheaven = 36^°0 - 63^°14' = 266^°56' The arc of the Midheaven = Sine 266^°46' × Sine 23^°27' = -Sine 63^°14' × Sine 23^°27' ∴ Longitude Arc = 6.6508 + 6.5666 = 6.5507 ∴ Southern Arc of Midheaven = 20^°46' North Latitude = 25^°20' ∴ Latitude of Midheaven = 46^°6' Difference between Midheaven and Tribhona Ascendant = 30^°03' - 266^°46' = 11^°17' ∴ Latitude of the Arc of Tribhona Ascendant = Sine 46^°6' / Sine 11^°17'

SOURCE PAGE 566

∴ Larintaanshcotijya = Lrikojya46° 6' — Lrikojya11° 17' = 6.8406 - 6.6615 = 6.1841

∴ Triangle of the Sun's Luminosity = 45 ° 3 ′

Dṛgagati = Tribhōnlagnakīntāṁśakōtijyā = sin^2 45° 3 therefore chēda = 1 / 4sin^2 45° 3'

At 14 ghaṭīs and 45.5 palas after sunrise, the true tropical position of the Sun is 6 signs, 23°14′55″; the Sun's motion per pala is 1′. Check the OCR numerals against the scan.

∴ 13 hours 50 minutes or apparent time Sun = 623 ° 13 ′ = 263 ° 13 ′

Sūrya Siddhānta

Scientific Commentary

Chapter on Mean Motions

Chapter on True Positions

Chapter on the Three Problems

The late

The eclipse will touch at 12 hours 58 minutes and some shadow on the sunrise in Kashi. Similarly, the true position of Moksha should also be known.

SOURCE PAGE 567

From this calculation it is clear that the touch and salvation of the solar eclipse in Kashi will be seen. However, this is not seen by any person gathered in Kashi as people experience. The reason for this is that according to Sūrya Siddhānta, the lunisolar positions are very coarse. For this reason, even though the method of knowing the latitude and so on has been completely changed, subtlety could not be achieved. The greatest error in these positions is in the position of Rahu, as stated in Chandragrahaadhikar.

What happens when taking the position of Rahu?

According to the Navik Panchang of 1626 AD, on Monday, January 11th, at the mean meridian time of Greenwich, the position of Sine Rahu was 1150.7505. At this time, 13 hours 50 minutes 31 seconds of the post-meridian occurred in Kashi (see page 251), which is 28 hours 50.5 minutes of the mean

morning time. From this time until the end of Maghi Amavasya, i.e., until 16 hours 54 minutes of the mean morning time of Thursday, it is 2 days 48 hours 305 minutes. The motion of Rahu in this time was as follows:-

1day's speed = 0 °.052652,, = 0 °.1056230hour's speed = 0 °.0264815,, = 0.013243,, = 0.002653minute's speed = 0.00004sum = 0.1483

This is the position of Rahu after subtraction, 11506022 = 115036' 1But the precession angle = 22041' ∴ Rahu's ecliptic portion (at the time of Amavasya) = 62055' Lunar ecliptic = 270033' ∴ The difference between Rahu and the Moon = 177038'

If the Moon's ultimate position is taken as 50842

SOURCE PAGE 568

Chandrajyā = sin 1977° 38' x sin 50° 42'' = sin 2022' x sin 50° 42'' = .0413 x .0867 = .0037

∴ Chandrashar = 12' 40" North = 12' .67 North = 34.62 South ∴ NatisanskritiChandrashar = 21' 65 South

∴ Graskaparamana = Manaikyakhand - NatisanskritiChandrashar = 34' .1 - 21' .65 = 12' .15

In this way it is also proven that if the portion of Rahu is taken exactly, then the magnitude of the eclipse is 12' .015, that is the touch and salvation of the eclipse can be seen in Kashi but this has not been experienced. Therefore, now it is to see what difference there is if the latitudes, longitudes, and diameters of the sun and moon, etc., are derived according to the principles and new methods.

According to the navigator's almanac:-

Sūrya Siddhānta

Scientific Commentary

Chapter on Mean Motions

Chapter on True Positions

Chapter on the Three Problems

61' 12" = 61' .2

SOURCE PAGE 569

=40.02 Chandrasar North = 7'·6 ∴ Nati Sanskriti Chandrasar = 32'·42 Manayakyakhand = 16'·68 + 16'·26 = 32'·97 ∴ Area of Grass = 32'·97 - 32'·42 = 0'·55

Thus it is proven that if the diameters, latitudes, and longitudes of Rahu, the Moon, and the Sun are taken according to the new calculations, the area is less than 1 art, which cannot be seen even if industry is done. This is also proven by experience. Therefore, it is extremely necessary to correct our Siddhanta texts according to trigonometry for the calculation of eclipses.

SOURCE PAGE 570

Chapter on Mean Motions (Brief Description)

[Verse 1—Purpose of the treatise. Verses 2-12—The method of drawing the treatise of the touch, liberation, and the eclipses of the mean time. Verse 13—How much part is covered when an eclipse is seen. Verses 14-16—The method of drawing the path of the observer. Verses 17-19—The method of drawing the treatise of an eclipse at a specific time. Verses 20-21—The method of drawing the treatise of the beginning time of a total eclipse. Verse 22—The method of drawing the treatise of the end time of a total eclipse. Verse 23—In what kind of lunar eclipse the moon is black, brown, etc. Verse 24—Whom should the method of drawing the treatise be followed.]

The name of this chapter is sometimes Chedyakaadhikar also. Both have the same meaning. Since Parilekha is simpler than Chedyaka, Parilekha is written here.

Purpose —

> न छेद्यकमृते यस्मात्क्षेपा ग्रहणयोः स्फुटाः ।
> ज्ञायन्ते यत्प्रवक्ष्यामि छेद्यरुज्ञानमुत्तभम् ॥१॥

(1) Without a diagram, inscription, or picture, there is no exact knowledge of the difference between the eclipses of the Sun and the Moon regarding which direction the eclipse begins, which direction it ends, and how much it will be. Therefore, I say that the best knowledge is that of making a diagram.

The method of drawing—

> सुसाधितायाभवनौ विन्दुं दत्वा ततो लिखेत् ।
> सप्तवर्गाङ्गुलेनादौ मण्डलं वलनाश्रितम् ॥२॥
> ग्राह्यग्राहकयोगार्धं सम्मितेन द्वितीयकम् ।
> मण्डलं तत्समासाख्यं ग्राह्यार्धेन तृतीयकम् ॥३॥
> याम्मोत्तरा प्राच्यपरा साधनं पूर्ववद् दिशाम् ।
> प्रागिन्दोग्रहणं पश्चान्मोक्षोऽर्कस्य विपर्ययात् ॥४॥

SOURCE PAGE 571

यथादिशं प्राग्ग्रहणं वलनं हिमदोधितेः ।
मौक्षिकं तु विपर्यस्तं विपरीतमिदं रवेः ॥५॥
वलनाग्रान्नयेन्यमध्यं सूत्रं तद्यत्र संस्पृशेत् ।
तत्समासे ततो देयौ विक्षेपौ ग्रासमौक्षिकौ ॥६॥
विक्षेपाग्रात् पुनस्सूत्रं मध्यविन्दुं प्रवेशयेत् ।
तद्ग्राह्यवृत्तसंस्पर्श ग्रासमोक्षौ विनिर्दिशेत् ॥७॥
नित्यशोऽर्कस्य विक्षेपाः परिलेखे यथादिशम् ।
विपरीतं शशाङ्कस्य तद्वशादथ मध्यमम् ॥८॥
वलनं प्राङ्मुखं नेयं तद्विक्षेपैकता यदि ।
भेदे पश्चान्मुखं नेयम् इन्दोर्भानो विपर्ययात् ॥६॥
वलनाग्रात्पुनस्सूत्रं मध्यविन्दुं प्रवेशयेत् ।
मध्यात्सूत्रेण विक्षेपं वलनाभिमुखं नयेत् ॥१०॥
विक्षेपाग्राल्लिखेद्वृत्तं ग्राहकार्धेन तेन यत् ।
ग्राह्यवृत्तं समाक्रान्तं तद्वस्तं तमसा भवेत् ॥११॥
छेद्यकं लिखितं भूमौ फलके वा विपश्चिता ।
विपर्ययो दिशां ग्राह्यः पूर्वापर कपालयोः ॥१२॥

(2) On a well-measured flat land, fix a point and draw a circle with a diameter of 46 angulas with this point as the center. This is called the involute circle. (3) From the same center, draw another circle whose diameter is equal to the sum of the diameters of the shadow and the apparent figures. This circle is called the composite circle. Similarly, from the same center, draw a third circle whose diameter is equal to the diameter of the figure of that planet on which the eclipse occurs. (4) From this point, draw the North-South line and the East-West line according to the manner described in the first (Chapter on the Three Problems verses 3, 4). In a lunar eclipse, the touch occurs from the east direction and the exit from the west direction, but in a solar eclipse, it is the opposite, that is, the touch occurs from the west and the exit from the east in a solar eclipse. (5) In a lunar eclipse, draw a line from the center to the involute circle at a distance equal to the sine of the touch time from the east point and in the same direction as the involute. Similarly, draw another line from the center to the involute circle at a distance equal to the sine of the exit time of the moon, but in the opposite direction to the direction of the involute, from the western point. In a solar eclipse, the order of the directions of the above lines is the opposite, which

SOURCE PAGE 572

It is explained in the lunar eclipse. These lines are called the lunar lines, and the points where these lines intersect the lunar circle are called the lunar points. (6) From the points that are made from the touch and demise times on the lunar circle (according to the 5th verse), the lines that go to the center from them, draw lines to the points where the compound circle intersects, and draw lines from the center to the compound circle on the difference between the lunar touch time and demise time of the moon. These lines are called the deviation points of the points where they intersect the compound circle. (7) The lines that go from these deviation points to the center and the points where they intersect the reflected image are called the touch points and demise points in sequence.

(8) In the diagram of solar eclipse, make the displacement point in the same direction as the direction of the Moon's shadow, but in the diagram of lunar eclipse, the direction of the displacement point is

opposite to the direction of the Moon's shadow. Accordingly, make the displacement point for the mean eclipse period as well.

(6) In the diagram of the middle period of lunar eclipse, if the directions of the declination and deviation are the same, the nodal point should be made in the east of the north-south line. But if the directions of the declination and deviation are different, the nodal point should be made in the west of the north-south line. If the direction of deviation is south, the nodal point should be made east or west of the north point. But if the direction of deviation is north, the nodal point should be made east or west of the south point. In the diagram of the middle period of solar eclipse, the opposite should be done, that is, if the directions of both the nodal and deviation are the same, the nodal point should be made towards the west of the north-south line and if the directions of both are different, the nodal point should be made towards the east of the north-south line. But if the direction of deviation is south, the nodal point should be from the south point and if north, from the north point towards the east or west.

(10) Draw a line from the nodal point to the center of the eclipse. On this line, make a point at the difference of deviation from the center in the direction of the nodal point; this is called the point of deviation of the middle period.

(11) Taking the point of deviation as the center, draw a circle with a radius equal to the semi-diameter of the client or the shadow. This circle is the part of the eclipse that is covered by the shadow of the moon (in lunar eclipse) and the shadow of the sun (in solar eclipse).

SOURCE PAGE 573

(12) The astrologer should draw a diagram on a flat surface or the celestial sphere (on a wooden board). The order of directions that exists in the anterior skull should be in the reverse order in the posterior skull, that is, where the directions of east, south, west, and north will be in the anterior skull, the directions of west, north, east, and south will be in the posterior skull respectively.

Scientific Commentary—in these verses, the method of knowing the distance and the direction in which the center of the observer is located at the touch, mean, and culmination of the reception, assuming the received image is fixed, is explained with the help of trigonometry. In a lunar eclipse, the moon is the received and the Earth's shadow is the observer. In a solar eclipse, the sun is the received and the moon is the observer. Now, the explanation of each method in the order of the verses is given:—

Verse 2—It is stated in verses 24-25 and pages 475-480 that the subtension is and how the knowledge of the ecliptic is obtained from it. It is also stated there that when the sine of the subtension is divided by 70, its sine value comes in angulas. Thus, the value of the radius is approximately 46 angulas because the radius is 3438 arts, which when divided by 70 gives a remainder of 46·1, which should be understood as 46 in integers. Therefore, in this verse, the method of drawing the circle with the semi-diameter of 46 angulas is explained. The line indicating the subtension can be easily drawn from this circle. Bhaskaracharya and other Acharyas have not stated the rule for drawing the semi-diameter circle. They have only written that on the circle of conjunction, the angles for east, west, north, and south should be made according to the direction for the magnitude of the subtension.

Verse 3—In this verse, the discussion is about the drawing of the circle of the image of the planet in which an eclipse occurs, that is, the circle of the apparent image. However, it is not made clear what its magnitude should be. If one considers one angula of 70 degrees, then the circle of the conjunction and the apparent image will be very small because the diameter of the apparent image circle is less than 16 degrees or one-fourth of an angula, and the diameter of the conjunction circle is about 1 angula.

Therefore, considering one angula of 70 degrees will not be convenient for these circles. In such a situation, the angula discussed in the 26th verse of the chapter on lunar eclipse is necessary.

SOURCE PAGE 574

But the value of the inch given in it is stated to change according to the advanced time (see page 481). But I understand that if the magnitude of the inch is always considered as 3 arts, there may be no special loss because as stated in page 482, due to the motion, a greater difference in the rise or set time of the sun or moon's images is seen in the time of rising or setting. At other times, this difference is so small that there is no need to consider it. Therefore, I find it easy to assume an inch of 3 arts here, no need to make any modifications in it.

Verse 4—In the first half of this, it is stated that the lines of North-South and East-West should be drawn from the point taken as the center, to draw the circle of the shadow, the circle of the mean motion, and the circle of the apparent image according to the three problems verses 2-4 and figure 44. In the latter half, it is stated that in a lunar eclipse, it occurs in the anterior part of the moon's image and in the posterior part of the moon; but in a solar eclipse, it occurs in the posterior part of the sun's image and in the anterior part of the sun's image. The reason for this is clear. The moon moves towards the east in the sky while revolving around the Earth, so at the time it begins to enter the shadow of the Earth, its eastern part enters the shadow first. Similarly, the posterior part of the lunar image is separated from the shadow at the time of the moon; but in a solar eclipse, the lunar image moves from the posterior to the anterior of the sun's image and covers the sun's image, so at the time of contact, the posterior part of the sun's image begins to be covered and at the time of eclipse, the anterior part of the sun's image is separated from the lunar image.

Verse 5—In the time of lunar eclipse, the direction of the radiation of the moon should be marked on the circle of the shadow in the direction of the radiation, same direction as the eastern point. But in the time of the eclipse, the direction of the radiation should be marked from the western point in the opposite direction. These marks are called the nodal-point. The reason for reversing the direction in the time of the eclipse becomes clear from Figure 101 of page 476. There it is shown that at the time when the ecliptic from the planet's arc, i.e., the eastern point, is towards the north, at the same time the ecliptic from the western point is towards the south. Therefore, the time when the direction of radiation is said to be north, at that time it is towards the eastern point, not the western point. But the direction of the radiation, which is proven by the 24-25 verses of the lunar eclipse, is understood from the eastern point, so that rule

SOURCE PAGE 575

According to which the direction of the motion at the time of salvation comes is according to the east point, but in a lunar eclipse, the motion is towards the west point, so to make the angle of the quadrature or the direction of the ecliptic, the direction of the quadrature is reversed.

What is written above should be done in the opposite direction in a solar eclipse. That is, in the time of touch, the direction of the quadrature should be made from the west point in the opposite direction, but in the time of salvation, the quadrature should be made in the direction of the east point. The reason is clear. In a solar eclipse, the shadow of the sun is towards the west and the salvation is towards the east. But the direction of the quadrature is reversed towards the west as stated above. Therefore, the direction of the quadrature at the time of touch has to be reversed in a solar eclipse, but no change is needed in the direction of the salvation time.

Verse 6-8—The line drawn from the nodal point to the center of the image that is inscribed in the circle, whether it is the circle of the shadow or the circle of the compound, only allows one to know the direction of the ecliptic. In a solar eclipse, the image is the Sun, and the Sun always remains on the ecliptic, so the line going from the nodal point to the center can be understood as the ecliptic. But in a lunar eclipse, the image is the Moon, and the Moon is uttara or dakṣiṇa at a distance equal to its distance from the ecliptic, so the line going from the center of the lunar image to the nodal point cannot be the ecliptic. This line is parallel to it. Whether it is a solar eclipse or a lunar eclipse, in both situations, the center of the occluder does not lie on the line joining the nodal point to the center because in a solar eclipse, the occluder is the Moon, which does not move in the Moon's orbit, so to find the center of the occluder in the time of contact or occultation, from the point where the nodal line intersects the compound circle, a line is drawn to the center of the compound circle in the difference of the lunar deflection. The point where this line intersects the compound circle is called the deflection point. At the time of contact or occultation, the center of the occluder is at this point. Therefore, if a circle is drawn with a radius equal to half the diameter of the occluder, taking this point as the center, where it touches the image, there will be the contact or occultation of the eclipse. The location of contact or occultation can also be known from the line drawn from the deflection point to the center because the point where the occluder and the image touch is the same point where the line from the deflection point to the center

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SOURCE PAGE 576

The line drawn also cuts the accepted image (see page 467 figure 100). In this figure, if C is taken as the center of the accepted image, the line drawn parallel to the declination circle D from C to the point V can be called the line going to the nodal point. The difference in this line from the earth shadow D is like the shadow of C. If C is taken as the center, the circle drawn with the radius of D will be the circle of conjunction. Where the line from C intersects the circle of conjunction, the difference of C from D will also be like the shadow of the moon. Thus, the rule stated in the seventh verse is proven.

This is not stated in the sixth verse in which the direction of the deviation line should be drawn from the declination line. This is stated in the eighth verse. In a solar eclipse, the deviation line should be drawn in the same direction as the direction of the moon's shadow, that is, if the direction of the lunar shadow is north, then the deviation line should also be north of the declination line, and if the lunar shadow is south, then the deviation line should be drawn south of the declination line. The reason for this is clear from figure 100 page 467. If we assume that 'ch' in this figure is the center of the sun's image and the ecliptic is the path of the moon, then if the lunar shadow is south, the moon touches the sun's image at a point to the south of the sun's image. That is, when the lunar shadow is south, the moon touches the sun's image to the south. Similarly, it can be proven that if the lunar shadow is north, then it will touch the sun's image to the north.

But in lunar eclipse it is opposite to this. This is also clear from this figure, if we assume 'cha' as the center of the umbral shadow. In the figure, the lunar arc is shown to the south. In such a situation, the umbral shadow touches the lunar image at a point which is to the north of the lunar image. Similarly, if the lunar arc is to the north, it can be proven that the umbral shadow will touch the lunar image to the south. Therefore, this rule is established that in a lunar eclipse, the direction of the point of contact should be opposite to the direction of the lunar arc, that is, in a lunar eclipse, the line of deviation and the line of declination should be drawn in the direction opposite to the direction of the lunar arc.

1. If 'ch' is taken as the center of the solar image and its circle is taken as the solar image, then all matters related to solar eclipse can be known from this figure.

SOURCE PAGE 577

The direction of the deflection of the nodal point in the time of salvation should also be determined according to this rule. If the direction of the lunar shadow is south, then the salvation of the moon in a lunar eclipse is in the latter part of the lunar image, as shown in the above figure where the moon is shown in the position of eclipse. But in a solar eclipse, the salvation of the sun is in the latter part of the solar image to the south. Similarly, if the direction of the lunar shadow is north, then the salvation of the moon is in the latter part of the lunar image to the south and the salvation of the sun is in the latter part of the solar image to the north.

The direction of deflection can also be determined by this rule even in the time of mean eclipse. From the same figure, it is revealed that when the lunar shadow is south, then in the middle period of a lunar eclipse, the center of the umbral shadow is to the north of the lunar image, but in the middle period of a solar eclipse, the moon is to the south of the center of the sun image. Similarly, when the lunar shadow is north, then in the middle period of a lunar eclipse, the center of the umbral shadow is to the south of the image and in the middle period of a solar eclipse, the moon is to the north of the center of the sun image.

Verse 6—In the time of the mean eclipse, if the direction of the lunar shadow and the direction of the epicycle are the same, then the nodal point should be made to the east of the north-south line. But if there is a difference in their directions, i.e., if the epicycle is north and the lunar shadow is south, or if the epicycle is south and the lunar shadow is north, then the nodal point should be made to the west of the north-south line. But it should be remembered that if the lunar shadow is south, then the nodal point should be made to the east or west of the north point, and if the lunar shadow is north, then the nodal point should be made to the east or west of the south point.

But for drawing the arc of the middle period of a solar eclipse, what is said in relation to the lunar eclipse should be opposite to it. That is, if the direction of the lunar shadow and the umbra are the same, then the umbra point should be towards the west from the north-south line, and if there is a difference in their directions, then the umbra point should be towards the east from the north-south line. Also, it should be noted that if the lunar shadow is south, then the umbra point should be made to the east or west from the south point, and if the lunar shadow is north, then the umbra point should be made to the east or west from the north point. This knowledge can be easily obtained from figure 102. The shadow on the earth in relation to the lunar eclipse can be taken as the sun's image in relation to the solar eclipse, so this picture can serve for both the lunar eclipse and the solar eclipse.

SOURCE PAGE 578

ch = earth shadow sun image center.

c = center of lunar image when the moon is south

cha = lunar reflection center when the moon is in the uttara is.

pu = the point before the image whose circumference is the letter

p = the point of the circle whose circumference is the letter.

U = the point of the celestial sphere whose circumference is the letter.

D = the point of the celestial sphere whose circumference is the letter.

KRAKA = ecliptic

KA = meridian circle

KRIKRI = circle parallel to the ecliptic passing through the center of the Moon

In this figure, the meridian is shown towards the north. Therefore, the ecliptic passing through each of the centers of the images is towards the north of the point of the meridian. The following statements written below this figure are self-evident:-

(1) When the Moon is in the shadow of the Sun and Earth on the Moon—

Lunar shadow center lunar north point declination

SOURCE PAGE 579

(2) At the time of lunar eclipse when the Moon is at C and the Earth's shadow is at S - the center of the Earth's shadow is to the north of the lunar subtension point, to the east of point D (3) At the time of solar eclipse when the Moon is at C and the Sun is at S - the center of the Moon is to the north of the Sun's image subtension point, to the east of point D (4) At the time of solar eclipse when the Moon is at D and the Sun is at S - the center of the Moon is to the north of the Sun's image subtension point, to the west of point U

Similarly, if a line is drawn from the center of each image to the point P to the south of the point P, then the direction of the subtension will be towards the south. In this case too, it will be clear that the rule of verse 6 is absolutely correct. When drawing the figure, it is necessary to keep in mind that the line joining C to P and P to C should be at a right angle to the ecliptic or the circle of declination.

Verse 10—When the subtension point of the mean eclipse period according to verse 6 is known, then it only remains to know at which point the center of the observer lies on the line joining the center of the image and the subtension point. It is obvious that in the mean eclipse period, the difference between the centers of the image and the observer images is the same as the arc of the Moon. Therefore, the observer's center should be measured and fixed on the difference of the lunar arc in the direction of the subtension point from the center of the image.

Verse 11—The circle drawn with the diameter of the observer image from the same center of the observer will indicate the observer's image. The part of the image that this circle covers will be the

covered part of the image. If the entire image of the moon is covered by the observer circle, then total eclipse will occur, otherwise partial eclipse will occur. Its proof has been explained in relation to figure 66 of page 461.

Verse 12—This verse states that a line can be drawn on a flat surface or on the plank of wood or any other object. Paper can also be used in place of the celestial sphere nowadays with ease.

SOURCE PAGE 580

In the latter part of this verse it is stated that the order of directions in the circumscription of the anterior skull should be opposite to that in the posterior skull. But this is not understood because if the touch of the eclipse is in the anterior skull and moksha in the posterior skull, as usually happens, then the circumscription of the time of eclipse or conjunction should be different from the circumscription of the time of inclination or moksha. But such a thing is neither convenient in practice nor very necessary. Apart from these, the methods for knowing the directions of conjunction and inclination mentioned in the following verses are only possible if the same circumscription is taken. Other Acharyas have not written anything in this regard. Only in the verse 1 26 of the chapter on eclipse response of Brahma Sphuta theory is written that if a circumscription is made on the celestial sphere, the directions marked on it will be opposite to the directions of the terrestrial circumscription. The reason for this is that the order of directions in the terrestrial circumscription is that which is stated in the verses 1-4 of the Chapter on the Three Problems. But there is also the facility in the circumscription of the celestial sphere that we can turn it towards the image of the grasped, and easily know the direction of the touch or moksha point. In such a situation, on the celestial sphere, east will be to the left hand, west to the right hand, north to the north, and south to the south. But in the terrestrial circumscription, east is to the right hand, west to the left hand, north to the north, and south to the south.

The commentators of Sūrya Siddhānta have written that one should make a difference in the order of directions by the difference of east or west skull. But I do not yet know the reason for this, so I do it against the method as explained in Brahma Sphut Siddhanta. I hope some learned will express their opinion on this and kindly explain its reason.

When is it possible to see an eclipse:-

> स्वच्छत्वारषोडशांशोऽपि ग्रस्तश्चन्द्रस्य दृश्यते ।
>
> लिप्तात्रयमपि ग्रस्तं तीक्ष्णत्वान्न विवस्वतः ॥१३॥
>
> १. प्राच्यपरे विपरीते विपरीतं मध्यवलनमर्केन्द्वोः ।
>
> पूर्ववदन्तत् सर्वं फलके स्वे ग्रहण परिलेखाः ॥ २६ ॥

As Sudhakarji does this commentary—the work is opposite to the fruit. On the earth, the point of the pole and the point of the west and the point of the fruit, the point of the pole is the work. The deviation is in the opposite direction.

SOURCE PAGE 581

Translation—(13) Even if the 12th part of the moon is also eclipsed, it can be seen due to clearness, but if the three aspects of the sun are also eclipsed, it is not seen due to the sharpness of the sun.

Scientific Commentary—Commentators have revealed a big difference of opinion in interpreting this. Acharya Ranganath ji, and his followers Madhav Purohit ji and Pandit Indra-Narayan Dwivedi ji interpret this that even if the 12th part of the moon is eclipsed, it is not seen due to clearness. But this meaning does not seem right to me. The word clearness cannot be taken as sharpness. From the word clearness, it is understood that the light of the moon is clean or clear, therefore even if the twelfth part is eclipsed, it can be seen clearly with clearness. The meaning as I have written above is the same as Shri Vignanananda Swami has done on page 203 of his Bengali translation.

In this regard, Bhaskaraacharya1, Brahmagupta etc. have written that if the eclipse of the moon is less than the 16th part, it cannot be seen and if the eclipse of the sun is less than the 12th part, it cannot be seen. From this, the meaning of the aforementioned verse of Sūrya Siddhānta is the same as what I have done. Brahmagupta ji has used the word clearness in this meaning as revealed by his quotations.

Drawing the path of the center of the shadow—

> स्वसंज्ञिताः त्रयः कार्या विक्षेपाग्रेषु विन्दवः ।
> तत्र प्राङ्मध्ययोर्मध्ये तथा मौक्षिक मध्ययोः ॥१४॥
> लिखेन्मत्स्यो तयोर्मध्यमुखपुच्छविनिर्गतम् ।
> प्रासार्य सूत्र द्वितयं तयोर्मंत्र मुलिर्भवेत् ॥१५॥
> सूत्रेण विलिखेद्वृत्तं तत्र विन्दुत्रयं स्पृशन् ।
> स पन्था ग्राहकस्योक्तः तेनाऽसौ सम्प्रयास्यति ॥१६॥

> १. इन्दोर्भागः षोडसः खण्डितोऽपि तेजः पुञ्जच्छन्नभावान्न लक्ष्यः ।
> तेजस्तैक्ष्यात् तीक्ष्णगोर्द्वादशांशो नादेश्योतोऽल्पोग्रहो बुद्धि मद्भिः ॥३७॥
> —सिद्धान्त शिरोमणि, गणिताध्याय चन्द्रग्रहणाधिकार
> वलनादि शशिवदन्यद् ग्रहणं तैक्षण्याद्रवेरनादेश्यम् ।
> द्वादशभागादूतं स्वच्छत्वात् षोडशादिन्दोः ॥२०॥
> —ब्राह्मस्फुट सिद्धान्त; सूर्यग्रहणाधिकार

SOURCE PAGE 582

Translation—(14) The locations where the center of the customer is in the touch, mean, and moksha times are determined by the dispersion points. Draw a fish with the touch and mean points and with the mean and moksha points (15). When the lines drawn by extending the lines that divide each fish into two equal parts and pass through its mouth and tail meet at a point (16), consider that point as the center and draw an arc that touches the aforementioned three points; this is the path of the center of the shadow during the eclipse.

Scientific Commentary - If two arcs are drawn by taking each of the two points as the center and drawing them at the distance of the other point, the area formed between them is fish-shaped. This shape is called Timi or Matsya (see page 223). The rule for making such a fish is stated in the 14th verse. When the common chords of the two fishes made from the centers of the shadow of the touch and mean times and the centers of the shadow of the mean and moksha times are extended, the point where they meet is considered the center of the path of the shadow centers, and the arc that touches the shadow centers from this center is considered the path of the shadow center. This is like the rule of drawing the imaginary line in the 41st verse of the Three Problems, and it is similarly tangible. The difference

between the path of the shadow center proved by this rule and the real path is very small. Therefore, the work taken according to the following verses is sufficient for practice.

Drawing the eclipse path at a certain time -

> ग्राह्यग्राहकयोगार्धात्प्रोज्झ्येष्टग्रासमागतम् ।
> अवशिष्टाङ्गुलसमां शलाकां मध्यविन्दुतः ॥१७॥
> तमोमार्गोन्मुखों दद्याद्भसतः प्रग्रहाश्रितम् ।
> विमुञ्चतो मोक्षदिशं ग्राहकाध्वानमेव वा ॥१८॥
> स्पृशेद्भत्र ततो वृत्तं ग्राहकार्धेन संलिखेत् ।
> तेन ग्राह्यं यदाक्रान्तं तत्तदा ग्रासमादिशेत् ॥१९॥

(17) Subtracting the measure of the auspicious time known from mathematics from the segment, the remaining part is taken as an angula, and with this, a similar string or straight stick is taken from the center of the meridian (18) If the auspicious time is before the middle time of the eclipse, then towards the point of contact and if the auspicious time is after the middle time, then the point of salvation

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SOURCE PAGE 583

Place it on the line of the center of the shadow and see that when one end of the sundial is at the center, where does its other end touch the line of the shadow, (16) where it touches, there will be the center of the shadow at the auspicious time. Taking this point as the center, the circle drawn with the semi-diameter of the shadow will be the image of the shadow at the auspicious time. The part of the shadow that this image covers will be the part afflicted at the auspicious time, and the circumference of this time will be the circumference of the auspicious grasp.

Sūrya Siddhānta—this work can be done easily with the help of Parakar. The essence of these three shlokas is that when the grasp of the lunar eclipse according to shlokas 18-20 of Chandra Graha Adhikar is known, then how to draw its perimeter. It is stated in Figure 66 of page 461 that the part of the moon in eclipse is j = chj + cz, where ch = maanakyaradh—the difference between the center of the umbral shadow and the center of the Earth's shadow. Therefore, if the part in eclipse is subtracted from maanakyaradh, the distance between the centers of the umbral and the umbrage can be found. When this distance is known and the center of the umbrage and the path of the umbrage are known, then the position of the umbrage is not very difficult to know. If the distance between the tips of both sides of Parakar is made equal to the distance between the centers of the umbrage and the umbrage, and a bow is drawn taking the center of the umbrage as the center, it will cut the path of the umbrage at two points. The point that is towards the point of tangency from the midpoint is the umbrage before the middle period, and the point that is towards the point of escape from the midpoint is the umbrage after the middle period. Knowing this point, the circle drawn with the semi-diameter of the umbrage will cover the umbrage up to where it covers the umbrage, which will be the eclipsed part. In this way, the perimeter of any auspicious time can be drawn easily.

The method of drawing the beginning or end of a total eclipse—

> मानान्तरार्धकमितां शलाकां ग्रासदिङ्मुखीम् ।
> निमीलनाख्यां दद्यात्सा तन्मार्गे यत्र संस्पृशेत् ॥२०॥
> तेतो ग्राहकखण्डेन प्राग्वन्मण्डलमालिखेत् ।
> तद्ग्राह्यमण्डलयुतिर्यत्र तत्र निमीलनम् ॥२१॥
> एवमुन्मीलने मोक्षदिङ्मुखं संप्रसारयेत् ।
> विलिखेन्मण्डलं प्राग्वदुन्मीलनमथोक्तवत् ॥२२॥

(20) from the center of the plate, that is, from the center of the accepted image, in a manner similar to the segment of the segment on the path of the shade-caster, place it towards the point of contact.

SOURCE PAGE 584

One end of the skylark touches the center and the other end touches the path of the shadow. At this place, the center of the shadow is at the time of conjunction. (21) If a circle is drawn by taking the radius equal to half the diameter of the customer's image, considering this place as the center, it will touch the point on the customer's image, and the conjunction will begin at that same place. (22) Similarly, if the skylark, which is equal to the segment of the other part, is placed towards the point of liberation, then considering the point where the end of the skylark will touch the path of the shadow as the center, the circle drawn with a radius equal to the diameter of the customer will touch the customer's image at that very place, and the revolution will occur there, meaning the end of the all-grasp will be from this point.

Sūrya Siddhānta—there is no much need to explain this, as it is clear by itself in the picture. The difference between the centers of the ecliptic and the ecliptic in the time of conjunction or opposition is like the segment of the mean motion. Therefore, when we know the center of the ecliptic, the path of the ecliptic, and the difference between the centers of the ecliptic and the ecliptic, it cannot be difficult to fix the center of the ecliptic. Yes, it should be kept in mind that when we want to draw the arc of the conjunction time, we should keep the direction of touch and when we want to draw the arc of the opposition time, we should keep the direction of release. This work can also be taken easily nowadays. By making the difference of both ends of the arc like the segment of the mean motion, by placing one end at the center and drawing an arc from the other end that will cut the path of the ecliptic at two points. The point that is towards the touch will be the center of the ecliptic in the time of conjunction, and the point that is towards the release will be the center of the ecliptic in the time of opposition. When the center of the ecliptic is fixed, it is not difficult to know the beginning and end of the total eclipse by drawing a circle with a radius equal to the diameter of the ecliptic.

What is the color of the image of the observer--

> अर्धादूनं च धूम्रं स्यात्कृष्णमर्धाधिकं भवेत् ।
>
> विमुञ्चतः कृष्णधूम्रं कपिल सकलग्रहे ॥२३॥

(23) When less than half of the lunar image is covered, the covered part has a smoky color. When more than half is covered, the covered part appears black. When a large part of the lunar image is covered and only a little remains, the color of the covered part is black with a reddish tinge. But

SOURCE PAGE 585

The color of the total eclipse is reddish-brown. (In a solar eclipse, the color of the part of the sun being eclipsed is always black.)

Scientific Commentary—As long as the moonlight remains bright, the eclipsed part appears smoky or black in comparison. But when only a small part of the moon remains, its light becomes without heat. Therefore, the color of the eclipsed part also appears somewhat red. The reason for this redness is that the subtle light of the sun, refracting through the atmosphere, falls on the lunar image, so some redness appears on the black eclipsed part. At the time when the entire lunar image falls into shadow, the lunar image does not appear black but brownish with redness. The reason for this is also the refracted light of the sun, which falls on the moon by passing through the Earth's atmosphere. If there were no atmosphere, the color of the eclipsed part of the moon would always be black, as the color of the eclipsed sun.

Due to the rotation of the atmosphere, sometimes another astonishing phenomenon is also seen. When an eclipse occurs during the rising or setting time, the eclipsed moon is seen in the presence of the shining sun, where one side of the moon is eclipsed and the other side of the sun keeps the Earth illuminated by its brightness. Such phenomena were seen in the years 1666, 1668, and 1750 AD.

The secret of drawing should be kept secret—

> रहस्यमेतद्देवानां न देयं यस्य कस्यचित् ।
> सुपरोक्षितशिष्याय दातव्यं ज्ञानमुत्तमम् ॥२४॥

Translation—(24) The knowledge of drawing is a secret of the gods. This knowledge should not be revealed to just any person. The disciple who has been well-tested should reveal this excellent knowledge.

Scientific Commentary—The essence of this is that the method of drawing cannot be understood easily, so there is no benefit in telling one who does not understand its elements well. The only one who can be qualified for this knowledge is the disciple who can understand its secrets.

The translation of Chapter 6 named Parilekhaadhikar has been completed.

SOURCE PAGE 586

Now, by giving an example of drawing the diagram of lunar eclipse, it will be shown how Western modern astronomers calculate solar eclipse and how they know in which places the total eclipse is seen and in which places how much eclipse is seen. After this, it will also be briefly explained how the eclipses of Khaldia and Greece were calculated. The example of drawing the diagram of a solar eclipse is omitted due to fear of expansion.

Example—Drawing the diagram of the lunar eclipse of the full moon of the year 1681 B.E.—

It is apparent that instantaneous refraction and knowledge of the Moon's shadow are required to draw the path of the center of the shadow planet, and the refractions of the sparśakāla, madhyakāla and mokṣakāla and the knowledge of the lunar shadows are required to draw the path of the center of the shadow. The calculations of the refractions of the sparśakāla and mokṣakāla are done in lunar eclipse chapter pages 466-505. Therefore, the calculation of the refractions of the middle period of the eclipse should also be done.

Calculation of Latitude

Moon's full moon shadow = 8.76 or 8.8 degrees (page 465) Lunar fraction = 268°34’ (page 465) Precession angle = 22°40’ (page 468-466)

Full moon instantaneous lunar sine 321014’

Full moon lunar sine = sine 23°27’ sine 321°14’ = .3676 X sine 38°46’ = .3676 X .6261 = 2461 ∴ Full moon lunar mean revolution = 14°25’.3 South ““Shar 8’ 8 North.. ““True revolution = 14°16’.5 South time from sunrise to touch time of the sun of Kashi = 45 hours 54 minutes shadow position = 4 “42 “ time from sunrise to middle of eclipse = 50 “36 “ time from sunrise to midnight (page 500) = 46 11 time from midnight to middle of eclipse = 4 “25 “

SOURCE PAGE 587

Therefore, the center of the Earth's shadow during the eclipse was 'west of the shadow time' 4 hours 25 minutes or 265 minutes or 1560 arcminutes. This was also the time of the moon during the eclipse because the portions of the umbra and the center of the moon are equal at this time. Therefore

The time of the moon during the eclipse = 1560 arcminutes = 1560' = 26°30' The sine of the mean position of the moon during the eclipse = 25°20' Sine 14°16'5 = 4734 x 2544 = 1204 ∴ The mean position during the eclipse = 6°55' According to the equations (g) of page 262, the meridian altitude of the moon during the eclipse = (sine 26°30' - sine 6°55') x sine 25°20' x sine 14°16'5 = (.8646 -.1204) x .6038 x .6661 = .7745 x .6038 x .6661 = .6784 ∴ The meridian altitude during the eclipse = 47°17' According to page 276, sine agra = sine 14°16'5 sine 47°17' x sine 25°20' + cosine 47°17' x sine 25°20

SOURCE PAGE 588

, x y g = 32031'

∴ The ntaansh of the mean time position of the moon = 32031'

$$\therefore \sin(\text{latitude}) = \sin 320^\circ 31' \times \sin 25^\circ 20' \cos 14^\circ 16'.5 = .5375 \times .4276.6661 = .2300.6661 = .2373$$

∴ Latitude = 13°44’ South, because the moon is in the western skull. The sine portion of the mean time moon = 321°14’

Adding 60° to this, the sine portion of the moon = 51014’ whose declination will be north, therefore the latitude will be north.

Sūrya Siddhānta

Scientific Commentary

Chapter on Mean Motions

Chapter on True Positions

SOURCE PAGE 589

Chapter on Mean Motions — Altitude = 4° 56' North

∴ sinAltitude = sin 4° 56' = 266' = 26670 = 4.23 inches; Lunar Distance = 8.83 = 2.63 inches (page 465)

Chapter on Mean Motions — Altitude = 3056' South (page 505)

∴ sin 3° 56' = 236' = 23670 = 3.4 inches; Lunar Distance = 3' North (page 500) = 1 inch

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SOURCE PAGE 590

The point P = circle, P, = north, east, south, and west points, S = the circle of intersection whose diameter is equal to the length of the segment, the smallest circle = lunar image C = lunar image, compound circle, and the center of the circle of intersection of the circle and the circle of intersection of the circle is the lunar ray or the radius of the circle whose angle is equal to the angle of the arc of the tangent at the east point, W = arc of the tangent, S = lunar ray, S = lunar ray and the intersection point of the compound circle. This is drawn to the south of the lunar ray because the moon is to the north and this is the lunar eclipse (Verse 8). V = point of the lunar deflection or the center of the lunar shadow. C = point of lunar deflection or the center of the lunar shadow. S = lunar deflection line, S = deflection line and the intersection point of the lunar image and the compound circle or the point of eclipse. D = the middle eclipse arc. V = middle eclipse lunar ray (Verse 6). C = middle eclipse lunar deflection (Verse 10). M = center of the shadow during the middle eclipse. The quantity of the middle or supreme shadow is known by drawing a circle with the diameter of the shadow, taking this as the center. P = moksha arc. W = moksha lunar ray. M = moksha lunar ray and the intersection point of the compound circle. M = moksha lunar deflection. This is also drawn to the south of the lunar ray. V = moksha deflection line. M = moksha deflection line and the intersection point of the compound circle or the moksha point of the eclipse. V = center of the moksha shadow.

SOURCE PAGE 591

mu, pu=the points of the fish's mouth and tail drawn on m and v by the center and end of the shadow of the middle eclipse and the time of salvation

mu pu = the points of the fish's mouth and tail drawn on m and v by the center and end of the shadow of the middle eclipse and the time of touch

k = the intersection point of mu pu and mu pu

v m v = the arc drawn by considering k as the center and k v as the radius, which is the path of the center of the shadow during the eclipse period (shlokas 14-16)

ch ni or ch u = the intermediate segment

ni= the center of the shadow in the time of diminution or culmination. The circle drawn from this center with the semi-diameter of the shadow touches the point where the lunar image begins. (shlokas 20-21)

U=center of the earth shadow in the time of motion. Taking this as the center, the circle drawn with the radius of the earth shadow touches the point where the lunar image touches, there is the end of the all-grasp. (Verse 22)

For all (see page 460) the semi-diameter of the Earth is = 43' · 67 = 43 · 67/3 = 14.66 inches. The semi-diameter of the lunar image is = 16' · 66 = 16 · 66 + 3 = 5.55 inches. The length of the meridian arc is = 60' · 63 = 60 · 63 div 3 = 20.21 inches. The length of the difference arc is = 27' · 31 = 27 · 31 div 3 = 9.1 inches.

Here I have divided each of the figures or arcs to know the finger measurement and explained three arts as one finger, as stated in the Scientific Commentary of verse 3 above. In this treatise, one finger of the lunar circle is taken as equal to 1 millimeter, and one finger for drawing other circles or arcs is considered as equal to one and a half millimeters.

Calculation of the direction of the touch point according to the new method—

Western astrologers do not calculate the direction of the point of contact from the direction of the pole circle, but calculate the direction of the touch or the point of motion from the direction of the polar circle. Therefore, they do not need to know the declination. The following method is also explained in brief:

SOURCE PAGE 592

ch = center of the Earth's shadow, dh = North Celestial Pole, ch = center of the Moon at the time of touch, ca = center of the Moon at the time of Moksha, sa = point of touch, m = point of Moksha, ch u dh = pole of the center of the Moon at the time of touch, u = point of the lunar image at the time of touch, u = point of Moksha...

SOURCE PAGE 593

l or l = perpendicular distance of the polar circle of the Moon from the center of the Moon

∠ Uchas = direction of the point before the moon's northern point to the point of contact ∠ Ucham = direction of the point after the moon's northern point

chadh = difference in declination of the center of the lunar image = 60° - moon's true time

chadha = difference of latitudes of the center of the shadow = 60° - declination of the shadow

Chad = difference in declination of the center of the lunar image = 60° - moon's nodal revolution

This is clear that in the time of contact or no return, the Moon remains very close to the Earth's shadow and the distance between these two is similar to the arc of a degree, the magnitude of which is approximately one degree, so in comparison, the difference in latitude of the Moon or the Earth's shadow is much larger than that of the Moon or the Earth's shadow. Therefore, the lines cha la, cha ca or cha lā, cha cā, the arc, and the straight lines and spherical triangles cha la or cā cha lā can be taken as simple triangles (Plane triangle) without any loss. From this reasoning, the difference in latitude can be considered equal to the difference in latitude because a perpendicular is drawn on the difference in latitude. Therefore, if the declination of the Moon and the declination of the Earth's shadow are the same,

chadh — chhad = chadh — chhad

If both the motion of the moon and the terrestrial shadow have a northern declination,

chadh - chhad = (60° - k) - (60° - ka) = ka - k

And if both have a southern declination,

chadh - chhad = (60° + k) - (60° + ka) = k - ka = = - ka - (- k)

That is, to know the magnitude of ch, in both cases, the declination of the terrestrial shadow should be subtracted from the declination of the moon. It should be remembered that northern declination is written as positive and southern declination as negative.

Therefore, jyauchas = jyalachcha = chalaka - kachchamanayakkhandbhuchayakikranti - chandramakikranti-manayakkhand

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SOURCE PAGE 594

In the same way in the time of moksha;

Kojyaucham = Kojyauchach = Chala / Chach = Ka - K / Manayakhand

It is necessary to remember here that when the moon is north of the umbra, the angle Uchasya or Uchacham will be greater than 60°, so its cotangent will be negative. But when the moon is south of the umbra, the angle Uchasya or Uchacham will be less than 60°, and its cotangent will be positive. The moon is north of the umbra when the north declination of the moon is greater than the north declination of the umbra, or the south declination of the moon is less than the south declination of the umbra. In the opposite case, the moon is south of the umbra.

Example - to know the directions of the touch and moksha points of the above lunar eclipse according to the recent method-

The touch-periods of the moon are known. Therefore, the touch-periods and moksha-periods of the center of the earth should be known.

Geographical touch time fraction = 268°26‘5

Precession angle = 22040 ′

∴ Earth's touch time sine fraction = 32106‘5

∴ Earth's true position of declination = sin23°27' X sin321°6' .5 = sin23°27' X sin(360° - 38°50'.5) = -sin23°27' X sin38°50'.5 = -sin.3676 X.6271 = -sin.2466

∴ Geographical touch inclination = 14°27'·3; South geographical motion-time fraction = 268°38‘·5

Precession angle = 22040 ′

Bhūbhākamokṣālikāsyānbhogāṁśa = 321018' 5

∴ Bhūbhākamokṣālikkrāntijyā = $\sin^2$ 23° 27' X sin 321018' 5 = $\sin^2$ 23° 27' X sin (360° - 38° 41'.5) = - $\sin^2$ 23° 27' X sin 38° 41'.5

SOURCE PAGE 595

= − 3676 X.6251 = −.2487

∴ The nodal revolution of the Earth = 14° 24’ South

Longitude = geoid shadow's rotation - moon's rotation fraction = - 14° 27'.3 - (-14° 31'.4) = 60.63 = + 4.160.63 = + 0.0676

∴ Uchas = 86 ° 7 ′

That is, the eclipse will touch from 86°7’ east of the uttara point of the lunar image.

sine - arcsegment = -14° 24' - (-14° 2') 60 · 63 = -22' 60 · 63 = -3.3626

This value is negative. Therefore, the altitude angle is more than 60°, so subtracting 180 from the co-latitude whose co-ordinate is 3626 will yield the value of the altitude.

Koti jya 68^°43’ = 3626

Therefore, height = 180° - 68° 43′ = 111° 17′

That is to say, the eclipse will be relieved from the north point of the lunar image by 111017' west. According to the navigator's almanac, the touch of the eclipse is stated to be 84° east and the relief is 110° west from the north point. The reason for this difference is that according to the Sūrya Siddhānta, the method of calculating the declination is somewhat crude.

After explaining the method of drawing the lunar eclipse, the thought was to briefly write the method of calculating the solar eclipse according to the recent method called the Besselian method, but due to the lack of two books and many obstacles at this time

1. The names of these books are (1) Chauvenet’s Manual of Spherical and Practical Astronomy Vol. I and (2) Loomi’s Introduction to Practical Astronomy. This subject is explained very well in the first book. Both books are in the Allahabad Public Library but are currently inaccessible due to annual inspection.

SOURCE PAGE 596

Because of this, this desire cannot be fulfilled even with the time difference. It is hoped that this topic will be well explained in the appendix when the book is finished.

A few more things are written regarding the eclipse at this time and this chapter will be completed.

It is stated in pages 457-58 that an eclipse of the Sun or Moon is possible only when the Sun or Moon is near a node of the Sun, Moon, Rahu, or Ketu. Therefore, it is proven that the cycle of an eclipse depends on the motions of the nodes of the Sun and Moon. If the Moon's node were stationary, the Sun would reach near both nodes in a year twice in the same month, keeping the month and date of the eclipse constant. However, the Moon's node moves towards the west by 3^ h10" 64 daily, while the Sun's mean daily motion is 56^ h8" 33 towards the east. Therefore, the Sun moves 62^ h18" 67 or 62^ h16" away from the Sun-Moon node each day. When the Sun moves this far each day, it reaches the same node in 360^ ^ extdegree / 62^ h16" = 1266000 / 3736 = 346.62 days. It takes half this time, 173.31 days, for it to reach the other node. If the cycles of Amavasya or Purnima also completed in such a short time, eclipses would be seen after every 346.62 or 173.31 days. However, the mean value of the lunar month is 26.53056 days, which is similar to 324.83647 days in 11 months and 354.36705 days in 12 months. Thus, it is proven that the cycle of an eclipse cannot occur in 346.62 days.

But there are 223 X 26053056 days or 6585032 days in 223 lunar months and 346062 days in 16 revolutions of 346062 days, so the cycle of eclipses, i.e., eclipse-cycle, is of 6585032 days. After these days, the same type of eclipses begin again. Therefore, this period can be called the eclipse-cycle. This cycle is not discussed in our ancient astrology. In Western astrology, it is called Saros and it was determined by the inhabitants of Khaldia about six hundred and five years before the beginning of the Vikram Samvat.

This eclipse cycle gave the people of Khaldia great facility in finding eclipses because without long calculations, they could easily know when eclipses will occur in the table of eclipses of 6585.32 days. But it should be remembered that this cycle (age) is calculated according to the mean motions of the Sun, Moon, and Rahu, so it has a little imprecision. Second,

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This era is not for the entire 6585 days, but rather seven to eight hours more. Its result is that in the same place and at the same time, the same eclipse will be seen sometimes and sometimes not. For example, when the lunar eclipse was seen at sunset in Prayag, after 6585 days, seven or eight hours after sunset, it will again be seen at 2 o'clock at night. But the third time this eclipse will occur when the sunrise has already happened in Prayag. Therefore, it will not be seen in Prayag, but in the place west of Prayag where the night will remain at the time of the eclipse.

There are 365.25876 days in a solar year. Therefore, in 18 years, 6574.6577 days have passed, which is only 10.66 days less than the eclipse cycle. Therefore, it is apparent that if the beginning of the eclipse cycle happened on the day of Aries transit, then the beginning of the second cycle will be 10.66 days after the Aries transit, and the beginning of the third cycle will be on the 21.32 days from the Aries transit.

How many eclipses can occur on one node—there are 26.53 days in one lunar month, so 14.765 days have passed on one side. It is stated above that in 1 day, the Sun is 62°16' away from Rahu. Therefore, on one side, 14.765 x 102°16' = 15°20' 6' away. If the day of the full moon is on the node, then a total lunar eclipse will surely occur on this day. At this time, the Sun will be on the second node, so it will be 15°20' ahead and behind the two new moons on both the preceding and succeeding days, which is less than the maximum limit of a solar eclipse of 18°5. Therefore, a partial solar eclipse may occur on these two new moons (see pages 463-64). Thus, a maximum of three eclipses can occur in one lunar month

while the Sun is 15°20' ahead and behind one node. However, these three eclipses are seen from the same place very little.

1. The discussion of two eclipses on one side in the Mahabharata is as follows:-

> चतुर्दशीं पंचदशीं भूतपूर्वा च षोडशीं । इमां तु नाभिजानेहममावस्यां त्रयोदशीं ॥
> चन्द्रसूर्यावुभौ ग्रस्तावेकमासीं त्रयोदशीं ॥३२॥ भीष्म पर्व अध्याय ३

Here is not just a discussion of two eclipses on one side, but also that one side has become 13 days. It has been expressed wonder that the sides of 14015 and 16 days have been seen, but the side of 13 days has not yet been heard. On this, Sw. Shankarbalakrishna Dikshit has given a good exposition on pages 114-115 of his Indian astrology and stated that the lunar

SOURCE PAGE 598

If the Sun is on the node on the day of amāvasyā, then there will definitely be a solar eclipse on this day. The Sun will be 15°20’ before or after this node before or after the full moon that comes before or after it. Therefore, the Moon will also be that much ahead or behind the second node on the full moon day. However, the maximum limit of lunar eclipse is 12036 (see page 108). Therefore, the Moon cannot be afflicted because it is farther than the maximum limit from the node on the full moon day. Thus, it is proven that in such a state, only one eclipse can occur on one node, and it is the solar eclipse of the entire. Therefore, at least one solar eclipse and a maximum of three eclipses (two solar eclipses and one lunar eclipse) can occur on one node.

How many eclipses can occur in a year—it is stated above that the Sun takes 273 days to go from one node to another and 177 days in 6 lunar months, so if the Sun is two degrees before and the Moon-

The solar eclipse on the day of Amavasya 13 days after the eclipse cannot be seen from the same place. My opinion on this is as follows:-

The wonder of the 13-day period is because at that time the values of the dates were known according to the mean calculation of Vedanga astrology, according to which there are 14 days 45 hours 26 minutes in one side. In this situation, the 13-day period is considered impossible, which is not surprising nowadays because according to the true calculation, the 13-day period has occurred many times and will continue to occur. In that ancient time, the 13-day period was known from the observation of eclipses. This is possible in this way:-

According to the true value, there are at least 13 days 50 hours in one side. Suppose that the full moon was one hour after the sunrise of the 11th date and the moon afflicted by this day set. In such a situation, it is evident that the beginning of the side will be considered on the 12th date. If the side is 13 days 57 hours, then the end of the new moon will be at 58 hours from the sunrise on the 24th date. If there is also an eclipse in the sun, the eclipsed sun will rise on the 25th date and the eclipse will be relieved shortly thereafter. From this, it is naturally apparent that the new moon ends on the night of the 24th date. In this way, the first day will be calculated on the 12th date and the new moon on the 24th date, and the side of 13 days will be seen. Such an event must have happened in the time of Mahā-Bhārata.

SOURCE PAGE 599

If an eclipse occurs, a solar eclipse may occur both before and after both new moons. From this lunar eclipse, the sun will be 2 degrees behind the second node 177 days later. Therefore, there will be a lunar eclipse at this time as well. Because the new moon before this lunar eclipse will be 13 degrees ahead of the second node from the sun, it will be afflicted, and the new moon behind it will also be afflicted because the maximum extent of a solar eclipse is about 18 degrees. In this way, according to three eclipses on both nodes, there are 6 eclipses. However, the sun will reach the first node in 346 days, so a solar eclipse may occur after 346 days. In this way, if 3 eclipses occur in the first month starting from a solar eclipse at the beginning of the year and three more eclipses occur in the middle of the year, then one more solar eclipse may occur at the end of the year. In such a situation, seven eclipses can occur within the same solar year. However, in a year of 12 lunar months or in the solar year starting from the Aries solstice, if there are no intercalary months, there will be only 6 eclipses because when the beginning of the year is considered from the first day of the waxing moon in Chaitra, the first lunar eclipse will be on the 15th day of the waxing moon in Chaitra. The solar eclipse before this will fall on the new moon of Chaitra, which will be counted in the previous year. In this way, although seven eclipses can occur in a year of 365 days, in a solar year starting from the Aries solstice or a lunar year starting from the waxing moon in Chaitra, at most only 6 eclipses will be seen. Of these 6 eclipses, 4 will be of the sun and 2 of the moon. If a maximum of 7 eclipses are considered in a year, there will be 5 solar eclipses and 2 lunar eclipses.

It has already been proven that if there is a solar or lunar eclipse at a node or three parts ahead or behind it, then there cannot be lunar eclipses of the full months coming before or after it. Therefore, there will be only one solar eclipse at this node. There may be only one solar eclipse at the second node as well. Therefore, at least 2 eclipses of the sun must occur within the year and these will be solar eclipses.

People will doubt this that solar eclipses are seen very little and lunar eclipses are more. The reason for this is that lunar eclipses are seen from most of the Earth's surface and solar eclipses, even though they occur many times, can be seen from very little part of the Earth's surface, so the number of solar eclipses seems less and the number of lunar eclipses seems more from the same place. But if the number of all eclipses in the world is considered, it proves that the number of solar eclipses is much more than the number of lunar eclipses.

SOURCE PAGE 600

In ten days, there are often 71 eclipses in the eclipse-cycle or eclipse-age of 18 years above, of which 41 are solar eclipses and 26 are lunar eclipses. The ratio of these two is the same as the ratio of the ultimate limits of solar and lunar eclipses.

From a place, the sarvagrāsa or kaṃkaṇa solar eclipse is seen very little, although in an eclipse-cycle, the number of sarvagrāsa and kaṃkaṇa solar eclipses of the whole world is about 28. According to the Western astronomer named Hallie, no sarvagrāsa solar eclipse was seen in London from March 20, 1140 AD to April 22, 1715 AD.

However, the total solar eclipse is an event of great importance and does not last more than seven and a half minutes or 16 pala in any place. Even for such a short time, modern Western astronomers travel to distant jungles, mountains, seas, or islands where there is a possibility of better viewing facilities by spending lakhs of rupees. This proves that the Sun is not a solid body. Flames are seen around it, and examinations prove that it also contains gaseous substances like hydrogen. However, this discussion has no special relation to the eclipse, so the chapter is concluded without writing anything more here.

SOURCE PAGE 601

Seventh Chapter

Chapter on Planetary Motions

Verse—1 The war, conjunction, and setting of the planets. Verse 2 and 3 The first half of the conjunction has already happened or is about to happen? The latter half of Verse 3, 4, 5, 6—when and where will the conjunction happen. Verses 7-10—the method of sighting. Verse 11—where the necessity of sighting is. Verse 12—what is the mutual difference of the Sanskrit planets at the time of their conjunction. Verses 13-14—the rules for knowing the mean and true values of the images of five stellar planets. Verses 15-17—the method of sighting to determine the directions of the planets in the era of conjunction. Verses from 18 to 22—definition of various types of wars. Verse 23—imagination of wars to know auspicious and inauspicious fruits.

In this chapter, the method of knowing when and where the planets appear very close to each other and what their auspicious and inauspicious fruits are is explained. The war, conjunction, and setting of the planets

> ताराग्रहाणामन्योन्यं स्यातां युद्धसमागमौ ।
>
> समागमः शशाङ्केन सूर्येणास्तमयस्सह ॥१॥

Translation(1)—The Earth, Mercury, Jupiter, Venus, and Saturn have a war and conjunction among the five stellar planets. When a stellar planet becomes with the Moon, then it has a conjunction with the Moon, and when the planet becomes with the Sun, it is said that the planet has set.

To know whether the conjunction has happened or is about to happen—

> शीघ्रं मन्दाधिकेऽतीतः संयोगो' भविताऽन्यथा ।
>
> तयोः प्राग्यायिनोरेवं वक्रिणोस्तु विपर्ययात् ॥२॥

Pragyāpinyadhike'tiḥ vakriṇyaiṣyasamāgamaḥ.

Translation(2)—If the portion of the planet whose motion is slow in the desired time is greater than the portion of the planet with fast motion and both planets are in the meridian, that is, east

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If they are moving towards each other, it should be understood that their conjunction has already happened before the auspicious time. But if the portion of the fast-moving planet is less than the portion of the slow-moving planet, then it should be understood that the conjunction is yet to happen. But if both planets are retrograde, i.e., moving towards the west, then the opposite of what is said above should be understood, i.e., if the portion of the fast-moving planet is more, then it should be understood that the conjunction is about to happen, and if it is less, then it should be understood that the conjunction has already happened. (3)

Scientific Commentary-Suppose in the given figure, M is the initial point of Aries and K, Kh are two planets. It is clear that its portion is more than the portion of K. If the motion of Kh is more than the

motion of K, then it is apparent that Kh will move further away from K and their conjunction has passed. But if the motion of Kh is slow

SOURCE PAGE 603

Then it is clear that the moving with great speed will reach near to X and both will coincide. These two events will occur when both planets are in the direct direction, that is, moving in the direction of the arrow. If both are retrograde, that is, moving in the opposite direction of the arrow, then if the retrograde motion of X is more, then there will be coincidence and if less, the coincidence has already happened. If X is direct and K is retrograde, then the coincidence has already happened, but if X is retrograde and K is direct, then the coincidence is going to happen.

To know at what time and in what place the planets will coincide—

> ग्रहान्तरकलास्स्वस्वभुक्तिलिप्तासमाहताः ॥३॥
> भुक्त्यन्तरेण विभजेदनुलोमविलोमयोः ।
> द्वयोर्वक्रिण्यथैकस्मिन्भुक्तियोगेन भाजयेत् ॥४॥
> लब्धं लिप्तादिकं शोध्यं गते देयं भविष्यति ।
> विपर्ययाद्वक्रगतावेकस्मिन् तद्धनक्षयौ ॥५॥
> समलिप्तौ भवेतां तो ग्रहौ भगणसंस्थितौ ।
> विवरं तद्वदुद्धृत्य दिनादिफलमिष्यते ॥६॥

Translation—(3) Find the difference of the portions of both planets of the auspicious time and multiply it separately by the arts of the daily motion of each planet. (4) Divide each product by the difference of the arts of the daily motions of both planets if both planets are direct and both planets are retrograde. But if one planet is retrograde and the other is direct, then divide the above product by the sum of the arts of the daily motions of both planets. (5) If the coincidence of both planets has already happened and both planets are direct, then subtract the portion multiplied by its daily motion from the portion of that planet; but if the coincidence is going to happen, then add the portion to the portion of the planet. If both planets are retrograde, then the opposite action should be done, that is, if the coincidence has happened, then add the portion to the portion of the planet and if it is going to happen, then subtract it. If one planet is retrograde and the other is direct, then according to these rules, where as much as needed, addition or subtraction should be done (6) By doing this, the portion of the circle is found where the coincidence of both planets has already happened or will happen. The difference of the portions of both planets is their

SOURCE PAGE 604

When the difference between the daily motions is divided, the resulting distance is the time before or after which the conjunction has already occurred or will occur.

Scientific Commentary—The two rules explained from the latter part of the 3rd verse to the end of the 6th verse are very similar to the rules of "time and distance" in arithmetic. An example of this is—a passenger train started from Prayag towards Patna at 6 o'clock in the morning at a speed of 25 miles per hour, and the mail train started in the same direction at 8 o'clock at a speed of 40 miles per hour. Then tell us where and when both will meet?

The same method is used to calculate the conjunction of the planets. In such questions, it is first necessary to know at what time the mail train started, what the difference was between the passenger train and that time, then it is necessary to know how long it will take for the mail train to cover this difference by moving 15 miles more per hour. Here, 15 miles is the difference in the hourly speeds of both trains because both trains are moving in the same direction.

If the passenger train starts from Prayag towards Patna and the mail train starts from Patna towards Prayag at 6 o'clock, then by adding the speeds of both and dividing this sum by the distance between Prayag and Patna, the time at which both trains will meet will be known. Here, the speeds are added because both trains are moving towards each other, so their speed of meeting is equal to the sum of their speeds.

Similarly, the conjunction time and conjunction position of the planets are calculated. Suppose that in an auspicious time, the fraction of planet a in planet b is k = b and the fraction of planet x in planet y is k = a. Also suppose that in the same auspicious time, the daily motions of a and x are respectively g and ga.

Difference of both planets k x = bh - bh

Difference of the daily motions of both planets = g - ga

Therefore, if the conjunction that has already occurred or will occur before or after the ecliptic is said to be s, then s = bh - bh / g - ga

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SOURCE PAGE 605

The proof of the second verse—if g is greater than a, then h will be positive, from which s will also be positive, in this case the conjunction will happen after so many days. But if g is less than a, then because h is negative, s will also be negative, which means that the conjunction of both planets has already happened so many days ago. Here, the motions of both planets are considered positive. For simplicity, it should be remembered here that the geocentric motion is considered positive and the retrograde motion is considered negative.

If both g and a are negative, that is, if both planets are retrograde, then the every (-g) -(-a) = a-g will be the exact opposite of the previous one, that is, if g is less than a, the time will be positive and there will be a conjunction, but if g is greater than a, the time will be negative and the conjunction has already happened. Thus, the proof of the second verse is proven.

Proof of the first half of the third verse—if k is geocentric and x is retrograde, then g will be positive and a will be negative, so the every of the equation will be the same as g-(-a), which will actually be g+a, i.e., positive, so the conjunction will happen after the same time.

But if x is geocentric and k is retrograde, then g will be negative and a will be positive, in this case the every of the equation g-a=a-g+(a)=a-g-a=(g+a) which is negative, so the conjunction has already happened at that time.

Here it is also proven that to know the time of conjunction, the difference of the portions of both planets should be divided by the difference of the motions of both planets if both planets are geocentric or both planets are retrograde; but if one of them is geocentric and the other is retrograde, then the division should be done by the sum of their motions.

Proof from the latter half of the third verse to the sixth verse—the essence of these verses is:-

Idéal time: Time from Utkal to Yuktal = Bhā-bh / Ga-gā

The motion of a planet from Ideal time to Yuktal = g X Bhā-bh / g-gā,,,, kh,, = gā X Bhā-bh / g-gā

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Therefore, if the portion of G in the auspicious time of K is added to X (Bhā)/(Ga) and the portion of G in the auspicious time of K is added to Ga × Bhā/Ga, then the portion of K in the mean time will be found, which will be the same for both because the portions of both planets are the same in the mean time. Here, the value of Ga × Bhā/Ga will change according to the regular and retrograde motions of the planets, as previously stated, because when both planets are regular, both G and Ga will be positive, and when both planets are retrograde, both G and Ga will be negative. In these two situations, the value of Ga × Bhā/Ga will be the difference between the two. But if one is retrograde and the other is regular, then the value of Ga × Bhā/Ga will be the sum of the two, but this sum will be negative if G is negative and positive if Ga is negative. Thus, the derivation of the fourth verse is proven.

It is already assumed that in the auspicious time, the portions of the planets K and Kh are respectively Bh and Bha, and their motions from the auspicious time to the current time are respectively G x Bha-Bha / G-Ga and Ga x Bha - Bha / G-Ga, therefore their portions in the current time

Sequentially, there are bha + ga x bhā—bha / ga—gā and bha + gā x bhā—bha / ga—gā. The plus sign of these two values will be positive or negative according to the sign of the second term of each value, as previously stated. In this way, the derivation of the first half of the 5th and 6th shlokas is proven. The derivation of the latter half of the 6th shloka has already been proven.

It should be remembered that the portion of the geocentric position found in this manner will be gross because the daily motion of a planet at a certain time does not remain the same every day; it varies, increasing and decreasing. Therefore, calculating according to the daily motions of the ecliptic leaves some grossness. For this reason, it is necessary that the time obtained from the above calculation be re-derived from the portion of the planet and its daily motions of that time by independent calculation and then taken to the geocentric position according to the rules given in the above four shlokas.

Method of observation—

> कृत्वा दिनक्षपामानं ततो विक्षेपलिप्तिकाः ।
> नतोन्नतं सार्धयित्वा स्वकाल्लग्नवशात्तयोः ॥७॥

SOURCE PAGE 607

विषुवच्छाययाऽभ्यस्ताद्विक्षेपाद्द्वादशोद्धृतात् ।
फलं स्वनतनाडीघ्नं स्वदिनार्धविभाजितम् ॥८॥
लब्धं प्राच्यामृणं सौम्याद्विक्षेपात्पश्चिमे धनम् ।
दक्षिणे प्राक्कपाले स्वं पश्चिमे तु तथा क्षयः ॥६॥
सत्रिभग्रहजक्रान्तिभागघ्नाः क्षेपलिप्तिकाः ।
विकलास्स्वमृणंक्रान्तिक्षेपयोभिन्नतुल्ययोः ॥१०॥
नक्षत्र ग्रहयोगेषु ग्रहास्तोदय साधने ।
शृङ्गोन्नतौ तु चन्द्रस्य दृक्कर्मादाविदं स्मृतम् ॥११॥
तात्कालिकौ पुनः कार्यो विक्षेपौ तु तयोरथ ।
दिक्तुल्यत्वेऽन्तरं भेदे योगश्शेषं ग्रहान्तरम् ॥१२॥

(7) One should know the diurnal and nocturnal motions and their deviations of the planets of the Yugakal, then in that time, one should know the sidereal time and the advance time of each planet from the position of the ascendant in that time. (8) The deviation should be multiplied by the equinoctial shadow of that place and divided by 12. If the obtained amount is not whole, it should be multiplied by half of the diurnal time of each planet and divided by it, and if it is night, it should be divided by half of the nocturnal time. (6) Now, if the obtained amount is such that the deviation is north, then subtract the portion of the planet in the eastern hemisphere and add it to the western hemisphere. But if the deviation is south, then add that amount to the portion of the planet in the eastern hemisphere and subtract it from the western hemisphere. (10) Add three quantities to the portion of the planet and find its declination, and multiply this declination by the arc of the deviation, and add the product to the portion of the planet, considering it as the obliquity. If the directions of the declination and the deviation are different, and if their directions are the same, then subtract. (11) In finding the rising and setting of the planet in the sum of the constellation and the planet, before knowing the elongation of the moon, this procedure should be performed. (12) Performing this procedure on the Yugakal and the deviations of the Sanskrit planets at this time, if the directions of the deviations are the same, then find the difference, and if they are different, then find the sum. By doing this, the difference between the two planets in the Yugakal will be obtained.

Scientific Commentary—the method of knowing the positions of the planets in the Yuktaka period, which is explained in 3-6 verses. From this it is known that at that time the planet is on the kadamba protvritta, but the true Yuktaka calls the time when both planets

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Planet conjunction 583

On the circumscribed circle, it means that it is on the circle that goes to the northern point of the horizon from both planets. Therefore, to know the true conjunction, two calculations are done in the portions of the planets that come in the given method, which are named akshadrikkarm and aynadrikkarm. These calculations are similar to akshavalan and aynavalan. Bhaskaracharya has described the method of deriving akshadrikkarm and aynadrikkarm from akshavalan and aynavalan according to Brahmagupta, which is mostly prevalent today, but Sūrya Siddhānta gives another method for this work. Here, the method of Sūrya Siddhānta is explained first, and briefly it is also stated what the method of Bhaskaracharya is.

From Figure 106 it appears that according to the method given for knowing the portions of the planets of this chapter up to the sixth verse, the portion of planets g and h will be the same as the portion of the point v of the ecliptic. But at this time, the circumscribed circles of these planets intersect the ecliptic at points j and p, so the difference of the circumscribed circles of these planets in the above conjunction will be the same as p j on the ecliptic. According to the principle, the time when this difference is equal to zero is called conjunction, meaning the time when the two planets are in the same circumscribed circle. To know this, it is necessary to first perform the action of when the portions of both planets will be equal, which is known according to 4-6 verses. After this, it is necessary to know what y p and y j are at that time. For convenience, I will call them respectively g and h of aksha-ayana-drikkarm-sanskrut-phala. It is revealed that—

The values of the two terms on the right side of each equation can be easily known, and in this way, the values of y and z can also be known. The method of knowing the first term is explained in 7-6 shlokas and it has been named akshadrikkarm by the Acharyas. The method of knowing the second term is explained in the 10th shloka and it has been named aynadrikkarm by the Acharyas. The first is called akshadrikkarm because its magnitude varies according to the observer's latitude, and the second is called aynadrikkarm because its magnitude changes according to the precession angles (see page 230) as will be proved later.

Akshadrikkarm—it is apparent that on the meridian, the northern point of the horizon u and the pole g become one, therefore the circum-orbit and pole-orbit of any planet here-

SOURCE PAGE 609

SOURCE PAGE 610

k = kadamba h = dhruva kh = khaswatik m = madhyalagna g, h = the place of two planets (in the picture, h has become dh) k g y h = kadambavritt l j ph y ch p m = krantivritt kh = the point of the krantivritt which is the ascendant in the eastern horizon. a y ch ra = the aurora circle of the point y dh ch ch h = the polar circle passing through the planet h dh g ph = the polar circle passing through the planet g dh u j h = the parallel circle passing through the planet h u g p = the parallel circle passing through the planet g j = the point of intersection of the parallel circle and the krantivritt of the planet h p = the point of intersection of the parallel circle and the krantivritt of the planet g pj = the difference between the parallel circles of both planets (on the krantivritt) ch j = the axis-direction of the planet h (the difference between the parallel and polar circles of the planet h on the krantivritt) ch y = the inclination-direction of the planet h (the difference between the kadamba and polar circles of the planet h on the krantivritt) y j = the axis-inclination-direction-Sanskrit result of the planet h (the difference between the parallel and kadamba circles of the planet h on the krantivritt) p ph = the axis-direction of the planet g (the difference between the parallel and polar circles of the planet g on the krantivritt) p ph = the inclination-direction of the planet g (the difference between the kadamba and polar circles of the planet g on the krantivritt) p y = the axis-inclination-direction-Sanskrit result of the planet g (the difference between the parallel and kadamba circles of the planet g on the krantivritt) p y = ph - pf and y j = ch j - ch y

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Ayaj = chaj - chay
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They remain in one circle. For this reason, the value of p or q or j is zero there, that is, the axis is zero there. As the latitude increases, that is, as 10

SOURCE PAGE 611

The pole moves upwards from the uttara vindu on the horizon circle, as the akṣadṛkkarma increases. At the time when the planet is on the meridian circle, its prime vertical circle and polar circle remain together because the meridian circle is at both the U and D points. Therefore, it is proven that the declination of the planet on the meridian circle of any place remains zero. Now only it remains to know what is the declination of the planet at other points of the sky. First, it should be seen what is the magnitude of the declination if the planet is on the horizon circle. It is clear that if the planet is on the horizon circle, the horizon circle is also its prime vertical circle. Figure 107 reveals that when the planet has its ascendant in the eastern horizon, its position on the declination circle is Y.

U L D = pūrva kṣitija vṛtta U = uttara vindu D = the position of the planet when rising L = rising ascendant K = kadamba Y = the position of the planet on the declination circle D = the intersection point of the polar circle and the declination circle of G and the declination circle.

SOURCE PAGE 612

The arc of the circumpolar circle intersects the polar circle at six points. A B C D is a spherical right-angled triangle because the circumpolar circle makes an angle of 60 degrees with the polar circle. The circumpolar circle is parallel to the equator circle, the angle between the equator circle and the horizon circle is equal to the meridian, so the angle C D is equal to the meridian. If C D are considered as a simple right-angled triangle, then the angle C D = 60 degrees and the angle D C = meridian. Therefore, the angle D C = latitude because latitude + meridian = 60 degrees. For this reason, the simple triangle C D has

' sin ∠ colatitude = sine of latitude / sine of colatitude = equinoctial shadow / 12 (see page 56 and 263)

If 6h is taken as equal to h and 6a is taken as equal to l, then

chala / ghaya = palabha / 12

∴ motion = gnomon × equinoctial shadow12 = planet shadow × equinoctial shadow12

But the axis-direction of the planet is. Therefore it is proven that at the time the planet is on the horizon, its axis-direction is obtained by multiplying it by the shadow and dividing by 12. This is the meaning of the first half of the 8th verse. In this way, when it is proven what the axis-direction of a planet on the horizon is and its value is zero at the meridian, then for other times its calculation is done from the trigonometrical way in such a way that when the value of the axis-direction in half a day of the planet is at least zero and no more than the horizon axis-direction, then what is its value at the auspicious time. That is, day: auspicious time:: horizon axis-direction: auspicious axis-direction. This is the meaning of the 8th verse.

In figures 106 and 107, shown in grah purvakpal, the shadow of gh is south. Then the luminous circle of gh intersects the circle of ja at a point which is to the east of ya. Therefore, by adding the axis of gh in the portion of ya in the south shadow, the portion of ja will be obtained. But when the shadow of ga is north, then the luminous circle of ga is the declination circle.

SOURCE PAGE 613

It cuts the place which is to the west of y, therefore the portion of y in the north hemisphere is obtained by subtracting the declination of g from the portion of p. In the western hemisphere, it is the opposite, that is, the declination of the planet in the southern hemisphere has to be subtracted from the portion of the planet and the declination of the planet in the northern hemisphere has to be added to the portion of the planet. This is also clear from figure 106 because if the figure is understood as the western hemisphere, then the point j is understood as west of y and the point p is understood as east of y because in the western hemisphere, the point below any point is west and the point above is east, but in the eastern hemisphere, the point below any point is east and the point above is west. Thus, the derivation of the addition and subtraction operation mentioned in the 6th verse is also proven.

It should be remembered that the method mentioned in the 8th verse is coarse because the imaginations from which this is proven are themselves coarse.

Ayan Drikkarma—

From figure 106 it is revealed that the Ayan Drikkarma of planet g is ch y. Now to see what is the method to know the value of ch y according to Sūrya Siddhānta.

The triangle ch y is so small that there is no harm in considering ch y as equal to ch y. There is no special harm in considering the triangle ch y g as a simple right-angled triangle because the arc of planet g is very small and the angle g ch y is a right angle because the arc of point a is the polar circle and the arc of d is the polar circle of g. Therefore, in the right-angled triangle ch y g

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ch y / gh = sin ∠ chgy / sin ∠ ghchy = sin ∠ chdhy / radius
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Since the arc of the planet is very small, there is no loss in considering angle cha and dha or angle dha and gha as equal to angle dha and ya. Therefore, it can be said that

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Shadow / Height = sine / tangent / radius ∴ Shadow = Height x sine angle tangent / radius
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But angle theta and psi is the motion of the point psi because it is between the circle parallel to the pole of psi and the circle parallel to the equator (see figure 101) and is equal to the precession of the portion beyond 60 degrees of psi (see page 480) therefore sine

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SOURCE PAGE 614

The sine of y + 60^ ° should be taken according to equation (2) on page 480. If this sine is taken equal to crā, then x = y imes crā / radius. All magnitudes in this equation should be understood in terms of angles.

It is stated (see Chapter on True Positions page 122) that the value of the true sine is 1367 degrees and is considered equal to 240. 24 times 58 is 1362, which is very close to 1367, so there is no harm if it is assumed that 24 times 58 is approximately equal to 1367. Therefore, when the sine of 24 degrees is equal to 24 times 58 degrees, it is not a great loss to understand that the sine of an angle is equal to 58 times its degree.

Therefore x = crā imes 58 / radius = 60458. Thus the form of the above equation will be:-

```
shadow =  y  imes  crā  imes 58 / 58  imes 60 =  x  imes  crā / 60  angle
```

The angle is multiplied by 60, so if the right side of the above equation is multiplied by 60, its value will be in angles. But when multiplied by 60, the lower 60 will be cut and the form of the equation will be:-

ch y = gh y x krantyaansh vikala

Here ch y = ch y = ayan drikarm, gh y grah dh ka shar ya vikshep kalaon mein hai aur kranti anshon mein hai. Is prakar yah siddh hota hai ki y ke aage ke 60 ansh ki kranti ko anshon mein likh kar isko vikshep ki kalaon se guna kar dene par jo aata hai vah viklaon mein gh grah ka ayan drikarm hai jaisa ki shlok 10 mein batlaya gaya hai. Is niyam ka doosra saral roop yah bhi ho sakta hai ki grah ke ayan valan ko anshon mein likh kar isko grah ki vikshep kala se guna kar dene se jo aata hai vah viklaon mein grah ka ayan drikarm hai.

Now it is to see when this ayan drikarm is positive and when it is negative, that is, when should this ayan drikarm be added and when should it be subtracted in the portion of the planet. Careful observation of picture 36 on page 200 of the Chapter on True Positions can show that as long as the planet remains in Uttaraayana,

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SOURCE PAGE 615

That is, as long as the sine of the beginning point of Capricorn is from U to the sine of the beginning point of Cancer, its arc is to the left of the arc of the pole, meaning the plane of the arc is above the plane of the arc of the pole, as shown in figure 36. But when the planet is in the south, meaning as long as the sine of the beginning point of Cancer is from D to the sine of the beginning point of Capricorn, its arc is to the right of the arc of the pole, meaning the plane of its arc is below the plane of the arc of the pole, as shown in figure 106.

From figure 36 it is revealed that when the planet G is in the Uttarayana and its Shar is North, then its arc of the pole is to the west of point P where its arc intersects the arc of declination. But if the Shar of the planet in Uttarayana is South, suppose it is at P, then it is clear that its arc of the pole will remain the same as that of G, but the arc of the arc will cut the arc of declination to its west (which is not shown in the figure), meaning the place on the arc of declination where the arc of G is will cut the arc of declination to the east of the pole. That is, in the first case, by subtracting the portion of the planet and in the second case, by adding, the position of the intersection of the arc of the pole and the arc of declination will be found.

Similarly, from figure 106 it is revealed that when the planets G and H are in the South direction, they are to the right of their arcs. In this situation, the arc of the pole of planet G with the south in the north cuts the arc of declination at point F which is to the east of Y, so by adding YF to the portion of B, the place of F will be found. But the arc of the pole of H with the south in the south cuts the arc of declination at point C which is to the west of Y, so by subtracting CYG from the portion of B, the place of C will be found.

It is evident that when the planet is in Uttarayana, adding 60 degrees to its portion results in a declination that is always north because when the sine remains between the sine of Capricorn and the sine of Cancer, then the portion 60 degrees ahead of it is ahead of the sine of Aries and before the sine of Libra, whose declination is north. Similarly, when the planet is in Dakshinayana, adding 60 degrees to its portion results in a declination that is always south. Therefore, what has been said about the relation between Uttarayana and Dakshinayana also applies to the relation between north declination and south declination, as stated in the latter part of the 10th verse.

SOURCE PAGE 616

The derivation of the 12th verse—upon performing the correction of the axis and the inclination, one should see the difference in the portions of the planets that arise, and know when the difference between both planets becomes zero. At the time when this difference is zero, the conjunction of both planets occurs on the prime vertical circle. At this time, if the poles of both planets are in the same direction, that is, both North or both South, then when the difference between them is calculated, and if the directions of the poles of both planets are different, that is, one North and the other South, then when the sum of both poles is taken, the planets appear to be on the prime vertical circle at the same distance.

Thus, according to the Sūrya Siddhānta, the derivation of the method of calculating the coordinates and inclination is proven, from which it is evident that this method is gross because it has been proven by many imaginations.

According to Bhaskaraacharya, the...

Bhaskaracharya says that at the time when the position of the planet's ecliptic is in the horizon, the planet is above or below the horizon due to its shadow. When the planet is above the horizon, it rises before the position of its ecliptic, and when it is below, it rises behind. How much earlier or later it rises is known by the apparent motion. This apparent motion has two parts. One part depends on the motion of the nodes, and the other depends on the motion of the axis. The part that depends on the motion of the nodes is called the nodal apparent motion, and the part that depends on the motion of the axis is called the axial apparent motion.

In Figure 108, the shadow of planet G is north. The great circle of G intersects the horizon at two points, so at the time when the meridian of G is on the horizon, the great circle of G has two points on the horizon, so G has risen earlier than the time it took for the arc of G's great circle to come above the horizon. But G = Gc + Cc, for which the value of each is known as follows:

Calculation of Gc—

The right angle triangle in Ga Cc has a right angle at Ga because the polar circle of Ga intersects the great circle of G at C place, so

SOURCE PAGE 617

U ch g a d = East horizon circle

U ch g a d=East horizon circle U=North point G=North pole K=latitude U G A V=meridian G=ecliptic G=ecliptic the place of G, at this time it is also the ascendant on the east horizon. The ecliptic is not shown so that the picture remains simple. A G C R=G's horizon circle; S P V=celestial circle; K G A=K's latitude circle; D C G P=G's latitude circle. D C F=F's latitude circle. G C=altitude; C S=azimuth; G A=G planet's deviation; C G=G planet's true deviation (see Ganitādhya GrahaChāyādhikāra shloka) < G A C=G's inclination < U G D=G's latitude

SOURCE PAGE 618

Sūrya Siddhānta as Scientific Commentary as Chapter on Mean Motions as Chapter on True Positions as Chapter on the Three Problems

But it is a section of the circle of the sun and the angle in front of it is the same as g dhc on the pole, which is the same as the section of the equator. Therefore, to know how long the g c section rises, we need to know the s p section, which is known by this ratio—

j yā g ca = j yā g dh = j yā g dh / t r i j yā j yā s p = j yā dh s p r n t u g dh = dh s - g s = 60° - g k i k r ā n t i (2)

∴ sine of declination = sine of inclination

therefore sin aps = radius × sin gachjagdh = radius × crantijya × gaga × aynavalanjya = gaga × aynavalanjya × gachikrantijya

Sūrya Siddhānta

The value of s that will come from this action will be in the arts if the calculations of the sine and koṭijyā are done in the Indian manner. This means that due to the inclination of only, the rising time of g will be ahead of the rising time of s as if of tears. If one wants to know which part of the ecliptic will rise in so much time, then the amount in which the planet is multiplied by eighteen hundred and divided by the longitudes of the zodiac sign should be divided. Because it is clear that when the longitudes of the zodiac sign have 30 degrees or 1800 art rise, then in the same time as s rises, how much part of the zodiac sign will rise. This is the essence of verse 4 of the planetary dominion.

Chapter on Mean Motions

Chapter on True Positions

SOURCE PAGE 619

It is a section of the polar circle, and it is a section of the azimuthal circle which is at a right angle to the polar circle. Gach to bhāskarācāryajī ne ga of true śara mānā is and bheda dikhalāne of lie ga gā to madhyama śara mānā hai। koṇa ca gā cha = koṇa dha gā u = akṣāvalana। if gā viṣavadvṛtta of pāsa ho then koṇa dha gā u akṣāṃśa of samāna mānā jā sakatā hai। aisī daśā in and if ca gā cha tribhuja samatala-tribhuja māna liyā jāya kyoṃki planet of true śara gā ca sādhāraṇataḥ bahuta choṭā hotā is taba ca cha gā koṇa lambāṃśa of samāna mānā jā sakatā is kyoṃki 60° - akṣāṃśa = lambāṃśa। aisī daśā in cūṃki golīya tribhuja ca gā cha in

Scientific Commentary

Chapter on Mean Motions

Chapter on True Positions

Chapter on the Three Problems

late

This is the meaning of the 7th verse of the chapter on planetary shadow. In this way, the value of p f which comes in the arts is the latitude.

The same rules for adding and subtracting the axis and the inclination should be applied for which the same rules are stated with respect to the earlier Sūrya Siddhānta.

A method to know the true position which is somewhat gross, Bhaskaraacharya has stated in the third verse of the chapter on planetary shadow:-

Multiplying the portion of the planet by three rāśiyoṃ which come to the portion, the arc of the declination, i.e., the sine of declination, by the mean motion and the product by the radius.

SOURCE PAGE 620

It should be divided. This is clear from the spherical right-angled triangle of ga ga ch in figure 108. Because adding 60^(°) to the portion of the planet results in the same as the declination (see Chandragrahaadhikar verse 25), which is the same as the angle ga ga ch here, therefore its sine of the meridian will be the same as the sine of the declination. If the ga ga ch triangle is assumed to be a plane right-angled triangle, then angle ga ga ch = 60^(°) - angle ga ga ch = 60^(°) - declination

jya ga = jya ga ch = jya ga ch · jya angle gch = jya angle ga gch = jya 60^(°) - declination therefore jya ga ch = jya ga ga × jya (60^(°) - declination) / jya60^(°) = jya madhyamshar × jya (60^(°) - declination) / radius

Images of the Planets—

> कुजाकिज्ञामरेज्यानां त्रिंशत्साघार्धवर्धिता ।
> विष्कम्भश्चन्द्रकक्ष्यायां भृगोष्षष्टिरुदाहृतः ॥१३॥
> चित्रतुःकर्णयोगाप्तास्ते द्विघ्नास्त्रिज्यया हताः ।
> स्फुटाः स्वकर्णास्तिथ्याप्ता भवेयुर्मानलिप्तिकाः ॥१४॥

Translation—(13) The diameters of the images of Mars, Saturn, Mercury, Jupiter, and Venus in the lunar orbit are 30, 37॥, 45, 52॥ and 60 yojanas. (14) To know the true diameter of the image of a planet, multiply the diameter written above that planet by the radius (3438) and divide the product by the sum of the radius and the fourth shadow hypotenuse of that planet; the resulting value is the true diameter of the image. If this is divided by 15, the magnitude of the image in terms of phases is known.

Scientific Commentary—In the 13th verse, it is stated that what are the diameters of the images of the planets in the lunar orbit. Based on this, according to verses 1-3 of Chandragrahaadhikar, it can be known in the reverse manner what is the diameter of the planet's image in its orbit. However, there is no need to know this regarding the conjunction. Here one should only know the phase of the planetary image in the conjunction period

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SOURCE PAGE 621

What is a measure. But the angular or artistic value of a body depends on its distance (see Chapter on True Positions page 85) and the distance of the planet from the Earth is not constant, it increases and decreases, so it is necessary to know what is the angular value of the planetary image. The magnitude of the planetary image in the Moon's orbit, which was explained in yojana, is given here. Also, it is stated in the following verses that the true artistic value of the planetary image in the desired time, which is in

yojana, becomes the true artistic value after dividing it by 15. It is stated on page 452 of Chapter on Lunar Eclipse that how the 15 yojana 1 kala of the lunar orbit is like. Therefore, it is clear that why the magnitudes of the images of the planets in the lunar orbit become in the art after dividing them by 15. Thus, the artistic values of the images of the planets in the lunar orbit are as follows:-

Mangalkābibmba = 30 yojana = 30 ÷ 15 = 2 kalashini,, = 37 llyojana = 37 ll ÷ 15 = 2 llkalabudha,, = 45 yojana = 45 ÷ 15 = 3kalaguru,, = = 52 llyojana = 52 ll ÷ 15 = 3 llkalashukra,, = 60 yojana = 60 ÷ 15 = 4kala

From this it is proven that our teachers considered the image of Mars to be the smallest. They understood the image of Saturn to be larger than this, etc. However, the table on page 66 of Chapter on True Positions reveals that if all planets are as far from the observer as the Sun is from the Earth, then the diameter of the image of Mercury is the smallest, i.e., 6.68 vikla. That of Mars is larger than this, i.e., 6.36 vikla. After this, the diameters of the images of Venus, Saturn, and Jupiter are 16.80, 16.5, and 164.72 vikla respectively. Thus, it is proven that the magnitudes of the images that our teachers calculated by gross instruments are very inaccurate, as written by M. M. Sudhakar Dwivedi ji as well.

1. The subtle distance of the Yantras, Budha, Shukra, and Shiva are all related to the increase of the moon and the growth of the sheep. By seeing the appearance of such Yantras, the increase of the moon and the image of the star are not obtained, and this is not the most clear. This is not all.

Commentary on the principles of the planets and the chapters on the rights of the planets, verses 3-4.

SOURCE PAGE 622

Now it is revealed that when the magnitudes of the images given in the 13th verse are inaccurate, then how the magnitudes of the true images can be precisely known according to the next verse.

Now it will be considered where the next verse is pure. The essence of the first line of this verse is:-

True image = mean image X 2 X radius / radius + fourth shadow hypotenuse

or clear image = midpoint × radius × radius + fourthshadowhypotenuse 2

This can be written in the form of the three problems as follows:-

radius + fourthepicycle: radius:: mean image: true image2

From this rule, it is proven that our Acharyas must have known that when the distance of the planet on the radius is equal to its mean value, then the value of the clear image will be less at a greater distance and the value of the clear image will be more at a lesser distance, as shown on page 85 of the *Sprechadhikar*. However, it is not possible to consider the radius as 3438. If instead of the radius, the distance from the Moon to the Earth is taken, and instead of radius + the fourth chord, the distance from the Earth to 2 * the planet in the ecliptic is taken, then this ratio may be correct. Some Acharyas take the third chord instead of the radius in the first term of this triangle. But it cannot yield the same purity as should be. To calculate what the distance from the Earth to any planet is in the ecliptic, one must first know what the distance from the Sun to that planet is according to the formula given on pages 176-80 of the *Sprechadhikar*. Then, according to the figure given on page 183 of the same authority, one should know the distance from the Earth to that planet, that is, the chord. Now, if the

mean image given on page 66 is multiplied by the distance between the Earth and the Sun and divided by this chord, the clear image of the planet can be known with purity.

SOURCE PAGE 623

According to Acharya Ketkar's Jyotigaritha, the smallest and largest values of the images of the five visible planets and the smallest and largest latitudes in Chapter on the Three Problems on page 410

Planet true image shadow hypotenuse gnomon revolution apogee epicycle apogee node precession angle prime vertical circle mean planet true planet Mercury 4.4 21.2 2524 524 Jupiter 31.6 46.7 6203 4203 Venus 6.6 60.0 1723 277 Saturn 15.8 19.5 10536 8536

It is a great pleasure that Acharya Bankatesh Bapu Ketkar is still alive and resides in Bijapur with his son, and both father and son are still engaged in the study of astrology. I mistakenly wrote 'Swargiya' on the first page of your name because I thought you had become celestial. However, it appears from the pages of Padam S. M. codrez that you are still alive. I am very grateful to these gentlemen for this information. It is proven by the report of the first session of the Maharashtra Panchangaikya Mandal of Pune of 1882 that you participate in astrological debates even while being old. Author

SOURCE PAGE 624

are given. From these it is revealed that the magnitude of the images changes according to the latitude, that is, if the latitude is more, then the true image is also more and if the latitude is less. So the true image is less. But the magnitude of the latitude changes in the inverse ratio of the distance, that is, when the distance becomes more, the latitude becomes less and when the distance becomes less, the latitude becomes more (see page 385).

It is revealed from Figure 34 (see page 183) that at the time the perihelion of the planet is zero, the distance of the planet from the Earth is very large, that is, at that time the perihelion of the planet is very large and it is equal to the sum of the distance from the Earth to the Sun and the distance from the Sun to the planet. But at the time the perihelion of the planet is 180 degrees, the distance of the planet from the Earth is very small and it is equal to the difference of the distance from the Earth to the Sun and the distance from the Sun to the planet. The relation between the perihelion of the planet and the images is well revealed in the table of the previous page.

Here, the distance from the Earth to the Sun or the perihelion of the Sun is taken as 1000. The speed of the planets in the period-

> छायां भूमौ विपर्यस्ते स शङ्कग्रे प्रदर्शयेत् ।
> ग्रहः स्वदर्पणान्तस्थ शंखवग्रे सम्प्रदृश्यते ॥ १५॥
> पच्चहस्तोच्छ्रितौ शङकु यथा दिग्भागसंस्थितौ ।
> ग्रहान्तरकलाक्षिप्तौ अघोहस्तनिखातितौ ॥१६॥
> कर्णं सूत्रे तथा दद्याच्छायाग्राच्छङ्कुमूर्धगे ।
> छायाकर्णाग्रसंयोगे संस्थितस्य प्रदर्शयेत् ॥१७॥
> स्वशङ्कुमूर्धगौ व्योम्नि ग्रहो दक्तुल्यतामितौ ।

Translation -(15) In the flat land on which the gnomon is planted and the shadow is measured, in the opposite direction to the direction where the planet is, the planet should be shown in the mirror placed in front of the shadow of the time of conjunction. In such a mirror, the planet is seen joined with the tip

of the gnomon. (16) Plant two gnomons of five hands high in the directions where the planets of the time of conjunction are. The difference in their meridians should be as much as the difference in those planets. To keep them firmly standing, dig a pit under the Earth with one hand.

SOURCE PAGE 625

Want. (17) Tie a rope that indicates the shadow hypotenuse from the foremost point of the planetary conjunction shadow to the peak of the gnomon. The viewer should place their eye on this same formula of the shadow hypotenuse. (18) By doing this, the planet will appear attached to the peak of the gnomon in the sky.

Sūrya Siddhānta—these three and a half shlokas are of great importance. From these it is proven that our Acharyas performed the subtle calculations of astrology because the direct position of the planets is the same as what is seen through the sight, because until the calculation of the planets is absolutely pure, we cannot see them in this way as stated in these shlokas. One more thing is also known from this that our Acharyas also knew the law of reflection of light.

To calculate the shadow of the planets, according to the method described in Chapter on the Three Problems, the sidereal time of the planets should be known according to the method stated in page 331 for their portions, declination, and variation. Upon knowing the sidereal time, the planetary positions should be known according to equations (b) and (c) on page 262. From the sidereal positions, the azimuth or altitude is necessary according to equation (b) on page 273. To know the shadow from the sidereal positions, the tangent of the sidereal position should be multiplied by the magnitude of the gnomon. If the magnitude of the gnomon is 12 angulas for the 15th verse, there is no objection, but for the 16th verse, the result of the gnomon should be 4 hands. By doing so, the observer can easily determine the positions of the planets while standing.

The essence of the 15th verse can be shown by the diagram as follows:-

When the eye of the observer is at any point on the line, they will see the planet G and the top K of the gnomon together in the mirror.

If the gnomon is four hands, then the line from the apex of the gnomon from k to d, which is the formula, will have the eye of a spectator at some place, then even the planet g will be seen joined with the apex of the gnomon from k. This is the essence of the first half of the 16th, 17th, and 18th shlokas.

It is necessary to understand here that nowadays this arc can only be precisely obtained when the planet's portion goes purely according to the sight. For this work, it is extremely necessary to make amendments in our Siddhanta texts according to new arcs.

SOURCE PAGE 626

g = position of the mean planet in the sidereal time, k x = gnomon fixed on the flat land, d = point where the shadow is cast where the mirror will be placed, n = eye of the observer

From these verses it also appears that the study of astrology-science should not be based on texts alone but should also involve observation. Therefore, it is proven that the study of astrology is possible only in the proper manner when there is a good observatory along with the astrology school. Only in such an observatory, instead of gnomon etc., should there be modern instruments like dūradarśaka etc., so that purity of observations can be achieved and the theoretical texts can be renovated by making appropriate amendments.

Five types of conjunctions—

> उल्लेखं तारकास्पर्शे भेदे भेदः प्रकीर्तितः ।१८।।
> आरादंशुविमर्दाख्यमशुयोगे परस्पम् ।
> अंशादूनेऽपसव्याख्यं युद्धमेकोऽत्र चेदणुः ॥१९॥
> समागमस्स्यादधिके भवतश्चेद्बलाधिकौ ।

SOURCE PAGE 627

Translation—(18) Latter part--If there is only a touch between the images of both planets in the conjunction, such conjunction is called Yuti. But if the image of one planet penetrates the image of the other, meaning it covers some, such conjunction is called Bheda. (16) If the images of both planets are some distance apart but their rays appear to meet, such conjunction is called Anshuvimard. If the difference between the images of both planets is less than one degree, such conjunction is called Apasvya Yuddha. In this war, if the image of one is small, it is expressed as Apasvya, otherwise Avyakta. (20) If the difference between both images is more than one degree, such conjunction is called Samagam. If both planets are Bali, meaning large, then it is Vyakta Samagam. Otherwise, it is Avyakta Samagam.

Scientific Commentary—Only definitions are given here which are clear. Therefore, there is no need to write more on this.

Characteristics of Defeated and Victorious Planets—

> अपसव्ये जितो युद्धे दूरेऽप्यणुरदीप्तिमान् ॥२०॥
> रुक्षो विवर्णो विध्वस्तौ मलिनो दक्षिणाश्रितः ।
> उदक्स्थो दीप्तिमान्स्थूलो जयी याम्येऽपि यो बली ॥२१॥

Translation—(20) In the war called Apasvya, the planet whose image is covered, is small and without brightness, (21) Rough, color-less or faint and towards the south is considered defeated. But the planet whose image is towards the north is bright and large is considered victorious. Even if the planet is Bali, meaning large and bright, and is towards the south, it is considered victorious.

Scientific Commentary—this is also clear.

> आसन्नायप्युभौ दीप्तौ भवतस्तौ समागमे ।
> स्वल्पो द्वावपि विध्वस्तौ भवेतां कूटविग्रहे ॥२२॥

Translation—(22) If both planets are affected even while being close, it is a war called conjunction and if both planets are small and faint, it is a war called occultation.

> उदक्स्थो दक्षिणस्थो वा भार्गवः प्रायशो जयी ।
> शशाङ्कैनैवमेतेषां कुर्यात्संयोगसाधनम् ॥२३॥

SOURCE PAGE 628

(23) Whether Venus is to the north or to the south, it is often victorious. Similarly, the conjunction of the five luminaries with the Moon should be achieved.

Scientific Commentary—from the table of the smallest and greatest apparent sizes of five star planets it is revealed that the smallest apparent size of Venus is greater than the smallest apparent sizes of Mars and Mercury, therefore at the time of their conjunction, Venus is more victorious due to its greater luminosity and bulk. At the time when the apparent size of Mars is supreme, it is 180 degrees ahead of the Sun. In such a situation, the conjunction of Venus with it cannot happen; the conjunction of Venus and Mars can only happen when Mars is also near the Sun. In such a situation, the apparent size of Mars will always be smaller than the apparent size of Venus. Therefore, Venus is always more luminous and victorious than Mars and Mercury. Yes, when Venus is in conjunction with Jupiter or Saturn, then Venus is smaller than them for some time before setting in the east and rising in the west. Therefore, it can be said that it is defeated by Saturn or Jupiter, but such a state occurs very rarely. That is why it is said in this verse that Venus is usually victorious.

> भावाभावाय लोकानां कल्पनेयं प्रदर्शिता ।
>
> स्वमार्गगाः प्रयान्त्येते दूरमन्योन्यमाश्रिताः ॥२४॥

(24) the conjunctions of planets, etc., for the auspicious and inauspicious fruits of people have been imagined. In reality, the planets move in their own orbits and are very far from each other, but they are interdependent or appear very close to each other.

Scientific Commentary—in this verse the Acharya has indicated some signs regarding predictive astrology but has not thought well about what kind of war or conjunction yields what result. The reason for this is that this is a text of astrology, so it does not have space to discuss predictive astrology with detail.

Thus the Scientific Commentary of the seventh authority named Grahyutyadhikara has ended.

SOURCE PAGE 629

Eighth Chapter

Chapter on Mean Motions

(Brief Description)

Verse 1—How are the fixed stars known from the enjoyment of the constellations. Verse 2-6- The value of the enjoyment and projections of the constellations. Verse 10, 11 and 12 first half—The enjoyment, fixed stars, and projections of the stars named Agastya, Mrigvyadha, Agni and Brahma-hridaya. Verse 12 second half—The method of testing the fixed stars and projections. Verse 13—When can Rohini-Shakta difference occur. Verse 14-15—The method of knowing the time and place of the conjunction of a star with a planet. Verse 16-16—Which star is the yogtara among the constellations. Verse 20-21—The fixed stars and projections of the stars prajāpati, Apamvatsa and Aap.

In this right it is stated that which constellations fall in the path of the sun, the moon, and the planets, where their positions are, and how the time of the conjunction of the planets with their main stars or yogtāras is known. Some such stars have also been discussed which are extremely talented and hold a special place in the literature of ancient times, but with which the planets do not conjoin. But why is the discussion of such stars or star clusters not mentioned here? I will also discuss such stars or star clusters in the appendix which are not given in this right but have come in ancient literature or have

special importance such as Saptarshi, Kashyap Mandala, etc. Whatever has been known about these stars by new observations nowadays will also be given briefly there.

> प्रोच्यते लिप्तिका भानां स्वभोगेन दशाहताः ।
>
> भवन्त्यतीतधिष्ण्यानां योगलिप्तायुता ध्रुवाः ॥१॥

Sūrya Siddhānta

Scientific Commentary

SOURCE PAGE 630

Vakra = Krantijyā = Tatārekādhruvaprotvṛtta

Vvi = Viśuvadvṛttakatti = Tatārekākadambaprotvṛtta

V = Vasantasampātatā = Takādhruvābhimukhabhogayadhruv

A = Aśvinīkāadibinduttatā = Takādhruvābhimukhvikṣep

T = Tārekāsthānati = Takākadambābhimukhabhogathavābhog

k = kadambati = takakambabhimukhavikshepathvavikshep

d = dhruv

SOURCE PAGE 631

It is neither a share nor is its magnitude in parts or aspects. The self-enjoyment of the star means the distance of the star from the beginning point of its constellation. This distance is in a unit that cannot be called a part nor an aspect. Therefore, it is stated that if this self-enjoyment is multiplied by ten, its magnitude is known in aspects. It seems that a unit different from the prevalent units has been used for brevity. What results from multiplying by ten is the angular distance of the star from the beginning point of its constellation in aspects. By adding this distance to the enjoyments of the preceding constellations, the pole of the star is known in aspects from the beginning point of Ashvini, that is, from the beginning point of the zodiac. It is stated earlier that the difference between any planet and the ecliptic is called Bhogansh, and the difference between the ecliptic and the meridian of that planet is called Vikshep. However, here instead of Bhogansh, Dhruvansh or Dhruv is mentioned. This is made clear by figure 110. If the meridian and the polar meridian are drawn while going from the star, they meet at two different points on the ecliptic. The difference from the beginning of Ashvini to the point where the meridian meets the meridian is called the enjoyment or Kadambabhimukh Bhog of the star. As stated in the earlier chapters, and from this point, the difference of the star is called Vikshep or Shar. It would be more appropriate to call this Kadambabhimukh Vikshep, but in this chapter, Bhog and

Vikshep are used in another sense. Bhog does not mean Kadambabhimukh Bhog, but rather Dhruvabhimukh Bhog, and the Vikshep discussed further does not mean Kadambabhimukh Vikshep, but rather Dhruvabhimukh Vikshep. This is further clarified by the description given below the figure. Confusion arises in expressing two different meanings with the same technical term, so it should be taken into good consideration.

In the Chapter on Planetary Motions, it is stated that two mental operations, āyana dṛkkarma and ākṣadṛkkarma, are required in the pleasures and displacements of the planets. The result obtained by performing the mental operation of āyana dṛkkarma in the pleasure of the planets is the polar-facing pleasure or pole of the planet. Therefore, when the polar-facing pleasure of the planets is written in this chapter, the need for āyanadṛkkarma with the constellations will not be necessary, only the need for ākṣadṛkkarma will be necessary, as stated in the 14th verse of this same chapter. This

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SOURCE PAGE 632

The fact is that in writing the declination of the stars, there is no need to do the inclination in it.

The self-motion and deviation of the stars—

> अष्टार्णवाः शून्यकृताः पञ्चषष्टिर्नगेश्ववः ।
> अष्टार्था गोऽब्धयोऽष्टागा षडगा मनवस्तथा ॥२॥
> कृतेषवो युगरसाः शून्यबाणा वियद्रसा ।
> खवेदास्सागरनगा अष्टागाः सागरर्तवः ॥३॥
> नवोऽथ रसा वेदा वैश्वमाप्याधभोगगम् ।
> आप्यस्यान्तेऽभिजित्तारा वैश्वान्ते श्रवणस्थितः ॥४॥
> त्रिचतुः पादयोः सन्धौ श्रविष्टा श्रवणस्य तु ।
> स्वभोगतो वियन्नागाः षट्कृतिर्यमलाश्विनः ॥५॥
> रन्ध्राद्रयः क्रमादेषां विक्षेपाः स्वादपक्रमात् ।
> दिङ्नासविषयास्सौम्ये याम्ये पञ्च दिशो भवाः ॥६॥
> सौम्ये रसाः खं याम्येऽगाः सौम्ये खार्का स्त्रयोदश ।
> दक्षिणे रुद्रयमलाः सप्तत्रिंशत्तथोत्तरे ॥७॥
> याम्येऽध्यर्धं त्रककृता नव सार्धशरेषवः ।
> उत्तरस्यां तथा षष्टिः त्रिंशत्षट्त्रिषदेव हि ॥८॥
> दक्षिणेऽतोर्धभागस्तु चतुर्विंशतिरुत्तरे ।
> भागाः शड्विंशतिः खञ्च दस्त्रादीनां यथाक्रमम् ॥९॥

The sequence of self-motion of the yoga-stars from Ashvini to Purvaashadh is 48, 40, 65, 57, 58, 4, 78, 76, 14, 54, 64, 50, 60, 40, 74; 78, 64, 14, 6, 4; Uttaraashadh's yoga-star is at half of Purvaashadh Nakshatra; the self-motion of the yoga-star of Abhijit is at the end of Purvaashadh Nakshatra, the yoga-star of Shravana is at the end of Uttaraashadh Nakshatra, the yoga-star of Dhanishtha is in the conjunction of the third and fourth stages of Shravana Nakshatra, that is at the end of the third stage. The self-motion of the yoga-stars of Shatabhishak, Purvabhadrapad, Uttara Bhadr Pad, and Revati is 80, 36, 22, and 76 respectively. The deviations of these Ashvinyadi yoga-stars from the ecliptic are 12, 12, 5 towards North; 5, 10, 6 towards South; 6, 0 towards North; 7 towards South; 0, 12, 13 towards

North; 11, 2 towards South; 37 towards North; 13, 3, 4, 6, 53, 5 towards South; 60, 30, 36 towards North; 3 towards South; 24, 26, and 0 degrees towards North.

SOURCE PAGE 633

Scientific Commentary—by multiplying the self-elongation of each star by 10 according to the first verse, the self-elongation art of the star will be obtained. By adding this to the courses of the past constellations, the pole of that star will be known. For example, the self-elongation of the Ashvini star is 48, multiplying it by 10 gives its self-elongation as 480 degrees. The Ashvini star is in the first constellation named Ashvini, so the past constellation is zero, hence the pole of the Ashvini star is 480 degrees or 8 arc minutes. Similarly, the self-elongation of the Rohini star is 570 degrees in arc minutes. The first three constellations of Rohini are Ashvini, Bharani, Krittika, so their elongation is 3 X 800 arc minutes because one constellation is equal to 800 arc minutes (see Chapter on True Positions verse 64). Therefore, the pole of the Rohini star = 570 + 3 + 800 arc minutes = 570 X 2400 arc minutes = 2670 arc minutes = 46 degrees 30 arc minutes.

Similarly, the declination of each star can be known. This is a characteristic of the elongations of the Uttaraashadha, Abhijita, Shravana, and Dhanishtha stars, so their declinations are stated as below:-

The star of Uttaraashadha is at half of the Purvashadha constellation, that is, at 400 arc minutes of the Purvashadha constellation. There are 16 constellations from the first Ashvini to the root of Purvashadha, whose elongations are equal to 16 X 800 arc minutes = 15200 arc minutes. Therefore, the declination of Uttaraashadha is 400 + 15200 arc minutes = 15600 arc minutes = 260 degrees.

The Abhijita star is stated at the end of Purvashadha, so its declination is 260 degrees + 400 arc minutes, that is, 266 degrees 40 arc minutes.

The pole of the Shravana star is at the end of the Uttaraashadha constellation. One constellation = 13 degrees 20 arc minutes. The end of the Purvashadha constellation is at 266 degrees 40 arc minutes, so the pole of the Shravana star at the end of Uttaraashadha is 280 degrees.

The Pole Star is at the end of the third quadrant of the Navagraha constellation. The three quadrants of the constellation are 600 degrees. Or they are equal to 10 degrees. Therefore, the pole of Dhinishtha is 280+10=260 degrees.

The deflection is given in degrees, so there is no need to write more on this.

It is necessary to state here that the poles of the stars given above are not the same in all Siddhanta texts. There can be many reasons for this—(1) the arcs

SOURCE PAGE 634

Difference (2) in determining the position of the initial point of Ashvini (3) in determining the stars and (4) the motion of the nodal point. The first reason is clear because there is a natural difference in the results of the observation due to the crudeness of the instruments. The second reason is also of special importance. It is understood from this that there was a difference of opinion among the old teachers in determining the initial point of Ashvini, as is the case today. However, this difference causes great difficulty in determining the conjunctions and occultations today, which prevents the stability of all-India dates and festivals. If unity is achieved in the astrology of all the provinces on this matter, a great work will be done, and those who dedicate their body, mind, and wealth to this industry will be the

recipients of great merit. Work has been done in Maharashtra and Gujarat provinces on this matter for a long time, but no determination has been made yet.

In this way, due to the motion of the nodal point, a difference occurs in the poles and deviations of the stars, although they remain stable in their courses and arcs.

In the 10-12 shlokas, the pole and deviations of the stars are explained, and an effort will be made in several tables to show what the opinions of ancient and modern commentators are regarding the relationship of the stellar polaritudes.

> अशीतिभागं र्यास्यायामगस्त्यो मिथुनान्तगः ।
> विंशे च मिथुनस्यांशे मृगव्याधो व्यवस्थितः ॥१०॥
> विक्षिप्तो दक्षिणे भागैः खार्णवैस्वादपक्रमात् ।
> हुतभुग्ब्रह्म हृदयौ वृषद्वाविंशभागगौ ॥११॥
> अष्टाभिः त्रिंशता चैव विक्षिप्तावुत्तरेण तौ ।
> गोलं बध्वोपरिक्षेत्रं विक्षेपध्रुवकान् स्फुटान् ॥१२॥

(10) The pole of the star Agastya is at the end of Gemini sign, that is, 60 degrees, and the southward deviation is 80 degrees. The pole of mṛgavyādha or lubdhaka star is at 20 degrees of Gemini, that is, 80 degrees. (11) Its deviation is 40 degrees south of the ecliptic. The poles of both Agni and Brahmhṛday stars are at 22 degrees of Taurus sign, that is, 52 degrees. (12) Their deviations are 8 degrees and 30 degrees towards the north from the ecliptic in sequence. These fine deviations and poles should be examined by the golayaṃtra.

Scientific Commentary—the latter part of the 12th verse is of great importance. This proves that our Acharyas were not desired to be fakirs of Lakor, therefore

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SOURCE PAGE 635

Names of stars sequence number popular constellation-theory Brahmagupta theory Sundara theory degrees arcminutes degrees arcminutes lalla dāmodarasya bhara tulya Sundara theory degrees arcminutes degrees arcminutes degrees arcminutes degrees arcminutes other stars' positions names 1 Ashvini 8 0 5 0 8 30 8 5 7 43 2 Bharani 20 0 20 0 21 50 21 21 51 3 Krittika 37 30 28 36 0 37 45 38 36 2 4 Rohini 46 30 46 0 46 0 50 46 47 37 5 Mrigashirsha 63 0 63 0 62 0 63 62 61 26 6 Ardra 67 0 67 0 66 0 67 66 75 43 7 Punarvasu 153 0 63 0 62 0 63 65 61 21

SOURCE PAGE 636

8 Punya 106 0 105 0 106 0 105 43 6 Aphalesha 106 0 108 0 107 15 108 38 10 Madha 126 0 128 0 126 0 126 56 11 Purva 144 0 136 20 148 144 31 12 Uttara 155 0 154 0 155 30 154 13 Phalgunī Hasta 170 0 173 0 170 0 165 6 14 Chittā 180 0 184 0 183 0 183 0 15 Swātī 166 0 167 0 119 30 168 28 16 Vishākha 213 0 212 0 212 15 20211 17 Anurādhā 224 0 224 0 224 15 224 8 18 Jyeshṭhā 226 0 228 0 226 30 230 56 19 Mūla 241 0 241 0 242 0 240 45

SOURCE PAGE 637

The names of the stars, the order of the constellations according to Brahmagupta Siddhanta Laltamanta Laltamanta Dyotaryartha Sundara Siddhanta part art part art 20 Pushyashada 254 0 254 0 255 32 255 253 23 21 Uttarashada 210 0 260 0 267 20 260 0 261 5 Abhijit 266 40 265 0 267 0 256 45 258 10

Shravana 280 0 278 0 283 10 275 15 278 275 58 Dhanishtha 260 0 260 0 296 20 387 30 260 286 47 Shataraka 320 0 320 0 313 20 320 0 320 318 43 Purvabhadrapad 326 0 326 0 327 0 325 0 326 325 322 3 Uttarabhadrapad 337 0 337 0 335 20 337 0 337 340 35

SOURCE PAGE 638

27 Revati 356 50 0 0 356 0 0 0 Canopus Augusty 60 0 87 0 87 0 0 356 1 Vyadh 80 0 86 0 86 0 80 80 Sirius Agni 52 0 52 52 53 □eta Tauri Brahma 52 0 52 52 56 alpha aurigae (capella) Prajapati 57 0 57 61 61 □eta aurigae Apamatsa 180 0 183 heta virginis* āpasa 180 0 delta virginis* Popular Hindu Astronomy Part I pp. 240-241

SOURCE PAGE 639

The sum of constellations and some other stars' polar direction or deflection (453) (See Indian Astrology Book page 453) The names of the stars of Nakshas the number of parts art lalla tantra sundar siddhanta sundar siddhanta prabala siddhanta shaar ki disha star names Brahmagupta Siddhanta parts art 1 Ashvini 10 10 10 90 5 North 2 Bharani 12 12 12 12 120 12 North 3 Krittika 5 4 4 4 5 4 North 4 Rohini 5 4 4 4 5 5 South 5 Mrigashira 10 10 10 100 13 South 6 Ardra 5 11 11 11 111 6 South 7 Punarvasu 6 6 6 6 6 6 North

SOURCE PAGE 640

Nakshatragrahayutyaadhikar 8 Pushya 0 0 0 0 0 05 Uttar 6 Ashlesha 7 77771124 Dakshin 10 Magha 0 0 0 0 0 026 Uttar 11 Po.Phalguni 12 12131245121031 Uttar 12 U.Phalguni 13 13131245131324 Uttar 13 Hasta 1181111111317 Dakshin 14 Chitra 214514522 Dakshin 15 Swati 37373715373256 Uttar 16 Vishakha 13012313011510 Dakshin 17 Anuradha 30144314521 Dakshin 18 Jyestha 433033437 Dakshin 19 Moola 6883081348 Dakshin

SOURCE PAGE 641

The order and number of the names of the stars, Brahmagupta Siddhanta, Lalla Tantra, Sundara Siddhanta, Mundarya Matdaly Ansha Kala Ansha Kala Shankre Balkrishna Shar ki Disha Purvashadh Uttarashadh Abhijit 20 5 30 5 20 5 30 5 30 5 2 7 Dakshin 21 5 30 5 5 5 5 5 5 1 28 U Dakshin 22 60 62 63 62 62 62 62 61 55 Uttar Uttar 23 30 36 36 25 30 36 34 15 0 25 Uttar 24 0 30 18 24 23 45 24 21 6 Uttar Uttar 25 24 26 26 26 26 26 26 27 13 45 Uttar Uttar

SOURCE PAGE 642

Revati Agastya byād Agni Brahma Prajapati Apānvats Āp 0 0 77 40 30 38 54 0 0 50 40 0 0 80 0 0 76 77 0 0 0 0 0

precession angle. From these things it is clear that our Acharyas desired that the mathematical calculations of astrology should be made in accordance with the direct sighting of the sky and appropriate corrections should be continuously made.

Here the armillary sphere will not be specially discussed because this subject is discussed in the 13th chapter of Jyotishopanishad-adhyaya where instruments are discussed, so it will be well explained there by giving a picture. Also, it will be stated that how very minute work can be done with some new instruments like the telescope, etc., and how much need there is for a new type of observatory with each astrological school.

The previous tables state what the pole and deviations of the above-mentioned stars are according to the opinions of different Acharyas. The poles and deviations of the Brahma Gupta Siddhanta are similar to those of the Dhruva and Vikshep in the Siddhanta Shiromani of Bhaskaraacharya. The poles and deviations of Lalitantra, Damodariyabhata Tulya, and Sundari-Siddanta are taken from the Indian astrology of Swargiya Shankar Balakrishna Dikshit. Dikshit ji, assuming the pole of the star Chitra to be 180 degrees, and the poles and deviations of the stars given in the Nautical Almanac of 1887 AD, which he had fixed from the latitudes and declinations, will also be given in this table. Dikshit ji has given two poles and two deviations for the star Revati. The reason for this is that according to their opinions, the sum of Revati may be Yata Pisium or Mu Pisium. Therefore, the first pole or deviation is of Jata Pisium and the second is of Mu Pisium.

When does the separation of the planet Rohini-Shakta occur—

> वृषे सप्तदशे भागे यस्य याम्योऽशकद्वयात् ।
> विक्षेपोऽभ्यधिको हन्याद् रोहिण्याश्शकटं तु सः ॥१३॥

(13) The planet situated at the 17th degree of the Taurus zodiac whose southward deviation is more than 2 degrees penetrates the shadow of the Rohini constellation.

SOURCE PAGE 644

Scientific Commentary—There are 5 stars in the Rohini constellation whose shapes are like a cart or the letter V of the English alphabet. The southern deviation of the northernmost star among these five is approximately 2 degrees 35 arc. This star is called Apellan star nowadays. And the southern deviation of the star conjunct Rohini is 5 arc 32 arc. The planet whose southern deviation or deviation falls between these two limits falls within the shadow of Rohini. This is called the penetration of the shadow of Rohini. It is apparent that the deviation of the planet depends on its node. The Moon completes one revolution in 18 years. In this one revolution, the Moon only passes through 5,6 years of shadow. If the southern deviation of the Moon is more than 2 degrees 35 arc and less than 5 arc 32 arc and at that time it is in the Rohini constellation, then it will certainly be within the shadow of Rohini, therefore the Moon will have the Rohini-shadow-penetration. Now it is to see that at the time the Moon is in the Rohini constellation, its southern deviation is less than 2 degrees 35 arc.

In page 75 of the Chapter on Mean Motions it is stated that the ultimate displacement of the Moon is 5 degrees 8 nakshas 42 vikalas. This means that when the Moon is 60 degrees ahead of Rahu, its south is 5 degrees 8 nakshas and 42 vikalas, and when it is 60 degrees ahead of Ketu, its south is the same. But when it is on Rahu or Ketu, its south is zero. Therefore, based on the verses of the Chapter on True Positions 28 and figure 25, it can be easily known that the south of the Moon is more than 2 degrees 35 nakshas. In this figure, if we assume the Moon's orbit, the ecliptic, the position of Rahu, the position of

the Moon, the Moon's south, and the Moon's ultimate displacement, then the relation between the Moon and the ecliptic can be easily known. Here, if we assume that the Moon and the ecliptic are 2 degrees 35 nakshas, then

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sine (Vasa) = sine (Sapa) / sine (Savpa) = sine2035' / sine506' = .0451 / .0868 = .5022
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,. Vasa = 30 degrees 6 arcminutes
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That is, when the moon is one amount ahead of its node, then its shadow is more than 2 degrees 35 arcminutes. But Rohini is to the south of the ecliptic and its polar-facing portion according to Sūrya Siddhānta is 46 degrees 30 arcminutes and according to Shankar Balakrishna Dikshit is 47 degrees 37 arcminutes and the Kadamba-facing portion

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SOURCE PAGE 645

From the calculation of Sūrya Siddhānta, there are 48 degrees 6 kala and Shankar Balakrishna Dikshit has 45 degrees 57 kala from the calculation. Therefore, if the portion of the star of Rohini facing Kadamba is assumed to be 46 degrees, then at the time when the portion of the moon is the same, the Rohini-Shakta-Bheda may occur if its south shadow is more than 2 degrees 35 kala. In such a situation, the moon should be at least 1 rashi ahead of Ketu, that is, the Rohini-Shakta-Bheda can only occur when the portion of Ketu is at least 16 degrees.

From the above calculation it is proved that when the Moon is 1 sign ahead of Ketu, its southern deviation is 2°35' art. After this, its southern deviation increases to 5°8' art. At that time it is 3 signs ahead of Ketu. Then its southern deviation begins to decrease and when it is 5 signs ahead of Ketu or 1 sign behind Rahu, its southern deviation is not less than 2°35' art. Within this limit, the Moon passes the shadow of Rohini. However, it is proved above that when Ketu's portion is 16 art, that is when Ketu is at the 16th degree of Aries, if the Moon's southern deviation is 2°35' art, then there will be a Rohini-shkat-bhed. After this, even when Ketu moves backward due to its retrograde motion, the Moon will still pass the shadow of Rohini because at that time its southern deviation in Rohini constellation will be more than 2°35' art. Thus, until Ketu moves 4 signs behind the 16th degree of Aries, the Moon's southern deviation in Rohini constellation will remain less than 2°35' art. This means that when Ketu comes to the 16th degree of Aries, the beginning of the Moon's Rohini-shkat bhed will occur, and until it does not come to the 16th degree of Sagittarius, the Moon's Rohini-shkat-bhed in Rohini constellation will be there. However, it remains 6 signs ahead of Rahu-Ketu. Therefore, it can also be said that until it remains in the range from the 16th degree of Gemini to the 16th degree of Libra, the Moon's Rohini-shkat-bhed occurs.

Similarly, the Rohini shadow-difference of other planets can also be calculated. However, it is revealed by the table given in Chapter on Mean Motions on page 75 that no planet, except for Venus and Mercury, has a true shadow-difference of more than 2 degrees 35 arcminutes, so only the Rohini shadow-difference of Mercury and Venus is possible. Saturn's true shadow-difference is 2 degrees 26 arcminutes 36 arcseconds, so Saturn's Rohini-shadow-difference is also considered impossible.

SOURCE PAGE 646

It happens. But Varaha Mihira1 and Grahalaghakar2 have written that the separation of Saturn or Mangala from Rohini-Shakta causes a greater misfortune.

The means of the Yukaal—

> ग्रहवद् द्युनिशेभानां कुर्याद् दृश्कर्म पूर्ववत् ।
> ग्रहमेलनविज्ञेयं ग्रहभुक्त्या दिनादिकम् ॥१४॥
> एष्यो होने ग्रहे योगो ध्रुवकादधिके गतः ।
> विषयंयाद्वक्रगतैः ग्रहैः ज्ञेयः समागमः ॥१५॥

(14) Just as the day and night of the planetary conjunctions have been asked to be known, similarly the day and night of the constellations should be ascertained by means of the nodal and axis-related calculations. After this, just as the mutual conjunction and conjunction position of the planets are known, similarly the conjunction and conjunction position of the planet and the constellation should be known by the motion of the planet alone. (15) If the nodal-axis-related-calculation of the planet is less than that of the constellation, then it should be understood that the constellation and the planet are about to be in conjunction, and if it is more, then it should be understood that the conjunction has already occurred. But if the planet is retrograde, it should be understood in reverse.

Scientific Commentary—the rules explained in these two shlokas have come into the Chapter on Mean Motions. Here, the inclination and latitude of the planet are asked to be determined, but only the latitude of the constellation is asked to be determined. The reason for this is clear. Because the portion of the planet that comes according to the Chapter on True Positions is towards Kadamba, therefore performing the inclination determination in it makes it towards the pole. Now if the determination of latitude is also performed in it, its portion comes in the prime vertical circle. But the poles of the constellations

> १. रोहिणी शकटमर्कनंदनी यदि भिनत्ति रुधिरोथवा शशी ।
> किं वदामि यदि नष्टसागरे जगदशेषमुपयाति संक्षये ॥३५॥
> —बृहत्संहिता ३४ अध्याय

> २. कभशकटमसौ भिनत्त्यसृक् शनिरुडुयो यदि चेज्जनक्षयः ॥७॥
> भौमकर्योः शकटभिदा युगान्तरे स्यात् सेदानीं न हि भवतीदृशि

> स्वपाते ॥८॥

—Planet brevity, constellation shadow

SOURCE PAGE 647

They are poleward, therefore only the correction of the axis is required in them. In this way, it is known by dividing the difference in the enjoyment of planets and constellations at a certain auspicious time by the daily motion of the planet, how much time it will take for the planet to be in conjunction with the constellation or will be. And all things should be understood according to the rules stated in the Chapter on Planetary Rights. Here the ease is that the constellations are fixed, so only the calculation of the relation of one planet is needed.

The method of recognizing the conjunctions of constellations—

फल्गुन्योः भाद्रपदयोः तथैवाऽऽषाढयोर्द्वयोः ।
विशाखाश्विनिसौम्यानां योगतारा तथोत्तरा ॥१६॥
पश्चिमोत्तरतारा या द्वितीया पश्चिमे स्थिता ।
हस्तस्य योगताराऽसौ श्रविष्ठायाश्च पश्चिमा ॥१७॥
ज्येष्ठाश्रवणमैत्राणां बार्हस्पत्यस्य मध्यमा ।
भरण्याग्नेयपित्याणां रेवत्याश्चापि दक्षिणा ॥१८॥
रोहिण्यादित्यमूलानां प्राची सार्पस्य चैव हि ।
यथाप्रधानं शेषाणां स्थूलास्स्युध्रुवतारकाः ॥१९॥
पूर्वस्यां ब्रह्महृदयादंशकैः पञ्चभिः स्थितः ।
प्रजापतिर्वृषान्तेऽसौ सौम्ये अष्टत्रिंशदंशकैः ॥२०॥
अपांवत्सस्तु चित्राया उत्तरेंऽशंच पञ्चभिः ।
बृहत्किञ्चिदतो भागैरापष्षड्भिस्तथोत्तरे ॥२१॥

The Eighth Chapter:

Translation—(16) The star that is the opposite star in the post-western direction of the Hasta constellation. The other star among them is the conjunction star of this constellation, and in the two opposite stars of the Hasta constellation, the opposite star is the conjunction star. (18) The stars between Jyeshtha, Shravana, Anuradha, and Pushya are the conjunction stars of each. The star to the south of Bharani, Krittika, Magha, and Revati is the conjunction star of each. (19) The star to the east of Rohini, Punarvasu, Moola, and Ashlesha is the conjunction star of each. Of the 28 constellations, the remaining ones are

SOURCE PAGE 648

That is to say, in the Nakshatras Ardra, Chitra, Swati, Abhijit, and Shatbhishak, the largest star of each Nakshatra is the star that is the sum of that Nakshatra. (20) The star named Prajapati is 5 degrees before the star Brahmahridaya. Its westward deviation is 38. (21) The star Apamatsa is 5 degrees north of the star Chitra, from which the star named Apa is some larger than 6 degrees north.

Scientific Commentary—in 16-16 shlokas it is stated that which star is considered the main star in each constellation, for which Dhruva and Shar are previously mentioned. Such main star is called Yogatara. Nowadays there are some differences of opinion among scholars regarding these Yogataras. A table will be given further to show which scholar considers which star to be a Yogatara nowadays. Sometimes the names of their deities have been used for the constellations, so for convenience it will also be stated which deity is the lord of which constellation and how many stars are in each constellation. There are also differences of opinion in the numbers of stars among the ancient Acharyas as will be seen from the table.

It is stated that the pole of the Pole Star of Brahmahridaya is 1 degree 22 minutes. From this, the Pole Star of Prajapati is 1 degree 5 minutes east. Therefore, the pole of Prajapati is 1 degree 27 minutes. It is stated in the verse that Prajapati is at the end of the Taurus zodiac, but the meaning should be taken that it is near the end of the Taurus zodiac. The southern shadow of the star Chitra is 2, and the star Apamatsa is 5 minutes north of Chinna, so the northern shadow of Apamatsa is 3 minutes. The star Aap tara is 6 minutes north of Apamatsa, so its northern shadow is 6 minutes.

Four sky-pictures will be given for the identification of stars and constellations from which it can be easily known which constellation appears where in the sky at which time.

These tables give the English names of the stars in a peculiar way, so it is necessary to state how these names have been kept. The names of the star clusters that are prevalent in English are mostly taken from the Latin and Greek languages. A Greek letter has been added before the name of each star cluster. The order of these letters is mostly kept in such a way that the brightest and largest star in that cluster is revealed by the first letter 'Alpha'. The star that is smaller than it is revealed by the second letter 'Beta', and so on. Some of the names of the principal stars are in this way

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SOURCE PAGE 649

Deity of the Nakshatra

(Muhurta Chintamani and Indian)

Serial number Names of the Nakshatras Lord or Deity Tittiriya Samhita Kalpa Vriddhargiya Samhita Narada Samhita Barah Mihir Khandkhadyak 1 Ashvini Ashvini Kumar 2 2 3 3 2 2 Bharani Yama 3 3 3 3 3 3 Krittika Agni 7 6 6 6 6 4 Rohini Brahma 1 5 5 5 5 5 Mrigashira Chandra 3 3 3 3 6 Ardra Rudra 1ya2 1 1 1 1 1 7 Punarvasu Aditi 2 2 2 4 5 2 8 Pushya Brihaspati 1 1 1 3 3 19 Ashlesha Sarpa 6 6 5 6 6 10 Magha Pitar 6 6 5 5 6 11 Purva Phalguni Bhag 2 2 2 8 2 12 Uttara Phalguni Aryama 2 2 2 2 2 13 Hasta Surya 5 5 5 5 5 14 Chitra Vishwakarma 1 1 1 1 1 15 Swati Pawan 1 1 1 1 1 1

SOURCE PAGE 650

and the number of stars

Astrology page 458)

The names of the stars according to Hindu Astronomy are 3 2 3 3 3 □eta, γ Arietis 5 3 3 3 35, 39, 41 Arietis 6 6 6 6 6 η Tauri, etc. 5 5 5 5 α, θ γ, δ, ∈ Taurus 3 3 3 3 3 λ, φ_1, φ_2 Orionis 1 1 1 1 α Orionis 4 2 4 4 □eta, α Geminorum 3 3 3 3 3 θ, δ, γ Cancri 5 5 5 5 ∈, δ, σ-, η, ρ Hydrae 5 5 5 5 α, η, γ, γ, l, μ, ∈ Leonis 2 2 2 2 δ, θ Leonis 2 2 2 2 □eta, 93 Leonis 5 5 5 5 5 δ, γ, ∈, α, β Corvi 1 1 1 1 α Virginis 1 1 1 1 α Bootis

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SOURCE PAGE 651

Serial number Names of Nakshatras the lords or deities of the constellations Taittiriya Samhita Nakshatrakal Vrddha Gargya Samhita Narada Samhita Barah Mihira Khand Khadyak 16 Vishakha Indra, Agni 2 2 2 2 5 27 Anuradha Mitra 4 4 4 4 18 Jyeshtha Indra 1 3 3 3 39 Mūla Rakṣa 1 2 6 11 11 20 Pūrva Ashāḍha Jal 4 4 4 2 4 21 Uttarā Ashāḍha Viśvadeva 4 4 2 8 4 Abhijit Brahma 1 3 3 3 32 Śravaṇa Viṣnu 1 3 3 3 33 Dhanishṭhā Vasu 4 5 4 5 24 Śatabhisha Varuna 1 1 1 100 100 1 25 Pūrva Bhādrapad Ajapāda 2 2 2 2 26 Uttar Bhādrapad Ahirbhu Dnya 4 2 2 2 8 2 27 Revati Puṣa 1 1 4 32 32 1

SOURCE PAGE 652

The names of the stars according to Hindu Astronomy are gamma, beta, alpha, l Librae 43444 delta, beta, π Scorpii 33333 alpha, gammȧ, L Scropii 11911111 lambda, mu, x, L theta, epsilon Scorpii 24442 delta, epsilon, Sagittarii 244326, zeta Sagittarii 33333 alpha, epsilon, zeta Lyrae 33333 alpha, beta, gamma, Aquilae 45444 beta, alpha, gamma, delta Delphini 1001001001000 lambda Aquarii, etc. 22222 alpha, beta Pegasi 22222 gamma Pegasi, alpha Andromedae 323232323232 gamma Pisceium, etc.

SOURCE PAGE 653

The theory of the stars according to Bentley and Keropant, according to Bapudev, according to Shankar Bal-Krishna Dikshit, according to Chandrashekhar Singh Samant, Siddhanta Darpanabhumika p. 56, 57 alpha Arietis beta Arietis 35 Arietis alpha Arietis beta Arietis beta Arietis alpha Arietis 41 Arietis 41 Arietis beta Arietis alpha Arietis 41 Arietis eta Tauri eta Tauri Aldebaran 1 Ashwini alpha Arietis beta Arietis 35 Arietis 35 Arietis 35 Arietis 41 Arietis Aldebaran Abedbaran Aldebaran lambda Orionis gamma Gemini- Pollux delta Cancer Proesepe 2 Bharani mu or 35, Arietis 35 Arietis 35 Arietis 35 Arietis 41 Arietis Aldebaran Abedbaran Pollux Pollux delta Cancer delta Cancer delta Cancer Proesepe Proesepe 3 Kritika eta Tauri eta Tauri eta Tauri eta Tauri eta Tauri Aldebaran Aldebaran Pollux Pollux Gemini- Gemini- Gemini- Gemini- 4 Rohini alpha Tauri that is Aldebaran alpha Tauri that is Aldebaran 116 Tauri 116 Tauri 133 Tauri alpha Orionis alpha Orionis alpha Orionis Pollux Pollux delta Cancer delta Cancer Proesepe Proesepe 5 Mrigshira lambda Orionis lambda Orionis lambda Orionis lambda Orionis lambda Orionis alpha Orionis alpha Orionis Pollux Pollux delta Cancer delta Cancer Proesepe Proesepe 6 Ardra alpha Orionis 133 Tauri alpha Orionis alpha Orionis alpha Orionis alpha Orionis alpha Orionis Pollux Pollux delta Cancer delta Cancer delta Cancer Proesepe Proesepe 7 Punarvasu Pollux that is beta Gemini Pollux Pollux Pollux Pollux Pollux Pollux Pollux Pollux delta Cancer delta Cancer delta Cancer Proesepe Proesepe 8 Pushya

SOURCE PAGE 654

6 Ashlesha alpha cancri 49 cancri = Hydrae alpha cancri zeta Hydrae 10 Magha alpha Leonis Regulus Regulus Regulus 11 Purva alpha Leonis delta Leonis delta Leonis theta Leonis 12 Uttara beta Leonis meaning Denebola Denebola Denebola 13 Hasta gamma or delta corvi delta corvi gamma or delta corvi delta corvi 14 Chitra Spica spica spica spica spica 15 Swati Arcturus Arcturus Arcturus Arcutus Arcutus 16 Vishakha alpha or K Librae 24 Librae a or K Librae 24 Librae alpha Librae 17 Anuradha delta Scorpii delta Scorpii delta Scorpii delta Scorpii 18 Jyeshta alpha Scorpii Antares Antares Antares 19 Moola nu or 34 scorpii 34 scorpii 34 scorpii 45 ophiuchi lambda Scorpii 20 Purvaashadha delta Saggittarii delta Sagittarii delta Sagittarii delta Sagittarii 21 Uttaraashadha t Sagittarius t Sagittarius t Sagittarius π Sagittarii phi Sagittarii

SOURCE PAGE 655

The theory of the stars according to Colbrick, the theory according to White and Burgess, the theory according to Shankar Bal-Krishna Dikshit, the theory of Chandrashekhar Singh Samant, Mirror Frame pages 56, 57, alpha Lyrae Aega Aega Aega Aega Aega Aega Aega Aega Aega Aega 22 śravaṇa alpha Aquiae Altair Altair beta Delphini alpha Delphini alpha Delphini alpha Delphini alpha Delphini alpha Aquarii lambda Aquarii 23 dhaniṣṭhā alpha Aquarii Altair Altair beta Delphini alpha Delphini alpha Delphini lambda Aquarii Markab Markab beta Pegasi alpha Piscium 24 śatabhiṣaka lambda aquarii Markab Markab Algenib alpha Andromedae alpha Andromedae gamma Pegasi gamma Pegasi (Algenib) alpha Piscim xi or mu Piscum gamma Piscium 25 Purva Bhadrapad Markab meaning Markab Markab Markab Algenib alpha Andromedae alpha Andromedae gamma Pegasi gamma Pegasi (Algenib) alpha Piscim xi or mu Piscum gamma Piscium 26 Uttara Bhadra- alpha Andromedac Alpherat alpha Pegasi alpha Andromedae alpha Andromedae alpha Andromedae alpha Andromedae alpha Piscim xi or mu Piscum xi or mu Piscum gamma Piscium gamma Piscium

SOURCE PAGE 656

Greek letter names pronunciation same pronunciation of Roman letters used in our sky picture the numbers used are alpha alpha a 1 beta Beta b 2 gamma Gamma g 3 delta Delta d 4 epsilon Epsilon e short 5 zeta Zeta z 6 eta Eta e long 7 theta Theta th 8 iota Iota i 9 kappa Kappa k 10 lambda Lambda l 11 mu Stet m 12 nu Nu n 13 xi xi x 14 omicron omicron o short 15 π Pi p 16 rho Rho r 17

SOURCE PAGE 657

The Greek letters names pronunciation the Roman letters used in our sky picture are the numbers sigma Sigma s 18 tau Tau t 19 upsilon upsilon u 20 phi Phi ph 21 chi chi ch 22 psi Psi ps 23 omega omega olog 24

SOURCE PAGE 658

1 Name prevalent in Sanskrit literature 2 Synonyms of Sanskrit 3 English names 4 Latin names Aries Ram Taurus Bull Gemini Twins Cancer Crab Leo Lion Virgo Maiden Libra Balance Scorpio Scorpion Sagittarius Archer Capricorn Capricornus Aquarius Water-bearer Pisces Fishes

SOURCE PAGE 659

They have remained, but the names prevalent in their literature also come into practice to this day.

If it is known what the name of a star is prevalent in Sanskrit literature and what the name is in English literature, then there is great facility in recognizing the stars. Therefore, first by stating what the letters and their names are in the Greek language, a table will also be shown that what are the names of star clusters in Sanskrit and English and Latin and Greek languages. Instead of letters, the Hindi numerals will be used in our sky picture in order, as shown in the last column.

All the names in Sanskrit, Latin, and English have the same meaning, but similarities are also found in the letters of the Greek names, which indicates that they originated from the same country. That country may be Bharatavarsha or Greece or any other country from which these two countries have taken. This fact can only be clarified by the linguists and philosophers of language as to the reason for this unity. There are also other words in astrology that have equality in their Sanskrit, Arabic, and Greek names. However, comparative consideration will not be made on this subject here because its content is rare at this time. If convenience allows, this subject will be raised again in the introduction.

In this chapter, it is necessary to provide pictures for the identification of the constellations that have been discussed. Therefore, the sky pictures of the Falgun months are given. The Greek names of the stars are not given in these pictures, so there may be some difficulty in recognizing the stars by yoga, but there can be no difficulty in understanding the constellations, i.e., star-groups, and their positions. In these pictures only

1. It is a pity that Greek names were not given due to the lack of the type of Greek letters.

2. For the period from the Kartik month of 1678 to the Bhadrapada month of 1676, for which it was illuminated by the knowledge circle of Kashi, the ten celestial maps were made by this author at the wish of the editor Babu Sampurnananda. Four of these pictures have been selected and given. They also show the positions of Mars, Jupiter, and other principal constellation groups of that time. The names of those among these which have been discussed in ancient Sanskrit texts have been taken from the Sanskrit texts, but the names of those which are not discussed in the ancient texts have been kept as they are nowadays.

SOURCE PAGE 660

Those stars are not given which are discussed in this chapter, but are found in the English works of the sky or for which Hindi equivalent-information words have been made. Like Kashyap Mandala for Cassiopeia, Sifius for Cepheus, Ajagar for Draco, Shashak etc. Acharya Venkatesh Babu Ketkar has named the names of several principal stars on the pages 324 of his astronomical calculations on the names of famous sages and gods like Kanva, Kuvera, Rudra, Yama, Parashara etc. But these names are not given in this picture because they have not yet been fixed by any assembly, so it will be more convenient for the readers if the names that have gained much fame in the literature of the world are given.

In these pictures, the views of the sky are shown which can be seen between 8 o'clock to 10 o'clock in the evening in the months shown in the pictures from all places of 25 latitudes. The beginning of the month is considered from the second day of the transit because a picture made according to the lunar month cannot be useful for more than a month, whereas a picture made according to the transit can be useful for hundreds of years. The idea of transit is also considered in three ways nowadays. Here the idea of transit is considered according to the method of Surya Siddhanta. It seems necessary to tell the readers which date of the English month the transit falls on. Some things are also given in the table of the next two pages on how the twelve months of the year can be used from these four pictures, the method of which will be explained further.

What is shown together for three months is that the sky of the middle month can be seen at the time shown in the 6th column on the first date of the months preceding and succeeding these three months. Or, in other words, the sky of the month shown in bold letters can be seen at the time shown in the 6th column on the first date of the months before and after this month.

This table only shows that the pictures of the planets and Saturn on the 1st day of the month were also given in their respective places, which cannot be removed from the block, so readers should note that they will no longer see the planets there because the positions of the planets keep changing like the stars, they do not remain the same. For the generosity shown by the administrator of the knowledge circle, Babu Shivprasad Guptji, in giving these blocks, both the Science Council and the author are indebted to Guptji.

SOURCE PAGE 661

636 Sūrya Siddhānta

The solar month begins from which declination, the date of the day when the first day of the solar month is considered, the sky is made at the meridian of the first day of the solar month, the sidereal time dagger The declination of the sun is at the meridian time, and the sidereal time is also the meridian time (see page 275 footnote). This is the declination of the meridian time of Prayag in 1685 Vikram Samvat. It decreases by one minute every year, but after 4 years it usually becomes the same again. However, this difference is negligible. dagger Sidereal time is known by the sidereal-ghatika-yantra, and the start of the sidereal day occurs at the time when it passes the vernal equinox point or the meridian (see pages 337-36). "

SOURCE PAGE 662

Nakshatragrahayadhikar śrāvaṇa Cancer 17 July 7 45 11 43 Capricorn 17 August 6 46 19 6 42 Virgo 17 September 11 38 7 51 Libra 18 October 13 31 11 57 Sagittarius 17 November 15 26 1 6 56 Capricorn 16

December 17 34 7 55 Aquarius 14 January 16 42 11 46 Pisces 13 February 21 46 7 30 6 43 Aries 15 March 23 36 7 50

SOURCE PAGE 663

Which time should one see the sky picture? If one needs to use the sky picture from any other date of the month, then it should be kept in mind that the view seen at 10 o'clock on the 1st date of the month will be seen 4 minutes before 10 o'clock on the 2nd date, 4x2=8 minutes before 10 o'clock on the 3rd date, 4x7=28 minutes before the 8th date (i.e., one week later), and 15x4=60 minutes or 1 hour before on the 16th date, meaning it will be seen at 6 o'clock. The reason for this is that the Earth moves forward by 1 degree in the orbit of the Sun during the day and night, causing the Sun to appear one degree towards the east of the stars. Therefore, the Sun takes 4 minutes longer to come into the meridian circle each day, or the Sun's declination increases by approximately 4 minutes per day. However, the celestial time for which the sky picture is made is fixed, so the time at which the sky is seen on any day is seen 4 minutes before at the same time on the next day (see pages 463-466). The direct rule is that the view shown in the sky picture appears in the sky at the same time as the celestial time of the sky picture is earlier or later than the Sun's declination at noon. For example, on the 1st date of Vaishakh, the noon declination of the Sun is approximately 1 hour 26 minutes, and the celestial time of the sky picture of Jyeshtha is 10 hours 30 minutes, which is 12 hours 1 minute behind the noon declination. Therefore, the sky picture of Jyeshtha will be seen at 12:01 at night on the 1st date of Vaishakh. But the reason why 11:56 is mentioned in the 6th column is that 12 hours 1 minute is in the celestial time and 11 hours 56 minutes is in the Sun-dial time for Savan. Because it has been stated that the Savan day is approximately 4 minutes larger than the celestial day (see pages 337-36), the 6 hours of celestial time are the same as 5 hours 56 minutes of the Savan time.

According to this rule, if you wish to see all the stars in the sky on the first day of the month of Magha in the same night, you can see them easily. On this date, the stars set before five and a half o'clock of the sun in the directions of the lines connecting Bombay and Jagannath Puri, that is, in all of North India. Therefore, the stars in the sky become clearly visible at the time of 6 o'clock in the evening. On this date, the obliquity of the meridian sun is 16 hours and 42 minutes, so from noon

SOURCE PAGE 664

The sky picture of Makara (Margarita) can be seen at 6 o'clock in the evening. This means that you can easily recognize the 13 constellations from Shravana to Punarvasu or the 12 constellations from Dhanishtha to Punarvasu by listening. If you look at the sky at 12 o'clock at night, 6 hours behind, then the Nakshatra time will be approximately 7 hours 42 minutes, which can be useful for the sky picture of Phalgun month because the sky picture of Phalgun month is the one when the Nakshatra time is 7 hours 30 minutes. From this picture, you can easily recognize the stars from Ashvini to Hasta. Similarly, if you look at the sky at approximately 6 o'clock in the morning or 10, 12 minutes before on the same night, the sky picture of Jyeshtha can be useful because the Nakshatra time at 6 o'clock in the morning will be approximately 13 hours 40 minutes, and the sky picture 12, 13 minutes before will be from the time of the Nakshatra time of 13 hours 30 minutes. From this sky picture, you can see the stars from Punarvasu to Moola or Purvashadh. Similarly, it can also be calculated what sky pictures of which month can be useful on which other night.

Simple description of the diagram—the round line drawn in the diagram is the horizon of 25 degrees latitude, so it is usually found with the horizon of Prayag or Kashi. A sign of wealth is shown in the center in this manner. From this, the point in the sky is revealed which is just above the head at 25

degrees latitude. This is called khswastik or khmadhya. North, South, East, West and the directions between them are visible near the round line. The straight line seen from North to South is the meridian circle. The sun remains on this line at noon. The crooked line seen from East to West is the equator. On the days of spring equinox and autumn equinox, the sun appears on this and rises exactly in the east and sets exactly in the west. There is another line that cuts the equator, which is called the ecliptic. The sun is seen moving on this every day. In reality, this is the path of our Earth on which the sun travels and completes one revolution throughout the year. This path is of great importance. The moon and planets revolve around it in the sky.

SOURCE PAGE 665

See the following. The ecliptic is divided into 27 equal parts which are called constellations. The names of the constellations are also given in the sky picture of Margashirsha, but in other pictures, only the serial number of the constellations is given. For example, where '1' is written on the ecliptic, the 1st constellation Ashvini ends, where '5' is written, the 5th constellation Mrigashirsha ends, etc. Where a mark is made within a small circle on the ecliptic, according to Sūrya Siddhānta, the end of the constellation Revati and the beginning of the constellation Ashvini are understood. The lines of the ecliptic, the equator, and the meridian are not visible in the sky. Astrologers have imagined them for convenience.

Even when there is a dark night in the clear sky, countless stars are visible, but in these pictures only those are shown which can be seen even in a moonlight night. Some such stars have also been taken to show the shape, which do not appear because of the greater light of the moon three or four days before-after the full moon. The Akash-Ganga, in which numerous tiny stars appear joined to each other, is not shown in these pictures. In the dark night, this Akash-Ganga surrounds the sky from north to south, nourishing the Scorpio and Sagittarius constellations, and separating the famous Agrahayana and Lubdhaka maṃḍala, which bathe the Akash-Ganga, with the Capricorn, Mizar, Kashyap, Rajahamsa, and Shravana constellations from north to south.

When looking some time before the picture was made, they will not see the stars near the eastern horizon because they have not risen, and they will see the stars near the western horizon a little higher, and the stars near the meridian circle will seem to have moved a little towards the east. But if the sky is seen some time behind the above time, the stars near the eastern horizon will seem to have risen a little higher, and some new stars will also have risen near the horizon; some stars will remain set in the western horizon, and the stars near the meridian circle will have inclined towards the west.

There will be some other stars seen in the north where the latitude is 25. But in the places where it is south, some other stars will be seen in the south and there will be some difference in the height and lowness of the stars, but this cannot cause any difficulty.

The method of viewing the picture—the direction from which one is to look at the sky, bring that same direction down and stand the picture up. The star at the very bottom is the one that will be seen near the horizon. The stars from below to the center in the picture

SOURCE PAGE 666

The same stars will be seen from the horizon to the zenith in the same order.

The sky of the jyeṣṭha month—

Above the head—there are some east and south of the lord Khastikas. In a quarter of an hour, it will come onto the meridian and at that time it will be 5 degrees south of Khastikas.

North—the first 5 stars of the Saptarshi have moved to the west of the meridian. The sixth star Vashistha is on the meridian. Its pair star Arundhati will also be seen when looking closely near it. The seventh star Marichi is a little east and will come onto the meridian in 15 minutes.

Below the Saptarshi, 4 dim stars appear spread in a line from east to west. These are the stars towards the tail of the serpent, whose mouth at this time appears at approximately the same height as the stars of the Lesser Saptarshi below the serpent's coil in the north. Three stars of Cyphus to the east of north appear near the horizon.

North-East—near the horizon in this direction, the stars of the Swan constellation are visible. From here to the corner of East-South, a shining road appears. This is called the Milky Way. Countless stars are in the initial state in it. Above the swan, a very bright star is Abhijit. This is the third star of the first class. Next to it is the face of the python towards the east.

East—near the horizon, there are three stars of the śravaṇa constellation shifted slightly towards the north, the star between which is very bright and of the first class. Above śravaṇa and between the horizon, there is a cluster of harikuleśa, all of whose stars are of dim light. Just above some of the stars of the harikuleśa cluster, 5, 7 stars appear in the shape of a crown. These stars are also of dim light. Above this and near khasvastika, there is the svātī cluster, of which the star named svātī is a bright star of the first class, somewhat reddish in color.

East-South—at this time the stars of the Scorpio sign are adorning the sky with their unique appearance in this direction. It seems as if a large heavy scorpion is hanging in the sky whose head is made up of three stars of the Anuradha constellation and in its belly, three stars of the jyeṣṭhā constellation are hanging. The middle star is also of the first class and somewhat red. The scorpion's stinger is spread towards the south.

SOURCE PAGE 667

which has many small stars shining. The stars of the original constellations also appear close to each other near the horizon. Some stars of the Purvaashadha constellation appear towards the east near the horizon. The stars of the original and Purvaashadha are in the Sagittarius constellation, which has not risen fully. Two positions of the planet Mangala are shown above Purvaashadha in the picture, but they will not be seen here now. Two stars of the Vishakha constellation are seen spreading to the left above Anuradha. These are not very bright but are of great importance.

South—in this direction, two or three stars of the Centauri group are of the first class near the horizon. These are so southern that we cannot see them for more than an hour to the residents of Kashi, Prayag. Lucknow residents will see them for even less time. Aligarh and Bareilly residents will see them with difficulty and those to the north will not see them. Four stars of the second class appear close to each other towards the west near the horizon. These also do not appear for more than an hour.

The first-class star is some distance above between the cross and the southern horizon, which is of great importance due to its position. It is usually on the ecliptic. One and a quarter thousand years ago, the autumn equinox was near this star, meaning when the sun reached here, it went to the southern hemisphere. Nowadays, the autumn equinox has shifted by about 22 degrees 50 minutes from this star to the west, and it is at the place where the letter 'Sh' is written near the 12th constellation. A big debate

is going on in the Maharashtra province regarding this star. Those who say that the beginning of Ashwini constellation or Aries sign should be considered from the point which is 180 degrees away from the citrā star are called the Chaitra pakṣa. The supporters of this side are Acharya Venkatesh Bapu Ji Ketkar and other gentlemen. There is another side against them, which Lokmanya Tilak also supported. Their opinion is that the beginning of Ashwini is the point which is about 184 degrees away from the citrā star. This point is in the Revati constellation (see the picture of the Bhadrapada month). That is why this side is called the Revati pakṣa.

From the picture, some of the 5 stars of the Hasta constellation appear spread downwards like the fingers of the hand. Above Hasta, several slow stars of the Virgo constellation are seen. The two or three stars below, which are usually in a straight line, appear near the declination circle, usually parallel to it. At the western end of this line

SOURCE PAGE 668

The star that is there has the Sharad sampat point nowadays, so when the sun comes here, it goes to the south pole. This is why the star Chitra is about 23 degrees away.

South-west—there are no important stars in the sky of this direction. A curve of very slow stars extends from below the constellations of Chitra and Hasta to the west, and at its western end, one star is somewhat bright.

West—near the horizon, a star named Prashva is seen. Several dim stars appear in a curved line to the north from it, at whose northern end are two first-class stars. These are the two stars of the Purnavasu constellation. The curved line formed by the dim stars from Prashva to Purnavasu is Gemini. Above Prashva, there is a curve of many dim stars called Cancer. This appears to look towards the west. Slightly above it, moving towards the west, the stars of Leo and the middle of the horizon show their unique appearance. The neck of Leo is hanging downwards, in which 6, 7 stars can be easily seen, which appear to be the shape of a sickle. The star on the south or left side is somewhat red and is of the first class. This is called the star of Magha or simply the star of Magha. It is usually on the ecliptic, so it is of great importance. To the right and north of it is another star, which is slightly less bright than Magha but is bright enough that it can be seen even on the full moon night! Two stars shine to the right and left above Magha. These are the stars of Purva Phalguni constellation and are in the waist of Siharashi. In the tail of Leo, there is the single star of Uttara Phalguni of Purva Phalguni a little higher. Thus, it is revealed that the stars of two constellations in the western direction can easily attract people with their brightness; only the stars of Cancer need to be carefully looked at towards the south between Gemini and Siharashi.

West — near the horizon in this direction, only a star named Prajapati of the Prajapati circle is seen. The star Brahmahridaya has set some time ago. Besides this, there is no other bright star or group of stars near the horizon. Very high up, the stars of the saptarṣimaṇḍala (Ursa Major) previously mentioned are seen. In the line of Krutu and Pulahki, the two polar-indicating stars of the saptarṣimaṇḍala, there is a star towards the south. Between this and the north of Purvafalguini, there are very faint, serpentine stars.

SOURCE PAGE 669

They remind of the famous king Nahusha in the Puranas who became a snake due to the curse of sage August.

Thus the description of the sky picture of the jyeṣṭha month is complete.

Sky Picture of the month of Bhadrapada

Above the head--at this time three famous constellations are seen near Khastrak. Three stars of the śravaṇamaṇḍala are usually seen a little south of Khastrak on the meridian. Near this, four stars of the Dhanishtha constellation are very close to each other but dim. This constellation is of great historical importance. When the sun reached here in the time of Vedanga-Jyotish-Kaal, the beginning of Uttarayana started.

Near Khastrak there is a dim star which is the last star of the swan's tail. There are two more stars in a straight line in the north-east direction from it which are brighter than it but the northern one is the brightest among them. Two or three stars are seen in a straight line next to the middle star, forming a right angle with the first line, which seem like the wings of a swan. It seems as if the swan is swimming with its wings spread in the galaxy. The southern constellation of the swan is whose brightest star is also famous by the name Abhijit. It is outside the Milky Way towards the west. In brightness, this star is in the third place.

Akashganga—it is not visible in the picture, but its view at this time is very pleasant. At this time it extends from the northeast horizon to the southwest horizon. In the northeast direction, the Parshu or Parsik circle is rising at this time. From there, the beginning of Akashganga is also seen, which, bathing the Kashyap circle in its path, happens beside Sifius and thoroughly intoxicates the swan. Its two branches are formed from the stars to the north of the swan, which seem to move in a parallel direction and meet near the southwest horizon. The eastern branch obscures the Shravana constellation while dipping the root of the Sagittarius sign and the Purvashadha constellations into the horizon. In the western branch, there are very few bright stars. It also disappears while submerging the stars of Scorpio near the southwest horizon. The constellation of Jyeshtha is seen on the western edge of this branch.

SOURCE PAGE 670

North—the stars of the little Ursa Major are spread towards the west from the pole. Seeing some more serpents of the little Ursa Major hanging to the west is seen, whose four stars of the mouth are seen spread near Abhijit. Near the tail of the serpent, the pole-indicating stars of the Ursa Major maṃḍala are seen near the horizon between the north and northwest directions. The other stars of this Ursa Major maṃḍala are seen in the northwest direction.

There are four maṃda stars above to the east of the Pole Star, and the stars of the Kashyap sphere to the east appear to form the basis of the English letter double-U (W). Below the Kashyap sphere, in the north-east direction, the stars of the Parshu or Parsik sphere are near the horizon.

East—near the horizon between the east and northeast directions, three stars of the Ashvini constellation appear as they rise. Above these, the arc of Andromedus is visible, which begins near the Parsec maṃḍala. On this arc are the stars of Purvabhadrapad and Uttarabhadrapad constellations. Two stars each of these two constellations together form a square, which is called Bhadrapadavarga or (square of Pegasus). The two stars below the square are in the Uttarabhadrapad constellation, and the upper stars are in the Purvabhadrapad constellation. When moving towards the south along the line of the stars of Uttarabhadrapad, there is a point of Spring-Equinox at approximately the same distance where these two stars are, where the ecliptic and the tropics appear to intersect each other. When the

sun appears here, the spring season begins, and the sun comes into the northern hemisphere. On this day, day and night are equal, and from this time, the day begins to become larger and the night smaller.

East-South - There are very few bright stars in this direction. The number of stars in this direction in the month of jyeṣṭha have all moved to the south-west direction in this month. A first-class star (Pleiades) is certainly visible near the horizon, which is appropriate to call Kumbha in Hindi, although Kumbha is very different from the star Augustya. I have named it Kumbha for two reasons. One reason is that it is near the constellation Aquarius, and the second reason is that it usually appears at 7, 8 o'clock in the evening during the month of Ashwin when the rainy season ends. Whereas the star named Augustya rises in the middle of the rainy season and is visible only for a short time in the morning. Some from Kumbha

SOURCE PAGE 671

And three stars to the south appear as three angle points of a right-angled triangle. They are named Saras because they are called Crane in English.

There are some dim stars of the Aquarius constellation that are shifted towards the east above Aquarius. Between Saras and Shravana constellation, there are dim stars of the Capricorn constellation.

South—there are no bright stars near the horizon in this direction at this time. The constellation Shravana is seen very high near Khastak.

South-West—just as in the month of jyeṣṭha, the south-east direction is adorned with the stars of the Scorpio and Sagittarius constellations, similarly in this month the south-west direction is twinkling with the stars of these two constellations. The special thing here is that at this time, all the stars of Sagittarius, as well as the stars of Purvashadha and Uttarashadha constellations, are visible. The stars of the Vishakha constellation also appear near the western horizon of Scorpio.

West—there are no stars of special importance in this direction at this time. The stars of Vishakha are some shifted south. The star of Swati is shifted towards the north, but it can be said that the star of Swati is usually in this direction. Above the Swati circle and above the crown, there are dim stars of the Kurita and the dim stars of the Harikulesh circle, which have been well described in the description of the direction to the east of the sky of the month of jyeṣṭha.

Sky Picture of Margashir Month

In this month the sky remains very clear. The dust of the months of Vaishakh and Jyeshtha and the clouds of Shravan and Bhadrapada are not seen anywhere, nor does the fog of Magha and Phagun reach the vision. Therefore, there is both knowledge and mental delight from the sky-picture of this month. In the sky of this month, many new stars and star groups are seen in the east direction, which have also been discussed in many places in ancient literature.

North—the stars of the small saptarṣi hang near the horizon. At this time, the Pole Star is at the very top among them. Three faint stars of Cygnus are seen spreading towards the west above the small saptarṣi. As much above the horizon as the Pole Star is, the constellation of Cassiopeia is seen in the shape of the English letter M. Its four large stars pass the meridian circle and go towards the west

SOURCE PAGE 672

They have gone, only one star is a little east of the meridian. There is a curve of Antarmada above the Kashyap circle, of which only one star is now east of the meridian and all have gone towards the west.

Above the head—the Ashvini constellation is seen exactly on the head.

North-East—some stars further east in this direction and further shifted have risen of Prajapati circle, from which five main stars appear forming a pentagonal region. The two stars in the north of this circle are very bright and appear below above. The lower stars are called Prajapati and the upper one is Brahmahridaya. Its place in the shine is the fourth. The brightest star in the sky is Lubdhaka, which is some south of the east direction at this time and is seen near the horizon. The second star is Agastya, which has not yet come above the horizon. The third star is Abhijit, which is seen near the north-west horizon, and the fourth star is Brahmahridaya. There is a star named Agni opposite to Brahmahridaya at the southern corner of the pentagonal region.

Above the Prajapati circle is the Parsik circle or Parashumanda, on whose southern end the 6 stars of Krittika constellation are seen close to each other. Above the Parsik circle, there are usually three stars of the Ashvini constellation on the head, of which two are large.

Before - A star named Prashva of the first class has risen in this direction, but it is very close to the horizon. A little further south near the horizon, Lubdhaka is shining with its divine light. Above Lubdhaka and Prashva is the famous constellation Orion, which is considered very important since ancient times due to its divine light, shape, and fame. Lokmanya Tilak has proven in his famous book Orion that the mantra in the Vedas in which it is discussed must have been published at least 6000 years ago. This is also called Kalpurush. It has been discussed in the Greek and Persian literature in very figurative language. In the middle of this constellation, three bright stars often appear close to each other in the same line, which are called Ilvak. The star at the very top is usually on the ecliptic, so the point on the horizon where this star rises is the exact east direction and where it sets is the west direction. Four stars show their extraordinary appearance on the four corners of Orion. The star at the bottom, the northern one, is somewhat red.

SOURCE PAGE 673

It appears. This is called the yoga star of the Ardra constellation. The star above it is called the yoga star of the mṛgaśirā constellation. The star above it towards the south is also of the first class. The villagers call this circle hannāhannī and use its position to determine the time in the winter night. Many stars in the southern part of the Agrahayana circle are dim, from which the shape of a shashka seems to be formed. That is why they can be called shashka (Leporis).

Above Agrahayana, slightly to the north, is the Rohini constellation, and the star below it is somewhat reddish of the first class. Because of this color, it was named Rohini. The shape formed by the 5 stars of the Rohini constellation resembles the letter (V) in English. To the north of the Rohini constellation is the Prajapati circle and above it is the Krittika cluster, which the villagers call Kachpachiya. Time is also determined from this at night. Above Krittika, there is usually the Ashvini constellation.

The star clusters discussed at this time and those visible in the east direction, which are visible all night in the winter season, are called the Winter constellations.

No bright stars are visible in this direction, East-South.

South—There is a cluster of three stars near the horizon in this direction, which is called Phoenix in English. A very dark region is visible above it, whose mouth seems to hang and spread downwards in

the shape of a whale. All its stars are dim.

South-West—In this direction, the crane and Kumbha or the second August are visible at this time. The discussion of the latter has already been done.

West — near the horizon between the south and west directions, the slow stars of Capricorn are spread out. The stars of Aquarius are also visible above them.

In this direction, near the horizon, the stars of the Virgo constellation are seen. Above Virgo, there are the stars of the Aquarius, a little north. Very above Virgo are the stars of the Pushya and Uttara Bhadrapada, whose square is also very clearly visible. The two stars of Pushya below the square area and the two stars of Uttara Bhadrapada above the square area are called the stars of Bhuj.

SOURCE PAGE 674

In this direction, the constellation visible in this direction appears near the horizon. Above the constellation, there are stars of the haṃsamaṃḍala. From this and the north horizon, some stars of the ajagara's face are seen.

From this and the north horizon, some stars of the ajagara's face are seen.

The ākāśa-gaṃgā – at this time the ākāśagaṃgā extends from the east horizon to the north-west horizon. In the east horizon, it divides in two branches near the haṃsamaṃḍala between the pārasīka maṃḍala and the kāśyapa maṃḍala, which leaves praśvā on the north shore and lubdhaka on the south shore on the north of āgrahāyaṇa, and between agni and brahmahṛdaya, and merges in the north-west and north-west horizons.

Sky map of the month of Phalgun

At the head—the Gemini sign is exactly at the head at this time. Both stars of Punarvasu are usually on the khastak and prashva is some south.

Answer - The little saptarṣi is spread to the east of the Pole Star. There are three stars of the Pisces in the west of the Pole Star, one of which is exactly on the horizon. The little saptarṣi is the wrap of the ajagara to the east, whose mouth is still below the horizon.

North-East - The seven stars of the saptarṣi constellation are visible in this direction. There are serpentine dim stars above the saptarṣi.

Near the horizon between the North-East and East directions, there is some red-colored Swati star.

East—Near the horizon in this direction, the stars of the Virgo constellation are visible. citrā has not yet risen. Above the Virgo constellation, the stars of the Leo constellation are visible. The single star below is the constellation Uttarafalgunī. Above it are two stars of the Purvaphalguni constellation. The stars of the Magha constellation appear in the shape of a swan above the Purvaphalguni. The two stars below this swan are very bright, among which the Dakshinwala star is a conjunction star of Magha. Some red-colored stars are also visible here.

Above the swan are many dim stars. The northern stars are called Pushyanakshatra and the southern stars are called Ashlesha Nakshatra. The Cancer constellation is also here. All the dim stars between Punarvasu and Magha can be said to be in the Cancer constellation.

SOURCE PAGE 675

4,5 stars are visible near the horizon between the east and east-south directions. These are the stars of Hasta Nakshatra.

The constellations visible near the ecliptic from the eastern horizon up to the head and some to the west are called the rain constellations. This is not because they are seen in the rainy season, but because rain occurs when the sun is in these constellations. The names of the rain constellations in order are:—Ardra, Punarvasu, Pushya, Ashlesha, Magha, Purva Phalguni, Uttara Phalguni, Hasta and Chitra.

No famous star is visible in this direction—east-south at this time.

South—There is a group of many bright stars near the horizon in this direction that looks like a ship, which is why it is called Argo Navis. The principal star of this group has moved to the west from the August azimuth and is visible near the horizon. Its place in brightness is second. The first place belongs to Lubdhaka, which is visible just above it. Above the Argo Navis is the Lubdhaka circle.

West-South—There is no attractive constellation near the horizon in this direction. Some above Shashaka and even higher are the famous Agrahayana circle. Above the Agrahayana circle are usually the stars of the Gemini constellation on the head.

West—Some Ashwini constellations are near the horizon, slightly north. Two or three very dim stars above them are called Bharani constellation. Three more stars to the north from Bharani form a triangle. To the west of Bharani is the Krittika constellation. Some above Krittika and to the west is the Rohini constellation. To the north of Krittika is the Parshika circle, and above these two constellations is the Prajapati circle, whose fire star is above Krittika and above Brahmahridaya Parshika. Above Brahmahridaya is the star of Prajapati. The stars to the north of Parshika and Prajapati circles, Brahmahridaya, Prajapati etc., are seen in the north-west direction.

Some stars of the triangle are visible near the horizon of the intermedium.

North-West—in this direction are the stars of the Parśika and Prajapati circles which have already been discussed. Some stars of the Kashyapa circle are near the horizon and away from this direction.

SOURCE PAGE 676

The ākāśa-gaṃgā is spread from the northwest corner to the southern horizon at this time. Starting from the northwest horizon, there are stars of Kashyap, Parsik, Prajapati, Agrahayan, Lubdhaka Mandala, and Nauka Punj and their surroundings.

From the celestial pictures and descriptions of these four months, the information of all the principal stars and star groups of the sky can be obtained. With their help, when the sky is clear at night, knowledge of direction, country, and time can be easily obtained.

Thus, the scientific commentary of the eighth chapter named Nakshatragrahayutyaadhikar has ended.

SOURCE PAGE 677

Chapter on the Right Ascension

(Brief Description)

[1 verse—The idea that planets and stars become invisible due to coming near the sun. 2-3 verses—The direction of the rising and setting of planets. 4-5 verses—The method of knowing the occultation of planets. 6-6 verses—The ultimate occultation of planets. 10-11 verses—The method of knowing how many days are remaining or have passed for a rising or setting at a certain time. 12-15 verses—Which star has what ultimate occultation. 16-17 verses—The method of knowing the day of visibility or disappearance of a star. 18 verses—The names of those stars that are never invisible]

In this chapter, it is explained when the rising and setting of planets and stars occur and how they are known. Here, the meaning of rising and setting is different from the ordinary meanings of rising and setting. Generally, when the sun, moon, etc., come above the eastern horizon, they are considered to have risen, and when they go below the western horizon, they are considered to have set. This is due to the daily motion of the Earth, which the old Acharyas called Pravah-gati. Besides this, when planets, the moon, or stars come very close to the sun, so that they come above the eastern horizon almost before sunrise and below the western horizon almost after sunset, they are also considered to have set. In such a situation, they cannot be seen due to the intense light of the sun. The time from which they become invisible because they come near the sun is considered as setting, and the time when they are so far from the sun that they begin to be seen a little before sunrise or a little after sunset is considered as rising. Similar discussions of rising and setting are mentioned in this right. Western astrologers call this heliacal rising and setting.

Purpose of the chapter—

> अथोदयास्तमययोः परिज्ञानं प्रकीर्त्यते ।
>
> दिवकरकराक्रान्त मूर्तीनामल्पतेजसाम् ॥१॥

SOURCE PAGE 678

Translation—(1) The method for knowing the rising and setting of bodies that have little light due to being affected by or obscured by the light of the sun is explained.

Scientific Commentary—Its explanation has been given above. Direction of rising and setting -

> सूर्यादभ्यधिकाः पश्चादस्तं जीवकुजार्कजाः ।
> ऊनाः प्रागुदयं यान्ति शुक्रज्ञौ वक्रिणौ तथा ॥२॥
> ऊनाः विवस्वतः प्राच्यामस्तं चन्द्रज्ञमार्गवाः ।
> व्रजन्त्यभ्यधिकाः पश्चादुदयं शीघ्रयायिनः ॥३॥

Translation—(2) When the portions of Jupiter, Mars, and Saturn are some more than the portion of the Sun, they set in the west, and when they are some less than their portion, they rise in the east. Similarly, the rising and setting of retrograde Venus and Mercury also occurs, that is, when the portions of retrograde Venus and Mercury are more than the portion of the Sun, they set in the west, and when they are less, they rise in the east. (3) The Moon, (emergent) the portions of Mercury and Venus are less than the portion of the Sun, then they set in the east, and when they move some ahead of the Sun due to rapid motion, they rise in the west.

Scientific Commentary—In these two verses, it is briefly stated how the positions of the portions of the Sun, Moon, Mars, Mercury, etc., or true positions, can be ascertained in a thick manner as to which planet will rise or set in which direction. For this work, the planets have been divided into two parts. In one part are Jupiter, Mars, and Saturn whose motion is slower than the motion of the Sun, and in the other part are Mercury, Venus, and the Moon whose motion is faster than the motion of the Sun. There is some difference in the motions of Mercury and Venus in these as well.

Because the Sun moves more than the Guru, Mangal, and Shani, the Sun appears to be further and further than the Guru, Mangal, and Shani. When the Sun gets so close to them that they become invisible, the portion of the Sun is greater than their portion because the measurement of the portion is from west to east. Before becoming invisible, these three planets are seen near the western horizon behind the sunset and there, due to the intensity of twilight, they become invisible, so it is said that these three planets set in the west. In some days when the Sun is ahead of them

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It increases and their portion becomes less than the portion of the sun, then they begin to appear a little before sunrise in the east. Therefore it is said that they rise in the east.

When the portions of retrograde Mercury and Venus are more than the portion of the sun, they appear in the western horizon after sunset and become invisible there. In a few days, due to their retrograde motion, they move very quickly to the other side of the sun and their portions become less than the portion of the sun. In such a situation, they begin to appear in the eastern horizon before sunrise. Therefore it is said that retrograde Mercury and Venus also set in the west and rise in the east.

But the motion of the Moon and the direct Mercury and Venus is more than the motion of the sun, so when they move towards the sun and reach close to it so much that they become invisible, their portions become less than the portion of the sun and they become invisible in the eastern horizon before sunrise. Therefore it is said that they set in the east. When they move ahead of the sun, their portions become more than the portion of the sun and begin to appear in the western horizon after sunset. Therefore it is said that the Moon and direct Mercury and Venus rise in the west.

Method of knowing the dark portion-

> सूर्यास्तकालिकौ पश्चात्प्राच्यामुदयकालिकौ ।
> दिवाकरग्रहौ कुर्यात् दृक्कर्माथ ग्रहस्य तु ॥४॥
> तयोलंग्नान्तरप्राणाः कालांशाः षष्टिभाजिताः ।
> प्रतीच्यां षड्भयुतयोस्तद्वल्लग्नान्तरासवः ॥५॥

(4) If one wants to know the time of rising or setting of a planet in the west, then clarify the Sun and the planet of the sunset time of the day estimated by estimation, and if one wants to know the time of rising or setting of a planet in the east, then clarify the Sun and the planet of the sunrise time of that day and perform the visual impression of the planet. Find the difference in the angles of the Sun and the planet of the rising signs of the Sanskrit planets and the Sun and divide it by 60, then the portion of time related to the rising or setting of the planet in the east is known. If one wants to know the portion of time related to the rising or setting of the planet in the west, then by adding 6 to the portions of the Sun and the planet in the shadow, find the portion of time by dividing the difference in the angles of the rising signs by 60.

SOURCE PAGE 680

Scientific Commentary—The time before the rising of the sun when any planet comes into the eastern horizon, that time is called the interval of that planet. The calculation of the ascendant time is done in asus for subtlety, and the rise of one aspect of the Vishnu-circle is one asu. Therefore, the rise of 60 kals is 60 asus, but 60 aspects are like a fraction. Therefore, the difference between the rising times of the sun and the planets, which usually occurs in asus and which is called lagnantara-prana or lagnantaras in the 5th verse, when divided by 60, should be understood in fractions, this is called the kālāṃśa of the planet.

It is stated in page 561 that to know at what time the planet is in the ascendant in the horizon, the axial and meridian coordinates should be calculated in the true portion because according to the true positions, only the portion of the planet that comes from the planet's orbit is known. However, the planet's orbit is different from the ecliptic, so at the time when the planet's ecliptic point comes to the horizon, the planet's image does not rise on the horizon but some ahead or behind according to its shadow (see figures 107, 108), the knowledge of which is obtained from the meridian coordinates. That is why in the fourth verse, it is said to perform the meridian coordinates first. When the exact position of the planet on the horizon is known after performing the meridian coordinates, then it can be known how much before sunrise the planet is in the ascendant in the eastern horizon.

But when the rising or setting of the planet is in the west, the time of the sun and the planet at sunset is clarified because then it is necessary to know how much time the planet sets behind the sunset. For this work, the *drikkarm sanskara* is also required in the planet, as is done at the time of the rising ascendant. Now, the difference between the *drikkarm sanskara* of the planet or in the words of Bhaskaracharya, *driggraha* and the *asatalagna* of the sun should be known, that is, to see how many degrees after the time the sun sets the image of the desired planet appears on the western horizon. Dividing these degrees by 60 gives the knowledge of the time or fraction of the setting time. But it is stated in the latter part of the 5th verse that the difference between the ascendants of the setting sun and the *driggraha* is obtained by adding 6 signs or 180 degrees to both and taking the difference of their ascendants. The reason for this is that at the time when the sun is setting, the point of the meridian of the ecliptic in the eastern horizon is

SOURCE PAGE 681

It is that which remains 180 degrees ahead of the sun. Similarly, when the planet is setting, that point remains in the eastern horizon which is 180 degrees ahead of the planet. Therefore, if it is known what the difference is in the ascensions of the positions of the eastern horizon in the settings of the sun and the planet, then the knowledge of the azaansh or kalansh can be obtained.

The ultimate azaansh of the planets—

> एकादशामरेड्यस्य तिथिसङ्ख्याऽर्कजस्य तु ।
> कालांशा भूमिपुत्रस्य दश सप्ताधिकास्तथा ॥६॥
> पश्चादस्तमयोऽष्टाभिः उदयः प्राङ्महत्तया ।
> प्रागस्तमुदयः पश्चादल्पत्वाद्दशभिः भृगोः ॥७॥
> एवं बुधो द्वादशभिः चतुर्दशभिरंशकैः ।
> वक्री शीघ्रगतिश्चार्कात्करोत्यस्तमयोदयौ ॥८॥
> एभ्योऽधिकैः कालभागैर्दृश्या न्यूनैरदर्शनाः ।
> भवन्ति लोके खचरा भानुभाग्रस्तमूर्तयः ॥६॥

Translation—(6) The ultimate azaansh of Jupiter is 11; that of Saturn is 15 and that of Mars is 17. (7) Because the image of Venus appears to be larger, the ultimate azaansh of it setting in the west and rising in the east is 8, but because the image is smaller, the azaansh of it setting in the east and rising in the west is 10. (8) Similarly, when Mercury, which has retrograde and quick motion, is at 12 azaansh from the sun, it sets in the west and rises in the east. But the azaansh of it setting in the east and rising in the west is 14. (9) Because it is affected by the light of the sun or is suppressed, if the azaansh of any planet at a certain time is more than its ultimate azaansh, then that planet is seen, and if it is less, it is not seen.

Scientific Commentary - In these verses, the limit of the azaans of the planets is indicated beyond which the planets are seen and below which they are not seen. Therefore, this limit can be called the ultimate azaansh. The ultimate azaansh of each planet is different. The reason for this is that the planet whose image is larger can be seen easily even when it is near the sun, and the planet whose image is smaller can be seen with some difficulty. Among the distant planets, Jupiter's image is the largest, so its ultimate azaansh is 11

SOURCE PAGE 682

It is considered that if Jupiter rises or sets 11 degrees or 110 pal 44 minutes before sunrise or 110 pal 44 minutes after sunset, then it can be seen even in the twilight of the morning or evening. Therefore, when the dark portion of Jupiter decreases until it becomes 11, then it becomes invisible in the western horizon. After this, when its dark portion decreases until it becomes zero, the knower knows that it rises or sets with the sun. From this time, its dark portion begins to increase, and until it becomes 11 degrees, it remains invisible because it cannot be seen in the intense light of the sun. This is also called Guru-Aditya or 'Guru-Badik' in common speech. This period is usually 1 month. During this period, Hindus do not perform any auspicious work like marriage, tonsure, etc.

The image of Saturn is smaller than the image of the Sun and larger than the image of the Sun, therefore the ultimate position of Saturn is considered to be 15 and that of the Sun is 17. However, the rising and setting of these are not considered in auspicious matters.

The supreme aspects of Venus are considered to be 8 and 10. The reason for this is that when Venus is retrograde and sets in the west and rises in the east, its distance from the Earth remains very small because it is between the Sun and the Earth (see Chapter on True Positions pages 101-103). Because of the closeness, its image appears very large, so it can be seen for a long time in the twilight. Its limit is considered to be 8 aspects 32 minutes or 80 shadows, meaning that if Venus sets in less than 32 minutes after sunset, it is not seen, and it is said that Venus has set in the west. After this, when Venus begins to rise 32 minutes before sunrise, it starts to be seen again, and it is said that Venus has risen in the east. This period is not more than a week because when Venus is retrograde, the distance between

Venus and the Sun decreases or increases by the sum of their motions daily, so Venus quickly goes behind the Sun.

But when Venus sets in the east and rises in the west, its absolute position is 10 because at this time it remains on the other side of the Sun, very far from the Earth (see, figure 21, 22). Because it is far, the image of Venus appears smaller, so it does not remain visible for as long during twilight as it does when retrograde.

SOURCE PAGE 683

When it happens, it is seen. When it sets in the east, it remains in the same direction, that is, its motion is in the direction in which the sun appears to move. Therefore, the difference between these two is the same as the difference in their motions per day. Therefore, this period of the setting of Venus is about two months.

Until Venus is set, no auspicious work like marriage, tonsure, etc., is done among Hindus.

Like Venus, when Mercury is retrograde, it appears larger due to being near the Earth and its ultimate degree is 12. But when it is in the other side of the sun, very far from the Earth, it appears smaller and its ultimate degree is 14.

The thought of Mercury setting is not done in marriage, tonsure, etc.

Up to here it has been stated how the rising and setting of the planets are calculated according to the Sūrya Siddhānta and what their ultimate positions are. Now here two questions arise, one is whether this method of knowing the positions is pure, and the other is whether these ultimate positions are correct. Answering this is easy because it can be checked by the direct observation of these planets. Because the definition of their rising and setting is such that they should be considered set until they cannot be seen without any instrument because they are near the sun, otherwise they should be considered risen. Applying this test proves that in the rising or setting times derived from the Sūrya Siddhānta or any other Indian astrological principle, there is sometimes a difference of ten or fifteen days. It is apparent that the exact calculation of the positions is only possible when the true portions and shadows of the planets are exact. However, the portions and shadows taken based on Indian principles are not correct as stated in many places in the previous chapters. For example (Pūrṇimānta) Chaitra Krishna 11 Bhauamvara Samvat 1683 according to the days, the midnight of 26 March 1627.

1. It is not based on the Indian astrological principles of astronomical calculation of Acharya Ketkar, but is based on Western theories, which also took the help of recent discoveries.

SOURCE PAGE 684

Sunrise Chapter Mars Mercury Jupiter Venus Saturn Ra An Ka Ra An Ka Ra An Ka Ra An Ka Ra An Ka 1 25 38 10 21 10 26 50 0 18 36 7 6 26 15 40 16 30 10 24 36 0 16 38 7 14 11 Bharatbhushan Panchang 1 Ganeshdatta Sharma's Panchang 1 1 25 52 10 21 47 10 24 44 0 15 48 Naval Kishore Press's Panchang 4 1 25 52 10 17 58 10 22 42 0 16 37 Vikram Vijay Panchang 5 1 26 53 10 22 20 10 23 54 0 15 35 according to astronomical calculations

SOURCE PAGE 685

The positions of the five celestial bodies are given according to the six Panchangs, from which it is also apparent how much difference there is in the calculation of the planets according to different opinions

in our place. (See page 656)

Comparing the portions of each planet reveals that they are very different from the portions calculated by Jyotirganita or Drigganita according to the pure Sūrya Siddhānta. The portions of Jupiter and Saturn differ by five-five six six degrees; in contrast, the portions known according to the Makaranda table are very similar to those from Drigganita. Therefore, the idea of the rising and setting of planets cannot be correct according to the Sūrya Siddhānta. Besides this, it has already been shown that the method of Driggkarma Sansara is also crude. Therefore, it is proven that to consider rising and setting, we should use only the Drigganita fundamental measures, and for this, we should either use Jyotirganita, which is based on the Western astrological principles, or prepare a new independent theory, because knowing the rising and setting of planets based on Navik Panchangs does not seem appropriate for determining one's religious acts, mutilations, marriages, etc.

SOURCE PAGE 686

The description of the place Panchang Shukastakal Shukrodayakal Guru ka astakal Guru ka udayakal Kashi Balakrishna Shastri ka jyeṣṭha Shukla 10, 85 26 May 1628 AD more Shravan Su 13, 85; 30 July 28 A.S. Shravan Su 13; 30 July Chaitra Shukla 3, 85 Vikram Samvat Vishakha Shukla 8, 85 VI 27,, Vishwapanchanga Kashi Vishwa Vidyalaya ka jyeṣṭha Shukla 10, 26 May A.S. Shravan Su 13; A.S. Shravan Su 12; 30 July 24 March 1628 AD Vishakha Shukla 8, 27 April Vishakha Shukla 7, 26 April Lucknow Ramprasad Siddhanti ka Naval Kishor Press ka jyeṣṭha Shukla 2, 21 May A.S. Shravan Su 3, 4 August Chaitra Shukla 4, 25 March Vishakha Shukla 7, 26 April Ound Shastrasuddha Aikya Vardhak Panchang A.S. Su 3, 6 June A.S. Shravan Su 11, 28 July Chaitra Shukla 3, 24 March Vishakha Shukla 1, 21 April Bagalkot Ketki Panchang,,,,,,,, Vishakha Su 1, 21 April Vishakha Su 1, 21 April Poona Chitrashala Panchang A.S. Su 3, 6 June A.S. Shravan Su 11, 28 July Chaitra Shukla 3, 24 March Vishakha Shukla 1, 21 April * Here it is written according to the calculation of Krishna Paksha Purnimanta. It is the jyeṣṭha Krishna which is located in the Maharashtra province.

SOURCE PAGE 687

Description of the Place Panchang Shukrastaka Shukrodayaka Guruastaka Guruudakaala Pune Panchang Pravartak Committee's Jya 13, 1 June Shra 15, 1 August Chaitra Shukla 3, 24 March Vaishakh Shra 3, 22 April,, Shankar Shastri's Jya 14, 2 June A. Shra 6, 26 July,, 2, 23 March Vaishakh Shra 4, 23 April Mumbai Balakrishna Tuka-Ram's,, 15, 3 June A. Shra 11, 28 July,, 1, 22,, Vaishakh Shra 2, 22 April,, Gurjarati Panna's News Press in Chaitri Panchang,, 3, 22 May A. Shra Kr 3, 4 August,, 4, 25,, Vaishakh Shra 6, 25 April

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SOURCE PAGE 688

Until here it has been stated that to know the rising and setting of planets, there is the ancient custom of knowing the aspect. Now it is also necessary to state how much difference there is nowadays in the magnitude of the ultimate aspect of the planets. For example, we have shown on the previous page the times of rising and setting of Jupiter and Venus of this year, and who considered how much ultimate aspect.

From this bracket it is clear that according to both calendars of Kashi, the days of Shukrast and Shukroday are the same, but there is a one-day difference on the day of the astakāla of Guru. Similarly, in the śāstraśuddha aikyavarddhaka calendar of Aundh, the Ketaki calendar of Bagalkot, and the Chitrasala calendar of Poona, the days of rise and setting of Shukra and Guru are the same. It is

apparent that the calendar makers of Kashi considered the ultimate portion of these planets in agreement and the calendar makers of Maharashtra considered something in agreement. It is proven by the viśvapaṃcāṃga of Kashi that the rise and setting of the planets in it have been fixed based on the Navik calendar of 1628 AD. The Ketaki calendar is made according to Jyotarganit which is similar to modern astrological principles, so it is easily known what the ultimate portion of Guru and Shukra was considered by Acharya Ketkar and his followers.

Chapter on True Positions

Date: Solstice of the Sun, Revolution of the Sun, Solstice of Venus, Revolution of Venus, Hour Minute Second Arc Degree Deviation Hour Minute Second Arc Degree Deviation 26 May 4 24 16.56 21 37 37.6 3 47 2.67 16 12 16.4 30 July 8 37 46.86 18 31 8.8 6 11 37.26 17 37 6.1 6 June 4 57 3.32 22 36 38.2 4 27 54.56 21 20 18.7 28 July 8 26 57.38 18 56 32.7 6 1 36.6 18 18 6.5

SOURCE PAGE 689

On May 26, the revolution of the sun is 21 degrees 37 arc minutes 37.6 arc minutes or 21038’ and the revolution of Venus is 16 degrees 12 arc minutes 16.4 arc minutes or 16012’. To know at what time the sun and Venus will appear on the horizon, it is necessary to first know their variable periods (see figure 60 page 308). The latitude of Kashi is 25020’.

```
rising sun's sine = 210° 38' X 25° 20' = 3666 x 4737 = 1879  therefore  arc = 10050'  therefore  time = 43 minutes 20 seconds
```

The revolution of the sun is to the north, therefore by subtracting this from the sun's obliquity, it will be known which point of the equinoctial circle is in the east at the time of sunrise (see figure 60).

```
time from sun's declination 42417 period 4320 difference 34057
```

Therefore, the point of the equinoctial motion in the time of sunrise which is in the east is 3 hours 40 minutes 57 seconds or 3 hours 41 minutes ahead of the vernal equinox.

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UdaykalikShukrakicharja = 16° 12' X 25° 20' = 3482 x.4737 = 1646 ∴ Charansh = 6° 30' ∴ Charakaal = 38min.h.m.Shukrakavishuvansh 347Charakaal 38Antar 36
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Therefore, at the time Venus will come on the eastern horizon, that point of the equant which is 3 hours 6 minutes ahead of the vernal equinox will be the ascendant in the east.

SOURCE PAGE 690

It is stated above that the time of the sun's ascendance in the Lagna period was 3 hours 41 minutes, so there will be a difference of 3 hours 6 minutes = 32 minutes in the ascensions of the sun and Venus. Therefore, according to the world calendar, the ultimate time of Venus at the time of setting of the sun is 32 minutes and the ultimate portion is 8. It is necessary to state here that the viṣuvāṃśas given in the navigator's calendar in the upper box are for the mean meridian time of May 26 of Greenwich, which is approximately five and a half o'clock in the evening of Kashi. In reality, the viṣuvāṃśas and declination from the sunrise time of Kashi on this day should be taken, but there is a very small difference in the

motions of Venus and the sun, so there is no harm in considering the relative difference of these two even in the morning as much as it is understood for the evening.

There is another thing to consider. It is stated in the Chapter on the Three Problems that there is a variation in light due to the atmosphere, due to which the sun appears two and a half minutes before the actual rising time (see page 378). Therefore, the ultimate time of Venus calculated from the above does not remain 30 minutes or even half a minute less in reality.

Now it should be seen what is the phase of Venus on June 6. For this, the viṣuvāṃśa and declination from the morning should be taken as this will have more accuracy. The calculations of seconds and declination will not be done here.

Puna Sun Venus solstice hour minute revolution degree art solstice hour minute revolution degree art solstice hour minute revolution degree art 5 June evening 4 53 22 33 4 23 21 6 6 June evening 4 57 22 40 4 28 21 20 6 June sunrise 4 55 22 36 4 25.5 21 13

therefore Pune mein Surya ki charjya = Spere18° 30' X Spere2236 = 3346 X 4163 = 1363

Puna's latitude is 18^(°)30'.

SOURCE PAGE 691

therefore arc = 8 ° therefore time = 32minutes

Therefore, in the time of sunrise, the Equinoctial ascendant = 4 hours 55 minutes - 32 minutes = 4 hours 23 minutes

Venus sine = 18° 30' X 21° 13' = 3346 X 3882 = 1266 therefore arc = 7° 28' therefore time = 30 minutes approximately

Therefore, at the time when Venus is on the horizon, the Equinoctial ascendant will be.

4h253 / 2minutes - 30minutes = 3h553 / 2minutes

But the time of the ecliptic longitude in sunrise = 4hours23minutes

Therefore, according to the astronomical tables or the Ketaki panchang, the ultimate time of Venus is established.

4hours23minutes - 3hours553 / 2minutes = 273 / 2minutes

If 23 minutes of hour is given from this, because due to the motion, sunrise happens 2 or two and a half minutes before the calculation time, then the ultimate time of Venus is 25 minutes, which is equal to two and a half degrees.

Thus, it is proven that there is a big difference in the sunrise time and the Venus-ast (Venus-ast) time even from visual calculation because some ultimate portion considers one thing and some another.

Therefore, there is a great need that the astrologers of the entire India must together decide what the ultimate portion of which planet should be considered, otherwise this confusion of the calendars can never end.

Now there is no need to expand by giving more examples. What Acharya Bankatesh Bapu Ketkar has written in page 333 of his astrological calculations regarding the ultimate portion of Venus is given here as is:-

In the pure atmosphere, between the summer and winter, the Sun is visible. It can be seen even between the time intervals of the five parts of the effort. In that relation, the light of the sun is generated, which only those who are knowledgeable in the light of the eye can see.

SOURCE PAGE 692

Childhood and Old Age

It is clear that the atmosphere is not always clear. In the days of summer, there is so much dust that the sun cannot be seen even a little distance above the horizon, therefore it is very difficult to see Venus or Jupiter in such a condition. The second thing is that due to the decline and intensity of the observer's vision, there can be a difference of two or three days in the visibility of the planets. Due to all these reasons, the days of the rising or setting of the planets may be two or three or four days before or after they become invisible. It seems for this reason that the old sages considered the childhood and old age of Jupiter and Venus, but there is no consensus in this either, as written in Muhurta Chintamani:-

> पुरःपश्चाद्धगोर्बाल्यं त्रिदशाहं च वार्धकम्
> पक्षं पंच दिनं ते द्वेगुरोः पक्षमुदाहृते ॥२७॥
> ते दशाहं द्वयोःप्रोक्तं कैश्चित्सप्तदिनं परैः ।
> त्यहं त्वात्ययिकेऽप्यन्यैरर्धाहं च त्यहं विधोः ॥२८॥१

Even in the childhood and old age of Jupiter and Venus, many auspicious acts are prohibited just as in their setting.2

Should the consideration of rising or setting be from kalaansh or unnataansh?

Another point is worth considering in this regard. From what has been said so far about the rising and setting of planets, it has become clear that they are said to have set only when they come so close to the sun that they cannot be seen due to the twilight of morning or evening. However, the intensity and limit of twilight are not the same in all seasons and places. Anyone can experience that the limit of twilight increases in the winter days and decreases in the summer days. The reason for this is that twilight is related to the portion of the sun below the horizon, which depends on the sun's declination and the latitude of the ecliptic (see page 261 sutra 1). It has been proven by experience that as long as the sun is not more than 180 degrees below the horizon, some part of its light still returns to the Earth through the atmosphere. From the time of the sun's setting until the time when it does not go more than 18 degrees below the horizon, when dim light is seen, that is called twilight.

1. Ritual Chapter

2. See the auspicious and inauspicious matters of Muhurta Chintamani, shlokas 46, 47

SOURCE PAGE 693

Similarly, from the time when the rising of the sun is eighteen degrees below the horizon, the commencement of the morning twilight begins. It is apparent that when the sun remains eighteen degrees below the horizon, it is 60+18=108 degrees below the cross. Therefore, it can be said that the twilight begins when the eastern part of the sun is 108 degrees, and the morning twilight remains until the time when its western part is 60 degrees. Similarly, as long as the western part of the sun remains between 60 and 108 degrees, the evening twilight remains.

In Sutra (1) of page 261, it is stated what is the relationship between nataansh and natkal and how it depends on latitude and declination. In this sutra, if the values of 108 for nataansh, and the latitude and the solar declination of the iststhan are raised, then if the solar rising or setting natkal is subtracted from the natkal that will arise, the magnitude of the saṃdhiprakāśa (saṃdhiprakāśa) of that day will be known. To know the natkal of rising or setting, the magnitude of nataansh of 60 degrees 35 art must be taken because the direct nataansh of the sun or any planet rising or setting is 60, but due to the rotation of the atmosphere, the actual nataansh increases by 35 art (see pages 378-380). Therefore, the nataansh of the solar rising or setting in reality is 60035'.

Thus it is proven that the meridian light is not the same in all seasons and all places, therefore for knowing the appearance and disappearance of planets, the ultimate portion of the same planet for all places and all seasons will have to be considered different, where the meridian light will remain for a long time, the same ultimate portion will not work, which can work for a short meridian light. Considering all these things, it seems reasonable that the consideration of the disappearance and appearance of planets should be done from their altitude rather than the declination, as written by Acharya Ketkar ji on page 333:-

All planets appear in the vicinity of the Sun by some measure. The opinion of the preceding four treatises does not agree that this wonder is due to the conjunction of the Sun and the planets. Because the twilight glow of the evening reflects the portion of the Sun's setting, and also the time intervals. In the country of 350 degrees of latitude, the light of the conjunction of the Sun and the twilight occurs 3 hours 40 minutes before the sunrise. On the day of the supreme inclination, it is 4 hours.

SOURCE PAGE 694

40 time passes. These parts of time are 220, 280 in sequence. Therefore, the demonstration is that if one observes the time with these parts, the mathematics mentioned by the predecessors is contrary to the gross.

Therefore, the observation of the planets is done only by the mathematical work of the precession angle.

According to Acharya Ketakar's view, the degrees of Venus is 60.4 and that of Jupiter is 11°. (See Jyotiganyat p. 351)

Example—What is the duration of the conjunction of light on the days of the conjunction of sign, Capricorn conjunction, and sign Cancer conjunction in Kashi?

Latitude of Kashi is 25°18'

The revolution of the sun on the days of the conjunction of the sign of Capricorn and the sign of Cancer is 23°27' (See page 306) and the revolution of the sun on the days of the conjunction of the sign of Aries or Libra is zero.

When the meridian light of the day is the transit of the sun north—

It is stated that the portion of the meridian light is 108 at the beginning or end of the sun, therefore according to the formula (1) of 261 pages

cojyanatkal = cojya108^ circ - sine25018' imes sine23027' cojya25018' imes cojya23027' = - sine18^ circ - sine25018' imes sine23027' cojya25018' imes cojya23027' = -0.3060 -0.4274 imes 0.3676.6041 imes 0.6175 = -0.3060 -0.1701.8265 = -0.4761.8265 = -0.5776

Here, cojyanatkal is negative, so the meridian angle is more than 60 degrees. If it is more than a degrees from 60 degrees, then

cojyanatkal = cojya (60 + a = - sine a = -5776 herefore a = 35^ circ 17' herefore meridian light start time = 60^ circ + 35^ circ 17' = 125^ circ 17'

SOURCE PAGE 695

According to page 376, the time of sunrise in Kashi

= 101° 50' + 43' = 102° 33'

Therefore, the time of twilight = 125° 17' - 102° 33' = 22044' = 1 hour 30 min 56 seconds

Twilight of the day of the sidereal Capricorn conjunction-

At this time, the revolution of the sun is south, therefore the minus sign will become a plus sign in the above formula (see page 263) and the time of the beginning of twilight will be proven by the equation written below—

Cozyantaka = -sin180 + sin25018 * sin23027Cozy25018 * Cozy23027 = -0.3060 + 0.1701.8265 = -0.1386.8265 = -0.1675

Sūrya Siddhānta

Scientific Commentary

Chapter on Mean Motions

Chapter on True Positions

Chapter on the Three Problems

SOURCE PAGE 696

Sūrya Siddhānta = - · 3066040 = - · 3418 therefore co-latitude = co-latitude (60 + a) = -sine a = - · 3418 therefore

a = 16° 56' therefore angle of commencement of twilight = 60 + 16° 56' = 106° 56'

According to page 380, the time of sunrise is 60^°38'7" or 60^°36'. Therefore

Sandiprakashkal = 106 ° 56 ′ - 60 ° 36 ′ = 16 ° 20 ′ = 1 hour 17 minutes 20 seconds

Thus it is proven that the twilight period at a place is not the same in all seasons. In the calculation above, the sun is understood to be on the horizon at the time when the center of the sun comes to the horizon, but the upper image of the sun touches the horizon about 1 minute before that because the apparent arc of the sun is about 16 degrees. Because of this, the twilight period becomes less than 1 minute.

How many days have passed or remain of the sunrise time—

> तत्कालांशान्तरकला भुक्त्यन्तर विभाजिताः ।
> दिनादितत्फलं लब्धैभुक्तियोगेन वक्रिणः ॥१०॥
> यल्लग्नासुहते भुक्ती अष्टादश शतोद्धृते ।
> स्यातां कालगतीताभ्यां दिनादि गत गम्ययोः ॥११॥

(10) The difference between the instantaneous time and the ultimate time of the planet, written in the arts, and dividing it by the difference in the daily timings of the sun and the planet (if the planet is direct) and by the sum (if the planet is retrograde), is the number of days. (11) Multiplying the ascendants of the sign in which the sun or planet is, by the true daily motion, and dividing the product by 1800, what is obtained is the planet's time motion. The days of rising or setting are known from the difference or sum of the time motions of the sun and the planet.

Scientific Commentary—if one wants to know on any day that the rise of a planet

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SOURCE PAGE 697

If there are how many days until setting or how many days have passed after rising or setting, then the zodiacal portion of the planet of that day should be known according to 4-5 shlokas so that it is known that the planet rises how many before sunrise or sets how many after sunset.

If the kālāṃśa is more than the paramakālāṃśa and the portion of the sun is more than the portion of the planet—and this planet is Mercury or Venus in the direct motion—then it should be understood that there are still some days remaining until its setting. But if this planet is Mars, Jupiter, or Saturn, or retrograde Mercury or Venus, then it should be understood that some days have passed since its rising. But if the kālāṃśa is more and the portion of the sun is less than the portion of the planet, and the planet is Mars, Jupiter, or Saturn, or retrograde Mercury or Venus, then it should be understood that there are still some days remaining until their setting. Conversely, if the kālāṃśa is less than the paramakālāṃśa and the portion of the sun is more than the portion of the planet—then it should be understood that some days have passed since the setting of the direct Mercury or Venus, and some days remain until the rising of Mars, Jupiter, Saturn, and retrograde Mercury or Venus. But if the portion of the sun is also less than the portion of the planet, then it should be understood that some days remain until the rising of the direct Mercury and Venus, and some days have passed since the setting of Mars, Jupiter, Saturn, and retrograde Mercury or Venus. In all these cases, to know the number of these days, the difference between the kālāṃśa and the paramakālāṃśa should be calculated and observed how

many days this difference will reduce to zero or how much it has increased. To do this, this difference should be divided by the difference in the daily motions of the sun and the auspicious planet if the planet is in direct motion, but if it is retrograde, their daily motions should be added as stated in the grahayutyadhikāra. However, the daily motion of the sun or planet is generally circumpolar, and the kālāṃśa is viṣuvad-circular, which is why in the 11th verse it is stated that to convert the circumpolar daily motions to the viṣuvad-circular, one should multiply the daily motions of the sun or planet in the zodiac sign in which it is, and divide by 1800 because the time of rising of the sign is the same as the time of rising of the zodiac signs; therefore, the time of rising of the planet should also be understood in the same proportion as its daily motion. Since the daily motion is small, it is generally written in degrees, which is why a sign is also written in 1800 degrees. This results in the daily motion of the planet becoming viṣuvad-circular, which is why

SOURCE PAGE 698

This is called time because from it the time is easily ascertained. The rule of 10-11 shlokas in the language of algebra can be written as follows:

The time of the house rightarrow time in the house = time of the sun × time of the house (1)/(1800) years of the number of days of the house × time of the house = time of the sun + time of the house (7) (R) (3)

If the planet is retrograde, the sign of the positive should be kept in the final equation, otherwise the difference of both should be calculated. Here the sign of difference is more appropriate than the sign of negative because the time of motion of a planet is more or less than the time of motion of the sun.

The rule given for knowing the time period of a planet is somewhat crude. The reason for this is that the motion of the planet does not occur on the ecliptic but on its orbital circle, which is somewhat different from the ecliptic, but this causes no particular loss. If the obliquity and declination of the planet are known, then the difference in the obliquity per day is the time period. Knowing the obliquity, the zodiacal position of the planet can also be known conveniently and accurately because then the need for the quadrant is not necessary. Therefore, in my opinion, more knowledge of the obliquity is required for calculating the rising, setting, and conjunction of planets or stars than the zodiacal positions, which can only be obtained with the help of modern astronomical principles and instruments. For this, a observatory is necessary, where our astrologers can observe the planets and stars and determine their positions and nodal points exactly.

The absolute portion of the stars

> स्वास्थ्यगस्त्यमृगव्याधचित्राज्येष्ठाःपुनर्वसुः।अभिजिद्ब्रह्महृदयंत्रयोदशभिरंशकैः॥१२॥

This is the sign of the difference and indicates that the smaller number should be subtracted from the larger number in its right and left.

SOURCE PAGE 699

> हस्तश्रवण फाल्गुन्यी घनिष्ठा रोहिणी मघाः ।
> चतुर्दशांशकैर्दस्या विशाखाऽश्विनिदैवतम् ॥१३॥
> कृत्तिकामैत्रमूलानि सार्परौद्रक्षमेव च ।
> दृश्यन्ते पञ्चदशभिराषाढाद्वितयं तथा ॥१४॥
> भरणीतिष्यसौम्यानि सौक्ष्म्यात्त्रिस्सप्तकांशकैः ।
> शेषाणि सप्तदशभिस्तथा दृश्यानि भानि तु ॥१५

(12) Swati, Augustya, Mrigvyadha or Lubdhaka, Chitra, Jyestha, Punarvasu, Abhijit, Brahmhridaya stars have the supreme angular positions of 13. (13) Hasta, Shravana, Purva Phalguni, Uttara Phalguni, Dhanishtha, Rohini, Magha, Vishakha and Ashvini have the supreme angular positions of 14. (14) Krittika, Anuradha, Moola, Ashlesha, Ardra, Purvashadha, and Uttarasadha constellations have the supreme angular positions of 15. (15) Because they are subtle, the supreme angular positions of Bharani, Pushya and Mrigshira are 21. If they are more than this, they are visible and if they are less, they are invisible. The supreme angular positions of the remaining constellations are 17.

Known Commentary—in the 15th verse, the constellations for which indications are given are those which have been discussed in the chapter on constellation planetary rights, but whose names are not given here. From these time intervals of the stars, it is also apparent which star is of which class according to the opinions of our teachers. The stars of the first class in brightness are given in the 12th verse, whose time intervals are 13. The stars of the second class come in the 13th verse and their time intervals are 14. The stars of the third class are written in the 14th verse, whose time intervals are 15. Except for these, the class of the stars that have come in the 15th verse cannot be precisely determined.

Nowadays, the division of stars according to brightness is done in a very subtle manner. In a dark night, without the aid of any instrument, a sharp-eyed man can see no more than 6000 stars in the entire sky. These 6000 stars are divided into 6 categories (magnitudes). The division of these categories is done in such a way that a star of the first category is 100 times brighter than a star of a particular star in the sixth category. It follows from this that a star of any category is 2.5116 times brighter than the star of the category below it, that is, a star of the first category is 2.5116 times brighter than a star of the second category, the star of the second category is 2.5116 times brighter than the star of the third category, but the star of the first category is...

SOURCE PAGE 700

From 2.5116 × 2.5116 = 6.3066. The brightness is multiplied, etc. These are the stars that can be seen by the sharp-eyed without any instrument. Stars up to the 15th order have been seen with the telescope. It is necessary to state here that the stars that are in the same order are not all of the same brightness. The number of stars placed in the first order is not more than 20, but among them, Lubdhaka is the brightest. After that comes the number of Agastya. There is such a difference in their brightness that anyone can see it easily. Therefore, for more subtle calculations, ten more divisions have been made in each order. It is apparent that the brighter the star's brightness, the smaller its order number is, so the brightest star of the first order, Lubdhaka, has a negative and— 1.4 order, and its brightness is considered to be 6.1. The relation between the order and brightness can be easily understood from the table below*:

The brightness of the star of the sixth class = 15th,,,,,, = 2.5 times 4th,,,,,, = 6.3,, 3rd,,,,,, = 15.8,, 2nd,,,,,, = 36.6,, The brightness of the star of the first class = 100 times the brightness of the brightest star of the

first class, Lubdhaka = 400,, The brightness of the sun = 2400000000000 times

Since the brightness of a star is not always the same, there has been a difference in the order of the first class of 20 stars in old and new books in two or four places. The reason for this difference is also that the standard for testing the brightness was earlier some gross and is now subtle. This will be revealed by the table on the next page2:-

1. Quoted from pages 10 of Sar Narman Lakyar's (Elementary Lessons in Astronomy).

2. See The Twentieth Century Atlas of Popular Astronomy by Heath, Second edition pp. 112

SOURCE PAGE 701

Name Category Brightness New Category1 Sun..... —26.5 Full Moon..... —12.0 Venus..... —3.0 36.8 Lobdhaka alpha canis majoris, sirius — 1.4 6.1 Argus alpha, canopus — 0.8 5.2 — 1.58 Brahmhriday alpha, Aurigae, caplela 0.1 2.3 —0.86 Swati alpha, Bootis, Arcturus 0.2 2.1 0.21 alpha centauri... 0.2 2.1 0.24 Abhijit alpha, Lyrae, Vega 0.2 2.1 0.06 beta Orionis, Regel... 0.3 1.6 0.14 alpha Eridani, Achernar... 0.4 1.7 0.34 Prashva alpha, canis minoris, Procyon 0.5 1.6 0.60 beta centauri..... 0.7 1.3 0.48 alpha orionis, Betelguese... 0.6 1.1 0.86 alpha cruxis..... 0.6 1.1 Parivartsheel Shravan alpha, aquilae, Altair... 0.6 1.1 0.86 Rohini alpha, tauri, Aldebaran 1.0 1.0 1.06 Chitra alpha, Virginis, spica... 1.1 0.6 1.21 Punarvasu alpha, Geminorum, Pollux..... 1.2 0.8 1.21 Jyeshta alpha, scorpii, Antares 1.2 0.8 1.22 Magha alpha, Leonis, Regulus 1.3 0.8 1.34 Kumbhaj alpha, Piscis Australis, Fomalhaut..... 1.3 0.8 1.26 alpha cygni, Deneb 1.4 0.7 1.33

1. This is according to the Nautical almanac of 1626 AD.

SOURCE PAGE 702

The Sun is 631000 times brighter than the full moon. What is revealed by the new series given in the fourth column is that there is a difference in the order of many stars. According to this, Centauri and Abhijit come after Augustya, not like Brahmahṛdaya as shown in the old series. Similarly, this should be understood regarding other stars.

The second method of knowing the navigable days—

> अष्टादशशताभ्यस्तान् दृश्यांशःस्वोदयासुभिः ।
> विभज्यलब्धाः क्षेत्रांशास्तैः दृश्यादृश्यताथवाः ॥१६॥

One can also know the day of being visible and invisible by multiplying the part (Bhogansh) obtained by dividing (16) or the apparent part (Kalansh) by 1800 by the lagnas.

Scientific Commentary—This is the opposite of the rule stated in the 10th and 11th shlokas. There, it is said to divide the apparent difference by the difference in daily motions, and it is stated here how the daily motion can be known from the daily time. Here, there is no need to know the daily motion, but rather it is stated to convert the apparent difference into the difference of the true positions. Therefore, its derivation is the same as that stated there. If this shloka had been given after the 11th shloka, it would have been more appropriate because its relation to the 15th shloka is very little.

Knowing the rising and setting of stars—

> प्रागेषामुदयः पश्चादस्तं दृक्कर्म पूर्ववत् ।
> गतैष्य दिवसप्राप्तिर्भानुभुक्त्या सदैव हि ॥१७॥

Translation—(17) The stars rise in the east and set in the west. The axis of the stars should be kept as before and to know the past-path of the rising and setting, one should use the motion of the Sun.

Scientific Commentary—Since no motion is seen in the constellations, the Sun is seen as reaching them. When the Sun becomes so close to them that they are obscured by its light, then their setting is understood. Therefore, their setting always happens in the west as with the slow-moving planets Mars, Jupiter, and Saturn. When the Sun moves so far ahead that they begin to be seen, then

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SOURCE PAGE 703

Their rising is understood and at this time they fall in the pūrva horizon before the sunrise.

It has already been said that the precession of the constellations does not change, therefore only the correction of the akṣajyā is required to know their kālāṃśa.

It has just been stated that to know the elapsed day of rising and setting, the kālāṃśa difference is divided by the difference in the time of the Sun and the planets. But the motion in the constellations is zero, so only the motion of the Sun needs to be used to divide the kālāṃśa difference.

Stars that never set—

> अभिजिद्ब्रह्महृदयं स्वातीवैष्णववासवाः ।
> अहिर्बुध्न्यमुदकस्थत्वान्न लुप्यन्तेऽर्करश्मिभिः ॥१८॥

Translation—(18) Abhijit, Brahmhṛdaya, Swātī, Shravaṇa, Dhanishṭha, Uttara Bhādrapad do not hide from the light of the Sun because they are very late.

Scientific Commentary—When the Sun comes to the declination of these stars or near it, their difference is so much to the north that they rise or set so much before the time of sunrise or sunset that they appear, so they can never be hidden from the light of the Sun. This is even clearer from the table of 676 pages*:-

From this it appears that the revolution of the sun is only towards the north in front of the Brahma-heart, otherwise it is to the south, when the revolution of the stars is always to the north. The difference in revolution between Brahma-heart and the sun is also about 23 degrees. Now to see what is the proper time of Brahma-heart in Kashi or Prayag?

sine = declination line X latitude line

∴ Brahmahridayacharya = 45° 56' X 25° 25' = 1.0331 X.4752 =.4606

∴ arc = 26 ° 24 ′

∴ time = 1 hour 58 minutes approximately

On this day, the ecliptic time of the sun is approximately 47 minutes. Both of them are north. Therefore, the rise of Brahmahṛdaya is 1 hour 58 minutes - 47 from the sunrise time.

SOURCE PAGE 704

Names of stars Vipuvansh Abhijit Kranti Uttar Surya ka Vishuvansh Surya ki Kranti Kranti Uttar Surya ka Vishuvansh Surya ki Kranti time minutes arc time minutes arc 38 43 0 45 55 49 20 36 38 20 18 34 26 1 18 36 16 23 11 18 South, 30 days 51 26 26 45 55 40 5 12 34 23 0 8 North. 10 June 14 12 25 16 33 5 14 13 5 13 23 6 days, 26 October 16 47 19 8 40 47 19 47 18 21 9, 41 days, 15 January 20 36 20 15 36 38 20 37 56 18 30 24 days, 27 January

SOURCE PAGE 705

Minute = 1 hour 11 minutes earlier and it will be so much behind the sunset that this day can be seen at both morning and evening times. The day when this star is in the ascendant in the eastern horizon during sunrise is then sixteen hours after the sunrise time, when the sun cannot set in more than 14 hours. Therefore, it can also be well seen in the evening on this day. Similarly, the day when this star is in the ascendant in the western horizon during sunset, it rises more than 2 hours before sunrise and gives sight to people. Therefore, it can be said that in the countries to the north of Kashi Prayag, it can never be invisible, yes, in those places whose latitude is less than 20 degrees, it will be invisible for some days, therefore it can be seen every day in Jagannath Puri but not in Bombay.

Among the remaining stars, there is a star whose northern precession is very little. Therefore, it should be seen where this rule is correct for it.

The time of listening in Kashi Prayag = approximately 17 minutes.

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Sūrya-carakaal = 43,,,,
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The revolutions of both are different, therefore the rising of Shravana will be 1 hour before the sunrise of this day, which is 17+43 minutes. But the phase of Shravana is 56 minutes, so this can be seen well. But this rule cannot be applied to the countries to the south of Kashi Prayag.

In this way, other stars can also be known.

Thus the Scientific Commentary of the sixth chapter named Udayāstāhikāra has ended.

SOURCE PAGE 706

Tenth Chapter Chapter on Mean Motions

(Brief Description)

[Verse 1—The method of knowing the rising and setting of the Moon is the same as before and the phase is 12. Verse 2-4—The method of knowing the daily setting of the Moon in the waxing phase. Verse 5—The method of knowing the daily rising of the Moon in the waning phase. Verse 6-8—The method of knowing the linear difference between the Sun and the Moon at sunset. Verse 9—The method of knowing the image of the waxing part of the Moon. Verse 10-14—The method of drawing the outline of

the waxing part of the Moon. Verse 15—The rule for drawing the outline of the lunar image in the waning phase.]

The method of knowing the rising and setting time of the Moon is explained in this chapter. In the preceding chapters, only the type of rising and setting is described in which the planet becomes invisible due to coming very close to the Sun. However, in this chapter, in addition to this type of rising and setting, there is also a method of knowing the daily rising and setting time of the Moon. Then the method of knowing is explained that what part of the lunar image is illuminated on which day and how its shape can be drawn. At the beginning of the waxing phase or at the end of the waning phase, the shape of the illuminated or waxing part of the Moon is like a snake and it remains raised towards the north or south, that is why the name of this chapter is Śr̥ṅgonnatyadhikāra

It is necessary to remind that the true position of the Moon is very different from the position known by the method of calculation of the Sūrya Siddhānta, as well as is well shown in pages 186-196 of the Spashtādhikāra. Besides this, the true declination of the Moon is also not correct according to the method of the Sūrya Siddhānta. For all these reasons, the calculations for this chapter should be taken from the principles of trigonometry, otherwise the time of the Moon's rising obtained by the principles of the Sūrya Siddhānta will be fifteen, sixteen minutes ahead or behind the actual time. Therefore, it is necessary to have a good observatory for revising Indian astrology in which the Moon, planets

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SOURCE PAGE 707

and the roots of the constellations should be re-fixed by taking the most minute arc of them. In such work, the navigator should not depend on the almanac in any way. The time of the rising and setting of the Moon—

Rising time of the Moon and its phase—

Method of rising: it is the duty of the fore-knowledge, even in winter.

The five visible directions are seen from the two directions. 1.1.

Translation—(1) The time of the rising and setting of the Moon should be known according to the method explained in verses 4, 5 of the Upadasha regarding the rising and setting of the Moon. When its phase is 12 degrees behind the Sun, it is visible in the west, and when it is earlier, it becomes invisible in the east.

Scientific Commentary—no special writing is required on this because just as the rising time of other planets is known, so is that of the Moon. The rising and setting of the Moon occurs only once in the lunar month. The motion of the Moon is very fast, so the setting of the Moon occurs in the west on the full moon of the Chaturdashi of the Krishna Paksha, and the rising occurs in the evening time after the Pratipada of the Shukla Paksha.

The method of knowing the time of daily sunrise—

> रवीन्द्वोः षड्भयुतयोः प्राग्वल्लग्नान्तरासवः ।
> एकराशौ रवीन्द्वोश्च कार्या विवरलिप्तिकाः ॥२॥
> तन्नाडिकाहते भुक्ती रवीन्द्वौः षष्टिभाजते ।
> तत्फलान्वित मो भूयः कर्तव्या विवरासव ॥३॥
> एवं यावत् स्थिरी भूता रवीन्द्वोरन्तरासवः ।
> तैः प्रणैरस्तमेतीन्द्रः शुक्लेऽर्कास्तमयात्परम् ॥४॥

Translation—(2) On the day when the time of sunset of the sun and moon needs to be known in the waxing phase, by clarifying the sun and moon and by performing the correction in the moon of the axis and the meridian, by adding six six zodiacs to the portions of the sun and moon, which are the intervals of the rising signs. If the sun and moon are in the same zodiac, it is sufficient to take the difference of their portions. (3) Make a clock of these intervals and multiply it by the daily motions of the sun and moon and divide the product by 60. The increment obtained from the motion of the sun should be in the portion of the sun and the increment obtained from the motion of the moon should be in the portion of the moon

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SOURCE PAGE 708

Add these and calculate the longitude again as before. (4) By doing this many times, the longitude becomes fixed. At the same time, the moon sets after sunset in the waxing phase.

Scientific Commentary—In some texts, instead of these three shlokas, there is only one shloka whose first half is the first half of the second shloka and the latter half is the latter half of the fourth shloka. Therefore, 4 lines from the latter half of the second shloka to the first half of the fourth shloka are inserted according to some opinion. Shri Indera Narayan Dwivedi, Shri Madhav Purohit, or Shri Baldev Prasad Mishra have written these four lines but have not given their meaning and have not written anything about them. Yes, in the Sanskrit commentary of Acharya Ranganath ji, which was also edited by Shri Baldev Prasad ji along with his Hindi commentary, it is well discussed where it is written—

> १. श्लोक मध्य एकराशावित्यादि रवीन्दोरित्यन्त रासव इत्यन्त श्लोक द्वयं
> केनचिन्मन्दमतिना समयोऽसकृदेव साध्य इति शिष्यधीवृद्धिद तन्त्रोक्तं सुबुद्धि मन्ये-
> नायुक्तमपि युक्तं मत्वानिक्षिप्तम् ।

These two verses are not given with the Sanskrit verses in the Surya-Siddhanta of Bengal, but are in the commentary of Bengal, and it is stated there why they are inserted. The third verse in Samant's Siddhanta-Darpan is quoted as is, and the meaning of the first half of the fourth verse has been elaborated in several verses, and the latter half has also been given. After this, this verse has been written—

> अत्रार्क साबनत्वं हि द्वयोस्तात्कालिकी कृतौ
> तत्कृतौ केवलस्येन्दोः प्राणानामाक्षता मता
> सूर्यास्तकालिकौ तौ चेद्ग्राह्यौ ते चंद्रसावना ॥११॥

From which it is proven that Chandrashekhar Singh Samant considered one and a half verses of the inserted verses of the Surya-Siddhanta to be very necessary. In reality, this is also necessary as will be shown now. Therefore, in my understanding, this

SOURCE PAGE 709

It is not appropriate to simply state that it is displaced and not to give its meaning because even if it is displaced, it is not improper, because according to this, not calculating may result in a difference of 1 hour or 24 minutes in the time of the moon's setting. Acharya Ranganathji had made his commentary in 1525 Shakas 1, so this dispute is several hundred years old whether it is displaced or not. I want to tell what the meaning of these shlokas is. The first half of shloka 2 alludes to the rule mentioned in the fourth and fifth shlokas of udayāstādhikāra, which is correct. The latter part states that if the sun and the moon are in the same zodiac sign, then the difference between their apparent-circumscribed-enjoyments should be taken as the angular portion to know how much time after sunset the moon will set. The reason for this seems to be that when the moon remains so little distance from the sun that they are in the same zodiac sign, the difference in their conjunctions is not very different from the difference in their enjoyments, so this gross rule is stated for ease.

After this, in verse 3, the method of knowing the setting time of the moon with subtlety from approximation is explained. The reason for this is that according to the first half of the preceding verses, the time of the moon's setting is not exact because the moon's motion is very fast. Therefore, the portion of the moon's setting in the setting of the sun increases by some amount, causing it to set a little later. The farther the moon is from the sun, the longer the delay in the moon setting. On the days of the thirteenth or fourteenth of the waxing moon, this delay becomes approximately 20 minutes because on these days, the moon sets more than 10 or 11 hours after sunset, and in such time, its motion is approximately 5 or 6 degrees, which can cause a delay of 20 to 24 minutes in its setting. To know this, it is said that by adding 6 signs to the Sun and the Moon, the resulting Lagnaantarasu can be made smaller by dividing the Asuas by 6 and the minutes by 60 to make a clock, and multiplying this by the daily motions of the Sun and the Moon and then dividing the product by 60, it will be known how much motion occurred in the Lagnaantarasus of the Sun and the Moon. Because when the Sun and the Moon in 60 minutes

SOURCE PAGE 710

If the motion is the same as the daily motion, then in the lunar intervals it will be in the same ratio. Upon knowing this motion, by adding it to the portions of the setting sun and the moon in the sunset time and adding 6 to the sum, the intervals of their signs are then found. In this way, when the difference becomes constant after performing the unobserved action 2 or 3 times, the moon sets at the same time after sunset.

One thing to consider here. When the portions of the setting sun and the moon are clarified once, and it is known for the first time that the moon will set after such a time from sunset, then the difference in the direct setting time of the moon will only be due to the motion of the moon, so for the unobserved action, only the motion of the moon should be added to the portion of the setting moon, not the motion of the sun as well. However, the rule asks to add the motions of both the sun and the moon. Adding the motion of the sun also results in a time that will not be the sidereal time but the Sawan time. But the first difference comes in the sidereal time, so the sum of sidereal time and Sawan time cannot be. Therefore, it is appropriate to perform the unobserved action only for the motion of the moon, but the difference from taking the motion of the sun can be at most 2 minutes because the 12 hours of sidereal time is only 2 minutes more than the 12 hours of Sawan time. Therefore, it does not seem wise to understand the rule as a mistake for such a error.

Knowing the rising time of the moon in the Krishna Paksha-

> भगणार्धं रवौ दत्वा कार्यास्तद्विवरासवः ।
>
> तै. प्राणैः कृष्णपक्षे तु शीतांशुस्तृणयं व्रजेत ॥५॥

Translation—(5) The time that comes by the unobserved action by adding 6 signs to the portion of the setting sun in the setting sun time and the interval of the rising time of the setting moon in the setting sun time, is the rising of the moon in the eastern horizon in the Krishna Paksha at the same time after sunset.

Scientific Commentary—The portion of the moon in the waning phase is more than 180 degrees from the portion of the sun at sunset, therefore the moon rises in the eastern horizon after sunset. To know this, one should only add 6 signs to the portion of the sun, knowing the portion of the sun and the moon at the time of sunset, because the moon's

SOURCE PAGE 711

The rising is in the eastern horizon, so it is only necessary to know which sign is in the eastern horizon at sunset and how much time it will take for the moon to reach the ascendant after that. The moon will not rise at that time because in such time the moon will move further east with its motion. To know how much difference will occur, one must perform an unaction according to the rule mentioned in the third verse. Here too, one should perform an unaction only by the motion of the moon.

Method of knowing the linear difference between the Sun and the Moon at sunset

> अर्केन्द्वोः क्रान्तिविश्लेषः दिक्साम्ये युतिरन्यथा ।
> तज्ज्येन्दुरर्काद्यत्रासौ विज्ञेया दक्षिणोत्तरा ॥६॥
> मध्याह्नेन्दुप्रभाकर्ण संगुणा यदि सोत्तरा ।
> तदाऽर्कघ्नाक्षजीवायाश्शोष्या योज्या तु दक्षिणे ॥७॥
> शेषो लम्बज्यया भक्तो लब्धं बाहुस्स्वदिङ्मुखः ।
> कोटिश्शङ्कुस्तयोर्वंगयुतेमूल श्रवो भवेत् ॥८॥

Translation—(6) If the directions of the Sun and the Moon are the same, then find the difference of their sines, and if they are different, then find the sum. The direction in which the Sun is, that direction should also understand this difference or sum; that is, if the Moon is south of the Sun, then understand the direction of the difference or sum as south, and if it is north, then understand it as north. (7) Multiply this sum or difference by the instantaneous shadow hypotenuse of the Moon. If the direction is north, then subtract this product from the product of 12 and the sine of latitude, and if it is south, then add it. (8) Divide this remainder or sum by the sine of colatitude and consider the result as the length of the desired direction. The result obtained by taking the square root of the sum of the squares of the length and the latitude is the hypotenuse. This hypotenuse is the formulaic or linear difference between the Sun and the Moon.

Scientific Commentary—The essence of these three verses is—

If the difference of the sines of the Sun and the Moon is taken as p, then according to the sixth verse

P = sine of lunar light ± sine of solar declination,

SOURCE PAGE 712

According to the first half of the seventh and eighth shlokas

Bhuj = p x shadow hypotenuse ± 12latitude / colatitude

Koti = lunar gnomon meaning lunar portion radius

∴ hypotenuse = √ side^2 + other^2

In the sixth shloka it is stated that one should work with the sine of the difference or sum of the revolutions of the Sun and the Moon according to the seventh shloka, but this rule can only be applied if the revolutions of the Sun and the Moon are very small because the difference and sine of an angle are very small only when the value of that angle is small. Therefore, in the translation, the sum or difference of the revolutions is referred to as the sum or difference of the sines of the declinations.

Similarly, in the first half of the seventh verse, it is called 'madhyahnainduprabhakarn', which means the shadow hypotenuse of the moon at noon, but this does not seem logical, so it is appropriate to consider the shadow of the moon at sunset or the moon at the time when the setting is to be known. Swami Vigyanananda ji and Acharya Ranganath ji have also interpreted it this way and stated that if the time between one sunrise and the next is considered as 1 day, then the time of sunset can be called noon. However, the word noon here is misleading because noon is generally taken to mean 12 o'clock in the day. Therefore, the word noon is not appropriate in the verse.

Proof—the linear difference between the sun and the moon in the time of sunset is called the hypotenuse here, and the method of knowing that is explained. The perpendicular distance of the moon in the sky at the time of sunset from the earth is called koti here, but according to Indian custom, it is the same as sine of latitude or sine of declination-altitude, and the other name for sine of declination-altitude is gnomon (see page 282). Therefore, the koti is called gnomon. The linear difference of the sun from this point is called arm, the method of knowing which is explained in the first half of verses 6, 7, and 8.

One more thing should be kept in mind here. This rule can only be used if the sunset time of the sun and moon is understood in the plane of the meridian (Yamatravritta), that is, the direction in which the moon is located from the observer is south

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SOURCE PAGE 713

One should understand the North direction and also the arc and hypotenuse of the moon in the plane of the meridian. All these things will be well understood from Figure 115.

P = the western point of the horizon, D = the southern point, P = the quadrant of the horizon from the western point to the southern point, X = the cross shape, R = the arc of the ecliptic of the sun which is between the meridian and the western horizon when the declination of the sun is north. Ra Ri = the arc of the ecliptic of the sun when the declination is south. Ri, Ri = the points of the western horizon where the sun sets. Cha = the arc of the ecliptic of the moon which is between the meridian and the western horizon. Vip = the quadrant of the equator which is between the meridian and the horizon.

* This figure is taken from the Bengal Sūrya Siddhānta of Swami Vigyanananda.

SOURCE PAGE 714

S = when the position of the moon in the Surya-siddhānta is west of this meridian.

Śaka = the perpendicular or lunar gnomon or arc on the horizon from the moon. Śari vā śarī = the linear difference or hypotenuse between the sun and the moon.

Kari ya kari = bhuja; The pat and chath are perpendicular to the circle of the sun's ecliptic.

In this figure, the portion of the celestial above the horizon on the plane of the meridian circle is shown which extends from the sunset point of the western horizon to the south point. Therefore, the position of the moon appears on the meridian circle even though it is west of the meridian circle, and the gnomon, hypotenuse of the moon are seen on the plane of the meridian circle. The fourth part of the circumpolar and equinoctial circles of the sun and moon is also shown on the plane of the meridian circle. In short, it can be said that the pole is the southern part and the projection of the meridian circle of the horizon and the meridian circle of the meridian. And the circumpolar circles of the sun and moon for the time of the three problems are. The arc of the sun is also the circumpolar circle of the sun when the declination is south. Therefore, the north declination of the sun, the south declination of the moon, and the south declination of the sun are. The latitude and longitude of the desired place are. The angles between the circumpolar circles and the horizon are also equal to the longitude.

From the picture it is revealed that the lunar radius is shari_2 = shak_2 + kari_2.

In this, there is the gnomon of the moon at the instant of the moon, which can be easily known according to the formula (a) of page 260 or formula (g) of page 262 of the Chapter on the Three Problems, and this is the arc of the moon, the method of knowing which is explained in the ḍhāī shlokas above.

Kari = Richa + Chak, in which Richa is dependent on the difference between the revolutions of the sun and the moon and Chak is on the latitude of the moon.

Right-angled triangle in the Indian manner,

```
4 / sine
```

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∴ four = chauth x radius / sine
```

But chath = chad + d tha = chandrakranti jya + suryakranti jya and jya charith = lambajya

SOURCE PAGE 715

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∴ four = ( sine of lunar motion + sine of solar motion ) sine of colatitude
```

Similarly, in a right-angled triangle

```
circle / sine of circle = arc / sine of circle
```

But the angle of the triangle = the complement of the angle of the arc, therefore it is equal to the latitude and the shadow of the moon is the gnomon, therefore,

gnomon X sine of latitude / sine of colatitude

Here the values of circle and four are in degrees because the values of sine are in degrees according to the Indian custom. However, for the inscription, the measurement is taken in fingers. Therefore, to convert this to fingers, it must be assumed that the shadow of the moon is 12 fingers and its instantaneous finger hypotenuse radius is equal to 3438. If it is assumed that the finger values of four and circle are T and Th respectively, then the three ratios written below are proven—

radius × 4 × shadow hypotenuse12

∴ t = 12 imes charishak = 12 imes chari imes shadow hypotenuse12 imes radius = 12 imes (sine of declination + sine of declination) radius imes shadow hypotenuse12 imes radius imes sine of latitude = shadow hypotenuse imes (sine of declination + sine of declination) sine of latitude

Similarly t = 12 imes chashak = 12 imes gnomon imes sine of latitude imes sine of colatitude = 12 sine of latitude sine of colatitude

Because shak and gnomon are the same object.

Here the sine of revolution of the Moon and the Sun are added because the directions of their revolutions are different. If the directions of both revolutions are the same, then the difference will have to be taken, as if the Sun is on Ra, then the difference will have to be taken because in this situation

circle = wheel - wheel

SOURCE PAGE 716

Thus, the derivation of 6-8 shlokas is proven.

The method to know the bright half of the lunar image—

> सूर्योणशीतगोलिप्ताः शुक्लं नवशतोद्धृताः ।
> चन्द्रबिम्बांगुलाभ्यस्तं हृतं द्वादशभिः स्फुटम् ॥६॥

Translation - The part of the moon obtained by subtracting the part of the sun from the part of the moon, which is obtained by making the art of the art by dividing by 600, is the bright half of the moon in fingers. Multiplying this by the instantaneous finger image of the moon and dividing by 12, the value of the bright part comes in fingers.

Scientific Commentary - The mean image of the full moon is considered to be 12 fingers. At the time the moon is full, it appears the whole bright part, and at the time of new moon, there is a lack of the bright part of the moon. As the moon moves further from the sun, its bright part also increases, and finally in the full moon period, its whole image appears bright. In such a situation, the difference between the moon and the sun is 180 degrees or 180 X 60 = 10800 art, so the magnitude of the bright part of the moon is such that when the moon moves 10800 art ahead of the sun, its bright part is equal to 12

fingers, so when the moon is 'a' art ahead of the sun in a certain period, its bright part = a X 12 / 10800 = a / 600 fingers

But this is based on the mean image. The true image is different from this, which should be calculated according to the lunar eclipse chapter (pages 481-82). When the value of the true image comes in fingers, then the ratio should be calculated that when the mean image is 12 fingers, then the desired bright part is a a 600 fingers, so when the true image is c then

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Bright part = true image × 600 div 12 = true image × 600 × 112
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SOURCE PAGE 717

This rule is gross because the increase in the waxing phase of the lunar phase does not increase in the ratio of the increase, as will be revealed. The cusps or horns of the waxing phase of the moon are called shringas. The line joining both shringas is the diameter of that circle of the lunar phase which separates its illuminated part from the unilluminated part. Therefore, it is at a right angle to the line joining the centers of the moon and the sun. It is also the diameter of the circle which separates the front part of the moon as seen by the observer from the rear part. Therefore, it is also at a right angle to the line joining the observer and the centers of the moon. When the line joining both shringas is at a right angle to the lines joining the observer and the centers of the moon and the sun, then it will also be at a right angle to the plane that passes through the observer, the center of the moon, and the center of the sun, that is, it cuts the diameter joining the shringas in half, which is the great circle joining the centers of the sun and the moon. The angle that this great circle makes with the horizon plane is very variable, so the moon's shringa varies in different ways in different months, that is, sometimes it is parallel to the horizon plane and sometimes in the direction of the perpendicular.

The bright half of the Moon's visible hemisphere is between two circles, one of which is the lunar image in front of the observer and the other is the lunar image in front of the Sun. The circle of the lunar image in front of the observer is towards the edge towards the Sun, but the circle of the lunar image in front of the Sun is inwards, and the observer sees it in an oblique direction, so it appears in the shape of an ellipse because the projection of a circle in an oblique line is an ellipse. Any person can easily check this with a circular rod and a lamp. The rod should be held between the wall and the lamp in such a way that the plane of the rod is parallel to the wall and the center of the lamp, the center of the rod, and the center of the shadow of the rod on the wall are at a right angle to the plane of the wall in the same line. In such a situation, the shadow of the rod will be round. If the rod is held in this place and tilted so that its plane is not parallel to the wall, or if the plane of the rod is kept parallel to the wall and the rod is moved down or up so that the line joining the centers of the three does not lie in the vertical direction of the wall, then the shadow of the rod on the wall

SOURCE PAGE 718

The shadow that falls will not be perfectly round but will be in the shape of an ellipse. In the first case, the long axis of the shadow's ellipse will be in the same direction (horizontal), and in the second case, vertical. Similarly, the inner limit of the waxing phase of the lunar image is an ellipse, whose long axis is the diameter of the lunar image

Figure 116 is taken from Hugh Godfray M. A.'s A Treatise on Astronomy.

SOURCE PAGE 719

It is the same and the small axis is always variable. Now it will be explained how the increase or decrease of the waxing phase according to the positions of the Sun and the Moon occurs.

Suppose that the center of the moon, the direction of the observer, and the plane of the lunar image which is at a right angle to the direction of the observer, and the direction of the sun, and the plane of the lunar image which is at a right angle to the direction of the sun. The part of the lunar surface that is between the arcs of kha ka gha and kha ja gha is the bright part of the lunar image that is seen by the observer. But since the observer sees the arc kha ja gha obliquely, it is projected onto the plane ka kha ga gha as a long arc in the form of kha ṭa gha. This long arc kha ṭa gha is also the inner limit of the bright part of the lunar image. Here, since the radius of the lunar circle is ja, it is the same as ka, and since ṭa is divisible by ja, therefore

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chata = chajkoja
chata = chakkoja
rachcha
```

Because the angle j is the angle between those latitudes of the moon which are at a right angle to the directions of the observer and the sun, therefore this angle is equal to the angle between the direction of the observer d and the direction of the sun ch as r and ch are equal. Therefore

```
tak = chak - chat = chak - cojyarchcha = chak ( 1 - cojyarchcha ) = chakutkramjyarchcha
```

The value of angle R can be easily known because the triangle D C R is the angle of this triangle and its three sides D C, C R and D R are the distances from the observer to the Moon, from the Moon to the Sun and from the observer to the Sun respectively, which can be known.

Thus it is proven that the value of the waxing part does not increase according to the ratio of tithi-vṛddhi as stated in the ninth verse because the value of the versed sine of any angle does not increase in the ratio of that angle's increase, as if the angle is doubled, its versed sine does not become double (see page 120).

The treatise to know the declination

> दत्त्वाऽर्कसंज्ञितं विन्दुं ततो बाहुं स्वदिङ्मुखम् ।
>
> ततः पश्चान्मुखं कोटि कर्णं कोट्यग्रमानुगम् ॥११॥

कोटिकणयुतेरिन्दोः बिम्बं तात्कालिकं लिखेत् ।
कणंसूत्रेण दिविसद्धि प्रथमं परिकल्पयेत् ॥१२॥
शुक्लं कर्णेन तद्बिम्बयोगावन्तमुखं नयेत् ।
शुक्लाग्रयाम्योत्तरयो मध्ये मत्स्यौ प्रसाधयेत् ॥१३॥
तन्मध्यसूत्रसंयोगाद् बिन्दु त्रिस्पृग्लिखेद्धनुः ।
प्राग्बिम्बं यादृगेव स्यात्तादृक् तत्र दिने शशी ॥१४॥
कोट्या दिवसाधनात्तियक् शुक्लं तच्छृङ्गमन्नतम् ।
दर्शयेदुन्नतां कोटिं कृत्वा चन्द्रस्य साऽऽकृतिः ॥१५॥
कृष्णे षड्भयुतं सूर्य विशोध्येन्दोस्तथाऽसितम् ।
दद्याद्वामं भुजं तत्र पश्चिमे मण्डलं विधोः ॥१६॥

Translation—(10) Write the point indicating the sun on a flat land, draw a line parallel to it in the direction of the wind, and draw a line of 12 inches towards the west from its front point and join the hypotenuse with the point indicating the sun. (11) Taking the point of intersection of the scale and the hypotenuse as the center, draw a circle similar to the instantaneous lunar image. On its circumference, make marks for directions based on the base of the hypotenuse. (12) From the point of intersection of the hypotenuse and the lunar image, make a mark for the waxing phase of the moon on the hypotenuse towards the center. Make two fish from this mark and the north-south points of the lunar image. (13) Taking the point of intersection of the lines passing through these fish as the center, draw an arc that touches all three points, i.e., the full moon point and the north-south points. As in the picture between this arc and the eastern part of the lunar image, the moon is seen that day. (14) Now, make marks for directions on the circumference of the lunar image based on the latitude. The part of the waxing phase that makes a right angle with the latitude line and the line passing through the lunar image will be seen as advanced, and the shape of the moon in the sky will be the same. (15) In the dark fortnight, the amount to be subtracted from the portion of the moon to find the unlit part of the lunar image by adding 6 zodiacs to the zodiac of the sun should be done in the same way. Here the direction of the wind is reversed and there is an increase in the black part of the lunar image.

Scientific Commentary—In these verses, it is stated how the waxing part of the moon is drawn. Suppose the page of the paper is flat

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SOURCE PAGE 721

There is a land or strip on which to make a writing, and the point is the position of the Sun (see figure 117). If the value of the distance traveled according to 6-8 shlokas is equal to r of b.

And if its direction is south, then from the point R, draw a line towards the south, and if it is north, draw a line towards the north similar to the line of B. From this line, draw a line of the same length as a degree, i.e., 12 fingers, towards the tail. This point is called Kotigra, and this should be considered the center of the instantaneous lunar image. The line R is called the hypotenuse, which is discussed in the eighth verse. Taking C as the center, draw a circle with a radius equal to half the diameter of the instantaneous lunar image on C, which indicates the lunar image in the periphery. Extend the hypotenuse line so that it reaches P on the other side of the lunar image. Draw a perpendicular line from C to P on the right, which reaches the U and D points of the lunar image. Consider these U, P, D, P points as the successive directions of the lunar image, south, north, and west. According to the ninth

verse, the amount of the waxing phase of the moon that has come, draw a point Ch at the same distance from P towards C on the right of U, D. What is drawn from the points U, D, and the line is

SOURCE PAGE 722

The inner edge of the waxing phase of the moon is and on that day the shape of the waxing phase of the moon will be the same as that between the arcs of U, Ch, and D. This method is explained for drawing the arcs of U, Ch, and D by considering U as the center and drawing an arc on Ch, and by considering Ch as the center and drawing an arc on U; draw the line joining the sum-points of these two arcs. Similarly, draw arcs on the points D and Ch and draw the line joining their sum-points. Where these two lines intersect within the lunar disc, the arc drawn on point Ch, considered as the center, will touch the points U, Ch, and D and will be the inner edge of the waxing phase of the moon.

The method of knowing the center of the circle to which U, Ch, and D belong is similar to the method of geometry because the lines joining the sum-points of the arcs are the bisecting perpendicular lines of the lines U, Ch, and D, whose intersection point is the center of the circle of U, Ch, and D. Point K is fixed in this manner. Now, considering K as the center, an arc of radius K to U, Ch, and D is drawn, and the shape of the area of U, Ch, and D is known, which is the shape of the waxing phase of the moon, in which U and D are the cusps of the moon.

To know which cusp is advanced (raised) in the directions of the lunar disc, it is said in the 14th verse, but I do not understand the need for it. Draw a line parallel to the side of K for the whole length of K-Agra. The cusp that lies above this line is called advanced. In the given figure, the upper cusp is advanced.

It is clear from the figure that if the direction of the side is South, then the northern cusp of the moon will be advanced, and if the direction of the side is North, then the southern cusp will be advanced. But if the side is zero, i.e., neither North nor South, then no cusp of the moon will be advanced, but it will be equal.

It has been stated that the method given to know the increase of the waxing phase is coarse and the arc of the dhanu is not a part of the circumference but a part of the ellipse. Therefore, the method written is coarse but sufficient for practical use.

The modification made in the rule for the dark fortnight gives knowledge of the lunar phase. But in my understanding, if the sunrise-time sun

SOURCE PAGE 723

It is better to calculate by subtracting the zodiac of the moon from the zodiac and draw a perimeter.

Now it will be appropriate to state in brief how the revolution is calculated by the method of pure mathematics.

The new method of calculating revolution—

The line joining the lunar arcs that pass through the great circle from the centers of the sun and moon images is perpendicular. The great circle passing through the center of the moon image and the center of the sun image, which makes an angle with the first great circle, is equal to the angle of revolution. Therefore, to know the revolution, it is necessary to know this angle, according to one formula of

spherical trigonometry, which has been discussed in several places in the Chapter on the Three Problems, naturally

u x d==yamottara circle x =khadya j =the observer's position u j d=north-south line u p d=western horizon c =position of the moon in the western circle r =position of the setting sun x c =the angle between the sun and the moon r x =the angle between the directions of the sun and the moon

SOURCE PAGE 724

It may be known. If the obliquity and declination of the Sun and Moon are known, the time of solstice and the time of night can be known from the obliquity, and the time of night can be known from the formula (1) of page 261 and the formula given on page 273, and the longitude can be known from these. From figure 118, it is known how the arc of the moon can be known based on these.

c = center of lunar image u = vertical circle of lunar center (meridian) and c = direction of the Sun k g h = waxing phase of the Moon angle u c r = arc of the moon = angle k c p

SOURCE PAGE 725

According to the formulas of spherical trigonometry, cos theta and sin theta = cos phi cos theta + sin phi sin theta cos omega

When this sutra comes,

cot angle x = (cot x - cot x × cot y × sin x)/(sin x)

The angle obtained by subtracting 180 degrees from x and y will be the same arc angle or angle because it is the same as z and y. If the Sun is north of the Moon, then the North arc will be advanced and if it is South, the South arc will remain advanced. If the directions of both the Sun and the Moon are the same, then the arcs will be equal. Knowing this much, the arc of the Moon's arc should be drawn as shown in figure 116.

It is clear that according to the Sūrya Siddhānta, it is not possible to calculate the declination of the Moon by clarifying the Sun and the Moon because there is some imprecision in the polar circles of the Sūrya Siddhānta. Therefore, it is appropriate to re-establish the polar circles of the planets by true observation.

SOURCE PAGE 726

Chapter Eleven Right (Brief Description)

[Shlokas 1-2—Definition of validity and deviation of nodes. Shlokas 3-5— Nature and effect of both nodes. Shloka 6—When the revolutions of the Sun and Moon are determined. Shlokas 7-8—Knowing whether the nodal period has passed or is about to happen. Shlokas 6-11—When the revolutions of the Sun and Moon are equal. Shlokas 12-13—Knowing the true nodal period from the true revolution. Shlokas 14-15—When the beginning, mean, and end of the nodal period occur. Shlokas 16-18—What should be done in the nodal period. Shloka 16—When do the nodes occur twice, and when do they not occur. Shloka 20—Knowing the deviation of the Panchang. Shloka 21—Definition of Sandhi and Gandant periods. Shloka 22—Why the node and Gandant period are prohibited. Shloka 23—Conclusion.]

This right includes not only mathematical astrology but also predictive astrology. This is its special feature. The second feature is that the three chapters that follow are not named 'rights' but 'chapters'. The nodes discussed in this right are also called Mahapata.

Definition of validity and deviation—

Ekayanagatau syatām sūryācandramaso yadā. Tadyutau maṇḍashe krāntyoh tulyatve vaidhritābhidhḥ॥ Viparītāyagatau candrākau krāntiliptikāḥ. Samastadā vyatīpāto bhagṇādhiṁ teyoruyutauḥ॥

Translation—(1) When the Sun and Moon are in the same inclination and when the sum of their portions is equal to 12 signs, then the revolutions of both are equal, this node is called validity. (2) When the Sun and Moon are in different inclinations and when the sum of their portions is equal to 6 signs, then the revolutions of them are equal, this node is called deviation.

SOURCE PAGE 727

Scientific Commentary—When the revolutions of the Sun and the Moon are the same, then the nodes and apogees exist, that is, when the distances of the Sun and the Moon from the equator are the same, then the nodes and apogees occur. But even when the revolutions of the Sun and the Moon are the same, both can be North or both South; or one North and the other South. Now we have to see when this condition occurs. When the Sun is on the equator, its revolution is zero. This phenomenon occurs twice a year—on the vernal equinox and the autumnal equinox. From Aries to Cancer, the revolution of the Sun increases from zero to about 23 degrees 27 signs. It begins to decrease from Cancer and decreases to Libra, becoming zero again. From Libra, the revolution is South, and it increases to Capricorn, becoming 23 degrees 27 signs. From Capricorn to Aries, it decreases and becomes zero. When the Sun moves ahead from Capricorn, it appears to move towards the North on the horizon at the time of rising or setting, and this motion is seen up to Cancer. That is why the time from the Aries equinox to the Cancer equinox is called Uttarayana. But after the Cancer equinox, the Sun appears to move towards the South on the horizon, that is why the time from the Cancer equinox to the Capricorn equinox is called Dakshinayana. The Moon also, like the Sun, remains in Uttarayana for half a month and Dakshinayana for half a month in its revolution of almost one month, but because its orbit is somewhat different from the ecliptic and due to the self-retrograde motion of its orbit and the ecliptic at the points of intersection with Rahu and Ketu, it is somewhat difficult to fix the time of its Uttarayana and Dakshinayana. But in broad terms, there is no harm in saying that when the Moon comes near the Capricorn sign, then it is Uttarayana, and when it comes near the Cancer sign, then it is Dakshinayana because the angle between the lunar orbit and the ecliptic, that is the Moon's true position, is only about 506. This is made clear by the given figure 120.

Suppose the given ellipse is the ecliptic and its V and Sh points are in sequence of vernal and autumnal equinoxes where the equinoctial circle meets the ecliptic. For simplicity, the equinoctial circle is not shown. If it is assumed that the orbit of the Moon is the ecliptic, then it is clear that when the Sun and the Moon are at the same distance from the V and Sh points, then their inclinations will be the same. Now to see that

SOURCE PAGE 728

How many times can this event of one revolution of the moon occur? Suppose the position of the sun is somewhere between the vernal equinox and the autumnal point. When the moon is also at the same place, i.e., on the day of Amavasya, then both of their revolutions will be the same. When the moon is at

C, Cha, and Chi, the revolutions will also be the same if Var = Ch = Shcha = Chiv. But when the moon is at the point Cha, it will be the full moon. According to Sūrya Siddhānta, for instantaneous revolution equality, the days of Amavasya and Purnima are not considered, so when the moon is at Ch and Chi, the sum of revolution equality will occur.

The first verse states that when the sum of the portions of the Sun and the Moon is 360 degrees, a node called Vaidhriti occurs. This condition can only happen when the Sun is at R, C, CA, or CI, and the Moon is in sequence at CI, CA, or R, because only then the sum of the portions of the Sun and the Moon can be 360 degrees during the vernal equinox. It is clear from the diagram that the position R and CI is between the Uttarayana point U, the vernal equinox V, and the Southayana point D, so both R and CI are in the Uttarayana. Similarly, C and CA are both in the Southayana. Therefore, it is stated in the first verse that a Vaidhriti node occurs when the Sun and Moon are in the same Ayan and the sum of their (sine) portions is 360 degrees. Conversely, when both are in different Ayans and the sum of the portions is 180 degrees, a Vyatipata occurs. In the diagram, if the Sun and Moon are at R, C, the sum of their portions will be 1800, and if they are at CA, CI, the sum of their portions will be 360 + 180 degrees or 180 degrees. However, since R and C or C and CI are in different Ayans, the sum of the Krantisamy is only when the Sun and Moon are in different Ayans.

SOURCE PAGE 729

And the amount by which the Sun is ahead or behind the vernal equinox, the Moon is behind or ahead of the autumnal equinox.

The nature and character of both nodes—

> तुल्यांशुजालसंपर्क्रत्तियोस्तु प्रवहाहतात्१ ।
> तद्दुक्क्रोधभवो वह्निः लोकाभावाय जायते ॥३॥
> विनाशयति पातोऽस्य लोकानामसकृद्यतः ।
> व्यतीपातः प्रसिद्धोऽत्र संज्ञाभेदेन वैधृतः ॥४॥
> स कृष्णो दारुणवपुः लोहिताक्षो महोदरः ।
> सर्वानिष्टकरो रौद्रः भूयोभूयः प्रजायते ॥५॥

(3) From the meeting of the same rays of the Sun and the Moon and from the fire ignited by the wrath of their sight, the flowing air produces inauspicious fruits for the world. (4) When the revolutions of the Sun and the Moon are equal, this node repeatedly destroys the world. This is called Vyatipata and Vaidhriti. (5) This node is black in color, having a hard body, red eyes, a large stomach, and is inauspicious to all, and is terrible and arises repeatedly.

Scientific Commentary—both nodes are drawn in these three shlokas, but even in the good almanacs of Kashi, their discussion is very little. Their discussion is not seen in the almanacs of the Bombay province either. Yes, the idea is present in the Gujarati 'Pratyaksha Panchang'. It is apparent that astrologers very little consider these great latitudes of the Sūrya Siddhānta.

There are also the yogas named Vyatipāt and Vaidyṛti. The first one has serial number 17 and the second one has 27. The yoga named Vyatipāt has nothing to do with the node named Vyatipāt, but the relation of the yoga named Vaidyṛti must have been with that node at the time when the spring-transit was in the initial place of the Ashvini constellation.

When should the revolution of the Sun and the Moon be determined—

> भास्करेन्दोभवक्रान्तश्चक्रार्धावधिसंस्थयोः ।
> दृक्तुल्यसाधितांशदि युक्तयोः स्वावपक्रमौ ॥६॥

1. It is the text circulated in the Venkateshwar Press and Bengali editions.

SOURCE PAGE 730

Translation—By the method explained in Chapter on the Three Problems, knowing the portion of the sun's shadow, subtract the portion of the apparent sun as known by the method of Chapter on True Positions, and add this to the portions of the sun and the moon. When the sum of the portions of the Sanskrit sun and the Sanskrit moon, i.e., the portions of the sine of the sun and the sine of the moon, is 12 signs or 6 signs, then the true motion of the declination of both should be determined.

Scientific Commentary—To know when the motions of the sun and the moon are equal, it is necessary to know the portions of the sine of the sun and the sine of the moon. Therefore, the method of adding the precession angle to the portions of the apparent sun and the moon is explained. The time of motion equality that will arise by this method will be real because the orbit of the moon is different from the ecliptic. More details on this subject are also given with Figure 120.

To know whether the node-time has passed or is yet to come—

> अथौजपदगस्येन्दोः क्रान्तिविक्षेपसंस्कृता ।
> यदि स्स्यादधिका भानोः क्रान्तेः पातो गतस्तदा ॥७॥
> ऊना चेत्स्यात्तदा भावी वामं युग्मपदस्य च ।
> पदान्यत्वं विधोः क्रान्तिविक्षेपाच्चेद्विशुध्यति ॥८॥

Translation —(7) After knowing the true motion of the sun and the moon, one should see whether the moon is in an odd position or an even position with respect to the vernal equinox. If the moon is in an odd position and its deviation-motion, i.e., the true motion, is more than the true motion of the sun, then it should be understood that the node-time has passed,(8) and if it is less, then it should be understood that the node-time is yet to come. But if the moon is in an even position, then the opposite should be understood, i.e., if the true motion of the moon in an even position is more than the motion of the sun, then it should be understood that the node-time is yet to come, and if it is less, then it should be understood that the node-time has passed. If the motion of the moon or the shadow is less and subtraction is required, then what is said about the odd position in the above rule should be understood about the even position, and what is said about the even position should be understood about the odd position.

Scientific Commentary—the discussion of Ojas and Yugmapada or viṣama and samapada has been well discussed in Chapter on True Positions pages 126-27. Here, from the vernal equinox point to the sine of Cancer point or the sine of Sagittarius point for the first part, from the sine of Sagittarius point to the autumn equinox

SOURCE PAGE 731

The second part up to the point, the third part up to the point of Capricorn or Uttarayana from the autumn equinox, and the fourth part from the Uttarayana point to the spring equinox is. The first and third parts are called odd or Oja parts, and the second and fourth parts are called even or pair parts.

From Figure 120 it is clear that when the Moon remains in an odd position, i.e., in V or Shu, then for the transit or occultation, the Sun should be in the Shu or U position in sequence. It is also clear that the revolution of the Sun or Moon increases in odd positions and decreases in mean positions. Therefore, when the Moon is in an odd position and the Sun is in a mean position, the revolution of the Moon continues to increase and the revolution of the Sun continues to decrease. Therefore, the gross time of the nodal period extracted from the sixth verse, if the revolution of the Moon is more than the revolution of the Sun at that time, the revolution of the Moon will increase and the revolution of the Sun will decrease. Therefore, both revolutions have become equal before this time and the nodal period has passed. Conversely, if the revolution of the Moon is less than the revolution of the Sun, due to the revolution of the Moon continuing to increase, the time is coming when both revolutions will be equal and then the nodal period will occur. Similarly, when the Moon is in mean positions, the Sun will be in odd positions. In such a situation, the revolution of the Moon will decrease and the revolution of the Sun will continue to increase. Therefore, if the Moon's revolution is more, it will become equal to the decreasing revolution of the Sun and will come after the time extracted from the nodal period verse 6. But if the revolution of the Moon is less, then the nodal period should be considered as passed.

In the latter part of the eighth verse it is stated that if the true motion is obtained by subtracting the mean motion from the epicycle motion, then the rule stated above must be used because if the mean motion and the direction of the pole are different, it cannot be directly stated whether the lunar motion is increasing or decreasing. In such a situation, it will be necessary to know the motion one day ahead and one day behind.

Determining the positions of the motions equal to the true motions from the unexecuted action—

> क्रान्तिज्ये विज्ञ्ययाऽभ्यस्ते परक्रान्तिज्ययोद्धुते ।
> तच्चापान्तरमर्धं वा योज्यं भाविनि शीतगौ ॥६॥
> शोध्यं चन्द्राद् गते पाते सूर्यस्य गतिताडितम् ।
> चन्द्रभुक्त्या हृतं भानो लिप्तादि शशिवत्फलम् ॥१०॥
> तद्वच्छगाङ्क पातस्य फलं देयं विपर्ययात् ।
> कर्मेतदसकृत्कुर्यात् यावत्क्रान्ती समे तयोः ॥११॥

SOURCE PAGE 732

Translation—(6) The sine of the revolution of the Sun and the Moon should be obtained by multiplying the sine of the ecliptic by the radius and dividing by the ultimate sine of the revolution. Find the difference by drawing arcs of the latitudes. If the nodal period is to come, add this difference or half of it to the Moon's portion. If the nodal period has passed, subtract this difference or half of it from the Moon's portion. Multiply the difference or half that is added or subtracted by the motion of the Sun on that day and divide by the motion of the Moon on that day. The resulting period should be added or subtracted from the Sun's portion in the same way as it is added or subtracted from the Moon. (11) Similarly, multiply this difference or half by the motion of the lunar node, i.e., Rahu's motion, and divide by the lunar motion, and give the resulting period in reverse order in the portion of Rahu. That is, if a difference is added to the Moon, it should be subtracted from Rahu, and if it is subtracted, it should be added. After these corrections, the true revolution of the Sun and the Moon should be known. If both are not equal, then the action described in 6-10 shlokas should be done. This approximation method should be done until the motion of the Sun and the Moon become equal.

Scientific Commentary—The rules stated in the first half of the ninth verse are the opposite of the rules stated in the twenty-eighth verse of the Chapter on True Positions (page 122-23). Here is the method to know the portion of the sine of declination, radius, and the supreme sine of declination. The portion obtained by this method will be less than 60 degrees. There is no need to know more than this because we only need to see how far the sun and the moon are from the vernal or autumnal equinox. Although the true declination will be different, these portions will also be close to each other. The difference between these portions will be the amount by which the moon will be ahead or behind the nodal time. If the nodal time is coming, this difference should be added to the moon's portion because by that time the moon will move only that much further, and if the nodal time has passed, this difference should be subtracted from the moon's portion because at the time of the passed nodal time, the moon will remain only that much behind. But the sun will also leave some place in such a time. Therefore, it is necessary to also determine the sun's position of the nodal time. For this, one should use the ratio that when the daily motion of the moon is this much, then the daily motion of the sun is this much, so when the motion of the moon is equal to that difference, what will be the motion of the sun, that is, the moon

SOURCE PAGE 733

Daily Motion: Moon difference :: Sun's daily: Sun difference. In this way, whatever difference arises should be added to the Sun's portion if the Moon's difference has been added, otherwise it should be subtracted. In this way, the true portions of the Sun and the Moon in the nodal period will be known. From this, the true revolution of the Sun and the Moon should be known. However, to know the true revolution of the Moon, it is necessary to know the Moon's shadow, which is dependent on the Moon's path of Rahu or Ketu, and in such a time, the lunar eclipse will also change its position due to retrograde motion, so similarly, the change of Rahu should also be known by the ratio. However, this change should be performed in the opposite manner in Rahu, that is, when adding the Moon and the Sun, one should subtract, and when subtracting, one should add because the motion of Rahu is opposite. When the true revolution is obtained by performing the operation of the Moon's shadow in the lunar revolution, then it will be seen that the Sun's revolution is still somewhat different. Repeat the procedure mentioned in 6-11 shlokas and repeat it until both revolutions are equal. This is called asakritkarma, which has been discussed in many places behind.

The explanation of the rule mentioned in 6-11 shlokas is sufficient. Here I only want to say that it is impossible to have exact knowledge of the nodal period even after doing all this trouble because the motion of the Moon is not as simple as stated in Sūrya Siddhānta. To know its true position, many operations have to be done, which have been well discussed in the Chapter on True Positions. Therefore, if the exact determination of the nodal period is to be made, one should use modern observations for which tables etc. should be prepared based on modern principles.

In the latter part of the ninth verse it is stated that the difference or half of this difference between the portions of the Sun and the Moon should be added or subtracted. Commentators have written that half should be taken when the difference is greater. This does not make any difference in calculation, only some simplicity comes because the aim is to know the time when the inclinations of both are equal by inaction.

The nodal time before or after midnight—

> क्रान्त्योस्समत्वे पातोऽथ प्रक्षिप्तांशोनिते विधौ ।
> हीनेऽर्धरात्रिकाद्यातो भावि तात्कालिकेऽधिके ॥१२॥

Translation—Nodal time occurs when the true inclinations of the Sun and the Moon are equal. According to the ninth verse, the nodal true Moon is known.

SOURCE PAGE 734

According to the portion of the true position, if it is less than the portion of the half-night time true moon, it should be understood that the nodal time has already passed before midnight, and if it is more, it should be understood that the nodal time will be after midnight.

Scientific Commentary—the portion of the moon is always increasing, so if the portion of the instantaneous true moon is less than the portion of the half-night time true moon, then it is certain that the nodal time has already passed before midnight, and if it is more, it will be after midnight.

How much before or after midnight is the nodal time—

> स्थिरीकृतार्धरात्रे द्वोःद्वयोविवरलिप्तिकाः ।
>
> षष्टि छनाश्चन्द्रभुक्त्याप्ताः पातकालस्य नाडिकाः ॥१३॥

Translation—multiplying the difference between the instantaneous lunar portion and the half-night time lunar portion, determined by the above rule, by writing it in degrees of arc and multiplying by sixty and dividing the product by the half-night time lunar motion, the resulting gain is the nodal time that has passed or will pass that many hours before or after.

Scientific Commentary—the difference between the portions of the instantaneous moon and the half-night time moon will show how much before or after the half-night time moon the instantaneous moon was. Then it should be calculated that if the daily motion of the half-night time moon is 60 hours in 60 degrees, then how many hours that difference has occurred. The nodal time should be this much further or behind.

If the calculation of the Sun and Moon is made very subtle by modern principles, then the resulting time of the node will be gross because the nodal calculation is very subtle and the daily motion of the Moon is so great that if the half-night motion is taken as nodal, as explained in this rule, subtlety cannot be achieved because if there is a great difference between the time of the node and the half-night, the lunar motions of both times will not be the same, therefore it would be better in my understanding to use the lunar motions two hours ahead and behind from that time according to this rule.

Knowing the time of the beginning and end of the nodal period

> रवीन्दोर्मानयोगार्धं षष्ट्या संगुण्य भाजयेत् ।
> तयोर्भुक्त्यन्यरेणाप्तं स्थित्यर्धं नाडिकादिकम् ॥१४॥

SOURCE PAGE 735

> पातकालस्फुटो मध्यः सोऽपि स्थित्यधंवर्जितः ।
>
> तस्य संभवकालस्तस्यात् संयुक्तश्चान्तसंज्ञितः ॥१५॥

Translation—(14) Add the values of the Sun and Moon images, divide by two, and multiply by 60 and then divide by the difference in their motions to get the arc of the time. (15) Subtracting the time that

comes from the mean time of the node from the time of the true node gives the time when the node begins, and adding the time that comes from it gives the time when the node ends.

Scientific Commentary—The definition of arc of time given in the Chapter on True Positions on pages 468-70 should also be understood here. In page 466, the formula 60 imes ch f cha is given. If ch f is replaced by the sum of the Sun and Moon images in it, then it will be the arc of the node, which is known in the same way as the touch of the eclipse and the time of salvation.

The essence of this is that the beginning of the node time starts when the inclinations of the edges of the images of the Moon and the Sun are equal, and the mean time of the node is when the inclinations of the centers of both images are equal, the method of knowing which is described up to 13 verses, and the end of the node time is when the inclinations of the other edges of both images also become equal.

Effect of the node time and its appropriate actions—

> आद्यन्तकालयोर्मध्ये कालो ज्ञेयोऽतिदारुणः ।
> प्रज्वलज्वलनाकारः सर्वकर्मसु गहितः ॥१६॥
> एककाष्ठां गतं यावदर्कैन्द्वोमण्डलान्तरम् ।
> संभवस्तावदेवास्य सर्वकर्मविनाशकृत् ॥१७॥
> स्नानदानजपश्राद्धव्रतहोमादिकर्मसु ।
> प्राप्यते सुमहच्छ्रेयः तत्कालज्ञानतस्तदा ॥१८॥

Translation—(16) The time from the beginning to the end of the node time is great, burning, and fire-like. All this is forbidden for auspicious works. (17) Until the inclination of any point of the Sun image is equal to the inclination of any point of the Moon image

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SOURCE PAGE 736

The state of this node remains that of destroying all actions until they are destroyed. (18) In this period, great merit is obtained from actions like bathing, donation, chanting, śrāddha, vows, home, etc., and also from the knowledge of this period.

Scientific Commentary—just as bathing, donation, chanting, etc., are considered good in the times of full moon, new moon, etc., so these actions are also considered good in the time of the node, and just as many auspicious actions are forbidden in the many yogas mentioned in Muhurta-Chintamani, so is it here too. However, astrologers do not really consider these great nodes, perhaps because their calculation is impossible based on old theories. That is why their discussion is negligible in the Panchangs. Even in the World Almanac of the Hindu University, they are left discussed in two places, although their writers find it very easy to know the time of the node with the help of the Navigational Almanac because they take help from it in other matters. The last point of the 18th verse is undoubtedly very beautiful. It states that knowing the time of the node also brings merit, that is, obtaining pure knowledge of the time of the node is also a meritorious act, which is only possible when the calculations of the Sun, Moon, etc., are done with exact visibility and the study of astrological principles is done in a new scientific manner. It is not wise to understand everything only from ancient theories and not to make amendments according to the country time, nor to consider the pure scientific method as inferior, nor is it in accordance with the method of ancient astrologers.

> रवीन्द्वोः तुल्यता क्रान्त्योविषुवत्सन्निधौ यदा ।

> द्विभवेच्च तदा पातः स्यादभावो विपर्ययात् ॥ १६ ॥

Translation - When the revolutions of the Sun and Moon are the same near the equinoxes, i.e., near the vernal or autumnal equinox, the node occurs twice. In the opposite situation, i.e., near the points of Aries or Capricorn, there is no node.

Sūrya Siddhānta — Scientific Commentary — when the Sun and the Moon are near the vernal or autumnal equinox, the motion of their revolutions is very rapid. Therefore, when the Moon is south of the ecliptic and the Sun is north, their revolutions are equal. After this, when the Moon becomes cause north with quick motion, its revolution also becomes the same as the revolution of the Sun. In this way, revolution-equality can occur twice between the same two days. But when the Sun and the Moon both

SOURCE PAGE 737

If the answer is from the solstice, then it will be the time of amāvasyā and in such a condition, the lunar motion is not considered as before and second shlokas prove it. Therefore, it is known that only shloka 16 is given to state that the sine of declination can happen twice, two in the difference of one day.

But if the sun sine is near the points of Cancer or Capricorn, then its motion remains near the supreme motion. If at this time the motion of the moon is less due to the direction of the direction of Shar being different, then motion of conjunction cannot happen and neither the conjunction of validity nor the conjunction of deviation can occur.

The method of knowing the third type of deviation--

> शशाङ्कर्कयुतेलिप्ता अभोगेन विभाजिताः ।
> लब्धं सप्तदशान्तोऽन्यो व्यतीपातः तृतीयकः ॥२०॥

Translation—Add the portions of the sun and the moon to make the art and divide it by 800. If it is at the end of 17, that is, near 17, then there is the third deviation.

Scientific Commentary—In shloka 65 of Spṣṭāadhikar, the method of knowing the 27 yogas is given. In these, the 17th yoga is stated as deviation (see page 216). The method of knowing this is repeated here as well. That is, so that it may be known that the deviation called by the conjunction of motion in this right has an equivalent yoga called deviation. From this reasoning it can be said that the validity of the 27th yoga should also be understood as a great motion called validity.

One thing to note here. The calculations of latitudes and longitudes are done from the ecliptic portions of the Sun and Moon, but the calculations of the great circles are done from the sine portions. Therefore, this question arises here whether the sine portions or the ecliptic portions should be used in the rule given in the 20th verse. In the commentary of the Gūḍhārtha Prakāśikā Sanskrit Tika, it is stated that the precession angle, i.e., the sine portion, should be used for calculation, and this has been followed by Pandit Mādhava Purohit and Pandit Indranārāyaṇa Dwivedi. However, Swami Vidyanānanda has not discussed this in his Bengali ṭīkā. It seems to me that this latitude is the latitude of the latitudes and so forth, it is no different from it. Therefore, just as these latitudes are calculated, so too the latitude mentioned in this verse.

SOURCE PAGE 738

It should be calculated from the portions of the lunar months only and the precession angles of the light rays should not be taken. There is an obstacle in taking the sine portions. That is, the declination or right ascension that will arise from this will be different from the declination and right ascension of the prime vertical circles. Thus, one will have to imagine four periods of declination and right ascension in one month, which seems improper even in the style of argumentation to the author.

When does the conjunction of Sandhi and Gandanta occur—

> सार्पेंद्रपौष्णधिष्ण्यानामन्त्याः पादा भसन्ध्यः ।
>
> तदग्रभेष्बाधपदो गण्डान्तं नाम कीर्त्यते ॥२५॥

Translation—The fourth quadrant of the constellations Ashlesha, Jyestha, and Revati are the constellation conjunctions, and the first quadrant of the following constellations Magha, Moola, and Ashvini are called Gandanta.

Scientific Commentary—In Muhurta-Chintamani and other Muhurta texts, their discussion is special. For the progeny in the constellation conjunction or Gandanta, it is called Sadha-ranta that it has occurred in Moola. It is considered inauspicious. On the 27th day when that same Gandanta or Sandhi period comes again, a special type of worship is done for peace in Moola. The meaning of discussing Gandanta here is that the inauspicious effects of the great misfortunes are also those of Gandanta, as revealed in the next verse. These Sandhis are the last parts of the fourth, eighth, and twelfth signs, and Gandanta are the initial parts of the fifth, ninth, and first signs.

> व्यतीपातत्रयं घोरं गण्डान्तं वितयं तथा ।
>
> एवं ससन्धित्रितयं सर्वं कर्मसु वजंयेत् ॥२॥

Translation—All three declinations, all three Gandantas, and all the constellation conjunctions are very terrible, so they are forbidden in auspicious deeds, that is, when these occur, no auspicious action should be done.

Scientific Commentary—this verse does not discuss valid expenditure, but from the style of argument and earlier verses it is known that validity is also included. Commentators have done so too.

Conclusion—

> इत्येवं परमं पुण्यं ज्योतिषां चरितं हितं ।
>
> रहस्यमिदमाख्यातं किमन्यच्छोतुमिच्छसि ॥२३॥

SOURCE PAGE 739

Translation—I have told the story of this most sacred, extremely mysterious, and beneficial জ্যোতি-science, what else do you want to hear?

Scientific Commentary — The Jyotish (astronomy) story which Surya (Sun) began in possession of from Mayur has ended here. The questions asked by Mayur on this will be discussed in three chapters further. Therefore, what has been said till here is called the first half of Sūrya Siddhānta. The three

chapters that follow are called the second half. Now we will conclude this first half by briefly telling how the great latitudes are calculated.

Determining the gross time of the great latitudes from the Panchangs—The calculation of the 27 yogoṃ of Viṣkambhādi must be in the Panchangs. The method to know these is explained in the 65th verse of the Chapter on True Positions, which is—add the portions of the Sun and the Moon to make the arts and divide it by 800. The number of yogoṃ that have passed since that obtained and the knowledge of the current conjunction is known from the remainder.

In this rule, the portions of the Sun and the Moon are measured from the initial point of the Ashwini constellation, and the portions are measured from the Vernal Equinox point for the calculation of the great latitudes. If the portions for both are measured from the Vernal Equinox, then knowing the time of the great latitudes would be much easier because at the time when half the time of the 14th conjunction passes, the sum of the portions of the Sun and the Moon is 180 degrees, and it is the midpoint of the nodal period called Vyatipata, and at the time when the Vaidyutī conjunction ends, the midpoint of the node called Vaidyuti occurs. But this is not the case. Therefore, a little calculation will be necessary. According to Sūrya Siddhānta, the initial point of Ashwini, where it is nowadays, is about 22 degrees 45 minutes west of the point corresponding to the vernal equinox as seen by the zenith. This difference is called the precession angle. If the portions of the Sun and the Moon are taken from here, the sum of both is 45 degrees 30 minutes more. The sum of the sines of the portions of the Sun and the Moon for Vyatipata is 180 degrees, so 180° - 45° 30' = 134° 30' = 8070 minutes. This is the sum of the portions of the Sun and the Moon for Vyatipata from the initial point of the Ashwini constellation. Dividing this by 800 minutes leaves 10 lekhas and 70 minutes. It is proven by 10 that the Gandha conjunction remains in the Vyatipata period and the growth conjunction begins. This-

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SOURCE PAGE 740

To determine the time of eclipse in a gross manner, one should calculate the subtle time of eclipse by knowing the true motion of the sun and moon at the time when the increase of conjunction begins.

To determine the nodal period named Vaidhriti, one should subtract 45 degrees 30 art from 360 degrees. Doing this leaves 314 degrees 30 art = 18870 art. Dividing this by 800 leaves 23 degrees and 470 art remaining, which shows that the 23rd conjunction in the nodal period named Vaidhriti remains auspicious and the half of the 24th conjunction of Shukla is also passed. Therefore, in a gross sense, Vaidhriti is at half the part of the Shukla conjunction. Therefore, to make finer calculations, the Sun, and the Moon and the true declination should be known at the same time. For this, the correction is made by knowing the true portions of the Sun, Moon and Rahu, the true declination of the Sun, the mean declination of the Moon and the shadow, to know the true declination of the Moon, the method of which is explained in Chapter on True Positions (spaRectaadhikar) pages 166-200. Therefore, there is no need to explain all these things in the example. Here, it is sufficient to show that calculating the time of the great conjunctions from the polar circles of the Sūrya Siddhānta is neither easy nor pure, whereas this can be known with purity from the polar circles known by the modern method. I have a Navik Panchang of 1626 AD at this time, so with its help, the time of the great conjunction named Vyateepaat of the Shukla Paksha of 1686 Vikram is calculated.

The Ganda Yoga ends on May 14 at 42 hours 40 minutes and the increase yoga begins after this, so there will be a great eclipse called Vyatipata on May 14 or 15: Now the navigator should see at what time the true inclinations of the Sun and the Moon will be the same on these dates. It is known from page 51 of the navigator's panchang that on May 14, the inclination of the Sun is 18 degrees 34 arc and 42

minutes, and on May 15, it is 18 degrees 46 arc and 6 minutes. But the inclination of the Moon is more than 22 degrees on May 14, so the Vyatipata period will not come on May 14, but this event may happen on the evening of May 15 because,

At the noon of May 15, the revolution of the sun is 16,,,,,, arc art 18 46 6.1 19 30.9

SOURCE PAGE 741

In 24 hours, the motion of the Moon is 14.4 (18 May evening 6 o'clock) the lunar motion is 18.5411.5,,,,7,,,,184233.4 in an hour, the speed of the lunar motion is 1138.1.

Here, the solar revolution is increasing and the moon's is decreasing, therefore it is certain from the motion of the moon's revolution that both revolutions will be equal around 6 o'clock. In 6 hours, the speed of the sun's revolution is 36 × (1 art 408 virat) = 3 art 3102 virat. Therefore, the solar revolution at 6 o'clock is 18 art 46 art 601 virat + 3 art 3102 virat = 18 art 52 art 3703 virat. This is less than the moon's revolution at six o'clock, and the moon-revolution is decreasing while the sun-revolution is increasing, therefore the revolutions of both will be equal only after a few minutes after six o'clock. To know this, the difference between the two revolutions should be divided by the difference in the revolutions of both.

Anshkalavikala6bjechandra - kranti = 185411 · 5,, Suryakranti = 185237 · 3Both difference = 134 · 2 = 64 · 2v. Surya ki1 ghante ki kranti - gati = 14kala4 · 8v.24 = 35 · 2vikalachandmaki1 ghante ki kranti gati = 11kala38 · 1vikala

Moon's 1 hour revolution motion = 11 lunar phases 3801 degrees

Since their directions are different, they should be added to know their difference. Therefore, the sum of both = 12 kala 13·3 vikala = 733·3 vikala.

When the difference of 733.3 degrees occurs in 1 hour, then in how much time will the difference of 6402 degrees occur.

Sūrya Siddhānta: Scientific Commentary

Chapter on Mean Motions: Chapter on True Positions

SOURCE PAGE 742

Therefore, the mean time of the eclipse will be at 6 o'clock 7 minutes 43 seconds on May 15. However, this calculation is done from Greenwich time which is 5 hours 30 minutes behind the railway time of India. Therefore, according to the railway time of India, the mean of the eclipse time will be at 11 o'clock 37 minutes 43 seconds on the night of May 15.

Now, if this is subtracted knowing the sthitirdha-time, the beginning time of the eclipse time will come, and if added, the end time will come. According to the 14th verse, this can be done easily, so there seems no need to give an example.

Thus, the Scientific Commentary of the 11th chapter named Patabikar has ended.

SOURCE PAGE 743

Twelve Chapter Geodesy (Brief Description)

[Shlokas 1-6—Many questions regarding the geography, astronomy, and seasons of Mayasur. Shlokas 10-11—Asking Suryaansh Purush to hear from Mayasur. Shlokas 12-23—The order of origin from Vasudeva to the five Mahabhutas. Shloka 24—The origin of five star-planets. Shloka 25—The origin of twelve zodiacs and 27 constellations. Shlokas 26-30—The origin of the terrestrial world. Shlokas 30-33—The order of the orbits of the planets in the cosmos and the position of the Earth. Shlokas 33-36—The positions of Patala, Sumeru, etc., in geography. Shlokas 37-42—The positions of the four cities located on the Equator. Shlokas 43-45—The relation between the Equator and the North and South Poles. Shloka 46—Why the rays of the sun are slow and intense in different seasons. Shlokas 47-50—The division of day and night for the inhabitants of the North Pole, i.e., the gods, and the inhabitants of the South Pole, i.e., the asuras. Shloka 51—The time of noon and midnight for the gods and asuras. Shlokas 52-53—Why one perceives each other as above and below at a distance of 180 degrees on the geography. Shloka 54—Why the geography appears like a wheel. Shlokas 55-58—The reason for the waxing and waning of day and night on the surface. Shloka 56—At what distance from the Equator the sun appears directly above at some time. Shlokas 60-61—At what distance from the Equator there are 60 hours of day and 60 hours of night. Shloka 62-60—Where is the day or night larger than an hour. Shlokas 63-67—Where are the days or nights of two months, four months, and six months. Shloka 68—Where the sun appears in the days of Uttarayana and Dakshinayana. Shloka 69—Where and in which direction the shadow of an object falls. Shlokas 70-71—When the sun rises at one place on the surface, where is noon and where is midnight or sunset. Shloka 72—By moving in the direction of the poles, the advancement of the celestial poles and the recession of the constellations are seen. Shloka 73—How the constellation-cycle moves by the flowing air. Shloka 74—The proof of day and night for the gods, ancestors, and humans. Shlokas 75-77—The relation between the orbits of the planets and their periods of revolution. Shloka

SOURCE PAGE 744

Geography Chapter 716

78-76—the relation of year-master, month-master, day-master and hour-masters. Verse 80—the extension of the constellation-circle. Verses 81-84—the proof of the celestial-circle and its relation to the orbits and motions of the planets. Verses 85-90—the magnitude of the circles in additions.

In this chapter, all matters of the origin, position, extension, etc., of geography are described, that is why it is named Geography Chapter. Along with that, proofs of the orbits of the planets, constellations, and the sky are also given.

The role of the questions of Mayasura and the answers of Suryansh Purusha-

अथाकाशसमुद्भूतं प्रणिपत्य कृताञ्जलिः ।
भक्त्या परमयाऽभ्यर्च्य पप्रच्छेदं मयोऽसुरः ॥१॥
मभवन् किंप्रमाणा भूः किमाकारा किमाश्रया ।
किविभागा कथं वाऽत्र सप्तपातालभूमयः ॥२॥
अहोरात्रव्यवस्थां च विद्रधाति कथं रविः ।
कथं पर्येति वसुषां भुवनानि विभावयन् ।३॥
देवासुराणामन्योन्यमहोरात्रं विपर्ययात् ।
किमर्थं तत्कथं वा स्याद् भानोः भगणपूरणात् ।।४।।
पित्यं मासेन भवति नाडीवष्ट्या तु मानुषम् ।
तदेव किल सर्वत्र न भवेत् केन हेतुना ॥५॥
दिनाब्दमाहहोराणामधिपा न समाः कुतः ।
कथं पर्येति भगणः सग्रहोऽयं किमाश्रयः ॥६॥
भूमेरुपर्युपर्युक्ताः किमुत्सेध्याः किमन्तराः ।
ग्रहर्क्षकक्ष्याः किम्मात्राः स्थिताः केन क्रमेण ताः ॥७॥
ब्रीष्मे तोत्राः करा मानोः न हेमन्ते तथाविधाः ।
क्यती तत्करव्याप्तिः मानानि कति किञ्च कैः ॥८॥
एतन्मे संशयं छिन्धि भगवन्भूतभावन ।
अन्यो न त्वामृते छेत्तां विद्यते सर्वदर्शिवान् ।।९।।
इति भक्त्योदितं श्रुत्वा मयेनार्कांशसंभवः ।
रहस्यतरमध्यायं पुनः प्राह यथाश्रुतम् ॥१०॥
शृणुम्वैकमना भूत्वा गुह्यमध्यात्मसंज्ञितम् ।
वक्ष्याम्यतीवभक्तानां नात्रेयं विद्यते मम ॥११॥

SOURCE PAGE 745

(1) Thereafter, Mayasura, joining his hands and worshipping with great devotion, asked after bowing to the man born from the portion of the sun.
(2) O Lord, what is the magnitude of this Earth, what is its shape, and on what is it based, how many divisions it has, and how are the lands of the seven paradises situated in it.
(3) How do the Sun and the Asuras arrange themselves and revolve around the Earth while illuminating the worlds.
(4) Why are the days and nights of the Devas and Asuras opposite to each other, and how does this happen when one revolution of the Sun is completed?
(5) Why is the day and night of the Pitrs (ancestors) one month and that of humans sixty hours? Why are not the days and nights the same everywhere?
(6) Why are the masters of day, year, month, and hour not the same, how do the constellations revolve with the planets, and what is their basis?
(7) What are the heights and the mutual differences of the orbits of the planets and constellations above the Earth, what are their values, and in what order are they situated?
(8) Why are the rays of the Sun very intense in the summer season and not in the winter season? How far do these rays extend; what are the values of the solar, lunar, etc., and what is the purpose of these?
(9) O one who feels the past, please remove these doubts from me because you are omniscient; no other human can remove my doubts except you.

(10) Hearing these words of Mayasura spoken with devotion, Suryansh Purusha called it the second chapter, the other mysterious form of the previous one.
(11) With a focused mind, listen to this element called spirituality which I say, because I do not consider any object unattainable for the devotees.

Scientific Commentary—All the elements know the desire to know the answers to the questions asked by Mayasura. Suryaansh Purush has stated on this that the mystery to be explained in the answer is related to spiritual knowledge. Many people will say this, but Suryaansh Purush has discussed the origin of this world, which can certainly be said to be related to spirituality because it is beyond physical science.

The order of creation—

> वासुदेवः परं ब्रह्म तन्मूर्तिः पुरुषः परः ।
> अव्यक्तो निर्गुणः शान्तः पञ्चविशात्परोऽव्ययः ॥१२॥
> प्रकृत्यन्तर्गतो देवः बोध्यमानश्च सर्वगः ।
> संकर्षणोऽपः सृष्टवादौ तासुवीर्यमवासृजत् ॥१३॥

SOURCE PAGE 746

Geography Chapter 721

> तदण्डमभवद्धं मं सर्वत्रतमसाऽऽवृतम् ।
> तत्रानिरुद्धः प्रथमं व्यक्तीभूतस्सनातनः ॥१४॥
> हिरण्यगर्भो भगवानेषच्छन्दसि पठ्यते ।
> आदित्यो ह्यादिभूतत्वात् प्रसूत्या सूर्य उच्यते ।।१५।।
> परं ज्योतिस्तमःपारे सूर्योऽयं सवितेति च ।
> पर्येति भुवनान्येव भावयन्भूतभावनः ॥१६॥
> प्रकाशात्मा तमोहन्ता महानित्येष विश्रुतः ।
> ऋचोऽस्य मण्डलं सामान्युस्रा मूर्तियंजूंषि च ॥१७॥
> त्रयीमयोऽयं भगवान् कालात्मा कालकृद्विभुः ।
> सर्वात्मा सर्वगस्सूक्ष्मः सर्वमस्मिन्प्रतिष्ठितम्:।१८।।
> रथे विश्वस्ये चक्रं कृत्वा संवत्सरात्मकम् ।
> छन्दांस्यश्वाः सप्तयुक्त्: पर्यटत्येष सर्वदा ॥१९॥
> त्रिपादममृत गुह्यं पादोऽयं प्रकटोऽभवत् ।
> सोऽहङ्कारं जगत्सृष्ट्यै ब्रह्माणमसृजद्विभुः ॥२०॥
> तस्मै वेदान्वरान्दत्वा सर्वलोकपितामहम् ।
> प्रतिष्ठाप्याण्डमध्येतु स्वयं पर्येति भावयन् ॥२१॥
> अथ सृष्ट्यां मनश्चक्रे ब्रह्माऽङ्कारमूर्तिभृत् ।
> मनसश्चन्द्रमा जज्ञे चक्षुषस्तेजसां निधिः ॥२२॥
> मनसः त्वं ततो वायुरग्निरापो घरा क्रमात् ।
> गुणैक वृद्धया पञ्चैव महाभूतानि अक्षिरे ॥२३॥

Translation—(12) Parambrahma Vasudeva. His idol is the Supreme Being who is beyond the unmanifest, without qualities, peaceful, and inexhaustible, and the twenty-five elements of the Samkhya Shastra. (13) The omniscient deity, inside and outside, entered nature and by the form of attraction created water in the beginning and placed the seed in it (14) which became the golden egg surrounded by darkness. In it, the eternal Aniruddha first appeared. (15) They are called Hiranyagarbha Bhagavan in the Vedas. Because they are first, they are called Surya and they are called the Sun because they caused the birth of Aditya and all animate beings. (16) Because they are full of supreme light and because they are at the end of the Sun and darkness, they are called Savita. They travel illuminating the worlds that create, sustain, and destroy the past-feeling or stationary and movable creation. (17) They are called the light-soul that destroys darkness and the great element in the Vedas.

SOURCE PAGE 747

Their sphere is the Rigveda, Samaveda, and Murti Yajurveda. (18) Therefore, they are called Vedatryātmaka. Time is calculated from them, so they are called Kalātmā and Kālakṛt. They are the soul of all, omnipresent, subtle, and all creation resides in them. (16) In the chariot of the world, they move forever with seven horses of seven chaṃdoṃ, having formed a wheel of saṃvatsara. (20) Their three stages are inaccessible because they are immortal, and this one stage has manifested. This Lord created Brahma in the form of ego for the creation of the world. (21) After this, by giving the superior Vedas to Brahma, the father of all realms, and establishing him among the eggs, and by illuminating the realms of aniruddha bhagavān themselves, they move. (22) After this, Brahma, embodied in ego, thought of creating the universe. From Brahma's mind and eyes, the radiance of the sun arose. (23) From the mind, space, from space, air, from air, fire, from fire, water, and from water, earth arose by the increase of one quality in a sequence of five great elements.

Scientific Commentary—the creation process described by Suryaansh Purush from Mayasura is a mixture of the creation order mentioned in Vedanta, Samkhya, Srimad Bhagavatam, etc. This order has been described in different texts in different ways, so it is not possible to explain all of them here. Lokmanya Tilak has thought well about this subject in the 6-6 chapters of his Gita-Rahasya and has compared it with the opinions of European scholars in some places, so readers should study that for this information. The essence of it can be given here.

According to Samkhya, the tree of the universe is given on page 723. See G.R.(page 176):-

The Brahman of Vedanta is beyond these 25 elements, which is discussed in the 12th verse of the Sūrya Siddhānta (see Gita Rahasya p. 203). The discussion in the Sūrya Siddhānta regarding Sankarsha and Aniruddha has come in Bhagvatadharmas as follows: 'From Vasudeva, the Supreme God, the soul in the form of Sankarsha arose; and then from Sankarsha, Pradyumna, i.e., mind, and from Pradyumna, Aniruddha, i.e., ego arose. Some people consider two, three, or one of these four views (see Gita Rahasya p. 426). The Sūrya Siddhānta does not discuss Pradyumna. Here, ego is called Brahma.

SOURCE PAGE 748

Bhugoladhyaya 723

The five great elements are called the five great elements: space, air, fire, water, and earth. In space there is one quality word, in air there are two quality words, touch, fire has three quality words, touch and form; in water there are four quality words, touch, form, and taste, and in earth there are five

quality words, touch, form, taste, and smell. That is why it is stated in the latter part of the 23rd verse that the origin of the five great elements occurred in sequence by the increase of each quality.

The lineage of the universe

Origin of five planets—

> अग्नीषोमो भानुचन्द्रौ ततस्त्वङ्गारकादयः ।
> तेजो भूरवाम्बुवातेभ्यः क्रमशः पञ्च जज्ञिरे ॥२४॥

Translation—After the origin of the Sun, which is the form of fire, and the Moon, which is the form of Soma, Mars from fire, Earth from Mercury, Jupiter from space, Venus from Jupiter, and Saturn from air were generated.

Origin of 12 zodiacs and 27 constellations—

> पुनर्द्विशषाऽऽत्मानं विभवे राशिसंशितम् ।
> नक्षत्ररूपिणं भूयः सप्तविशात्मकं वशी ॥२५॥

SOURCE PAGE 749

Translation—then Jitatma Brahma divided the mind-created desires into the first 12 zodiacs and 27 constellations.

Origin of the universe of the moving and unmoving—

> ततश्चराचरं विश्वं निर्ममे देवपूर्वकम् ।
> ऊर्ध्वमध्याधरेभ्योऽथ गात्रेभ्यः प्रकृतिं सृजन् ॥२६॥
> गुणकर्मविभागेन सृष्ट्वा प्राग्वदनुक्रमात् ।
> विभागं कल्पयामास यथास्वं वेददर्शनात् ॥२७॥
> ग्रहक्षत्रताराणां भूमेः विश्वस्य वा विभुः ।
> देवानां च मनुष्याणां सिद्धानां च यथाक्रमम् ॥२८॥
> ब्रह्माण्डमेतत् सुषिरं यत्रेदं भूर्भुवादिकम् ।
> कटाहद्वितयस्यैव संपुटं गोलकाकृतिः ॥२६॥

Translation—(26) Thereafter, by creating the nature of Sattva, Rajas, and Tamas by the superior, mean, and inferior sources, deities, humans, demons, etc., created the moving world. (27) According to the order of creation by qualities and actions, they divided it according to the method described in the Vedas and according to the country and time. (28) The almighty Brahma established the planets, constellations, stars, Earth, the world, the gods, humans, and Siddhas in that order, (26) just as two mouths of equal bowls joined together form a hollow sphere, similarly the worlds such as Bhur Bhuvah etc. are situated in this cosmic space.

Order of the orbits of the planets in the universe—

> ब्रह्माण्डमध्यपरिधि व्योमकक्ष्याऽभिधीयते ।
> तन्मध्ये भ्रमणं भानां तदधोऽधः क्रमादथ ॥३०॥
> मन्दामरेज्यभूपुत्रसूर्यशुक्रेन्दुजेन्दवः ।
> परिश्रमन्त्यधोऽधस्तात्सिद्धविद्याधरा घनाः ॥३१॥
> मध्ये समन्तादण्डस्य भूगोलो व्योम्नि तिष्ठति ।
> विभ्राणः परमां शक्तिं ब्रह्मणो धारणात्मिकाम् ॥३२॥

Translation —(30) The circumference of the universe is called the celestial orbit within which the constellations move; then in succession (31) Saturn, Jupiter, Mars, Sun, Venus, Mercury, and the Moon move below it. Below these are Siddhas, Vidyadharas

SOURCE PAGE 750

Geography Chapter 725

And there are clouds. (32) In the very center of this universe, this geography is held in zero by the power of the concept of Brahma, the supreme power.

Scientific Commentary—it is stated in these three shlokas that the orbits of the constellations and planets within the ultimate perimeter of the universe are in what order. The position of our Earth is considered to be in the exact center of this universe, that is, this geography is in the center of the entire universe. This is contrary to the modern astronomical principles. In modern astrology, the Sun is considered the center of the world. The orbit of the planet Mercury, closest to the Sun, is followed by Venus, Earth, Mars, Jupiter, and Saturn in sequence.

Akash Nakshatra Shani Guru Mangal Ravi Shukra Budha Chandra Bhū

According to Indian astrology, the orbits have moved away (in the center of Earth). The orbit of the Moon is around the Earth. The orbit of the constellations

SOURCE PAGE 751

According to modern astrology, the fixed cannot be established because all stars are not at the same distance. The limit of the celestial sphere also cannot be fixed because nowadays the distance of some stars is considered so great that the limit of the celestial sphere is negligible in front of it. The differences of views of Hindu astrology and modern astrology will be well understood from figures 121 and 122.

There are realms of clouds, constellations, and Siddhas between the Earth and the Moon's orbit, which cannot be shown in this figure.

According to modern astrology, the order of the orbits of the planets (here the sun is at the center)

SOURCE PAGE 752

Geography Chapter 727

The orbit of the moon is not shown in this picture because the moon revolves around the Earth and also around the Sun along with the Earth. Many moons have been seen revolving around the moon, Mars, Jupiter, and Saturn. The distances of the orbits in Figure 122 appear to be approximately the same and

the shape is round, but this is not actually the case. This will be considered further; here only the order is shown.

The concept of the force of dhāraṇātmikā śakti discussed in Shloka 32 is what is called gravitation nowadays. From this shloka it becomes clear that according to Sūrya Siddhānta, our Earth is considered to be in a vacuum. It is not held by any living being. The power of God by which this Earth remains stationary in a vacuum is called dhāraṇā-tmikāśakti. Nowadays, it is believed that the Earth, the moon, planets, etc., are bound by the gravitational pull of the Sun, and this gravity is also the cause of the motions of the planets and satellites.

Places of Patala, Sumeru, etc. in Geography:-

> तदन्तरपुटास्सप्त नागासुरसमाश्रयाः ।
> दिव्यौषधिरसोपेता रम्याः पातालभूमयः ॥३३॥
> अनेकरत्ननिचयो जाम्बूनदमयो गिरिः ।
> भूगोलमध्यगो मेरुः उभयत्र विनिगंतः ॥३४॥
> उपरिष्टात्स्थियास्तस्य सेन्द्रा देवा महर्षयः ।
> अघस्तादसुरास्तद्वत् द्विषन्तोऽन्योन्यमाश्रिताः ॥३५॥
> ततस्समन्तात्परिधिः क्रमेणायं महार्णवः ।
> मेखलावत्स्थितो घाट्या देवासुरविभागकृत् ॥३६॥

Translation—(33) In the inner layers of this geography, there are seven beautiful subterranean lands where snakes and demons reside and where there are light-giving and juicy trees. (34) Sumeru mountain, filled with various types of gems, adorned with the golden Jambu river, extends on both sides of the geography. (35) Above this Sumeru mountain, gods and sages reside with Indra and other deities and demons. These gods and demons are enemies of each other. (36) This ocean (salt sea) surrounding this Sumeru mountain is situated like the rope of the Earth and divides the gods and demons.

SOURCE PAGE 753

Science commentary—seven nether-worlds are considered within geography, whose names are atal, vital, sutal, rasatal, talatal, mahatal, and patal. Nagas and asuras reside here. Jambunadi is near Mount Sumeru. This mountain is considered to have emerged on both sides of the center of geography, that is, the northern and southern poles. Gods reside at the northern pole and asuras reside at the southern pole, who are mutually hostile. Around this Meru mountain is a salt sea of the Earth which separates the land of the gods and asuras and is like the rope of the Earth.

In this description, many things seem to have arisen from imagination, so their existence cannot be stated. The north and south poles should be considered above and below Mount Sumeru. Between them, the saline sea near the equator is considered, which is still in this position nowadays. Description of the four cities located on the equator:

> समन्तान्मेरुमध्यात्त तुल्यभागेषु तोयधेः ।
> द्वीपेषु विक्षु पूर्वादिनगर्यो देवनिर्मिताः ॥३७॥
> भूवृत्तपादे पूर्वस्यां यमकोटीति विश्रुता ।
> भद्राश्ववर्षे नगरी स्वर्णप्राकारतोरणा ॥३८॥
> याम्यायां भारते वर्षे लंका तद्वन्महापुरी ।
> पश्चिमे केतुमालाख्ये रोमकाख्या प्रकीर्तिता ॥३६॥
> उदक्सिद्धपुरी नाम कुरुवर्षे प्रतिष्ठिता ।
> तस्यां सिद्धा महात्मानो निवसन्ति गतव्यथाः ॥४०॥
> भूवृत्तपादविवराः ताश्चान्योन्यं प्रतिष्ठिताः ।
> ताभ्यश्चोत्तरतो मेरुः तावानेवासुराश्रयः ॥४१॥
> तासामुपरिगो याति विषुवस्थो दिवाकरः ।
> न वासु विषवच्छाया नाक्षस्योन्नतिरिष्यते ॥४२॥

> न तासु विषुवच्छाया नाक्षस्योन्नतिरिष्यते ॥४२॥

Translation—(37) Around the middle part of Meru, there are four cities made by the gods, in the east, south, and north directions of Jambudvipa. (38) In the east, the city of Yamakoti is famous in the fourth part of Bhuparidhi in the year of Bhadrasva, where there are walls and gates of gold; (36) In the south, in Bharatavarsha, Lankapuri is famous, and in the west, in Ketumala country, Romakpuri is famous; (40) In the north, in Kuru country, there is Siddapuri where Siddhas, Mahatma people, who are free from all kinds of sorrows, live.

SOURCE PAGE 754

are. (41) These cities are situated in the quadrant difference of the earth's circumference from each other, and in the same difference in the north direction, Meru resides. (42) When the sun comes to the equator, it is directly above these cities, therefore there is neither the equatorial shadow nor the latitude there.

Sūrya Siddhānta – in these six shlokas there is a very clear description of the position of four cities located on the Tropic of Cancer. These cities are in the fourth quadrant of the Earth's circumference from each other, that is, they are separated by 60 degrees and North Meru (North Pole) is also at the same distance from them to the north. The directions of these cities are considered from the Indian subcontinent. Lanka is at the Tropic of Cancer of the south of the Indian subcontinent, and its position should be considered as the longitude of Ujjain according to the 62nd shloka of the Chapter on Mean Motions (madhyamādhikāra) (page 65). The longitude of Ujjain from Greenwich is approximately 76 degrees. Therefore, if Lanka is considered at this longitude and the Tropic of Cancer, there is a sea here nowadays. The place 60 degrees east of this is at a longitude of 166 degrees east of Greenwich. Therefore, the place of Yamakoti city is also a sea nowadays. Since there are no land names on the Tropic of Cancer 60 degrees west of Lanka or 14 degrees west of Greenwich, the location of Romak city also cannot be determined. This Romak city would have been about 50 miles south of the Free Town of Western Africa nowadays. Similarly, Siddapuri would have been more than 1000 miles south of Mexico.

If the existence of these four spheres ever existed, then that time would be very ancient because nowadays the difference has become so much that there is no mark of that time in the present. It is also possible that the existence of these four spheres remained only in the poet's imagination and this has been described in figurative language that these four places on the equator are such that when noon is in Lanka, then sunrise is in Romaka, midnight in Siddhapuri, and sunset in Yamakoti.

It is clear that when the sun is on the cross on the equator, then at noon there is no shadow of any standing object. There are the north and south poles on the horizon of this line, so the height of the pole star here is zero. Therefore, the latitude is also zero. For this reason, the equator is called the meridian country. Its further and clearer description is in the next three verses.

Sūrya Siddhānta is stated to be emerging from the middle of the Earth on both sides, therefore it should be understood as the axis of the Earth whose northern end is the North Pole.

SOURCE PAGE 755

and the southern end is called the southern pole. A parallel is considered to be the equator around the center of this axis, i.e., the geocenter, which was understood to be the boundary of the Jambudvipa and the saline sea.

Relation between the Equator and the North and South Poles

> मेरोरुभयतो मध्ये ध्रुवतारे नभःस्थिते ।
> निरक्षदेशसंस्थानामुभये क्षितिजाधिये ॥४३॥

> अतो नाक्षोच्छ्रयस्तासु ध्रुवयोः क्षितिजस्थयोः ।

> नयतिलम्बकांशास्तु मेरावक्षांशकास्तथा ॥४४॥

> मेषादौ देवभागस्थे देवानां याति दर्शनम् ।

> असुराणां तुलादौ तु सूर्यस्तद्भागसञ्चरः ॥४५ ।

Translation—(43) The pole star located towards the North and South poles on both sides of Meru is exactly in the middle; for those living in the meridian, these two stars appear on the horizon. (44) Therefore, due to the presence of both pole stars on the horizon of these cities, the axis of these poles is not high, i.e., their latitude is zero, but the longitude is 60. Similarly, the latitude of Merus, i.e., the poles, is 60. (45) When the sun is in the deva-bhag, i.e., in the northern hemisphere, the gods get their first sight in the beginning of Aries, and when the sun is in the asur bhag, i.e., in the southern hemisphere, the asuras see them first in the beginning of Libra.

Scientific Commentary—Here it is stated that the North and South Poles have the Pole Star in the middle, which is on the horizon of the meridian. It can be estimated from this that when the Sūrya Siddhānta was said in ancient times, there were two Pole Stars. It can also be said that just as there is a star in the middle of the North Pole, a star was also understood to be in the middle of the South Pole. But it is certain that the star seen in the middle of the North Pole at this time was not at this place in ancient times because its position is also changing due to precession (see pages 240-42). Therefore, the Pole Stars described here are the positions of the celestial poles which are in the middle of the North and South Poles. They have no eternal relationship with any star. When a star comes near them due to

precession, it also begins to be called the Pole Star in direct view. It has been stated in many places that the latitude on the equator is zero and

It has been stated many times that the latitude on the equator is zero and

SOURCE PAGE 756

Geography Chapter 731

How the length 60° and latitude 60 and longitude zero occur at the northern and southern poles (see pages 58, 56, 256, 257 etc).

Verse 45 is of great importance. It is stated that when the sun is at the beginning of the Aries sign, the gods see it first, that is, when the sun rises for the inhabitants of the North Pole, and when it is at the beginning of the Libra sign, the demons see it first, that is, when it rises for the inhabitants of the South Pole. It appears that the beginning of the Aries sign should be understood as the place where the ecliptic and the tropics are in conjunction and where the sun becomes in the northern hemisphere. This place is called the Vernal Equinox point. Similarly, the beginning point of Libra is the Autumnal Equinox point where the sun becomes in the southern hemisphere. When the sun comes to the tropics at the beginning of Aries, sunrise occurs for the northern pole dwellers, and they see the sun equally for 6 months. This time is called the day of the gods. And the night of the demons because until the sun is seen by the northern pole dwellers, it remains invisible for the southern pole dwellers and there is night. When the sun reaches the beginning of Libra, sunset occurs at the northern pole and sunrise at the southern pole. For 6 months from this time, the sun is seen equally at the southern pole and there is the day of the month. These characteristics on the northern and southern poles and the equator occur because the poles are 60 degrees apart from the equator (see page 62).

Why are the rays of the sun slow and intense?

> अत्यासन्नतया तेन ग्रीष्मे तीव्रकरा रवेः ।
> देवभागे सुराणां तु हेमन्ते मन्दतान्यथा ॥४६॥

Translation—When the Sun God is in the part, that is, in the northern hemisphere, due to being very close to the gods, its rays are very intense in the summer season and slow due to being far away in the winter season.

Scientific Commentary—this verse states that the rays of the sun are more intense in the summer season because the sun is near and less intense in the winter season because the sun is far, but this is not correct. Nowadays in reality, the sun is farther from the Earth in the summer season and nearer in the winter season as apparent from the size of its images (see page 85). The real reason is

SOURCE PAGE 757

Because the rays of the sun come in a vertical form in the summer season, their intensity is more, and in the winter season, because the rays come under the sun, they come crooked, so their intensity is reduced. This is seen every day. In the noon, the sun is high, so its rays are usually vertical and the heat also increases. But in the morning and evening, its rays remain very slanted, so there is not that much heat. This condition occurs on the entire earth in a period of one year. In the countries near the equator, the sun appears on the head all year, so its rays come vertically and are very intense.

SOURCE PAGE 758

Geography Chapter 733

But the sun's rays become very oblique at the north and south poles, which is why there is always cold there. This will be clarified by figure 123. This figure shows that the rays coming from the sun fall perpendicular to the plane of g and h, and those same rays become oblique on the plane of k and h. It is clear that the plane k h is larger than the plane g h because it is the hypotenuse of the right-angled triangle k g h. Therefore, when the rays spread out over a larger space, their force is reduced, and the heat in the plane g h cannot be the same as in the plane k h. All, whether read or not, experience this, because when the sun's rays come obliquely, then

SOURCE PAGE 759

People place an object on a plane in such a way that the rays fall perpendicularly to dry it.

Figure 124 shows that the rays of the sun that come around the equator spread over a larger area and become weaker due to the rays being slanted in countries farther from the equator. It is clearly seen from this figure that the rays falling on the North and South parts of the Earth near the equator are as many as the rays falling on the North and South parts of the Earth near the North Pole, which is much larger in area, and due to spreading, their intensity decreases.

The division of day and night of the gods and asuras—

> देवासुरा विषुवति क्षितिजस्थं दिवाकरम् ।
> पश्यन्त्यन्योन्यमेतेषां वाम सव्ये दिनक्षपे ॥४७॥
> मेषादावुदितस्सूर्यः त्रीन् राशिनुदगुत्तरे ।
> संचरन्प्रागहर्मध्यं पूरयेन्मेरुवासिनाम् ॥४८॥
> कर्क्यादिस चरंस्तद्वद् अह्नः पश्चाधंमेव सः ।
> तुलादीन् त्रीन्मृगादींश्च तद्वदेव सुरद्विषाम् ॥४६॥
> अतो दिनक्षपे तेषामन्योन्यं हि विपर्ययात् ।
> अहोरात्र प्रमाणं च भानोर्भगण पूरणात् ॥५०॥

Translation—(47) On the day when the sun is on the equatorial circle, both gods and asuras see it on the horizon; their day and night are opposite to each other. (48) Rising at the beginning of the Aries sign, the sun moves towards the north in the three signs Aries, Taurus, and Gemini, completing the first half of the day of the North Meru-inhabitants, i.e., the gods. (46) Similarly, moving further from the beginning of the Cancer sign, it completes the second half of the day of the gods in the three signs Cancer, Leo, and Libra. Similarly, moving into Libra, Scorpio, and Sagittarius signs, it completes the first half of the day of the asuras, and moving into Capricorn, Aquarius, and Pisces signs, it completes the second half of the day of the asuras. (50) Therefore, the day and night of the gods and asuras are opposite to each other, and after one revolution (turn) of the sun, they have one day and night.

Scientific Commentary—The day when the sun comes to the vernal equinox point is called the equinoctial day. On this day the horizon from the north and south poles

SOURCE PAGE 760

Geography Chapter 735

Therefore, the inhabitants of the North Pole see the gods on the horizon and the inhabitants of the South Pole see the demons. However, because the sun's motion is towards the North, it appears to rise for the gods and set for the demons. That is, from this day the day of the gods and the night of the demons begins. This position of the sun, that is the Vernal Equinox point, is called the beginning of Aries. After this, the sun moves towards the North every day. When it reaches 60 degrees from the Vernal Equinox point, its northward movement stops. On this day, the gods see it at its highest point. This height is like the sun's ultimate revolution. Therefore, on this day is the noon of the gods and the midnight of the demons. From the Vernal Equinox point up to 60 degrees, there are three zodiacs: Aries, Taurus, and Gemini. When the sun passes from the beginning of Cancer to Cancer, Leo, and Virgo, and reaches the beginning of Libra, it comes onto the ecliptic. At this time, the gods see it setting. Therefore, from this time the night of the gods and the day of the demons begins. This position of the sun is called the Autumnal Equinox point, and this day is also called the Equinox day. After this, as long as the sun remains in the Libra, Scorpio, and Sagittarius zodiacs, there is the afternoon of the demons and the pre-night of the gods. When the sun reaches the Capricorn zodiac, it is the midnight of the gods and the noon of the demons. When the sun is in the Capricorn, Aquarius, and Pisces zodiacs, it is the afternoon of the demons. In this way, the time taken for one revolution of the sun is the same time for one night of the gods or one day of the demons. But the day of the gods is the night of the demons, and the night of the gods is the day of the demons.

From this description it is clear that the beginning of the zodiac signs such as Aries, Taurus, etc., is considered from the vernal equinox, not from the retrograde Aries which is currently 23 degrees ahead of the vernal equinox and which is always moving further ahead of the vernal equinox. This difference is called precession angle. More than 1400 years ago, when the vernal equinox and retrograde Aries were together, the position of Aries was the same as what is called retrograde Aries today, but this is not the case now. For this reason, two schools have emerged among astrologers today, the siderealists and the retrogradists. Those called siderealists consider the vernal equinox as the beginning of Aries. But the retrogradists consider retrograde Aries as the starting point of the zodiac signs.

SOURCE PAGE 761

There is no difference between sine and declination in Sūrya Siddhānta. It is known that at the time the current Sūrya Siddhānta was inscribed, the vernal equinox was at the same place as the beginning of Aries in the current Nirayana. After this, in the treatises like Siddhānta Shiromani, both are discussed.

From the description of the Ahoratra of the gods or asuras, which appears in many places in the Sūrya Siddhānta, it is proven that their Ahoratra is the same as the sidereal year and this is the natural value of the year. However, the proof of this Ahoratra is also stated to be similar to the period of revolution of the sun, which according to the verses of the Chapter on Mean Motions (madhyamādhikāra) 26 and 37 is the same as the mean Sawan day of 36502658756 and the mean Sawan day is 016540 greater than the sidereal year. This period of revolution is also 002382 days greater than the pure sidereal-solar year. (See the footnote on page 245). Therefore, it seems that in the Sūrya Siddhānta, the value of the sidereal year is taken as similar to the period of revolution of the sun in a gross sense.

When is the noon time of the Devasuras and what is the meaning of up and down-

> अतो दिनक्षपे तेषामन्योन्यं हि विपर्ययात् ।
> उपर्यात्मानमन्योन्यं कल्पयन्ति सुरासुराः ॥५१॥
> अन्यऽपि समसूत्रस्था मन्यन्तेऽधः परस्परम् ।
> भद्राश्वकेतुमालस्था लंकासिद्धपुराश्रिताः ॥५२॥
> सबंत्रैव महीगोले स्वस्थानमुपरिस्थितम् ।
> मन्यन्ते खे यतो गोलस्तस्य क्वोर्ध्वं क्व वाऽप्यधः ॥५३॥

(51) The meridian and midnight progression of the gods and asuras are opposite to each other. Both gods and asuras consider themselves above the other. (52) Those who live in the direction of the terrestrial longitude also consider the other below them, just as the Ketumala region (of Yamakoti Nagar) residents consider the Romak Nagar residents and the Siddapur residents of Lanka Nagar to be below them in the Bhadrasva year. (53) On this geography, people everywhere consider their place to be above because this geography is situated in the sky, so where is its above and below?

Scientific Commentary—the first half of the 51st verse is related to the 50th verse and the latter states that both gods and demons relate themselves to each other

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Understand this. The proof of this is shown in the next two verses with examples.

The reason for the opposition of the noon and midnight of the gods and demons at the end of the ecliptic is clear. Because at the time when the sun enters the sign of Cancer, it appears highest to the inhabitants of the North Pole and remains lowest and invisible to the inhabitants of the South Pole, therefore at this time there is the noon of the gods and the midnight of the demons. Similarly, when the sun enters the sign of Capricorn, there is the noon of the demons and the midnight of the gods.

It is not difficult to understand what is above and what is below because everyone considers the direction in the middle of the sky to be up and its opposite direction to be down. The Earth is round and there is the sky around it, so all the inhabitants of everywhere consider themselves to be above and the one at the other end of their meridian to be below.

In Figure 125, the round line is the Earth. For those living at the North Pole, the direction in which the letter 'ka' is shown is upwards, and its opposite direction is where the mid-latitude is. However, in the line of this direction, the South Pole is on the other side of geography, so the South Pole appears below the North Pole. But for those at the South Pole, the direction in which the letter 'kha' is shown is upwards, and the direction of the mid-latitude or the North Pole is below. This fact can be easily understood by reading the figure in reverse. Similarly, for the place 'ca', the direction of 'ga' is upwards and the direction of 'cha' or 'gha' is downwards, but for the place 'cha', the direction of 'gha' is upwards and the direction of 'ca' or 'ga' is downwards.

The reason for the Earth appearing flat—

> अल्पकायतया लोकाः स्वत्स्थानात्सर्वतो दिशम् ।

> पश्यन्ति वृत्तामप्येतां चक्राकारां वसुन्धराम् ॥५४॥

Translation—Because man is very small compared to the Earth, he sees the round Earth in all directions from his place.

Scientific Commentary—The arc and its sine in a very small part of a circle have such little difference that both are considered equal, that is, even if the arc is curved, it is equal to the sine and not equal to the curvature of the arc. Therefore

SOURCE PAGE 763

So the sine of 225 art is also understood as 225 art (see Chapter on True Positions verse 15). Similarly, it appears to be the same when a very small part of the surface of a spherical body is curved.

It can also be seen. It can be calculated that by standing on a flat land or the bottom of a large lake, one cannot see the surface of the ground more than 3 or 4 miles away all around.

Suppose there is a place on the Earth, let kh be the height of a man, gh be the center of the geography, and kg be the straight line that touches the Earth at point g. From geometry it is proven that

$kg^2 = kh \times kch = kh(kh + khch)$ Suppose $kh = u$, $khgh = gch = tr$, $kg = ks$ then $ks^2 = u \times (u + 2\,tr) = u^2 + 2\,u \times tr$

SOURCE PAGE 764

Geography Chapter 736

Here, u^2 is so small compared to u that it can be considered negligible because the radius of the Earth is 3660 miles, and u is the height of man, which is approximately one thousandth of a mile, so it can be assumed that

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Sūrya Siddhānta = 2uv (1)
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In this equation all measurements are in miles. If it is assumed that the measurement of u is f in feet then

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f = u x 1760 x 2
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yau = f / 3 x 1760
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By raising this value in equation (1) and keeping the place of t as 3660

SOURCE PAGE 765

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Sūrya Siddhānta
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Scientific Commentary

Chapter on Mean Motions

Chapter on True Positions

Chapter on the Three Problems

late

The reason for the waxing and waning of day and night on the Earth—

> सव्यं भ्रमति देवानामपसव्यं सुरद्विषाम् ।
> उपरिष्टाद्भूगोलोऽयं व्यक्षे पश्चान्मुखं सदा ॥५५॥
> अतस्तत्र दिनं त्रिंशन्नाडिकं शर्वरी तथा ।
> हानिवृद्धी सदा बामं सुरासुर विभागयोः ॥५६॥
> मेषादौ प्रत्यहं वृद्धिः उदगुत्तरतोऽधिका ।
> देवभागे क्षपाहानिः विपरीतं तथाऽऽसुरे ॥५७॥
> तुलादौ द्युनिशोर्वामं क्षयवृद्धी तयोरुभे ।
> देशक्रान्तिवशान्नित्यं तद्विज्ञानं पुरोदितम् ॥५८॥

Translation—(55) This constellation circle moves in the direction of the divine quadrant, that is from left to right and from the inauspicious direction to the right and from the west above the head of the Niraksh people. (56) Therefore, in the Niraksh country, there are 30 hours of day and 30 hours of night, but in the divisions of the gods and asuras, that is, in the countries to the north and south of the tropics, the increase of day and night and the loss of night are in mutual opposition. (57) After entering the Aries sign, the sun moves towards the north, in the countries to the north of the tropics, the increase of day and night and the loss of night is the same, but in the countries to the south of the tropics

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SOURCE PAGE 766

Geography Chapter 741

It is the opposite, that is, there is a decrease in day and an increase in night. (58) After entering Tularashi, as the sun moves towards the south, there is a loss of day and an increase of night in the northern part and a gain of day and a loss of night in the southern part. The decrease and increase of day and night depend on the latitude of the place and the precession of the sun, which has been considered before.

In the 55th verse of the Scientific Commentary, it is stated that the inhabitants of the North Pole see the constellation cycle moving in the clockwise direction and the inhabitants of the South Pole see it in the counter-clockwise direction. The explanation of the words clockwise and counter-clockwise is given on page 126 of the Scientific Commentary. In countries near the equator, the constellation cycle moves above the head from east to west. The magnitude of day and night at the equator is always 30 hours for the day and 30 hours for the night. In the countries of the North and South, the magnitude of day or night is 30 hours only for the vernal day when the precession of the sun is zero. In other times when the precession of the sun is north, the day in the northern countries is larger than 30 hours and the night is as little as 30 hours, but in the southern countries, the day is smaller than 30 hours and the night is as large as 30 hours. And when the precession of the sun is south, the day is larger and the night is smaller in the southern countries, and the night is larger and the day is smaller in the northern countries. The consideration of the decrease or increase of day or night is done according to the precession of the sun and the latitude of the place, as stated in verses 60-61 of the Chapter on True Positions and their Scientific Commentary.

The reason for this movement of the constellation cycle is the flow of air according to the ancient view and the daily motion of the Earth according to the new view, which will be considered in the Scientific Commentary of the 74th verse.

> इन श्लोकों में मेष और तुला का अर्थ सायन मेष और सायन तुला समझना
> चाहिए क्योंकि दिन रात की क्षयवृद्धि सायन राशियों के ही अनुसार होती है ।
> विषुवत्रेखा से कितने योजन उत्तर या दक्षिण सूर्य ठीक ऊपर होता है ।

> भूवृत्तं क्रान्तिभागघ्नं भगणांशविभाजितम् ।
> अवाप्तयोजनैरर्को व्यक्षाच्चेदुपरिस्थितः ॥५६॥

When the sums of the latitudes of the Earth are multiplied by the parts of the sun's instantaneous declination and divided by 360, the resulting distance is the same as the distance the sun is above the equator.

SOURCE PAGE 767

Scientific Commentary - The latitude is exactly above at the same declination of the sun. If the declination is north, the latitude should be understood as north, and if the declination is south, the latitude should be understood as south (see page 261 of the Three Problems). The method of calculating how many yojanas a latitude is from the equator is the same as the method of calculation shown in this verse. The value of the circumference in yojanas is the same as 360 degrees, so the latitude desired is how many yojanas it is from the equator, this ratio is shown in it.

360 degrees: Declination: Circumference yojanas: Desired yojanas.

Where is the day of 60 watches or the night of 60 watches-

> परमापक्रमादेवं योजनानि विशोधयेत् ।
> भूवृत्तपादाच्छेषाणि यानि स्युर्योजनानि तैः ॥६०॥
> अयनान्ते विलोमेन देवासुरविभागयोः ।
> नाडीषष्ट्या सकृदहनिशाऽन्यस्मिन्सकृत्तथा ॥६१॥

Translation—(60) In this way, by knowing the value of the plan from the supreme declination of the sun and subtracting it from the fourth part of the circumference, it is found that it is at the same distance from the equator in yojanas (61) at the end of the Aries, i.e., on the day of the Tropic of Cancer, there is one day of 60 watches to the north and one night of 60 watches to the south, and on the day of the Capricorn declination, there is one day of 60 watches to the south and one night of 60 watches to the north.

Scientific Commentary—to understand the meaning of these verses, one should repeat the verses of Chapter on True Positions 60-61 and figures 42, 43 and their descriptions. With the help of these figures, it is easy to know that when the sun's precession is complete, at what latitude its arc is exactly above the horizon line. Figure 127 is made in the style of Figure 42, the only difference being that the center of this figure indicates the place whose declination is equal to the sun's complete precession and the latitude is its complement. Here, the meridian, the khxswastika, the north-south line of the horizon, a point of the prime vertical circle, and the sun are the same when its precession is complete, that is, the

sun of the day of the solstice of the Tropic of Cancer. The latitude here is the latitude, so Viu=Vir. It is clear that R is the arc of the sun of that day which is the horizon

SOURCE PAGE 768

Geography Chapter 743

It is directly above, therefore the sun will not go below the horizon or set on this day, and it will be a 60-hour day. Conversely, at the same southern latitude, the diameter of the sun's arc will be zero on this day, meaning it will be a 60-hour night because the sun will remain on the horizon there for 60 hours. The latitude of the place being discussed is nowadays 60° - 23° 27’ = 66° 33’. Because the sun's ultimate declination is approximately 23° 27’. The northern place is nowadays called the North Polar Circle and the southern place is called the South Polar Circle.

As on the day of the Cancer solstice, the North Polar region has a day of 60 *ghadi* and the South Polar region has a night of 60 *ghadi*, similarly on the day of the Capricorn solstice, the South Polar region has a day of 60 *ghadi* and the North Polar region has a night of 60 *ghadi*. This occasion comes only once a year.

> श्लोकों में अक्षांश को अंशों में न लिख कर योजनों में विषुवत् रेखा से दूरी
> बतलायी गयी है ।

> दिन-रात का प्रमाण ६० घड़ी का कहाँ होता है—
> तदन्तरेऽपि षष्ट्यन्तं क्षयवृद्धी अहर्निशोः ।
> परतो विपरीतोऽयं भगोलः परिवर्तते ॥६२॥

SOURCE PAGE 769

Translation—In the countries between these latitudes, the proof of Ahuratra is 60 hours, and the increase of day and night occurs within this time, but this rule changes in other places because the position of the constellation-circle changes there.

Where are the two months of day or night—

> ऊने भूवृत्तापादे तु द्विज्यापक्रमयोजनैः ।
> धनुर्मृगस्थः सविता देवभागे न दृश्यते ॥६३॥
> तथा चासुरभागे तु मिथुने कर्कते स्थितः ।
> नष्टच्छाया महीवृत्तापादे दर्शनमादिशेत् ॥६४॥

Translation—(62) Subtracting the sum of the revolutions of two zodiacs from the quadrant of the circumference, the sun of Sagittarius and Capricorn is not seen to the north and (64) the sun of Gemini and Cancer is not seen to the south from the equator. Because where the shadow is zero at local noon, the sun is seen up to the quadrant of the circumference from that place.

Scientific Commentary—Swami Vignanananda Ji has done this in his Bangla ṭīkā on the latter part of verse 64 that where there is no umbra, the sun is seen. In the commentary of Prakashika Sanskrit ṭīkā, its meaning is given as 'Absence of shadow, the shadow is there, the sun is always seen in the quadrant of the circumference'. In the Hindi commentary of Pan. Indranarayan Dwivedi and Madhav Purohit, this meaning is not. I have interpreted it as the day when the sun is seen up to the quadrant of the

circumference from the place where the shadow of an object is zero at local noon. Because where the local noon shadow is zero, the sun is on the kh-cross on that place and from here the sun is seen all around for 60 degrees at this time. Besides this, the meaning of 'shadow' is not umbra, local shadow is appropriate. Therefore, 'nashtachhaya' means the place where the local noon shadow is zero.

These two verses explain where the two months of night occur when the Sun's sine is in the Sagittarius and Capricorn signs. When the Sun's sine enters Sagittarius, its south revolution is 2010', (see page 316), and until it remains in Sagittarius and Capricorn signs, its south revolution is more than 2010'. Now we have to see when the Sun's south revolution 2010'

SOURCE PAGE 770

Geography Chapter 745

Then it can be seen on which part of the Earth. It is clear that at this time the sun is on the khastak of that place, whose south latitude is 20° 10'. Therefore, the meridian shadow at this place will also be zero, and the sun can be seen up to the fourth part of the Earth's circumference, i.e., 60 degrees North-South. The latitude of the place 60 degrees North of 20° 10' South latitude is 60° - 20° 10 = 66° 50'. Therefore, the sun's rays can reach here. After this, until the sun is more than 20° 10' South of the south revolution, it will not be seen at 66° 50' North latitude, i.e., there will be a two-month night at this place. In its opposite, at 66° 50' South latitude, there will be a two-month day. The rule to know what the planimetric difference of this place will be from the equator is explained in these two shlokas, according to the rule stated in shloka 56 and which is applied in shlokas 60-61.

Similarly, when the sun is in the signs of Gemini and Cancer, its north revolution is more than 20° 10', due to which the sun is seen equally at the places of 66° 50' North latitude for these two months, so there is a two-month day here, and due to the sun being invisible for two months at such a south latitude, there is night.

Where are the four months of day or night-

> एकज्यापक्रमानीतयोजनैः परिवर्जितैः ।
> भूमिकक्ष्या चतुर्थांशे व्यक्षाच्छेषंस्तु योजनैः ॥६५॥
> धनुर्मृगालिकुम्भेषु संस्थितोऽर्को न दृश्यते ।
> देवभागेऽसुराणां तु वृषाद्ये भचतुष्टये ॥६६॥

Translation—(65) Subtracting the sums of the revolutions of a sign from the fourth part of the Earth's circumference, the sun is not seen in the signs of Sagittarius, Capricorn, Aquarius, and Pisces on the side of the equator, and in the signs of Taurus, Gemini, Cancer, and Leo on the south side.

Scientific Commentary—when the sun is in the zodiac signs of Sagittarius, Capricorn, Aquarius, and Pisces, then its south revolution is more than the revolution of one sign, that is, more than 11° 26', therefore in these four months the sun does not appear at the place whose north latitude is 60° - 11° 26' = 78° 31'. The effect of this is that there are four months of night here, but at the south latitude of 78° 31'.

SOURCE PAGE 771

There is a 4-month day. In the same way, when the sun is more than 110 26' of north precession, that is, when it remains in the zodiac signs of Taurus, Gemini, Cancer, and Leo, then there is a 4-month night at

78° 31' South latitude and a 4-month day at the north latitude.

In the shlokas, the method of knowing the distance from the equator is given in terms of the lines of latitude, as in the earlier shlokas.

Where is the 6-month day or night?

> मेरौ मेषादिचक्रार्धे देवाः पश्यन्ति भास्करम् ।
>
> सकृदेवोदितं तद्वदसुराश्च तुलादिगम् ॥६७॥

Translation—When the sun remains in 6 zodiac signs from Aries to Virgo, then the deities residing at the North Pole see it rise once, that is, it does not set for 6 months, and when the sun remains in the zodiac signs of Libra to Pisces, then the demons at the South Pole see it rise equally.

Scientific Commentary—When the sun enters Aries, it comes into the northern hemisphere and remains in the northern hemisphere for 6 months, so it is always visible at the North Pole during these months and invisible at the South Pole. Therefore, in these 6 months, there is one day for the gods and one night for the demons. But when the sun enters Libra, it becomes in the southern hemisphere and remains in the southern hemisphere for 6 months, so the demons see the sun equally during these 6 months, and there is a day of 6 months here, and because it is invisible from the North Pole, there is a 6-month night for the gods.

Where does the sun appear exactly above on the day of the Cancer or Capricorn solstice and what is the special feature here--

> भूमण्डलात् पञ्चदशे भागे दैवे तथाऽऽसुरे ।
> उपरिष्टाद्व्रजत्यर्कः सौम्य याम्यायनान्तगः ॥६८॥
> तदन्तरालयोश्छाया याम्योदक् संभवत्यपि ।
> मेरोरभिमुख यातः परतश्च स्वभागयोः ॥६९॥

(68) The sun travels at the time of the end of Uttarayana or Dakshinayana directly above the north or south position at the distance of the 15th part of the circumference of the Earth from the tropics. (66) Between these lines, the meridian shadow

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It can be south or north. In the places outside these, the meridian shadow lies towards the Meru of its respective division.

Scientific Commentary—The end of the Uttarayana occurs at the time of the Karka Sankranti when the northward revolution of the sun is equal to the supreme, according to the view of Sūrya Siddhānta, 24

Therefore, on this day, the sun is exactly above the 24° North latitude at the meridian and the meridian shadow is zero. Similarly, at the end of the South season, the southward revolution of the sun is 24°. Therefore, on this day, the sun is exactly above the 24° South latitude, but the 24° portion of the Earth is the 15th part of the entire circumference. Nowadays, this is approximately 23°27' of the zodiac. Therefore, on the day of the Cancer solstice, the sun is directly above the 23°27' North latitude and on

the same latitude, the sun is directly above the Capricorn solstice, 23°27'. Therefore, the land between these two latitudes is called the Tropic of Cancer because the great heat arises from the sun remaining above for twelve months here.

In this land, the shadow at the meridian of every place can be north or south because the latitude of any place here is less than the supreme revolution of the sun

SOURCE PAGE 773

Therefore, when the latitude of a place and the ecliptic of the sun are in the same direction and the ecliptic of the sun is less, then the meridian shadow will be towards the pole in that direction, but if the ecliptic is more, then the direction of the shadow will be opposite (see Chapter on the Three Problems figures 55, 56). But in the countries north of the Tropic of Cancer, the direction of the meridian-shadow will always be towards the north, and in the countries south of the Tropic of Capricorn, the meridian-shadow will always be towards the south.

In Chitta 127, the area within the circle of the meridian reveals the hemisphere of the Earth. U and D are the North and South poles, respectively. Roll is the equator. Yamakoti, Lanka, and Ro Rochaka are cities. Siddapuri cannot be seen on this hemisphere because it is in the other hemisphere in the same circle as Lanka. From the equator, the North Tropic and South Capricorn lines are 23° 27' North and South. These lines are parallel to the equator. On the land between these two lines, the meridian shadow can be North or South. The North and South polar circles and their parallels are 66° 33' North and South of the equator. On these lines, the proof of day is once a year, 60 hours or 24 hours, and the proof of night is also once in the same way as stated in 60-61 shlokas. The proof of Ahuratra in the land between these lines is 60 hours. The proof of day and night in the land outside these lines is strange. On the parallel line 66° 50' north of the North polar circle and the equator, there are 2 months of day and 2 months of night once in a year. On the line 78° 31' north of the equator, there are 4 months of day and 4 months of night. This also happens in the South polar circle. On the North poles, there are 6 months of day and 6 months of night.

When does sunrise and sunset occur in the four cities of the Tropic of Cancer-

> भद्रारवोपरिगः कुर्याद् भारते तूदयं रविः ।
> रात्र्यर्धं केतुमालाख्ये कुरुष्वस्तमयं तथा ॥७०॥
> भारतादिषु वर्षेषु तद्वदेव परिभ्रमात्
> मध्योदयार्धरात्रास्तकालात् कुर्यात्प्रदक्षिणम् ॥७१॥

(70) When the sun is directly overhead in the cities of Yamakoti in the year Bhadrasva, then it rises in the city of Lanka of Bharatavarsha, there is midnight in the city of Romak of Ketumala desh, and it sets in the city of Siddapuri of Kurudesh.

SOURCE PAGE 774

Geography Chapter 746

It is. (71) In this way, in countries like India, the mean day, sunrise, midnight, and sunset occur in sequence.

The relationship between these four cities has been explained in 38-40 shlokas in the Scientific Commentary. The relationship between their times is explained here. When there is mean day in Yamakoti, then sunrise in Lanka is 60 degrees west of it, midnight in Romak is 60 degrees west of Lanka, and sunset in Siddpuri is 60 degrees west of Romak. Similarly, when there is mean day in Lanka, then sunrise in Romak, half-night in Siddpuri, and sunset in Yamakoti occurs.

The mutual difference between the Pole Star and the constellation circle—

> ध्रुवोन्नतिभंचक्रस्य नतिर्मैरुं प्रयास्यतः ।
> निरक्षाभिमुखं यातुर्विपरीते नतोन्नते ॥७२॥

Translation-moving towards the poles, the declination of the Pole Star and the declination of the constellation circle increase, but moving towards the equator, it is the opposite, that is, the declination of the Pole Star and the declination of the constellation circle increase.

Scientific Commentary—the constellation circle is near the prime vertical circle, so the constellation circle appears directly above the equator and the pole star appears on the horizon. Moving towards the poles from here, the latitude of the poles increases, and the latitude of the prime vertical circle decreases or the declination increases. At the poles, the latitude of the pole star is 60 and the latitude of the prime vertical circle is zero or the declination is 60 because the prime vertical circle becomes the horizon at the poles. In contrast, moving towards the equator, the declination of the pole star and the latitude of the constellation circle increase.

Cause of the motion of the nakshatra cycle—

> भचक्रं ध्रुवयोर्बद्धम् आक्षिप्तं प्रवहानिलैः ।
> पर्यैत्यजस्रं तद्बद्धा ग्रहकक्ष्या यथाक्रमम् ॥७३॥

Translation—the nakshatra cycle rotates continuously by being bound to both pole stars and receiving the push of the flow of air. The planetary orbits bound in sequence also rotate with it.

SOURCE PAGE 775

Sūrya Siddhānta—Scientific Commentary—the Sun, Moon, planets, and stars all rise above the eastern horizon, move towards the west and set, and come to rise again above the eastern horizon in 24 hours. The reason for this is understood in ancient times that the entire celestial sphere is revolving by the flowing air bound in both celestial poles, and the orbits of the planets are also revolving from east to west in the same celestial sphere. Some scholars of Bharatavarsha still support this view and are ready to debate. However, it has now been proven by irrefutable evidence that the cause of the motion of the celestial sphere is not the flowing air but the motion of the Earth itself. With one motion, the Earth moves from west to east on its axis once every 24 hours; this daily motion is called the rotation of the Earth's axis. From this, all objects in the sky appear to move from east to west. From this, the origin of day and night occurs. With the second motion, the Earth completes one orbit of the Sun in a year, which causes the origin of the seasons, and in the sky, the Sun appears to move from west to east while orbiting the Earth in a year. This motion is called the annual motion of the Earth. Both these motions occur simultaneously in the Earth, as a ball thrown upwards dances on its axis and changes its position, or as a top played by boys dances and changes its position.

In our ancient religious texts, the Earth is considered motionless, therefore the motion of the Earth is not accepted by some scholars of Sanatana Dharma; however, in the debate, they present the same arguments that Acharya Varahamihira, Brahmagupta, etc., presented. Therefore, it must first be considered where these arguments are compatible with mathematics. After this, it will be proven by many proofs of mathematics and physical science that there are two motions in the Earth, daily and annual, and because of these, the constellations appear to move from east to west once a day, and the change of seasons, etc., occurs, and the motion of the planets appears to be of a certain type. The methods by which Acharya Varahamihira and Brahmagupta refuted the motion of the Earth are these ()^1. It seems necessary to state here that our Acharyas, Aryabhata, consider the movement of the Earth in their Aryabhatiya text and its

> १. भ्रमति भ्रमस्थितेव क्षितिरित्यपरेत्वदन्ति नोडुगणः ।
> यद्येव श्येनाद्यान खात्पुनः स्वनिलयमुपेयुः ॥६॥

SOURCE PAGE 776

Geography Chapter 751

They support this example with the fact that just as a person sitting on a moving boat sees the boat and the trees, houses, etc., appear to move in the opposite direction, similarly, humans living on a revolving Earth see the constellations moving towards the west even when the celestial circle is stationary, but no one accepted this against tradition, and Varahamihira etc. presented these arguments.

One argument of Acharya Varahamihira is that if the Earth revolves towards the east, then why do the birds that fly away from their nests reach back to the nest? Because due to the Earth's rotation, the nest on the Earth is very far to the east, and the bird remains very behind when it stays in the sky. The second argument he made is that if the Earth revolves towards the east, then the flag and so on always appear to fly towards the west, because this is a common experience that if a person runs with a handkerchief in hand, due to the speed, the handkerchief starts flying backward. And if it is said that the Earth moves very slowly, then the flag etc. do not appear to fly to the west, then how does the Earth complete a revolution in a day at such a slow speed?

The reason for the birds reaching their nests is that when the bird flies in the sky, the velocity of the geostationary motion remains in that nest, so the bird also moves with the same velocity, though it does not realize it. Also, it can generate its own motion, by which it can go wherever it wants away from the nest. Like a person on a train does not experience the velocity with which the train is moving, but that velocity remains with him.

SOURCE PAGE 777

is. Even with this velocity, it can move here and there in the box with its will, jump, and play with a ball. Because all the objects kept in the cart have the same velocity as the cart, this velocity is not apparent in all of them because it remains the same in all. This can also be easily ascertained. If a stone is thrown straight outwards while sitting in a very fast-moving cart, it is seen to move forward along with the cart until it touches the Earth. This fact is seen even more clearly when the stone is thrown at the time when the cart starts moving on a river bridge because in such a situation, it will take some time for the stone to reach the ground, so it will be seen moving forward with the cart for some time and will not fall at the place where it was aimed but will fall in a straight line where it is aimed after moving forward. From this, it can be known that when an object is separated from a moving cart, airplane, etc. with a certain

velocity, that velocity remains in it which was in the cart, and its velocity is not destroyed until the object stops on another object. This is why if someone jumps from a moving cart, they fall forward due to the velocity of the cart.

The second test of this can be done in this way. Everyone knows that if a thing is dropped from a certain height, it falls to the earth directly below it. The height of a large train car is about 10 feet from the floor. Therefore, if a piece of stone is dropped from near the ceiling, it will take it 6, 10 feet to reach the floor and it will take about a quarter of a second. In this time, if the train is moving at a speed of 30 miles per hour, it moves 33 feet ahead. Therefore, if Varahamihira's argument is correct, the stone should not fall at the place that is directly below the place from which the stone is dropped, but it should fall 33 feet behind. But this is not seen. In sight, it falls from the place from which it is dropped directly above. The reason for this is that at the time the stone is dropped from the ceiling, the motion of the train that is currently present in it remains present even at the time of falling, so the thing moves forward with the train and falls exactly where it is dropped from. Jumping over the back of a horse while running in a circus game and then landing on its back is a result of this rule.

SOURCE PAGE 778

Geography Chapter 753

Now the matter of the flag. In every particle of the flag, the velocity of the Earth remains. Similarly, the part of the Earth in the air also has the current velocity, which is why the cloth of the flag does not appear to fly to the west due to the motion of the Earth. If a flag is attached outside a vehicle, motor, or train, it appears to fly backward because there is no connection between the outside air and the motion of the train or motor; it moves by cutting the air, so it moves backward and blows the flag and so on backward. Yes, if all doors of the train or motor are closed, the connection of the air inside is broken from the outside air, and the velocity of the vehicle remains in it, so it does not have the power to blow the flag backward. Similarly, the Earth's atmosphere is also unable to blow the flag backward because the Earth does not move by cutting the atmosphere but by being carried along with it, so the same velocity remains in it.

The argument of Acharya Brahmagupta that the peaks of high houses, mountains, etc., sometimes go up and sometimes down due to the rotation of the Earth, and when they go down, they must fall, but this does not happen, therefore the Earth does not rotate, but it is exactly the Earth. The imagination of height and lowliness is made from the thought of the Earth. The direction towards the Earth is called the direction of down, and the direction opposite to it towards the sky is called the direction of up, and the objects fall on Earth due to the Earth's attractive force, so if a sphere is rotated in the air above the Earth and an object is attached to it, then when it is towards the Earth, it falls on the Earth, so this is absolutely correct. But where the question of the Earth's rotation is, what is below it, by whose attraction do the high houses or mountains of the surface fall towards it? The sky is the sky all around the Earth, so no matter how much it rotates, the peaks of the houses and mountains on it will always remain towards the sky, and the foundation towards the Earth, so where can they fall.

The doubts have been resolved up to this point. Now, by giving examples, it will be proven on the basis of mathematics that there is motion in the Earth.

Proofs of the axial rotation of the Earth from recent science—

This is a matter of common experience that the point of the wheel which is away from the axle

SOURCE PAGE 779

it moves more from the point near the axis and there is more velocity in the points on the edge of the wheel than those in between. If the Earth is considered to revolve on such an axis whose one end is at the North Pole and the other at the South Pole, then it is clear that a tall tree, house, or minaret is farther from the Earth's axis than its base, so the motion of the top will be greater than the motion of the base. Therefore, if an object is dropped from the top of a tall minaret, it will not fall straight down due to greater velocity but will fall towards the east because the point directly below it

the speed of which is slower. Suppose S is the top of a minaret from where the object is dropped and P is the base of the minaret which is directly below S, so when the line connecting S and P is drawn, it will reach C at the center of the Earth. If it is assumed that the time taken for the object to reach the Earth's surface is the time taken for the top S to move from S to A, then the base of the minaret P will move from P to A because the line joining the top and the base will always pass through the center of the Earth. It is clear that P A is less than S A. It is also clear that the motion of P is less than the motion of S. However, the object dropped from point S will have the same motion as the motion of S, so it will retain its upper motion even while falling, therefore

SOURCE PAGE 780

Geography Chapter 755

It will not fall on Pa but will fall on Pi where Pi is equal to Sa, that is, the object will fall some towards the east of the meridian line.

Therefore, if it is proven by examination that an object falling from above moves towards the east in reality while reaching the Earth, then it can also be imagined that the Earth moves towards the east. But this examination is difficult because a place cannot be made so high that the difference in the motions of its peak and base is so great that it can be clearly seen, because the radius of the Earth is about 4000 miles and the top of the tower cannot be even 1000 feet. From the experiments done in Bolon and Hemburg on this matter, it has been proven that an object falling from a height of 250 feet moves three-tenths of an inch to the east from the meridian. By calculation, it can be seen that the difference in the motions of the peak and base or the points S and P is the same amount of time as the time it takes for an object to fall 250 feet. Therefore, it is proven that there is a motion in the Earth. The entire calculation of this experiment cannot be given here because it cannot be understood without knowledge of higher mathematics. Therefore, only the essence is given here.

Newton first imagined this experiment. The picture given on page 754 is suitable for countries located on the equator. Some changes have to be made in its calculation for other places because in other places, the object falling from the equator has two motions whose directions are different. One motion is the daily revolution of the Earth, which the falling object obtains from the top of the mast and is in the perpendicular plane to the axis of the Earth, and the other motion is due to the Earth's gravity, by which the object falls towards the center of the Earth. Therefore, the object moves towards the east from the meridian, along with some south or north also. While falling, there is also some effect of the friction of the air on the object, but despite all these things, there is no difference in the original principle.

This experiment is also done in the deep coal mines because more depth can be obtained for falling. An object dropped from a height of 500 feet moves approximately 1 inch to the east from the north direction. This is the mean of several experiments, and the calculation also proves the same thing.

SOURCE PAGE 781

(2) But this is even more intuitive and clear in Foucault's pendulum experiment. Mathematics proves that if a pendulum only oscillates due to the effect of gravity or swings, its plane of oscillation will remain the same, and the motion of the base of the pendulum will not be affected by the direction of this plane because there is no other force that can deflect it from this plane. It can be seen intuitively that if a heavy pendulum is hung by a thin string and swung like a clock pendulum, and if the base in which the pendulum is hung is rotated, its rotation does not cause any change in the plane of oscillation of the pendulum because the slight twisting of the string or cord to which the pendulum is attached is more natural than the change of the pendulum's plane of oscillation by a heavy pendulum when it is already swinging in one plane. Therefore, it is proven that if the Earth is stationary, the direction of the pendulum's oscillation will also remain stationary with respect to surrounding objects and the base, and if there is a motion in it, then there will be a change in the directions of the ground rather than the plane of the pendulum's oscillation, and the plane of the pendulum's oscillation will be seen as changing. Therefore, from this pendulum experiment, not only the motion of the Earth will not be known, but its direction will also be known.

Foucault performed this experiment in Paris in the year 1851 or 1608. He hung his pendulum from the dome of a large building named Panthéon. The wire was 200 feet long and the weight of the sphere was about 1 mina (80 pounds). At the time the pendulum swung, the needle that came out from below the sphere continued to make the mark of its swinging on the plane, and it was seen that the plane of the sand was rotating in the direction of the apse, that is, from right to left, from west to east.

Two things must be kept with great care in this experiment. The pendulum should remain swinging for as long as its string is long, otherwise it will stop quickly by rubbing against the air with its high speed. Secondly, it is good if its sphere is as heavy as possible because this reduces the effects of hanging defects and air friction greatly.

The endeavor to perform this experiment very successfully was done by a scientist from America, Russell Debelue Porter. He made about 12

1. See the Scientific American of July 1628, pages 14, 15.

SOURCE PAGE 782

Geography Chapter 757

A sphere of iron weighing 40 pounds or 20 seers was hung from the roof support by a long wire. It was observed that the motion of the pendulum gradually slows down, but if the pendulum is pulled and tilted by three feet from the meridian direction, it continues to swing two feet here and there even after half an hour.

SOURCE PAGE 783

Yes, it should be kept in mind that the pendulum should not have any oscillation in the ceiling where it is hung and there should be no breeze in the room. When the pendulum hangs, it tends to rotate, which causes a strain in the wire or string. This generates another motion in the pendulum. Therefore, to prevent this, they hung the wire in a brass hook which was shaped like a question mark and the tip of the hook was given a pillar in a shallow bowl which was well tightened on the thread. The bottom of the bowl was made very smooth.

Before swinging the pendulum, it should be kept completely still. Therefore, a thread was tied to the sphere and the thread was pulled so that when tied to the wall, the sphere moves 2, 3 feet away from the line perpendicular to the Earth-point. Now if the thread is burned, the sphere will start to move and will continue to swing in the same plane. If this is not done, the sphere starts to swing in a long ellipse and if there is even a little disturbance at the beginning, it takes a very large shape in a while.

No matter the direction of the pendulum's swing, but if it is started from the North-South direction, it is good. The needle sticking out from below the sphere should be so close to the table that it is just barely touched by the paper sheet placed on it. After swinging the sphere, draw a straight line with a pencil on the paper sheet and move the sheet on the table in such a way that the needle does not touch it and the drawn line exactly coincides with the plane of the needle's swing. Now you should carefully look at the southern edge of the sheet. In two or three minutes, the southern end of the sheet will be seen rotating from west to east, i.e., in the opposite direction to the clock or in the opposite direction of the clock. The reason is that the sheet keeps moving from west to east with the Earth. If this experiment is done in countries south of the equator, then the line of the sheet will be seen rotating in the clockwise direction.

Now we have to see where the result of the experiment is obtained from the calculation.

If it is possible that the pendulum is hung on the North Pole, then the meridian of the pendulum and the axis of the Earth will be in the same direction. Therefore, as the Earth rotates from west to east, the plane of the observer's standing will also rotate from west to east, and the plane of the pendulum's oscillation will seem to shift from east to west because the observer does not see the Earth's rotation.

SOURCE PAGE 784

Geography Chapter 756

It can. Therefore, one will see it revolving at the speed of the oscillation-plane of the pendulum in the opposite direction in 23 hours 56 minutes 4 seconds in one revolution.

SOURCE PAGE 785

It is not possible that a pendulum swung once continues to swing for 24 hours. But the longer it swings, in the same amount of time, its plane of oscillation will have rotated so much that by the ratio, it can be easily known what the time taken for one revolution might be.

Porter had hung his pendulum in this manner. A brass hook of thickness one inch is placed in a steel cup in which there is no strain. (See figure 131)

If the pendulum is swung on the equator, the line formed by its tip will be above each other because the east-west motion of both its ends is the same here, so the plane of oscillation will not appear to rotate but will continue to move on the same line.

But this will not be the case in places different from the equator because the speed of the Earth's rotation in the part of the surface directly below the pendulum is greater than that near the pole, so the result will be that the line formed by the tip of the pendulum on the sand will rotate faster than the edge that is towards the equator than the edge towards the pole in the east.

SOURCE PAGE 786

But you will see that the orbit of the edge that moves and the one towards the pole will take more than 23 hours 56 minutes 4 seconds to complete, as proven by the calculation below.

Imagine that the northern latitude of the place of examination is a. The equator, the center of the Earth, the axis of the Earth, and the North Pole are V. The angular velocity of the Earth revolving on the axis can be divided into two parts according to kinematics, one part revolving on a and the other part revolving on b.

This part of the velocity which is on a is equal to the co-latitude (60°-a) or sin a, and the part which is on b is equal to cos a. But the velocity revolving on b will be parallel to a, so its effect on the pendulum will be the same as on the equator, that is, there will be no change in the direction of the line formed by the pendulum due to it, but the velocity revolving on a will change the direction of the line formed by the pendulum by swinging with the pendulum's needle, so the southern end of the line will appear to move from west to east and it will be known that the plane of oscillation of the pendulum is also rotating from east to west because the second line will be formed towards the west from the first line.

Now it is to see how much time it will take for the pendulum's oscillation to complete one revolution if it keeps swinging continuously. It is assumed that the velocity of rotation on the axis of the Earth is v and the component velocity at that place is v sin theta. Therefore, it is easy to know that when one revolution is completed in 24 hours with velocity v, it will take more time than one revolution in velocity v sin theta. Therefore, the entire revolution of the lines formed by the pendulum will be completed in v × 24 hours = 24 × sin theta sin theta time. Where theta is the latitude of the place. If this experiment is done in Prayag, then one revolution will take 24 hours sin 250 sin 25' = 24 hours. 4262 = 55 hours 55 minutes or approximately 56 hours. Therefore, if the pendulum swings for even half an hour, a considerable change in the direction of the oscillation plane will be seen because when the revolution is completed in 56 hours, in half an hour, (1)/(2) × 360 × (1)/(56) degrees = (45)/(14) = 3 arc minutes will be approximately changed.

SOURCE PAGE 787

which can be seen easily because if the pendulum is swung by even 2 feet from the plumb line, it will move more than 1 inch in 3 degrees of change.

In this way, different scientists have made these experiments on different latitudes, and all the experiments prove that the time taken for the line formed by the pendulum to complete a full revolution = 24 hours / sine of latitude and the magnitude of rotation in one hour.

In this way, 24 hours / sine of latitude: 1 hour:: 360 degrees: desired magnitude

∴ desired magnitude = 360 X sine of latitude / 24 = 15 sine of latitude

The results of different experiments are given in the table on the next page—

From this table it becomes evident that the change in the direction of the plane of oscillation of the pendulum happens due to the motion of the Earth itself. All these experiments are for the countries to the north of the equator. The change in the countries to the south of the equator also happens by this rule.

(3) There is also a third method to prove the motion of the Earth, which Foucault himself derived. If the edge of a circle is very heavy and its axis makes a right angle with the surface from which it passes through its center, and such a circle can rotate with great velocity on its axis, such a circle is called a gyroscope. If it also has a base with which it remains still, then it is called a gyroscope. A diagram of a simple gyroscope is 133—

K, Kh, Chakra Sam Dharatal Aksh Ga Gh Par Ghoom Sakta Hai Aur Jis Chakra Par Ga Gh Aksh He Vah Cha Ch Sam Paratal Aksh Par Ghoom Sakta Hai (Ch Akshar Chitre Mein Spasht Nahin Hai. Yah Gh Ke Paas Aur Yantra Ke Kuch Poche Hain). Cha Ch Aksh Kul Koleta Hua Ja M Lamb Aksh Par Ghoom Sakta Hai. Yah Yantra Aisa Banana Chahiye Ki Iske Ghoomte Samay Ragad Kam Se Kam Ho. Ye Teeno Aksh Ek Doosre Se Samkon Par Hone Hain, Ga, Gh Aur Cha Ch Aksh Sam Dharatal Mein Aur Ja M Aksh Lamb Disha Mein. Yadi Ragad

1. Based on the Movements of the earth by Norman Lockyer F.R.S., taken from the Urdu scientific monthly journal 'Roshni' of April 1616 AD, pages 280-81.

SOURCE PAGE 788

The place of experiment latitude change in the direction of the plane of oscillation in 1 hour by experiment change in the direction of the plane of oscillation in 1 hour by calculation name of experimenter Ceylon 656 1.870 arc 1.815 arc Schaw and Lamprey New York 40 44 9.733 6.814 Loomis Provinces R. I. 40 46.5 9.655 6.833 Carowell and Norton Newhaven 41 185 6.670 6.627 Geneva 46 12 10.522 10.856 Dufair and Wartman Paris 48 50 11.500 11.763 Bunt Bristol 51 27 11.788 11.763 Galbraith & Houghton Dublin 53 20 11.615 12.065 Gerard Aberdeen 57 6 12.700 12.628

SOURCE PAGE 789

If very little is so that the motion of each axis is completely independent, then many wonderful properties are found in the revolving instrument when the circle of k k chakra is spinning very fast. One important property is that if the k k chakra is made to run fast, the direction of the g th axis always remains the same when the j m is taken from one place to another by removing it from one place. When the direction of the axis of the revolving circle is kept parallel to the axis of the Earth, then its direction remains stable with respect to nearby objects, but if its axis is turned in another direction and it is rotated, then the axis changes direction just as the stars. If the circle is rotated by making its axis in the direction of a particular star, then as long as the circle keeps revolving, the axis will always remain in the direction of that star. This proves that the directions of the stars

SOURCE PAGE 790

Geography Chapter 765

The direction is fixed and their daily movement from east to west is due to the daily motion of the Earth.

Besides these experiments, there are many other phenomena from which the rotation of the Earth's axis is proven. The line formed by the pole of the pendulum in the northern hemisphere rotates in the adverse direction of the clock, similarly here there is a direction of rotation of the latitudes. But in the southern hemisphere, the line formed by the pole of the pendulum and the direction of the latitudes rotate in the favorable direction of the clock. The winds that blow from the equator towards the pole

move towards the east in the northern hemisphere, i.e., to the right, and in the southern hemisphere also towards the east, i.e., they turn to their left. The reason for this and what else could it be?

SOURCE PAGE 791

it is that when the air above the equator is warm and light, it rises, so to fill its place, the cold air near the poles flows towards the equator. However, the motion of the Earth on the equator is very rapid to the east, and as you go towards the poles, this motion slows down, so the motion of the air that flows from the equator also remains rapid to the east, so it reaches the countries towards the poles whose eastward motion is slow. Then it turns to the east. Similarly, the air that flows from the poles towards the equator turns to the west.

The direction of the ocean currents is also the same. The warm current near the Equator from the Gulf of Mexico flows north in the Atlantic Ocean, it then turns to the east and reaches countries like France, England, Norway, etc., on the other side of the Atlantic Ocean in the northeast direction, and the cold current from Greenland flows towards North America. The result of this is that the harbor of Harfangst in Norway, which is at 703 North latitude, remains free of ice for twelve months, whereas the western edge of North America remains covered with ice for more than 40 degrees of latitude due to cold and heat.

Similarly, the warm current flowing north from the Indian Ocean island group turns to the east and keeps the eastern part of Japan warm, and the cold current from the north flows in the exact opposite direction in the China Sea from the western edge of Japan.

This is briefly explained what changes happen in the directions of winds and currents due to the daily motion of the Earth. If you want to know more about this topic, you should use good books of geography.

Except for the motion of this axis, there is another motion in the Earth, by which it completes the orbit of the sun in a year, but it seems as if the sun orbits the Earth. The proof of this motion of the Earth is even more subtle, the consideration of which will be done elsewhere. At this time, it will be sufficient to remind that due to this motion of the Earth, eight types of motions are seen in the planets (see Chapter on True Positions pages 84-64, 67-105).

It is stated in the latter part of 73 shlokas that the planetary orbits are also in a cycle

SOURCE PAGE 792

Geography Chapter 767

From the bound east to the west are going. But the cause of all these motions is the daily motion of the Earth.

u = North Pole d = South Pole v = Equator

> सकृदुद्गतमब्दार्धं पश्यन्त्यर्क सुरासुराः ।
>
> पितरः शशिगाः पक्षं स्वदिनं च नराभुवि ॥७४॥

Translation—Sur and Asura keep watching the sun that rises once and see it for half a year continuously, the inhabitants of the lunar world, the ancestors, watch it for one side, and the inhabitants of the Earth, humans, watch it for one day.

Scientific Commentary—The meaning of the first half of this verse is the same as stated in the 67th verse. There is some special meaning in the meaning of the first word of the latter part which needs to be explained. The Sanatana Dharma Hindus believe that the ancestors reside in the upper part of the lunar sphere. This part is not in front of the Earth. Western astrologers also say that the moon revolves around the Earth in such a way that its lower part remains in front of the Earth and the upper part always remains behind.

SOURCE PAGE 793

Thus, the Moon moves on its axis in the same number of days as it takes to orbit the Earth. Its proof is not difficult. By looking carefully at the lunar image, it is known that its black spots always appear in the same position relative to the edge of the image, which shows that the part of the lunar image facing the Earth always remains in the same condition, that is, the period of the Moon's axis rotation is the same as its orbital period. It can be said that the Moon does not have any axis rotation. But this is not correct. This will be made clear by an example. Place a lamp in the middle and look at it. Suppose the lamp is to your north. Now, while looking at the lamp, move around it in the clockwise direction. When you complete a quarter revolution, the lamp will be to your east. After completing half a revolution, the lamp will be to your south, after three-quarter revolution it will be to your west, and after a full revolution, coming back to the place from where the rotation began, the lamp will be to your north again. This proves that in such a revolution, your face always remains towards the lamp and your back always remains behind it. Also, your body also moves once because while rotating, your face is always towards north, east, south, and west.

When the upper part of the moon always remains opposite to the Earth, what is its relation to the sun? On the day of Amavasya, the moon is between the sun and the Earth, so its upper part remains directly in front of the sun. The Pitris reside in the upper part, so on the day of Amavasya, the sun is directly above the ancestors, meaning it is their noon on this day. That is why obituaries and offerings are done for the ancestors during the noon of Amavasya. On the day of Purnima, it is their midnight. After half of the Krishna Paksha passes, the sun appears to rise while seeing the ancestors, and for half of the Shukla Paksha, it appears to see them equally, meaning the ancestors' morning is on the eighth day of the Krishna Paksha and the evening is on the eighth day of the Shukla Paksha.

Relation between planetary orbits and planetary motions

> उपरिस्थस्य महती कक्षाऽल्पाधः स्थितस्यच ।
>
> महत्याकक्षया भागा महान्तोऽल्पास्तथाल्पया ॥७५॥

SOURCE PAGE 794

Geography Chapter 766

> कालेनाल्पेन भगणं भुङ्क्तेऽल्प भ्रमणाश्रितः ।
> ग्रहः कालेन महता मण्डले महति भ्रमन ॥७६॥
> स्वल्पयातो बहून भुङ्क्ते भगणांश्छीतदीधितिः ।
> महत्या कक्षया गच्छंस्ततः स्वल्पं शनैश्चरः ॥७७॥

Translation—(75) The magnitude of a planet whose orbit is above, i.e., farther from the Earth, is greater, and the magnitude of a planet whose orbit is below, i.e., nearer to the Earth, is smaller. The fractions of the larger orbit are larger and the fractions of the smaller orbit are smaller. (76) Planets moving on the smaller orbit complete their revolution in a short time, and planets moving on the larger orbit complete their revolution in a long time. (77) The Moon's orbit is very small, so the Moon completes many revolutions, whereas Saturn, being in the larger orbit, completes only a few revolutions.

Scientific Commentary—Some explanation has already been given in Chapter on Mean Motions verses 26, 27, and its Scientific Commentary pages 14-17, so no further elaboration is needed here. The proof of how the fractions of the larger orbit are larger and the fractions of the smaller orbit are smaller can be easily found in the picture 1 of page 14. As much as 1/24 of the larger circle is as much as 1/36 of the smaller circle, meaning one part of the larger circle is greater than one part of the smaller circle. It is also clear that the time period for the revolution of the planets that move in the larger orbit is larger, and the time period for the revolution of the planets that move in the smaller orbit is smaller. However, there is no such simple relationship between the revolution period of the planet and its distance as understood by Indian astrologers and as further explained in this chapter. This relationship is according to Kepler's third law, which is determined by the subtle consideration of the motions and distances of the planets (see pages 84-61).

Method of knowing the Sun, Moon, etc.

> मन्दादधः क्रमेण स्युश्चतुर्था दिवसाधिपः ।
> वर्षाधिपतयस्तद्वत्त तीयाश्चे प्रकीर्तिता ॥७८॥
> ऊर्ध्वं क्रमेण शशिनो मासानमधिपाः स्मृताः ।
> होरेशा सूर्यतनयादधोधः क्रमशस्तथा ॥७६॥

Translation -(78) The fourth planet below Saturn in sequence is the Sun, and the third planet is the Year-Lord. (76) The planets above the Moon are respectively the Month-Lord, and the planets below Saturn are respectively the Hora-Lord.

SOURCE PAGE 795

Scientific Commentary—the full explanation of these two shlokas is given in the Chapter on Mean Motions on pages 40-45, so there is no need to write more here.

The relation between the orbital paths of the constellations, the celestial sphere, and the motions of the planets—

> भवेद्भूकक्षा तिग्मांशोर्भ्रमणं षष्टि ताडितम् ।
> सर्वोपरिष्टाद्भ्रमति योजनैस्तैर्भमण्डलम् ॥८०॥
> कल्पोक्त चन्द्रभगणा गुणिताः शशिकक्षया ।
> आकाशकक्षा सा ज्ञेया कर व्याप्तिस्तथा रवेः ॥८१॥
> सैव यत्कल्पभगणैर्भक्ता तद्भ्रमणं भवेत् ।
> कुवासरैविभज्याह्नः सर्वेषां प्राग्गतिः स्मृता ॥८२॥
> भुक्तियोजनजा संख्या सेन्दोर्भ्रमण संगुणा ।
> स्वकक्षाप्तातु सा तस्य तिथ्याप्ता गति लिप्तिकाः ॥८३॥

Translation—(80) Multiplying the sums of the solar orbits by 60 gives the value of the sums of the constellation orbits. The constellation circle revolves in the same amount above all the planets. (81) Multiplying the sums of the ecliptic by the number of revolutions of the lunar months gives the value of the celestial sphere. The sun's rays travel up to that point. (82) The value of the celestial sphere, when divided by the number of revolutions of the planet, the value of that planet's orbit will be found in the sums. Dividing the celestial sphere by the number of Sawan days of the kalpa gives the daily motion of all the planets in the sums. (83) Multiplying this planimetric planetary motion by the lunar orbit and dividing by the orbit of the planet, and then dividing the result by 15, the daily motion of that planet will be found in the arts.

Scientific Commentary—what is stated in these shlokas has been discussed in many places (see page 15-17; 451-53). In short, the essence of it is:—

(1) Constellation orbit = Solar orbit X 60

(2) Celestial orbit = Lunar revolutions X Celestial orbit (3) Number of revolutions of any planet in the kalpa = Orbit of that planet's celestial orbit (4) Sawan days of the kalpa = Daily planimetric motion of each planet (5) Planetary planimetric motion X Lunar orbit / Planetary orbit X 15 = Daily artistic motion of the planet

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SOURCE PAGE 796

Bhugoladhyaya 771

From the second and third equations it is clear that the extension of the celestial orbit is considered as much as each planet moves in conjunctions in a Kalpa. This proves that our Acharyas understood the planetary motion of each planet to be the same, which is inaccurate according to modern observations. The principle for knowing the daily art motion of a planet is the same which is well explained in pages 451-53.

The orbits of the constellations and the extension of the celestial orbit are hypothetical. There is no limit to the distance of the constellations or stars. Modern observations prove that some stars are so far from the Earth that it takes millions of years for their light to reach us.

Method of knowing the distance of a planet

> कक्ष्या भुकर्णगुणिता महीमण्डलभाजिता ।
>
> तत्कर्णा भूमिकणत्स्यु ग्रहोच्चे स्वे दलीकृताः ॥८४॥

Translation—When the orbit of a planet is multiplied by the geodiameter and divided by the earth's circumference, the diameter of that planet's orbit is obtained. By subtracting the geodiameter from the remainder and halving it, the height or distance of that planet from the earth is found.

Scientific Commentary—This is a rule for knowing the diameter from the circumference. The relationship between the geodiameter and the earth's circumference is the same relationship that exists in the diameters and circumferences of the orbits of each planet. The simple meaning of the first part of this verse is that dividing the planet's orbit by 301416 gives its orbital diameter.

What is stated in the latter part of the shloka is clear from figure 78 of page 408. In this figure, if the line BD is extended to D so that it reaches the lunar orbit and the solar orbit, then the distance from the point of the lunar orbit to D should be understood as the altitude of the Moon and the distance from the point of the solar orbit to D as the altitude of the Sun. Similarly, one should also understand the altitude of other planets.

In the extensions of the planetary orbits

> खत्रयाब्धिद्विदहनाः कक्ष्या तुहिनदीधितेः ।
> ज्ञशीघ्रस्याष्टखद्वित्रिक्रशून्येन्दवस्तथा ॥८५॥
> शुक्रशीघ्रस्य सप्ताग्नि रसाब्धि रसषड्यमाः ।
> ततोऽकंबुधशुक्राणाँ खत्वार्थे कसुरार्णवाः ॥८६॥
> कुजस्यातोऽष्टशून्याङ्कषडवेदैकभुजङ्गमाः ।
> चन्द्रोच्चस्य रसार्थाष्टमुनिद्वित्वष्टह्लयः ॥८७।
> कृतर्तुमुनिपञ्चाद्रिगुणेन्दुविषया गुरोः ।
> स्वर्भानोर्दस्रतत्त्वाब्धिशैलार्थाकाशकुञ्जराः ॥८८॥

SOURCE PAGE 797

> Sukra 8.2242
> Busi 3.2198 पञ्चपञ्चाश्विनागर्तु रसाद्रर्काश्शनेस्ततः ।
> Mangala 25.1448 भानां खखखशून्याङ्कवसुरन्ध्रशराश्विनः ॥८६॥
> Guru 158.5672 खव्योमखत्रयखसागरषट्कनाग-
> Sani 394.0378 व्योमाष्टशून्ययमरूपनगाष्टचन्दाः ।
> Sum 13.3688 ब्रह्माण्डसंपुटपरिस्रमणं समन्ता-
> दभ्यन्तरा दिनकरस्य कर प्रसाराः ॥६०॥

Compared to Sun

Translation—(85) The orbit of the Moon is 324000 yojanas, the orbit of Mercury is 1043206 yojanas; (86) The orbit of Venus is 2664637 yojanas, the orbits of the Sun, Mercury and Venus are 4331500; (87) The orbit of Mars is 8146606 yojanas, the orbit of the Moon is 38328484 yojanas; (88) The orbit of Jupiter is 51375764 yojanas; the orbit of Rahu is 80572864 yojanas; (89) The orbit of Saturn is 127668255 yojanas; the orbit of the constellations is 256860012 yojanas and (90) the circumference of the sky or universe is 18,71,20-80,864,00,000 yojanas to which the rays of the Sun spread.

Scientific Commentary—If the calculations of the orbits are done according to the verses 26-33 of the Chapter on Mean Motions and according to verse 82 of the Chapter on Mean Motions, then there is a

slight difference in the digits of the units of the above numbers. The reason for this is that to write the whole number, the fractional part has either been omitted or taken as 1 because it is more than half. It seems that the orbits of the Moon and Rahu are given to show the similarity of the rule because they are not independent bodies in the sky; they are just two special points of the lunar orbit. The constellation orbit also does not seem to have special importance.

Nowadays, observations prove that the orbits of the planets are not round but elliptical, with the Sun at one focus, and all planets revolve around the Sun. The distance of any planet from the Earth is not always the same, as is clear from the table of latitudes in page 410 and the table of true positions given in page 568. The values of these true positions are given in units where 1000 units are considered the mean distance from the Earth to the Sun. Such 1000 units are equal to 6260000 miles (6 crore 26 lakh miles) because the mean distance from the Earth to the Sun is the same (see page 452). If this distance is to be known in miles, then the miles should be divided by 5 (see page 54).

SOURCE PAGE 798

Thirty Chapters of Astrology Upanishad (Brief Description)

Shloka 1-3—How should the instruction for the creation of Bhugol be given? Shloka 3-13—The method of making Bhugol. Shloka 13-15—Determining the positions of Lagna, Antya, etc. Shloka 16-17—How Bhugol can move by itself. Shloka 18-24—Discussion of other instruments that tell the time. Shloka 25—The greatness of astrology.

Preparation for making Bhugol—

> अथ गुप्ते शुचौ देशे स्नातश्शुचिरलङ्कृतः ।
> संपूज्य भास्करं भक्त्या ग्रहान्मान्यथ गुह्यकान् ॥१॥
> पारंपर्योपदेशेन यथा ज्ञातं गुरोर्मुखात् ।
> आचार्यः शिष्यबोधार्थं सर्वं प्रत्यक्षदर्शिवान् ॥२॥
> भूभगोलकस्य रचनां कुर्यादाश्चर्यकारिणीम् ।

Translation—The teacher should create a composition that causes wonder for the Bhugol by worshipping the Sun, planets, constellations, and yakshas with devotion after bathing and wearing ornaments, in a secluded and pure place, and by the teachings received from tradition, and the knowledge heard from the teacher's mouth and seen directly by oneself, to fully explain to the disciple.

Scientific Commentary—Many commentators have taken the meaning of Acharya as Suryansh Purush and the meaning of disciple as Mayasur; but in my understanding, this is a simple instruction for all Acharyas. The word 'Kuryat' also reveals this. From these shlokas it appears that the Acharya should not be satisfied with oral instruction alone, but should also impart practical and action-oriented knowledge for which he should also possess practical knowledge and should be the one who directly shows the knowledge, not as if he were to comment on such esoteric texts as Surya Siddhanta, but not even recognize the main stars of the sky.

In the worship of the planets, the worship of the Sun also comes in, but here the word Sun appears separately with the planets, which indicates that the Sun in a special way

SOURCE PAGE 799

Worship should be done because the original Acharya of this principle is considered to be Suryadev. The Sun does not come in the modern definition of planets, but the Sun-Siddhantaka did not separate the name of the Sun from this idea because this opinion is not revealed anywhere else.

The Yaksha people are servants of the god of wealth Kubera, they guard his treasury and gardens. They are probably considered skilled in craftsmanship as well because the Pushpak Vimana belonged to Kubera. Therefore, they are also ordered to be worshipped to obtain divine help in the meaning of Yantra construction.

The word Bhugol is made from three words: Bhu, Bha, and Gol, so its meaning is a sphere in which the sky is the sphere with which the geography moves, in which planets and constellations etc are considered to revolve.

Method of making geography-

> अभीष्टं पृथिवीगोलं कारयित्वा तु दारवम् ॥३॥
> दण्डं तन्मध्यगं मेरोरुभयत्र विनिर्गतम् ।
> आधारकक्ष्याद्वितयं कक्ष्यां वैषुवतीं तथा ॥४॥
> भगणांशाङ्गुलैः कार्या दलितास्तिस्र एव ताः ।
> स्वाहोरात्रार्धकर्णेश्च तत्प्रमाणानुपाततः ।।५।।
> क्रान्तिविक्षेपभागैश्च दलिता दक्षिणोत्तरा ।
> स्वैस्स्वैरपक्रमैः कार्या मेषादीनामंपक्रमात् ॥६॥
> कक्ष्याः प्रकल्पयेत्ताश्च कर्य्यादीनां विपर्ययात् ।
> तद्वत्तिस्त्रस्तुलादीनां मृगादीनां विलोमतः ॥७॥
> याम्यगोलाशिताः कुर्यात् कक्ष्याधारद्वयोपरि ।
> याम्योदग्भागसंस्थानां भानामभिजितस्तथा ।।८।।
> सप्तर्षीणामगस्त्यस्य ब्रह्मादीनां प्रकल्पयेत् ।
> मध्ये वैषुवती कक्ष्या सर्वासामेव संस्थिता ।।९।।
> तदाधारयुतेः भार्धमयने विषुवद्वये ।
> विषुवत्स्थानतो भागैः स्फुटैर्भगणसंचरात् ।।१०।।
> क्षेत्राण्येवमजादीनां तिर्यग्ज्याभिः प्रकल्पयेत ।
> अयनादयनं चैव कक्ष्या तिर्यक्तथाऽपरा ।।११।।
> क्रान्तिसंज्ञा तया सूर्यः सदा पर्येति मासयन् ।
> चन्द्राद्याश्च स्वकैः पातैरपमण्डलमाश्रितैः ।।१२।।
> ततोऽपकृष्टा दृश्यन्ते विक्षेपाग्रेष्वपक्रमात् ।

SOURCE PAGE 800

Jyotishopanishad Chapter 775

Translation—(3) A straight rod should be made by making a circle (geography) of the desired shape of wood (4) in it by making a hole so that it remains equal on both sides from the center of geography and acts as the Meru rod. Fix two base circles in this rod (at right angles to each other) whose midpoint is the ecliptic. (5) Divide these three circles into 360 degrees by an inch. In accordance with the diameter of the equator circle (6) of the Ahoratra circle, the differences in the inclinations of the final points of

the Aries, Taurus, and Gemini signs marked on it by the degrees of the ecliptic and Sagittarius, these three signs' Ahoratra circles are fixed in the opposite manner as the Ahoratra circles of Cancer, Leo, and Virgo. Similarly, three Ahoratra circles (8) of the Libra, Scorpio, and Sagittarius and in the opposite manner of Capricorn, Aquarius, and Pisces signs are fixed in the South Celestial Sphere above both base circles. Similarly, the Ahoratra circles of the constellations, stars like Abhijit (9) Saptarshi, Agatsya, Brahmhridya located in the North and South Celestial Spheres are fixed. The equator circle is between all these Ahoratra circles. (10) Both the arc points and both the latitudes are at the conjunction points of the equator circle and both base circles. Start the zodiac cycle from the place of the latitude. (11) In this way, the divisions of signs like Aries, Taurus, etc., are done by the obliquity of the sines. The line of inclination which is fixed from one arc point to another arc point and from the second to the first (2) is called the ecliptic on which the sun always travels shining. The planets like the Moon, Mars, etc., appear at the end of their inclinations by their own latitudes (13) due to their own inclinations.

Sūrya Siddhānta - here it is not stated what the basis of the base orbit and the equatorial orbit should be made of. In other texts, thin rods of bamboo have been used because they are so flexible that they can be bent in a circle. Nowadays, this work can be easily done with iron or brass wire. The geoid formed by the method mentioned above becomes very incomprehensible due to many circles (circles), so nowadays, if the geoid is made according to figure 136, it will be easier to make and easier to understand.

The smallest circle in the middle of this picture indicates geography (Earth-sphere), which is why 'Bhū' is written in the middle. 'Dhadhā' is the line that passes through the center of the Earth-sphere and extends on both sides of it. The distance from Bhū to Dh and Dha is equal. Two meridians, 'Dhashdhāv' and 'Dhapdhāpū', are at right angles to each other from these places.

SOURCE PAGE 801

They are tied. On these two circles, a circle of 'P V P' is tied at the same distance from 'D' and 'DH', which is called the equatorial orbit. These three circles are divided into 360 equal parts, and marks are made on them, which are called degrees. After this, the order is to make the circumpolar circles of the twelve zodiac signs, such as Aries. But in my understanding, this is not necessary. After tying the three upper circles, it will be easier to tie the circumpolar circle directly. This should also be taken as equal to any of the upper circles. It should be tied at the junction of the equatorial circle and the meridian circle at the 'V' and 'SH' places first, and then tied at the second meridian circles at the 'K' and 'M' places. The difference between 'K' or 'M' should be the same as the ultimate revolution of the sun, which is nowadays approximately twenty-three and thirty-plus degrees (23° 30’). This revolution circle should also be divided into 360 equal parts. The 'V' place is called the vernal equinox or spring equinox, and the 'SH' place is called the autumnal equinox. The 'K' and 'M' places are called the sine of Cancer and the sine of Libra or the points of South and North transit respectively (see p. 230). With regard to these places, the 'DH' base orbit is called the equinoctial circle and the 'DPMDHPUK' base orbit is called the solstitial circle, because the first has both equinoctial points and the second has both solstitial points. In the picture, a circumpolar circle 'AI' is shown parallel to the equatorial circle at the 'A' place. This is found at the second place 'A' of the circumpolar circle. The point 'A' is at the same difference as the vernal equinox from 'V', but

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SOURCE PAGE 802

Sūrya Siddhānta

In the opposite direction, the position of 'sha' in the autumn conjunction is either 'sha' from 'aa' or 'k' from 'a' and 'aa' are at the same distance. If 'a' sine is at the beginning of the Taurus sign, then 'aa' sine will be at the beginning of the Virgo sign, and if the first sine is at the beginning of Gemini, then the second sine will be at the beginning of Leo. Similarly, an apothem circle of any point of the ecliptic can be drawn. This is stated in shlokas 6-7. Besides this, an apothem circle can be understood near the position of 'dha'.

In the 12th shloka, the orbits of the classes of the planets like the Moon, Mars, etc., are also discussed, but no such orbit is shown in the picture. To draw such an orbit, the nodal point of the first planet should be fixed on the ecliptic. The orbit of that planet should be drawn such that its second conjunction is at 180 degrees on the ecliptic from that position. These two places are the nodal positions of the planet. Then this circle should also be drawn at the north and south positions equal to the ultimate deviation of that planet with a difference of 60 degrees from the ecliptic (the discussion of ultimate deviation is in Chapter on Mean Motions, pages 74-76).

Determination of rising sign, mean sign, setting, etc.

> उदयं क्षितिजे लग्नमस्तं गच्छति तद्वशात् ॥१३॥
> लङ्कोदयैस्तथा सिद्धं खमध्योपरि मध्यगम् ।
> मध्यक्षितिजयोर्मध्ये या ज्या साऽन्त्याऽभिधीयते ॥१४॥
> ज्ञेया चरदलज्या च विषुवत्क्षितिजान्तरम् ।
> कृत्वोपरि स्वकं स्थानं मध्ये क्षितिजमण्डलम् ॥१५॥

Translation—(13) The point of the ecliptic that lies in the eastern horizon is the rising sign. According to this rising sign, the point of the ecliptic that lies in the western horizon is the setting sign. (14) The mean sign is on the meridian circle, which is calculated from the rising ascensions of Lanka. The sine that lies between the apothem circle and the meridian circle is called setting. (15) The horizon of the tropics, which is called the circumcircle, and the difference between the horizon of one's place and the horizon of its place is called latitude. When placing one's place at the top of geography, the horizon circle is in the middle of geography.

Scientific Commentary—the same word is used for several definitions in these verses, therefore their meaning is not easily understood. In the first half of the 12th verse, 'madhyam' has come for the mean ascendant. In the latter half, the word 'madhya'

The circle and the arc of the meridian have come for the place of the location. In the first half of the 15th verse, 'vishuvat' has been used for the horizon or the meridian for the celestial sphere, and the word horizon has been used for the circle of the horizon in its place. In the latter half, the word 'madhye' has come for the middle of the vertical meridian circle in its place. This type of usage is very misleading and is considered a defect for scientific texts.

All these definitions can be well understood on the 286-60 pages of the Chapter on the Three Problems and the figures 36, 42, 43 of the Chapter on True Positions and the figures 63 of the Chapter on the Three Problems. With the above description, knowledge of all these definitions can be easily obtained from the geodetic instrument.

The device by which the geodetic instrument always rotates—

> वस्त्रच्छन्नं बहिश्चापि लोकालोकेन वेष्टितम् ।
> अमृतस्रावयोगेन कालभ्रमणसाधनम् ॥१६॥
> गुणबोजसमाकृष्ट गोलयंत्र प्रकल्पयेत् ।
> गोप्यमेतत्प्रकाश्योक्त सर्वगम्यं भवेद्यतः ॥१७॥

Translation—(16) Make a sphere surrounded by the horizon, i.e., the horizon circle, cover it with cloth and arrange it by the flow of water so that it rotates by itself and indicates the time of the stars. (17) Or make this spherical instrument by the combination of mercury so that it rotates by itself. It should be kept secret. If it is stated clearly, then everyone in this world will know it.

Scientific Commentary—the language of these two shlokas is very obscure. There is not even a little understanding of how the armillary sphere proved the daily motion of the sky by revolving by itself. It is apparent that the meridian of the armillary sphere was fixed in such a way that it remained towards the pole. Then some device must have been made in it so that a collision from the flow of water occurred in it, so that it would revolve once in a nakṣatra day like a water wheel. Mercury was also used instead of water, but it is not known how. Finally, it is stated that this device should not be false at all. Perhaps this is why it is given only as a hint. From this people can deduce that the author himself did not know this action well. He only had the impression that such a machine could be made that moves by itself, that is why he kept all things round. It is also possible that in ancient times there was such advancement in craftsmanship that such self-moving instruments were made by the combination of mercury and water.

SOURCE PAGE 804

Jyotishopanishad Chapter 776

> से उसी प्रकार बनते थे कि जैसे आजकल घड़ी आदि अपने आप चलने वाले यन्त्र
> बनते हैं, परन्तु बीच में समय के फेर से सब ज्ञान नष्ट हो गया हो ।

> तस्माद्गुरूपदेशेन रचयेद्गोलमुत्तमम् ।
> युगे युगे समुत्पन्ना रचनेयं विवस्वतः ॥१८॥
> प्रसादात्कस्यचिद्भूयः प्रादुर्भवति कालतः ।
> कालसंसाधनार्थाय तथा यन्त्राणि कारयेत् ॥१६॥
> एकाकी योजयेद्बीजं यन्त्रे विस्मयकारणम् ।

Translation—(18) Therefore, according to the teachings of the Guru, the construction of the excellent circle should be done. This construction is destroyed in every age and is obtained again by the grace of the Sun God as per his will. Similarly, other instruments should also be constructed to know the time. (20) In the instrument that causes wonder (to operate it), mercury should be used in isolation.

Scientific Commentary—These verses reveal that the act of making self-sustaining instruments is destroyed when time is obtained, which is then told to someone by the Sun God as per his will. Besides this, it is also said to make other instruments that tell time, and finally it is stated that mercury should be used in isolation, not telling everyone.

Names of other instruments that tell time—

> शङ्कुयष्टिधनुश्चक्रैश्छाया यन्त्रैरनेकधा ॥२०॥
> गुरूपदेशाद्विज्ञाय कालज्ञानमतन्द्रितः ।
> तोययन्त्रैः कपालाख्यैर्मयूरनरवानरैः ।
> सूत्रैश्च वेणुगर्भस्थैः सम्यक्कालं प्रसाधयेत् ॥२१॥
> पारतालाबुसूत्राणि गुञ्जातैलजलानि च ।
> बीजानि पांसवश्चैषां प्रयोगास्ते सुदुर्लभाः ॥२२॥

Translation—(20) By means of many types of shadow instruments named gnomon, yasti, dhanu, and chakra (21), clever and diligent men obtain knowledge of time from the teachings of the Guru. From skull-like water instruments, from peacock, man, and monkey instruments in which sand remains in the belly which comes out with the help of thread, the exact knowledge of time is obtained

SOURCE PAGE 805

should be done. (22) Use of mercury, water, thread, plumb, oil and water, mercury and sand, all these things should be used. But this is also difficult.

Scientific Commentary—many names of instruments for knowing time are mentioned in these verses, but the method of making them is not told anywhere. Since time can be known by observing the shadow of the sun using the first four instruments, they can work during the day and are called shadow instruments. The discussion of the gnomon among these is greatly discussed in the third chapter named Chapter on the Three Problems. It is stated in that chapter how the latitude, declination, local time, direction, etc. are known from a 12-finger gnomon. To know the time by this, a complex operation of multiplication and division has to be done; time does not come directly. The measurement of the shadow also cannot be taken very accurately, so the time obtained from this instrument can be four or five minutes more or less. Therefore, there is no need to use it nowadays.

Yasti instrument—it is not discussed anywhere else in this book, so it cannot be told how it was used. Bhaskaracharya ji has specially discussed it in the instrument chapter of Siddhanta Shiromani, which shows that even with this, complex calculations are needed to know the time. Therefore, even though good means are available nowadays, this instrument is not necessary.

Knowledge of time can be easily obtained from the dhanuṣa and cakra-yantra. These two instruments are actually the same. The cakra-yantra has a round circle whose edges are marked in equal parts. If there are 60 equal parts, each part indicates the time of a clock. If a straight spike of iron or brass is fixed from the center of this circle and the circle is fixed on the earth in two pillars in such a way that the head of the spike is in the direction of the pole of the sky, then the time of the sun (Hour Angle) can be easily known from this instrument. In the মান-mandiras of Delhi and Kashi, etc., circular instruments of stone have been made which are fixed on the earth in such a way that their planes are parallel to the equator and their center is about 6 inches in the direction of the poles extending on both sides of the spike. The edges on both sides of this circle-instrument are marked in equal parts. When the sun is in the northern hemisphere (from the vernal equinox to the autumnal equinox), the shadow of the spike falls on the northern plane of the instrument and tells the Hour Angle, and when the sun is in the southern hemisphere, the shadow of the spike falls on the southern plane and indicates the Hour Angle. As the sun rises, the shadow

SOURCE PAGE 806

It is below. At noon, the shadow of the spike falls exactly below. If desired, its edges can also be marked in hours. This is also a type of sundial.

Bhaskaracharya has described another type of wheel-instrument. It is also circular but is not fixed. There is a hole some distance from the edge in which a chain is attached. It can be hung with this chain. Its edges are marked in 360 degrees. When hung, its center remains directly below the hole. The line perpendicular to the line joining these two points, and which also remains at the center of the wheel, has a mark of zero on one edge and 180 on the other. When time is to be known, hang it and rotate it so that the shadow of the tip of the notch fixed at its center falls on the edge parts. If its edges are also divided in the rising points of the auspicious positions of the signs of the sine, then the knowledge of the ascendant can also be obtained from it. There is no special advantage from this instrument in front of modern instruments.

Arc or bow instrument—if half of the wheel-instrument is taken, then the time of the sun can be known in the same way.

Nowadays, the shadow of the sun at any time of the day can easily be known by a simple arc instrument made of cardboard, and the time can also be known from it. This author has discussed such an instrument in Science Part 43 number 1 page 138 in Figure 138. Many tables are also given in it from which the time of day can be known in places from 28 latitudes to 22 latitudes. Time is known according to the shadow of the sun with this instrument, and there is no more than four or five minutes of difference. When the author's clock stops or does not give the exact time, he matches it with this. The latitude of a place can also be easily known from this instrument. This table is prepared according to the Gur (1) of page 261 of Chapter on the Three Problems. The brief method of using it is given at the end of this chapter where a sundial is also discussed.

Besides these shadow instruments, work was also done with water-clocks and sand-clocks, etc. The water-clock was called the skull instrument because it was half-spherical like the skull and had a small hole in it. If it were floated in water, water would slowly fill the bowl through the hole and fill it so much that the bowl sinks. The shape of this bowl and the hole was such

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that the sun sinks and rises 60 times in a day and night. The time it took to sink once was called a clock. For this reason, it got the name ghatiyantra, which later became famous as the clock. Rules were made in ancient times to make such instruments from which it was clear how large the skull or bowl should be and what the hole should be. This is specially described in the 23rd verse of the Vigyan Bhashya.

There is no detailed writing about the Mayur and Vanar instruments. These were probably instruments made in the shape of a peacock and a monkey, in which time was measured with the help of thread and sand. In these instruments, a device was done by means of mercury, oil, water, etc., so that they would move by themselves and indicate time, but there is no clear description of how they were made; it is only indicated that their use is very rare.

It is unknown whether such self-moving instruments were actually made or were only in imagination. Nowadays, all kinds of self-moving clocks can be used in work, but there is no need to explain their principle here.

Skull instrument—

> ताम्रपात्रमघश्छिद्रं न्यस्तं कुम्भेऽमलाम्भसि ।
> षष्टिर्मज्जत्यहोरात्रे स्फुटं यन्त्रं कपालकम् ॥२३॥

Translation—When a copper bowl with holes is placed in a pool of pure water, it sinks and rises 60 times in a day and night, then it is a pure skull instrument.

Vigyan Bhashya—Acharya Shripati gave a special description of such a skull instrument in Siddhanta Shekhar as follows:-

The six-sided figure is made by doubling the arc. With this, a vessel is filled, which is half the size of the pot. The water is contained in it.

That is to say, taking ten *pal* of copper, a hemispherical bowl is made whose height is 6 *angulas* and whose mouth width is double that, in which 60 *pal* of water enters, and in which there should be a hole large enough for 33 *masha* of gold thread to pass through, so that in one hour, that bowl is filled with water.

From this it is proven that this Acharya is for the pure measurement of time

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How much effort it was. But making such an instrument was not easy. For this reason, Bhaskaracharya 1 has neglected it.

Narayantra—

> नरयन्त्रं तथा साधु विवा च विसले रवौ ।
>
> छाया संसाधनैः प्रोक्तं कालसाधनमुत्तमम् ॥२४॥

Translation—In this way, Narayantra or gnomon is better when the sun is clear during the day. The method of knowing the time exactly from the exact measurement of the shadow is explained.

Vigyan-bhashya—In the Chapter on the Three Problems, it is explained in detail how the direction, country, and time are calculated from the shadow of a gnomon of 12 angulas.

Conclusion—

> ग्रहनक्षत्रचरितं ज्ञात्वा गोलं च तत्त्वतः ।
>
> ग्रहलोकमवाप्नोति पर्यायेणात्मवान् नरः ॥२५॥

Thus Surya Siddhanta Jyotishopanishad Chapter

Translation—The person who knows the motion of planets and constellations and the elements of spherical mathematics attains the planet realm and is self-knowing in other lives.

Scientific Commentary—From this verse it is revealed how much our teachers understood the elements of astrology and what kinds of instruments they constructed to bring direct knowledge of it. Nowadays, there has been so much advancement in instruments that the motion of celestial bodies can be

measured with great subtlety by them. In this, a brief discussion of such an instrument is made in the chapter. Those who wish to know about them in a special manner should read the books of light-science.

Telescope—From this instrument, the reflection of distant objects appears very large and clear. Its principle is briefly:—

On one end of a brass tube there is such a groove on which the reflection of a distant object is formed on its navel or near it. Near this reflection

1. How is the deduction made when the characteristics of ten lines are reduced? (Explanation of verse 8 of the Yantra chapter in the Gola chapter).

SOURCE PAGE 809

There is a small scale at the other end from which this reflection appears larger. This small scale is fixed at the end of another tube, which can move in the larger tube. The larger scale is towards the object, so it is called 'object scale', and the small scale is towards the observer's eye, so it is called 'eye scale'. The diameter of the object scale of the rangefinder of Mount Wilson is 100 inches or 8 feet 4 inches. A scale with a diameter of 200 inches is being made every ten years. Its thickness is 25 inches and its weight is something above 33000 pounds or 402 min. The reflection formed by this scale can be magnified ten lakh times. The moon is 25000 miles away from here, but when viewed through this scale, it will seem as if it is at a distance of only 20 miles. The length of the tube in which this scale will be fixed is 55 feet and its weight is 250000 pounds or 3046 min.1.

If this instrument is fixed in such a way that it rotates in the plane of the meridian or the meridian circle in the east-west direction, then its name becomes the meridian instrument. From this, the time of meridian passage, declination, and transit of any planet or star can be measured. From figure 137, the method of fixing this instrument can be roughly known. There are two circles at both ends of the 'pūrva-paścima' axis on whose edges marks of aṃśoṃ and kalāoṃ are inscribed. When the distant observer holds the circle in the line of a planet or star, the amount by which the circle rotates is the same as the planet's

Nataṁśa is known. The latitude of the place is always known. With the help of these two, the revolution of that planet can be easily known.

SOURCE PAGE 810

Sūrya Siddhānta Chapter 785

SOURCE PAGE 811

OCR table: month, date, sunrise (hours and minutes), and time corrections. The column boundaries are unclear in the supplied OCR. Numerical entries follow in their source order: 1 2 3 4 5 6 7 8 9 ma 5 11|| 31|| 34||| 43 44 66. 12|| 10 10|| 26|| 32||| 41||| 52||| 65. 78. 8 16 21|| 27|| 26|| 31 40 51|| 78. 9 21 27|| 25|| 26. 38||| 50|| 63. 76|| 10||| 26 5||| 23|| 27|| 46. 62. 75|| 86 11|| 31 4|| 21|| 25||| 36 48 61. 74||| 12 5 2||| 16|| 24. 34||| 47. 60|| 74||| 13 10 11|| 17|| 22|| 33|| 46. 56|| 73 14 16 0 15|| 21 21 32|| 45|| 72. 15 22 -1|| 13|| 16|| 31|| 44|| 58 71|| 16 28 2|| 11|| 16 30|| 43|| 57 70||| 17 4 3. 6|| 16|| 26|| 43 56|| 70 18 12 7|| 15|| 15|| 28|| 42. 55||| 66. 19 20 5|| 14|| 14|| 28|| 40||| 54||| 67||| 20 1 2|| 3||..... 21 2 +1|| 2|| 13|| 27. 41. 54|| 50||| 22

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Looking at the table will show that where 12 is written indicates the meridian time. In the next column, 1 is written below and 11 is written above. This means that the portion of the sun is as much as 11 o'clock as it is as 1 o'clock. However, the meridian portion does not differ much from these. In January and December, this difference is not more than two and a quarter degrees. In February and October, it becomes up to 2 degrees. In March and September, it becomes approximately 4. In April and August, it becomes more than 6 degrees. Therefore, this time from the sun's hour cannot be taken very accurately from 1 to 1 o'clock in the winter days.

Time from 1 to 2 o'clock or from 10 to 11 o'clock can be taken easily. Additionally, the knowledge of time in other times can be known with subtlety.

Time from the sun's hour is the pure local time. The time known from the star house clock is different from that. To know the time of the star house clock from the local time, two calculations in local time are required. One is called the time equation and the other is called the longitude calculation. The time equation is positive from January 16 to April 14, after which it remains negative until June 14. After June 14, it becomes positive again and remains so until August. From September to December, it is usually negative. When it is positive, it has to be added to the time of the sun's hour, and when it is negative, it has to be subtracted. Doing this calculation makes the pure local time equal to the mean local time.

Longitude calculation—After knowing the mean-local time, if the position is 1 degree before the longitude line by 82° degrees, then 4 minutes, if 2 degrees before, then eight minutes, and if 10 degrees before, then 40 minutes must be subtracted. But if the position is west of the longitude line by 82° degrees, then it has to be added accordingly. Doing this gives the time which is the time of the telegraph or railway clock.

Sunrise and sunset time—The sunrise time is also given in hours and minutes in the table. If the sunrise time is subtracted from 12 hours, the sunset time will come. This is the time when the center of the sun's image is on the horizon

According to mathematics, it should be. But in reality, due to the refraction of light, the image of the sun appears even while remaining below the horizon. Due to this motion, sunrise occurs approximately 2 minutes before the given time and sunset 2 minutes after.

The image of the sun is not like a point, therefore its upper edge often rises one minute before and sets one minute after. Therefore, by subtracting 3 minutes from the sunrise time, the time will come when the upper edge of the sun's image begins to be seen. Similarly, by adding 3 minutes in the sunset time, the time will come when the entire image of the sun disappears. But this time is the apparent time of space. To know the time of a railway clock, the time equation and the longitude calculation should also be done. Looking at the equation shows that the sunrise time is the same for two dates. For example, on January 10th and December 3rd, the sunrise time is 6 hours and 44 minutes in Allahabad or in all the places of 25 degrees latitude. However, in these dates, a difference in the railway clock time for the sunrise time is seen. The reason is clear. On December 3rd, 10 minutes must be subtracted from the time equation. And on January 10th, 7 minutes must be added. The other calculations are the same in

both. For example, the sunrise time of these two dates as it comes from the railway clock is shown below —

3 December 10 January clear sunrise time 6 hours 44 minutes 6 hours 44 minutes ritual- correction — 2॥ minutes —2॥ time equation ritual —10.” +7॥ country difference ritual (for Allahabad) + 2” + 2 train clock sunrise time 6 hours 33.min 6 hours 51 min

If one wants to know the time of rising of the upper edge of the sun's image, then one should subtract 1 minute. In these dates, by subtracting the true sunrise time of sunset from the time of sunset using a railway clock, it is 5 hours 16 minutes. In this, the motion, time equation, and longitude-version should be done in this manner.

3 December 10 January clear sunset 5 hours 16 minutes 5 hours 16 minutes motion-process +2॥ min +2॥ min

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Sūrya Siddhānta Chapter 786

Time-equation - 10” + 7॥ M. Desantar + 2 M. + 2 " Railway clock time 5 hours 10 minutes 5 hours 28 minutes

Note—The calculation of the mathematical principles should be subtracted in the sunrise time and added in the sunset time.

Knowing the time by measuring the portion of the Sun—

Example 1—On February 17, the portion of the Sun before noon is 50. What is the time? The portion at 10 o'clock is 47।।। And at 6 o'clock is 57।।। Therefore, the portion of the Sun between 6 and 10 o'clock will be 50. It is also apparent that the portion decreases by 10 degrees between 6 and 10 o'clock. Therefore, in this hour, the portion is decreasing at a rate of 10 degrees per hour, which means it decreases by 1 degree in 6 minutes. There are 7।।। degrees from 57।।। to 50. Therefore, the decrease of 7।।। degrees is in 6 X 7।।। minutes or 46।। mi0. Therefore, the true local time is 6 hours 46 minutes. On this day, the time-equation is + 14 minutes. Therefore, upon giving this calculation, the mean local time is 6 hours 46 minutes + 14 minutes = 10 hours 1 minute. If the place is Allahabad, then 2 minutes should be added. In this way, the railway time is 10 hours 3 minutes. If the place is Kashi, then 2 minutes should be subtracted from ten hours 1 minute, because Kashi is 2 minutes earlier. Therefore, at the time when the portion of the Sun is 50 in Kashi on this date, it would be 6 hours 56 minutes.

Example 2—On March 23, the nataṃśa of the Sun after noon in Patna city is 74 degrees. What time does the railway clock show at this time?

There is no 23 March in the table. It gives the nataṃśa and natakaal for the 21st and 26th of March. For 21 March, the nataṃśa at 4 o'clock is 63. And the nataṃśa at 5 o'clock is 76. For 26 March, the nataṃśa at 4 o'clock is 62. And the nataṃśa at 4 o'clock is 75. In 5 days, the nataṃśa at 4 o'clock decreases by 1 degree and the nataṃśa at 4 o'clock decreases by a quarter degree. Therefore, in 2 days, the nataṃśa at 4 o'clock will decrease by almost half a degree and the nataṃśa at 5 o'clock will decrease by almost a quarter degree. Therefore, for 23th, the nataṃśa at 4 o'clock will be 62. And the nataṃśa at 5 o'clock will be 76. The difference between these two is 13 degrees. This increase happens in 1 hour. Therefore, the speed of increase of the nataṃśa is approximately 4 minutes per degree. But the time of the 74 nataṃśa needs to be known.

SOURCE PAGE 815

If there is a deficit of 2. degrees from 76. to 2., it will be completed in about 10 minutes at the rate of 2. degrees per minute. Therefore, the local true time is 5 o'clock plus 10 minutes, which means it is 4 o'clock 50 minutes. This is the time of the sun-dial.

One should see what the time-equation for this day is. The time-equation for March 21st is 7. minutes and the 26th is 5.. Therefore, in 5 days, there is a deficit of 1.. minute in the time-equation, and in 2 days, there is a quarter minute. Therefore, on March 23rd, the time-equation is 6.. minutes. This is positive, so adding it results in the local mean time of 4 o'clock 56.. minutes or 4 o'clock 57 minutes.

The longitude of Patna is approximately 85 degrees 30 minutes from Greenwich, which is 3 degrees west of the prime meridian of India-Year 820 30'. Therefore, the longitude of Patna is 12 minutes before. Subtracting 12 minutes from the above time gives 4 hours 45 minutes. This is the time of the rail-clock.

To know the latitude of a place—know the meridian altitude (12 o'clock) of the desired day from a table. Then look in the same table what is the meridian altitude of March 21st. Know the difference between both. This is the precession of the sun on that day. Now look at the meridian altitude of your place from the meridian reflection. If the precession is north, adding it to this altitude and if it is south, subtracting it will give the latitude of that place.

Example—The meridian altitude of Raebareli on February 17th is 38.. In the table, the meridian altitude of February 17th is 37॥, and that of March 21st is 25; the difference between these two altitudes is 12. Therefore, the precession of the sun on this day is 120 South. Subtracting this precession from 38॥ gives 26॥, which is the latitude of Raebareli. This is 26. in reality. Therefore, there is an error of a quarter degree in this.

Maharaja Sawai Jai Singh II of Jaipur built observatories in Delhi, then Jaipur, Mathura, Kashi, and Ujjain in the first phase of the 18th century or the fourth phase of the same century of Vikram to properly study the science of astrology. In each observatory, seven or eight instruments are usually of the same type but different in size, which are still visible. Their names are:--

1- Samrat Yantra, 2- Shashṭāṁśa Yantra, 3-Rāśivalaya Yantra, 4-Jaiprakash

SOURCE PAGE 816

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Yantra, 5- Kapala Yantra, 6- Rama Yantra, 7- Digansh Yantra, 8- Nadivalaya Yantra, 9- Dakshnottarbhitti Yantra, 10- Unnatansh Yantra, 11- Chakra Yantra, 12- Krantivritta Yantra

There is no need to discuss these instruments in detail here. From these, it appears how much effort and how much money our kings and emperors spent in the ancient times for astrological discoveries. This chapter will be concluded by writing down the method for making a simple sundial. For such a sundial, a sheet of brass with a width of about 3 inches, a length of 15 to 20 inches, and a width of ten or twelve inches will suffice. From this sheet, a right-angled triangle should be cut such that the angle formed on the longer side is equal to the latitude of the place where the sundial is to be installed. A similar right-angled triangle will also be formed in the remaining sheet. Suppose there is a square piece of brass sheet, let's call it T. From this, a right-angled triangle is cut such that the angle of T is equal to the latitude of Prayag, 25° 25'. The remaining piece is T, in which a line parallel to T is drawn, and the

angle U of this line will also be equal to 25° 25'. Now, from this piece, one should see what kind of circular segment can be cut from it. The center of such a circular segment should be determined, and two parallel lines should be drawn near it, whose distance between them is equal to the width of the sheet and which are perpendicular to the side of T. Then, drawing two equal arcs with the tips of each line as the center. The angle at the center of these arcs should not be less than 60 degrees and not more than 105 degrees because the day length in Prayag is not more than 14 hours. The edges of this circular segment should first be divided at intervals of fifteen-fifteen marks, and then these parts should be divided into three or four equal parts (only three parts are shown in the figure). The parts with fifteen-fifteen marks will indicate the hour, and if there are three parts in each hour, then from each one, twenty minutes can be known, and if there are four parts, then fifteen minutes can be known from each. Smaller parts are not shown in the figure, but in reality, three or four equal parts can also be made from every fourth part, from which five minutes can be known. Such sundials have been installed in the government high schools of Ballia and Raebareli. The sundial of Ballia was prepared under the supervision of a student of the ninth class by this author, but the sundial of Raebareli was made by a mechanic.

SOURCE PAGE 817

If you look carefully at Figure 136, you will understand these things. You should cut out the half part between two parallel radii of the arc and take it towards the circumference, and draw two parallel lines making a right angle with the base and the ground in such a way that the distance between them is like the fold of a sheet. You should cut out the part between these lines whose length is equal to the part of the arc revealed by the cut lines. Now, you should join this arc to the base and ground in such a way that the center of the arc comes to the base and ground, and its open part rests on the part of the triangle revealed by the cut lines. Doing this, the shape of these two will be the same as the part of Figure 140 shown on the pillar.

You should install this sundial in a place where the sun remains throughout the day; it is good if the shade of a house or tree is less. Construct a solid pillar four and a half feet high and fix this on it in such a way that its top part is below the base of the pillar, on the true north-south line, and the base of the triangle is perfectly vertical, not tilted to the east or west. In this condition, the base will remain in the direction of the celestial pole. This pillar should be firmly fixed, so it will be good to make holes in C, G, places and fix two iron rods, each one foot long, in them, with their other ends inclined at two inches. By these, this sundial will be fixed in the pillar to a depth of one and a half feet (see Figure 140).

SOURCE PAGE 818

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The time of this moment is also the same as local time. To know the railway time, the time equation and deśāṃtara correction should be done in the same way as previously stated.

Thus, the Scientific Commentary of the chapter named Jyotishopanishad, which is the 13th chapter, has ended.

SOURCE PAGE 819

Fourteenth Chapter Mean Chapter (Brief Description)

[Verse 1-2—Number and behavior of Manus. Verse 3-11—Ahoratra, Shadshitimukha, Ayana, viṣuva conjunctions from solar values; puṇyakāla and season calculation of conjunctions. Verse 12-14—Determination of Tithi, Karan, Vivaha etc., of lunar months from Chandramaan. Verse 15-16—Names of lunar months from Nakshatras and Nakshatras. Verse 17 — Names of years of Jupiter. Verse 18-19—Determination of Yajna period and Sutaka etc. from Sawan days. Verse 20-21—Divya, Prajapatya and Brahmaman. Verse 22-27—Conclusion. At the end, there are 21 verses of Bijo- Panayana chapter which are considered Kshepaka.

Names of Nine Values

> ब्राह्मं पित्र्यं तथा दिव्यं प्राजापत्यं च गौरवम् ।
>
> सौरं च सावनं चान्द्रमार्क्षं मानानि वै नव ॥१॥

Translation—Brahma, Divya, Pitya, Prajapatya, Varahaspatya, Solar, Sawan, Chandra and Nakshatra are nine time-values.

Scientific Commentary—The explanation of these words is given in the following verses, so there is no need to say more at this time.

Values in Practice

> चतुर्भिव्यवहारोऽत्र सौरचान्द्रार्क्ष सावनै ।
> बार्हस्पत्येन षष्ट्यब्दा ज्ञेया नान्यैस्तु नित्यशः ॥२॥

Translation—This verse involves the measurements of solar, lunar, stellar, and monsoon values. The calculation of sixty centuries is done with the value of Jupiter, the other four values do not have to be fixed.

Scientific Commentary—The discussion of these five values has also come in the Chapter on Mean Motions.

SOURCE PAGE 820

Chapter 765

Sormān

> सौरेण द्युतिशौर्मानं बडशीतिसुखानि च ।
> अयनं विषुवच्चैव सङ्क्रान्तेः पुण्यकालता ॥३॥

Translation—The period of day and night, sextile faces, Uttarayana and Dakshina-yana, solstice, and other solstices are determined from Sormān.

Sextile faces

> तुलादेः षडशीत्यंशैः षडशीतिमुखं दिनम् ।
> भचतुष्टयमेवं स्याद् द्विस्वभावेषु राशिषु ॥४॥
> षड्विंशे धनुषो भागे द्वाविंशेऽतिमिनस्य च ।
> मिथुनेऽष्टादशे भागे कन्यायां च चतुर्दशे ॥५॥

Translation—(4) From the tulā solstice, sixty days of the sextile face occur in sequence. There are four and they are in the dual signs. (5) Up to the 26th degree of Sagittarius, 22nd degree of Pisces, 18th degree of Gemini, and 14th degree of Virgo.

Scientific Commentary—The meaning of day in these verses is not Sawan day, but the time when the sun moves by one degree. Such a year of 360 days is more than 365 days and 6 hours according to Sawan. But the motion of the sun is not always the same, so the values of the four sextile faces are also not the same in Sawan days. Starting from the Libra sign, and adding thirty-thirty degrees of Libra and Scorpio signs and 26 degrees of Sagittarius, they become 86 degrees, so it ends at the 26 degrees of Sagittarius. The second sextile face starts from the 27th degree of Sagittarius and ends at the 22nd degree of Pisces. Similarly, it ends at the 18th degree of Gemini and the 14th degree of Virgo. The four signs in which the sextile faces end are those of the dual signs, which have been discussed in predictive astrology.

In some texts, the place of Timina is written as Nimisha, which seems incorrect because Nimisha does not mean the Pisces sign.

Pitru Paksha

> ततश्शेषे तु कन्याया यान्यहानि तु षोडश ।
> क्रतुभिस्तानितुल्यानि पितृणांदत्तमक्षयम् ॥६॥

SOURCE PAGE 821

Translation—After this, the remaining 16 days of the Virgo zodiac are like the Yagya period. Performing the rituals of ancestors in this yields eternal fruit. -

Scientific Commentary—It appears that the rituals of ancestors should be performed when the sun is between 15 to 30 degrees in the Virgo zodiac. Nowadays, based on the full moon calculation, the Pitru Paksha is considered in the Ashvin Krishna Paksha, and based on the lunar calculation, in the Bhadra Krishna Paksha, i.e., according to the lunar months.

Names of the Nodes—

> भचक्रनाभौ विषुवद्वितीयं समसूत्रगम् ।
> अयनद्वितयं चैव चतस्रः प्रथितास्तु ताः ॥७॥
> तदन्तरेषु सङ्क्रान्तिद्वितयं द्वितयं पुनः ।
> नैरन्तर्यात्त सङ्क्रान्त्यो ज्ञेयं विष्णुपदीद्वयम् ॥८॥

Translation—(7) There are two equinoctial nodes and two nodal nodes on the same diameter in the middle of the ecliptic. (8) There are two nodes between them, and among these, the nodes that come after these four are called Vishnu-padi.

Scientific Commentary—From the fourth verse to the eighth verse, the names of 12 nodes have been mentioned. A node occurs at the time the sun enters a zodiac. There are twelve zodiacs, of which four are called the sextile-faced (shadshiti-timuth). The remaining two are called equinoctial, two are called nodal, and four are called Vishnu-padi.

Names of Zodiacal Nodes: 1 Aries equinoctial 2 Taurus Vishnu-padi 3 Gemini sextile-faced 4 Cancer nodal 5 Leo Vishnu-padi 6 Virgo sextile-faced 7 Libra equinoctial 8 Scorpio Vishnu-padi 9 Sagittarius

sextile-faced 10 Capricorn nodal

SOURCE PAGE 822

Chapter 767 is assumed

11 Kumbh Vishnu-padi 12 Meen Shadshitimukha "Spring"

Uttarayana and Dakshinayana and seasons—

> तयोर्मकरसङ्क्रान्तेः षण्मासेषूत्तरायणम् ।
> कर्क्यादिस्तु तथैव स्यात् षण्मासा दक्षिणायनम् ॥६॥
> द्विराशिमानाद्दतवः षडुक्ताशिशिरादयः ।
> मेषादयो द्वादशैते मासास्तैरेव वत्सरः ॥१०॥

Translation—(6) The time from which the sun enters the Capricorn sign for 6 months is Uttarayana, and the time from which it enters the Cancer sign for 6 months is Dakshinayana. (10) The seasons experience two signs; the cycle of seasons like Shishir begins with the Capricorn solstice; the beginning of 12 solar months begins with the Aries solstice, which is one year.

Scientific Commentary—These verses show the mutual relationship between the signs, solstices, and seasons. The signs are considered fixed, and their beginning is from the beginning point of Ashwini according to Sūrya Siddhānta, according to which the portion of the celestial star is 180. However, the order of the seasons follows according to the equinoxes, which is moving, so the relationship between the signs, ayanas, and seasons is gradually slipping. There was a time when the beginning of Uttarayana occurred at the same time when the sun's motion also began in the north and it moved towards the north for 6 months. Similarly, the beginning of Dakshinayana occurred at the time when the sun's motion was towards the south in the Cancer sign. But now these two events do not happen together; the sun's northward motion begins 23 days before the Capricorn solstice. In five hundred years, this difference will be about one month. Special discussion has been made on this subject in Triprashnādhikāra. It is an established view of Sūrya Siddhānta that the equinox remains at the 27 degrees of Ashwini, there is no more difference, but this is neither proven by modern science nor accepted by Bhaskaracharya etc. The positions of the Nakshatras mentioned in Brahmanic texts, such as Krittika, prove that the view of Sūrya Siddhānta is not correct (Triprashnādhikāra page 240).

Some scholars also have the opinion, which is well supported on the basis of Brahmanic texts, that the commencement of Uttarayana is not considered from that time.

SOURCE PAGE 823

When the tendency of the sun is towards the north, it was considered that time when the sun comes into the north in the North latitude from the equator. This also resolves the days and nights of the gods well because the gods are considered to reside in the North Pole and the beginning of the day or sunrise on the North Pole occurs at the same time when the sun starts moving north from the equator. Therefore, the Uttaraayana is considered the day of the gods and Dakshinayana their night. This is also logical. If Bhaskaraacharya ji were to consider this, he would not have to make this imagination regarding Uttaraayana.1

Auspicious Period of Sankranti

> अर्कमानकलाषष्ट्या गुणिता भुक्तिभाजिताः ।
> तदर्धनाड्यस्सङ्क्रास्तेरर्वाक्पुण्यास्तथापराः ॥११॥

Translation - The auspicious period of Sankranti is half an hour before and after the Sankranti when the result of dividing the art of the image of the sun by the art of the value of the sun is multiplied by sixty of its daily motion.

Scientific Commentary—Sankranti occurs at the time when the center of the sun's image enters the zodiac, but the value of the sun's image is approximately 32 arts, so the auspicious period of Sankranti begins at the time when the eastern edge of the sun's image touches the zodiac while entering and remains until the western edge of the image does not cross the beginning point of the zodiac. This time is approximately 32 hours, which is half 16 hours. Therefore, the auspicious period begins approximately 16 hours before Sankranti and lasts until 16 hours after. For fine calculation, one should use the ratio shown in the shlokas. If the daily motion of the sun during the Sankranti period is 60 hours, then how many hours will be the motion of the sun's image? That is

Sūrya Siddhānta X 60-year auspicious period = — — — — — — / Sun's daily motion

> १. दिनं सुराणामयनं यदुत्तरं निशेतत् साहितिकैः प्रकीर्तितम् ।
> दिनोन्मुखेऽर्के दिनमेव तन्मतं निशा तथा तत् फल कीर्तनाय तत् ॥११॥

(Goladhyāya Three Problems)

SOURCE PAGE 824

Chapter 766

The beginning of the virtuous period is known by subtracting half of the result from this, and the time of its end is found by adding it. Chandra and Pitriman

> अर्काद्विनिस्सृतः प्राचीं यद्यात्यहरहश्शशी ।
> तच्चान्द्रमानमंशस्तु ज्ञेया द्वादशभिस्तिथिः ॥१२॥
> तिथिः करणमुद्वाहक्षौरकर्मादिसत्क्रियाः ।
> व्रतोपवासयात्राणां क्रिया चान्द्रेण गृह्यते ॥१३॥
> त्रिंशता तिथिभिर्मासश्चान्द्रः पितृमहस्स्मृतम् ।
> निशा च मासपक्षान्ते तयोर्मध्ये विभागतः ॥१४॥

Translation—(12) The moon, separating from the sun, which moves towards the east day by day, is the lunar month. There is a tithi of 12 degrees of this difference. (13) All actions and vows, such as tithi, karana, marriage, kshaura karma, tonsure, etc., and fasts, pilgrimages, etc., are determined by the lunar month. (14) There is a lunar month of thirty tithis which is considered an all-night for the ancestors. At the end of the lunar month, that is at the end of Amavasya, the midday of the ancestors and the midnight of the ancestors occur at the end of the fortnight, i.e., Purnima. In this way, the beginning of the ancestors' night is from half of the Ashtami of the Shukla Paksha, and the beginning of their day is from half of the Ashtami of the Krishna Paksha.

Scientific Commentary—special discussion has been made in Chapter on Mean Motions on page 8 and Chapter on True Positions on page 217. Also, much has been written about Karana in page 218. See page 767-68 of the Geography chapter regarding the midday and midnight of the ancestors.

Names of months according to Nakshatras and Nakshatras—

> भचक्रभ्रमणं नित्यं नाक्षत्रं दिनमुच्यते ।

> नाक्षत्रनाम्ना मासास्तु ज्ञेयाः पर्वान्तियोगतः ॥१५॥
> कार्तिकादिषु संयोगे कृत्तिकादि द्वयं द्वयम् ।

> अन्त्योपान्त्यौ पञ्चमश्च त्रिधा मासास्त्रयस्स्मृताः ॥१६॥

Translation—(15) The time taken for one revolution of the constellation cycle is called a Nakshatra day. The names of the months are based on the constellation in which the moon is at the end of Purnima. (16) The conjunction of months like Kartik etc. with Krittika etc.

SOURCE PAGE 825

Mass pūrṇimā of nakṣatra krama- saṃkhyā sahita pūrṇimāntakāla in nakṣaatron of vāstavika sthiti 1965 vikramī 1661 1662 1663 1964 caita 14—citrā 15—svātī 16—viśākhā hasta + (viśākhā anurādhā) citrā viśākhā citrā viśākhā citrā viśākhā jyeṣṭha 17—anurādhā 18—jyeṣṭhā 19—mūla mūla jyeṣṭhā mūla jyeṣṭhā āṣāḍha 20—pūrvāṣāḍhā uttarāṣāḍhā pūrvāṣāḍhā uttarāṣāḍhā pūrvāṣāḍhā śrāvaṇa 21—uttarāṣāḍhā dhaniṣṭhā śravaṇa dhaniṣṭhā dhaniṣṭhā bhādrapada 22—śravaṇa 23—dhaniṣṭhā 24—śatabhiṣā 25—pūrvābhādrapada uttarābhādrapada (śatabhiṣā u0 bhādrapada) u0 bhādrapada pūrvābhādrapada

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SOURCE PAGE 826

Chapter 801

The actual position of the constellations in the lunar time including the sequence number: 1661 1662 1663 1664 1965 Vikram Samvat Ashwin 26—Uttarabhadrapada Ashwini Revati Bharani Ashwini Revati Kartik 27—Revati Ashwini Revati Bharani + Krittika Bharani + Margashirsha 28—Bharani Krittika Bharani + Rohini Krittika Rohini + Paush 29—Mrigashira Mrigashira Mrigashira Ardra Mrigashira Rohini + Magha 30—Punarvasu Pushya Punarvasu Punya Punarvasu Punarvasu Phalguna 31—Ashlekh Magha Ashlekh Po. Phalguni Hasta U. Phalguni 12—Uttara Phalguni 32—Purva Phalguni Uttara Phalguni Hasta U. Phalguni Purva Phalguni 13—Hasta Purva Phalguni

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SOURCE PAGE 827

The conjunction of two with two happens with the constellations, only the last month and the month just before it and the five months happen with three three constellations.

Scientific Commentary—is explained on pages 7, 300-301, and 336 of the Nakshatra Day. The names of the lunar months are kept on the names of the constellations on which the moon remains during the full moon. The alliance made by this principle of tithi, month, and constellations is unparalleled in the history of world astrology. This proves well that the Hindu astrologers in ancient times were very talented and the accusation of copying the astrology of other countries by them is very shameless and biased. Now, by sūkṣma gaṇanā (subtle calculation), it is certainly proven that this mutual relationship between constellations and months is sometimes missed, but here it must also be considered that the

fact that the rule which has been running for more than three thousand years has become loose somewhere is not a wonder and is proof of the ignorance of the rule makers. It can be easily seen from the table given on pages 557-58 how much difference has occurred at this time.

The constellations in this table are written from the pure principles-panchika of Bangla of 1664-65 VI, which is made based on modern astrology, in which the year is 365 days 6 hours 6 minutes 6-504 seconds, and the passage of the star Chitta is considered to be exactly 180 degrees. The constellations of the remaining three years are taken from the ordinary Panchangs of Lucknow and Kashi. Those constellations on which the sign of wealth is marked have become somewhat different according to the above rule. Where two constellations are given together, they are indicators of the major divisions. It is revealed that this rule is still working well.

Names of the Varhaspati year—

> वैशाखादिषु कृष्णेः च योगः पञ्चदशे तिथौ ।
> कार्तिकादीनि वर्षेषु गुरोर्युक्तोदयास्तभात् ॥१७॥

(just as the names of the Nakshatras of the full moon of the moon are named after the names of the lunar months) by the combination of the setting and rising of Jupiter on the fifteenth day of the dark fortnight of the months of Vaishakha, etc., names of the years of Kartik, etc., were given.

SOURCE PAGE 828

Chapter 803

Science commentary—at the time when Jupiter comes very close to the Sun, because of the sunlight, it cannot be seen, so it is considered set. Then when it moves so far away from the Sun that it becomes visible, it is considered risen. (See Udaya-stadhikar p. 656-57). This event happens approximately when there is conjunction of the Sun and Jupiter, which used to occur at an interval of about 366 days or 13 months. This period is called 'Varhaspati year'. The names of such years are kept according to the constellations on which the Sun and the Moon are both present at the time of Jupiter's rising or setting. It is stated in the 16th verse that the names of the lunar months are based on the names of those constellations on which the Moon remains in the full moon period, therefore it is proven that the Sun is in the 14th constellation of those lunar months' full moon constellations. For example, in the month of Vaishakh, the full moon falls on the Vishakha or Anuradha constellations, so in this month, the Sun will remain in the 14th constellation of Vishakha or Anuradha, which is Krittika or Rohini. If Jupiter rises or sets at this time, then it is certain that it will also remain on these constellations or one or two constellations ahead or behind, and the new moon will also be on these constellations, therefore Jupiter's 'Kartika year' will begin from this time. That is, if Jupiter rises or sets in the month of Vaishakh, then Jupiter's 'Kartika year' will be applicable, if it rises in the month of Jyeshtha, then 'Varhaspati marg shirsh' year will be applicable, etc. To remove the dilemma between the lunar months and the Varhaspati years, this difference is also made that a 'Maha' is added before the names of the Varhaspati years. However, these Kartik etc. years are not promoted nowadays.

Upon careful observation, it will be apparent that this rule of Sūrya Siddhānta is very loose. The difference between the setting time and the rising time of Jupiter is approximately one month, during which the sun moves more than two constellations. It is possible that at the time of setting, the sun is in the constellation Swati, and at the time of rising, it is in Anuradha. In such a situation, which Varahaspati year should be considered, 'Maha Chaitra' or 'Maha Vaishakh'? Perhaps to resolve this

dilemma, Acharya Varahamihira has given this rule in the Brihat Samhita that the name of Jupiter's year should be the name of the constellation in which Jupiter is at the time of rising. Varahamihira has discussed the different results of these years.

> १. नक्षत्रेण सहोदयमुपगच्छति येन देवपति मन्त्री ।
> तत्संज्ञं वक्तव्यं वर्षं मास क्रमेणैव ।।१।।
> वर्षाणि कार्तिकादीन्याग्नेयाद्भद्वयानुयोगीनि ।
> क्रमशस्त्रिभंतु पञ्चममुपांत्यमंत्य च यद्वर्षम् ॥२॥(गुरुचाराध्याय)

SOURCE PAGE 829

The year of Jupiter also has another type called Samvatsara (Chapter on Mean Motions verse 55 and its commentary). These Samvatsaras are discussed in the Panchangs. They are also useful in the mantras of Sankalpa (resolve). There is a cycle of these 60 Samvatsaras. Besides these, there is a cycle of 5 Samvatsaras whose names in order are—(1) Samvatsara, (2) Parivatsara, (3) Idavatsara, (4) Anuvatsara, (5) Idvatsara. They are discussed in Vedanga Jyotisha and Vṛhatsaṃhitā where their results are also stated.

Sawanman-

> उदयादुदयं भानोः सावनं तत्प्रकीर्त्यते ।
> सावनानि स्युरेतानि यज्ञकालविधिस्तु तैः ॥१८॥
> सूतकादि परिच्छेदो दिनमासाब्दपास्तथा ।
> मध्यमाग्रह भुक्तिश्च सावनेन प्रकीर्त्यते ॥१९॥

(18) The time between one sunrise and the next sunrise of the sun is called Sawan days. The regulation for performing yajnas is made from the Sawan days. (16) The limit, day, month, and year masters of birth and the vows of Sutak and Chandranyana, the calculation of the mean motion of the planets is done from the Sawan days.

Scientific Commentary — this subject has been well considered in the Chapter on Mean Motions. The periods for which certain actions are to be performed in the Sawan days are counted here. The house where a child is born, or the house where someone dies, is considered impure for ten, twelve, or fifteen days. This is the sutak whose duration is determined according to the Sawan days.

Divyamāna (daylight measure).

> सुरासुराणामन्योन्यमहोरात्रं विपर्ययात् ।
> यत्प्रोक्तं तद्भवेद्विद्वयं भानोर्भगणपूरणात् ॥२०॥

That which is stated in contradiction by the gods and the asuras is divine. And there is a time for the completion of one revolution of the sun.

Scientific Commentary—Its discussion has been made in detail in the commentary of the Geography chapter verses 47-50.

SOURCE PAGE 830

Chapter 805

Prajapatya and Brahmamāna

> मन्वन्तरव्यवस्था च प्राजापत्यमुदाहृतम् ।
> न तत्र द्युनिशोश्छेदः ब्राह्मं कल्पं प्रकीर्तितम् ॥२१॥

Translation- The system of Manvantara is an example of Prajapatya, where there is no difference between day and night. Kalpa is called Brahmamāna.

Scientific Commentary—The explanation of Kalpa, Manvantara, etc., has been done in the verses 18-20 of Chapter on Mean Motions and in its scientific commentary.

Glory

> एतत् ते सर्वमाख्यातं रहस्यं परमाद्भुतेम् ।
> ब्रह्मतत्परमं पुण्यं सर्वपापप्रणाशनम् ॥२२॥
> दिव्यं चाक्षं ग्रहाणाञ्च दर्शित ज्ञानमुत्तमम् ।
> विज्ञायार्कादि लोकस्य स्थानं प्राप्नोति तादृशम् ॥२३॥

Translation—(22) From you this second section is told which is mysterious and very wonderful, it is the form of Brahman, excellent, pure, and destroyer of all sins. (23) Excellent knowledge of the celestial, stellar, and planets has been shown, by which a person attains the same place in the worlds of Sun, etc.

Scientific Commentary—the first half of Sūrya Siddhānta had ended with the right of succession, which was concluded in its final verse. After that, the latter half of the questions of Mayasura was begun in the beginning of the chapter on geography, which ends here, so the conclusion of this section is also given here.

Conclusion

> इत्युक्त्वा मयमामन्त्र्य सम्यक्तेनापि पूजितः ।
> दिवमुद्गत्य सूर्यशांः प्रविवेश स्वमण्डलम् ॥२४॥
> मयोऽथ दिव्यं तज्ज्ञानं ज्ञात्वा साक्षाद्विवस्वतः ।
> कृतकृत्यमिवात्मानं मेने निर्धू तकल्मषम् ॥२५॥
> ज्ञात्वा तमृषयश्चाथ सूर्यलन्धवरं मयम् ।
> परिवब्रू रुपेत्याथो ज्ञानं पप्रच्छुरावरात् ॥२६॥

SOURCE PAGE 831

Of the planets given to you, the course is great. The secret of the world is joined with Brahman.||27||

(24) Having said this, Suryaansh departed from Surya Purusha Maya and, being well worshipped by him, went to heaven and entered his circle. (25) Then Mayasur, having obtained this divine knowledge directly from the Lord, considered himself free of sin and grateful. (26) Then the sages, knowing that Maya had received this boon from the Sun God, came to him and asked with reverence while surrounding him. (27) On this, he told them about the great character of the planets, which is like the secret and knowledge of Brahman, and which causes great wonder in this world.

Scientific Commentary—From this it is clear that this astrological text named Sūrya Siddhānta was received by the sages through Maya, who obtained this knowledge from the Sun God or the Suryaansh Purusha through penance. Some scholars draw the conclusion that this text was received from the Yavanas astrologers. In this regard, some thoughts will be written in detail in the introduction without further discussion here.

Thus the Sūrya Siddhānta, Chapter on Scientific Commentary, Chapter on Mean Motions, Chapter on True Positions, Chapter on the Three Problems

In some texts, a short chapter of 21 shlokas named Vijopanayanadhyāya is found, on which commentators have not commented that the need for Vija-samskara might not have been necessary during the time of the text's composition, so this chapter has been added from the back. In my short opinion, it is possible that what has been written so far has been combined from Suryansh Purusha to Mayasura in this form, but after that Mayasura must have lived for many years and found some difference from his experience, according to which he finally added Vijopanayanadhyāya. Or perhaps some other astrologer after Mayasura extended it. An effort will be made to shed light on this in the introduction. At this time, I also dare to add this chapter.

> यथा शिखा मयूराणां नागानां मणयो यथा
> तद्वद्वेदाङ्ग शास्त्राणां गणितं मूर्धनि स्थितम् ॥१॥
> न देयं तत् कृतघ्नाय वेदविप्लवकाय च ।
> अर्थलुब्धाय मूर्खाय साहङ्काराय पापिने ॥२॥
> एवं विधाय पुत्लायाप्यदेयं सहजाय च ।
> दत्तेन वेद मार्गस्य समुच्छेदः कृतो भवेत् ॥३॥

Chapter 807

ब्रजेतामन्धतामिस्रं गुरु शिष्यौ सुदारुणम् ।
ततः शान्ताय शुचये ब्राह्मणायैव दापयेत् ॥४॥
चक्रानुपातजो मध्यो मध्यवृत्तांशजः स्फुटः ।
कालेन दृक्समो न स्यात् ततो बीजक्रियोच्यते ।।५।।
राश्यादिरिन्दुरङ्कघ्नो भक्तो नक्षत्रकक्षया ।
शेषं नक्षत्रकक्षायास्त्यजेच्छेषकयोस्तयोः ॥६॥
यदल्पं तद्भजेद्वानां कक्षया तिथिनिघ्नया ।
बीजं भागादिकं तत् स्यात् कारयेत् तद्धनं रवौ ॥७॥
त्रिगुणं शोधयेदिन्दौ जिनघ्नं भूमिजे क्षिपेत् ।
दृग्यमघ्नऋणं ज्ञोच्चे खरामघ्नं गुरावृणम् ।।८।।
ऋणं व्योमनवघ्नं स्याद्दानवेज्यचलोच्चके ।
धनं सप्ताहतं मन्दे परिधीनामथोच्यते ॥६॥
युग्मान्तोक्ताः परिधयो ये ते नित्यं परिस्फुटाः ।
ओजान्तोक्तास्तु ते ज्ञेयाः परबीजेन संस्कृताः ॥१०॥
वच्मि निर्वीजकानोजपदान्ते वृत्त भागकान् ।
सूर्येन्द्वोर्मनवो दन्ता धृतितत्वकलोनिताः ॥११॥
बाणतर्का महीजस्य सौम्यस्याचल बाहवः ।
वाक्पतेरष्टनेत्राणि व्योमशीतांशवो भृगोः ॥१२॥
शून्यर्त्तवोऽर्कं पुत्रस्य बीजमेतेषु कारयेत् ।
बीजं खाग्न्युद्धृतं शोध्यं परिध्यंशेषु भास्वतः ॥१३॥
इनाप्तं योजयेदिन्दोः कुजस्याश्वहृतं क्षिपेत् ।
विदश्वन्द्रहृतं योज्यं सूरेरिन्द्रहतं धनम् ॥१४॥
धनं भृगोर्भुवा निघ्नं रविघ्नं शोधयेच्छने ।
एवं मान्दाः परिध्यंशाः स्फुटाः स्युर्वच्मि शीघ्रकान् ।।१५।।
भौमस्याभ्रगुणाक्षीणि बुधस्याब्धि गुणेन्दवः ।
बाणाक्षा देव पूज्यस्य भार्गवस्येन्दु षड्यमाः ॥१६॥
शनिश्चन्द्राब्धयः शीघ्रा ओजान्ते वीजवर्जिताः ।
द्विघ्नं स्वं कुज भागेषु बीजं द्विघ्नमृणं विदः ॥१७॥
अत्यष्टिघ्नं धनं सूरेरिन्दुघ्नं शोधयेत्कवेः ।
चन्द्रघनमृणमार्कस्य स्यूरेभिर्दृक्समा ग्रहाः ॥१८॥
'एतद्बोजं मया ख्यातं प्रीत्या परमया तव ।
गोपनीयमिदं नित्यं नोपदेश्यं यतस्ततः ॥१९॥

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परीक्षिताय शिष्याय गुरुभक्ताय साधवे ।
देयं विप्राय नान्यस्मै प्रतिकञ्चुककारिणे ॥२०॥
बीजं निःशेष सिद्धान्त रहस्यं परमं स्फुटम् ।
यात्रापाणिग्रहादीनां कार्याणां शुभसिद्धिदम् ॥२१॥

Scientific Commentary—these are found after the 23rd verse and the first 4 verses of the last chapter of the mānādhyāya and are at least 400 years old because Ranganath ji has mentioned it in his gūḍhārtha prakāśikā ṭīkā written in 1525. This proves that the need to perform seed correction in Sūrya Siddhānta also existed, and our old Acharyas were in favor of making the astrology flawless and comprehensive, not like some pundits nowadays, not the fakirs of the line.

No commentator from here has commented on these verses, so I also consider it useless to translate it because these old seeds are no longer useful. Now it is necessary to fix all calculations with new observations.

Thus, the Scientific Commentary of the fourteenth and last chapter named mānādhyāya of Sūrya Siddhānta has ended.

SOURCE PAGE 834

Appendix

Numerical terms used in Sūrya Siddhānta

0—Ambar, Kha, Viyat, Vyom, Shunya Vasu, Serp 1—Indu, Ku, Nishakar, Rupa 6—Angka, Go, Chhidra, Nav, Randhra 2—Akshi, Ashvi, Ashwin, Dastra, 10—Dik, Ding Dastrak, Dvi, Netra, Yam, Yamal, Lochan 11—Ish, Ishwar, Rudra, Shankar 3—Agni, Guna, Jvalanaḥ, Tri, Trik, 12—Arka, Maas, Surya Pavak, Vahni, Shikhi, Hutash 13—Vishwa 4—Abadhi, Arnav, Krut, Chatuh, Chatushk, 14—Manu Yuga, Ved, Samudra, Sagar 15—Tithi 5—Artha, Ishu, Panch, Bana, Margana 18—Dhriti Vishaya, Shara 16—Atidhriti 6—Anga, Ritu, Rasa, Shat, Shad 20—Kriti, Nakha 7—Ag, Adri, Naga, Parvat, Bhudhar, 25—Tatva Bhumidhar, Muni 32—Rada 8—Ashta, Kunjar, Gaj, Naga, Bhujanga, 33 —Sur

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List of Texts

(List of those texts, almanacs, or leaves in Sanskrit, Hindi, or Urdu, which have been taken in the Scientific Commentary.)

The Sanskrit commentary by Aryabhatiya Parameshvaracharya and the Hindi Toka by Uday Narayan Singh, printed by Brahma Press Etawah in the year 1663.

Khandkhadya Marathi based on Indian Astrology.

Ganak Tarangini written by M.M. Sudhakar Dwivedi and printed by Medical Hall Press Banaras in the year 1646.

Hindi translation of the Marathi book Gita Rahasya by Lokmanya Bal Gangadhar Tilak by Madhavrao Sapre, which was printed in the year 1673 in the Chitrasala Steam Press in Pune.

Grahalaghava Mallari, with the Sanskrit commentary of Vishwanath and Sudhakar Dwivedi and edited by M.M. Sudhakar Dwivedi.

Authored by M.M. Sudhakar Dwivedi in Chalanakala. Printed in 1812 by Jyotirganit Benchatesh Bapuji Ketkar. Based on the Tittiriya Samhita on the basis of Indian Jyotishastra.

Published in 1826 by Dharma-Sindhu Nirnay Sagar Press. Based on the Nakshatra Kalpa of Indian Jyotishastra. Narada Samhita. Published by Nirnay Sindhu Nirnay Sagar Press.

Published in 1886 AD by Medical, Hal Press Banaras, with the English and Sanskrit translations by Varahmihir and Dr. Thivo and M.M. Sudhakar Dwivedi. Panchasiddantika by Varahamihira and Doctor Thivo and M.M. Sudhakar Dwivedi with English, Sanskrit translations. Panchang (Ketaki of 1685, from Chitrashala Press, Gujarati, from Naval Kishore Press, from Panchang Pravartak Committee, from Balakrishna Shastri, from Balakrishna Tukaram, World Panchang of Hindu Vishwa-vidyalaya, Shastra Shuddha Aikya Vardhak and

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List of books 811

Ganeshdatta Sharma of 1683 B.E., of Naval Kishore Press, Bharat Bhushan, Vijay and Vishwa Panchang; Navik Panchang of 1622 and 1628 AD

Published in Naval Kishore Press with the Hindi commentary of M.M. Durgaprasad Dwivedi of Barahi Samhita (Brihatsamhita).

Published in Medical Hall Press, Banaras in 1602 AD with the Sanskrit commentary of M.M. Sudhakar Dwivedi and edited by them.

Bhagavt Gita Bhagavt Gita Rahasya (Dekho Gita Rahasya) written in Marathi by Shankar Balakrishna Dikshit of Indian Jyotishastra and published in 1818 by Ayan Bhushan Press of Pune.

Manusmriti by Janardan Jha with the language commentary, published in 1681 by the Hindi Tak agency of Calcutta.

Mahasiddanta II written by Aryabhata, printed by Chandraprabha Press, Banaras in 1610 AD with the commentary of M.M. Sudhakar Dwivedi.

Madhuri Khand 2 number 4 monthly magazine published from Naval Kishore Press. Monthly magazine published from Kashi Gyan Mandal of 1678 and 1676.

Muhurta Chintamani with the commentary of Rameshwar Bhatt, volume 1674, printed by Nirnay Sagar Press.

Muhurta Tattva based on the basis of Indian Jyotish Shastra. Ratnamal by Shripati, Ratnakosh by Lalla, Roshni Urdu ka Vigyanik Masik par of April 1616 AD. Laghu Manas based on Indian Jyotish Shastra. Lalla Tantra based on Indian Jyotish Shastra. Lilavati Vigyan Masik Patra Vishuddha Siddhanta Panjika Bangla.

SOURCE PAGE 837

The commentary of Vedanga Jyotish by Somakar and with the commentary of M.M. Sudhakar Dwivedi and edited by them.

Based on the vṛhattithi ciṃtāmaṇi gaṇaka taraṃgiṇī. Based on the śatapatha brāhmaṇa. Based on the basis of vṛhattithi ciṃtāmaṇi gaṇaka taraṃgiṇī. Based on the śatapatha brāhmaṇa.

Published in 1624 AD from Banaras by the company of Shakaly Brahma Siddhanta, including the comments of M.M. Sudhakar Dwivedi and M.M. Muralidhar Sharma.

Written by Siddhanta Darpan by Chandrashekhar Singh Samant and edited by Pro. Yogeshchandra Rai and published in 1866 AD.

Siddanta Shiromani(1) printed in 1915 AD of the Vachaspati Yantra of Calcutta and the printed in 1866 AD of the Narayan Yantra of Goladhyaya, (2) with the comments of M.M. Bapudev Shastri, printed in 1866 AD by the Medical Hall Press of Banaras, (3) edited with the Prabhā Bhashya by P.G. Prasad Dwivedi and printed in 1911 AD in Naval Kishore Press.

Based on the Siddhanta Shekhar Sudha Varshini commentary.

Based on Sundari Siddhanta of the science.

Sūrya Siddhānta (1) translated and edited by Indra Narayan Dwivedi and published in 1975 by Hindi Sahitya Sammelan Prayag.

And compiled by Hindi Sahitya Sammelan (2) with the commentary published with the hidden meaning and translated and edited into Hindi by Baldev Prasad Mishra and printed in 1866 AD in Benchteshwar Press. (3) Edited by Girija Prasad Dwivedi with the commentary of Madhav Purohit and printed in 1604 AD in Naval Kishor Press. (4) Translated and edited into Bengali by Vignanananda Swami and printed in 1606 AD by Bharat Mihir Yantralaya. (5) Printed for the second time in 1625 AD by the Asiatic Society of Bengal with the Sudhavarshini commentary of M.M. Sudhakar Dwivedi, but the author received it in 1634 AD. (6) Edited by Doctor Thivo and Sudhakar Dwivedi of Panch Siddhantika.

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List of English books used in the Scientific Commentary

Askwith's pure Geometry. Asutosh Mukhopadhyaya's Geometry of Conics. Ball's Spherical Astronomy. Berry's Short History of Astronomy. Brennand's History of Hindu Astronomy. Burgess Suryasiddhanta published by Calcutta University in 1935 Encyclopaedia Brittanica. Godfray's Treatise of Astronomy. Hall and knight's Trigonometry. Heath's Popular Astronomy. Heroes of Science(Astronomer). Herschel's Outlines of Astronomy. Imperial Gazetteer. Kaye's Hindu Astronomy. L.D. Swami kannu Pillai's Indian Chronology. Leader 25th Oct. 1939. Lockyer's Elementary Lessons on Astronomy. Loney's Elements of Coordinate Geometry. Loomi's Practical Astronomy. Nautical Almanac of 1922 and 1928. Parkers Elements of Astronomy. Popular Hindu Astronomy Part I(by Kalidas Mukherji). Scientific American July 1928. Todhunter and Leathern Spherical Trigonometry. Williamson Differential Calculus.